

IEEE P802.3dj™/D2.0

Draft Standard for Ethernet Amendment: Media Access Control Parameters for 1.6 Tb/s and Physical Layers and Management Parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Operation

Prepared by the

LAN/MAN Standards Committee
of the
IEEE Computer Society

This draft is an amendment of IEEE Std 802.3-2022 as amended by IEEE Std 802.3dd-2022, IEEE Std 802.3cs-2022, IEEE Std 802.3db-2022, IEEE Std 802.3ck-2022, IEEE Std 802.3de-2022, IEEE Std 802.3cx-2023, IEEE Std 802.3cz-2023, IEEE Std 802.3cy-2023, IEEE Std 802.3df-2024, and IEEE Std 802.3-2022/Cor 1-2024. This amendment includes Media Access Control parameters for 1.6 Tb/s and Physical Layers and management parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s operation. Draft D2.0 is prepared for Working Group ballot. This draft expires 6 months after the date of publication or when the next version is published, whichever comes first.

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PDF: ISBN 978-0-7381-xxxx-x STDxxxxx
Print: ISBN 978-0-7381-xxxx-x STDPDxxxxx

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Abstract: This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 174 through Clause 187. The amendment adds MAC parameters for 1.6 Tb/s, and Physical Layers and management parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s operation.

Keywords: 200GAUI-1, 200GBASE-CR1, 200GBASE-DR1, 200GBASE-DR1-2, 200GBASE-KR1, 400GAUI-2, 400GBASE-CR2, 400GBASE-DR2, 400GBASE-DR2-2, 400GBASE-KR2, 800GAUI-4, 800GBASE-CR4, 800GBASE-DR4, 800GBASE-DR4-2, 800GBASE-ER1, 800GBASE-ER1-20, 800GBASE-FR4, 800GBASE-FR4-500, 800GBASE-KR4, 800GBASE-LR1, 800GBASE-LR4, 1.6TAUI-16, 1.6TAUI-8, 1.6TBASE, 1.6TBASE-CR8, 1.6TBASE-DR8, 1.6TBASE-DR8-2, 1.6TBASE-KR8, 1.6TBASE-R, 1.6 Terabit Ethernet, 1.6TMII, 1.6TXS

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Introduction

This introduction is not part of IEEE Std 802.3dj-20xx, IEEE Draft Standard for Ethernet—Amendment: Media Access Control Parameters for 1.6 Tb/s and Physical Layers and Management Parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Operation.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3dj-20xx publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

IEEE Std 802.3ck™-2022

Amendment 4—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 161 through Clause 163, Annex 120F, Annex 120G, Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

IEEE Std 802.3de™-2022

Amendment 5—This amendment includes changes to IEEE Std 802.3-2022 to add 10 Mb/s Single-Pair Ethernet point-to-point PHYs to the PHYs supporting the MAC Merge function and the Time Synchronization Service Interface (TSSI).

IEEE Std 802.3cx™-2023

Amendment 6—This amendment to IEEE Std 802.3-2022 modifies Clause 30, Clause 45, and Clause 90, and adds Annex 90A to enhance support for time synchronization protocols by providing options for sub-nanosecond reporting of the transmit and receive path data delays, selection of the data delay measurement point, and dynamic reporting of path data delay variation.

IEEE Std 802.3cz™-2023

Amendment 7—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 166. This amendment adds 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, and 50 Gb/s Physical Layer specifications and management parameters for optical automotive Ethernet using graded-index glass optical fiber.

IEEE Std 802.3cy™-2023

Amendment 8—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 165 and Annex 165A. This amendment adds Physical Layer specifications and management parameters for operation at 25 Gb/s over a single balanced pair of conductors.

IEEE Std 802.3df™-2024

Amendment 9—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 169 through Clause 173, Annex 172A, and Annex 173A. This amendment includes Physical Layer specifications and management parameters for 400 Gb/s and 800 Gb/s operation.

IEEE Std 802.3-2022/Cor 1-2024

Corrigendum 1—This corrigendum includes changes to IEEE Std 802.3-2022 to correct the MDI return loss specifications in Clause 149 and Clause 165.

IEEE Std 802.3dj™-202x

This amendment includes changes to IEEE Std 802.3-2022, and adds Clause 174 through Clause 187 and <annexes>. This amendment includes Physical Layer specifications and management parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s operation.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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Draft Standard for Ethernet

Amendment:

Media Access Control Parameters for 1.6 Tb/s and Physical Layers and Management Parameters for 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Operation

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NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~striketrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in ***green***.

TO BE REMOVED PRIOR TO FINAL PUBLICATION: Reviewers and the publication editor should note that editing instructions have been written to minimize the probability of changes being lost at publication from other IEEE 802.3 amendment projects running in parallel that modified the same text and tables.

1. Introduction

1.1 Overview

1.1.3 Architectural perspectives

1.1.3.2 Compatibility interfaces

Change item t) (as modified by IEEE Std 802.3ck-2022) as follows:

- t) *200 Gb/s Attachment Unit Interface (200GAUI-n).* ~~The A 200GAUI-n is an optional physical instantiation of the PMA service interface to extend the connection between 200 Gb/s capable PMAs. While conformance with implementation of this interface is not necessary to ensure communication, it is recommended, since it allows maximum flexibility in intermixing PHYs and DTEs at 200 Gb/s speeds. The 200GAUI-n is intended for use as a chip-to-chip or a chip-to-module interface. The 200GAUI-n C2C is intended for use as a chip-to-chip interface and 200GAUI-n C2M as a chip-to-module interface. For chip-to-chip interfaces and for chip-to-module interfaces, three widths of 200GAUI-n are defined: an eight-lane version (200GAUI-8) in Annex 120B and Annex 120C, a four-lane version (200GAUI-4) in Annex 120D and Annex 120E, and a two-lane version (200GAUI-2) in Annex 120F and Annex 120G. Four widths of 200GAUI-n are defined: eight-lane 200GAUI-8 C2C in Annex 120B and 200GAUI-8 C2M in Annex 120C, four-lane 200GAUI-4 C2C in Annex 120D and 200GAUI-4 C2M in Annex 120E, two-lane 200GAUI-2 C2C in Annex 120F and 200GAUI-2 C2M in Annex 120G, and one-lane 200GAUI-1 C2C in Annex 176C and 200GAUI-1 C2M in Annex 176D. No mechanical connector is specified for use with the 200GAUI-n. The 200GAUI-n is optional.~~

Change item v) (as modified by IEEE Std 802.3ck-2022) as follows:

- v) *400 Gb/s Attachment Unit Interface (400GAUI-n).* ~~The A 400GAUI-n is an optional physical instantiation of the PMA service interface to extend the connection between 400 Gb/s capable PMAs. While conformance with implementation of this interface is not necessary to ensure communication, it is recommended, since it allows maximum flexibility in intermixing PHYs and DTEs at 400 Gb/s speeds. The 400GAUI-n is intended for use as a chip-to-chip or a chip-to-module interface. The 400GAUI-n C2C is intended for use as a chip-to-chip interface and 400GAUI-n C2M as a chip-to-module interface. For chip-to-chip interfaces and for chip-to-module interfaces, three widths of 400GAUI-n are defined: a sixteen-lane version (400GAUI-16) in Annex 120B and Annex 120C, an eight-lane version (400GAUI-8) in Annex 120D and Annex 120E, and a four-lane version (400GAUI-4) in Annex 120F and Annex 120G. Four widths of 400GAUI-n are defined: 16-lane 400GAUI-16 C2C in Annex 120B and 400GAUI-16 C2M in Annex 120C, 8-lane 400GAUI-8 C2C in Annex 120D and 400GAUI-8 C2M in Annex 120E, 4-lane 400GAUI-4 C2C in Annex 120F and 400GAUI-4 C2M in Annex 120G, and 2-lane 400GAUI-2 C2C in Annex 176C and 400GAUI-2 C2M in Annex 176D. No mechanical connector is specified for use with the 400GAUI-n. The 400GAUI-n is optional.~~

Change item x (inserted by IEEE Std 802.3df-2024) as follows:

- x) ***800 Gb/s Attachment Unit Interface (800GAUI-n).*** An 800GAUI-n is an optional physical instantiation of the PMA service interface to extend the connection between 800 Gb/s capable PMAs. 800GAUI-n C2C is intended for use as a chip-to-chip interface and 800GAUI-n C2M as a chip-to-module interface. One-Two widths of 800GAUI-n is/are defined: eight-lane 800GAUI-8 C2C in Annex 120F and 800GAUI-8 C2M in Annex 120G, and four-lane 800GAUI-4 C2C in Annex 176C and 800GAUI-4 C2M in Annex 176D. No mechanical connector is specified for use with an 800GAUI-n.

Insert the following compatibility interfaces after “x) 800 Gb/s Attachment Unit Interface ...” (inserted by IEEE Std 802.3df-2024):

- y) ***1.6 Tb/s Media Independent Interface (1.6TMII).*** The 1.6TMII is designed to connect a 1.6 Tb/s capable MAC to a 1.6 Tb/s PHY. The 1.6TMII is a logical interconnection intended for use as an intra-chip interface. No mechanical connector is specified for use with the 1.6TMII. The 1.6TMII is optional.
- z) ***1.6 Gb/s Attachment Unit Interface (1.6TAUI-n).*** A 1.6TAUI-n is an optional physical instantiation of the PMA service interface to extend the connection between 1.6 Tb/s capable PMAs. 1.6TAUI-n C2C is intended for use as a chip-to-chip interface and 1.6TAUI-n C2M as a chip-to-module interface. Two widths of 1.6TAUI-n are defined: 16-lane 1.6TAUI-16 C2C in Annex 120F and 1.6TAUI-16 C2M in Annex 120G, and 8-lane 1.6TAUI-8 C2C in Annex 176C and 1.6TAUI-8 C2M in Annex 176D. No mechanical connector is specified for use with a 1.6TAUI-n.

1.3 Normative references

Insert the following references in alphanumeric order:

IEC 61753-021-02, Fibre optic interconnecting devices and passive components – Performance standard – Part 021-02: Single-mode fibre optic connectors terminated as pigtails and patchcords for category C – Controlled environment.

IEC 61753-1, Fibre optic interconnecting devices and passive components - Performance standard - Part 1: General and guidance.

IEEE Std 1241TM-2010, IEEE Standard for Terminology and Test Methods for Analog-to-Digital Converters

OSFP Octal Small Form Factor Pluggable Module, Rev 5.1, September 12, 2024.⁶

QSFP-DD/QSFP-DD800/QSFP-DD1600 Hardware Specification for QSFP Double Density 8x Pluggable Transceivers, Rev 7.1, June 25, 2024.⁷

SFF-8665, Rev 1.9.4, April 1, 2022, QSFP+ 4X Pluggable Transceiver Solutions.

SFF-TA-1011 Rev 1.1, April 19, 2024, Cross Reference to Select SFF Connectors and Modules.

⁶ OSFP1600 specifications are available from OSFP MSA (<http://osfpmsa.org/>).

⁷ SFP-DD224, QSFP224, and QSFP-DD1600 specifications are available from QSFP-DD MSA (<http://www.qsfp-dd.com/>).

SFF-TA-1027, Rev 1.0, April 16, 2024, QSFP2 Connector, Cage, & Module Specification.

SFF-TA-1031, Rev 1.0, June 11, 2023, SFP2 Cage, Connector, & Module Specification.

1.4 Definitions

Insert the following new definitions between 1.4.91 and 1.4.92:

1.4.92a 1.6 Tb/s Attachment Unit Interface (1.6TAUI-n): A physical instantiation of the PMA service interface to extend the connection between 1.6 Tb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module electrical interfaces. For chip-to-module interfaces and for chip-to-chip interfaces, two widths of 1.6TAUI-n are defined: 16-lane 1.6TAUI-16 C2C and 1.6TAUI-16 C2M, and eight-lane 1.6TAUI-8 C2C and 1.6TAUI-8 C2M. (See IEEE Std 802.3, Annex 120F, Annex 120G, Annex 176C, and Annex 176D.)

1.4.92b 1.6 Tb/s Media Independent Interface (1.6TMII): The interface between the Reconciliation Sublayer (RS) and the Physical Coding Sublayer (PCS) for 1.6 Tb/s operation. (See IEEE Std 802.3, Clause 170.)

1.4.92c 1.6TMII Extender: The 1.6 Tb/s Media Independent Interface Extender extends the reach of the 1.6TMII and consists of two 1.6TXS sublayers with one or two 1.6TAUI-n between them. (See IEEE Std 802.3, Clause 171.)

1.4.92d 1.6TXS: The 1.6TMII Extender Sublayer (1.6TXS) is part of the 1.6TMII Extender. In functionality, it is almost identical to the 1.6TBASE-R PCS defined in Clause 175. Two types of 1.6TXS are defined: the DTE 1.6TXS adjacent to the Reconciliation Sublayer (RS) and the PHY 1.6TXS adjacent to the PHY. (See IEEE Std 802.3, Clause 171.)

1.4.92e 1.6TBASE-CR8: IEEE 802.3 Physical Layer specification for 1.6 Tb/s using 1.6TBASE-R encoding over eight lanes of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 179.)

1.4.92f 1.6TBASE-DR8: IEEE 802.3 Physical Layer specification for 1.6 Tb/s using 1.6TBASE-R encoding and 4-level pulse amplitude modulation over eight single-mode fibers in each direction, with reach up to at least 500 m. (See IEEE Std 802.3, Clause 180.)

1.4.92g 1.6TBASE-DR8-2: IEEE 802.3 Physical Layer specification for 1.6 Tb/s using 1.6TBASE-R encoding and 4-level pulse amplitude modulation over eight single-mode fibers in each direction, with reach up to at least 2 km. (See IEEE Std 802.3, Clause 181.)

1.4.92h 1.6TBASE-KR8: IEEE 802.3 Physical Layer specification for 1.6 Tb/s using 1.6TBASE-R encoding over eight lanes of an electrical backplane. (See IEEE Std 802.3, Clause 178.)

1.4.92i 1.6TBASE-R: An IEEE 802.3 family of Physical Layer devices using the physical coding sublayer defined in Clause 175 for 1.6 Tb/s operation. (See IEEE Std 802.3, Clause 174.)

Insert the following new definition between 1.4.24 and 1.4.24a (as inserted by IEEE Std 802.3ck-2022):

1.4.24aa 200GBASE-CR1: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding over one lane of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 179.)

Insert the following new definition between 1.4.102 and 1.4.103:

1.4.102a 200GBASE-DR1: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding and 4-level pulse amplitude modulation over one single-mode fiber in each direction, with reach up to at least 500 m. (See IEEE Std 802.3, Clause 180.)

Insert the following new definition between 1.4.104 and 1.4.105:

1.4.104a 200GBASE-DR1-2: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding and 4-level pulse amplitude modulation over one single-mode fiber in each direction, with reach up to at least 2 km. (See IEEE Std 802.3, Clause 181.)

Insert the following new definition between 1.4.105 and 1.4.106:

1.4.105a 200GBASE-KR1: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding over one lane of an electrical backplane. (See IEEE Std 802.3, Clause 178.)

Change 1.4.110 (as modified by IEEE Std 802.3ck-2022) as follows:

1.4.110 200 Gb/s Attachment Unit Interface (200GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 200 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module interconnections. For chip-to-module interconnections and for chip-to-chip interconnections, ~~three-four~~ widths of 200GAUI-n are defined: ~~an eight-lane version (200GAUI-8), a four-lane version (200GAUI-4), and a two-lane version (200GAUI-2)~~ eight-lane 200GAUI-8 C2C and 200GAUI-8 C2M, four-lane 200GAUI-4 C2C and 200GAUI-4 C2M, two-lane 200GAUI-2 C2C and 200GAUI-2 C2M, and one-lane 200GAUI-1 C2C and 200GAUI-1 C2M. (See IEEE Std 802.3, [Annex 120B](#) and [Annex 120C](#) for 200GAUI-8, [Annex 120D](#) and [Annex 120E](#) for 200GAUI-4, ~~or~~ [Annex 120F](#) and [Annex 120G](#) for 200GAUI-2, or Annex 176C and Annex 176D for 200GAUI-1.)

Insert the following new definition between 1.4.134 and 1.4.134a (as inserted by IEEE Std 802.3ck-2022):

1.4.134aa 400GBASE-CR2: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding over two lanes of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 179.)

Insert the following new definitions between 1.4.134a (as inserted by IEEE Std 802.3ck-2022) and 1.4.135:

1.4.134b 400GBASE-DR2: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding and 4-level pulse amplitude modulation over two single-mode fibers in each direction, with reach up to at least 500 m. (See IEEE Std 802.3, Clause 180.)

1.4.134c 400GBASE-DR2-2: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding and 4-level pulse amplitude modulation over two single-mode fibers in each direction, with reach up to at least 2 km. (See IEEE Std 802.3, Clause 181.)

Insert the following new definition between 1.4.138 and 1.4.138a (as inserted by IEEE Std 802.3ck-2022):

1.4.138aa 400GBASE-KR2: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding over two lanes of an electrical backplane. (See IEEE Std 802.3, Clause 178.)

Change 1.4.145 (as modified by IEEE Std 802.3ck-2022) as follows:

1.4.145 400 Gb/s Attachment Unit Interface (400GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 400 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module interconnections. For chip-to-module interconnections and for chip-to-chip interconnections, three-four widths of 400GAUI-n are defined: a sixteen-lane version (400GAUI-16), an eight-lane version (400GAUI-8), and a four-lane version (400GAUI-4) ~~16-lane 400GAUI-16 C2C and 400GAUI-16 C2M, 8-lane 400GAUI-8 C2C and 400GAUI-8 C2M, four-lane 400GAUI-4 C2C and 400GAUI-4 C2M, and 2-lane 400GAUI-2 C2C and 400GAUI-2 C2M.~~ (See IEEE Std 802.3, [Annex 120B](#) and [Annex 120C](#) for 400GAUI-16, Annex 120D and [Annex 120E](#) for 400GAUI-8, ~~or Annex 120F and Annex 120G for 400GAUI-4,~~ or Annex 176C and Annex 176D for 400GAUI-2.)

Insert the following new definition between 1.4.184 and 1.4.184a (as inserted by IEEE Std 802.3df-2024):

1.4.184aa 800GBASE-CR4: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding over four lanes of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 179.)

Insert the following new definition between 1.4.184a and 1.4.184b (as inserted by IEEE Std 802.3df-2024):

1.4.184ba 800GBASE-DR4: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding and 4-level pulse amplitude modulation over four single-mode fibers in each direction, with reach up to at least 500 m. (See IEEE Std 802.3, Clause 180.)

Insert the following new definition between 1.4.184b and 1.4.184c (as inserted by IEEE Std 802.3df-2024):

1.4.184ca 800GBASE-DR4-2: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding and 4-level pulse amplitude modulation over four single-mode fibers in each direction, with reach up to at least 2 km. (See IEEE Std 802.3, Clause 181.)

Insert the following new definitions between 1.4.184c and 1.4.184d (as inserted by IEEE Std 802.3df-2024):

1.4.184da 800GBASE-ER1: IEEE 802.3 Physical Layer specification for 800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection with reach up to at least 40 km. (See IEEE Std 802.3, Clause 186 and Clause 187).

1.4.184db 800GBASE-ER1-20: IEEE 802.3 Physical Layer specification for 800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection with reach up to at least 20 km. (See IEEE Std 802.3, Clause 186 and Clause 187).

1.4.184dc 800GBASE-FR4: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding and 4-level pulse amplitude modulation over four WDM lanes on single-mode fiber, with reach up to at least 2 km. (See IEEE Std 802.3, Clause 183.)

1.4.184dd 800GBASE-FR4-500: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding and 4-level pulse amplitude modulation over four WDM lanes on single-mode fiber, with reach up to at least 500 m. (See IEEE Std 802.3, Clause 183.)

1.4.184de 800GBASE-KR4: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding over two lanes of an electrical backplane. (See IEEE Std 802.3, Clause 178.)

Insert the following new definitions between 1.1.184d and 1.4.184e (as inserted by IEEE Std 802.3df-2024):

1.4.184ea 800GBASE-LR1: IEEE 802.3 Physical Layer specification for 800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection, over single-mode fiber, with reach up to at least 10 km. (See IEEE Std 802.3, Clause 185.)

1.4.184eb 800GBASE-LR4: IEEE 802.3 Physical Layer specification for 800 Gb/s using 800GBASE-R encoding and 4-level pulse amplitude modulation over four WDM lanes on single-mode fiber, with reach up to at least 10 km. (See IEEE Std 802.3, Clause 183.)

Change 1.4.184h (as inserted by IEEE Std 802.3df-2024) as follows:

1.4.184h 800 Gb/s Attachment Unit Interface (800GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 800 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module electrical interfaces. For chip-to-module interfaces and for chip-to-chip interfaces, one two widths of 800GAUI-n is are defined: eight-lane 800GAUI-8 C2C and 800GAUI-8 C2M, and four-lane 800GAUI-4 C2C and 800GAUI-4 C2M. (See IEEE Std 802.3, Annex 120F ~~and~~, Annex 120G, Annex 176C and Annex 176D.)

Change 1.4.205 as follows:

1.4.205 BASE-R: An IEEE 802.3 family of Physical Layer devices using the 64B/66B encoding defined in Clause 49, Clause 82, Clause 107, Clause 119, Clause 129, ~~or~~ Clause 172, or Clause 175 of IEEE Std 802.3.

Change 1.4.461 as follows:

1.4.461 PCS lane (PCSL): In 40GBASE-R, 50GBASE-R, 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~and 800GBASE-R, and 1.6TBASE-R,~~ the PCS distributes encoded data to multiple logical lanes that are called PCS lanes. One or more PCS lanes can be multiplexed and carried together on a physical lane at the PMA service interface. (See IEEE Std 802.3, [Clause 83](#), Clause 120, [Clause 135](#), ~~and Clause 173, and Clause 176.~~)

1.5 Abbreviations

Insert the following new abbreviations into the list, in alphanumeric order:

1.6TAUI-n	1.6 Tb/s Attachment Unit Interface over n lanes
1.6TMII	1.6 Tb/s Media Independent Interface
1.6TXS	1.6 Tb/s Extender Sublayer
BM-PMA	bit-multiplexing Physical Medium Attachment
DP-16QAM	dual polarization 16-state quadrature amplitude modulation
ILT	inter-sublayer link training
ILdd	differential-mode to differential-mode insertion loss
ISL	inter-sublayer link
MLSD	maximum likelihood sequence detection
RLcc	common-mode to common-mode return loss
RLcd	differential-mode to common-mode return loss
RLdc	common-mode to differential-mode return loss
RLdd	differential-mode to differential-mode return loss
SM-PMA	symbol-multiplexing Physical Medium Attachment

4. Media Access Control

4.4 Specific implementations

4.4.2 MAC parameters

Change Table 4–2 (as amended by IEEE Std 802.3df-2024) as follows:

Table 4–2—MAC parameters

Parameters	MAC data rate			
	Up to and including 100 Mb/s	1 Gb/s	2.5 Gb/s, 5 Gb/s, 25 Gb/s, 40 Gb/s, 50 Gb/s, 100 Gb/s, 200 Gb/s, 400 Gb/s, and 800 Gb/s, and 1.6 Tb/s	10 Gb/s
slotTime	512 bit times	4096 bit times	not applicable	not applicable
interPacketGap ^a	96 bits	96 bits	96 bits	96 bits
attemptLimit	16	16	not applicable	not applicable
backoffLimit	10	10	not applicable	not applicable
jamSize	32 bits	32 bits	not applicable	not applicable
maxBasicFrameSize	1518 octets	1518 octets	1518 octets	1518 octets
maxEnvelopeFrameSize	2000 octets	2000 octets	2000 octets	2000 octets
minFrameSize	512 bits (64 octets)	512 bits (64 octets)	512 bits (64 octets)	512 bits (64 octets)
burstLimit	not applicable	65 536 bits	not applicable	not applicable
ipgStretchRatio	not applicable	not applicable	not applicable	104 bits

^a References to interFrameGap or interFrameSpacing in other clauses (e.g., [Clause 13](#), [Clause 35](#), and [Clause 42](#)) shall be interpreted as interPacketGap.

Change Note 7 in 4.4.2 (as amended by IEEE Std 802.3df-2024) as follows:

NOTE 7—For 40 Gb/s, 50 Gb/s, 100 Gb/s, 200 Gb/s, 400 Gb/s, ~~and 800 Gb/s,~~ and 1.6 Tb/s operation, the received interpacket gap (the spacing between two packets, from the last bit of the FCS field of the first packet to the first bit of the Preamble of the second packet) can have a minimum value of 8 BT (bit times), as measured at the XLGMII, 50GMII, CGMII, 200GMII, 400GMII, ~~or 800GMII,~~ or 1.6TMII receive signals at the DTE due to clock tolerance and lane alignment requirements.

21. Introduction to 100 Mb/s baseband networks, type 100BASE-T

21.5 State diagrams

21.5.4 Operators

Change Table 21–1 as follows:

Table 21–1—State diagram operators

Character	Meaning
*	Boolean AND
+	Boolean OR
$\wedge$	Boolean XOR
!	Boolean NOT
<	Less than
$\leq$	Less than or equal to
=	Equals (a test of equality)
$\neq$	Not equals
$\geq$	Greater than or equal to
>	Greater than
()	Indicates precedence
$\leftarrow$	Assignment operator
$\in$	Indicates membership
$\notin$	Indicates nonmembership
	Catenate
ELSE	No other state condition is satisfied
<u>++</u>	<u>Increment value of preceding variable by 1</u>

30. Management

30.3 Layer management for DTEs

30.3.2 PHY device managed object class

30.3.2.1 PHY device attributes

30.3.2.1.2 aPhyType

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.3.2.1.2 after the entry for 400GBASE-R:

APPROPRIATE SYNTAX:
800GBASE-ER1 Clause 186 type 800GBASE-ER1 PCS

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.3.2.1.2 after the entry for 800GBASE-R (inserted by IEEE Std 802.3df-2024):

APPROPRIATE SYNTAX:
1.6TBASE-R Clause 175 1.6 Tb/s multi-PCS lane 64B/66B

30.3.2.1.3 aPhyTypeList

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.3.2.1.3 after the entry for 400GBASE-R:

APPROPRIATE SYNTAX:
800GBASE-ER1 Clause 186 type 800GBASE-ER1 PCS

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.3.2.1.3 after the entry for 800GBASE-R (inserted by IEEE Std 802.3df-2024):

APPROPRIATE SYNTAX:
1.6TBASE-R Clause 175 1.6 Tb/s multi-PCS lane 64B/66B

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

APPROPRIATE SYNTAX:

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 200GBASE-CR2 (inserted by IEEE Std 802.3ck-2022):

200GBASE-CR1	200GBASE-R PCS/PMA over shielded balanced copper cable PMD as specified in Clause 179
--------------	---

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 200GBASE-DR4:

200GBASE-DR1	200GBASE-R PCS/PMA over single-mode fiber PMD with a reach of up to at least 500 m as specified in Clause 180
--------------	---

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 200GBASE-ER4:

200GBASE-DR1-2	200GBASE-R PCS/PMA over single-mode fiber PMD with a reach of up to at least 2 km as specified in Clause 182
----------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 200GBASE-KR2 (inserted by IEEE Std 802.3ck-2022):

200GBASE-KR1	200GBASE-R PCS/PMA over an electrical backplane PMD as specified in Clause 178
--------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 400GBASE-CR4 (inserted by IEEE Std 802.3ck-2022):

400GBASE-CR2	400GBASE-R PCS/PMA over 2-lane shielded balanced copper cable PMD as specified in Clause 179
--------------	--

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 400GBASE-DR4:

400GBASE-DR2	400GBASE-R PCS/PMA over 2-lane single-mode fiber PMD with a
--------------	---

400GBASE-DR2-2 reach of up to at least 500 m as specified in Clause 180
400GBASE-R PCS/PMA over 2-lane single-mode fiber PMD with a
reach of up to at least 2 km as specified in Clause 182

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 400GBASE-KR4 (inserted by IEEE Std 802.3ck-2022):

400GBASE-KR2 400GBASE-R PCS/PMA over an electrical backplane PMD as specified
in Clause 178

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 800GBASE-CR8 (inserted by IEEE Std 802.3df-2024):

800GBASE-CR4 800GBASE-R PCS/PMA over 4-lane shielded balanced copper cable
PMD as specified in Clause 179

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 800GBASE-DR8 (inserted by IEEE Std 802.3df-2024):

800GBASE-DR4 800GBASE-R PCS/PMA over 4-lane single-mode fiber PMD with a
reach of up to at least 500 m as specified in Clause 180
800GBASE-DR4-2 800GBASE-R PCS/PMA over 4-lane single-mode fiber PMD with a
reach of up to at least 2 km as specified in Clause 182

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 800GBASE-DR8-2 (inserted by IEEE Std 802.3df-2024):

800GBASE-ER1 800GBASE-ER1 PCS/PMA over single-mode fiber PMD with a reach of
up to at least 40 km as specified in Clause 187
800GBASE-ER1-20 800GBASE-ER1 PCS/PMA over single-mode fiber PMD with a reach of
up to at least 20 km as specified in Clause 187
800GBASE-FR4 800GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD
with a reach of up to at least 2 km as specified in Clause 183
800GBASE-FR4-500 800GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD
with a reach of up to at least 500 m as specified in Clause 181

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 800GBASE-KR8 (inserted by IEEE Std 802.3df-2024):

800GBASE-KR4 800GBASE-R PCS/PMA over an electrical backplane PMD as specified
in Clause 178

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 before the entry for 800GBASE-R (inserted by IEEE Std 802.3df-2024):

800GBASE-LR1	800GBASE-R PCS/PMA over single-mode fiber PMD with a reach of up to at least 10 km as specified in Clause 185
800GBASE-LR4	800GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with a reach of up to at least 10 km as specified in Clause 183

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 800GBASE-VR8 (inserted by IEEE Std 802.3df-2024):

1.6TBASE-CR8	1.6TBASE-R PCS/PMA over 8-lane shielded balanced copper cable PMD as specified in Clause 179
1.6TBASE-DR8	1.6TBASE-R PCS/PMA over 8 single-mode fibers in each direction PMD with reach up to at least 500 m as specified in Clause 180
1.6TBASE-DR8-2	1.6TBASE-R PCS/PMA over 8 single-mode fibers in each direction PMD with reach up to at least 2 km as specified in Clause 182
1.6TBASE-KR8	1.6TBASE-R PCS/PMA over an electrical backplane PMD as specified in Clause 178
1.6TBASE-R	1.6TBASE-R Multi-lane PCS as specified in Clause 175 over undefined PMA/PMD

30.6 Management for link Auto-Negotiation

30.6.1 Auto-Negotiation managed object class

30.6.1.1 Auto-Negotiation attributes

30.6.1.1.5 aAutoNegLocalTechnologyAbility

APPROPRIATE SYNTAX:

Change the items in 30.6.1.1.5 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows (some unchanged items not shown):

200GR2	200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as specified in Clause 163
<u>200GR1</u>	<u>200GBASE-CR1 as specified in Clause 179 or 200GBASE-KR1 as specified in Clause 178</u>
400GR4	400GBASE-CR4 as specified in Clause 162 or 400GBASE-KR4 as specified in Clause 163
<u>400GR2</u>	<u>400GBASE-CR2 as specified in Clause 179 or 400GBASE-KR2 as specified in Clause 178</u>
800GR8	800GBASE-CR8 as specified in Clause 162 or 800GBASE-KR8 as specified in Clause 163
<u>800GR4</u>	<u>800GBASE-CR4 as specified in Clause 179 or 800GBASE-KR4 as specified in Clause 178</u>
<u>1.6TR8</u>	<u>1.6TBASE-CR8 as specified in Clause 179 or 1.6TBASE-KR8 as specified in Clause 178</u>

30.13 Management for oTimeSync entity

30.13.1 TimeSync entity managed object class

Change 30.13.1.1 to 30.13.1.12 (as amended by IEEE Std 802.3cx-2023) as follows:

30.13.1.1 aTimeSyncCapabilityNsTX

ATTRIBUTE

APPROPRIATE SYNTAX:
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with nanosecond resolution, is supported in the transmit path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.1, see 45.2.1.175
- For Inner FEC: 1.1800.5, see 45.2.1.175
- For WIS: 2.1800.1, see 45.2.2.20
- For PCS: 3.1800.1, see 45.2.3.67
- For PHY XS: 4.1800.1, see 45.2.4.28
- For DTE XS: 5.1800.1, see 45.2.5.28
- For TC: 6.1800.1, see 45.2.6.14.;

30.13.1.2 aTimeSyncCapabilityNsRX

ATTRIBUTE

APPROPRIATE SYNTAX:
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with nanosecond resolution, is supported in the receive path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.0, see 45.2.1.175
- For Inner FEC: 1.1800.4, see 45.2.1.175
- For WIS: 2.1800.0, see 45.2.2.20
- For PCS: 3.1800.0, see 45.2.3.67
- For PHY XS: 4.1800.0, see 45.2.4.28
- For DTE XS: 5.1800.0, see 45.2.5.28
- For TC: 6.1800.0, see 45.2.6.14.;

30.13.1.3 aTimeSyncDelayNsTXmax

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the maximum transmit path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1801 and 1.1802, see 45.2.1.176
- For Inner FEC: 1.1813 and 1.1814, see 45.2.1.177a
- For WIS: 2.1801 and 2.1802, see 45.2.2.21
- For PCS: 3.1801 and 3.1802, see 45.2.3.68
- For PHY XS: 4.1801 and 4.1802, see 45.2.4.29
- For DTE XS: 5.1801 and 5.1802, see 45.2.5.29
- For TC: 6.1801 and 6.1802, see 45.2.6.15.;

30.13.1.4 aTimeSyncDelayNsTXmin

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the minimum transmit path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1803 and 1.1804, see 45.2.1.176
- For Inner FEC: 1.1815 and 1.1816, see 45.2.1.177a
- For WIS: 2.1803 and 2.1804, see 45.2.2.21
- For PCS: 3.1803 and 3.1804, see 45.2.3.68
- For PHY XS: 4.1803 and 4.1804, see 45.2.4.29
- For DTE XS: 5.1803 and 5.1804, see 45.2.5.29
- For TC: 6.1803 and 6.1804, see 45.2.6.15.;

30.13.1.5 aTimeSyncDelayNsRXmax

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the maximum receive path data delay as specified in 90.7,

expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1805 and 1.1806, see 45.2.1.177
- For Inner FEC: 1.1819 and 1.1820, see 45.2.1.177b
- For WIS: 2.1805 and 2.1806, see 45.2.2.22
- For PCS: 3.1805 and 3.1806, see 45.2.3.69
- For PHY XS: 4.1805 and 4.1806, see 45.2.4.30
- For DTE XS: 5.1805 and 5.1806, see 45.2.5.30
- For TC: 6.1805 and 6.1806, see 45.2.6.16.;

30.13.1.6 aTimeSyncDelayNsRXmin

ATTRIBUTE

APPROPRIATE SYNTAX:
INTEGER

BEHAVIOUR DEFINED AS:

The integer nanosecond portion of the minimum receive path data delay as specified in 90.7, expressed in units of ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1807 and 1.1808, see 45.2.1.177
- For Inner FEC: 1.1821 and 1.1822, see 45.2.1.177b
- For WIS: 2.1807 and 2.1808, see 45.2.2.22
- For PCS: 3.1807 and 3.1808, see 45.2.3.69
- For PHY XS: 4.1807 and 4.1808, see 45.2.4.30
- For DTE XS: 5.1807 and 5.1808, see 45.2.5.30
- For TC: 6.1807 and 6.1808, see 45.2.6.16.;

30.13.1.7 aTimeSyncCapabilitySubNsTX

ATTRIBUTE

APPROPRIATE SYNTAX:
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with sub-nanosecond resolution, is supported in the transmit path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.3, see 45.2.1.175
- For Inner FEC: 1.1800.7, see 45.2.1.175

- For WIS: 2.1800.3, see 45.2.2.20
- For PCS: 3.1800.3, see 45.2.3.67
- For PHY XS: 4.1800.3, see 45.2.4.28
- For DTE XS: 5.1800.3, see 45.2.5.28
- For TC: 6.1800.3, see 45.2.6.14.;

30.13.1.8 aTimeSyncCapabilitySubNsRX

ATTRIBUTE

APPROPRIATE SYNTAX:
BOOLEAN

BEHAVIOUR DEFINED AS:

True if the TimeSync path data delay provisioning capability, with sub-nanosecond resolution, is supported in the receive path and false otherwise.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute is equal to the logical OR operation over the values stored in the following instantiated MDIO registers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1800.2, see 45.2.1.175
- For Inner FEC: 1.1800.6, see 45.2.1.175
- For WIS: 2.1800.2, see 45.2.2.20
- For PCS: 3.1800.2, see 45.2.3.67
- For PHY XS: 4.1800.2, see 45.2.4.28
- For DTE XS: 5.1800.2, see 45.2.5.28
- For TC: 6.1800.2, see 45.2.6.14.;

30.13.1.9 aTimeSyncDelaySubNsTXmax

ATTRIBUTE

APPROPRIATE SYNTAX:
INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the maximum transmit path data delay as specified in 90.7, expressed in units of 2^{-16} ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1809, see 45.2.1.176
- For Inner FEC: 1.1817, see 45.2.1.177a
- For WIS: 2.1809, see 45.2.2.21
- For PCS: 3.1809, see 45.2.3.68
- For PHY XS: 4.1809, see 45.2.4.29
- For DTE XS: 5.1809, see 45.2.5.29
- For TC: 6.1809, see 45.2.6.15.;

30.13.1.10 aTimeSyncDelaySubNsTXmin

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the minimum transmit path data delay as specified in 90.7, expressed in units of 2^{-16} ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum transmit path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1810, see 45.2.1.176
- For Inner FEC: 1.1818, see 45.2.1.177a
- For WIS: 2.1810, see 45.2.2.21
- For PCS: 3.1810, see 45.2.3.68
- For PHY XS: 4.1810, see 45.2.4.29
- For DTE XS: 5.1810, see 45.2.5.29
- For TC: 6.1810, see 45.2.6.15.;

30.13.1.11 aTimeSyncDelaySubNsRXmax

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the maximum receive path data delay as specified in 90.7, expressed in units of 2^{-16} ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the maximum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1811, see 45.2.1.177
- For Inner FEC: 1.1823, see 45.2.1.177b
- For WIS: 2.1811, see 45.2.2.22
- For PCS: 3.1811, see 45.2.3.69
- For PHY XS: 4.1811, see 45.2.4.30
- For DTE XS: 5.1811, see 45.2.5.30
- For TC: 6.1811, see 45.2.6.16.;

30.13.1.12 aTimeSyncDelaySubNsRXmin

ATTRIBUTE

APPROPRIATE SYNTAX:

INTEGER

BEHAVIOUR DEFINED AS:

The sub-nanosecond portion of the minimum receive path data delay as specified in 90.7,

expressed in units of 2^{-16} ns.

If a Clause 45 MDIO Interface to PMA/PMD, Inner FEC, WIS, PCS, PHY XS, DTE XS, and/or TC is present, then the value stored in this attribute represents the minimum receive path data delay values, consisting of the sum of the values of the register sets in the instantiated sublayers (for each MMD, in case of multiple instances):

- For PMA/PMD: 1.1812, see 45.2.1.177
- For Inner FEC: 1.1824, see 45.2.1.177b
- For WIS: 2.1812, see 45.2.2.22
- For PCS: 3.1812, see 45.2.3.69
- For PHY XS: 4.1812, see 45.2.4.30
- For DTE XS: 5.1812, see 45.2.5.30
- For TC: 6.1812, see 45.2.6.16.;

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

Change the first sentence of the first paragraph as follows:

For devices operating at 25 Gb/s or higher speeds, the PMA may be instantiated as multiple sublayers (see [83.1.4](#), [109.1.4](#), ~~and [120.1.4](#)~~, [173.1.4](#), and [176.1.5](#) for how MMD addresses are allocated to multiple PMA sublayers for the respective speeds).

Change Table 45–3 (as amended by IEEE Std 802.3cx-2023, IEEE Std 802.3cz-2023 and IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–3—PMA/PMD registers

Register address	Register name	Subclause
...		
1.72	BASE-AU PMA/PMD extended ability	45.2.1.60a
1.73	800G PMA/PMD ability	45.2.1.60b
1.74	800G PMA/PMD extended ability	45.2.1.60c
1.75	400G PMA/PMD extended ability 2	45.2.1.60d
1.76	1.6T PMA/PMD ability	45.2.1.60e
1.77 through 1.79 1.74 through 1.79	Reserved	
...		
1.1458 through 1.1459 1.1458 through 1.1499	Reserved	
1.1460 through 1.4475	PMA/PMD per lane PRBS31 seeds used for training	45.2.1.168a
1.1476	PMA/PMD interface training status	45.2.1.168b
1.1477	PMA/PMD interface training control	45.2.1.168c
1.1478	RS-FEC symbol multiplexing PMA status	45.2.1.168d
1.4479 through 1.1499	Reserved	
...		
1.1811 and 1.1812	TimeSync PMA/PMD receive path data delay in sub-ns	45.2.1.177
1.1813 through 1.1818	TimeSync inner FEC transmit path data delay registers	45.2.1.177a
1.1819 through 1.1824	TimeSync inner FEC receive path data delay registers	45.2.1.177b
1.1825	TimeSync FEC configuration	45.2.1.177c
1.1826 through 1.1899 1.1813 through 1.1899	Reserved	

Table 45–3—PMA/PMD registers (*continued*)

Register address	Register name	Subclause
...		
<u>1.2207 through 1.2209</u>	Reserved <u>IFEC codeword counter</u>	<u>45.2.1.220a</u>
<u>1.2210 through 1.2241</u> 1.2210 through 1.2217	IFEC symbol error counter, lane 0 to 15 <u>3</u>	45.2.1.222
<u>1.2242 through 1.2245</u> 1.2218 through 1.2245	Reserved	
...		
<u>1.2318 through 1.2400</u> 1.2318 through 1.32767	Reserved	
<u>1.2401</u>	<u>Inner FEC PRBS13Q testing</u>	<u>45.2.1.251</u>
<u>1.2402</u>	<u>Inner FEC PRBS31 testing</u>	<u>45.2.1.252</u>
<u>1.2403</u>	<u>Inner FEC SSPRO testing</u>	<u>45.2.1.253</u>
<u>1.2404</u>	<u>Inner FEC square wave testing</u>	<u>45.2.1.254</u>
<u>1.2405</u>	<u>PRBS31 encoded by Inner FEC</u>	<u>45.2.1.255</u>
<u>1.2406 through 1.2409</u>	<u>Reserved</u>	
<u>1.2410</u>	<u>Inner FEC alignment status</u>	<u>45.2.1.256</u>
<u>1.2411</u>	<u>Inner FEC status 1</u>	<u>45.2.1.257</u>
<u>1.2412 through 1.2413</u>	<u>Inner FEC corrected codewords counter</u>	<u>45.2.1.258</u>
<u>1.2414 through 1.2415</u>	<u>Inner FEC uncorrected codewords counter</u>	<u>45.2.1.259</u>
<u>1.2416 through 1.2419</u>	<u>Inner FEC total bits counter</u>	<u>45.2.1.260</u>
<u>1.2420 through 1.2423</u>	<u>Inner FEC corrected bits counter</u>	<u>45.2.1.261</u>
<u>1.2424 through 1.2433</u>	<u>Inner FEC error bin counters</u>	<u>45.2.1.262</u>
<u>1.2434 through 1.2587</u>	<u>Inner FEC counter registers for lanes 1 through 7</u>	<u>45.2.1.263</u>
<u>1.2588 through 1.2599</u>	<u>Reserved</u>	
<u>1.2600 through 1.3007</u>	<u>PMA test block error bin counters</u>	<u>45.2.1.264</u>
<u>1.3007 through 1.3099</u>	<u>Reserved</u>	
<u>1.3100 through 1.3131</u>	<u>IFEC lane mapping</u>	<u>45.2.1.265</u> <u>45.2.1.266</u>
<u>1.3132 through 1.3133</u>	<u>IFEC alignment status</u>	<u>45.2.1.267</u>
<u>1.3134 through 1.3163</u>	<u>IFEC codeword error bin counters</u>	<u>45.2.1.268</u>
<u>1.3164 through 1.3169</u>	<u>Reserved</u>	
<u>1.3170 through 1.3199</u>	<u>IFEC symbol error counter, lane 16 to 31</u>	<u>45.2.1.222</u>
<u>1.3200 through 1.3999</u>	<u>Reserved</u>	
<u>1.4000 through 1.5499</u>	<u>Duplication of ILT training registers for the upper AUI component</u>	<u>45.2.1.269</u>
<u>1.5500 through 1.32767</u>	<u>Reserved</u>	
...		

45.2.1.1 PMA/PMD control 1 register (Register 1.0)

Change Table 45–4 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–4—PMA/PMD control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = 1.6 Tb/s Reserved 1 0 1 0 = 800 Gb/s 1 0 0 1 = 400 Gb/s 1 0 0 0 = 200 Gb/s 0 1 1 1 = 5 Gb/s 0 1 1 0 = 2.5 Gb/s 0 1 0 1 = 50 Gb/s 0 1 0 0 = 25 Gb/s 0 0 1 1 = 100 Gb/s 0 0 1 0 = 40 Gb/s 0 0 0 1 = 10PASS-TS/2BASE-TL 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only

45.2.1.1.3 Speed selection (1.0.13, 1.0.6, 1.0.5:2)

Change the last paragraph of 45.2.1.1.3 as follows:

When bits 5 through 2 are set to 0010 the use of a 40G PMA/PMD is selected; when set to 0011 the use of a 100G PMA/PMD is selected; when set to 0100 the use of a 25G PMA/PMD is selected; when set to 0101 the use of a 50G PMA/PMD is selected; when set to 0110 the use of a 2.5G PMA/PMD is selected; when set to 0111 the use of a 5G PMA/PMD is selected; when set to 1000 the use of a 200G PMA/PMD is selected; when set to 1001 the use of a 400G PMA/PMD is selected; when set to 1010 the use of an 800G PMA/PMD is selected; when set to 1011 the use of a 1.6T PMA/PMD is selected. More specific selection is performed using the PMA/PMD control 2 register (register 1.7) (see 45.2.1.6.3).

45.2.1.6 PMA/PMD control 2 register (Register 1.7)

Change Table 45–7 (as amended by IEEE Std 802.3df-2024) as follows (unchanged table rows and bit description lines not shown):

Table 45–7—PMA/PMD control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.7.7:0	PMA/PMD type selection	7 6 5 4 3 2 1 0 1 1 x x x x x = reserved 1 0 1 x x x x x = reserved 1 0 0 1 x x x x = reserved 1 0 0 0 1 x x x = reserved 1 0 0 0 0 1 x x = reserved 1 0 0 0 0 1 1 x = reserved 1 0 1 0 0 1 x x = reserved <u>1 0 1 1 x x x x = reserved</u> <u>1 0 1 0 1 x x x = reserved</u> <u>1 0 1 0 0 1 x x = reserved</u> <u>1 0 1 0 0 1 1 = 1.6TBASE-DR8-2 PMA/PMD</u> <u>1 0 1 0 0 1 0 = 1.6TBASE-DR8 PMA/PMD</u> <u>1 0 1 0 0 0 1 = 1.6TBASE-CR8 PMA/PMD</u> <u>1 0 1 0 0 0 0 = 1.6TBASE-KR8 PMA/PMD</u> <u>1 0 0 1 1 1 x x = reserved</u> <u>1 0 0 1 1 0 1 1 = 400GBASE-DR2-2 PMA/PMD</u> <u>1 0 0 1 1 0 1 0 = 400GBASE-DR2 PMA/PMD</u> <u>1 0 0 1 1 0 0 1 = 400GBASE-CR2 PMA/PMD</u> <u>1 0 0 1 1 0 0 0 = 400GBASE-KR2 PMA/PMD</u> <u>1 0 0 1 0 1 x x = reserved</u> <u>1 0 0 1 0 0 1 1 = 200GBASE-DR1-2 PMA/PMD</u> <u>1 0 0 1 0 0 1 0 = 200GBASE-DR1 PMA/PMD</u> <u>1 0 0 1 0 0 0 1 = 200GBASE-CR1 PMA/PMD</u> <u>1 0 0 1 0 0 0 0 = 200GBASE-KR1 PMA/PMD</u> <u>1 0 0 0 1 1 1 1 = 800GBASE-ER1 PMA/PMD</u> <u>1 0 0 0 1 1 1 0 = 800GBASE-ER1-20 PMA/PMD</u> <u>1 0 0 0 1 1 0 1 = 800GBASE-LR1 PMA/PMD</u> <u>1 0 0 0 1 1 0 0 = 800GBASE-LR4 PMA/PMD</u> <u>1 0 0 0 1 0 1 1 = 800GBASE-FR4 PMA/PMD</u> <u>1 0 0 0 1 0 1 0 = 800GBASE-FR4-500 PMA/PMD</u> <u>1 0 0 0 1 0 0 1 = 800GBASE-DR4-2 PMA/PMD</u> <u>1 0 0 0 1 0 0 0 = 800GBASE-DR4 PMA/PMD</u> <u>1 0 0 0 0 1 1 1 = 800GBASE-CR4 PMA/PMD</u> <u>1 0 0 0 0 1 1 0 = 800GBASE-KR4 PMA/PMD</u> ...	R/W

^a R/W = Read/Write, RO = Read only

45.2.1.7 PMA/PMD status 2 register (Register 1.8)

45.2.1.7.4 Transmit fault (1.8.11)

Change Table 45–9 as follows (most unchanged rows not shown):

Table 45–9—Transmit fault description location

PMA/PMD	Description location
...	
400GBASE-FR4, 400GBASE-LR4-6	151.5.10
<u>200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, 1.6TBASE-KR8</u>	<u>178.8.8</u>
<u>200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8</u>	<u>179.8.8</u>
<u>200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, 1.6TBASE-DR8</u>	<u>180.5.10</u>
<u>800GBASE-FR4-500</u>	<u>181.5.9</u>
<u>200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, 1.6TBASE-DR8-2</u>	<u>182.5.10</u>
<u>800GBASE-FR4, 800GBASE-LR4</u>	<u>183.5.10</u>
<u>800GBASE-LR1</u>	<u>185.5.9</u>
<u>800GBASE-ER1, 800GBASE-ER1-20</u>	<u>187.5.9</u>

45.2.1.7.5 Receive fault (1.8.10)

Change Table 45–10 as follows (most unchanged rows not shown):

Table 45–10—Receive fault description location

PMA/PMD	Description location
...	
400GBASE-FR4, 400GBASE-LR4-6	151.5.11
200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, 1.6TBASE-KR8	178.8.8
200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8	179.8.8
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, 1.6TBASE-DR8	180.5.11
800GBASE-FR4-500	181.5.10
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, 1.6TBASE-DR8-2	182.5.11
800GBASE-FR4, 800GBASE-LR4	183.5.11
800GBASE-LR1	185.5.10
800GBASE-ER1, 800GBASE-ER1-20	187.5.10

45.2.1.8 PMD transmit disable register (Register 1.9)

Change Table 45–12 as follows (most unchanged rows not shown):

Table 45–12—Transmit disable description location

PMA/PMD	Description location
...	
400GBASE-FR4 and 400GBASE-LR4-6	151.5.7
200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8	178.8.6
200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8	179.8.6
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8	180.5.7
800GBASE-FR4-500	181.5.7
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2	182.5.7
800GBASE-FR4 and 800GBASE-LR4	183.5.7
800GBASE-LR1	185.5.7
800GBASE-ER1 and 800GBASE-ER1-20	187.5.7

45.2.1.21 200G PMA/PMD extended ability register (Register 1.23)

Change Table 45–24 (as amended by IEEE Std 802.3ck-2022) as follows (most unchanged rows not shown):

Table 45–24—200G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.23.15	200G PMA remote loopback ability	1 = 200G PMA has the ability to perform a remote loopback function 0 = 200G PMA does not have the ability to perform a remote loopback function	RO
1.23.14:11	Reserved	Value always 0	RO
<u>1.23.14</u>	<u>200GBASE-DR1-2 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-DR1-2</u> <u>0 = PMA/PMD is not able to perform 200GBASE-DR1-2</u>	<u>RO</u>
<u>1.23.13</u>	<u>200GBASE-DR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-DR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-DR1</u>	<u>RO</u>
<u>1.23.12</u>	<u>200GBASE-CR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-CR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-CR1</u>	<u>RO</u>
<u>1.23.11</u>	<u>200GBASE-KR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-KR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-KR1</u>	<u>RO</u>
1.23.10	200GBASE-SR2 ability	1 = PMA/PMD is able to perform 200GBASE-SR2 0 = PMA/PMD is not able to perform 200GBASE-SR2	RO
...			

^a RO = Read only.

Insert 45.2.1.21.1a to 45.2.1.21.1d after 45.2.1.21.1 as follows:

45.2.1.21.1a 200GBASE-DR1-2 ability (1.23.14)

When read as a one, bit 1.23.14 indicates that the PMA/PMD is able to operate as a 200GBASE-DR1-2 PMA/PMD type. When read as a zero, bit 1.23.14 indicates that the PMA/PMD is not able to operate as a 200GBASE-DR1-2 PMA/PMD type.

45.2.1.21.1b 200GBASE-DR1 ability (1.23.13)

When read as a one, bit 1.23.13 indicates that the PMA/PMD is able to operate as a 200GBASE-DR1 PMA/PMD type. When read as a zero, bit 1.23.13 indicates that the PMA/PMD is not able to operate as a 200GBASE-DR1 PMA/PMD type.

45.2.1.21.1c 200GBASE-CR1 ability (1.23.12)

When read as a one, bit 1.23.12 indicates that the PMA/PMD is able to operate as a 200GBASE-CR1 PMA/PMD type. When read as a zero, bit 1.23.12 indicates that the PMA/PMD is not able to operate as a 200GBASE-CR1 PMA/PMD type.

45.2.1.21.1d 200GBASE-KR1 ability (1.23.11)

When read as a one, bit 1.23.11 indicates that the PMA/PMD is able to operate as a 200GBASE-KR1 PMA/PMD type. When read as a zero, bit 1.23.11 indicates that the PMA/PMD is not able to operate as a 200GBASE-KR1 PMA/PMD type.

45.2.1.23 PMA/PMD extended ability 2 (Register 1.25)

Change Table 45–26 (as amended by IEEE Std 802.3df-2024) as follows:

Table 45–26—PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.25.15:3 1.25.15:2	Reserved	Value always 0	RO
<u>1.25.2</u>	<u>1.6T capable</u>	<u>1 = PMA/PMD is capable of operating at 1.6 Tb/s and has the abilities listed in register 1.76</u> <u>0 = PMA/PMD is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
1.25.1	800G capable	1 = PMA/PMD is capable of operating at 800 Gb/s and has the abilities listed in register 1.73 0 = PMA/PMD is not capable of operating at 800 Gb/s	RO
1.25.0	50G extended abilities	1 = PMA/PMD has 50G extended abilities listed in register 1.20 0 = PMA/PMD does not have 50G extended abilities	RO

^a RO = Read only

Insert 45.2.1.23.aa before 45.2.1.23.a as follows:

45.2.1.23.aa 1.6T capable (1.25.2)

When read as a one, bit 1.25.2 indicates that the PMA/PMD is capable of operating at 1.6 Tb/s and has the abilities listed in register 1.76. When read as a zero, bit 1.25.2 indicates that the PMA/PMD is not capable of operating at 1.6 Tb/s.

45.2.1.60b 800G PMA/PMD ability (Register 1.73)

Change Table 45–58b (as inserted by IEEE Std 802.3df-2024) as follows:

Table 45–58b—800G PMA/PMD ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.73.15	800G PMA remote loopback ability	1 = 800G PMA is able to perform a remote loopback function 0 = 800G PMA is not able to perform a remote loopback function	RO
1.73.14 1.73.14:6	Reserved	Value always 0	RO
<u>1.73.13</u>	<u>800GBASE-LR1 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-LR1</u> <u>0 = PMA/PMD is not able to perform 800GBASE-LR1</u>	<u>RO</u>
<u>1.73.12</u>	<u>800GBASE-LR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-LR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-LR4</u>	<u>RO</u>
<u>1.73.11</u>	<u>800GBASE-FR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-FR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-FR4</u>	<u>RO</u>
<u>1.73.10</u>	<u>800GBASE-DR4-2 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-DR4-2</u> <u>0 = PMA/PMD is not able to perform 800GBASE-DR4-2</u>	<u>RO</u>
<u>1.73.9</u>	<u>800GBASE-FR4-500 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-FR4-500</u> <u>0 = PMA/PMD is not able to perform 800GBASE-FR4-500</u>	<u>RO</u>
<u>1.73.8</u>	<u>800GBASE-DR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-DR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-DR4</u>	<u>RO</u>
<u>1.73.7</u>	<u>800GBASE-CR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-CR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-CR4</u>	<u>RO</u>
<u>1.73.6</u>	<u>800GBASE-KR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-KR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-DR8-2</u>	<u>RO</u>
1.73.5	800GBASE-DR8-2 ability	1 = PMA/PMD is able to perform 800GBASE-DR8-2 0 = PMA/PMD is not able to perform 800GBASE-DR8-2	RO
1.73.4	800GBASE-DR8 ability	1 = PMA/PMD is able to perform 800GBASE-DR8 0 = PMA/PMD is not able to perform 800GBASE-DR8	RO
1.73.3	800GBASE-SR8 ability	1 = PMA/PMD is able to perform 800GBASE-SR8 0 = PMA/PMD is not able to perform 800GBASE-SR8	RO
1.73.2	800GBASE-VR8 ability	1 = PMA/PMD is able to perform 800GBASE-VR8 0 = PMA/PMD is not able to perform 800GBASE-VR8	RO
1.73.1	800GBASE-CR8 ability	1 = PMA/PMD is able to perform 800GBASE-CR8 0 = PMA/PMD is not able to perform 800GBASE-CR8	RO
1.73.0	800GBASE-KR8 ability	1 = PMA/PMD is able to perform 800GBASE-KR8 0 = PMA/PMD is not able to perform 800GBASE-KR8	RO

^a RO = Read only

Insert 45.2.1.60b.1a to 45.2.1.60b.1i after 45.2.1.60b.1 as follows:

45.2.1.60b.1a 800GBASE-LR1 ability (1.73.13)

When read as a one, bit 1.73.13 indicates that the PMA/PMD is able to operate as an 800GBASE-LR1 PMA/PMD type. When read as a zero, bit 1.73.13 indicates that the PMA/PMD is not able to operate as an 800GBASE-LR1 PMA/PMD type.

45.2.1.60b.1b 800GBASE-LR4 ability (1.73.12)

When read as a one, bit 1.73.12 indicates that the PMA/PMD is able to operate as an 800GBASE-LR4 PMA/PMD type. When read as a zero, bit 1.73.12 indicates that the PMA/PMD is not able to operate as an 800GBASE-LR4 PMA/PMD type.

45.2.1.60b.1c 800GBASE-FR4 ability (1.73.11)

When read as a one, bit 1.73.11 indicates that the PMA/PMD is able to operate as an 800GBASE-FR4 PMA/PMD type. When read as a zero, bit 1.73.11 indicates that the PMA/PMD is not able to operate as an 800GBASE-FR4 PMA/PMD type.

45.2.1.60b.1d 800GBASE-DR4-2 ability (1.73.10)

When read as a one, bit 1.73.10 indicates that the PMA/PMD is able to operate as an 800GBASE-DR4-2 PMA/PMD type. When read as a zero, bit 1.73.10 indicates that the PMA/PMD is not able to operate as an 800GBASE-DR4-2 PMA/PMD type.

45.2.1.60b.1e 800GBASE-FR4-500 ability (1.73.9)

When read as a one, bit 1.73.9 indicates that the PMA/PMD is able to operate as an 800GBASE-FR4-500 PMA/PMD type. When read as a zero, bit 1.73.9 indicates that the PMA/PMD is not able to operate as an 800GBASE-FR4-500 PMA/PMD type.

45.2.1.60b.1f 800GBASE-DR4 ability (1.73.8)

When read as a one, bit 1.73.8 indicates that the PMA/PMD is able to operate as an 800GBASE-DR4 PMA/PMD type. When read as a zero, bit 1.73.8 indicates that the PMA/PMD is not able to operate as an 800GBASE-DR4 PMA/PMD type.

45.2.1.60b.1g 800GBASE-CR4 ability (1.73.7)

When read as a one, bit 1.73.7 indicates that the PMA/PMD is able to operate as an 800GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.73.7 indicates that the PMA/PMD is not able to operate as an 800GBASE-CR4 PMA/PMD type.

45.2.1.60b.1h 800GBASE-KR4 ability (1.73.6)

When read as a one, bit 1.73.6 indicates that the PMA/PMD is able to operate as an 800GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.73.6 indicates that the PMA/PMD is not able to operate as an 800GBASE-KR4 PMA/PMD type.

Insert 45.2.1.60c, 45.2.1.60d, and 45.2.1.60e after 45.2.1.60b (inserted by IEEE Std 802.3df-2024) as follows:

45.2.1.60c 800G PMA/PMD extended ability register (Register 1.74)

The assignment of bits in the 800G PMA/PMD extended ability 2 register is shown in Table 45–58c.

Table 45–58c—800G PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.74.15:2	Reserved	Value always 0	RO
1.74.1	800GBASE-ER1 ability	1 = PMA/PMD is able to perform 800GBASE-ER1 0 = PMA/PMD is not able to perform 800GBASE-ER1	RO
1.74.0	800GBASE-ER1-20 ability	1 = PMA/PMD is able to perform 800GBASE-ER1-20 0 = PMA/PMD is not able to perform 800GBASE-ER1-20	RO

^a RO = Read only.

45.2.1.60c.1 800GBASE-ER1-20 ability (1.74.0)

When read as a one, bit 1.74.0 indicates that the PMA/PMD is able to operate as a 800GBASE-ER1-20 PMA/PMD type. When read as a zero, bit 1.74.0 indicates that the PMA/PMD is not able to operate as a 800GBASE-ER1-20 PMA/PMD type.

45.2.1.60c.2 800GBASE-ER1 ability (1.74.1)

When read as a one, bit 1.74.1 indicates that the PMA/PMD is able to operate as a 800GBASE-ER1 PMA/PMD type. When read as a zero, bit 1.74.1 indicates that the PMA/PMD is not able to operate as a 800GBASE-ER1 PMA/PMD type.

45.2.1.60d 400G PMA/PMD extended ability 2 register (Register 1.75)

The assignment of bits in the 400G PMA/PMD extended ability 2 register is shown in Table 45–58d.

Table 45–58d—400G PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.75.15:4	Reserved	Value always 0	RO
1.75.3	400GBASE-DR2-2 ability	1 = PMA/PMD is able to perform 400GBASE-DR2-2 0 = PMA/PMD is not able to perform 400GBASE-DR2-2	RO
1.75.2	400GBASE-DR2 ability	1 = PMA/PMD is able to perform 400GBASE-DR2 0 = PMA/PMD is not able to perform 400GBASE-DR2	RO
1.75.1	400GBASE-CR2 ability	1 = PMA/PMD is able to perform 400GBASE-CR2 0 = PMA/PMD is not able to perform 400GBASE-CR2	RO
1.75.0	400GBASE-KR2 ability	1 = PMA/PMD is able to perform 400GBASE-KR2 0 = PMA/PMD is not able to perform 400GBASE-KR2	RO

^a RO = Read only.

45.2.1.60d.1 400GBASE-DR2-2 ability (1.75.3)

When read as a one, bit 1.75.3 indicates that the PMA/PMD is able to operate as a 400GBASE-DR2-2 PMA/PMD type. When read as a zero, bit 1.75.3 indicates that the PMA/PMD is not able to operate as a 400GBASE-DR2-2 PMA/PMD type.

45.2.1.60d.2 400GBASE-DR2 ability (1.75.2)

When read as a one, bit 1.75.2 indicates that the PMA/PMD is able to operate as a 400GBASE-DR2 PMA/PMD type. When read as a zero, bit 1.75.2 indicates that the PMA/PMD is not able to operate as a 400GBASE-DR2 PMA/PMD type.

45.2.1.60d.3 400GBASE-CR2 ability (1.75.1)

When read as a one, bit 1.75.1 indicates that the PMA/PMD is able to operate as a 400GBASE-CR2 PMA/PMD type. When read as a zero, bit 1.75.1 indicates that the PMA/PMD is not able to operate as a 400GBASE-CR2 PMA/PMD type.

45.2.1.60d.4 400GBASE-KR2 ability (1.75.0)

When read as a one, bit 1.75.0 indicates that the PMA/PMD is able to operate as a 400GBASE-KR2 PMA/PMD type. When read as a zero, bit 1.75.0 indicates that the PMA/PMD is not able to operate as a 400GBASE-KR2 PMA/PMD type.

45.2.1.60e 1.6T PMA/PMD ability register (Register 1.76)

The assignment of bits in the 1.6T PMA/PMD ability register is shown in Table 45–58e.

Table 45–58e—1.6T PMA/PMD ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.76.15	1.6T PMA remote loopback ability	1 = 1.6T PMA has the ability to perform a remote loopback function 0 = 1.6T PMA does not have the ability to perform a remote loopback function	RO
1.76.14:4	Reserved	Value always 0	RO
1.76.3	1.6TBASE-DR8-2 ability	1 = PMA/PMD is able to perform 1.6TBASE-DR8-2 0 = PMA/PMD is not able to perform 1.6TBASE-DR8-2	RO
1.76.2	1.6TBASE-DR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-DR8 0 = PMA/PMD is not able to perform 1.6TBASE-DR8	RO
1.76.1	1.6TBASE-CR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-CR8 0 = PMA/PMD is not able to perform 1.6TBASE-CR8	RO
1.76.0	1.6TBASE-KR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-KR8 0 = PMA/PMD is not able to perform 1.6TBASE-KR8	RO

^a RO = Read only.

45.2.1.60e.1 1.6T PMA remote loopback ability (1.76.15)

When read as a one, bit 1.76.15 indicates that the 1.6T PMA is able to perform the remote loopback function. When read as a zero, bit 1.76.15 indicates that the 1.6T PMA is not able to perform the remote loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

45.2.1.60e.2 1.6TBASE-DR8-2 ability (1.76.3)

When read as a one, bit 1.76.3 indicates that the PMA/PMD is able to operate as a 1.6TBASE-DR8-2 PMA/PMD type. When read as a zero, bit 1.76.3 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-DR8-2 PMA/PMD type.

45.2.1.60e.3 1.6TBASE-DR8 ability (1.76.2)

When read as a one, bit 1.76.2 indicates that the PMA/PMD is able to operate as a 1.6TBASE-DR2 PMA/PMD type. When read as a zero, bit 1.76.2 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-DR2 PMA/PMD type.

45.2.1.60e.4 1.6TBASE-CR8 ability (1.76.1)

When read as a one, bit 1.76.1 indicates that the PMA/PMD is able to operate as a 1.6TBASE-CR8 PMA/PMD type. When read as a zero, bit 1.76.1 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-CR8 PMA/PMD type.

45.2.1.60e.5 1.6TBASE-KR8 ability (1.76.0)

When read as a one, bit 1.76.0 indicates that the PMA/PMD is able to operate as a 1.6TBASE-KR8 PMA/PMD type. When read as a zero, bit 1.76.0 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-KR8 PMA/PMD type.

45.2.1.139 PMA precoder control Tx output (Register 1.600)

Change Table 45–109 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–109—PMA precoder control Tx output register bit definitions

Bit(s)	Name	Description	R/W ^a
1.600.15:8	Reserved	Value always 0	RO
<u>1.600.15</u>	<u>Lane 15 Tx output precoder enable</u>	<u>1 = Lane 15 Tx output precoder enabled</u> <u>0 = Lane 15 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.14</u>	<u>Lane 14 Tx output precoder enable</u>	<u>1 = Lane 14 Tx output precoder enabled</u> <u>0 = Lane 14 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.13</u>	<u>Lane 13 Tx output precoder enable</u>	<u>1 = Lane 13 Tx output precoder enabled</u> <u>0 = Lane 13 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.12</u>	<u>Lane 12 Tx output precoder enable</u>	<u>1 = Lane 12 Tx output precoder enabled</u> <u>0 = Lane 12 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.11</u>	<u>Lane 11 Tx output precoder enable</u>	<u>1 = Lane 11 Tx output precoder enabled</u> <u>0 = Lane 11 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.10</u>	<u>Lane 10 Tx output precoder enable</u>	<u>1 = Lane 10 Tx output precoder enabled</u> <u>0 = Lane 10 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.9</u>	<u>Lane 9 Tx output precoder enable</u>	<u>1 = Lane 9 Tx output precoder enabled</u> <u>0 = Lane 9 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.8</u>	<u>Lane 8 Tx output precoder enable</u>	<u>1 = Lane 8 Tx output precoder enabled</u> <u>0 = Lane 8 Tx output precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.139.a to 45.2.1.139.4 with 45.2.1.139.1 as follows:

45.2.1.139.1 Tx output precoder lane enable

The bits in PMA register 1.600 enable the output precoder on individual lanes in the transmit direction.

45.2.1.140 PMA precoder control Rx input (Register 1.601)

Change Table 45–110 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–110—PMA precoder control Rx input register bit definitions

Bit(s)	Name	Description	R/W ^a
1.601.15:8	Reserved	Value always 0	RO
<u>1.601.15</u>	<u>Lane 15 Rx input precoder enable</u>	<u>1 = Lane 15 Rx input precoder enabled</u> <u>0 = Lane 15 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.14</u>	<u>Lane 14 Rx input precoder enable</u>	<u>1 = Lane 14 Rx input precoder enabled</u> <u>0 = Lane 14 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.13</u>	<u>Lane 13 Rx input precoder enable</u>	<u>1 = Lane 13 Rx input precoder enabled</u> <u>0 = Lane 13 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.12</u>	<u>Lane 12 Rx input precoder enable</u>	<u>1 = Lane 12 Rx input precoder enabled</u> <u>0 = Lane 12 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.11</u>	<u>Lane 11 Rx input precoder enable</u>	<u>1 = Lane 11 Rx input precoder enabled</u> <u>0 = Lane 11 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.10</u>	<u>Lane 10 Rx input precoder enable</u>	<u>1 = Lane 10 Rx input precoder enabled</u> <u>0 = Lane 10 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.9</u>	<u>Lane 9 Rx input precoder enable</u>	<u>1 = Lane 9 Rx input precoder enabled</u> <u>0 = Lane 9 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.8</u>	<u>Lane 8 Rx input precoder enable</u>	<u>1 = Lane 8 Rx input precoder enabled</u> <u>0 = Lane 8 Rx input precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.140.a to 45.2.1.140.4 with 45.2.1.140.1 as follows:

45.2.1.140.1 Rx input precoder lane enable

The bits in PMA register 1.601 enable the input precoder on individual lanes in the receive direction.

45.2.1.141 PMA precoder control Rx output (Register 1.602)

Change Table 45–111 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–111—PMA precoder control Rx output register bit definitions

Bit(s)	Name	Description	R/W ^a
1.602.15:8	Reserved	Value always 0	RO
<u>1.602.15</u>	<u>Lane 15 Rx output precoder enable</u>	<u>1 = Lane 15 Rx output precoder enabled</u> <u>0 = Lane 15 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.14</u>	<u>Lane 14 Rx output precoder enable</u>	<u>1 = Lane 14 Rx output precoder enabled</u> <u>0 = Lane 14 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.13</u>	<u>Lane 13 Rx output precoder enable</u>	<u>1 = Lane 13 Rx output precoder enabled</u> <u>0 = Lane 13 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.12</u>	<u>Lane 12 Rx output precoder enable</u>	<u>1 = Lane 12 Rx output precoder enabled</u> <u>0 = Lane 12 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.11</u>	<u>Lane 11 Rx output precoder enable</u>	<u>1 = Lane 11 Rx output precoder enabled</u> <u>0 = Lane 11 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.10</u>	<u>Lane 10 Rx output precoder enable</u>	<u>1 = Lane 10 Rx output precoder enabled</u> <u>0 = Lane 10 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.9</u>	<u>Lane 9 Rx output precoder enable</u>	<u>1 = Lane 9 Rx output precoder enabled</u> <u>0 = Lane 9 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.8</u>	<u>Lane 8 Rx output precoder enable</u>	<u>1 = Lane 8 Rx output precoder enabled</u> <u>0 = Lane 8 Rx output precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.141.a to 45.2.1.141.4 with 45.2.1.141.1 as follows:

45.2.1.141.1 Rx output precoder lane enable

The bits in PMA register 1.602 enable the output precoder on individual lanes in the receive direction.

45.2.1.142 PMA precoder control Tx input (Register 1.603)

Change Table 45–112 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–112—PMA precoder control Tx input register bit definitions

Bit(s)	Name	Description	R/W ^a
1.603.15:8	Reserved	Value always 0	RO
<u>1.603.15</u>	<u>Lane 15 Tx input precoder enable</u>	<u>1 = Lane 15 Tx input precoder enabled</u> <u>0 = Lane 15 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.14</u>	<u>Lane 14 Tx input precoder enable</u>	<u>1 = Lane 14 Tx input precoder enabled</u> <u>0 = Lane 14 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.13</u>	<u>Lane 13 Tx input precoder enable</u>	<u>1 = Lane 13 Tx input precoder enabled</u> <u>0 = Lane 13 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.12</u>	<u>Lane 12 Tx input precoder enable</u>	<u>1 = Lane 12 Tx input precoder enabled</u> <u>0 = Lane 12 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.11</u>	<u>Lane 11 Tx input precoder enable</u>	<u>1 = Lane 11 Tx input precoder enabled</u> <u>0 = Lane 11 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.10</u>	<u>Lane 10 Tx input precoder enable</u>	<u>1 = Lane 10 Tx input precoder enabled</u> <u>0 = Lane 10 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.9</u>	<u>Lane 9 Tx input precoder enable</u>	<u>1 = Lane 9 Tx input precoder enabled</u> <u>0 = Lane 9Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.8</u>	<u>Lane 8 Tx input precoder enable</u>	<u>1 = Lane 8 Tx input precoder enabled</u> <u>0 = Lane 8 Tx input precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.142.a to 45.2.1.142.4 with 45.2.1.142.1 as follows:

45.2.1.142.1 Tx input precoder lane enable

The bits in PMA register 1.603 enable the input precoder on individual lanes in the transmit direction.

45.2.1.144 PMA precoder request Rx input status (Register 1.605)

Change Table 45–114 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–114—PMA precoder request Rx input status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.605.15:8	Reserved	Value always 0	RO
<u>1.605.15</u>	<u>Lane 15 Rx input precoder request status</u>	<u>1 = Lane 15 precoder Rx input requested</u> <u>0 = Lane 15 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.14</u>	<u>Lane 14 Rx input precoder request status</u>	<u>1 = Lane 14 precoder Rx input requested</u> <u>0 = Lane 14 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.13</u>	<u>Lane 13 Rx input precoder request status</u>	<u>1 = Lane 13 precoder Rx input requested</u> <u>0 = Lane 13 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.12</u>	<u>Lane 12 Rx input precoder request status</u>	<u>1 = Lane 12 precoder Rx input requested</u> <u>0 = Lane 12 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.11</u>	<u>Lane 11 Rx input precoder request status</u>	<u>1 = Lane 11 precoder Rx input requested</u> <u>0 = Lane 11 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.10</u>	<u>Lane 10 Rx input precoder request status</u>	<u>1 = Lane 10 precoder Rx input requested</u> <u>0 = Lane 10 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.9</u>	<u>Lane 9 Rx input precoder request status</u>	<u>1 = Lane 9 precoder Rx input requested</u> <u>0 = Lane 9 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.8</u>	<u>Lane 8 Rx input precoder request status</u>	<u>1 = Lane 8 precoder Rx input requested</u> <u>0 = Lane 8 precoder Rx input not requested</u>	<u>RO</u>

^aRO = Read only

Replace 45.2.1.144.a to 45.2.1.144.4 with 45.2.1.144.1 as follows:

45.2.1.144.1 Rx input precoder request status

The bits in PMA register 1.605 indicate the input precoder request status on individual lanes in the receive direction.

45.2.1.145 PMA precoder request Tx input status (Register 1.606)

Change Table 45–115 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–115—PMA precoder request Tx input status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.606.15:8	Reserved	Value always 0	RO
<u>1.606.15</u>	<u>Lane 15 Tx input precoder request status</u>	<u>1 = Lane 15 precoder Tx input requested</u> <u>0 = Lane 15 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.14</u>	<u>Lane 14 Tx input precoder request status</u>	<u>1 = Lane 14 precoder Tx input requested</u> <u>0 = Lane 14 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.13</u>	<u>Lane 13 Tx input precoder request status</u>	<u>1 = Lane 13 precoder Tx input requested</u> <u>0 = Lane 13 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.12</u>	<u>Lane 12 Tx input precoder request status</u>	<u>1 = Lane 12 precoder Tx input requested</u> <u>0 = Lane 12 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.11</u>	<u>Lane 11 Tx input precoder request status</u>	<u>1 = Lane 11 precoder Tx input requested</u> <u>0 = Lane 11 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.10</u>	<u>Lane 10 Tx input precoder request status</u>	<u>1 = Lane 10 precoder Tx input requested</u> <u>0 = Lane 10 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.9</u>	<u>Lane 9 Tx input precoder request status</u>	<u>1 = Lane 9 precoder Tx input requested</u> <u>0 = Lane 9 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.8</u>	<u>Lane 8 Tx input precoder request status</u>	<u>1 = Lane 8 precoder Tx input requested</u> <u>0 = Lane 8 precoder Tx input not requested</u>	<u>RO</u>

^aRO = Read only

Replace 45.2.1.145.a to 45.2.1.145.4 with 45.2.1.145.1 as follows:

45.2.1.145.1 Tx input precoder request status

The bits in PMA register 1.606 indicate the input precoder request status on individual lanes in the transmit direction.

45.2.1.161 BASE-R PAM4 PMD training LP control, lane 0 through lane 7 registers (Register 1.1120 through 1.1127)

Change Table 45–129 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1120.15:14	Reserved	Value always 0	RO
1.1120.13:11	Initial condition request ^b	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control	R/W
1.1120.10	Reserved Continue training	Value always 0 1 = Continue training 0 = Switch to data when training is completed	RO R/W
1.1120.9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	R/W
1.1120.7:5	Reserved	Value always 0	RO
<u>1.1120.6:5</u>	<u>Training pattern request</u>	6 5 1 1 = Free-running PRBS31 1 0 = Reserved 0 1 = Free-running PRBS13 0 0 = Synchronous PRBS13	<u>R/W</u>
1.1120.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1120.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^a R/W = Read/Write, RO = Read only.

^b Support for a given preset is PHY dependent.

^c Support for a given coefficient is PHY dependent.

45.2.1.163 BASE-R PAM4 PMD training LP status, lane 0 through lane 7 registers (Register 1.1220 through 1.1227)

Change Table 45–130 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–130—BASE-R PAM4 PMD training LP status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1220.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1220.14:12	Reserved <u>ILT</u>	Value always 0 1 = <u>ILT is supported</u> 0 = <u>ILT is not supported</u>	RO
<u>1.1220.13:12</u>	<u>Training pattern status</u>	<u>13 12</u> <u>1 1 = Free-running PRBS31</u> <u>1 0 = Reserved</u> <u>0 1 = Free-running PRBS13</u> <u>0 0 = Synchronous PRBS13</u>	<u>RO</u>
1.1220.11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	RO
1.1220.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1220.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1220.7	Parity	Even parity bit	RO
1.1220.6	Reserved	Value always 0	RO
1.1220.5:3	Coefficient select echo ^b	5 4 3 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	RO
1.1220.2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated	RO

^a RO = Read only.

^b Support for a given coefficient is PHY dependent.

45.2.1.165 BASE-R PAM4 PMD training LD control, lane 0 through lane 7 registers (Register 1.1320 through 1.1327)

Change Table 45–131 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–131—BASE-R PAM4 PMD training LD control, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1320.15:14	Reserved	Value always 0	RO
1.1320.13:11	Initial condition request ^b	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control	R/W
1.1320.10	Reserved Continue training	Value always 0 1 = Continue training 0 = Switch to data when training is completed	RO R/W
1.1320.9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	R/W
1.1320.7:5	Reserved	Value always 0	RO
<u>1.1320.6:5</u>	<u>Training pattern request</u>	6 5 1 1 = Free-running PRBS31 1 0 = Reserved 0 1 = Free-running PRBS13 0 0 = Synchronous PRBS13	<u>R/W</u>
1.1320.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1320.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^a R/W = Read/Write, RO = Read only.

^b Support for a given preset is PHY dependent.

^c Support for a given coefficient is PHY dependent.

45.2.1.167 BASE-R PAM4 PMD training LD status, lane 0 through lane 7 registers (Register 1.1420 through 1.1427)

Change Table 45–132 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–132—BASE-R PAM4 PMD training LD status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1420.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1420.14:12	Reserved <u>ILT</u>	Value always 0 1 = <u>ILT is supported</u> 0 = <u>ILT is not supported</u>	RO
<u>1.1420.13:12</u>	<u>Training pattern status</u>	<u>13 12</u> <u>1 1 = Free-running PRBS31</u> <u>1 0 = Reserved</u> <u>0 1 = Free-running PRBS13</u> <u>0 0 = Synchronous PRBS13</u>	<u>RO</u>
1.1420.11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	RO
1.1420.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1420.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1420.7	Parity	Even parity bit	RO
1.1420.6	Reserved	Value always 0	RO
1.1420.5:3	Coefficient select echo ^b	5 4 3 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	RO
1.1420.2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated	RO

^a RO = Read only.

^b Support for a given coefficient is PHY dependent.

Insert 45.2.1.168a through 45.2.1.168d after 45.2.1.168 as follows:

45.2.1.168a PMA/PMD PRBS31 seed value lanes 0 through 7 (Register 1.1460 through 1.1475)

The assignment of bits in the PRBS seed value lane 0 register is shown in Table 45–133a. The assignment of bits in the PMA/PMD training pattern lanes 1 through 7 registers are defined similarly to lane 0. Registers 1.1460 and 1.1461 define the seed for PMA/PMD lane 0; registers 1.1462 and 1.1463 define the seed for lane 1; etc.

Table 45–133a—PRBS31 seed value lane 0 bit definitions

Bit(s)	Name	Description	R/W ^a
1.1460.15:0	PRBS31 seed lower	Bits 15:0 of the PRBS31 binary seed used for lane 0 training pattern	R/W
1.1461.15	Reserved	Value always 0	RO
1.1461.14:0	PRBS31 seed upper	Bits 30:16 of the PRBS31 binary seed used for lane 0 training pattern	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.168b PMA/PMD interface training status (Register 1.1476)

The assignment of bits in the PMA/PMD training status register is shown in Table 45–133b.

Table 45–133b—PMA/PMD interface training status bit definitions

Bit(s)	Name	Description	R/W ^a
1.1476.15:9	Reserved	Value always 0	RO
1.1476.8	isl ready	1 = local_rx_ready and remote_rx_ready are true for all lanes of the interface 0 = local_rx_ready or remote_rx_ready is false on any of the lanes of the interface	RO
1.1476.7	Polarity inversion lane 7	1 = polarity inversion detected on lane 7 0 = polarity inversion not detected on lane 7	RO
1.1476.6	Polarity inversion lane 6	1 = polarity inversion detected on lane 6 0 = polarity inversion not detected on lane 6	RO
1.1476.5	Polarity inversion lane 5	1 = polarity inversion detected on lane 5 0 = polarity inversion not detected on lane 5	RO
1.1476.4	Polarity inversion lane 4	1 = polarity inversion detected on lane 4 0 = polarity inversion not detected on lane 4	RO
1.1476.3	Polarity inversion lane 3	1 = polarity inversion detected on lane 3 0 = polarity inversion not detected on lane 3	RO
1.1476.2	Polarity inversion lane 2	1 = polarity inversion detected on lane 2 0 = polarity inversion not detected on lane 2	RO
1.1476.1	Polarity inversion lane 1	1 = polarity inversion detected on lane 1 0 = polarity inversion not detected on lane 1	RO
1.1476.0	Polarity inversion lane 0	1 = polarity inversion detected on lane 0 0 = polarity inversion not detected on lane 0	RO

^a R/W = Read/Write, RO = Read only.

45.2.1.168c PMA/PMD interface training control (Register 1.1477)

The assignment of bits in the PMA/PMD interface training control register is shown in Table 45–133c.

Table 45–133c—PMA/PMD interface training control bit definitions

Bit(s)	Name	Description	R/W ^a
1.1476.15:9	Reserved	Value always 0	RO
1.1477.8	Uses recovered clock	1 = Use recovered clock for transmission 0 = Use local clock for transmission	R/W
1.1477.7:0	Maximum recovery events	Maximum allowed number of transitions into the RECOVERY state in the Training control state diagram	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.168d RS-FEC symbol multiplexing PMA status (Register 1.1478)

The assignment of bits in the RS-FEC symbol multiplexing PMA status register is shown in Table 45–133d.

Table 45–133d—RS-FEC symbol multiplexing PMA status bit definitions

Bit(s)	Name	Description	R/W ^a
1.1478.15:14	Reserved	Value always 0	RO
1.1478.13	align_status_mux	1 = PCS lane synchronization is complete. This bit indicates that all_locked_mux is true and deskewed 0 = local_rx_ready or remote_rx_ready is false on any lane of the interface	RO
1.1478.12	all_locked_mux	1 = all_locked as defined in 119.2.6.2.2 is true 0 = all_locked is not true	RO
1.1478.11	deskew_done_mux	1 = deskew is complete 0 = deskew is not complete	RO
1.1478.10	pcs_lanes_identified_mux	1 = all input lanes are locked to a unique alignment marker sequence 0 = not all input lanes are locked to a unique alignment marker sequence	RO
1.1478.9	all_locked_demux	1 = pmal_locked_demux is true for all lanes 0 = pmal_locked_demux is not true for all lanes	RO
1.1478.8	pcs_lanes_identified_demux	1 = all demultiplexed PCS lanes are locked to a unique alignment marker sequence 0 = not all demultiplexed PCS lanes are locked to a unique alignment marker sequence	RO
1.1478.7:0	pmal_locked_demux[7:0]	1 = amps_lock is true for all PCSs within the indicated input lane in the demultiplexing direction 0 = amps_lock is not true for all PCSs within the indicated input lane in the demultiplexing direction	RO

^a R/W = Read/Write, RO = Read only.

Change 45.2.1.175 (as amended by IEEE Std 802.3cx-2023) as follows:

45.2.1.175 TimeSync PMA/PMD capability (Register 1.1800)

The TimeSync PMA/PMD capability register (see Table 45–139) indicates the capability of the PMA/PMD and inner FEC to report the transmit path data delay (with ns resolution in registers 1.1801 through 1.1804 and, separately, with sub-ns resolution in registers 1.1809 and 1.1810) and receive path data delay (with ns

resolution in registers 1.1805 through 1.1808 and, separately, with sub-ns resolution in registers 1.1811 and 1.1812). See 90.7.

Table 45–139—TimeSync PMA/PMD capability

Bit(s)	Name	Description	R/W ^a
<u>1.1800.15:9</u> 1.1800.15:4	Reserved	Value always 0	RO
<u>1.1800.8</u>	<u>FEC alignment marker location</u>	<u>1 = FEC supports alignment marker location</u> <u>0 = FEC does not support alignment marker location</u>	<u>RO</u>
<u>1.1800.7</u>	<u>TimeSync inner FEC transmit path data delay ability, in sub-ns</u>	<u>1 = Inner FEC provides information on transmit path data delay with sub-ns resolution in registers 1.1817 and 1.1818</u> <u>0 = Inner FEC does not provide information on transmit path data delay with sub-ns resolution</u>	<u>RO</u>
<u>1.1800.6</u>	<u>TimeSync inner FEC transmit path data delay ability, in ns</u>	<u>1 = Inner FEC provides information on transmit path data delay with ns resolution in registers 1.1813 through 1.1816</u> <u>0 = Inner FEC does not provide information on transmit path data delay with ns resolution</u>	<u>RO</u>
<u>1.1800.5</u>	<u>TimeSync inner FEC receive path data delay ability, in sub-ns</u>	<u>1 = Inner FEC provides information on receive path data delay with sub-ns resolution in registers 1.1823 and 1.1824</u> <u>0 = Inner FEC does not provide information on receive path data delay with sub-ns resolution</u>	<u>RO</u>
<u>1.1800.4</u>	<u>TimeSync inner FEC receive path data delay ability, in ns</u>	<u>1 = Inner FEC provides information on receive path data delay with ns resolution in registers 1.1819 through 1.1823</u> <u>0 = Inner FEC does not provide information on receive path data delay with ns resolution</u>	<u>RO</u>
1.1800.3	TimeSync transmit path data delay ability, in sub-ns	1 = PMA/PMD provides information on transmit path data delay with sub-ns resolution in registers 1.1809 and 1.1810 0 = PMA/PMD does not provide information on transmit path data delay with sub-ns resolution	RO
1.1800.2	TimeSync receive path data delay ability, in sub-ns	1 = PMA/PMD provides information on receive path data delay with sub-ns resolution in registers 1.1811 and 1.1812 0 = PMA/PMD does not provide information on receive path data delay with sub-ns resolution	RO
1.1800.1	TimeSync transmit path data delay ability, in ns	1 = PMA/PMD provides information on transmit path data delay with ns resolution in registers 1.1801 through 1.1804 0 = PMA/PMD does not provide information on transmit path data delay with ns resolution	RO
1.1800.0	TimeSync receive path data delay ability, in ns	1 = PMA/PMD provides information on receive path data delay with ns resolution in registers 1.1805 through 1.1808 0 = PMA/PMD does not provide information on receive path data delay with ns resolution	RO

^a RO = Read only.

Insert 45.2.1.177a through 45.2.1.177c after 45.2.1.177 as follows:

45.2.1.177a TimeSync inner FEC transmit path data delay (Registers 1.1813 through 1.1818)

The TimeSync inner FEC transmit path data delay registers contain the maximum and minimum values of the inner FEC transmit path data delay. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum inner FEC transmit path data delay value is provided in two components. The first component (registers 1.1813 and 1.1814, see Table 45–141a) provides the integer nanoseconds portion of the maximum inner FEC transmit path data delay, in units of nanoseconds. The second component (register 1.1817, see Table 45–141a) provides the fractional nanoseconds portion of the maximum inner FEC transmit path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.6 and 1.1800.7, see Table 45–139), the maximum inner FEC transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum inner FEC transmit path data delay.

The minimum inner FEC transmit path data delay value is provided in two components. The first component (registers 1.1815 and 1.1816, see Table 45–141a) provides the integer nanoseconds portion of the minimum inner FEC transmit path data delay, in units of nanoseconds. The second component (register 1.1818, see Table 45–141a) provides the fractional nanoseconds portion of the minimum inner FEC transmit path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.6 and 1.1800.7, see Table 45–139), the minimum inner FEC transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum inner FEC transmit path data delay.

Table 45–141a—TimeSync inner FEC transmit path data delay register

Bit(s)	Name	Description	R/W ^a
1.1813.15:0	Maximum inner FEC transmit path data delay in ns, lower	inner_FEC_delay_ns_TX_max [15:0]	RO, MW
1.1814.15:0	Maximum inner FEC transmit path data delay in ns, upper	inner_FEC_delay_ns_TX_max [31:16]	RO, MW
1.1815.15:0	Minimum inner FEC transmit path data delay in ns, lower	inner_FEC_delay_ns_TX_min [15:0]	RO, MW
1.1816.15:0	Minimum inner FEC transmit path data delay in ns, upper	inner_FEC_delay_ns_TX_min [31:16]	RO, MW
1.1817.15:0	Maximum inner FEC transmit path data delay in sub-ns	inner_FEC_delay_subns_TX_max [15:0]	RO
1.1818.15:0	Minimum inner FEC transmit path data delay in sub-ns	inner_FEC_delay_subns_TX_min [15:0]	RO

^a RO = Read only, MW = Multi-word.

45.2.1.177b TimeSync inner FEC receive path data delay (Registers 1.1819 through 1.1824)

The TimeSync inner FEC receive path data delay registers contain the maximum and minimum values of the inner FEC receive path data delay. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum inner FEC receive path data delay value is provided in two components. The first component (registers 1.1819 and 1.1820, see Table 45–141b) provides the integer nanoseconds portion of the maximum inner FEC receive path data delay, in units of nanoseconds. The second component (register 1.1823, see Table 45–141b) provides the fractional nanoseconds portion of the maximum inner FEC receive path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.4 and 1.1800.5, see Table 45–139), the maximum inner FEC receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum inner FEC receive path data delay.

The minimum inner FEC receive path data delay value is provided in two components. The first component (registers 1.1821 and 1.1822, see Table 45–141b) provides the integer nanoseconds portion of the minimum inner FEC receive path data delay, in units of nanoseconds. The second component (register 1.1824, see Table 45–141b) provides the fractional nanoseconds portion of the minimum inner FEC receive path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.4 and 1.1800.5, see Table 45–139), the minimum inner FEC receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum inner FEC receive path data delay.

Table 45–141b—TimeSync inner FEC receive path data delay register

Bit(s)	Name	Description	R/W ^a
1.1819.15:0	Maximum inner FEC receive path data delay in ns, lower	inner_FEC_delay_ns_RX_max [15:0]	RO, MW
1.1820.15:0	Maximum inner FEC receive path data delay in ns, upper	inner_FEC_delay_ns_RX_max [31:16]	RO, MW
1.1821.15:0	Minimum inner FEC receive path data delay in ns, lower	inner_FEC_delay_ns_RX_min [15:0]	RO, MW
1.1822.15:0	Minimum inner FEC receive path data delay in ns, upper	inner_FEC_delay_ns_RX_min [31:16]	RO, MW
1.1823.15:0	Maximum inner FEC receive path data delay in sub-ns	inner_FEC_delay_subns_RX_max [15:0]	RO
1.1824.15:0	Minimum inner FEC receive path data delay in sub-ns	inner_FEC_delay_subns_RX_min [15:0]	RO

^a RO = Read only, MW = Multi-word.

45.2.1.177c TimeSync FEC configuration (Register 1.1825)

The assignment of bits in the TimeSync FEC configuration register is shown in Table 45–141c.

Table 45–141c—TimeSync FEC configuration

Bit(s)	Name	Description	R/W ^a
1.1825.15	Reserved	Value always 0	RO
1.1825.14	alignment marker location enable	0 = alignment marker location is not enabled 1 = alignment marker location is enabled	R/W
1.1825.13:0	Reserved	Value always 0	RO

^a RO = Read only, R/W = Read/Write.

45.2.1.177c.1 FEC alignment marker location enable (1.1825.14)

Bit 1.1825.14 enables and disables alignment marker location, see 186.2.3.1.5.

45.2.1.216 IFEC control register (Register 1.2200)

Change Table 45–180 as follows:

Table 45–180—IFEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
<u>1.2200.15:5</u> <u>1.2200.15:3</u>	Reserved	Value always 0	RO
<u>1.2200.4</u>	<u>IFEC degraded SER enable</u>	<u>1 =IFEC decoder indicates degraded SER</u> <u>0 = IFEC decoder does not indicate degraded SER</u>	<u>R/W</u>
<u>1.2200.3</u>	<u>Reserved</u>	<u>Value always 0</u>	<u>RO</u>
1.2200.2	IFEC enable	1 = Inverse RS-FEC is enabled 0 = Inverse RS-FEC is disabled	R/W
1.2200.1	IFEC bypass indication enable	1 = IFEC decoder does not indicate errors 0 = IFEC decoder indicates errors	R/W
1.2200.0	IFEC bypass correction enable	1 = IFEC decoder performs error detection without error correction 0 = IFEC decoder performs error detection and error correction	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.217 IFEC status register (Register 1.2201)

Change Table 45–181 as follows:

Table 45–181—IFEC status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2201.15	PCS align status	1 = IFEC encoder has locked and aligned all PCS lanes 0 = IFEC encoder has not locked and aligned all PCS lanes	RO
1.2201.14	IFEC align status	1 = IFEC receive lanes locked and aligned 0 = IFEC receive lanes not locked and aligned	RO
1.2201.13:12	Reserved	Value always 0	RO
1.2201.11	IFEC AM lock 3	1 = IFEC lane 3 locked and aligned 0 = IFEC lane 3 not locked and aligned	RO
1.2201.10	IFEC AM lock 2	1 = IFEC lane 2 locked and aligned 0 = IFEC lane 2 not locked and aligned	RO
1.2201.9	IFEC AM lock 1	1 = IFEC lane 1 locked and aligned 0 = IFEC lane 1 not locked and aligned	RO
1.2201.8	IFEC AM lock 0	1 = IFEC lane 0 locked and aligned 0 = IFEC lane 0 not locked and aligned	RO
1.2201.7:6 1.2201.7:3	Reserved	Value always 0	RO
<u>1.2201.5</u>	<u>Received local degraded</u>	<u>1 = IFEC has received local degraded status indication</u> <u>0 = IFEC has not received local degraded status indication</u>	<u>RO</u>
<u>1.2201.4</u>	<u>Received remote degraded</u>	<u>1 = IFEC has received remote degraded status indication</u> <u>0 = IFEC has not received remote degraded status</u>	<u>RO</u>
<u>1.2201.3</u>	<u>IFEC degraded SER ability</u>	<u>1 = IFEC decoder has the ability to indicate the presence of a degraded SER</u> <u>0 = IFEC decoder does not have the ability to indicate the presence of a degraded SER</u>	<u>RO</u>
1.2201.2	IFEC high SER	1 = IFEC errors have exceeded threshold 0 = IFEC errors have not exceeded threshold	RO/LH
1.2201.1	IFEC bypass indication ability	1 = IFEC decoder has the ability to bypass error indication 0 = IFEC decoder does not have the ability to bypass error indication	RO
1.2201.0	IFEC bypass correction ability	1 = IFEC decoder has the ability to bypass error correction 0 = IFEC decoder does not have the ability to bypass error correction	RO

^a RO = Read only, LH = Latching high.

Insert 45.2.1.217.6a through 45.2.1.217.6c after 45.2.1.217.6 as follows:

45.2.1.217.6a IFEC received local degraded (1.2201.5)

Bit 1.2201.4 is set to one when the IFEC receiver detects the value 1 in the local degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames. Bit 1.2201.4 is set to zero on reset or when the IFEC receiver detects the value 0 in the local degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames.

45.2.1.217.6b IFEC received remote degraded (1.2201.4)

Bit 1.2201.4 is set to one when the IFEC receiver detects the value 1 in the remote degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames. Bit 1.2201.4 is set to zero on reset or when the IFEC receiver detects the value 0 in the remote degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames.

45.2.1.217.6c IFEC degraded SER ability (1.2201.3)

This bit is set to one to indicate that the decoder has the ability to indicate the presence of a degraded SER. The bit is set to zero if this ability is not supported.

Insert 45.2.1.220a after 45.2.1.220 as follows:

45.2.1.220a IFEC codeword counter register (Register 1.2207, 1.2208, 1.2209)

The assignment of bits in the IFEC codeword counter register is shown in Table 45–184a. It is a 48-bit counter that counts once for each IFEC codeword received when align status is set to one. Its bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Registers 1.2207, 1.2209, and 1.2209 are used to read the value of the 48-bit counter. When using these registers to read the 48-bit counter value, the register 1.2207 is read first, the values of the registers 1.2208 and 1.2209 are latched when (and only when) register 1.2207 is read, and reads of registers 1.2208 and 1.2209 return their latched values rather than the current value of the counter.

Table 45–184a—IFEC codeword counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2207.15:0	IFEC codewords lower	IFEC_cw_counter[15:0]	RO, NR
1.2208.15:0	IFEC codewords middle	IFEC_cw_counter[31:16]	RO, NR
1.2209.15:0	IFEC codewords upper	IFEC_cw_counter[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

Change 45.2.1.222 as follows:

45.2.1.222 IFEC symbol error counter, lane 1 through ~~31~~ 3 (Register 1.2212, ~~1.2243, 1.2244, 1.2245, 1.2246, 1.2247~~ through 2241, and 1.3170 through 1.3199)

The behavior of the IFEC symbol error counters, lane 1 through ~~31~~ 3 is identical to that described for FEC lane 0 in 45.2.1.221. Errors detected in each FEC lane are counted and shown in the corresponding register. FEC lane 1, lower 16 bits are shown in register 1.2212; FEC lane 1, upper 16 bits are shown in register 1.2213; FEC lane 2, lower 16 bits are shown in register 1.2214; through register 1.2217 for FEC lane 3, upper 16 bits; and so on.

Insert 45.2.1.251 through 45.2.1.269 after 45.2.1.250 as follows:

45.2.1.251 Inner FEC PRBS13Q testing control register (Register 1.2401)

The Inner FEC PRBS13Q testing control register is used for PHY types that implement PRBS13Q testing in the inner FEC. The assignment of bits in the PRBS13Q testing control register is shown in Table 45–212a.

Table 45–212a—Inner FEC PRBS13Q testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2401.15:8	Reserved	Value always 0	RO
1.2401.7	Lane 7 PRBS13Q enable	1 = Enable PRBS13Q on lane 7 0 = Disable PRBS13Q on lane 7	R/W
1.2401.6	Lane 6 PRBS13Q enable	1 = Enable PRBS13Q on lane 6 0 = Disable PRBS13Q on lane 6	R/W
1.2401.5	Lane 5 PRBS13Q enable	1 = Enable PRBS13Q on lane 5 0 = Disable PRBS13Q on lane 5	R/W
1.2401.4	Lane 4 PRBS13Q enable	1 = Enable PRBS13Q on lane 4 0 = Disable PRBS13Q on lane 4	R/W
1.2401.3	Lane 3 PRBS13Q enable	1 = Enable PRBS13Q on lane 3 0 = Disable PRBS13Q on lane 3	R/W
1.2401.2	Lane 2 PRBS13Q enable	1 = Enable PRBS13Q on lane 2 0 = Disable PRBS13Q on lane 2	R/W
1.2401.1	Lane 1 PRBS13Q enable	1 = Enable PRBS13Q on lane 1 0 = Disable PRBS13Q on lane 1	R/W
1.2401.0	Lane 0 PRBS13Q enable	1 = Enable PRBS13Q on lane 0 0 = Disable PRBS13Q on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.252 Inner FEC PRBS31 testing control register (Register 1.2402)

The Inner FEC PRBS31 testing control register is used for PHY types that implement PRBS31 testing in the inner FEC. The assignment of bits in the PRBS31 testing control register is shown in Table 45–212b.

Table 45–212b—Inner FEC PRBS31 testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2402.15:8	Reserved	Value always 0	RO
1.2402.7	Lane 7 PRBS31 enable	1 = Enable PRBS31 on lane 7 0 = Disable PRBS31 on lane 7	R/W
1.2402.6	Lane 6 PRBS31 enable	1 = Enable PRBS31 on lane 6 0 = Disable PRBS31 on lane 6	R/W
1.2402.5	Lane 5 PRBS31 enable	1 = Enable PRBS31 on lane 5 0 = Disable PRBS31 on lane 5	R/W
1.2402.4	Lane 4 PRBS31 enable	1 = Enable PRBS31 on lane 4 0 = Disable PRBS31 on lane 4	R/W
1.2402.3	Lane 3 PRBS31 enable	1 = Enable PRBS31 on lane 3 0 = Disable PRBS31 on lane 3	R/W
1.2402.2	Lane 2 PRBS31 enable	1 = Enable PRBS31 on lane 2 0 = Disable PRBS31 on lane 2	R/W
1.2402.1	Lane 1 PRBS31 enable	1 = Enable PRBS31 on lane 1 0 = Disable PRBS31 on lane 1	R/W
1.2402.0	Lane 0 PRBS31 enable	1 = Enable PRBS31 on lane 0 0 = Disable PRBS31 on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.253 Inner FEC SSPRQ testing control register (Register 1.2403)

The Inner FEC SSPRQ testing control register is used for PHY types that implement SSPRQ testing in the inner FEC. The assignment of bits in the SSPRQ testing control register is shown in Table 45–212c.

Table 45–212c—Inner FEC SSPRQ testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2403.15:8	Reserved	Value always 0	RO
1.2403.7	Lane 7 SSPRQ enable	1 = Enable SSPRQ on lane 7 0 = Disable SSPRQ on lane 7	R/W
1.2403.6	Lane 6 SSPRQ enable	1 = Enable SSPRQ on lane 6 0 = Disable SSPRQ on lane 6	R/W
1.2403.5	Lane 5 SSPRQ enable	1 = Enable SSPRQ on lane 5 0 = Disable SSPRQ on lane 5	R/W
1.2403.4	Lane 4 SSPRQ enable	1 = Enable SSPRQ on lane 4 0 = Disable SSPRQ on lane 4	R/W
1.2403.3	Lane 3 SSPRQ enable	1 = Enable SSPRQ on lane 3 0 = Disable SSPRQ on lane 3	R/W

Table 45–212c—Inner FEC SSPRQ testing control register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.2403.2	Lane 2 SSPRQ enable	1 = Enable SSPRQ on lane 2 0 = Disable SSPRQ on lane 2	R/W
1.2403.1	Lane 1 SSPRQ enable	1 = Enable SSPRQ on lane 1 0 = Disable SSPRQ on lane 1	R/W
1.2403.0	Lane 0 SSPRQ enable	1 = Enable SSPRQ on lane 0 0 = Disable SSPRQ on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.254 Inner FEC Square Wave testing control register (Register 1.2404)

The Inner FEC Square Wave testing control register is used for PHY types that implement Square Wave testing in the Inner FEC. The assignment of bits in the Square Wave testing control register is shown in Table 45–212d.

Table 45–212d—Inner FEC Square Wave testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2404.15:8	Reserved	Value always 0	RO
1.2404.7	Lane 7 Square Wave enable	1 = Enable Square Wave on lane 7 0 = Disable Square Wave on lane 7	R/W
1.2404.6	Lane 6 Square Wave enable	1 = Enable Square Wave on lane 6 0 = Disable Square Wave on lane 6	R/W
1.2404.5	Lane 5 Square Wave enable	1 = Enable Square Wave on lane 5 0 = Disable Square Wave on lane 5	R/W
1.2404.4	Lane 4 Square Wave enable	1 = Enable Square Wave on lane 4 0 = Disable Square Wave on lane 4	R/W
1.2404.3	Lane 3 Square Wave enable	1 = Enable Square Wave on lane 3 0 = Disable Square Wave on lane 3	R/W
1.2404.2	Lane 2 Square Wave enable	1 = Enable Square Wave on lane 2 0 = Disable Square Wave on lane 2	R/W
1.2404.1	Lane 1 Square Wave enable	1 = Enable Square Wave on lane 1 0 = Disable Square Wave on lane 1	R/W
1.2404.0	Lane 0 Square Wave enable	1 = Enable Square Wave on lane 0 0 = Disable Square Wave on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.255 PRBS31 encoded by Inner FEC control register (Register 1.2405)

The PRBS31 encoded by Inner FEC control register is used to enable PRBS31 test patterns from the PMA layer above the Inner FEC on a per lane basis. The assignment of bits in the PRBS31 encoded by Inner FEC control register is shown in Table 45–212e.

Table 45–212e—PRBS31 encoded by Inner FEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2405.15:8	Reserved	Value always 0	RO
1.2405.7	Lane 7 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 7 0 = Disable PRBS31 encoded by Inner FEC on lane 7	R/W
1.2405.6	Lane 6 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 6 0 = Disable PRBS31 encoded by Inner FEC on lane 6	R/W
1.2405.5	Lane 5 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 5 0 = Disable PRBS31 encoded by Inner FEC on lane 5	R/W
1.2405.4	Lane 4 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 4 0 = Disable PRBS31 encoded by Inner FEC on lane 4	R/W
1.2405.3	Lane 3 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 3 0 = Disable PRBS31 encoded by Inner FEC on lane 3	R/W
1.2405.2	Lane 2 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 2 0 = Disable PRBS31 encoded by Inner FEC on lane 2	R/W
1.2405.1	Lane 1 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 1 0 = Disable PRBS31 encoded by Inner FEC on lane 1	R/W
1.2405.0	Lane 0 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 0 0 = Disable PRBS31 encoded by Inner FEC on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.256 Inner FEC alignment status register (Register 1.2410)

The assignment of bits in the Inner FEC alignment status register is shown in Table 45–212f.

Table 45–212f—Inner FEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2410.15	Inner FEC alignment 7	1 = Lane 7 transmit direction is aligned 0 = Lane 7 transmit direction is not aligned	RO
1.2410.14	Inner FEC alignment 6	1 = Lane 6 transmit direction is aligned 0 = Lane 6 transmit direction is not aligned	RO
1.2410.13	Inner FEC alignment 5	1 = Lane 5 transmit direction is aligned 0 = Lane 5 transmit direction is not aligned	RO
1.2410.12	Inner FEC alignment 4	1 = Lane 4 transmit direction is aligned 0 = Lane 4 transmit direction is not aligned	RO

Table 45–212f—Inner FEC alignment status register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.2410.11	Inner FEC alignment 3	1 = Lane 3 transmit direction is aligned 0 = Lane 3 transmit direction is not aligned	RO
1.2410.10	Inner FEC alignment 2	1 = Lane 2 transmit direction is aligned 0 = Lane 2 transmit direction is not aligned	RO
1.2410.9	Inner FEC alignment 1	1 = Lane 1 transmit direction is aligned 0 = Lane 1 transmit direction is not aligned	RO
1.2410.8	Inner FEC alignment 0	1 = Lane 0 transmit direction is aligned 0 = Lane 0 transmit direction is not aligned	RO
1.2410.7	Inner FEC lock 7	1 = Lane 7 receive direction is locked 0 = Lane 7 receive direction is not locked	RO
1.2410.6	Inner FEC lock 6	1 = Lane 6 receive direction is locked 0 = Lane 6 receive direction is not locked	RO
1.2410.5	Inner FEC lock 5	1 = Lane 5 receive direction is locked 0 = Lane 5 receive direction is not locked	RO
1.2410.4	Inner FEC lock 4	1 = Lane 4 receive direction is locked 0 = Lane 4 receive direction is not locked	RO
1.2410.3	Inner FEC lock 3	1 = Lane 3 receive direction is locked 0 = Lane 3 receive direction is not locked	RO
1.2410.2	Inner FEC lock 2	1 = Lane 2 receive direction is locked 0 = Lane 2 receive direction is not locked	RO
1.2410.1	Inner FEC lock 1	1 = Lane 1 receive direction is locked 0 = Lane 1 receive direction is not locked	RO
1.2410.0	Inner FEC lock 0	1 = Lane 0 receive direction is locked 0 = Lane 0 receive direction is not locked	RO

^a RO = Read only.

45.2.1.257 Inner FEC status 1 register (Register 1.2411)

The assignment of bits in the Inner FEC status 1 register is shown in Table 45–212g. All the bits in the Inner FEC status 1 register are read only; therefore, a write to the Inner FEC status 1 register shall have no effect.

45.2.1.258 Inner FEC corrected codewords counter (Register 1.2412, 1.2413)

Table 45–212h shows the assignment of bits for the lane 0 Inner FEC corrected codewords counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2412 and 1.2413 are used to read the value of a 32-bit counter. When registers 1.2412 and 1.2413 are used to read the 32-bit counter value,

Table 45–212g—PMA/PMD status 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2411.15:2	Reserved	Value always 0	RO
1.2411.1:0	DSP lock	Set to 1 when the 800GBASE-LR1 Inner FEC receive function has detected the location of the PS for a given polarization symbol stream (see 184.7.2.2)	RO

^a RO = Read only, LL = Latching low.

register 1.2412 is read first, the value of the register 1.2413 is latched when (and only when) register 1.2412 is read, and reads of register 1.2413 return the latched value rather than the current value of the counter.

Table 45–212h—Inner FEC corrected codewords counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2412.15:0	FEC corrected codewords lower	FEC_corrected_cw_counter[15:0]	RO, NR
1.2413.15:0	FEC corrected codewords upper	FEC_corrected_cw_counter[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.259 Inner FEC uncorrected codewords counter (Register 1.2414, 1.2415)

Table 45–212i shows the assignment of bits for the lane 0 Inner FEC uncorrected codewords counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2414 and 1.2415 are used to read the value of a 32-bit counter. When registers 1.2414 and 1.2415 are used to read the 32-bit counter value, register 1.2414 is read first, the value of the register 1.2415 is latched when (and only when) register 1.2414 is read, and reads of register 1.2415 return the latched value rather than the current value of the counter.

Table 45–212i—Inner FEC uncorrected codewords counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2414.15:0	FEC uncorrected codewords lower	FEC_uncorrected_cw_counter[15:0]	RO, NR
1.2415.15:0	FEC uncorrected codewords upper	FEC_uncorrected_cw_counter[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.260 Inner FEC total bits register (Register 1.2416, 1.2417, 1.2418, 1.2419)

Table 45–212j shows the assignment of bits for the lane 0 Inner FEC total bits register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2416, 1.2417, 1.2418, and 1.2419 are used to read the value of a 64-bit counter. When registers 1.2416, 1.2417, 1.2418, and 1.2419 are used to read the 64-bit

counter value, register 1.2416 is read first, the values of registers 1.2417, 1.2418, and 1.2419 are latched when (and only when) register 1.2416 is read, and reads of registers 1.2417, 1.2418, and 1.2419 return the latched value rather than the current value of the counter.

Table 45–212j—Inner FEC total bits register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2416.15:0	FEC total bits lowest	FEC_total_bits[15:0]	RO, NR
1.2417.15:0	FEC total bits	FEC_total_bits[31:16]	RO, NR
1.2418.15:0	FEC total bits	FEC_total_bits[47:32]	RO, NR
1.2419.15:0	FEC total bits highest	FEC_total_bits[64:48]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.261 Inner FEC corrected bits register (Register 1.2420, 1.2421, 1.2422, 1.2423)

Table 45–212k shows the assignment of bits for the lane 0 Inner FEC corrected bits register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2420, 1.2421, 1.2422, and 1.2423 are used to read the value of a 64-bit counter. When registers 1.2420, 1.2421, 1.2422, and 1.2423 are used to read the 64-bit counter value, register 1.2420 is read first, the values of registers 1.2421, 1.2422, and 1.2423 are latched when (and only when) register 1.2420 is read, and reads of registers 1.2421, 1.2422, and 1.2423 return the latched value rather than the current value of the counter.

Table 45–212k—Inner FEC corrected bits register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2420.15:0	FEC corrected bits lowest	FEC_corrected_bits[15:0]	RO, NR
1.2421.15:0	FEC corrected bits	FEC_corrected_bits[31:16]	RO, NR
1.2422.15:0	FEC corrected bits	FEC_corrected_bits[47:32]	RO, NR
1.2423.15:0	FEC corrected bits highest	FEC_corrected_bits[64:48]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.262 Inner FEC codeword error bin registers 0 through 4 (Registers 1.2424 through 1.2433)

Table 45–212l shows the assignment of bits for the lane 0 Inner FEC codeword error bin registers. The Inner FEC codeword error bin registers increment depending upon the error signature of a corrected codeword. Their bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Two registers are used to read the value of each 32-bit counter, the value of the second register is latched when the first register is read.

The bin 0 register (1.2424, 1.2425) keeps a count of codewords with no bit errors, the bin 1 register (1.2426, 1.2427) keeps a count of codewords with 1 bit error corrected, the bin 2 register (1.2428, 1.2429) keeps a count of codewords with 2 bits corrected, the bin 3 (1.2430, 1.2431) register keeps a count of codewords with 3 bits corrected, and the bin 4 (1.2432, 1.2433) register keeps a count of codewords with 4 or more bits corrected.

Table 45–212I—Inner FEC codeword error bin register definitions

Bit(s)	Name	Description	R/W ^a
1.2424.15:0	Inner FEC codeword error bin 0 lower	inner_FEC_codeword_error_bin_0[15:0]	RO, NR
1.2425.15:0	Inner FEC codeword error bin 0 upper	inner_FEC_codeword_error_bin_0[31:16]	RO, NR
1.2426.15:0	Inner FEC codeword error bin 1 lower	inner_FEC_codeword_error_bin_1[15:0]	RO, NR
1.2427.15:0	Inner FEC codeword error bin 1 upper	inner_FEC_codeword_error_bin_1[31:16]	RO, NR
1.2428.15:0	Inner FEC codeword error bin 2 lower	inner_FEC_codeword_error_bin_2[15:0]	RO, NR
1.2429.15:0	Inner FEC codeword error bin 2 upper	inner_FEC_codeword_error_bin_2[31:16]	RO, NR
1.2430.15:0	Inner FEC codeword error bin 3 lower	inner_FEC_codeword_error_bin_3[15:0]	RO, NR
1.2431.15:0	Inner FEC codeword error bin 3 upper	inner_FEC_codeword_error_bin_3[31:16]	RO, NR
1.2432.15:0	Inner FEC codeword error bin 4 lower	inner_FEC_codeword_error_bin_4[15:0]	RO, NR
1.2433.15:0	Inner FEC codeword error bin 4 upper	inner_FEC_codeword_error_bin_4[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.263 Inner FEC counter registers for lanes 1 through 7 (Registers 1.2434 through 1.2587)

The twenty two counter registers 1.2412 through 1.2433 are repeated for each Inner FEC lane present, with registers 1.2412 through 1.2433 being for lane 0, registers 1.2434 through 1.2455 being for lane 1, etc.

45.2.1.264 PMA test block error bin counter registers (Registers 1.2600 through 1.3007)

The PMA test block error bin counter registers provide emulation of FEC error statistics from a PRBS data stream. These registers are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow. Three registers are used to read the value of each 48-bit counter. The values of all registers for a given PMAL are latched when the first register of bin 0 is read.

There are 17 bin counters for eight PMALs. Each 48-bit counter consumes three registers, so 17 counters for eight PMALs spans a total of 408 registers. The bin 0 register keeps a count of test blocks with no test symbol errors. The bin 1 register keeps a count of test blocks with 1 test symbol error. The bin 2 register

keeps a count of test blocks with 2 test symbol errors, and so on up to 15 test symbol errors. The bin 16p register counts test blocks with 16 or more test symbol errors.

Table 45–212m shows the assignment of bits for the lane 0 PMA test block error bin 0 to bin 3 registers. Lane 0's error bin registers end at 1.2650 with test_block_error_bin_0_16p upper. Lane 1's error bin registers start at 1.2651 with test_block_error_bin_1_0 lower.

Table 45–212m—Lane 0 PMA test block error bin register definitions (for 4 of the 17 bins)

Bit(s)	Name	Description	R/W ^a
1.2600.15:0	PMA test block error bin 0 lower	test_block_error_bin_0_0[15:0]	RO, NR
1.2601.15:0	PMA test block error bin 0 middle	test_block_error_bin_0_0[31:16]	RO, NR
1.2602.15:0	PMA test block error bin 0 upper	test_block_error_bin_0_0[47:32]	RO, NR
1.2603.15:0	PMA test block error bin 1 lower	test_block_error_bin_0_1[15:0]	RO, NR
1.2604.15:0	PMA test block error bin 1 middle	test_block_error_bin_0_1[31:16]	RO, NR
1.2605.15:0	PMA test block error bin 1 upper	test_block_error_bin_0_1[47:32]	RO, NR
1.2606.15:0	PMA test block error bin 2 lower	test_block_error_bin_0_2[15:0]	RO, NR
1.2607.15:0	PMA test block error bin 2 middle	test_block_error_bin_0_2[31:16]	RO, NR
1.2608.15:0	PMA test block error bin 2 upper	test_block_error_bin_0_2[47:32]	RO, NR
1.2609.15:0	PMA test block error bin 3 lower	test_block_error_bin_0_3[15:0]	RO, NR
1.2610.15:0	PMA test block error bin 3 middle	test_block_error_bin_0_3[31:16]	RO, NR
1.2611.15:0	PMA test block error bin 3 upper	test_block_error_bin_0_3[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.265 IFEC lane 0 mapping register (Register 1.3100)

The assignment of bits in the IFEC lane 0 mapping register is shown in Table 45–212n. When the IFEC detects and locks the RS-FEC for IFEC service interface lane 0, the detected RS-FEC lane number is recorded in this register. The contents of the Lane 0 mapping register is valid when the Inverse RS-FEC align status (1.2201.14) is set to one and are invalid otherwise.

Table 45–212n—IFEC lane 0 mapping register bit definitions

Bit(s)	Name	Description	R/W ^a
1.3100.15:6	Reserved	Value always 0	RO
1.3100.5:0	Lane 0 mapping	PCS lane received in service interface lane 0	RO

^a RO = Read only.

45.2.1.266 IFEC lanes 1 through 31 mapping registers (Registers 1.3101 through 1.3131)

The definition of Lanes 1 through 31 mapping registers is identical to that described for lane 0 in 45.2.1.265. The lane mapping for lane 1 is in register 1.3101; lane 2 is in register 1.3102; etc.

45.2.1.267 IFEC alignment status register (Register 1.3132, 1.3133)

The assignment of bits in the IFEC alignment status register is shown in Table 45–212o and Table 45–212p. All the bits in the IFEC alignment status register are read only; a write to the IFEC PCS alignment status register shall have no effect.

Table 45–212o—IFEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.3132.15	Lane 15 aligned	1 = Lane 15 alignment marker is locked 0 = Lane 15 alignment marker is not locked	RO
1.3132.14	Lane 14 aligned	1 = Lane 14 alignment marker is locked 0 = Lane 14 alignment marker is not locked	RO
1.3132.13	Lane 13 aligned	1 = Lane 13 alignment marker is locked 0 = Lane 13 alignment marker is not locked	RO
1.3132.12	Lane 12 aligned	1 = Lane 12 alignment marker is locked 0 = Lane 12 alignment marker is not locked	RO
1.3132.11	Lane 11 aligned	1 = Lane 11 alignment marker is locked 0 = Lane 11 alignment marker is not locked	RO
1.3132.10	Lane 10 aligned	1 = Lane 10 alignment marker is locked 0 = Lane 10 alignment marker is not locked	RO
1.3132.9	Lane 9 aligned	1 = Lane 9 alignment marker is locked 0 = Lane 9 alignment marker is not locked	RO
1.3132.8	Lane 8 aligned	1 = Lane 8 alignment marker is locked 0 = Lane 8 alignment marker is not locked	RO
1.3132.7	Lane 7 aligned	1 = Lane 7 alignment marker is locked 0 = Lane 7 alignment marker is not locked	RO
1.3132.6	Lane 6 aligned	1 = Lane 6 alignment marker is locked 0 = Lane 6 alignment marker is not locked	RO
1.3132.5	Lane 5 aligned	1 = Lane 5 alignment marker is locked 0 = Lane 5 alignment marker is not locked	RO
1.3132.4	Lane 4 aligned	1 = Lane 4 alignment marker is locked 0 = Lane 4 alignment marker is not locked	RO
1.3132.3	Lane 3 aligned	1 = Lane 3 alignment marker is locked 0 = Lane 3 alignment marker is not locked	RO

Table 45–212o—IFEC alignment status register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.3132.2	Lane 2 aligned	1 = Lane 2 alignment marker is locked 0 = Lane 2 alignment marker is not locked	RO
1.3132.1	Lane 1 aligned	1 = Lane 1 alignment marker is locked 0 = Lane 1 alignment marker is not locked	RO
1.3132.0	Lane 0 aligned	1 = Lane 0 alignment marker is locked 0 = Lane 0 alignment marker is not locked	RO

^a RO = Read only.

Table 45–212p—IFEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.3133.31	Lane 31 aligned	1 = Lane 31 alignment marker is locked 0 = Lane 31 alignment marker is not locked	RO
1.3133.30	Lane 30 aligned	1 = Lane 30 alignment marker is locked 0 = Lane 30 alignment marker is not locked	RO
1.3133.29	Lane 29 aligned	1 = Lane 29 alignment marker is locked 0 = Lane 29 alignment marker is not locked	RO
1.3133.28	Lane 28 aligned	1 = Lane 28 alignment marker is locked 0 = Lane 28 alignment marker is not locked	RO
1.3133.27	Lane 27 aligned	1 = Lane 27 alignment marker is locked 0 = Lane 27 alignment marker is not locked	RO
1.3133.26	Lane 26 aligned	1 = Lane 26 alignment marker is locked 0 = Lane 26 alignment marker is not locked	RO
1.3133.25	Lane 25 aligned	1 = Lane 25 alignment marker is locked 0 = Lane 25 alignment marker is not locked	RO
1.3133.24	Lane 24 aligned	1 = Lane 24 alignment marker is locked 0 = Lane 24 alignment marker is not locked	RO
1.3133.23	Lane 23 aligned	1 = Lane 23 alignment marker is locked 0 = Lane 23 alignment marker is not locked	RO
1.3133.22	Lane 22 aligned	1 = Lane 22 alignment marker is locked 0 = Lane 22 alignment marker is not locked	RO
1.3133.21	Lane 21 aligned	1 = Lane 21 alignment marker is locked 0 = Lane 21 alignment marker is not locked	RO
1.3133.20	Lane 20 aligned	1 = Lane 20 alignment marker is locked 0 = Lane 20 alignment marker is not locked	RO
1.3133.19	Lane 19 aligned	1 = Lane 19 alignment marker is locked 0 = Lane 19 alignment marker is not locked	RO

Table 45–212p—IFEC alignment status register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.3133.18	Lane 18 aligned	1 = Lane 18 alignment marker is locked 0 = Lane 18 alignment marker is not locked	RO
1.3133.17	Lane 17 aligned	1 = Lane 17 alignment marker is locked 0 = Lane 17 alignment marker is not locked	RO
1.3133.16	Lane 16 aligned	1 = Lane 16 alignment marker is locked 0 = Lane 16 alignment marker is not locked	RO

^a RO = Read only.

45.2.1.268 IFEC codeword error bin registers 1 through 15 (Registers 1.3134 through 1.3163)

The assignment of bits in the IFEC codeword error bin 1 register is shown in Table 45–212q. The assignment of bits in the other IFEC codeword error bin registers is identical to that of bin 1. The IFEC codeword error bin registers increment depending upon the error signature of a corrected codeword. Their bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Two registers are used to read the value of each 32-bit counter, the value of the second register is latched when the first register is read. For example when registers 1.3134 and 1.3135 are used to read the 32-bit counter value of FEC_codeword_error_bin_1, the register 1.3134 is read first, the value of the register 1.3135 is latched when (and only when) register 1.3134 is read, and reads of register 1.3135 return the latched value rather than the current value of the counter.

Table 45–212q—IFEC codeword error bin 1 bit definitions

Bit(s)	Name	Description	R/W ^a
1.3134.15:0	IFEC codeword error bin 1 lower	IFEC_codeword_error_bin_1[15:0]	RO, NR
1.3135.15:0	IFEC codeword error bin 1 upper	IFEC_codeword_error_bin_1[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.269 Duplication of ILT training registers for the upper AUI component (Registers 1.4000 through 1.5449)

Inter sublayer training requires control registers for the upper and lower AUI components. The upper AUI component has the same control functionality as the lower AUI component so the relevant registers are duplicated with an address offset of 4000.

45.2.3 PCS registers

45.2.3.1 PCS control 1 register (Register 3.0)

Change Table 45–234 (as amended by IEEE Std 802.3cs-2022) as follows (unchanged rows not shown):

Table 45–1—PCS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.0.5:2	Speed selection	5 4 3 2 1 1 1 * 1 1 0 1 1 1 1 1 1 1 1 0 1 1 0 1 1 0 1 1 1 0 1 0 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 0 1 0 1 0 1 0 0 0 0 1 1 0 0 1 0 0 0 0 1 0 0 0 0 = Reserved = Reserved = Reserved = 1.6 Tb/s = 800 Gb/s = 10/2.5 Gb/s = 25/10 Gb/s = 400 Gb/s = 200 Gb/s = 5 Gb/s = 2.5 Gb/s = 50 Gb/s = 25 Gb/s = 100 Gb/s = 40 Gb/s = 10/1 Gb/s = 10PASS-TS/2BASE-TL = 10 Gb/s	R/W
...			

^a RO = Read only, R/W = Read/Write, SC = Self-clearing.

Editor's note (to be removed after first working group ballot): 800G selection is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1437.pdf

45.2.3.2 PCS status 1 register (Register 3.1)

Change 45.2.3.2.7 as follows

45.2.3.2.7 PCS receive link status (3.1.2)

When read as a one, bit 3.1.2 indicates that the PCS receive link is up. When read as a zero, bit 3.1.2 indicates that the PCS receive link is down. When a 10/25/40/50/100/200/400/800GBASE-R, 1.6TBASE-R, 10GBASE-W, or any MultiGBASE-T mode of operation is selected for the PCS using the PCS type selection field (3.7.4:0 ~~3.7.3:0~~), this bit is a latching low version of bit 3.32.12. When a 10GBASE-X mode of operation is selected for the PCS using the PCS type selection field (3.7.4:0 ~~3.7.3:0~~), this bit is a latching low version of bit 3.24.12. The receive link status bit shall be implemented with latching low behavior.

Editor's note (to be removed after first working group ballot): 800G status is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1438.pdf

45.2.3.4 PCS speed ability (Register 3.4)

Change Table 45–236 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–236—PCS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
3.4.15:12 3.4.15:11	Reserved for future speeds	Value always 0	RO
<u>3.4.11</u>	<u>1.6T capable</u>	<u>1 = PCS is capable of operating at 1.6 Tb/s</u> <u>0 = PCS is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
...			

^a RO = Read only

Insert 45.2.3.4.12 after 45.2.3.4.11 as follows:

45.2.3.4.12 1.6T capable (3.4.11)

When read as a one, bit 3.4.11 indicates that the PCS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 3.4.11 indicates that the PCS is not able to operate at a data rate of 1.6 Tb/s.

45.2.3.6 PCS control 2 register (Register 3.7)

Change Table 45–237 (as amended by IEEE Std 802.3cs-2022) as follows:

Table 45–237—PCS control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.7.15:5	Reserved	Value always 0	RO
3.7.4:0	PCS type selection	4 3 2 1 0 1 1 x x x = reserved 1 0 1 1 x = reserved 1 0 1 1 1 = Select 1.6TBASE-R PCS type 1 0 1 1 0 = Select 800GBASE-R PCS type 1 0 1 0 1 = Select 10GBASE-SP PCS type 1 0 1 0 0 = Select 10/2.5GBASE-SP PCS type 1 0 0 1 1 = Select 25GBASE-PQ PCS type 1 0 0 1 0 = Select 25/10GBASE-PQ PCS type 1 0 0 0 1 = Select 25GBASE-PQ PCS type, Tx only 1 0 0 0 0 = Select 25GBASE-PQ PCS type, Rx only 0 1 1 1 1 = Select 5GBASE-R PCS type 0 1 1 1 0 = Select 2.5GBASE-X PCS type 0 1 1 0 1 = Select 400GBASE-R PCS type 0 1 1 0 0 = Select 200GBASE-R PCS type 0 1 0 1 1 = Select 5GBASE-T PCS type 0 1 0 1 0 = Select 2.5GBASE-T PCS type 0 1 0 0 1 = Select 25GBASE-T PCS type 0 1 0 0 0 = Select 50GBASE-R PCS type 0 0 1 1 1 = Select 25GBASE-R PCS type 0 0 1 1 0 = Select 40GBASE-T PCS type 0 0 1 0 1 = Select 100GBASE-R PCS type 0 0 1 0 0 = Select 40GBASE-R PCS type 0 0 0 1 1 = Select 10GBASE-T PCS type 0 0 0 1 0 = Select 10GBASE-W PCS type 0 0 0 0 1 = Select 10GBASE-X PCS type 0 0 0 0 0 = Select 10GBASE-R PCS type	R/W

^a RO = Read only, R/W = Read/Write.

Editor's note (to be removed after first working group ballot): 800G selection is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1437.pdf

45.2.3.8 PCS status 3 register (Register 3.9)

Change Table 45–239 (as amended by IEEE Std 802.3cs-2022) as follows (unchanged rows not shown):

Editor's note (to be removed after first working group ballot): 800G capability is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1439.pdf

Table 45–239—PCS status 3 register bit definitions

Bit(s)	Name	Description	R/W ^a
<u>3.9.15:12</u> 3.9.15:10	Reserved	Value always 0	RO
<u>3.9.11</u>	<u>1.6TBASE-R capable</u>	<u>1 = PCS is able to support 1.6TBASE-R PCS type</u> <u>0 = PCS is not able to support 1.6TBASE-R PCS type</u>	<u>RO</u>
<u>3.9.10</u>	<u>800GBASE-R capable</u>	<u>1 = PCS is able to support 800GBASE-R PCS type</u> <u>0 = PCS is not able to support 800GBASE-R PCS type</u>	<u>RO</u>
...			

^a RO = Read only.

Insert 45.2.3.8.1a and 45.2.3.8.1b before 45.2.3.8.1 as follows:

45.2.3.8.1a 1.6TBASE-R capable (3.9.11)

When read as a one, bit 3.9.11 indicates that the PCS is able to support the 1.6TBASE-R PCS type. When read as a zero, bit 3.9.11 indicates that the PCS is not able to support the 1.6TBASE-R PCS type.

45.2.3.8.1b 800GBASE-R capable (3.9.10)

When read as a one, bit 3.9.10 indicates that the PCS is able to support the 800GBASE-R PCS type. When read as a zero, bit 3.9.10 indicates that the PCS is not able to support the 800GBASE-R PCS type.

45.2.3.15 BASE-R and MultiGBASE-T PCS status 1 register (Register 3.32)

Change 45.2.3.15.1 as follows:

45.2.3.15.1 BASE-R and MultiGBASE-T receive link status (3.32.12)

When read as a one, bit 3.32.12 indicates that the PCS is in a fully operational state. When read as a zero, bit 3.32.12 indicates that the PCS is not fully operational. This bit is a reflection of the PCS_status variable defined in [49.2.14.1](#) for 5/10/25GBASE-R, in [126.3.7.1](#) for 2.5GBASE-T and 5GBASE-T, in [55.3.7.1](#) for 10GBASE-T, in [113.3.7.1](#) for 25GBASE-T and 40GBASE-T, in [82.3.1](#) for 40/50/100GBASE-R, ~~and~~ in [119.3](#) for 200/400GBASE-R, in [172.3](#) for 800GBASE-R, and in [175.8](#) for 1.6TBASE-R.

Editor's note (to be removed after first working group ballot): 800G receive link status is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1440.pdf

45.2.3.19 BASE-R PCS test-pattern control register (Register 3.42)

Change the fourth and fifth sentences of 45.2.3.19 (as amended by IEEE Std 802.3df-2024) as follows:

Scrambled idle test patterns are defined for 25/40/50/100/200/400/800GBASE-R and 1.6TBASE-R PCS only. The test-pattern methodology is described in 49.2.8, 82.2.11, 119.2.4.9, and 172.2.4.11, and 175.2.4.11.

45.2.4 PHY XS registers

45.2.4.1 PHY XS control 1 register (Register 4.0)

Change Table 45–315 as follows (unchanged rows not shown):

Table 45–315—PHY XS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
4.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = Reserved 1 1 1 1 = Reserved 1 1 1 0 = 1.6 Tb/s 1 1 0 1 = 800 Gb/s 1 1 0 0 = Reserved 1 0 1 1 = Reserved 1 0 1 0 = 400 Gb/s 1 0 0 1 = 200 Gb/s 1 0 0 0 = Reserved 0 1 x x = Reserved 0 0 1 x = Reserved 0 0 0 1 = Reserved 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only.

Editor's note (to be removed after first working group ballot): 800G selection is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1437.pdf

45.2.4.4 PHY XS speed ability (Register 4.4)

Change Table 45–317 (as amended by IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–317—PHY XS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
4.4.15:12 4.4.15:11	Reserved	Value always 0	RO
4.4.11	1.6T capable	1 = PHY XS is capable of operating at 1.6 Tb/s 0 = PHY XS is not capable of operating at 1.6 Tb/s	RO
4.4.10	800G capable	1 = PHY XS is capable of operating at 800 Gb/s 0 = PHY XS is not capable of operating at 800 Gb/s	RO
...			

^a RO = Read only

Insert 45.2.4.4.aa before 45.2.4.4.a as follows:

45.2.4.4.aa 1.6T capable (4.4.11)

When read as a one, bit 4.4.11 indicates that the PHY XS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 4.4.11 indicates that the PHY-XS is not able to operate at a data rate of 1.6 Tb/s.

45.2.4.12 BASE-R PHY XS status 1 register (Register 4.32)

Change 45.2.4.12.1 as follows:

45.2.4.12.1 BASE-R PHY XS receive link status (4.32.12)

When read as a one, bit 4.32.12 indicates that the PHY XS is in a fully operational state. When read as a zero, bit 4.32.12 indicates that the PHY XS is not fully operational. This bit is a reflection of the PHY XS equivalent of the PCS_status variable defined in 119.3 for 200/400GBASE-R, in 172.3 for 800GBASE-R, and in 175.8 for 1.6TBASE-R.

Editor's note (to be removed after first working group ballot): 800G receive link status is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1440.pdf

Change 45.2.4.13 as follows

45.2.4.13 BASE-R PHY XS test-pattern control register (Register 4.42)

The assignment of bits in the BASE-R PHY XS test-pattern control register is shown in Table 45–338. This register is only required when the 200/400/800GBASE-R or 1.6TBASE-R capability is supported. The test-pattern methodology is described in 119.2.4.9 for 200/400GBASE-R, in 172.2.4.11 for 800GBASE-R, and in 175.2.4.11 for 1.6TBASE-R.

Editor's note (to be removed after first working group ballot): 800G capability is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1441.pdf

45.2.5 DTE XS registers

45.2.5.1 DTE XS control 1 register (Register 5.0)

Change Table 45–340 as follows (unchanged rows not shown):

Table 45–340—DTE XS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
5.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = Reserved 1 1 1 1 = Reserved 1 1 1 0 = 1.6 Tb/s 1 1 0 1 = 800 Gb/s 1 1 0 0 = Reserved 1 0 1 1 = Reserved 1 0 1 0 = 400 Gb/s 1 0 0 1 = 200 Gb/s 1 0 0 0 = Reserved 0 1 x x = Reserved 0 0 1 x = Reserved 0 0 0 1 = Reserved 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only.

Editor's note (to be removed after first working group ballot): 800G selection is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1437.pdf

45.2.5.4 DTE XS speed ability (Register 5.4)

Change Table 45–342 (as amended by IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–342— DTE XS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
5.4.15:12 5.4.15:11	Reserved	Value always 0	RO
<u>5.4.11</u>	<u>1.6T capable</u>	<u>1 = DTE XS is capable of operating at 1.6 Tb/s</u> <u>0 = DTE XS is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
5.4.10	800G capable	1 = DTE XS is capable of operating at 800 Gb/s 0 = DTE XS is not capable of operating at 800 Gb/s	RO
...			

^a RO = Read only

Insert 45.2.5.4.aa before 45.2.5.4.a as follows:

45.2.5.4.aa 1.6T capable (5.4.11)

When read as a one, bit 5.4.11 indicates that the DTE XS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 5.4.11 indicates that the DTE XS is not able to operate at a data rate of 1.6 Tb/s.

45.2.5.12 BASE-R DTE XS status 1 register (Register 5.32)

Change 45.2.5.12.1 as follows:

45.2.5.12.1 BASE-R DTE XS receive link status (5.32.12)

When read as a one, bit 5.32.12 indicates that the DTE XS is in a fully operational state. When read as a zero, bit 5.32.12 indicates that the PHY XS is not fully operational. This bit is a reflection of the DTE XS equivalent of the PCS_status variable defined in 119.3 for 200/400GBASE-R, in 172.3 for 800GBASE-R, and in 175.8 for 1.6TBASE-R.

Editor's note (to be removed after first working group ballot): 800G receive link status is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1440.pdf

Change 45.2.5.13 as follows

45.2.5.13 BASE-R DTE XS test-pattern control register (Register 5.42)

The assignment of bits in the BASE-R DTE XS test-pattern control register is shown in Table 45–338. This register is only required when the 200/400/800GBASE-R or 1.6TBASE-R capability is supported. The test-pattern methodology is described in 119.2.4.9 for 200/400GBASE-R, in 172.2.4.11 for 800GBASE-R, and in 175.2.4.11 for 1.6TBASE-R.

Editor's note (to be removed after first working group ballot): 800G capability is added in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1441.pdf

45.2.7 Auto-Negotiation registers

Change Table 45–378 as follows (most unchanged rows not shown):

Table 45–378—Auto-Negotiation MMD registers

Register address	Register name	Subclause
...		
7.50 through 7.51 7.59	Reserved	
<u>7.52 through 7.53</u>	<u>Backplane Ethernet extended technology ability advertisement</u>	<u>45.2.7.13a</u>
<u>7.54 through 7.55</u>	<u>Backplane Ethernet link partner extended technology ability</u>	<u>45.2.7.13b</u>
<u>7.56 through 7.59</u>	<u>Reserved</u>	
7.60	EEE advertisement 1	45.2.7.14
...		

45.2.7.12 Backplane Ethernet, BASE-R copper status (Register 7.48)

Change the title of 45.2.7.12.3 (as amended by IEEE Std 802.3df-2024) as follows:

45.2.7.12.3 Negotiated Port Type (7.48.1, 7.48.2, 7.48.3, 7.48.5, 7.48.6, 7.48.8, 7.48.9, 7.48.10, 7.48.11, 7.48.12, 7.48.13, 7.48.14, 7.48.15, 7.49.0, 7.49.1, 7.49.2, 7.49.3, 7.49.4, 7.49.5, 7.49.7, 7.49.8, 7.49.9, 7.49.10, 7.49.11)

45.2.7.13 Backplane Ethernet, BASE-R copper status 2 (Register 7.49)

Change Table 45–388 (as amended by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

**Table 45–388—Backplane Ethernet, BASE-R copper status 2 register (Register 7.49)
bit definitions**

Bit(s)	Name	Description	RO ^a
<u>7.49.15:12</u> <u>7.49.15:8</u>	Reserved	Value always 0	RO
<u>7.49.11</u>	<u>1.6TBASE-KR8 or</u> <u>1.6TBASE-CR8</u>	<u>1 = PMA/PMD is negotiated to perform 1.6TBASE-KR8 or</u> <u>1.6TBASE-CR8</u> <u>0 = PMA/PMD is not negotiated to perform 1.6TBASE-KR8</u> <u>or 1.6TBASE-CR8</u>	<u>RO</u>
<u>7.49.10</u>	<u>800GBASE-KR4 or</u> <u>800GBASE-CR4</u>	<u>1 = PMA/PMD is negotiated to perform 800GBASE-KR4 or</u> <u>800GBASE-CR4</u> <u>0 = PMA/PMD is not negotiated to perform 800GBASE-KR4</u> <u>or 800GBASE-CR4</u>	<u>RO</u>
<u>7.49.9</u>	<u>400GBASE-KR2 or</u> <u>400GBASE-CR2</u>	<u>1 = PMA/PMD is negotiated to perform 400GBASE-KR2 or</u> <u>400GBASE-CR2</u> <u>0 = PMA/PMD is not negotiated to perform 400GBASE-KR2</u> <u>or 400GBASE-CR2</u>	<u>RO</u>
<u>7.49.8</u>	<u>200GBASE-KR1 or</u> <u>200GBASE-CR1</u>	<u>1 = PMA/PMD is negotiated to perform 200GBASE-KR1 or</u> <u>200GBASE-CR1</u> <u>0 = PMA/PMD is not negotiated to perform 200GBASE-KR1</u> <u>or 200GBASE-CR1</u>	<u>RO</u>
7.49.7	800GBASE-KR8 or 800GBASE-CR8	1 = PMA/PMD is negotiated to perform 800GBASE-KR8 or 800GBASE-CR8 0 = PMA/PMD is not negotiated to perform 800GBASE-KR8 or 800GBASE-CR8	RO
...			

^a RO = Read only

Insert 45.2.7.13a and 45.2.7.13b after 45.2.7.13 as follows:

45.2.7.13a Backplane Ethernet extended technology ability advertisement register (7.52 and 7.53)

Backplane Ethernet extended abilities are advertised in the 32-bit unformatted code field of next pages with message code 2 (see Table 73A–1). The unformatted code field of transmitted message code 2 next pages are saved in the read only Backplane Ethernet extended technology ability advertisement register. Bits 7.53.15:0 map to U31:U16 and bits 7.52.15:0 map to U15:U0.

45.2.7.13b Backplane Ethernet link partner extended technology ability register (7.54 and 7.55)

Backplane Ethernet extended abilities are indicated in the 32-bit unformatted code field of next pages with message code 2 (see Table 73A–1). The unformatted code field of received message code 2 next pages are saved in the read only Backplane Ethernet link partner extended technology ability register. Bits 7.55.15:0 map to U31:U16 and bits 7.54.15:0 map to U15:U0.

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69. Introduction to Ethernet operation over electrical backplanes

69.1 Overview

69.1.1 Scope

Change the second paragraph of 69.1.1 (as modified by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

Backplane Ethernet supports the IEEE 802.3 full duplex MAC operating at 1000 Mb/s, 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, 40 Gb/s, 50 Gb/s, 100 Gb/s, 200 Gb/s, 400 Gb/s, ~~or~~ 800 Gb/s, or 1.6 Tb/s. The following Physical Layers are supported:

Change the list after the second paragraph of 69.1.1 (as modified by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

- 1000BASE-KX for 1 Gb/s operation over a single lane
- 2.5GBASE-KX for 2.5 Gb/s operation over a single lane
- 5GBASE-KR for 5 Gb/s operation over a single lane
- 10GBASE-KX4 for 10 Gb/s operation over four lanes
- 10GBASE-KR for 10 Gb/s operation over a single lane
- 25GBASE-KR and 25GBASE-KR-S for 25 Gb/s operation over a single lane
- 40GBASE-KR4 for 40 Gb/s operation over four lanes
- 50GBASE-KR for 50 Gb/s operation over a single lane
- 100GBASE-KR4 and 100GBASE-KP4 for 100 Gb/s operation over four lanes
- 100GBASE-KR2 for 100 Gb/s operation over two lanes
- 100GBASE-KR1 for 100 Gb/s operation over one lane
- 200GBASE-KR4 for 200 Gb/s operation over four lanes
- 200GBASE-KR2 for 200 Gb/s operation over two lanes
- 200GBASE-KR1 for 200 Gb/s operation over one lane
- 400GBASE-KR4 for 400 Gb/s operation over four lanes
- 400GBASE-KR2 for 400 Gb/s operation over two lanes
- 800GBASE-KR8 for 800 Gb/s operation over eight lanes
- 800GBASE-KR4 for 800 Gb/s operation over four lanes
- 1.6TBASE-KR8 for 1.6 Tb/s operation over eight lanes

73. Auto-Negotiation for backplane and copper cable assembly

73.3 Functional specifications

Change the last sentence of the third paragraph of 73.3 as follows:

Technology-Dependent PHYs are those supported by the Auto-Negotiation process (see Table 73–4 and Table 73A–1).

Change 73.4 as follows:

73.4 Transmit function requirements

The Transmit function provides the ability to transmit DME pages (see 73.5.2). The first DME pages exchanged by the local device and its link partner after Power-On, link restart, or renegotiation is referred to as the Base Page and contains the base Base Page link codeword defined in Figure 73–6 (see 73.6.2).

To exchange information beyond what is available in the Base Page, additional DME pages may be transmitted using the Next Page function (see 73.4.2).

73.4.1 Technology ability

Multiple technologies may be advertised by the Auto-Negotiation process simultaneously. The advertisement of technologies the device supports is transmitted in the Base Page (see 73.6.2) and a specific Next Page (see 73A.1a).

The local device may modify the link codeword (see 73.6) to disable an ability it possesses, but will not transmit an ability it does not possess. This makes possible the distinction between local abilities and advertised abilities so that multi-ability devices may Auto-Negotiate to a mode lower in priority than the highest common ability.

It is the responsibility of the Arbitration function (see 73.7.4) to determine the common mode of operation shared by a link partner and to resolve multiple common modes.

NOTE—Previous editions of this standard prohibited simultaneous advertisement of PHYs that support operation over electrical backplanes with PHYs that support operation over copper cable assemblies.

73.4.2 Next Page function

The Next Page function uses the Auto-Negotiation arbitration mechanisms to allow for the exchange of information beyond what is available in the Base Page. DME pages exchanged after the Base Page are referred to as Next Pages. Next page exchange occurs after the Base Page has been exchanged if either end of the link segment has set the Next Page bit to logical one in the link codeword (see 73.6.1.2) indicating that it has at least one Next Page to send. There are two encodings of the Next Page link codeword that may be sent during Next Page exchanges, specifically: Message code (see 73.6.3) and Unformatted (see 73.6.4).

Multiple Next Pages, using appropriate link codewords (Message code and Unformatted), are transmitted to send extended messages. Each series of Next Pages begins with the transmission of a Message code Next

Page. Each Message code Next Page has its own specific format and requirements for whether any Unformatted Next Pages will follow (see Annex 73A).

Any number of Next Pages may be sent in any order, with the exception that if a Message code 2 Next Page (see 73A.1a) is to be sent it shall be the first Next Page transmitted following the Base Page exchange.

It is recommended that the total number of Next Pages sent be kept small to minimize the link startup time.

Next page transmission ends when both ends of a link segment set their Next Page bits to logical zero, indicating that neither has anything additional to transmit. It is possible for one device to have more pages to transmit than the other device. Once a device has completed transmission of its Next Page information, it shall transmit Message code 1 Next Pages (Null message code, see 73A.1) with the NP bit set to logical zero in the Null message link codeword while its link partner continues to transmit valid Next Pages. An Auto-Negotiation able device shall recognize reception of Message Pages with Null message codes as the end of its link partner's Next Page information.

73.4.3 Transmit Switch function

During Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state (see Figure 73–11), the Transmit Switch function shall connect only the DME page generator controlled by the Transmit State Diagram to the MDI.

Upon entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect the transmit path from a single technology-dependent (highest common denominator) PHY to the MDI.

When a PHY is connected to the MDI through the Transmit Switch function, the signals at the MDI shall conform to all of the PHY specifications within 20 ms.

73.5 DME transmission

73.5.1 DME electrical specifications

Change the second paragraph of 73.5.1 (as amended by IEEE Std 802.3ck-2022) as follows:

For any multi-lane PHY, DME pages shall be transmitted only on lane 0. The transmitters on other lanes should be disabled as specified in 71.6.7, 84.7.7, 85.7.7, 92.7.7, 93.7.7, 94.3.6.7, 136.8.7, 137.8.7, 162.8.7, ~~or~~ 163.8.7, 178.8.7, or 179.8.7.

Replace Table 73–1 as follows:

Table 73–1—DME electrical characteristics

Parameter	Min Value	Max Value Case 1 ^a	Max Value Case 2 ^b	Units
Transmit differential peak-to-peak output voltage	600	1200	1000	mV
Receive differential peak-to-peak input voltage	200	1200	1200	mV

^a When not advertising any of the PHY types defined in Clause 178 or Clause 179

^b When advertising any of the PHY types defined in Clause 178 or Clause 179

73.6 Link codeword encoding

Change 73.6 and renumber its subclauses (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

The 48-bit Auto-Negotiation page takes on three different encodings referred to as link codewords. The three different encodings of link codewords are as follows:

- ~~link codeword~~ Base Page link codeword
- ~~link codeword~~ Message code Next Page link codeword
- ~~link codeword~~ Unformatted Next Page link codeword

The mappings of the Auto-Negotiation page data bits (D[47:0]) to specific fields, for each type of link codeword, are shown in Figure 73–6, Figure 73–7 and Figure 73–8. Within a link codeword, D0 shall be the first bit transmitted after the Manchester violation (see Figure 73–5).

All link codeword encodings contain the Ack and NP fields described in 73.6.1. The encoding for the Base Page is described in 73.6.2. The encodings for Message code and Unformatted Next Pages are described in 73.6.3 and 73.6.4 respectively.

The mapping of all possible technologies available for backplane and copper cable assembly by Clause 73 Auto-Negotiation are allocated over two link codeword encodings. The mappings are defined in Table 73–4 and Table 73A–1 and are transmitted in the Base Page (see 73.6.2) and Message code 2 Next Page link codewords (see 73.6.3 and 73A.1a).

73.6.1 Common link codeword fields

The Ack and NP fields are present in all link codeword encodings and are used to handshake the flow of the Auto-Negotiation process between the two partners.

73.6.1.1 Acknowledge

Acknowledge (Ack) is used by the Auto-Negotiation function to indicate that a device has successfully received its link partner’s link codeword. The Acknowledge Bit is encoded in bit D14 of the link codeword. If no Next Page information is to be sent, this bit shall be set to logical one in the link codeword after the reception of at least three consecutive and consistent DME pages (ignoring the Acknowledge bit value). If

Next Page information is to be sent, this bit shall be set to logical one after the device has successfully received at least three consecutive and matching DME pages (ignoring the Acknowledge bit value), and will remain set until the Next Page information has been loaded into the AN XNP transmit register (registers 7.22, 7.23, 7.24). In order to save the current received link codeword, it has to be read from the AN LP XNP ability register (register 7.25, 7.26, 7.27) before the Next Page of transmit information is loaded into the AN XNP transmit register. After the COMPLETE ACKNOWLEDGE state has been entered (see Figure 73–11), the link codeword will be transmitted at least six times.

73.6.1.2 Next Page

Next Page (NP) is encoded in bit D15 of the link codeword. Support of Next Pages is mandatory. If the device does not have any Next Pages to send, the NP bit shall be set to logical zero. If a device wishes to engage in Next Page exchange, it shall set the NP bit to logical one. If a device has no Next Pages to send and its link partner has set the NP bit to logical one, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner transmits valid Next Pages. Next page exchanges will occur if either the device or its link partner sets the Next Page bit to logical one. The Next Page function is defined in 73.4.2.

73.6.2 Base Page link codeword

The base link codeword (Base Page) transmitted within a DME page shall convey the encoding shown in Figure 73–6. The Auto-Negotiation function supports additional pages using the Next Page function. Encoding for the link codeword(s) used in the Next Page exchange are defined in 73.7.7. In a DME page, D0 shall be the first bit transmitted.

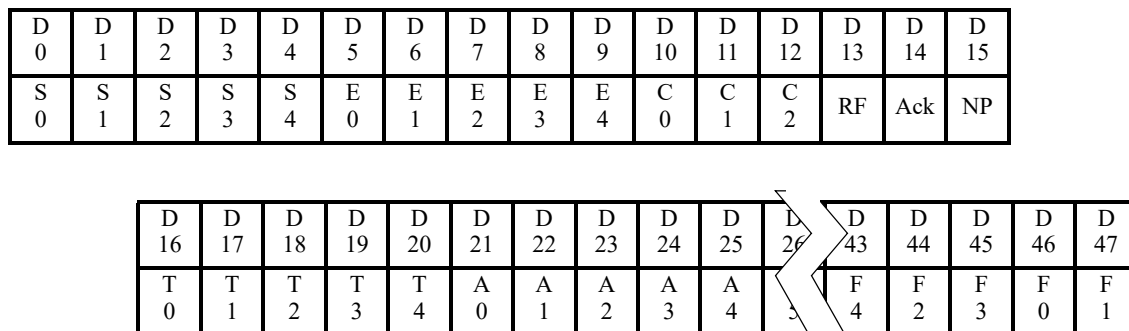


Figure 73–6— Link codeword Base Page link codeword

D[4:0] contains the Selector Field. D[9:5] contains the Echoed Nonce field. D[12:10] contains capability bits to advertise capabilities not related to the PHY. C[1:0] is used to advertise pause capability. The remaining capability bit C[2] is reserved. D[15:13] contains the RF, Ack, and NP bits. These bits shall function as specified in 28.2.1.2. D[20:16] contains the Transmitted Nonce field. D[42:21] contains the Technology Ability Field. D[47:43] contains FEC capability (see 73.6.5).

73.6.2.1 Selector Field

Selector Field (S[4:0]) is a five-bit wide field, encoding 32 possible messages. Selector Field encoding definitions are shown in Annex 28A. Combinations not specified are reserved for future use. Reserved combinations of the Selector Field shall not be transmitted.

The Selector Field for IEEE Std 802.3 is shown in Table 73–3.

Table 73–3—Selector Field Encoding

S4	S3	S2	S1	S0	Selector description
0	0	0	0	1	IEEE Std 802.3

73.6.2.2 Echoed Nonce Field

Echoed Nonce Field (E[4:0]) is a 5-bit wide field containing the nonce received from the link partner. When Acknowledge is set to logical zero, the bits in this field shall contain logical zeros. When Acknowledge is set to logical one, the bits in this field shall contain the value received in the Transmitted Nonce Field from the link partner.

73.6.2.3 Transmitted Nonce Field

Transmitted Nonce Field (T[4:0]) is a 5-bit wide field containing a random or pseudo-random number. A new value shall be generated for each entry to the Ability Detect state. The method of generating the nonce is left to the implementer. The transmitted nonce should have a uniform distribution in the range from 0 to $2^5 - 1$. The method used to generate the value should be designed to minimize correlation to the values generated by other devices.

73.6.2.4 Technology Ability Field

Technology Ability Field (A[21:0]) is a 22-bit wide field containing information indicating supported technologies specific to the selector field value when used with the Auto-Negotiation for backplane and copper cable assembly. These bits are mapped to individual technologies such that abilities are advertised in parallel for a single selector field value. The Technology Ability Field encoding for the IEEE 802.3 selector with Auto-Negotiation for backplane and copper cable assembly is described in Table 73–4.

~~Multiple technologies may be advertised in the link codeword. A device shall support the data service ability for a technology it advertises. It is the responsibility of the Arbitration function to determine the common mode of operation shared by a link partner and to resolve multiple common modes.~~

~~NOTE—Previous editions of this standard prohibited simultaneous advertisement of PHYs that support operation over electrical backplanes with PHYs that support operation over copper cable assemblies.~~

25GBASE-KR-S abilities are a subset of 25GBASE-KR abilities, and likewise 25GBASE-CR-S abilities are a subset of 25GBASE-CR abilities. To allow interoperation between 25GBASE-KR-S and 25GBASE-KR PHY types, and between 25GBASE-CR-S and 25GBASE-CR PHY types, a device that supports 25GBASE-KR or 25GBASE-CR should advertise both A9 and A10 ability bits during auto-negotiation.

The fields A[21:20] are reserved for future use. Reserved fields shall be sent as zero and ignored on receive.

73.6.2.5 FEC capability

~~FEC capability (F4, F2, F3, F0, F1) is encoded in bits D43:D47 of the base link codeword. The FEC bits are used as follows:~~

- ~~F0 is 10 Gb/s per lane FEC ability~~
- ~~F1 is 10 Gb/s per lane FEC requested~~

Table 73–4—Technology Ability Field encoding

Bit	Technology
A0	1000BASE-KX
A1	10GBASE-KX4
A2	10GBASE-KR
A3	40GBASE-KR4
A4	40GBASE-CR4
A5	100GBASE-CR10
A6	100GBASE-KP4
A7	100GBASE-KR4
A8	100GBASE-CR4
A9	25GBASE-KR-S or 25GBASE-CR-S
A10	25GBASE-KR or 25GBASE-CR
A11	2.5GBASE-KX
A12	5GBASE-KR
A13	50GBASE-KR or 50GBASE-CR
A14	100GBASE-KR2 or 100GBASE-CR2
A15	200GBASE-KR4 or 200GBASE-CR4
A16	100GBASE-KR1 or 100GBASE-CR1
A17	200GBASE-KR2 or 200GBASE-CR2
A18	400GBASE-KR4 or 400GBASE-CR4
A19	800GBASE-KR8 or 800GBASE-CR8
A20 through A21	Reserved

- c) F2 is 25G RS-FEC requested
- d) F3 is 25G BASE-R FEC requested
- e) F4 is 100GBASE-P RS-FEC-Int requested

Bits F0 and F1 are used for 10 Gb/s per lane operation PHYs. F2 and F3 are used for resolving FEC operation for 25G PHYs. F4 is used by 100GBASE-P PHYs where RS-FEC-Int (see [Clause 161](#)) is an alternative to the default RS-FEC (see [Clause 91](#)).

73.6.2.5.1 FEC resolution for 100GBASE-P PHYs that include RS-FEC-Int

For 100GBASE-P PHYs that include RS-FEC-Int (see [Clause 161](#)) in addition to RS-FEC (see [Clause 91](#)), the F4 field is used to negotiate which FEC sublayer is to be used. If either PHY requests RS-FEC-Int operation then the RS-FEC-Int sublayer is enabled and the RS-FEC sublayer is disabled. Otherwise, the RS-FEC-Int sublayer is disabled and the RS-FEC sublayer is enabled.

73.6.2.5.2 FEC resolution for 25G PHYs

For 25G PHYs if neither PHY requests FEC operation in bits F2 or F3 then FEC is not enabled.

For 25GBASE-KR and 25GBASE-CR PHYs if either PHY requests RS-FEC then RS-FEC operation is enabled, otherwise if either PHY requests BASE-R FEC then BASE-R operation is enabled.

For 25GBASE-KR-S and 25GBASE-CR-S PHYs, if either PHY requests RS-FEC or BASE-R FEC then BASE-R operation is enabled. This is because 25GBASE-KR-S and 25GBASE-CR-S PHYs do not support RS-FEC operation.

73.6.2.5.3 FEC resolution for 10 Gb/s per lane PHYs

For 10 Gb/s per lane operation, when the FEC ability bit F0 is set to logical one, it indicates that the PHY has FEC ability (see [Clause 74](#)). When the FEC requested F1 bit is set to logical one, it indicates a request to enable FEC on the link.

Since the local device and the link partner may have set the FEC capability bits differently, the priority resolution function is used to enable FEC in the respective PHYs. The FEC function shall be enabled on the link if 10GBASE-KR, 40GBASE-KR4, 40GBASE-CR4, or 100GBASE-CR10 is the HCD technology (see 73.7.6), both devices advertise FEC ability on the F0 bits, and at least one device requests FEC on the F1 bits; otherwise FEC shall not be enabled.

~~The variable `an_rs_fec_int_negotiated_control` indicates that RS-FEC-Int operation has been negotiated. If the value is false, then RS-FEC-Int has not been negotiated. If the value is true, then RS-FEC-Int has been negotiated.~~

73.6.2.5.4 FEC control variables

The variable `an_baser_fec_control` indicates that BASE-R FEC operation has been negotiated. If the value is false, then BASE-R FEC has not been negotiated. If the value is true, then BASE-R FEC has been negotiated.

The variable `an_rs_fec_control` indicates that RS-FEC operation has been negotiated. If the value is false, then RS-FEC has not been negotiated. If the value is true, then RS-FEC has been negotiated.

~~The variable `an_rs_fec_int_negotiated_control` indicates that RS-FEC-Int operation has been negotiated. If the value is false, then RS-FEC-Int has not been negotiated. If the value is true, then RS-FEC-Int has been negotiated.~~

The mapping of these variables to MDIO register bits is defined in Table 73–6.

If `mr_autoneg_enable` (see 73.10.1) is false, the FEC function is controlled by implementation-dependent means.

73.6.2.6 Pause Ability

Pause (C0:C1) is encoded in bits D11:D10 ~~of the base link codeword~~. The two-bit Pause is encoded as follows:

- a) C0 is the same as PAUSE as defined in [Annex 28B](#)
- b) C1 is the same as ASM_DIR as defined in [Annex 28B](#)

The Pause encoding is defined in [Clause 28B.2, Table 28B–2](#). The PAUSE bit indicates that the device is capable of providing the symmetric PAUSE functions as defined in [Annex 31B](#). The ASM_DIR bit indicates that asymmetric PAUSE is supported. The value of the PAUSE bit when the ASM_DIR bit is set indicates the direction the PAUSE frames are supported for flow across the link. Asymmetric PAUSE configuration results in independent enabling of the PAUSE receive and PAUSE transmit functions as defined by [Annex 31B](#). See [28B.3](#) regarding PAUSE configuration resolution.

73.6.2.7 Remote Fault

Remote Fault (RF) is encoded in bit D13 of the base link codeword. The default value is logical zero. The Remote Fault bit provides a standard transport mechanism for the transmission of simple fault information. When the RF bit in the AN advertisement register (register 7.16.13) is set to logical one, the RF bit in the transmitted base link codeword is set to logical one. When the RF bit in the received base Base Page link codeword is set to logical one, the Remote Fault bit in the AN LP Base Page ability register (register 7.19.13) will be set to logical one, if the management function is present.

73.6.3 Message code Next Page link codeword encoding

The Message code Next Page transmitted within a DME page shall convey the encoding shown in Figure 73–7.

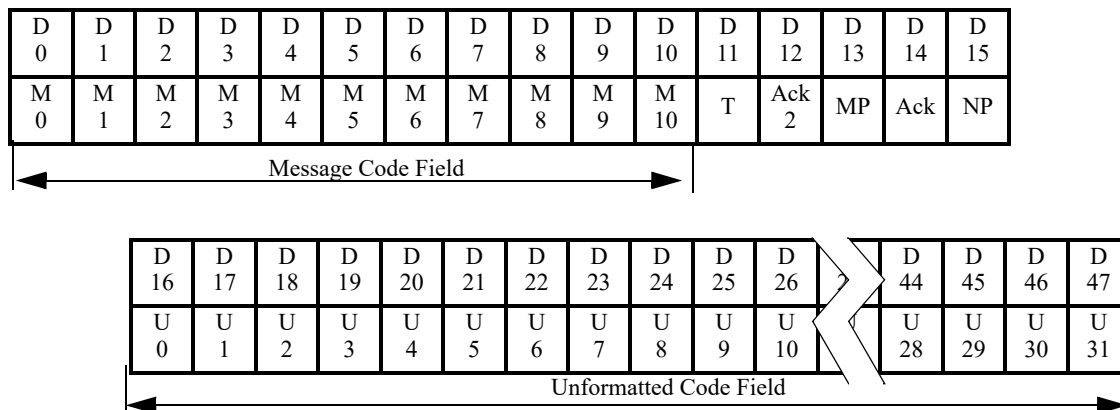


Figure 73–7— Message code Next Page link codeword

D[10:0] contains the Message Code Field, D[15:11] contains the T, Ack2, MP, Ack and NP bits. D[47:16] contains the Unformatted Code Field.

73.6.3.1 Message Code Field

Message Code Field (M[10:0]) is an eleven bit wide field, encoding 2048 possible messages. Message Code Field definitions are shown in Annex 73A. Combinations not specified are reserved for future use. Reserved combinations of the Message Code Field shall not be transmitted.

73.6.3.2 Toggle

Identical to that specified in 28.2.3.4.7.

73.6.3.3 Acknowledge 2

Identical to that specified in 28.2.3.4.6.

73.6.3.4 Message page

Identical to that specified in 28.2.3.4.5.

73.6.3.5 Unformatted code field

The Unformatted code field (U31:0) is a 32-bit generic container. The definition for each bit depends upon the Message code Field of the link codeword. See Annex 73A for details.

73.6.4 Unformatted Next Page link codeword encoding

The Unformatted Next Page transmitted within a DME page shall convey the encoding shown in Figure 73–8.

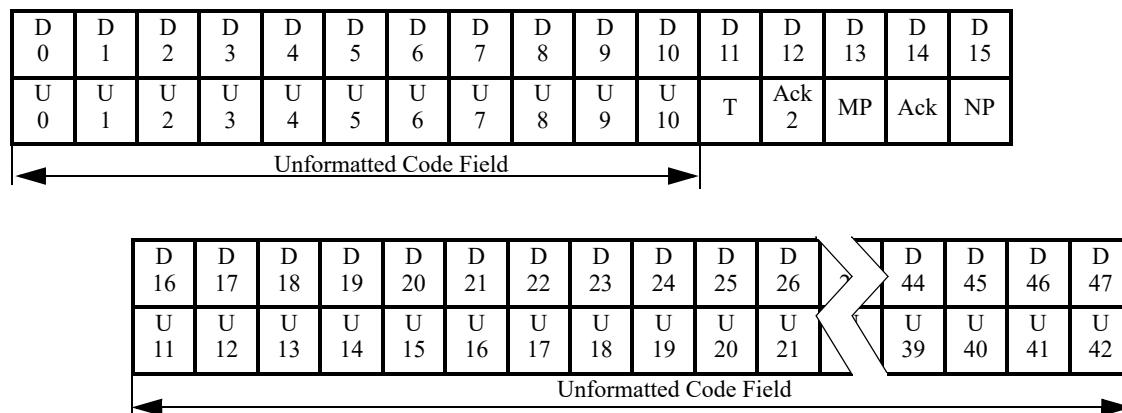


Figure 73–8—Unformatted Next Page link codeword

D[10:0] and D[47:16] contain the Unformatted Code Field, D[15:11] contains the T, Ack2, MP, Ack and NP bits.

73.6.4.1 Toggle

Identical to that specified in 28.2.3.4.7.

73.6.4.2 Acknowledge 2

Identical to that specified in 28.2.3.4.6.

73.6.4.3 Message page

Identical to that specified in 28.2.3.4.5.

73.6.4.4 Unformatted code field

The Unformatted code field (U42:0) is a 43-bit generic container. The definition for each bit depends upon the Message code Next Page that preceded the Unformatted Next Page. See Annex 73A for details.

73.6.8 Acknowledge

Acknowledge (Ack) is used by the Auto-Negotiation function to indicate that a device has successfully received its link partner's link codeword. The Acknowledge Bit is encoded in bit D14 of link codeword. If no Next Page information is to be sent, this bit shall be set to logical one in the link codeword after the reception of at least three consecutive and consistent DME pages (ignoring the Acknowledge bit value). If Next Page information is to be sent, this bit shall be set to logical one after the device has successfully received at least three consecutive and matching DME pages (ignoring the Acknowledge bit value), and will remain set until the Next Page information has been loaded into the AN_XNP transmit register (registers 7.22, 7.23, 7.24). In order to save the current received link codeword, it has to be read from the AN_LP_XNP ability register (register 7.25, 7.26, 7.27) before the Next Page of transmit information is loaded into the AN_XNP transmit register. After the COMPLETE ACKNOWLEDGE state has been entered, the link codeword will be transmitted at least six times.

73.6.9 Next Page

Next Page (NP) is encoded in bit D15 of link codeword. Support of Next Pages is mandatory. If the device does not have any Next Pages to send, the NP bit shall be set to logical zero. If a device wishes to engage in Next Page exchange, it shall set the NP bit to logical one. If a device has no Next Pages to send and its link partner has set the NP bit to logical one, it shall transmit Next Pages with Null message codes and the NP bit set to logical zero while its link partner transmits valid Next Pages. Next page exchanges will occur if either the device or its link partner sets the Next Page bit to logical one. The Next Page function is defined in 73.7.7.

73.6.10 Transmit Switch function

During Auto-Negotiation and prior to entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect only the DME page generator controlled by the Transmit State Diagram to the MDI.

Upon entry into the AN_GOOD_CHECK state, the Transmit Switch function shall connect the transmit path from a single technology dependent (highest common denominator) PHY to the MDI.

When a PHY is connected to the MDI through the Transmit Switch function, the signals at the MDI shall conform to all of the PHY's specifications within 20 ms.

Change 73.7 title as follows:

73.7 Receive function requirements

73.7.6 Priority Resolution function

Change the first paragraph of 73.7.6 as follows:

Since a local device and a link partner may have multiple common abilities, a mechanism to resolve which mode to configure is required. The mechanism used by Auto-Negotiation is a Priority Resolution function that predefines the hierarchy of supported technologies. The single PHY enabled to connect to the MDI by Auto-Negotiation shall be the technology corresponding to the bit in the Technology Ability Field (see 73.6.2.4) and the Extended Technology Ability Field (see 73A.1a) that is common to the local device and link partner that has the highest priority as defined in Table 73–5 (listed from highest priority to lowest pri-

ority). The Priority Resolution function shall only select technologies from the Extended Technology Ability Field when a Message code 2 Next Page is both transmitted and received.

Change the second sentence of the second paragraph as follows:

If the local device receives a Technology Ability Field or Extended Technology Ability Field with a bit set that is reserved, the local device shall ignore that bit for priority resolution.

Change Table 73–5 (as inserted by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

Table 73–5—Priority Resolution

Technology	Capability
<u>1.6TBASE-KR8 or 1.6TBASE-CR8</u>	<u>1.6 Tb/s 8 lane, highest priority</u>
<u>800GBASE-KR4 or 800GBASE-CR4</u>	<u>800 Gb/s 4 lane</u>
800GBASE-KR8 or 800GBASE-CR8	800 Gb/s 8 lane, highest priority
<u>400GBASE-KR2 or 400GBASE-CR2</u>	<u>400 Gb/s 2 lane</u>
400GBASE-KR4 or 400GBASE-CR4	400 Gb/s 4 lane
<u>200GBASE-KR1 or 200GBASE-CR1</u>	<u>200 Gb/s 1 lane</u>
200GBASE-KR2 or 200GBASE-CR2	200 Gb/s 2 lane
200GBASE-KR4 or 200GBASE-CR4	200 Gb/s 4 lane
100GBASE-KR1 or 100GBASE-CR1	100 Gb/s 1 lane
...	

Delete subclause “73.7.7 Next Page Function”, as the contents of this section are now included within 73.4.

73.8 Management register requirements

Change Table 73–6 (as amended by IEEE Std 802.3ck-2022) as follows (some unchanged rows not shown):

**Table 73–6—Backplane Ethernet Auto-Negotiation
variable to MDIO register mapping**

Variable	Description
mr_adv_ability[48:1]	{7.18.15:0, 7.17.15:0, 7.16.15:0} AN advertisement registers
mr_adv_extended_ability[32:1]	<u>{7.52.15:0, 7.53.15:0} extended technology advertisement registers</u>
mr_autoneg_complete	7.1.5 Auto-Negotiation Complete
mr_autoneg_enable	7.0.12 Auto-Negotiation Enable
mr_lp_adv_ability[48:1]	For Base Page: {7.21.15:0, 7.20.15:0, 7.19.15:0} AN LP Base Page ability registers For Next Page(s): {7.27.15:0, 7.26.15:0, 7.25.15:0} AN LP XNP ability registers
mr_lp_adv_extended_ability[32:1]	<u>{7.54.15:0, 7.55.15:0} LP extended technology ability registers</u>
...	

73.10 State diagrams and variable definitions

73.10.1 State diagram variables

Change the variable list in 73.10.1 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows (some unchanged list entries not shown):

200GR2;	represents the 200GBASE-KR2 or 200GBASE-CR2 PMD.
<u>200GR1;</u>	<u>represents the 200GBASE-KR1 or 200GBASE-CR1 PMD.</u>
400GR4;	represents the 400GBASE-KR4 or 400GBASE-CR4 PMD.
<u>400GR2;</u>	<u>represents the 400GBASE-KR2 or 400GBASE-CR2 PMD.</u>
800GR8;	represents the 800GBASE-KR8 or 800GBASE-CR8 PMD.
<u>800GR4;</u>	<u>represents the 800GBASE-KR4 or 800GBASE-CR4 PMD.</u>
<u>1.6TR8;</u>	<u>represents the 1.6TBASE-KR8 or 1.6TBASE-CR8 PMD.</u>

Insert new variable after mr_adv_ability variable as follows:

mr_adv_extended_ability[32:1]

A 32-bit array that contains the unformatted code field of transmitted message code 2 next pages.
How this status variable is updated is implementation specific and beyond the scope of this

standard.

For each element within the array:

Values: zero; data bit is logical zero.
one; data bit is logical one.

Insert new variable after `mr_lp_adv_ability` variable as follows:

`mr_lp_adv_extended_ability[32:1]`

A 32-bit array that contains the unformatted code field of received message code 2 next pages.
How this status variable is updated is implementation specific and beyond the scope of this
standard.

For each element within the array:

Values: zero; data bit is logical zero.
one; data bit is logical one.

Change the entry for `single_link_ready` in 73.10.1 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows:

`single_link_ready`

Status indicating that an `_receive_idle` = true and only one of the following indications is being
received:

- 1) `link_status_[1GKX]` = OK
- 2) `link_status_[2.5GKX]` = OK
- 3) `link_status_[5GKR]` = OK
- 4) `link_status_[10GKX4]` = OK
- 5) `link_status_[10GKR]` = OK
- 6) `link_status_[25GR]` = OK
- 7) `link_status_[40GKR4]` = OK
- 8) `link_status_[40GCR4]` = OK
- 9) `link_status_[50GR]` = OK
- 10) `link_status_[100GCR10]` = OK
- 11) `link_status_[100GKP4]` = OK
- 12) `link_status_[100GKR4]` = OK
- 13) `link_status_[100GCR4]` = OK
- 14) `link_status_[100GR2]` = OK
- 15) `link_status_[100GR1]` = OK
- 16) `link_status_[200GR4]` = OK
- 17) `link_status_[200GR2]` = OK
- 18) `link_status_[200GR1]` = OK
- ~~19~~18) `link_status_[400GR4]` = OK
- ~~20~~19) `link_status_[400GR2]` = OK
- ~~21~~20) `link_status_[800GR8]` = OK
- ~~22~~21) `link_status_[800GR4]` = OK
- ~~23~~22) `link_status_[1.6TR8]` = OK

73.10.2 State diagram timers

Change Table 73–7 (as amended by IEEE Std 802.3ck-2022 and IEEE Std 802.3df-2024) as follows
(most unchanged rows not shown):

Table 73–7—Timer min/max value summary

Parameter	Min	Value and tolerance	Max	Units
...				
link fail inhibit timer (when the link is 200GBASE-KR1, 200GBASE-CR1, 400GBASE-KR2, 400GBASE-CR2, 800GBASE-KR4, 800GBASE-CR4, 1.6TBASE-KR8, or 1.6TBASE-CR8)	60		61	s
link fail inhibit timer (when the link is 100GBASE-KR1, 100GBASE-CR1, 200GBASE-KR2, 200GBASE-CR2, 400GBASE-KR4, 400GBASE-CR4, 800GBASE-KR8, or 800GBASE-CR8)	12.3		12.4	s
...				

73.11 Protocol implementation conformance statement (PICS) proforma for Clause 73, Auto-Negotiation for backplane and copper cable assembly¹

73.11.4 PICS proforma tables for Auto-Negotiation for backplane and copper cable assembly

73.11.4.4 Receive function requirements

Change 73.11.4.4 as follows (most unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
RF16	Priority resolution with no common technology	73.7.6	HCD takes on value of NULL and link_status=FAIL	M	Yes []
<u>RF17</u>	<u>Priority resolution with MsgPg2 technology</u>	<u>73.7.6</u>	<u>Ignores message code 2 next page technologies unless sent by both ends of the link</u>	<u>M</u>	<u>Yes []</u>

73.11.4.5 Next Page function

Change 73.11.4.5 as follows (most unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
NP12	Transmission of Null Message Code field	73.7.7.2	Sent if device has no other information to transmit	M	Yes []
<u>NP13</u>	<u>Next page order</u>	<u>73.4.2</u>	<u>Message code 2 next pages are transmitted first</u>	<u>M</u>	<u>Yes []</u>

¹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

90. Ethernet support for time synchronization protocols

90.1 Introduction

Change the second paragraph of 90.1 (as amended by IEEE Std 802.3de-2022 and IEEE Std 802.3df-2024) as follows:

The TSSI is defined for 10BASE-T1S (see [Clause 147](#)) in full-duplex and point-to-point half-duplex modes of operation, and for other PHY types in full-duplex mode. It supports MAC operation at various data rates. The MII ([Clause 22](#)), GMII ([Clause 35](#)), XGMII ([Clause 46](#)), 25GMII ([Clause 106](#)), XLGMII ([Clause 81](#)), CGMII ([Clause 81](#)), 50GMII ([Clause 132](#)), 200GMII ([Clause 117](#)), 400GMII ([Clause 117](#)), ~~and 800GMII (Clause 170), and 1.6TMII (Clause 170)~~ specifications are all compatible with the generic Reconciliation Sublayer (gRS) defined in [90.5](#).

116. Introduction to 200 Gb/s and 400 Gb/s networks

116.1 Overview

116.1.2 Relationship of 200 Gigabit and 400 Gigabit Ethernet to the ISO OSI reference model

Change the lettered list in 116.1.2 (as modified by IEEE Std 802.3db-2022, IEEE Std 802.3ck-2022, and IEEE Std 802.3df-2024) as follows (items a to e not shown):

- f) The PMA service interface, which, when physically implemented as 200GAUI-2 (200 Gb/s two-lane Attachment Unit Interface) or 400GAUI-2 (400 Gb/s two-lane Attachment Unit Interface) at an observable interconnection port, uses a 2-lane data path as specified in Annex 120F, ~~or Annex 120G, Annex 176C, or Annex 176D.~~
- g) The PMA service interface, which, when physically implemented as 200GAUI-1 (200 Gb/s one-lane Attachment Unit Interface) at an observable interconnection port, uses a 1-lane data path as specified in Annex 176C or Annex 176D.
- h) ~~g)~~The MDI as specified in **Clause 123** for 400GBASE-SR16 uses a 16-lane data path.
- i) ~~h)~~The MDIs as specified in:
 - **Clause 122** for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8
 - **Clause 138** for 400GBASE-SR8
 - **Clause 150** for 400GBASE-SR4.2all use an 8-lane data path.
- j) ~~i)~~The MDIs as specified in:
 - **Clause 121** for 200GBASE-DR4
 - **Clause 122** for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4
 - **Clause 124** for 400GBASE-DR4 and 400GBASE-DR4-2
 - **Clause 136** for 200GBASE-CR4
 - **Clause 137** for 200GBASE-KR4
 - **Clause 138** for 200GBASE-SR4
 - **Clause 151** for 400GBASE-FR4 and 400GBASE-LR4-6
 - **Clause 162** for 400GBASE-CR4
 - **Clause 163** for 400GBASE-KR4
 - **Clause 167** for 400GBASE-VR4 and 400GBASE-SR4all use a 4-lane data path.
- k) ~~j)~~The MDIs as specified in:
 - **Clause 162** for 200GBASE-CR2
 - **Clause 163** for 200GBASE-KR2
 - **Clause 167** for 200GBASE-VR2 and 200GBASE-SR2
 - Clause 178 for 400GBASE-KR2
 - Clause 179 for 400GBASE-CR2
 - Clause 180 for 400GBASE-DR2
 - Clause 182 for 400GBASE-DR2-2all use a 2-lane data path.
- l) The MDIs as specified in:
 - Clause 178 for 200GBASE-KR1
 - Clause 179 for 200GBASE-CR1
 - Clause 180 for 200GBASE-DR1
 - Clause 182 for 200GBASE-DR1-2all use a 1-lane data path

116.1.3 Nomenclature

Change Table 116–1 and Table 116–2 (as modified by IEEE Std 802.3db-2022, IEEE Std 802.3ck-2022, and IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

Table 116–1—200 Gb/s PHYs

Name	Description
<u>200GBASE-KR1</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over one lane of an electrical backplane (see Clause 178)</u>
200GBASE-KR2	200 Gb/s PHY using 200GBASE-R encoding over two lanes of an electrical backplane (see Clause 163)
200GBASE-KR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of an electrical backplane (see Clause 137)
<u>200GBASE-CR1</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over one lane of twinaxial copper cable (see 1.4.559 and Clause 179)</u>
200GBASE-CR2	200 Gb/s PHY using 200GBASE-R encoding over two lanes of twinaxial copper cable (see 1.4.559 and Clause 162)
...	
200GBASE-SR2	200 Gb/s PHY using 200GBASE-R encoding over two multimode fibers in each direction, with reach up to at least 100 m (see Clause 167)
<u>200GBASE-DR1</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over one single-mode fiber in each direction, with reach up to at least 500 m (see Clause 180)</u>
200GBASE-DR4	200 Gb/s PHY using 200GBASE-R encoding over four single-mode fibers in each direction, with reach up to at least 500 m (see Clause 121)
<u>200GBASE-DR1-2</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over one single-mode fiber in each direction, with reach up to at least 2 km (see Clause 182)</u>
200GBASE-LR4	200 Gb/s PHY using 200GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 10 km (see Clause 122)
200GBASE-ER4	200 Gb/s PHY using 200GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 40 km (see Clause 122)

Table 116–2—400 Gb/s PHYs

Name	Description
<u>400GBASE-KR2</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over two lanes of an electrical backplane (see Clause 178)</u>
400GBASE-KR4	400 Gb/s PHY using 400GBASE-R encoding over four lanes of an electrical backplane (see Clause 163)
<u>400GBASE-CR2</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over two lanes of twinaxial copper cable (see 1.4.559 and Clause 179)</u>
400GBASE-CR4	400 Gb/s PHY using 400GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 162)
...	
400GBASE-SR4.2	400 Gb/s PHY using 400GBASE-R encoding over eight multimode fibers in a bidirectional WDM format, with reach up to at least 150 m (see Clause 150)
<u>400GBASE-DR2</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over two single-mode fibers in each direction, with reach up to at least 500 m (see Clause 180)</u>
400GBASE-DR4	400 Gb/s PHY using 400GBASE-R encoding over four single-mode fibers in each direction, with reach up to at least 500 m (see Clause 124)
<u>400GBASE-DR2-2</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over two single-mode fibers in each direction, with reach up to at least 2 km (see Clause 182)</u>
400GBASE-DR4-2	400 Gb/s PHY using 400GBASE-R encoding over four single-mode fibers in each direction, with reach up to at least 2 km (see Clause 124)
...	

Insert new Table 116–3aa after Table 116–3 as follows:

Table 116–3aa—PHY type and clause correlation (200GBASE copper with 1 lane)

PHY type	Clause ^a																											
	Auto-Negotiation		73		90		117		118		119		120		120B		120D		120F		176		176C		178		179	
	Time Synchronization																											
	RS																											
	200GMII																											
200GBASE-KR1	M	O	M	O	O	M	C	O	O	O	O	M	O	O	M	O	M											
200GBASE-CR1	M	O	M	O	O	M	C	O	O	O	O	M	O														M	

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Change Table 116–3a (as inserted by IEEE Std 802.3ck-2022) as follows:

Table 116–3a—PHY type and clause correlation (400GBASE copper with 4 lanes)

PHY type	Clause ^a													
	<u>Time Synchronization</u>	<u>90</u>												
		116												
		117												
		Auto-Negotiation	RS	400GMII	400GMII Extender	400GBASE-R PCS	400GBASE-R BM-PMA	400GAUI-16 C2C	400GAUI-8 C2C	400GAUI-4 C2C	400GBASE-CR4 PMD	400GBASE-KR4 PMD	400GBASE-R SM-PMA	400GAUI-1 C2C
400GBASE-KR4	<u>O</u>	M	M	O	O	M	M	O	O	O		M	<u>C</u>	<u>O</u>
400GBASE-CR4	<u>O</u>	M	M	O	O	M	M	O	O	O	M		<u>C</u>	<u>O</u>

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Insert new Table 116–3b after Table 116–3a as follows:

Table 116–3b—PHY type and clause correlation (400GBASE copper with 2 lanes)

PHY type	Clause ^a													
	Auto-Negotiation	73				90								
	Time Synchronization	117				118								
	RS	400GMII				119								
	400GMII Extender	400GBASE-R PCS				120								
400GBASE-KR2	M	O	M	O	O	M	C	O	O	O	M	O	M	
400GBASE-CR2	M	O	M	O	O	M	C	O	O	O	M	O		M

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Change Table 116–4 (as modified by IEEE Std 802.3db-2022) as follows:

Table 116–4—PHY type and clause correlation (200GBASE-R optical with 2 or 4 lanes)

PHY type	Clause ^a															
	78	90	117	118	119	120	120B	120C	120D	120E	120F	120G	176	176C	176D	167
	EEE	Time Synchronization	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R BM-PMA	200GAUI-8 C2C	200GAUI-8 C2M	200GAUI-4 C2C	200GAUI-4 C2M	200GAUI-2 C2C	200GAUI-2 C2M	200GBASE-R SM-PMA	200GAUI-1 C2C	200GAUI-1 C2M
200GBASE-VR2	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-SR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-SR2	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-DR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-FR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-LR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q
200GBASE-ER4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Insert Table 116–4a after Table 116–4 as follows:

Table 116–4a—PHY type and clause correlation (200GBASE-R optical with 1 lane)

PHY type	Clause ^a																		
	Time Synchronization			90															
	RS	200GMII		117															
200GBASE-DR1	O	M	O	O	M	C	O	O	O	O	O	O	O	M	O	O		M	
200GBASE-DR1-2	O	M	O	O	M	C	O	O	O	O	O	O	O	M	O	O	M		M

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Change Table 116–5 (as modified by IEEE Std 802.3db-2022, IEEE Std 802.3ck-2022, and IEEE Std 802.3df-2024) as follows:

Table 116–5—PHY type and clause correlation (400GBASE optical with 4, 8, or 16 lanes)

PHY type	Clause ^a																											
	EEE	Time Synchronization	RS	400GMII	400GMII Extender	400GBASE-R PCS	400GBASE-R BM-PMA	400GAUI-16 C2C	400GAUI-16 C2M	400GAUI-8 C2C	400GAUI-8 C2M	400GAUI-4 C2C	400GAUI-4 C2M	400GBASE-R SM-PMA	400GAUI-2 C2C	400GAUI-2 C2M	400GBASE-VR4 PMD	400GBASE-SR4 PMD	400GBASE-SR16 PMD	400GBASE-SR8 PMD	400GBASE-SR4.2 PMD	400GBASE-DR4 PMD	400GBASE-DR4-2 PMD	400GBASE-FR8 PMD	400GBASE-FR4 PMD	400GBASE-LR4-6 PMD	400GBASE-LR8 PMD	400GBASE-ER8 PMD
	78	90	117		118	119	120	120B	120C	120D	120E	120F	120G	176	176C	176D	167		123	138	150	124	124	122	151		122	
400GBASE-VR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q	M											
400GBASE-SR16	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q			M									
400GBASE-SR8	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q				M								
400GBASE-SR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q		M										
400GBASE-SR4.2	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q					M							
400GBASE-DR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q						M						
400GBASE-DR4-2	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q							M					
400GBASE-FR8	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q								M				
400GBASE-FR4	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q									M			
400GBASE-LR4-6	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q										M		
400GBASE-LR8	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q											M	
400GBASE-ER8	O	Q	M	O	O	M	M	O	O	O	O	O	O	C	Q	Q												M

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Insert new Table 116–5a after Table 116–5 as follows:

Table 116–5a—PHY type and clause correlation (400GBASE-R optical with 2 lanes)

PHY type	Clause ^a																		
	Time Synchronization			90															
	RS			117															
		400GMII		118															
				119															
				120	120B	120C	120D	120E	120F	120G	176	176C	176D	177	180	182			
400GBASE-DR2	O	M	O	O	M	C	O	O	O	O	O	M	O	O		M			
400GBASE-DR2-2	O	M	O	O	M	C	O	O	O	O	O	M	O	O	M		M		

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

116.2 Summary of 200 Gigabit and 400 Gigabit Ethernet sublayers

116.2.4 Physical Medium Attachment (PMA) sublayer

Change the second paragraph in 116.2.4 as follows:

The 200GBASE-R and 400GBASE-R PMAs that use bit multiplexing (BM-PMA) are specified in Clause 120.

200GBASE-R and 400GBASE-R PMAs that use symbol multiplexing (SM-PMA) are specified in Clause 176.

116.2.5 Auto-Negotiation

Change the second paragraph in 116.2.5 as follows:

Clause 73 Auto-Negotiation is used by the ~~200 Gb/s and 400 Gb/s backplane PHYs (200GBASE-KR4, 200GBASE-KR2, and 400GBASE-KR4) and the 200 Gb/s and 400 Gb/s copper PHYs (200GBASE-CR4, 200GBASE-CR2, and 400GBASE-CR4).~~ following PHY types:

- 200GBASE-KR4
- 200GBASE-KR2
- 200GBASE-KR1
- 200GBASE-CR4
- 200GBASE-CR2
- 200GBASE-CR1
- 400GBASE-KR4
- 400GBASE-KR2
- 400GBASE-CR4
- 400GBASE-CR2

Insert new subclause 116.2.8 and 116.2.9 after 116.2.7 “Management” as follows:

116.2.8 Time Synchronization

A 200 Gb/s or 400 Gb/s Physical Layer may optionally support time synchronization protocols that require knowledge of packet egress and ingress time. Ethernet support for time synchronization protocols is defined in Clause 90.

If time synchronization is supported, then the following capabilities are provided:

- The 200 Gb/s and 400 Gb/s Reconciliation Sublayer (RS) provides a Time Synchronization Service Interface (TSSI) that connects to a TimeSync client (see 90.4).
- The path data delay values of each sublayer in a Physical Layer are reported as status variables, which may be mapped to MDIO status registers (see 90.6).

116.2.9 Inter-sublayer link training (ILT)

Inter-sublayer link training (ILT) facilitates the orderly start-up of an inter-sublayer link (ISL) and coordinates the start-up of a series of ISLs along a network path.

For each ISL, ILT provides a mechanism for a receiver to control peer transmitter states (e.g., equalization, modulation, and precoding), to indicate the receiver state, and to coordinate the transition to DATA mode. ILT is specified in Annex 178B.

ILT is supported by the following PHY types:

- 200GBASE-KR1
- 200GBASE-CR1
- 200GBASE-DR1
- 200GBASE-DR1-2
- 400GBASE-KR2
- 400GBASE-CR2
- 400GBASE-DR2

— 400GBASE-DR2-2

116.3 Service interface specification method and notation

116.3.1 Inter-sublayer service interface

Change the text in 116.3.1 as follows:

The inter-sublayer service interface is described in an abstract manner and does not imply any particular implementation. The inter-sublayer service interface primitives are defined as follows:

IS_UNITDATA_*i*.request
IS_UNITDATA_*i*.indication
IS_SIGNAL.indication
IS_SIGNAL.request

The IS_UNITDATA_*i*.request (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next lower (closer to the medium) sublayer. The IS_UNITDATA_*i*.indication (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next higher (closer to the MAC) sublayer. The IS_SIGNAL.indication primitive is used to define the transfer of signal status from a sublayer to the next higher sublayer. The IS_SIGNAL.request primitive is used to define the transfer of signal status from a sublayer to the next lower sublayer.

116.3.2 Instances of the Inter-sublayer service interface

Change the text in 116.3.2 as follows:

The inter-sublayer interface can be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

- a) PMD:—for primitives issued on the interface between the PMD sublayer and the PMA sublayer called the PMD service interface.
- b) PMA:—for primitives issued on the interface between the PMA sublayer and the PCS, DTE 200GXS, DTE 400GXS, or PMA sublayer above called the PMA service interface.
- c) FEC:—for primitives issued on the interface between the Inner FEC and the PMA above called the FEC service interface
- d) PHY XS:—for primitives issued on the interface between the PMA sublayer and the PHY 200GXS sublayer or PHY 400GXS sublayer called the PHY XS service interface.

Examples of inter-sublayer service interfaces for 200GBASE-R and 400GBASE-R with their corresponding instance names are illustrated in Figure 116–2 ~~and through~~ Figure 116–3a, respectively. For example, the primitives for one instance of the inter-sublayer service interface, named the PMD service interface, are identified as follows:

PMD:IS_UNITDATA_0.request
PMD:IS_UNITDATA_0.indication
PMD:IS_SIGNAL.indication-
PMD:IS_SIGNAL.request

Primitives for other instances, of inter-sublayer interfaces, are represented in a similar manner as described above.

Replace Figure 116-2 with the following figure:

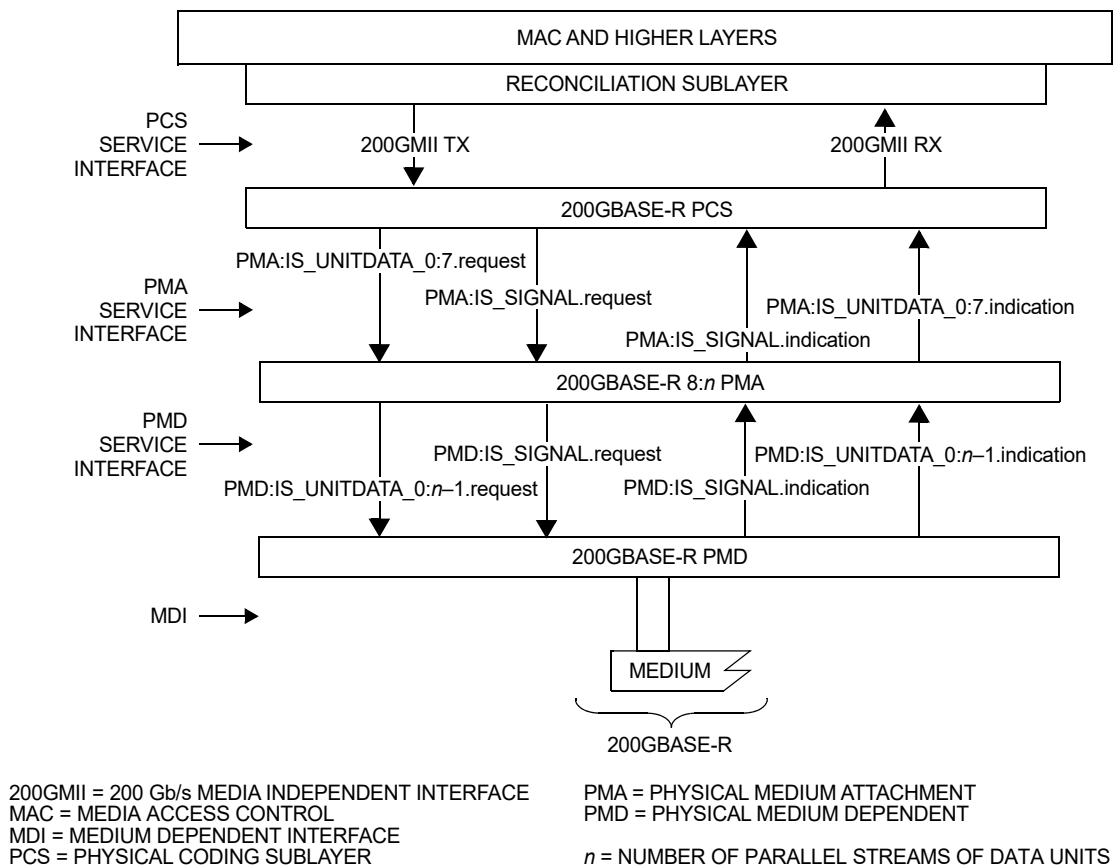


Figure 116-2—200GBASE-R inter-sublayer service interfaces

Insert new Figure 116–2a after Figure 116–2 as follows:

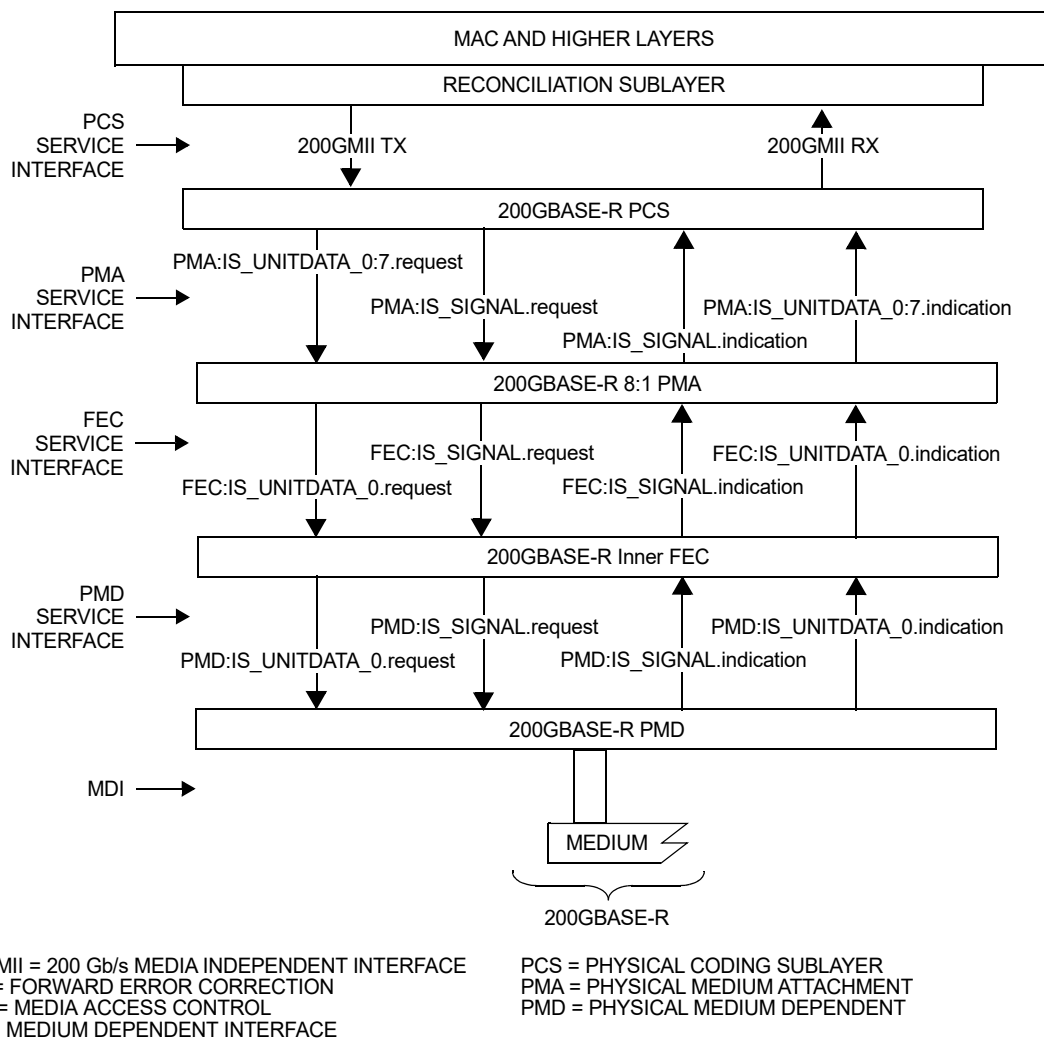


Figure 116–2a—200GBASE-R inter-sublayer service interfaces including 200GBASE-R Inner FEC

Replace Figure 116–3 with the following figure:

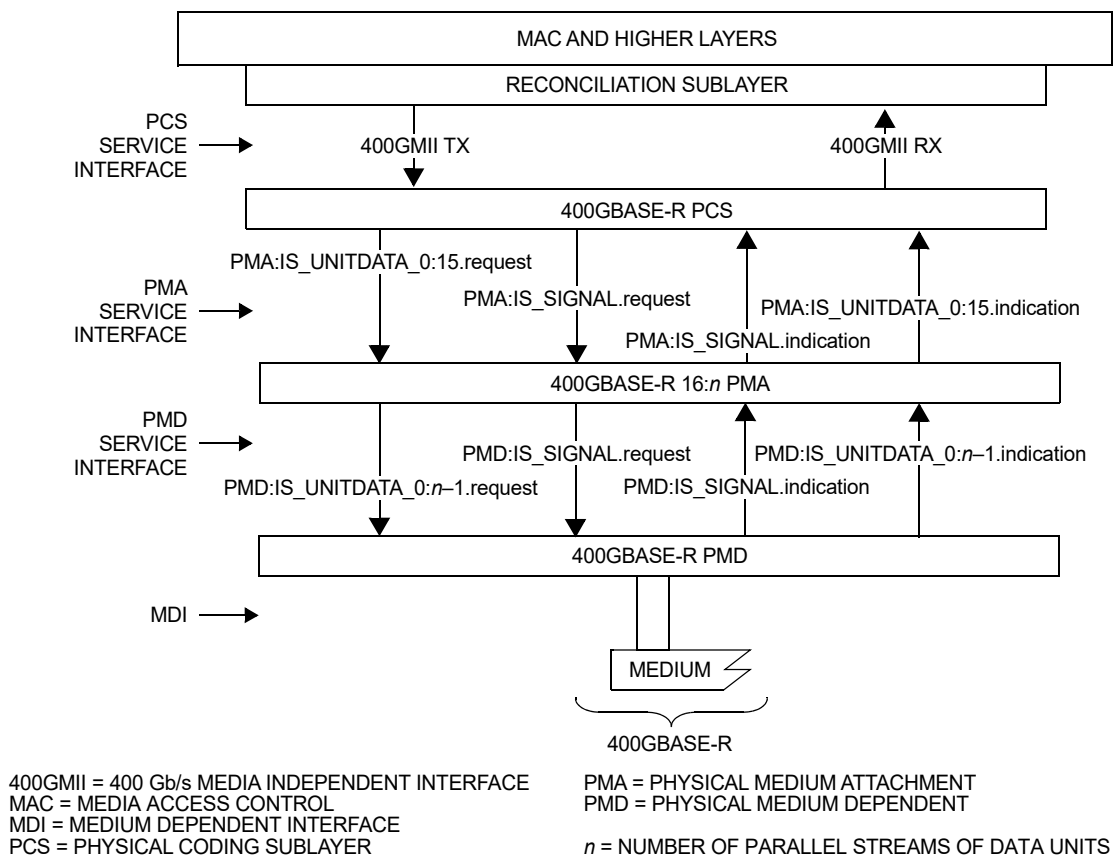


Figure 116–3—400GBASE-R inter-sublayer service interfaces

Insert new Figure 116–3a after Figure 116–3 as follows:

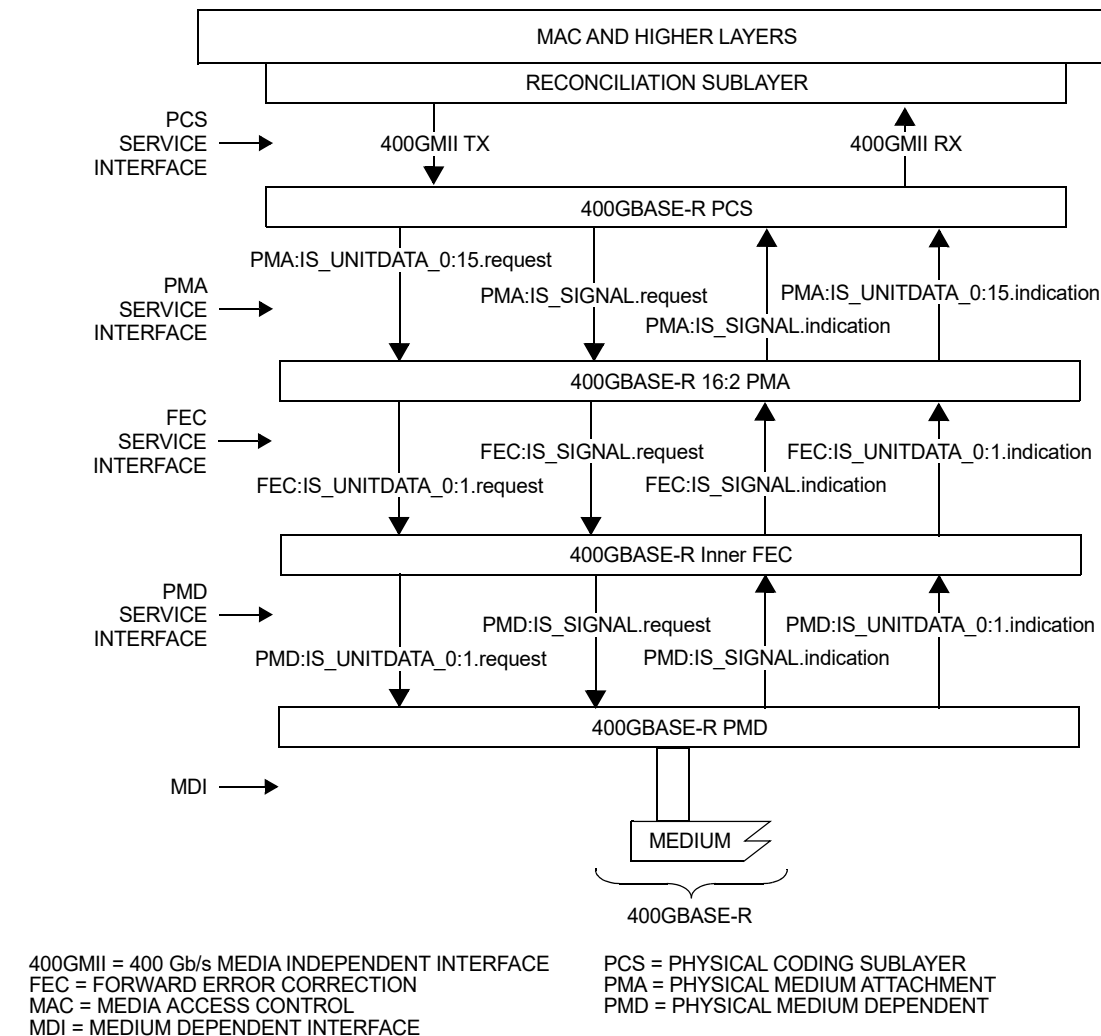


Figure 116–3a—400GBASE-R inter-sublayer service interfaces including 400GBASE-R Inner FEC

116.3.3 Semantics of inter-sublayer service interface primitives

116.3.3.3 IS_SIGNAL.indication

Change the text in 116.3.3.3 as follows:

The IS_SIGNAL.indication primitive is generated by the sublayer to the next higher sublayer to indicate the status of the receive process. This primitive is generated by the receive process to propagate the detection of severe error conditions (e.g., no valid signal being received by the sublayer that generates this primitive on IS_UNITDATA.indication in the receive direction) to the next higher sublayer and to indicate the ILT status for Physical Layer implementations that use the ILT function defined in Annex 178B.

116.3.3.3.1 Semantics of the service primitive

Change the text 116.3.3.3.1 as follows:

IS_SIGNAL.indication(SIGNAL_OK)

~~The SIGNAL_OK parameter can take on one of two values: OK or FAIL. A value of FAIL denotes that invalid data is being presented (rx_symbol parameters undefined) by the sublayer to the next higher sublayer. A value of OK does not guarantee valid data is being presented by the sublayer to the next higher sublayer.~~

The SIGNAL_OK parameter takes on one of four values: OK, FAIL, IN_PROGRESS, or READY. The values IN_PROGRESS and READY are defined only for Physical Layer implementations that use the ILT function defined in Annex 178B.

A value of OK indicates that communication with the next lower sublayer is established. If the service interface supports the values IN_PROGRESS and READY, then a value of OK indicates that valid data is being presented by the sublayer to the next higher sublayer in the rx_symbol parameters.

A value of FAIL indicates the sublayer has not established communication with the next lower sublayer. Data is not being presented by the sublayer to the next higher sublayer (the rx_symbol parameters are unspecified). If the service interface supports the values IN_PROGRESS and READY, then a value of FAIL indicates that an attempt to communicate with the next lower sublayer has failed.

A value of IN_PROGRESS indicates that the sublayer is establishing communication with the next lower sublayer. Data is not being presented by the sublayer to the next higher sublayer (the rx_symbol parameters are unspecified) but it is considered a temporary state, and the sublayer does not require management intervention.

A value of READY indicates that communication with the next lower sublayer is established but communication with the peer interface is not fully established yet. The rx_symbol parameters presented to the next higher sublayer do not represent traffic data and might be invalid. It is considered a temporary state, and the sublayer does not require management intervention.

Insert new subclause 116.3.3.4 as follows:

116.3.3.4 IS_SIGNAL.request

The IS_SIGNAL.request primitive is generated by the transmit process to propagate the detection of severe error conditions (e.g., no valid signal being received by the sublayer on IS_UNITDATA.request in the transmit direction) to the next lower sublayer and to indicate the ILT status for Physical Layer implementations that use the ILT function defined in Annex 178B.

116.3.3.4.1 Semantics of the service primitive

IS_SIGNAL.request(SIGNAL_OK)

The SIGNAL_OK parameter takes on one of four values: OK, FAIL, IN_PROGRESS, or READY. The values IN_PROGRESS and READY are defined only for Physical Layer implementations that use the ILT function defined in Annex 178B.

A value of OK indicates that communication with the next higher sublayer is established. If the service interface supports the values IN_PROGRESS and READY, then a value of OK indicates that valid data is being presented by the sublayer to the next lower sublayer in the tx_symbol parameters.

A value of FAIL indicates the sublayer is not functional or has not established communication with the next higher sublayer. Data is not being presented by the sublayer to the next lower sublayer (the tx_symbol parameters are unspecified). If the service interface supports the values IN_PROGRESS and READY, then a value of FAIL indicates that an attempt to communicate with the next higher sublayer has failed.

A value of IN_PROGRESS indicates that the sublayer is establishing communication with the next higher sublayer. Data is not being presented by the sublayer to the next lower sublayer (the tx_symbol parameters are unspecified) but it is considered a temporary state, and the sublayer does not require management intervention.

A value of READY indicates that communication with the next higher sublayer is established but communication with an upper ISL has not completed. The tx_symbol parameters presented to the next lower sublayer are valid but do not represent traffic data. It is considered a temporary state and the sublayer does not require management intervention.

116.3.3.4.2 When generated

The next higher sublayer generates the IS_SIGNAL.request primitive to the sublayer whenever there is a change in the value of the SIGNAL_OK parameter.

116.3.3.4.3 Effect of receipt

The effect of receipt of this primitive is defined by the sublayer that receives this primitive.

116.4 Delay constraints

Change Table 116–6 (as modified by IEEE Std 802.3db-2022 and IEEE Std 802.3ck-2022) as follows (some unchanged rows not shown):

Table 116–6—Sublayer delay constraints (200GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause quanta) ^b	Maximum (ns)	Notes ^c
...				
200GBASE-R BM-PMA	18 432	36	92.16	See 120.5.4.
200GBASE-R 8:1 or 1:8 SM-PMA	13 312	26	66.56	See 176.8.
200GBASE-R 1:1 SM-PMA	3 072	6	15.36	See 176.8.
200GBASE-R Inner FEC	59 392	116	296.96	See 177.8.

Table 116–6—Sublayer delay constraints (200GBASE) (*continued*)

Sublayer	Maximum (bit time) ^a	Maximum (pause quanta) ^b	Maximum (ns)	Notes ^c
<u>200GAUI-1 C2C or C2M component</u>	<u>9 216</u>	<u>18</u>	<u>46.08</u>	<u>Includes allocation of 5 ns for one direction through AUI channel. See 176C.5 and 176D.5.</u>
<u>200GBASE-KR1 PMD</u>	<u>14 848</u>	<u>29</u>	<u>74.24</u>	<u>Includes allocation of 14 ns for one direction through backplane medium. See 178.6.</u>
200GBASE-KR2 PMD	8 192	16	40.96	Includes allocation of 14 ns for one direction through backplane medium. See 163.5.
200GBASE-KR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through backplane medium. See 137.5.
<u>200GBASE-CR1 PMD</u>	<u>14 848</u>	<u>29</u>	<u>74.24</u>	<u>Includes allocation of 14 ns for one direction through cable medium. See 179.6.</u>
200GBASE-CR2 PMD	8 192	16	40.96	Includes allocation of 14 ns for one direction through cable medium. See 162.5.
...				
200GBASE-SR2 PMD	4 096	8	20.48	Includes 2 m of fiber. See 167.3.1.
<u>200GBASE-DR1 PMD</u>	<u>14 848</u>	<u>29</u>	<u>74.24</u>	<u>Includes 2 m of fiber. See 180.4.1.</u>
200GBASE-DR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 121.3.1.
<u>200GBASE-DR1-2 PMD</u>	<u>14 848</u>	<u>29</u>	<u>74.24</u>	<u>Includes 2 m of fiber. See 182.4.1.</u>
200GBASE-FR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 122.3.1.
...				

^a For 200GBASE-R, 1 bit time (BT) is equal to 5 ps. (See 1.4.235 for the definition of bit time.)

^b For 200GBASE-R, 1 pause_quantum is equal to 2.56 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 116–7 (as modified by IEEE Std 802.3db-2022 and IEEE Std 802.3ck-2022) as follows (some unchanged rows not shown):

Table 116–7—Sublayer delay constraints (400GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause quanta) ^b	Maximum (ns)	Notes ^c
...				
400GBASE-R <u>BM-PMA</u>	36 864	72	92.16	See 120.5.4.
400GBASE-R 16:2 or 2:16 SM- <u>PMA</u>	16 384	32	40.96	See 176.8.
400GBASE-R 2:2 SM-PMA	6 144	12	15.36	See 176.8.
400GBASE-R Inner FEC	70 656	138	176.64	See 177.8.
400GAUI-2 C2C or C2M component	18 432	36	46.08	Includes allocation of 5 ns for one direction through AUI channel. See 176C.5 and 176D.5.
400GBASE-KR2 PMD	29 696	58	74.24	Includes allocation of 14 ns for one direc- tion through backplane medium. See 178.6.
400GBASE-KR4 PMD	8 192	16	20.48	Includes allocation of 14 ns for one direc- tion through backplane medium. See 163.5.
400GBASE-CR2 PMD	29 696	58	74.24	Includes allocation for 14 ns for one direc- tion through cable medium. See 179.6.
400GBASE-CR4 PMD	8 192	16	20.48	Includes allocation for 14 ns for one direc- tion through cable medium. See 162.5.
...				
400GBASE-SR4.2 PMD	8 192	16	20.48	Includes 2 m of fiber. See 150.3.1.
400GBASE-DR2 PMD	29 696	58	74.24	Includes 2 m of fiber. See 180.4.1.
400GBASE-DR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 124.3.1.
400GBASE-DR2-2 PMD	29 696	58	74.24	Includes 2 m of fiber. See 182.4.1.
400GBASE-FR8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 122.3.1.
...				

^a For 400GBASE-R, 1 bit time (BT) is equal to 2.5 ps. (See 1.4.235 for the definition of bit time.)

^b For 400GBASE-R, 1 pause_quantum is equal to 1.28 ns. (See 31B.2 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

116.5 Skew constraints

Change the fourth and fifth paragraph in 116.5 as follows:

In the transmit direction, the Skew points are defined in the following locations:

- SP1 on the 200GAUI-n/400GAUI-n interface closest to the PMD, at the input of the PMA ~~closest to the PMD~~;

- SP2 on the PMD service interface, at the input of the PMD;
- SP3 at the output of the PMD, at the MDI.

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD;
- SP5 on the PMD service interface, at the output of the PMD;
- SP6 on the 200GAUI-n/400GAUI-n interface closest to the 200GBASE-R/400GBASE-R PCS or DTE 200GXS/400GXS, at the output of the PMA closest to the 200GBASE-R/400GBASE-R PCS or DTE 200GXS/400GXS.

Replace Figure 116–4 with the following figure:

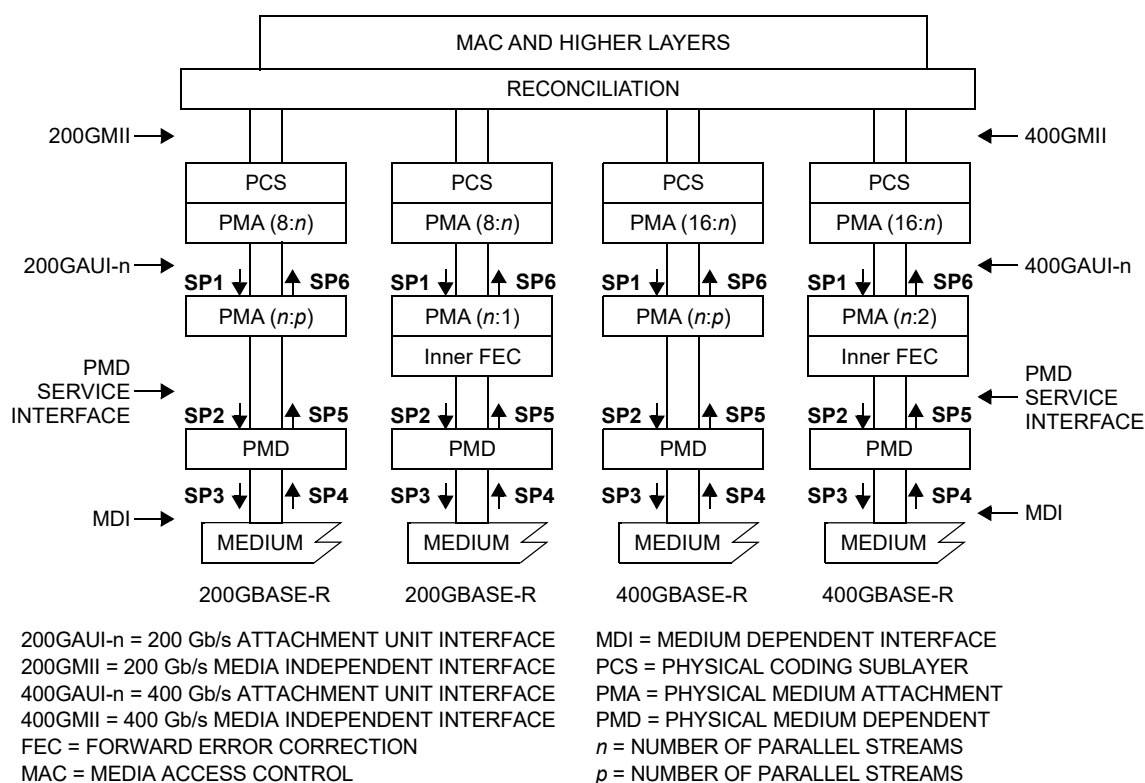


Figure 116–4—200GBASE-R and 400GBASE-R Skew points for single 200GAUI-n or 400GAUI-n

Replace Figure 116–5 with the following figure:

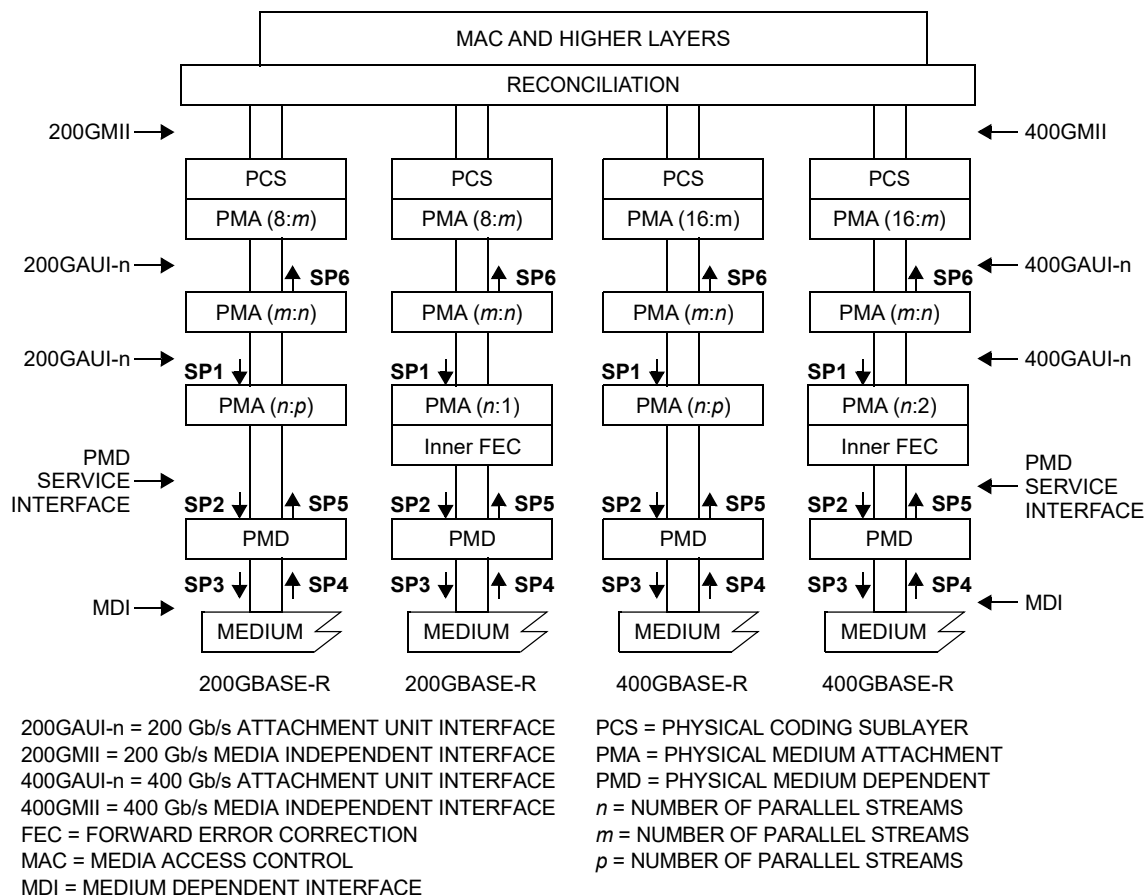


Figure 116–5—200GBASE-R and 400GBASE-R Skew points for multiple 200GAUI-n or 400GAUI-n

Change Table 116–8 (as modified by IEEE Std 802.3db-2022 and IEEE Std 802.3ck-2022) as follows:

Table 116–8—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 200GBASE-R or 400GBASE-R PCS lane (UI) ^b	Notes ^c
SP1 ^d	29	≈ 770	See 120.5.3.1 , 176.9
SP2 ^d	43	≈ 1142	See 120.5.3.3 , 121.3.2 , 122.3.2 , 123.3.2 , 124.3.2 , 136.6.2 , 137.6.2 , 138.3.2.2 , 150.3.2 , 151.3.2 , 162.6.2 , 163.6.2 , 167.3.2 , 178.7 , 179.7 , 180.4.2 , 182.4.2
SP3 ^d	54	≈ 1434	See 121.3.2 , 122.3.2 , 123.3.2 , 124.3.2 , 136.6.2 , 137.6.2 , 138.3.2.2 , 150.3.2 , 151.3.2 , 162.6.2 , 163.6.2 , 167.3.2 , 178.7 , 179.7 , 180.4.2 , 182.4.2
SP4 ^d	134	≈ 3559	See 121.3.2 , 122.3.2 , 123.3.2 , 124.3.2 , 136.6.2 , 137.6.2 , 138.3.2.2 , 150.3.2 , 151.3.2 , 162.6.2 , 163.6.2 , 167.3.2 , 178.7 , 179.7 , 180.4.2 , 182.4.2
SP5 ^d	145	≈ 3852	See 121.3.2 , 122.3.2 , 123.3.2 , 124.3.2 , 136.6.2 , 137.6.2 , 138.3.2.2 , 150.3.2 , 151.3.2 , 162.6.2 , 163.6.2 , 167.3.2 , 178.7 , 179.7 , 180.4.2 , 182.4.2
SP6 ^d	160	≈ 4250	See 120.5.3.5 , 176.9
At PCS receive	180	≈ 4781	See 119.2.5.1

^a The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^b The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

^d The maximum Skew specified at this Skew point excludes the deliberate delay of two RS-FEC codewords applied to odd PCS lanes (see [176.4.2.4](#)) by the 200GBASE-R 1:8 or 8:1 PMA or 400GBASE-R 2:16 or 16:2 PMA if the PHY includes any of these PMA types.

Change Table 116–9 (as modified by IEEE Std 802.3db-2022 and IEEE Std 802.3ck-2022) as follows:

Table 116–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 26.5625 GBd PMD-lane (UI) ^a	Maximum Skew Variation for 53.125 GBd PMD-lane (UI) ^b	Maximum Skew Variation for 106.25 GBd PMD-lane (UI) ^c	Maximum Skew Variation for 113.4375 GBd PMD-lane (UI) ^d	Notes ^e
SP1	0.2	≈ 5	N/A	≈ 21	N/A	See 120.5.3.1, 176.9
SP2	0.4	≈ 11	≈ 21	≈ 43	≈ 45	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2, 178.7, 179.7, 180.4.2, 182.4.2
SP3	0.6	≈ 16	≈ 32	≈ 64	≈ 68	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2, 178.7, 179.7, 180.4.2, 182.4.2
SP4	3.4	≈ 90	≈ 181	≈ 361	≈ 386	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2, 178.7, 179.7, 180.4.2, 182.4.2
SP5	3.6	≈ 96	≈ 191	≈ 383	≈ 408	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2, 178.7, 179.7, 180.4.2, 182.4.2
SP6	3.8	≈ 101	N/A	≈ 404	N/A	See 120.5.3.5, 176.9
At PCS receive	4	≈ 106	N/A	N/A	N/A	See 119.2.5.1

^a The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.

^b The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^c The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 9.41176 ps at PMD lane signaling rate of 106.25 GBd.

^d The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 8.31543 ps at PMD lane signaling rate of 113.4375 GBd.

^e Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

118. 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)

118.1 Overview

Insert a new paragraph, Table 118–a, and Table 118–b after the third paragraph of 118.1 and before Figure 118–1 as follows:

Physical Layer clauses associated with the 200GMII Extender and 400GMII Extender are listed in Table 118–a and Table 118–b, respectively.

Table 118–a—Physical Layer clauses associated with the 200GMII Extender

Associated clause	200GMII Extender
117—200GMII ^a	Optional
118—DTE 200GXS	Required
118—PHY 200GXS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120C—200GAUI-8 C2M ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120E—200GAUI-4 C2M ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
120G—200GAUI-2 C2M	Optional ^d
176—200GBASE-R SM-PMA	Conditional ^e
176C—200GAUI-1 C2C	Optional ^d
176D—200GAUI-1 C2M	Optional ^d
78—Energy-Efficient Ethernet	Optional

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b A 200GBASE-R BM-PMA is required if the 200GMII Extender includes a 200GAUI-8, 200GAUI-4, or 200GAUI-2.

^c When implemented in a 200GMII Extender the signaling rate range is constrained as specified in 120.1.4.

^d A 200GMII Extender includes at least one 200GAUI-n.

^e A 200GBASE-R SM-PMA is required if the 200GMII Extender includes a 200GAUI-1.

Table 118–b—Physical Layer clauses associated with the 400GMII Extender

Associated clause	400GMII Extender
117—400GMII ^a	Optional
118—DTE 400GXS	Required
118—PHY 400GXS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120C—400GAUI-16 C2M ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120E—400GAUI-8 C2M ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
120G—400GAUI-4 C2M	Optional ^d
176—400GBASE-R SM-PMA	Conditional ^e
176C—400GAUI-2 C2C	Optional ^d
176D—400GAUI-2 C2M	Optional ^d
78—Energy-Efficient Ethernet	Optional

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b A 400GBASE-R BM-PMA is required if the 400GMII Extender includes a 400GAUI-16, 400GAUI-8, or 400GAUI-4.

^c When implemented in a 400GMII Extender the signaling rate range is constrained as specified in 120.1.4.

^d A 400GMII Extender includes at least one 400GAUI-n.

^e A 400GBASE-R SM-PMA is required if the 400GMII Extender includes a 400GAUI-2.

118.1.3 200GAUI-n/400GAUI-n

Change 118.1.3 (as amended by IEEE Std 802.3ck-2022) as follows:

A 200GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (200GAUI-n), as shown in Figure 118–1:

- 200GAUI-8 chip-to-chip (Annex 120B)
- 200GAUI-8 chip-to-module (Annex 120C)
- 200GAUI-4 chip-to-chip (Annex 120D)
- 200GAUI-4 chip-to-module (Annex 120E)
- 200GAUI-2 chip-to-chip (Annex 120F)
- 200GAUI-2 chip-to-module (Annex 120G)

- 200GAUI-1 chip-to-chip (Annex 176C)
- 200GAUI-1 chip-to-module (Annex 176D)

The 200GAUI-n may be instantiated using any form described in 176B.4.2.

A 400GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (400GAUI-n), as shown in [Figure 118–1](#):

- 400GAUI-16 chip-to-chip ([Annex 120B](#))
- 400GAUI-16 chip-to-module ([Annex 120C](#))
- 400GAUI-8 chip-to-chip (Annex 120D)
- 400GAUI-8 chip-to-module ([Annex 120E](#))
- 400GAUI-4 chip-to-chip (Annex 120F)
- 400GAUI-4 chip-to-module (Annex 120G)
- 400GAUI-2 chip-to-chip (Annex 176C)
- 400GAUI-2 chip-to-module (Annex 176D)

The 400GAUI-n may be instantiated using any form described in 176B.5.2.

119. Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R

119.1 Overview

119.1.4 Inter-sublayer interfaces

119.1.4.2 Physical Medium Attachment (PMA) service interface

Change the text in 119.1.4.2 as follows:

The PMA service interface for the PCS is described in an abstract manner and does not imply any particular implementation. The PMA Service Interface supports the exchange of encoded data between the PCS and PMA sublayer. The PMA service interface is ~~defined in 120.3 and is an instance of the inter-sublayer service interface definition defined in 116.3.~~

119.1.5 Functional block diagram

Replace Figure 119–2 with the following figure:

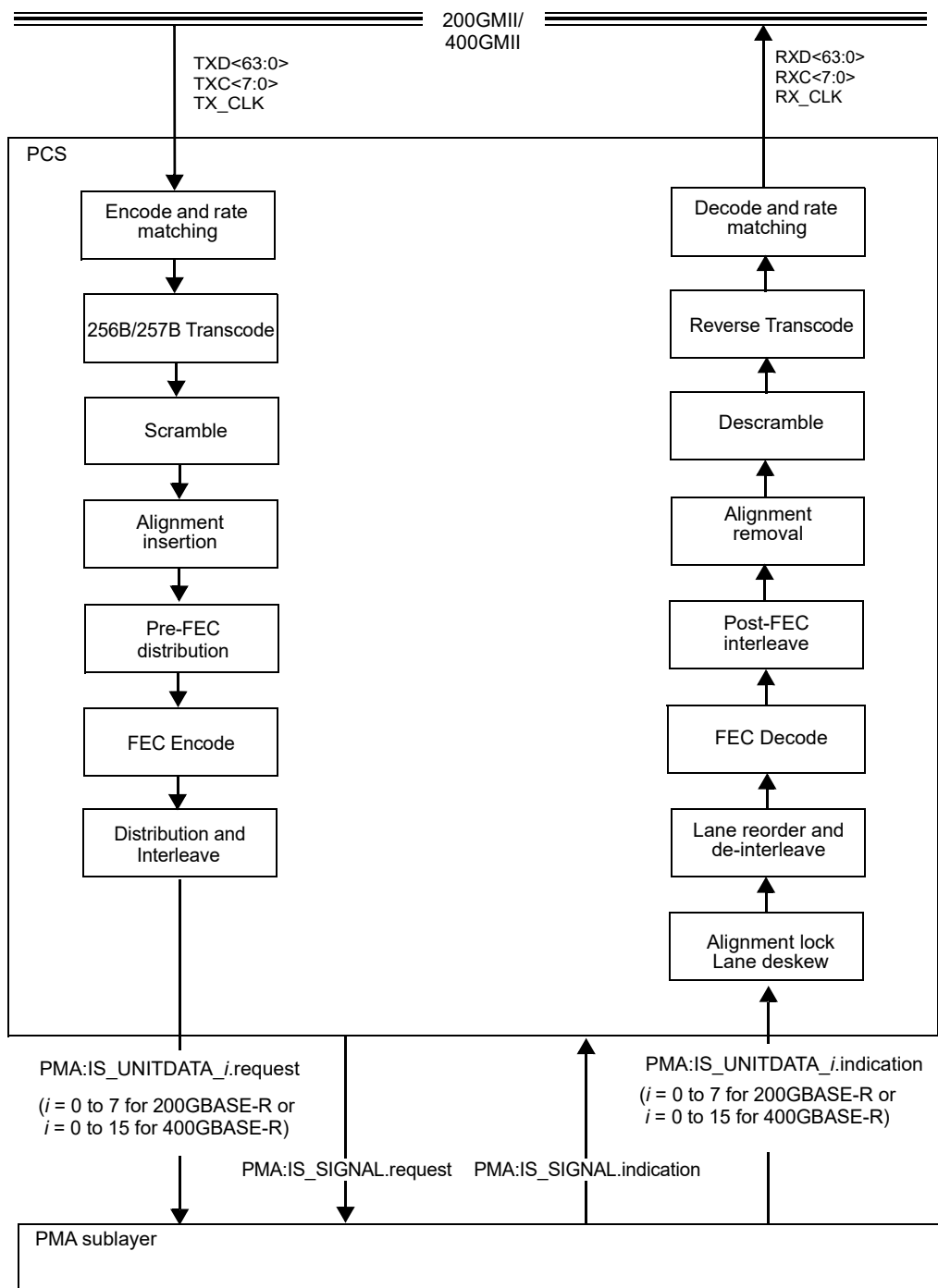


Figure 119–2—Functional block diagram

119.2 Physical Coding Sublayer (PCS)

119.2.1 Functions within the PCS

Change the text of the fourth paragraph of 119.2.1 as follows:

Transmit data-units are sent to the service interface via the PMA:IS_UNITDATA_1.request primitive. The SIGNAL_OK parameter of the PMA:IS_SIGNAL.request primitive is set to OK when the transmit data units are valid and is set to FAIL otherwise. The PMA:IS_SIGNAL.request primitive is only required if the sublayer below the PCS is an SM-PMA (see Clause 176).

119.2.4 Transmit

119.2.4.1 Encode and rate matching

Change the text in 119.2.4.1 as follows:

The PCS generates 66-bit blocks based upon the TXD<63:0> and TXC<7:0> signals received from the 200GMII/400GMII. One 200GMII/400GMII data transfer is encoded into one 66-bit block. If the transmit PCS spans multiple clock domains, it may also perform clock rate compensation via the deletion of idle control characters or sequence ordered sets or the insertion of idle (or LPI) control characters.

Idle (or LPI) control characters or sequence ordered sets are removed, if necessary, to accommodate the insertion of the alignment markers. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules, and 119.2.4.4 for more details on alignment markers.

~~The transmit PCS generates blocks as specified in the transmit state diagram as shown in Figure 119-14. The contents of each block are contained in a vector tx_coded<65:0>, which is passed to the transcoder. tx_coded<1:0> contains the sync header and the remainder of the bits contain the block payload.~~

The transmit PCS generates 66-bit blocks using the state-diagram encoder defined in 119.2.4.1.1 for any 200GBASE-R or 400GBASE-R PHY type, or using the alternative stateless encoder defined in 119.2.4.1.2 for the following PHY types:

- 200GBASE-KR1
- 200GBASE-CR1
- 200GBASE-DR1
- 200GBASE-DR1-2
- 400GBASE-KR2
- 400GBASE-CR2
- 400GBASE-DR2
- 400GBASE-DR2-2

Editor's note (to be removed prior to publication): The alternative stateless decoder is only specified to be used with the specific PHY types listed above because these are the only PHY types that are within scope for this project.

The contents of each 66-bit block are contained in a vector tx_coded<65:0>, which is passed to the transcoder. tx_coded<1:0> contains the sync header and the remainder of the bits contain the block payload

NOTE—The stream of 66-bit blocks generated by this process, together with the FEC_degraded_SER and rx_local_degraded bits should be used as the reference signal for mapping to OTN.

Insert 119.2.4.1.1 and 119.2.4.1.2 after 119.2.4.1 as follows:

119.2.4.1.1 State-diagram encoder

The state-diagram encoder generates 66-bit blocks based on the TXD<63:0> and TXC<7:0> signals received from the 200GMII/400GMII as specified by the transmit state diagram shown in Figure 119–14.

119.2.4.1.2 Stateless encoder

The stateless encoder generates 66-bit blocks based only on the current and preceding 200GMII/400GMII transfers. Each 200GMII/400GMII transfer is mapped into a 72-bit vector tx_raw<71:0> (see 119.2.6.2.2). The encoder shall encode each tx_raw<71:0> to a 66-bit block tx_coded<65:0> according to the rules in Table 172–1. Constants LBLOCK_T and EBLOCK_T are defined in 119.2.6.2.1. Variables reset, tx_raw, and tx_coded are defined in 119.2.6.2.1. Functions T_TYPE and ENCODE, and the block types are defined in 119.2.6.2.3

119.2.5 Receive function

119.2.5.8 Decode and rate matching

Change the text in 119.2.5.8 as follows:

The receive PCS decodes blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 200GMII/400GMII. One 200GMII/400GMII data transfer is decoded from each block. The receive PCS may insert idle control characters to compensate for the removal of alignment markers. If the receive PCS spans multiple clock domains, it may also perform clock rate compensation via the deletion of idle control characters or sequence ordered sets or the insertion of idle control characters (see 82.2.3.6 and 82.2.3.9 for insertion and deletion rules).

~~The PCS receive decodes blocks as specified in the receive state diagram shown in Figure 119–15.~~

The receive PCS decodes 66-bit blocks using the state-diagram decoder defined in 119.2.5.8.1 for any 200GBASE-R or 400GBASE-R PHY type, or using the alternative stateless decoder defined in 119.2.5.8.2 for the following PHY types:

- 200GBASE-KR1
- 200GBASE-CR1
- 200GBASE-DR1
- 200GBASE-DR1-2
- 400GBASE-KR2
- 400GBASE-CR2
- 400GBASE-DR2
- 400GBASE-DR2-2

Editor's note (to be removed prior to publication): The alternative stateless decoder is only specified to be used with the specific PHY types listed above because these are the only PHY types that are within scope for this project.

Insert 119.2.5.8.1 and 119.2.5.8.2 after 119.2.5.8 as follows:

119.2.5.8.1 State-diagram decoder

The state-diagram decoder decodes 66-bit blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 200GMII/400GMII as specified by the receive state diagram shown in [Figure 119–15](#).

119.2.5.8.2 Stateless decoder

The stateless decoder generates 200GMII/400GMII transfers based only on the current and preceding 66-bit blocks. The decoder shall decode each 66-bit block rx_coded<65:0> to a 72-bit vector rx_raw<71:0> (see [119.2.6.2.2](#)) according to the rules in [Table 172–4](#). Constants LBLOCK_R and EBLOCK_R are defined in [119.2.6.2.1](#). Variables reset, rx_raw, and rx_coded are defined in [119.2.6.2.2](#). Functions R_TYPE and DECODE, and the block types are defined in [119.2.6.2.3](#).

119.2.6 Detailed functions and state diagrams

119.2.6.2 State variables

119.2.6.2.2 Variables

Change the definition of the signal_ok variable in 119.2.6.2.2 as follows:

signal_ok

Boolean variable that is set based on the most recently received value of PMA:IS_SIGNAL.indication(SIGNAL_OK). It is true if the value was OK and false ~~if the value was FAIL~~ otherwise.

119.3 PCS management

119.3.1 PCS MDIO function mapping

Change Table 119–5 as follows (some unchanged rows not shown):

Table 119–1—MDIO/PCS status variable mapping

MDIO status variable	PCS register name	Register/bit number	PCS status variable
...			
PCS FEC symbol errors, PCS lanes 0 to x	PCS FEC symbol error counter register, lanes 0 to x	3.600 to 3.631	FEC_symbol_error_counter_ i
<u>FEC codeword counter</u>	<u>PCS FEC codeword counter register</u>	<u>3.300 to 3.302</u>	<u>FEC_cw_counter</u>
<u>FEC codeword error bin counter, 1 to 15</u>	<u>PCS FEC codeword error bin, 1 to 15</u>	<u>3.340 to 3.369</u>	<u>FEC_codeword_error_bin_<1:15></u>
Tx LPI indication	PCS status 1	3.1.9	Tx LPI indication
...			

Insert new subclauses 119.3.4a and 119.3.4b after 119.3.4 as follows:

119.3.4a FEC_cw_counter

The following optional counter may be implemented for these PHY types:

- 200GBASE-KR1
- 200GBASE-CR1
- 200GBASE-DR1
- 200GBASE-DR1-2
- 400GBASE-KR2
- 400GBASE-CR2
- 400GBASE-DR2
- 400GBASE-DR2-2

FEC_cw_counter

A 48-bit counter that counts all FEC codewords received by the PCS when align_status is true.

This variable is mapped to the registers defined in 45.2.3.48a (3.300 to 3.302).

119.3.4b FEC_codeword_error_bin_ i

The following optional counters may be implemented for these PHY types:

- 200GBASE-KR1
- 200GBASE-CR1
- 200GBASE-DR1

- 200GBASE-DR1-2
- 400GBASE-KR2
- 400GBASE-CR2
- 400GBASE-DR2
- 400GBASE-DR2-2

FEC_codeword_error_bin_*i*

A set of fifteen 32-bit counters where counter *i* counts once for each codeword received with exactly *i* correctable 10-bit symbols when align_status is true, *i* = 1 to 15. These variables are mapped to registers defined in 45.2.3.48b (3.340 to 3.369).

119.6 Auto-Negotiation

Change the text in 119.6 (as modified by IEEE Std 802.3ck-2022) as follows:

The following requirements apply to a PCS used with a 200GBASE-CR4, 200GBASE-CR2, 200GBASE-CR1, 200GBASE-KR4, 200GBASE-KR2, 200GBASE-KR1, 400GBASE-CR4, 400GBASE-CR2, or 400GBASE-KR4, or 400GBASE-KR2 PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN_LINK.indication(link_status) primitive (see 73.9). The parameter link_status shall take the value FAIL when PCS_status = false and the value OK when PCS_status = true. The primitive shall be generated when the value of link_status changes.

119.7 Protocol implementation conformance statement (PICS) proforma for Clause 119, Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R

119.7.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 119, Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

119.7.2 Identification

119.7.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

119.7.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-202x, Clause 119, Physical Coding Sublayer (PCS) 64B/66B, type 200GBASE-R and 400GBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-202x.)	

Date of Statement	
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119.7.4 PICS proforma tables for Physical Coding Sublayer (PCS) 64B/66B, type 200GBASE-R and 400GBASE-R

119.7.4.1 Transmit function

Change the table in 119.7.4.1 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	119.2.4.1.2		<u>O</u>	Yes [] No []
TF1	Encode and rate matching	119.2.4.1		<u>M</u>	Yes []
TF2	Transmit 64B/66B encoder uses state diagram in Figure 119-14	119.2.4.1.1		<u>!SE:M</u>	Yes [] N/A []
TF3	Transmit 64B/66B encoder uses stateless rules in Table 172-1	119.2.4.1.2		<u>SE:M</u>	Yes [] N/A []
TF4	64B/66B to 256B/257B transcoder	119.2.4.2	tx_xcoded<256:0> constructed per 119.2.4.2	M	Yes []
TF5	Transmission bit ordering	119.2.4.8	First bit transmitted is bit 0	M	Yes []
TF6	Pad value	119.2.4.4	PRBS9	M	Yes []
TF7	Alignment marker insertion	119.2.4.4		M	Yes []
TF8	Pre-FEC distribution	119.2.4.5	Distribute the data to two FEC codewords	M	Yes []
TF9	Reed-Solomon encoder	119.2.4.6	RS(544,514)	M	Yes []
TF10	Symbol distribution	119.2.4.7	Distribution is based on 10b symbols	M	Yes []

119.7.4.2 Receive function

Change the table in 119.7.4.2 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
<u>*SD</u>	<u>Uses stateless decoder</u>	<u>119.2.5.8.2</u>		<u>O</u>	<u>Yes []</u> <u>No []</u>
RF1	Skew tolerance	119.2.5.1	Maximum Skew of 180 ns between PCS lanes and a maximum Skew Variation of 4 ns	M	Yes []
...					
RF6	256B/257B to 64B/66B transcoder	119.2.5.7	rx_coded_j<65:0>, j=0 to 3 constructed per 119.2.5.7	M	Yes []
<u>RF7</u>	<u>Receive 64B/66B decoder uses state diagram in Figure 119-15</u>	<u>119.2.5.8.1</u>		<u>!SD:M</u>	<u>Yes []</u> <u>N/A []</u>
<u>RF8</u>	<u>Receive 64B/66B decoder uses stateless rules in Table 172-4</u>	<u>119.2.5.8.2</u>		<u>SD:M</u>	<u>Yes []</u> <u>N/A []</u>
<u>RF9</u>	<u>Decode and rate matching</u>	<u>119.2.5.8</u>		<u>M</u>	<u>Yes []</u>

119.7.4.12 Auto-Negotiation for Backplane Ethernet functions

Change the table in 119.7.4.12 (as modified by IEEE Std 802.3ck-2022) as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Support for use with a 200GBASE-CR4, 200GBASE-CR2, <u>200GBASE-CR1</u> , 200GBASE-KR4, 200GBASE-KR2, <u>200GBASE-KR1</u> , 400GBASE-CR4, or <u>400GBASE-CR2</u> , 400GBASE-KR4 PMD , or <u>400GBASE-KR2 PMD</u>	119.6	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN1	AN_LINK.indication primitive	119.6	Support of the AN_LINK.indication(link_status) primitive	AN:M	Yes [] N/A []
AN2	link_status parameter	119.6	Takes the value OK or FAIL, as described in 119.6	AN:M	Yes [] N/A []
AN3	Generation of AN_LINK.indication primitive	119.6	Generated when the value of link_status changes	AN:M	Yes [] N/A []

120. Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R

120.1 Overview

120.1.1 Scope

Change the first paragraph in 120.1.1 as follows:

~~This clause specifies the Physical Medium Attachment sublayer (PMA) for the families of 200 Gb/s and 400 Gb/s Physical Layer implementations, known as 200GBASE-R and 400GBASE-R. The PMA allows the PCS (specified in Clause 119) to connect in a media-independent way with a range of physical media, or for the DTE XS to connect to the PHY XS (specified in Clause 118). The 200GBASE-R PMA(s) can support any of the 200 Gb/s PMDs in Table 116-1, and the 400GBASE-R PMA(s) can support any of the 400 Gb/s PMDs in Table 116-2. The terms 200GBASE-R and 400GBASE-R are used when referring generally to Physical Layers using the PMA defined in this clause.~~

This clause specifies forms of the Physical Medium Attachment (PMA) sublayer that use bit-multiplexing for 200GBASE-R and 400GBASE-R Physical Layer implementations.

When necessary to differentiate the bit-multiplexing PMA (BM-PMA) types defined in this clause from the symbol-multiplexing PMA (SM-PMA) types defined in Clause 176, the term BM-PMA is used. Within this clause the term PMA refers specifically to the BM-PMA.

120.1.1a Introduction

The PMA allows the PCS (specified in Clause 119) to connect in a media-independent way with a range of physical media, or for the DTE XS to connect to the PHY XS (specified in Clause 118). The 200GBASE-R PMA(s) supports any of the PMDs listed in Table 116-3 and Table 116-4. The 400GBASE-R PMA(s) supports any of the PMDs listed in Table 116-3a and Table 116-5.

120.1.4 PMA sublayer positioning

In the fourth paragraph of 120.1.4 (as modified by IEEE Std 802.3ck-2022), change item b) as follows:

- b) 200GAUI-n is a physical instantiation of the connection between two adjacent 200GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface. 400GAUI-n is a physical instantiation of the connection between two adjacent 400GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface.
 - 1) As physical instantiations, these define electrical and timing specification as well as requiring a receive re-timing function.
 - 2) 200GAUI-8 is a 26.5625 GBd by 8 lane NRZ physical instantiation of the 200 Gb/s connection. 400GAUI-16 is a 26.5625 GBd by 16 lane NRZ physical instantiation of the 400 Gb/s connection.
 - 3) 200GAUI-4 is a 26.5625 GBd by 4 lane PAM4 physical instantiation of the 200 Gb/s connection. 400GAUI-8 is a 26.5625 GBd by 8 lane PAM4 physical instantiation of the 400 Gb/s connection.

- 4) 200GAUI-2 is a 53.125 GBd by 2 lane PAM4 physical instantiation of the 200 Gb/s connection. 400GAUI-4 is a 53.125 GBd by 4 lane PAM4 physical instantiation of the 400 Gb/s connection.
- 5) For a PHY that includes a 200GAUI-2 interface, it is recommended that the signaling rate range for a 200GAUI-8 or 200GAUI-4 PMA output that is in the same package as the PCS be limited to $26.5625 \text{ GBd} \pm 50 \text{ ppm}$.
- 6) For a PHY that includes a 400GAUI-4 interface, it is recommended that the signaling rate range for a 400GAUI-16 or 400GAUI-8 PMA output that is in the same package as the PCS be limited to $26.5625 \text{ GBd} \pm 50 \text{ ppm}$.
- 7) For a PHY that includes a 200GAUI-1 interface or a 200GBASE-KR1, 200GBASE-CR1, 200GBASE-DR1, or 200GBASE-DR1-2 PMD, the signaling rate range for a 200GAUI-8, 200GAUI-4, or 200GAUI-2 PMA output that is in the same package as the PCS shall be limited to $\pm 50 \text{ ppm}$, instead of $\pm 100 \text{ ppm}$.
- 8) For a 200GMII Extender that includes a 200GAUI-1 interface, the signaling rate range for a 200GAUI-8, 200GAUI-4, or 200GAUI-2 PMA output that is in the same package as the DTE 200GXS or PHY 200GXS shall be limited to $\pm 50 \text{ ppm}$, instead of $\pm 100 \text{ ppm}$.
- 9) For a PHY that includes a 400GAUI-2 interface or a 400GBASE-KR2, 400GBASE-CR2, 400GBASE-DR2, or 400GBASE-DR1-2 PMD, the signaling rate range for a 400GAUI-16, 400GAUI-8, or 400GAUI-4 PMA output that is in the same package as the PCS shall be limited to $\pm 50 \text{ ppm}$, instead of $\pm 100 \text{ ppm}$.
- 10) For a 400GMII Extender that includes a 400GAUI-2 interface, the signaling rate range for a 400GAUI-16, 400GAUI-8, or 400GAUI-4 PMA output that is in the same package as the DTE 400GXS or PHY 400GXS shall be limited to $\pm 50 \text{ ppm}$, instead of $\pm 100 \text{ ppm}$.

Clause 169 was added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024:

169. Introduction to 800 Gb/s networks

169.1 Overview

169.1.2 Relationship of 800 Gigabit Ethernet to the ISO OSI reference model

In 169.1.2, insert items e) through g) at the end of the lettered list as follows:

- e) The PMA service interface, when physically implemented as 800GAUI-4 (800 Gb/s 4-lane Attachment Unit Interface) at an observable interconnection port, as specified in Annex 176C or Annex 176D, uses a 4-lane data path.
- f) The MDI for each of the following PMDs uses a 4-lane data path:
 - 800GBASE-KR4 as specified in Clause 178
 - 800GBASE-CR4 as specified in Clause 179
 - 800GBASE-DR4 as specified in Clause 180
 - 800GBASE-FR4-500 as specified in Clause 181
 - 800GBASE-DR4-2 as specified in Clause 182
 - 800GBASE-FR4 and 800GBASE-LR4 as specified in Clause 183
- g) The MDI for each of the following PMDs uses a 1-lane data path:
 - 800GBASE-LR1 as specified in Clause 185
 - 800GBASE-ER1-20 and 800GBASE-ER1 as specified in Clause 187

169.1.3 Nomenclature

Change Table 169–1 as follows:

Table 169–1—800 Gb/s PHYs

Name	Description
800GBASE-KR8	800 Gb/s PHY using 800GBASE-R encoding over eight lanes of an electrical backplane (see Clause 163)
<u>800GBASE-KR4</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four lanes of an electrical backplane (see Clause 178)</u>
800GBASE-CR8	800 Gb/s PHY using 800GBASE-R encoding over eight lanes of twinaxial copper cable (see 1.4.559 and Clause 162)
<u>800GBASE-CR4</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 179)</u>
800GBASE-VR8	800 Gb/s PHY using 800GBASE-R encoding over eight multimode fibers in each direction with reach up to at least 50 m (see Clause 167)
800GBASE-SR8	800 Gb/s PHY using 800GBASE-R encoding over eight multimode fibers in each direction with reach up to at least 100 m (see Clause 167)
800GBASE-DR8	800 Gb/s PHY using 800GBASE-R encoding over eight single-mode fibers in each direction with reach up to at least 500 m (see Clause 124)
<u>800GBASE-DR4</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four single-mode fibers in each direction with reach up to at least 500 m (see Clause 180)</u>
<u>800GBASE-FR4-500</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 500 m (see Clause 181)</u>
800GBASE-DR8-2	800 Gb/s PHY using 800GBASE-R encoding over eight single-mode fibers in each direction with reach up to at least 2 km (see Clause 124)
<u>800GBASE-DR4-2</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four single-mode fibers in each direction with reach up to at least 2 km (see Clause 182)</u>
<u>800GBASE-FR4</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 2 km (see Clause 183)</u>
<u>800GBASE-LR4</u>	<u>800 Gb/s PHY using 800GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 10 km (see Clause 183)</u>
<u>800GBASE-LR1</u>	<u>800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection, over single-mode fiber, with reach up to at least 10 km (see Clause 185)</u>
<u>800GBASE-ER1-20</u>	<u>800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection with reach up to at least 20 km (see Clause 187)</u>
<u>800GBASE-ER1</u>	<u>800 Gb/s PHY using 800GBASE-R encoding, dual polarization 16-state quadrature amplitude modulation (DP-16QAM), and coherent detection with reach up to at least 40 km (see Clause 187)</u>

169.1.4 Physical Layer signaling systems

Change the text in 169.1.4 as follows:

This standard specifies a family of Physical Layer implementations. Table 169–2 and Table 169–3 through Table 169–3a specify the correlation between PHY types and clauses. Implementations conforming to one or more PHY types meet the requirements of the corresponding clauses.

Change Table 169–2 and Table 169–3 as follows:

Table 169–2—PHY type and clause correlation (800GBASE copper)

PHY type	Clause ^a													
	73	90	170		171	171	173	120F	162	163	176	176C	178	179
	Auto-Negotiation	Time Synchronization	RS	800GMII	800GMII Extender	800GBASE-R PCS	800GBASE-R BM-PMA	800GAUI-8 C2C	800GBASE-CR8 PMD	800GBASE-KR8 PMD	800GBASE-R SM-PMA	800GAUI-4 C2C	800GBASE-KR4 PMD	800GBASE-CR4 PMD
800GBASE-KR4	M	O	M	O	O	M	C	O	==	==	M	O	M	==
800GBASE-KR8	M	O	M	O	O	M	M	O	==	M	C	O	==	==
800GBASE-CR4	M	O	M	O	O	M	C	O	==	==	M	O	==	M
800GBASE-CR8	M	O	M	O	O	M	M	O	M	==	C	O	==	==

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Table 169–3—PHY type and clause correlation (800GBASE optical PAM4)

PHY type	Clause ^a																				
	Time Synchronization		90																		
	RS		170																		
		800GMII		171	172	173	120F	120G		124		167		176	176C	176D	177	180	181	182	183
		800MII Extender		800GBASE-R PCS	800GBASE-R BM-PMA	800GAUI-8 C2C	800GAUI-8 C2M	800GBASE-DR8 PMD	800GBASE-DR8-2 PMD	800GBASE-VR8 PMD	800GBASE-SR8 PMD	800GBASE-R SM-PMA	800GAUI-4 C2C	800GAUI-4 C2M	800GBASE-R Inner FEC	800GBASE-DR4 PMD	800GBASE-FR4-500 PMD	800GBASE-DR4-2 PMD	800GBASE-FR4 PMD	800GBASE-LR4 PMD	
800GBASE-VR8	Q	M	O	O	M	M	O	O			M		C	Q	Q						
800GBASE-SR8	Q	M	O	O	M	M	O	O				C	Q	Q							
800GBASE-DR4	Q	M	Q	Q	M	C	Q	Q				M	Q	Q		M					
800GBASE-FR4-500	Q	M	Q	Q	M	C	Q	Q				M	Q	Q			M				
800GBASE-DR8	Q	M	O	O	M	M	O	O	M			C	Q	Q							
800GBASE-DR4-2	Q	M	Q	Q	M	C	Q	Q				M	Q	Q	M			M			
800GBASE-DR8-2	Q	M	O	O	M	M	O	O		M		C	Q	Q							
800GBASE-FR4	Q	M	Q	Q	M	C	Q	Q				M	Q	Q	M				M		
800GBASE-LR4	Q	M	Q	Q	M	C	Q	Q				M	Q	Q	M					M	

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

Insert new Table 169–3a after Table 169–3 as follows:

Table 169–3a—PHY type and clause correlation (800GBASE optical coherent)

PHY type	Clause ^a														
	90	170		171	172	173	120F	120G	176	176C	176D	184	185	186	187
	Time Synchronization	RS	800GMII	800MII Extender	800GBASE-R PCS	800GBASE-R BM-PMA	800GAUI-8 C2C	800GAUI-8 C2M	800GBASE-R SM-PMA	800GAUI-4 C2C	800GAUI-4 C2M	800GBASE-LR1 Inner FEC	800GBASE-LR1 PMD	800GBASE-ER1 FEC/PMA	800GBASE-ER1-20 PMD
800GBASE-LR1	O	M	O	O	M	C	O	O	C	O	O	M	M	—	—
800GBASE-ER1-20	O	M	O	O	M	C	O	O	C	O	O	—	—	M	M
800GBASE-ER1	O	M	O	O	M	C	O	O	C	O	O	—	—	M	M

^a O = Optional, M = Mandatory, C = Conditional (refer to PMD clause for details).

169.2 Summary of 800 Gigabit Ethernet architecture

169.2.4 Physical Medium Attachment (PMA) sublayer

Change the second sentence in 169.2.4 as follows:

The 800GBASE-R PMA that uses bit multiplexing (BM-PMA) is specified in Clause 173.

The 800GBASE-R PMA that uses symbol multiplexing (SM-PMA) is specified in Clause 176.

The 800GBASE-ER1 PMA is specified in Clause 186.

Insert 169.2.4a and 169.2.4b after 169.2.4 as follows:

169.2.4a Attachment Unit Interface (800GAUI-n)

The 800 Gb/s Attachment Unit Interface (800GAUI-n) provides an electrical interface within an 800GBASE-R PHY or 800GMII Extender. The 800GAUI-n is defined for chip-to-chip (C2C) and chip-to-module (C2M) implementations.

The 800GAUI-n C2C is specified in Annex 120F and Annex 176C.

The 800GAUI-n C2M is specified in Annex 120G and Annex 176D.

The 800GAUI-n instantiations are described in 176B.6.

169.2.4b FEC sublayer

The 800GBASE-R Inner FEC and 800GBASE-LR1 Inner FEC sublayers provide error correction, in addition to that provided by the 800GBASE-R PCS, for the PMD. The 800GBASE-ER1 FEC sublayer terminates the FEC provided by the 800GBASE-R PCS and replaces it with a separate FEC for use with the 800GBASE-ER1 and 800GBASE-ER1-20 PMDs.

The 800GBASE-R Inner FEC sublayer is specified in Clause 177. It is used for 800GBASE-DR4-2, 800GBASE-FR4, and 800GBASE-LR4 PHY types.

The 800GBASE-LR1 Inner FEC sublayer is specified in Clause 184. It is used for the 800GBASE-LR1 PHY type.

The 800GBASE-ER1 FEC sublayer is specified in Clause 186. It is used for 800GBASE-ER1 and 800GBASE-ER1-20 PHY types.

Insert new subclauses 169.2.9 and 169.2.10 after 169.2.8 as follows:

169.2.9 Time Synchronization

An 800 Gb/s Physical Layer may optionally support time synchronization protocols that require knowledge of packet egress and ingress time. Ethernet support for time synchronization protocols is defined in Clause 90.

If time synchronization is supported, then the following capabilities are provided:

- The 800 Gb/s Reconciliation Sublayer (RS) provides a Time Synchronization Service Interface (TSSI) that connects to a TimeSync client (see 90.4).
- The path data delay values of each sublayer in a Physical Layer are reported as status variables, which may be mapped to MDIO status registers (see 90.6).

169.2.10 Inter-sublayer link training (ILT)

Inter-sublayer link training (ILT) facilitates the orderly start-up of an inter-sublayer link (ISL) and coordinates the start-up of a series of ISLs along a network path. For each ISL, ILT provides a mechanism for a receiver to control peer transmitter states (e.g., equalization, modulation, and precoding), to indicate the receiver state, and to coordinate the transition to DATA mode. ILT is specified in Annex 178B.

ILT is supported by the following PHY types:

- 800GBASE-KR4
- 800GBASE-CR4
- 800GBASE-DR4
- 800GBASE-FR4-500
- 800GBASE-DR4-2
- 800GBASE-FR4
- 800GBASE-LR4

169.3 Service interface specification method and notation

169.3.2 Instances of the inter-sublayer service interface

Change the text in 169.3.2 as follows:

The inter-sublayer service interface may be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

- PMA:—for primitives issued on the interface between the PMA and the PCS or DTE 800GXS above called the PMA service interface
- PHY_XS:—for primitives issued on the interface between the PHY 800GXS and the PMA above called the PHY 800GXS service interface
- PMD:—for primitives issued on the interface between the PMD and the PMA above called the PMD service interface
- FEC:—for primitives issued on the interface between the Inner FEC or Segmented FEC, and the PMA, PCS, or PHY 800GXS above called the FEC service interface

As an example, the primitives for the PMD service interface are identified as follows:

PMD:IS_UNITDATA_*i*.request
PMD:IS_UNITDATA_*i*.indication
PMD:IS_SIGNAL.indication
PMD:IS_SIGNAL.request

Examples of inter-sublayer service interfaces for 800GBASE-R with their corresponding instance names are illustrated in Figure 169–2, ~~and Figure 169–3,~~ Figure 185–2, and Figure 187–2.

Primitives for other instances of inter-sublayer interfaces are represented in a similar manner.

Replace Figure 169–2 with the following figure:

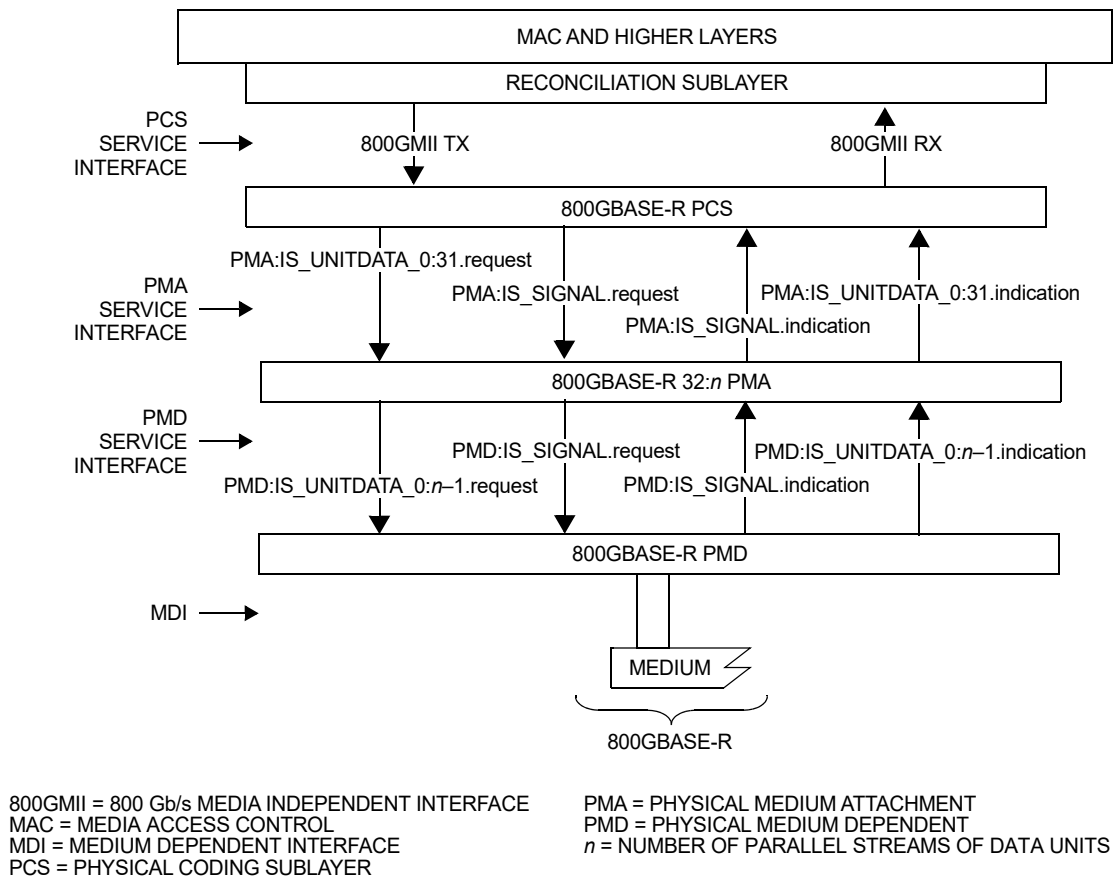


Figure 169–2—800GBASE-R inter-sublayer service interfaces not including 800GMII Extender

Insert new Figure 169–2a after Figure 169–2 as follows:

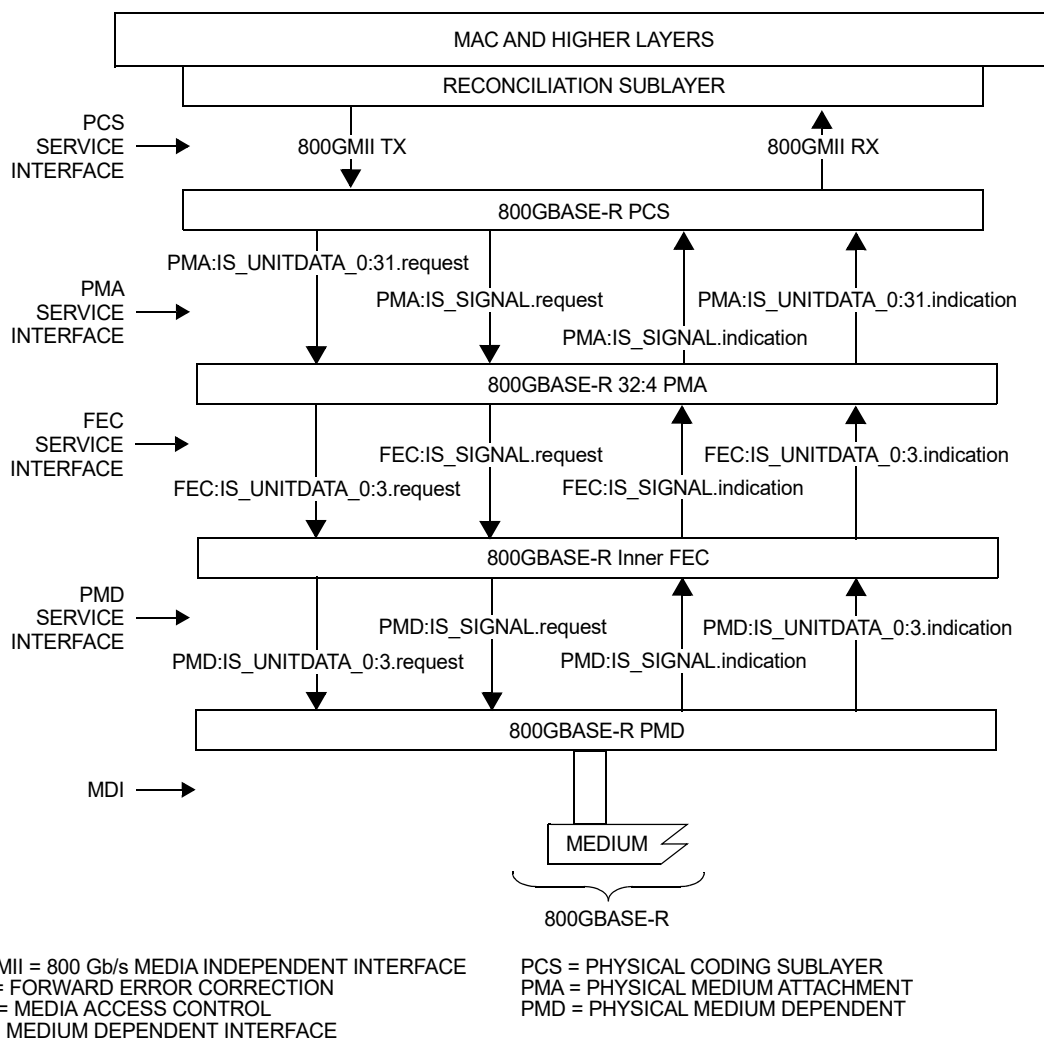


Figure 169–2a—800GBASE-R inter-sublayer service interfaces including 800GBASE-R Inner FEC

Replace Figure 169–3 with the following figure:

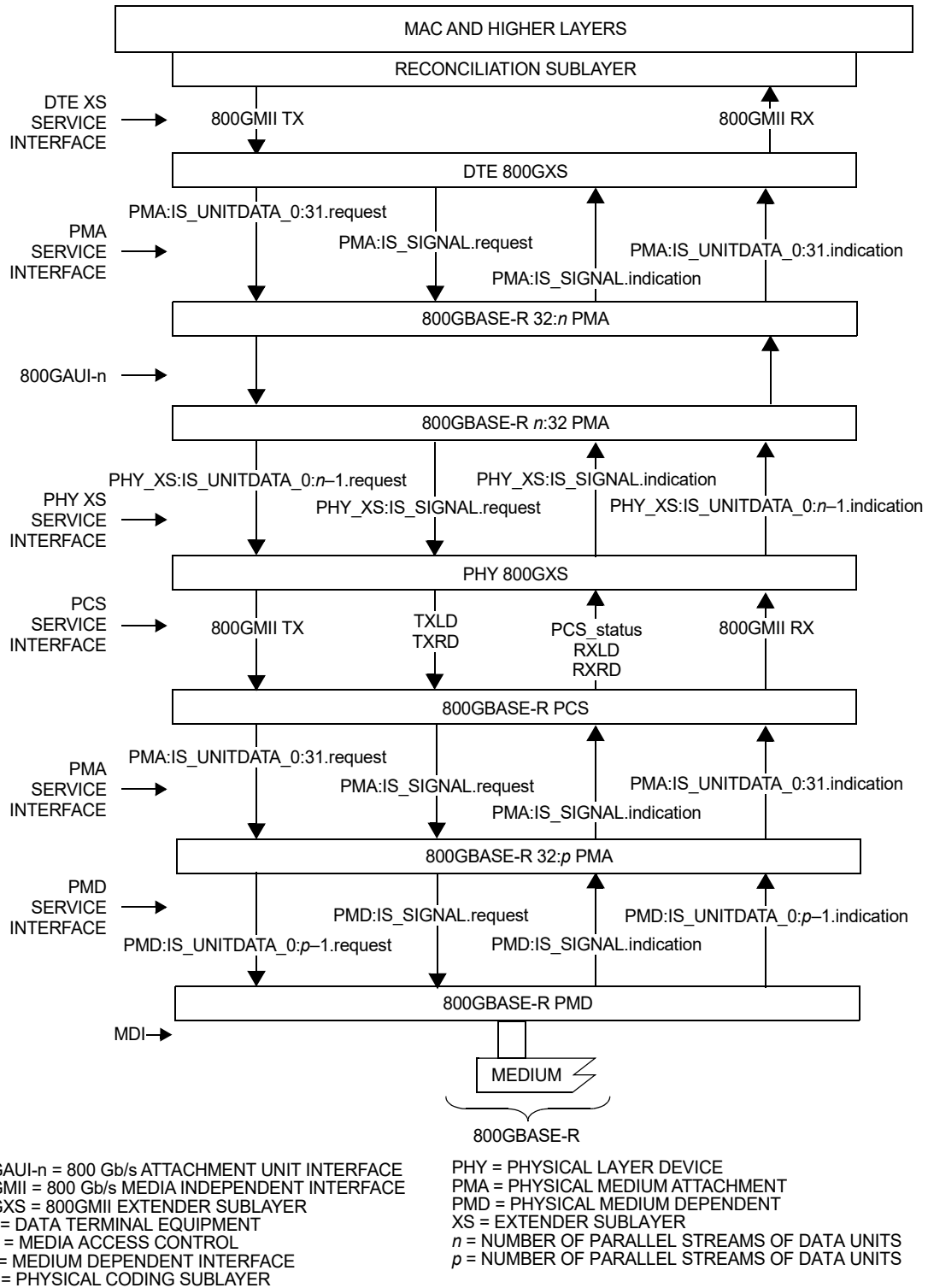


Figure 169–3—800GBASE-R inter-sublayer service interfaces including 800GMII Extender

169.3.3 Semantics of inter-sublayer service interface primitives

Change 169.3.3 as follows:

The semantics of the inter-sublayer service interface primitives for the 800GBASE-R sublayers are described in ~~116.3.3.1 through 116.3.3.3~~ 116.3.3.

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169.4 Delay constraints

Change Table 169–4 as follows:

Table 169–4—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantas) ^b	Maximum (ns)	Notes ^c
800G MAC, RS, and MAC Control	196 608	384	245.76	See 170.1.4.
800GBASE-R PCS or 800GXS ^d	640 000	1250	800	See 172.5.
800GBASE-R 32:8 PMA or 8:32 BM -PMA	36 864	72	46.08	See 173.5.4.
800GBASE-R 8:8 BM -PMA	72 728	144	92.16	See 173.5.4.
800GBASE-R 32:4 or 4:32 SM -PMA	12 288	24	15.36	See 176.8.
800GBASE-R 4:4 SM -PMA	12 288	24	15.36	See 176.8.
800GBASE-R Inner FEC	93 184	182	116.48	See 177.8.
800GBASE-LR1 Inner FEC	131 072	256	163.84	See 184.8.
800GBASE-ER1 FEC and PMA	4 000 256	7813	5000.32	See 186.5.
800GAUI-4 C2C or C2M component	36 864	72	46.08	Includes allocation of 5 ns for one direction through AUI channel. See 176C.5 and 176D.5.
800GBASE-KR8 PMD	32 768	64	40.96	Includes allocation of 14 ns for one direction through backplane medium. See 163.5.
800GBASE-KR4 PMD	59 392	96	74.24	Includes allocation of 14 ns for one direction through backplane medium. See 178.6.
800GBASE-CR8 PMD	32 768	64	40.96	Includes allocation of 14 ns for one direction through cable medium. See 162.5.
800GBASE-CR4 PMD	59 392	96	74.24	Includes allocation of 14 ns for one direction through cable medium. See 179.6.
800GBASE-VR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 167.3.1.
800GBASE-SR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 167.3.1.
800GBASE-DR8 PMD	16 384	32	20.48	Includes 2 m of fiber. See 124.3.1.
800GBASE-DR4 PMD	59 392	96	74.24	Includes 2 m of fiber. See 180.4.1.
800GBASE-FR4-500 PMD	59 392	96	74.24	Includes 2 m of fiber. See 181.4.1.
800GBASE-DR8-2 PMD	16 384	32	20.48	Includes 2 m of fiber. See 124.3.1.
800GBASE-DR4-2 PMD	59 392	96	74.24	Includes 2 m of fiber. See 182.4.1.
800GBASE-FR4 PMD	59 392	96	74.24	Includes 2 m of fiber. See 183.4.1.

Table 169–4—Sublayer delay constraints (*continued*)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
<u>800GBASE-LR4 PMD</u>	<u>59 392</u>	<u>96</u>	<u>74.24</u>	<u>Includes 2 m of fiber. See 183.4.1.</u>
<u>800GBASE-LR1 PMD</u>	<u>16 384</u>	<u>32</u>	<u>20.48</u>	<u>Includes 2 m of fiber. See 185.4.</u>
<u>800GBASE-ER1-20 PMD</u>	<u>16 384</u>	<u>32</u>	<u>20.48</u>	<u>Includes 2 m of fiber. See 187.4.</u>
<u>800GBASE-ER1 PMD</u>	<u>16 384</u>	<u>32</u>	<u>20.48</u>	<u>Includes 2 m of fiber. See 187.4.</u>

^a For 800GBASE, 1 bit time (BT) is equal to 1.25 ps. (See 1.4.215 for the definition of bit time.)

^b For 800GBASE, 1 pause_quantum is equal to 640 ps. (See 1.4.459 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^d If an implementation includes the 800GMII Extender, the delay associated with the 800GMII Extender includes two 800GXS sublayers.

169.5 Skew constraints

Change the fourth and fifth paragraphs in 169.5 as follows:

In the transmit direction, the Skew points are defined in the following locations:

- SP1 on the 800GAUI-n interface closest to the PMD; at the input of the PMA ~~closest to the PMD~~
- SP2 on the PMD service interface, at the input of the PMD
- SP3 at the output of the PMD, at the MDI

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD
- SP5 on the PMD service interface, at the output of the PMD
- SP6 on the 800GAUI-n interface closest to the PCS or DTE 800GXS; at the output of the PMA ~~closest to the PCS or DTE 800GXS~~

Replace Figure 169–4 with the following figure:

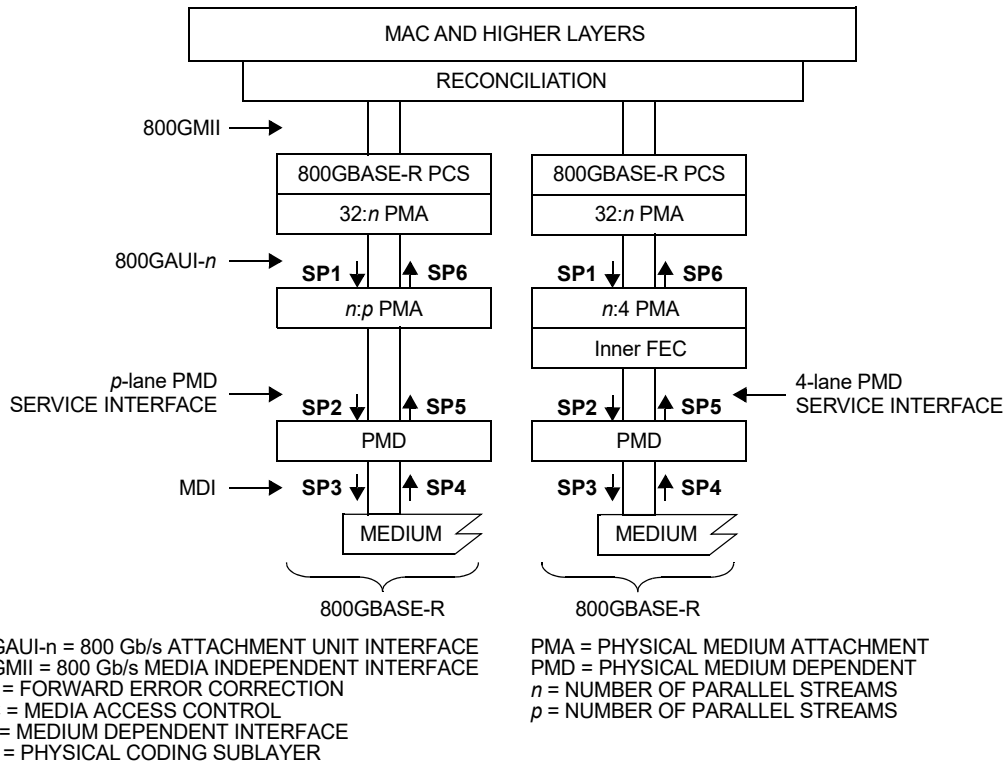


Figure 169–4—800GBASE-R Skew points for a PHY with one 800GAUI-n

Replace Figure 169–4 with the following figure:

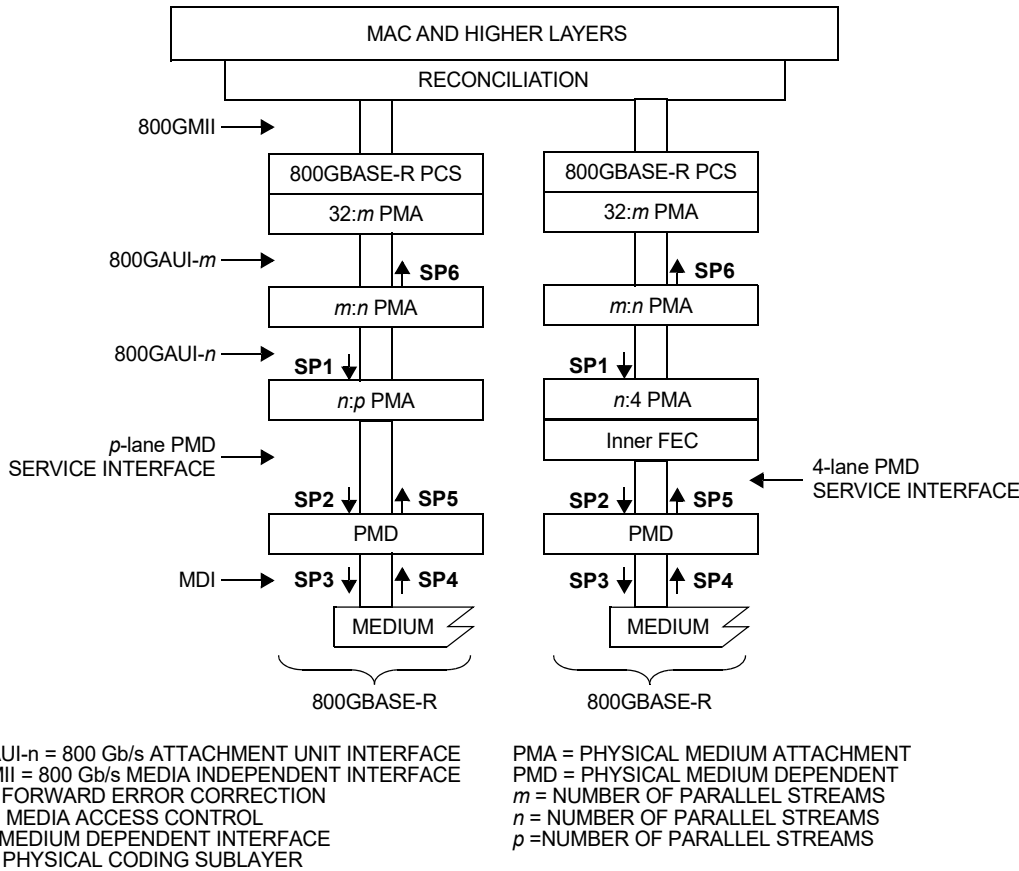


Figure 169–5—800GBASE-R Skew points for a PHY with two 800GAUI-n

Change Table 169–5 as follows:

Table 169–5—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 800GBASE-R PCS lane (UI) ^b	Notes ^c
SP1	16	≈ 425	See 173.5.3 , 176.9
SP2	25	≈ 664	See 173.5.3 , 124.3.2.2 , 162.6.3 , 163.6.3 , 167.3.2.2 , 178.7 , 179.7 , 180.4.2 , 181.4.2 , 182.4.2 , 183.4.2
SP3	36	≈ 956	See 173.5.3 , 124.3.2.2 , 162.6.3 , 163.6.3 , 167.3.2.2 , 178.7 , 179.7 , 180.4.2 , 181.4.2 , 182.4.2 , 183.4.2
SP4	116	≈ 3081	See 173.5.3 , 124.3.2.2 , 162.6.3 , 163.6.3 , 167.3.2.2 , 178.7 , 179.7 , 180.4.2 , 181.4.2 , 182.4.2 , 183.4.2
SP5	127	≈ 3373	See 173.5.3 , 124.3.2.2 , 162.6.3 , 163.6.3 , 167.3.2.2 , 178.7 , 179.7 , 180.4.2 , 181.4.2 , 182.4.2 , 183.4.2
SP6	145	≈ 3852	See 173.5.3 , 176.9
At PCS receive	152	≈ 4038	See 172.2.5.1

^a The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^b The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 169–6 as follows:

Table 169–6—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 26.5625 Gb/s PCS lane (bit) ^a	Maximum Skew Variation for 53.125 GBd PMD-lane (UI) ^b	Maximum Skew Variation for 106.25 GBd PMD lane (UI) ^c	Maximum Skew Variation for 113.4375 GBd PMD lane (UI) ^d	Notes ^e
SP1	0.2	N/A	≈ 11	≈ 21	N/A	See 173.5.3, 176.9
SP2	0.4	N/A	≈ 21	≈ 43	≈ 45	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2, 178.7, 179.7, 180.4.2, 181.4.2, 182.4.2, 183.4.2
SP3	0.6	N/A	≈ 32	≈ 64	≈ 68	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2, 178.7, 179.7, 180.4.2, 181.4.2, 182.4.2, 183.4.2
SP4	3.4	N/A	≈ 181	≈ 361	≈ 386	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2, 178.7, 179.7, 180.4.2, 181.4.2, 182.4.2, 183.4.2
SP5	3.6	N/A	≈ 191	≈ 383	≈ 408	See 173.5.3, 124.3.2.2, 162.6.3, 163.6.3, 167.3.2.2, 178.7, 179.7, 180.4.2, 181.4.2, 182.4.2, 183.4.2
SP6	3.8	N/A	≈ 202	≈ 404	N/A	See 173.5.3, 176.9
At PCS receive	4	≈ 106	≈ 213 N/A	N/A	N/A	See 173.5.3, 176.9

^a The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^b The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^c The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 9.41176 ps at PMD lane signaling rate of 106.25 GBd.

^d The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 8.31543 ps at PMD lane signaling rate of 113.4375 GBd.

^e Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Change the title of Clause 170 (added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024) as follows:

170. Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s (800GMII) and 1.6 Tb/s (1.6TMII)

170.1 Overview

Change the text in 170.1 as follows:

This clause defines the characteristics of the Reconciliation Sublayer (RS), ~~and the 800 Gb/s Media Independent Interface (800GMII), and the 1.6 Tb/s Media Independent Interface (1.6TMII) between Ethernet media access controllers and various 800 Gb/s PHYs.~~ Figure 170–1 shows the relationship of the RS ~~and 800GMII, and 1.6TMII~~ to the ISO/IEC OSI reference model.

The 800GMII and 1.6TMII ~~is an~~ optional logical interfaces between the MAC sublayer and the Physical Layer device (PHY). The 800GMII/1.6TMII Extender may optionally be used to extend the 800GMII/1.6TMII (see Clause 171).

The RS adapts the bit serial protocols of the MAC to the parallel format of the PCS service interface. Though the 800GMII and 1.6TMII ~~is an~~ are optional interfaces, it is they are used in this standard as a basis for specification. The Physical Coding Sublayer (PCS) is specified to the 800GMII/1.6TMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and 800GMII/1.6TMII were implemented.

~~The 800GMII has the following characteristics:~~

- ~~a) It supports a speed of 800 Gb/s.~~
- ~~b) Data and delimiters are synchronous to a clock reference.~~
- ~~c) It provides independent 64-bit wide transmit and receive data paths.~~
- ~~d) It supports full duplex operation only.~~

The 800GMII/1.6TMII have the following characteristics:

- The 800GMII supports a speed of 800 Gb/s.
- The 1.6TMII supports a speed of 1.6 Tb/s.
- Data and delimiters are synchronous to a clock reference.
- They provide independent 64-bit wide transmit and receive data paths.
- They support full duplex operation only.

Replace Figure 170–1 with the following figure:

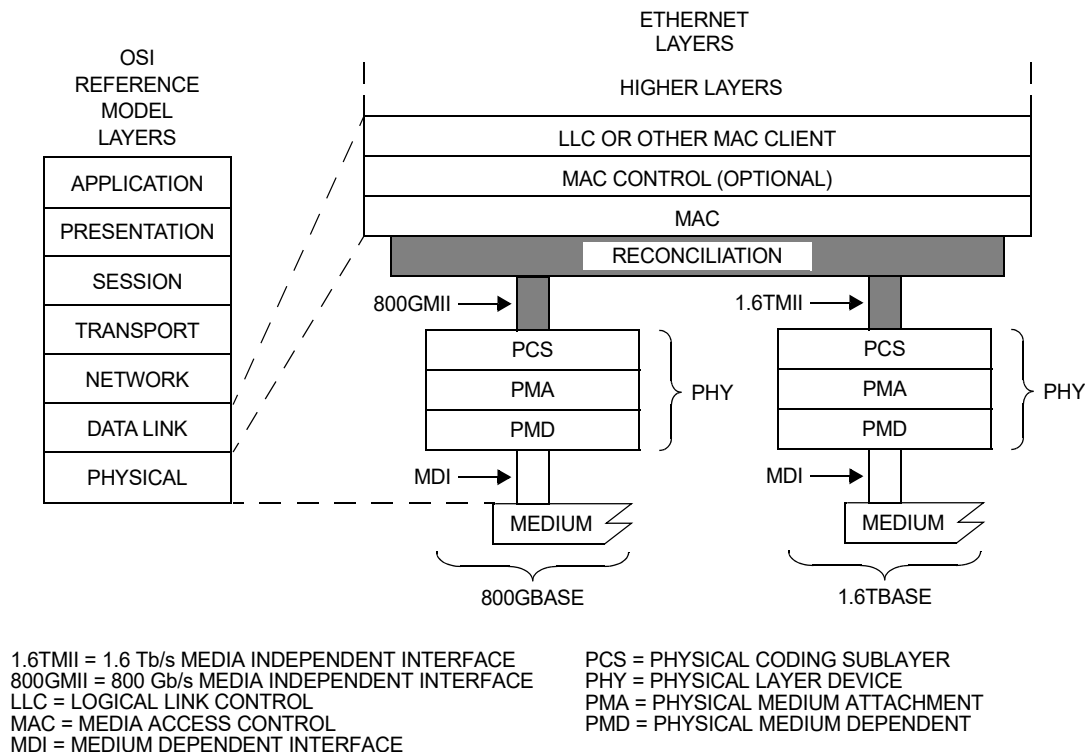


Figure 170–1—RS, 800GMII, and 1.6TMII relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

170.1.1 Summary of major concepts

Change the text in 170.1.1 as follows:

The following are the major concepts of the 800GMII/1.6TMII and RS:

- The 800GMII/1.6TMII are is-functionally similar to other media independent interfaces that have been defined for lower speeds, as they all define an interface allowing independent development of MAC and PHY logic.
- The RS converts between the MAC serial data stream and the parallel data paths of the 800GMII/1.6TMII.
- The RS maps the signal set provided at the 800GMII/1.6TMII to the PLS service primitives provided at the MAC.
- Each direction of data transfer is independent and serviced by data, control, and clock signals.
- The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.

- The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.

170.1.2 Application

Change the text in 170.1.2 as follows:

~~This clause applies to the interface between the MAC and PHY, or between the PHY 800GXS and the PHY when the optional 800GMII Extender is used.~~ This clause applies to the interface between the MAC and PHY. An 800GMII Extender or 1.6TMII Extender (see Clause 171) may be used to extend the reach of the 800GMII or 1.6TMII, respectively. This logical interface is used to provide media independence so that an identical media access controller may be used with supported PHY types.

170.1.3 Rate of operation

Change the text in 170.1.3 as follows:

The 800GMII is specified to support 800 Gb/s operation. The 1.6TMII is specified to support 1.6 Tb/s operation.

170.1.4 Delay constraints

Change the text and table in 170.1.4 as follows:

The maximum cumulative MAC Control, MAC, and RS delay (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 170–1. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references for 800 Gb/s, and in 174.4 and its references for 1.6 Tb/s.

Table 170–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
800 Gb/s MAC, RS, and MAC Control	196 608	384	245.76
<u>1.6T Tb/s MAC, RS, and MAC Control</u>	<u>393 216</u>	<u>768</u>	<u>245.76</u>

170.1.5 Allocation of functions

Change the text in 170.1.5 as follows:

The allocation of functions at the 800GMII/1.6TMII balances the need for media independence with interface simplicity. The 800GMII/1.6TMII provides media independence by separating the Data Link and Physical Layers of the OSI seven-layer reference model.

Change the title and text of 170.1.6 as follows:

170.1.6 800GMII/1.6TMII structure

The 800GMII/1.6TMII structure is identical to the CGMII structure specified in 81.1.6.

Change the title and text of 170.1.7 as follows:

170.1.7 Mapping of 800GMII/1.6TMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the 800GMII/1.6TMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at 800 Gb/s, therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the 800GMII/1.6TMII. This behavior and restrictions are the same as described in 22.2.1, with the details of the signaling described in 170.3.

The RS maps all primitives in an identical manner as the CGMII does and as specified in 81.1.7 except that EEE and LPI signaling are not supported.

Change the title and text of 170.2 as follows:

170.2 800GMII/1.6TMII data stream

The 800GMII/1.6TMII data stream has the same characteristics as the CGMII data stream described in 81.2.

Change the title and text of 170.3 as follows:

170.3 800GMII/1.6TMII functional specifications

The 800GMII/1.6TMII functions identically to the CGMII specified in 81.3 except that EEE and LPI signaling are not supported.

Change the title of 170.4 as follows:

170.4 Protocol implementation conformance statement (PICS) proforma for Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII)²

170.4.1 Introduction

Change the text in 170.4.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

170.4.2 Identification

170.4.2.2 Protocol summary

Change the table in 170.4.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-202x IEEE Std 802.3dj-202x, Clause 170, Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s <u>and 1.6 Tb/s</u> operation (800GMII <u>and 1.6TMII</u>)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-202x IEEE Std 802.3dj-202x.)	

Date of Statement	
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²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

170.4.3 Major capabilities/options

Change the table in 170.4.3 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*MII	PHY support of <u>either</u> 800GMII <u>or</u> 1.6TMII	170.2, 170.3		O	Yes [] No []

Change the title of 170.4.4 as follows:

170.4.4 PICS proforma tables for Reconciliation Sublayer (RS) and Media Independent Interface for 800 Gb/s and 1.6 Tb/s operation (800GMII and 1.6TMII)

170.4.4.3 Data stream structure

Change the table in 170.4.4.3 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	170.2	Within 800GMII/ <u>1.6TMII</u> data stream	MII:M	Yes [] N/A []
...					

Change the title of 170.4.4.4 as follows:

170.4.4.4 800GMII/1.6TMII signal functional specifications

Change the title of Clause 171 (added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024) as follows:

171. 800GMII Extender, 1.6TMII Extender, ~~and 800GMII Extender Sublayer (800GXS)~~, and 1.6TMII Extender Sublayer (1.6TXS)

171.1 Overview

Change the first three paragraphs in 171.1 as follows:

This clause defines the functional characteristics for the optional 800GMII Extender and 800GMII Extender Sublayer (800GXS), and also for the optional 1.6TMII Extender and 1.6TMII Extender Sublayer (1.6TXS).

The 800GMII/1.6TMII Extender allows the extension of the 800GMII/1.6TMII to the PCS via a physical instantiation. The 800GMII Extender is composed of a DTE 800GXS at the RS end and a PHY 800GXS at the PHY end, with one or two 800GAUI-n between them. The 1.6TMII Extender is composed of a DTE 1.6TXS at the RS end and a PHY 1.6TXS at the PHY end, with one or two 1.6TAUI-n between them.

Physical Layer clauses associated with the 800GMII Extender are listed in Table 171–1. Physical Layer clauses associated with the 1.6TMII Extender are listed in Table 171–1a.

Change Table 171–1 as follows:

Table 171–1— Physical Layer clauses associated with 800GMII Extender

Associated Clause	800GMII Extender
170—800GMII ^a	Required <u>Optional</u>
171—DTE 800GXS	Required
171—PHY 800GXS	Required
173—800GBASE-R <u>BM-PMA</u>	Required <u>Conditional</u> ^b
176—800GBASE-R <u>SM-PMA</u>	<u>Conditional</u> ^c
120F—800GAUI-8 C2C	Optional ^{ad}
120G—800GAUI-8 C2M	Optional ^{ad}
176C—800GAUI-4 C2C	<u>Optional</u> ^d
176D—800GAUI-4 C2M	<u>Optional</u> ^d
90—Time Synchronization	<u>Optional</u>

^a ~~An 800GMII Extender includes at least one 800GAUI-8.~~ The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^b ~~An 800GBASE-R BM PMA is required if the 800GMII Extender includes an 800GAUI-8.~~

^c ~~An 800GBASE-R SM PMA is required if the 800GMII Extender includes an 800GAUI-4.~~

^d ~~An 800GMII Extender includes at least one 800GAUI-n.~~

Insert Table 171–1a below Table 171–1 as follows:

Table 171–1a— Physical Layer clauses associated with 1.6TMII Extender

Associated Clause	1.6TMII Extender
170—1.6TMII ^a	Optional
171—DTE 1.6TXS	Required
171—PHY 1.6TXS	Required
176—1.6TBASE-R SM PMA	Required
120F—1.6TAUI-16 C2C	Optional ^b
120G—1.6TAUI-16 C2M	Optional ^b
176C—1.6TAUI-8 C2C	Optional ^b
176D—1.6TAUI-8 C2M	Optional ^b
90—Time Synchronization	Optional

^a The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^b A 1.6TMII Extender includes at least one 1.6TAUI-n.

Change the fourth paragraph in 171.1 as follows:

Figure 171–1 shows the relationship of the 800GMII/1.6TMII Extender and 800GXS/1.6TXS with other sublayers to the ISO Open System Interconnection (OSI) reference model.

Replace Figure 171–1 with the following figure:

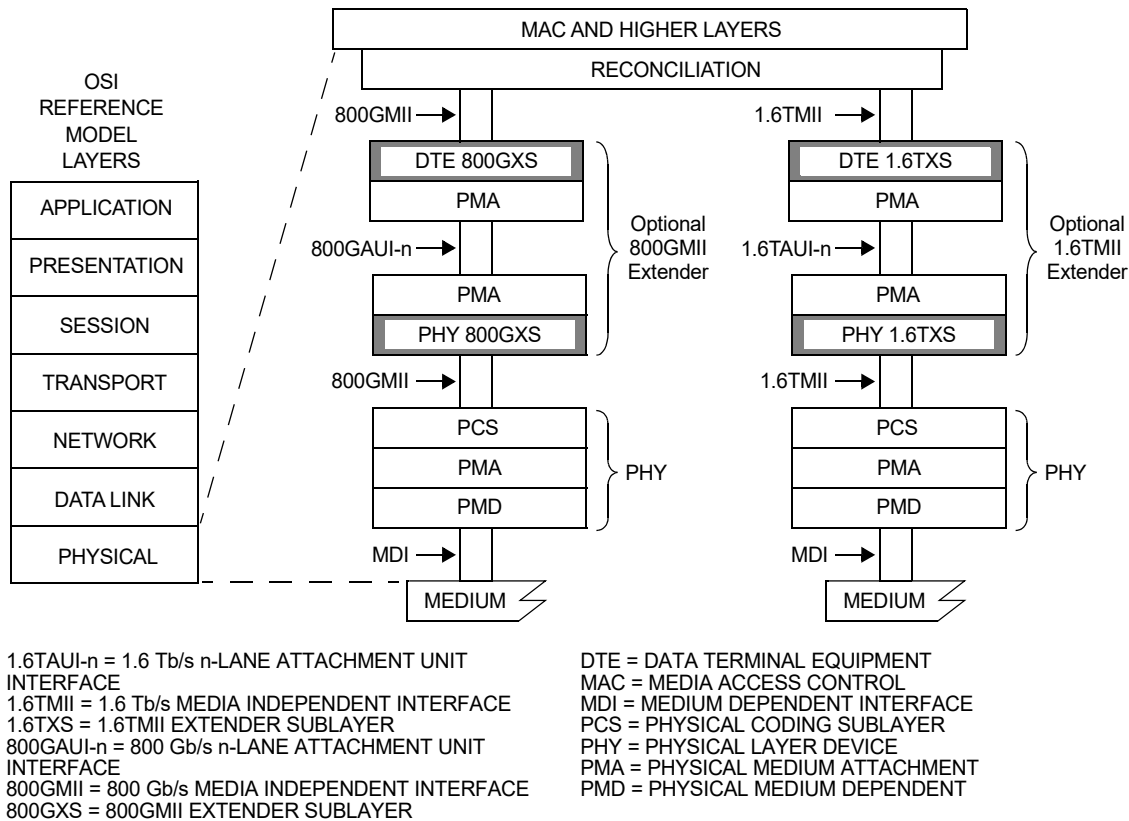


Figure 171-1—800GXS and 1.6TXS relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

171.1.1 Summary of functions

Change the text in 171.1.1 as follows:

The following is a list of the major functions of the 800GMII Extender:

- Simple signal mapping to the 800GMII
- The optional 800GMII Extender can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 800GMII
- Independent transmit and receive data paths
- Each 800GXS has the same functions as the 800GBASE-R PCS (see Clause 172)
- Each 800GXS connects to an 800GAUI-n as shown in Figure 171–1

The following is a list of the major functions of the 1.6TMII Extender:

- ### Simple signal mapping to the 1.6TMII

- The optional 1.6TMII Extender can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 1.6TMII
- Independent transmit and receive data paths
- Each 1.6TXS has the same functions as the 1.6TBASE-R PCS (see Clause 175)
- Each 1.6TXS connects to a 1.6TAUI-n as shown in Figure 171–1

Insert new subclause 171.1a after 171.1.1 as follows:

171.1a Error ratio allocation

An 800GMII/1.6TMII Extender is expected to meet the frame loss ratio specifications in 174A.4.

Change the title and text of 171.2 as follows:

171.2 DTE 800GXS and DTE 1.6TXS

The DTE 800GXS shall be identical in function to the 800GBASE-R PCS (see Clause 172) with the exception that the FEC degrade signaling is defined in 171.6. Figure 172–2 provides a functional block diagram of the DTE 800GXS.

The DTE 1.6TXS shall be identical in function to the 1.6TBASE-R PCS (see Clause 175) with the exception that the FEC degrade signaling is defined in 171.6. Figure 175–2 provides a functional block diagram of the DTE 1.6XS.

Change the title and text of 171.2.1 as follows:

171.2.1 DTE 800GXS and DTE 1.6TXS inter-sublayer interfaces

The service interfaces used by the DTE 800GXS are identical to those defined in 172.1.5.

The service interfaces used by the DTE 1.6TXS are identical to those defined in 175.1.4.

Change the title and text of 171.3 as follows:

171.3 PHY 800GXS and PHY 1.6TXS

The PHY 800GXS shall be identical in function to an 800GBASE-R PCS (see Clause 172) with the following exceptions:

- The PCS is inverted with the transmit function used for the receive direction and vice versa.
- The service interface signals are remapped as defined in 171.3.2 and 171.3.3.
- FEC degrade signaling is defined in 171.6.

The PHY 1.6TXS shall be identical in function to a 1.6TBASE-R PCS (see Clause 175) with the following exceptions:

- The PCS is inverted with the transmit function used for the receive direction and vice versa.
- The service interface signals are remapped as defined in 171.3.2a and 171.3.3a.
- FEC degrade signaling is defined in 171.6.

Figure 171–2 provides a functional block diagram of the PHY 800GXS. Figure 171–2a provides a functional block diagram of the PHY 1.6TXS.

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Insert Figure 171–2a below Figure 171–2 as follows:

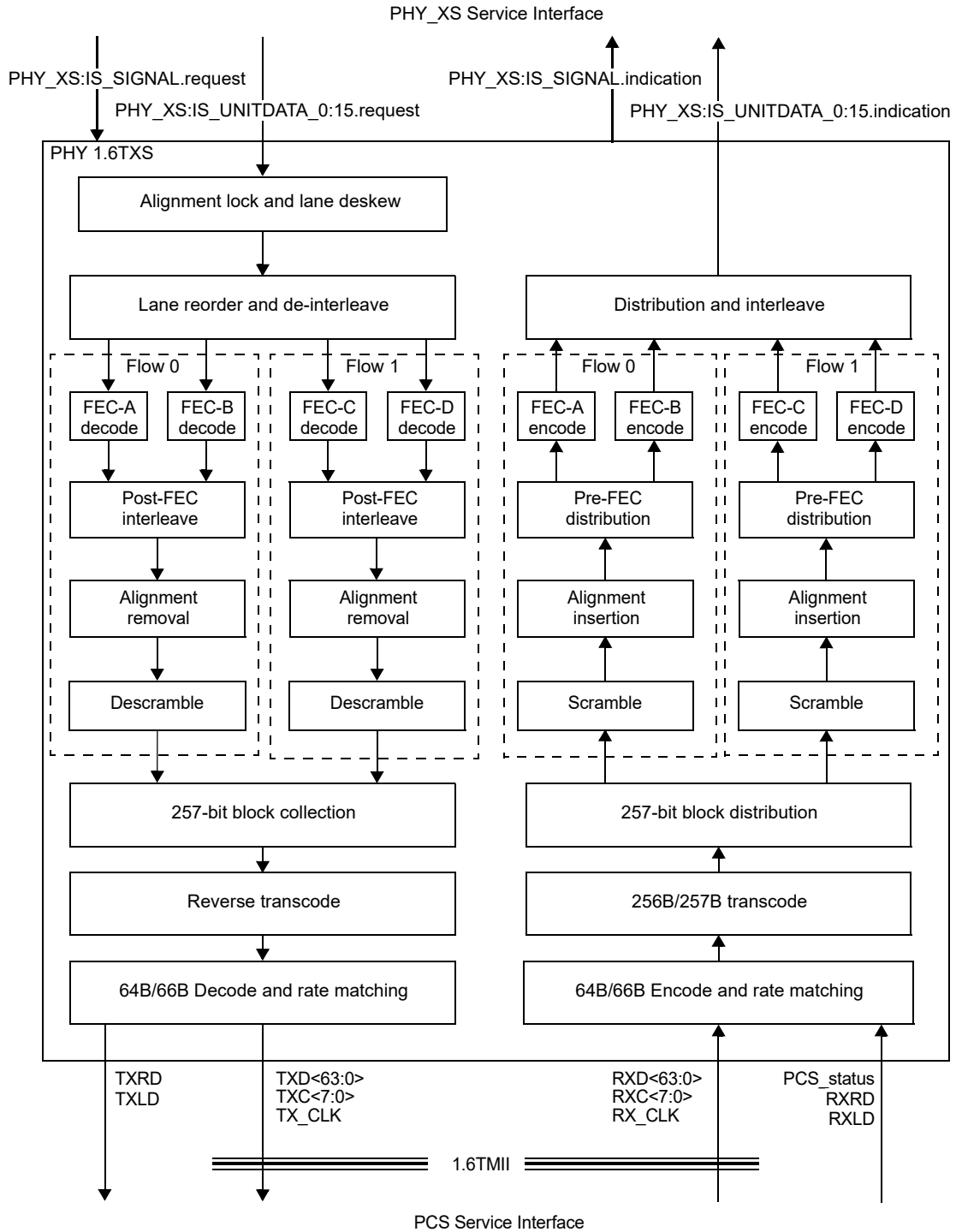


Figure 171–2a—Functional block diagram for the PHY 1.6TXS

Change the title and text of 171.3.1 as follows:

171.3.1 PHY 800GXS and PHY 1.6TXS inter-sublayer interfaces

The PHY 800GXS inter-sublayer service interfaces are shown in Figure 169–3.

The PHY 1.6TXS inter-sublayer service interfaces are shown in Figure 174–3.

Insert 171.3.2a below 171.3.2 as follows:

171.3.2a PHY 1.6TXS service interface

The PHY 1.6TXS service interface primitives are summarized as follows:

```
PHY_XS:IS_UNITDATA_i.request(tx_symbol)
PHY_XS:IS_UNITDATA_i.indication(rx_symbol)
PHY_XS:IS_SIGNAL.request(SIGNAL_OK)
PHY_XS:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to 15. In the transmit direction, the PHY 1.6TXS receives 16 parallel bit streams, each at a nominal signaling rate of 106.25 Gb/s, from the PMA above. In the receive direction, the PHY 1.6TXS sends 16 parallel bit streams, each at a nominal signaling rate of 106.25 Gb/s, to the PMA.

The PHY 1.6TXS receives signal status information from the PMA using the PHY_XS:IS_SIGNAL.request primitive.

The PHY 1.6TXS provides signal status information to the PMA using the PHY_XS:IS_SIGNAL.indication(SIGNAL_OK) primitive. The SIGNAL_OK parameter is set to OK when PCS_status (see 175.1.4.1) is true and FAIL when PCS_status is false

171.3.3 Service interface below the PHY 800GXS

Change the text of 171.3.3 as follows:

The service interface below the PHY 800GXS allows the PHY 800GXS to transfer information to and from the PCS below.

The service interface below the PHY 800GXS is the 800GMII defined in Clause 170, with the following exceptions and additional signals:

- TX_CLK is used in place of RX_CLK, and vice versa
- TXD<63:0> is used in place of RXD<63:0>, and vice versa
- TXC<7:0> is used in place of RXC<7:0>, and vice versa
- Additional signals RXRD, RXLD, and PCS_status are provided in the receive direction (see 172.1.5.1)
- An additional signal TXRD indicates the state of the rx_rm_degraded variable (see 171.6.2 172.2.6.2.2) as detected by the PHY 800GXS in the transmit direction
- An additional signal TXLD indicates the state of the FEC_degraded_SER variable (see 171.6.2 172.2.6.2.2) as detected by the PHY 800GXS in the transmit direction

NOTE—In the transmit direction, if no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 800GMII will be reduced by approximately 49 ppm.

Insert 171.3.3a below 171.3.3 as follows:

171.3.3a Service interface below PHY 1.6TXS

The service interface below the PHY 1.6TXS allows the PHY 1.6TXS to transfer information to and from the PCS below.

The service interface below the PHY 1.6TXS is the 1.6TMII defined in Clause 170, with the following exceptions and additional signals:

- TX_CLK is used in place of RX_CLK, and vice versa
- TXD<63:0> is used in place of RXD<63:0>, and vice versa
- TXC<7:0> is used in place of RXC<7:0>, and vice versa
- Additional signals RXRD, RXLD, and PCS_status are received from the PCS below (see 175.1.4.1)
- Additional signals TXRD and TXLD are sent to the PCS below. TXRD indicates the state of the rx_rm_degraded variable (see 175.2.5.5) as detected by the PHY 1.6TXS in the transmit direction. TXLD indicates the state of the FEC_degraded_SER variable (see 175.2.5.3) as detected by the PHY 1.6TXS in the transmit direction.

NOTE—In the transmit direction, if no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 1.6TMII will be reduced by approximately 12 ppm.

Change the title and text of 171.4 as follows:

171.4 800GAUI-n and 1.6TAUI-n

An 800GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers, as shown in Figure 171–1:

- 800GAUI-8 chip-to-chip (Annex 120F)
- 800GAUI-8 chip-to-module (Annex 120G)
- 800GAUI-4 chip-to-chip (Annex 176C)
- 800GAUI-4 chip-to-module (Annex 176D)

The 800GAUI-n may be instantiated using any form described in 176B.6.2.

A 1.6TMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers, as shown in Figure 171–1:

- 1.6TAUI-16 chip-to-chip (Annex 120F)
- 1.6TAUI-16 chip-to-module (Annex 120G)
- 1.6TAUI-8 chip-to-chip (Annex 176C)
- 1.6TAUI-8 chip-to-module (Annex 176D)

The 1.6TAUI-n may be instantiated using any form described in 176B.7.2.

171.5 Link fault signaling in the receive direction

Change the text of 171.5 as follows:

Link fault signaling (see 81.3.4) generated by the local PHY or by the link partner is sent toward the Reconciliation Sublayer as ordered sets through the 800GMII/1.6TMII Extender.

~~The PHY 800GXS provides signal status from the local PHY to the client PMA using the PHY_XS:IS_SIGNAL.indication(SIGNAL_OK) primitive (see 171.3.2 and Figure 171-2).~~

171.6 FEC degrade

Change the text of 171.6 as follows:

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in 116.6.

FEC degrade detection is specified in 172.2.5.3 for 800 Gb/s and in 175.2.5.3 for 1.6 Tb/s. FEC degrade detection is optional.

~~FEC degrade signaling using the PCS alignment marker status field is specified for the DTE 800GXS and PHY 800GXS in 171.6.1 and 171.6.2, respectively.~~

FEC degrade signaling using the PCS alignment marker status field is specified for the DTE 800GXS and DTE 1.6TXS in 171.6.1. FEC degrade signaling using the PCS alignment marker status field is specified for the PHY 800GXS and PHY 1.6TXS in 171.6.2.

FEC degrade signaling across the service interface below the PHY 800GXS is specified in 171.3.3. FEC degrade signaling across the service interface below the PHY 1.6TXS is specified in 171.3.3a.

Change the title and text of 171.6.1 as follows:

171.6.1 DTE 800GXS and DTE 1.6TXS FEC degrade signaling

The variable tx_am_sf is set as follows:

tx_am_sf<2:0> = {FEC_degraded_SER + rx_local_degraded, 0, 0}

where FEC_degraded_SER and rx_local_degraded are defined in 172.2.6.2.2 for DTE 800GXS and in 175.2.5.3 and 175.2.5.5 for DTE 1.6TXS, and + denotes logical OR

Change the title and text of 171.6.2 as follows:

171.6.2 PHY 800GXS and PHY 1.6TXS FEC degrade signaling

The variable tx_am_sf is set as follows:

tx_am_sf<2:0> = {RXRD, RXLD, 0}

where RXRD and RXLD are signals from the PCS on the PCS service interface (see 172.1.5.1 for the PHY 800GXS and 175.1.4.1 for the PHY 1.6TXS).

Insert new subclause 171.6a after 171.6 as follows:

171.6a Path data delay for time synchronization

The 800GXS and 1.6TXS optionally provide the ability to report path data delay information for time synchronization (see Clause 90).

171.6a.1 PHY 800GXS and PHY 1.6TXS path data delay

The ability to report PHY 800GXS or PHY 1.6TXS path data delays is indicated by the PHY_XS_delay_ns_TX_ability, PHY_XS_delay_subns_TX_ability, PHY_XS_delay_ns_RX_ability, and PHY_XS_delay_subns_RX_ability status variables (see Table 171–3 and Table 171–5b).

The PHY 800GXS and PHY 1.6TXS transmit and receive path data delays are reported as if the DDMP (data delay measurement point) is at the start of the set of interleaved RS-FEC codewords (see 90.7).

Four separate delays are reported in the following eight path data delay status variables:

- PHY_XS_delay_ns_TX_max and PHY_XS_delay_subns_TX_max
- PHY_XS_delay_ns_TX_min and PHY_XS_delay_subns_TX_min
- PHY_XS_delay_ns_RX_max and PHY_XS_delay_subns_RX_max
- PHY_XS_delay_ns_RX_min and PHY_XS_delay_subns_RX_min

171.6a.2 DTE 800GXS and DTE 1.6TXS path data delay

The ability to report DTE 800GXS or DTE 1.6TXS path data delays is indicated by the DTE_XS_delay_ns_TX_ability, DTE_XS_delay_subns_TX_ability, DTE_XS_delay_ns_RX_ability, and DTE_XS_delay_subns_RX_ability status variables (see Table 171–5 and Table 171–5d).

The DTE 800GXS and DTE 1.6TXS transmit and receive path data delays are reported as if the DDMP (data delay measurement point) is at the start of the set of interleaved RS-FEC codewords (see 90.7).

Four separate delays are reported in the following eight path data delay status variables:

- DTE_XS_delay_ns_TX_max and DTE_XS_delay_subns_TX_max
- DTE_XS_delay_ns_TX_min and DTE_XS_delay_subns_TX_min
- DTE_XS_delay_ns_RX_max and DTE_XS_delay_subns_RX_max
- DTE_XS_delay_ns_RX_min and DTE_XS_delay_subns_RX_min

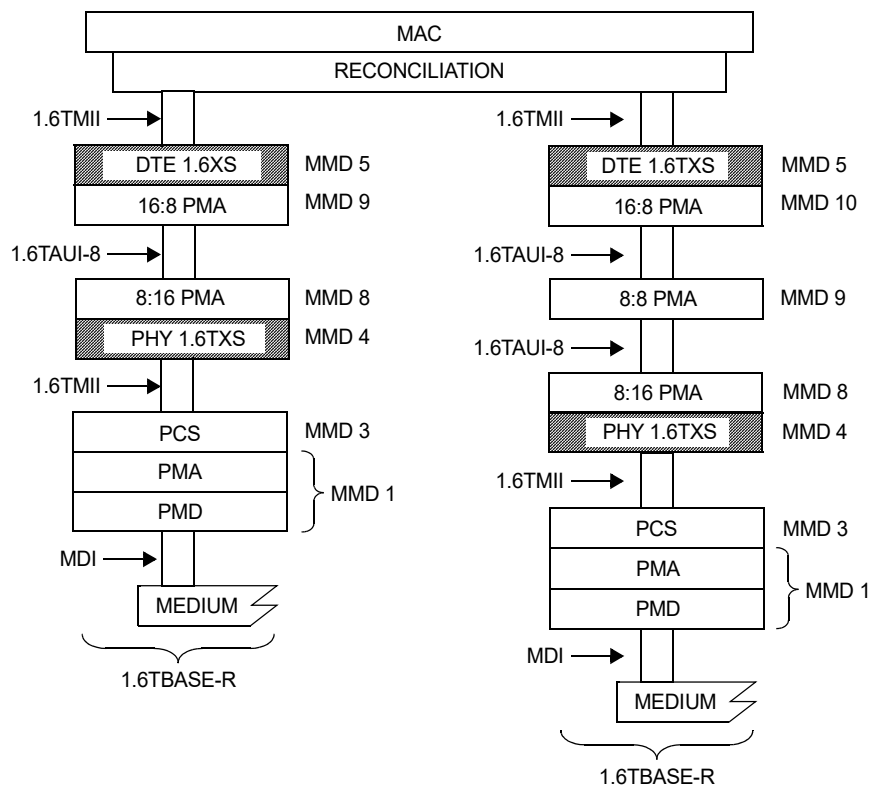
Change the title and text of 171.7 as follows:

171.7 800GXS and 1.6TXS partitioning examples

Two partitioning examples and MMD numbering using the 800GXS are shown in Figure 171–3. ~~Annex 173A and Annex 176B shows additional examples of 800GXS partitioning and MMD numbering.~~ Annex 173A shows additional examples of 800GXS partitioning and MMD numbering using the BM PMA. 176B.6.2 shows additional examples of 800GXS partitioning using both the BM PMA and SM PMA.

Two partitioning examples and MMD numbering using the 1.6TXS are shown in Figure 171–3a. ~~Annex 176B shows additional examples of 1.6TXS partitioning and MMD numbering. 176B.7.2 shows additional examples of 1.6TXS partitioning~~

Insert new Figure 171–3a after Figure 173–3 as follows:



1.6TAUI = 1.6 Tb/s ATTACHMENT UNIT INTERFACE
1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
1.6TXS = 1.6Tb/s MEDIA INDEPENDENT EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

MMD = MDIO MANAGEABLE DEVICE
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 171–3a—Example 1.6TBASE-R PMA layering with 1.6TXS

Change the title and text of 171.8 as follows:

171.8 800GXS and 1.6TXS MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the 800GXS. If MDIO is implemented, it shall map MDIO PHY 800GXS and DTE 800GXS control bits to Clause 172 control variables as shown in Table 171–2 and Table 171–4, respectively. Similarly, if MDIO is implemented, it shall map MDIO PHY 800GXS and DTE 800GXS status bits to Clause 172 status variables as shown in Table 171–3 and Table 171–5, respectively. MDIO registers relevant to the PHY 800GXS and DTE 800GXS are described in 45.2.4 and 45.2.5, respectively.

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for the 800GXS or 1.6TXS.

If MDIO is implemented:

- MDIO PHY 800GXS and DTE 800GXS control bits are mapped to Clause 172 control variables as shown in Table 171–2 and Table 171–4, respectively.
- MDIO PHY 800GXS and DTE 800GXS status bits are mapped to Clause 172 status variables as shown in Table 171–3 and Table 171–5, respectively.
- MDIO PHY 1.6TXS and DTE 1.6TXS control bits are mapped to Clause 175 control variables as shown in Table 171–5a and Table 171–5c, respectively.
- MDIO PHY 1.6TXS and DTE 1.6TXS status bits are mapped to Clause 175 status variables as shown in Table 171–5b and Table 171–5d, respectively.

MDIO registers relevant to the PHY 800GXS and PHY 1.6TXS are described in 45.2.4. MDIO registers relevant to the DTE 800GXS and DTE 1.6TXS are described in 45.2.5.

Change Table 171–3 as follows (some unchanged rows not shown):

Table 171–3—MDIO PHY 800GXS to Clause 172 status variable mapping

MDIO status variable	PHY XS register name	Register/bit number	Clause 172 status variable
...			
PHY XS FEC symbol errors, lane 0 to lane 31	PHY XS FEC symbol error counter, lane 0 to lane 31	4.600 to 4.663	FEC_symbol_error_counter_<0:31>
...			
FEC uncorrected codewords	PHY XS FEC uncorrected codewords counter	4.804, 4.805	FEC_uncorrected_cw_counter
<u>PHY_XS_delay_ns_TX_max</u> ^a	<u>TimeSync PHY XS transmit path data delay in ns</u>	<u>4.1801, 4.1802</u>	<u>PCS_delay_ns_RX_max</u>
<u>PHY_XS_delay_subns_TX_max</u> ^a	<u>TimeSync PHY XS transmit path data delay in sub-ns</u>	<u>4.1809</u>	<u>PCS_delay_subns_RX_max</u>
<u>PHY_XS_delay_ns_TX_min</u> ^a	<u>TimeSync PHY XS transmit path data delay in ns</u>	<u>4.1803, 4.1804</u>	<u>PCS_delay_ns_RX_min</u>
<u>PHY_XS_delay_subns_TX_min</u> ^a	<u>TimeSync PHY XS transmit path data delay in sub-ns</u>	<u>4.1810</u>	<u>PCS_delay_subns_RX_min</u>
<u>PHY_XS_delay_ns_RX_max</u> ^a	<u>TimeSync PHY XS receive path data delay in ns</u>	<u>4.1805, 4.1806</u>	<u>PCS_delay_ns_TX_max</u>
<u>PHY_XS_delay_subns_RX_max</u> ^a	<u>TimeSync PHY XS receive path data delay in sub-ns</u>	<u>4.1811</u>	<u>PCS_delay_subns_TX_max</u>
<u>PHY_XS_delay_ns_RX_min</u> ^a	<u>TimeSync PHY XS receive path data delay in ns</u>	<u>4.1807, 4.1808</u>	<u>PCS_delay_ns_TX_min</u>
<u>PHY_XS_delay_subns_RX_min</u> ^a	<u>TimeSync PHY XS receive path data delay in sub-ns</u>	<u>4.1812</u>	<u>PCS_delay_subns_TX_min</u>
<u>PHY_XS_delay_ns_RX_ability</u> ^a	<u>TimeSync PHY XS capability</u>	<u>4.1800.0</u>	<u>PCS_delay_ns_TX_ability</u>
<u>PHY_XS_delay_ns_TX_ability</u> ^a	<u>TimeSync PHY XS capability</u>	<u>4.1800.1</u>	<u>PCS_delay_ns_RX_ability</u>
<u>PHY_XS_delay_subns_RX_ability</u> ^a	<u>TimeSync PHY XS capability</u>	<u>4.1800.2</u>	<u>PCS_delay_subns_TX_ability</u>
<u>PHY_XS_delay_subns_TX_ability</u> ^a	<u>TimeSync PHY XS capability</u>	<u>4.1800.3</u>	<u>PCS_delay_subns_RX_ability</u>

^a In the PHY XS MDIO status variable name, TX and RX are transposed from the corresponding Clause 172 MDIO status variable because a PHY XS is essentially an upside down PCS.

Change Table 171–5 as follows (some unchanged rows not shown):

Table 171–5—MDIO DTE 800GXS to Clause 172 status variable mapping

MDIO status variable	DTE XS register name	Register/bit number	Clause 172 status variable
...			
DTE XS FEC symbol errors, lane 0 to lane 31	DTE XS FEC symbol error counter, lane 0 to lane 31	5.600 to 5.663	FEC_symbol_error_counter_<0:31>
...			
FEC uncorrected codewords	DTE XS FEC uncorrected codewords counter	5.804, 5.805	FEC_uncorrected_cw_counter
<u>DTE_XS_delay_ns_TX_max</u>	<u>TimeSync DTE XS transmit path data delay in ns</u>	<u>5.1801, 5.1802</u>	<u>PCS_delay_ns_TX_max</u>
<u>DTE_XS_delay_subns_TX_max</u>	<u>TimeSync DTE XS transmit path data delay in sub-ns</u>	<u>5.1809</u>	<u>PCS_delay_subns_TX_max</u>
<u>DTE_XS_delay_ns_TX_min</u>	<u>TimeSync DTE XS transmit path data delay in ns</u>	<u>5.1803, 5.1804</u>	<u>PCS_delay_ns_TX_min</u>
<u>DTE_XS_delay_subns_TX_min</u>	<u>TimeSync DTE XS transmit path data delay in sub-ns</u>	<u>5.1810</u>	<u>PCS_delay_subns_TX_min</u>
<u>DTE_XS_delay_ns_RX_max</u>	<u>TimeSync DTE XS receive path data delay in ns</u>	<u>5.1805, 5.1806</u>	<u>PCS_delay_ns_RX_max</u>
<u>DTE_XS_delay_subns_RX_max</u>	<u>TimeSync DTE XS receive path data delay in sub-ns</u>	<u>5.1811</u>	<u>PCS_delay_subns_RX_max</u>
<u>DTE_XS_delay_ns_RX_min</u>	<u>TimeSync DTE XS receive path data delay in ns</u>	<u>5.1807, 5.1808</u>	<u>PCS_delay_ns_RX_min</u>
<u>DTE_XS_delay_subns_RX_min</u>	<u>TimeSync DTE XS receive path data delay in sub-ns</u>	<u>5.1812</u>	<u>PCS_delay_subns_RX_min</u>
<u>DTE_XS_delay_ns_RX_ability</u>	<u>TimeSync DTE XS capability</u>	<u>5.1800.0</u>	<u>PCS_delay_ns_RX_ability</u>
<u>DTE_XS_delay_ns_TX_ability</u>	<u>TimeSync DTE XS capability</u>	<u>5.1800.1</u>	<u>PCS_delay_ns_TX_ability</u>
<u>DTE_XS_delay_subns_RX_ability</u>	<u>TimeSync DTE XS capability</u>	<u>5.1800.2</u>	<u>PCS_delay_subns_RX_ability</u>
<u>DTE_XS_delay_subns_TX_ability</u>	<u>TimeSync DTE XS capability</u>	<u>5.1800.3</u>	<u>PCS_delay_subns_TX_ability</u>

Insert Table 171–5a, Table 171–5b, Table 171–5c and Table 171–5d after Table 171–5 as follows:

Table 171–5a—MDIO PHY 1.6TXS to Clause 175 control variable mapping

MDIO control variable	PHY XS register name	Register/bit number	Clause 175 control variable
Loopback	PHY XS control 1 register	4.0.14	Loopback
Reset	PHY XS control 1 register	4.0.15	reset
Transmit test-pattern enable	BASE-R PHY XS test-pattern control register	4.42.3	tx_test_mode
PHY XS FEC bypass indication enable	PHY XS FEC control register	4.800.1	FEC_bypass_indication_enable
PHY XS FEC degraded SER enable	PHY XS FEC control register	4.800.2	FEC_degraded_SER_enable
PHY XS FEC degraded SER activate threshold	PHY XS FEC degraded SER activate threshold register	4.806, 4.807	FEC_degraded_SER_activate_threshold
PHY XS FEC degraded SER deactivate threshold	PHY XS FEC degraded SER deactivate threshold register	4.808, 4.809	FEC_degraded_SER_deactivate_threshold
PHY XS FEC degraded SER interval	PHY XS FEC degraded SER interval	4.810, 4.811	FEC_degraded_SER_interval

Table 171–5b—MDIO PHY 1.6TXS to Clause 175 status variable mapping

MDIO status variable	PHY XS register name	Register/bit number	Clause 175 status variable
BASE-R PHY XS receive link status	BASE-R PHY XS status 1	4.32.12	PCS_status
PHY XS lane alignment status	Multi-lane BASE-R PHY XS alignment status 1	4.50.12	align_status
Lane 0 to 15 aligned	Multi-lane BASE-R PHY XS alignment status 3 and 4	4.52.7:0 4.53.7:0	amps_lock<0:15>
FEC codeword counter	PHY XS RS-FEC codeword counter	4.300 to 4.302	FEC_cw_counter
FEC codeword error bin counter, 1 to 15	PHY XS RS-FEC codeword error bin, 1 to 15	4.340 to 4.369	FEC_codeword_error_bin_<1:15>
Lane 0 to 15 mapping	PHY XS lane mapping, lane 0 through lane15	4.400 through 4.415	pcs_lane_mapping<0:15>

Table 171–5b—MDIO PHY 1.6TXS to Clause 175 status variable mapping (*continued*)

MDIO status variable	PHY XS register name	Register/bit number	Clause 175 status variable
PHY XS FEC symbol errors, lane 0 to lane 15	PHY XS FEC symbol error counter, lane 0 to lane 15	4.600 to 4.631	FEC_symbol_error_counter_<0:15>
PHY XS FEC bypass indication ability	PHY XS FEC status	4.801.1	FEC_bypass_indication_ability
PHY XS FEC high SER	PHY XS FEC status register	4.801.2	hi_ser_combined
PHY XS FEC degraded SER ability	PHY XS FEC status register	4.801.3	FEC_degraded_SER_ability
PHY XS FEC degraded SER	PHY XS FEC status register	4.801.4	FEC_degraded_SER
Remote degraded SER received	PHY XS FEC status register	4.801.5	rx_rm_degraded
Local degraded SER received	PHY XS FEC status register	4.801.6	rx_local_degraded
FEC corrected codewords	PHY XS FEC corrected codewords counter	4.802, 4.803	FEC_corrected_cw_counter
FEC uncorrected codewords	PHY XS FEC uncorrected codewords counter	4.804, 4.805	FEC_uncorrected_cw_counter
PHY_XS_delay_ns_TX_max ^a	TimeSync PHY XS transmit path data delay in ns	4.1801, 4.1802	PCS_delay_ns_RX_max
PHY_XS_delay_subns_TX_max ^a	TimeSync PHY XS transmit path data delay in sub-ns	4.1809	PCS_delay_subns_RX_max
PHY_XS_delay_ns_TX_min ^a	TimeSync PHY XS transmit path data delay in ns	4.1803, 4.1804	PCS_delay_ns_RX_min
PHY_XS_delay_subns_TX_min ^a	TimeSync PHY XS transmit path data delay in sub-ns	4.1810	PCS_delay_subns_RX_min
PHY_XS_delay_ns_RX_max ^a	TimeSync PHY XS receive path data delay in ns	4.1805, 4.1806	PCS_delay_ns_TX_max
PHY_XS_delay_subns_RX_max ^a	TimeSync PHY XS receive path data delay in sub-ns	4.1811	PCS_delay_subns_TX_max
PHY_XS_delay_ns_RX_min ^a	TimeSync PHY XS receive path data delay in ns	4.1807, 4.1808	PCS_delay_ns_TX_min
PHY_XS_delay_subns_RX_min ^a	TimeSync PHY XS receive path data delay in sub-ns	4.1812	PCS_delay_subns_TX_min
PHY_XS_delay_ns_RX_ability ^a	TimeSync PHY XS capability	4.1800.0	PCS_delay_ns_TX_ability

Table 171–5b—MDIO PHY 1.6TXS to Clause 175 status variable mapping (*continued*)

MDIO status variable	PHY XS register name	Register/bit number	Clause 175 status variable
PHY_XS_delay_ns_TX_ability ^a	TimeSync PHY XS capability	4.1800.1	PCS_delay_ns_RX_ability
PHY_XS_delay_subns_RX_ability ^a	TimeSync PHY XS capability	4.1800.2	PCS_delay_subns_TX_ability
PHY_XS_delay_subns_TX_ability ^a	TimeSync PHY XS capability	4.1800.3	PCS_delay_subns_RX_ability

^a In the PHY XS MDIO status variable name, TX and RX are transposed from the corresponding Clause 175 MDIO status variable because a PHY XS is essentially an upside down PCS.

Table 171–5c—MDIO DTE 1.6TXS to Clause 175 control variable mapping

MDIO control variable	DTE XS register name	Register/bit number	Clause 175 control variable
Loopback	DTE XS control 1 register	5.0.14	Loopback
Reset	DTE XS control 1 register	5.0.15	reset
Transmit test-pattern enable	BASE-R DTE XS test-pattern control register	5.42.3	tx_test_mode
DTE XS FEC bypass indication enable	DTE XS FEC control register	5.800.1	FEC_bypass_indication_enable
DTE XS FEC degraded SER enable	DTE XS FEC control register	5.800.2	FEC_degraded_SER_enable
DTE XS FEC degraded SER activate threshold	DTE XS FEC degraded SER activate threshold register	5.806, 5.807	FEC_degraded_SER_activate_threshold
DTE XS FEC degraded SER deactivate threshold	DTE XS FEC degraded SER deactivate threshold register	5.808, 5.809	FEC_degraded_SER_deactivate_threshold
DTE XS FEC degraded SER interval	DTE XS FEC degraded SER interval	5.810, 5.811	FEC_degraded_SER_interval

Table 171–5d—MDIO DTE 1.6TXS to Clause 175 status variable mapping

MDIO status variable	DTE XS register name	Register/bit number	Clause 175 status variable
BASE-R DTE XS receive link status	BASE-R DTE XS status 1	5.32.12	PCS_status
DTE XS lane alignment status	Multi-lane BASE-R DTE XS alignment status 1	5.50.12	align_status
Lane 0 to 15 aligned	Multi-lane BASE-R DTE XS alignment status 3 and 4	5.52.7:0 5.53.7:0	amps_lock<0:15>
FEC codeword counter	DTE XS RS-FEC codeword counter	5.300 to 5.302	FEC_cw_counter
FEC codeword error bin counter, 1 to 15	DTE XS RS-FEC codeword error bin, 1 to 15	5.340 to 5.369	FEC_codeword_error_bin_<1:15>
Lane 0 to 15 mapping	DTE XS lane mapping, lane 0 through lane 15	5.400 through 5.415	pcs_lane_mapping<0:15>
DTE XS FEC symbol errors, lane 0 to lane 15	DTE XS FEC symbol error counter, lane 0 to lane 15	5.600 to 5.631	FEC_symbol_error_counter_<0:15>
DTE XS FEC bypass indication ability	DTE XS FEC status	5.801.1	FEC_bypass_indication_ability
DTE XS FEC high SER	DTE XS FEC status register	5.801.2	hi_ser_combined
DTE XS FEC degraded SER ability	DTE XS FEC status register	5.801.3	FEC_degraded_SER_ability
DTE XS FEC degraded SER	DTE XS FEC status register	5.801.4	FEC_degraded_SER
Remote degraded received	DTE XS FEC status register	5.801.5	rx_rm_degraded
Local degraded received	DTE XS FEC status register	5.801.6	rx_local_degraded
FEC corrected codewords	DTE XS FEC corrected codewords counter	5.802, 5.803	FEC_corrected_cw_counter
FEC uncorrected codewords	DTE XS FEC uncorrected codewords counter	5.804, 5.805	FEC_uncorrected_cw_counter
DTE_XS_delay_ns_TX_max	TimeSync DTE XS transmit path data delay in ns	5.1801, 5.1802	PCS_delay_ns_TX_max
DTE_XS_delay_subns_TX_max	TimeSync DTE XS transmit path data delay in sub-ns	5.1809	PCS_delay_subns_TX_max
DTE_XS_delay_ns_TX_min	TimeSync DTE XS transmit path data delay in ns	5.1803, 5.1804	PCS_delay_ns_TX_min

Table 171–5d—MDIO DTE 1.6TXS to Clause 175 status variable mapping (*continued*)

MDIO status variable	DTE XS register name	Register/bit number	Clause 175 status variable
DTE_XS_delay_subns_TX_min	TimeSync DTE XS transmit path data delay in sub-ns	5.1810	PCS_delay_subns_TX_min
DTE_XS_delay_ns_RX_max	TimeSync DTE XS receive path data delay in ns	5.1805, 5.1806	PCS_delay_ns_RX_max
DTE_XS_delay_subns_RX_max	TimeSync DTE XS receive path data delay in sub-ns	5.1811	PCS_delay_subns_RX_max
DTE_XS_delay_ns_RX_min	TimeSync DTE XS receive path data delay in ns	5.1807, 5.1808	PCS_delay_ns_RX_min
DTE_XS_delay_subns_RX_min	TimeSync DTE XS receive path data delay in sub-ns	5.1812	PCS_delay_subns_RX_min
DTE_XS_delay_ns_RX_ability	TimeSync DTE XS capability	5.1800.0	PCS_delay_ns_RX_ability
DTE_XS_delay_ns_TX_ability	TimeSync DTE XS capability	5.1800.1	PCS_delay_ns_TX_ability
DTE_XS_delay_subns_RX_ability	TimeSync DTE XS capability	5.1800.2	PCS_delay_subns_RX_ability
DTE_XS_delay_subns_TX_ability	TimeSync DTE XS capability	5.1800.3	PCS_delay_subns_TX_ability

Change the title of 171.9 as follows:

171.9 Protocol implementation conformance statement (PICS) proforma for Clause 171, 800GMII Extender, ~~1.6TMII Extender~~, and 800GMII Extender Sublayer (800GXS), and ~~1.6TMII Extender Sublayer (1.6TXS)~~³

171.9.1 Introduction

Change the text in 171.9.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 171, 800GMII Extender, ~~1.6TMII Extender~~, and 800GMII Extender Sublayer (800GXS), and ~~1.6TMII Extender Sublayer (1.6TXS)~~, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

171.9.2 Identification

171.9.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

171.9.2.2 Protocol summary

Change the text in 171.9.2.2 as follows:

171.9.3 Major capabilities/options

Change the table in 171.9.3 as follows:

³Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Identification of protocol standard	IEEE Std 802.3df-202x, Clause 171, 800GMII Extender, <u>1.6TMII Extender, and 800GMII Extender Sublayer (800GXS), and 1.6TMII Extender Sublayer (1.6TXS)</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-202x*Std 802.3dj-202x.)	

Date of Statement	
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Item	Feature	Subclause	Value/Comment	Status	Support
MII	800GMII <u>or 1.6TMII</u> logical interface is supported	172.1.5.1, <u>175.1.4.1</u>		O	Yes ☐ No ☐
*800GXS	<u>Using the XS for 800GBASE-R</u>	<u>171.1</u>	<u>See 171.9.4</u>	<u>O.1</u>	<u>Yes ☐</u> <u>No ☐</u>
*1.6TXS	<u>Using the XS for 1.6TBASE-R</u>	<u>171.1</u>	<u>See 171.9.5</u>	<u>O.1</u>	<u>Yes ☐</u> <u>No ☐</u>
*DTEXS	DTE 800GXS <u>or DTE 1.6TXS</u>	171.2		O/2	Yes ☐ No ☐
*PHYXS	PHY 800GXS <u>or PHY 1.6TXS</u>	171.3		O/2	Yes ☐ No ☐
*FDD	Support for FEC degrade detection	171.6		O	Yes ☐ No ☐
*MD	MDIO	171.8		O	Yes ☐ No ☐

171.9.4 PICS proforma tables for 800GMII Extender and 800GMII Extender Sublayer (800GXS)

171.9.4.1 Transmit function

Change the table in 171.9.4.1 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	<u>172.2.4.1.2</u>		O	Yes ☐ No ☐

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Transmit 64B/66B encoder uses state diagram in Figure 119–14.	172.2.4.1 <u>172.2.4.1.1</u>		!SE:M	Yes [] N/A []
TF2	Transmit 64B/66B encoder uses stateless rules in Table 172–1.	172.2.4.1.2		SE:M	Yes [] N/A []
...					

171.9.4.4 Management

Change the table in 171.9.4.4 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to XS management objects is provided	172.3		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits for PHY 800GXS	171.8	See Table 118–1 and Table 118–2 <u>See Table 171–2 and Table 171–3</u>	MD*PHYXS:M	Yes [] N/A []
M3	Mapping of MDIO control bits and MDIO status bits for DTE 800GXS	171.8	See Table 118–3 and Table 118–4 <u>See Table 171–4 and Table 171–5</u>	MD*DTEXS:M	Yes [] N/A []

Insert the following subclause after 171.9.4.6:

171.9.4.6a Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	171.6a		O	Yes [] No []
TS1	PHY 800GXS Path data delays	171.6a.1	Path data delays reported as if DDMP is at start of a set of four interleaved RS-FEC codewords.	TS*PHYXS:M	Yes [] N/A []
TS2	PHY 800GXS path data delay nanosecond resolution	171.6a.1	Path data delays reported with nanosecond or better resolution.	TS*PHYXS:M	Yes [] N/A []
TS3	PHY 800GXS path data delay sub-nanosecond resolution	171.6a.1	Path data delays reported with sub-nanosecond resolution.	TS*PHYXS:O	Yes [] No [] N/A []
TS4	DTE 800GXS Path data delays	171.6a.2	Path data delays reported as if DDMP is at start of a set of four interleaved RS-FEC codewords.	TS*DTEXS:M	Yes [] N/A []
TS5	DTE 800GXS path data delay nanosecond resolution	171.6a.2	Path data delays reported with nanosecond or better resolution.	TS*DTEXS:M	Yes [] N/A []
TS6	DTE 800GXS path data delay sub-nanosecond resolution	171.6a.2	Path data delays reported with sub-nanosecond resolution.	TS*DTEXS:O	Yes [] No [] N/A []

Insert the following subclauses after 171.9.4.6a:

171.9.5 PICS proforma tables for 1.6TMI Extender and 1.6TMI Extender Sublayer (1.6TXS)

171.9.5.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	175.2.4.1		O	Yes [] No []
TF1	Transmit 64B/66B encoder uses state diagram in Figure 119–14.	172.2.4.1.1		!SE:M	Yes [] N/A []
TF2	Transmit 64B/66B encoder uses stateless rules in Table 172–1.	172.2.4.1.2		SE:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
TF3	64B/66B to 256B/257B transcoder	175.2.4.3		M	Yes []
TF4	Block Distribution	175.2.4.4		M	Yes []
TF5	Scrambler	175.2.4.5		M	Yes []
TF6	Pad value	175.2.4.6		M	Yes []
TF7	Alignment marker insertion	175.2.4.6		M	Yes []
TF8	FEC degrade signaling	175.2.4.6		M	Yes []
TF9	Pre-FEC distribution	175.2.4.7		M	Yes []
TF10	Reed-Solomon encoder	175.2.4.8		M	Yes []
TF11	Symbol distribution	175.2.4.9		M	Yes []
TF12	Transmission bit ordering	175.2.4.10		M	Yes []
TF13	Scrambled idle transmit test-pattern generator	175.2.4.11		M	Yes []

171.9.5.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
*BI	Bypass indication supported	175.2.5.3		O	Yes [] No []
*SD	Uses stateless decoder	175.2.5.9		O	Yes [] No []
RF1	Alignment marker lock	175.2.5.1		M	Yes []
RF2	PCS lane number is captured to an MDIO register	119.2.6.3		MD:M	Yes [] N/A []
RF3	Deskew	175.2.5.1		M	Yes []
RF4	Lane reorder and de-interleave	175.2.5.2		M	Yes []
RF5	Reed-Solomon decoder corrects any combination of up to $t=15$ symbol errors in a codeword	175.2.5.3		M	Yes []
RF6	Reed-Solomon decoder is capable of indicating when a codeword was not corrected.	175.2.5.3		M	Yes []
RF8	Error monitoring and assertion of hi_ser , while error indication is bypassed	175.2.5.3		BI:M	Yes [] N/A []
RF9	Detect FEC degrade	175.2.5.3		FDD:M	Yes [] N/A []
RF10	Alignment marker removal	175.2.5.5		M	Yes []
RF11	Descrambler	175.2.5.6		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
RF12	256B/257B to 64B/66B transcoder	175.2.5.8		M	Yes []
RF13	Receive 64B/66B decoder uses state diagram in Figure 119–15.	172.2.5.9.1		!SD:M	Yes [] N/A []
RF14	Receive 64B/66B decoder uses stateless rules in Table 172–4.	172.2.5.9.2		SD:M	Yes [] N/A []

171.9.5.3 64B/66B coding rules

Item	Feature	Subclause	Value/Comment	Status	Support
C1	Encoder (and ENCODE function) implements the code as specified	175.2.3, 175.2.6.2.3		M	Yes []
C2	Decoder (and DECODE function) implements the code as specified	175.2.3, 175.2.6.2.3		M	Yes []
C3	Only valid block types are transmitted	175.2.3		M	Yes []
C4	Invalid block types are treated as an error	175.2.3		M	Yes []
C5	Only valid control characters are transmitted	175.2.3		M	Yes []
C6	Invalid control characters are treated as an error	175.2.3		M	Yes []
C7	Idles do not interrupt data	175.2.3		M	Yes []
C8	IDLE control code insertion and deletion	175.2.3	Insertion or deletion in groups of 8 /I/s	M	Yes []
C9	Sequence ordered set deletion	175.2.3	Only one whole ordered set of two consecutive sequence ordered sets may be deleted	M	Yes []

171.9.5.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to XS management objects is provided	171.8		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits for PHY 1.6TXS	171.8	See Table 171–5a and Table 171–5b	MD*PHYXS:M	Yes [] N/A []
M3	Mapping of MDIO control bits and MDIO status bits for DTE 1.6TXS	171.8	See Table 171–5c and Table 171–5d	MD*DTEXS:M	Yes [] N/A []

171.9.5.5 Loopback

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Supports loopback	175.4		M	Yes []
L2	When in loopback, transmits what it receives from the 1.6TMII	175.4		M	Yes []
L3	When in loopback, ignore all data presented by the PMA	175.4		M	Yes []

171.9.5.6 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1	1.6TXS delay constraint	175.5		M	Yes [] N/A []

171.9.5.7 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	171.6a		O	Yes [] No []
TS1	PHY 1.6TXS Path data delays	171.6a.1	Path data delays reported as if DDMP is at start of a set of four interleaved RS-FEC codewords.	TS*PHYXS:M	Yes [] N/A []
TS2	PHY 1.6TXS path data delay nanosecond resolution	171.6a.1	Path data delays reported with nanosecond or better resolution.	TS*PHYXS:M	Yes [] N/A []
TS3	PHY 1.6TXS path data delay sub-nanosecond resolution	171.6a.1	Path data delays reported with sub-nanosecond resolution.	TS*PHYXS:O	Yes [] No [] N/A []
TS4	DTE 1.6TXS Path data delays	171.6a.2	Path data delays reported as if DDMP is at start of a set of four interleaved RS-FEC codewords.	TS*DTEXS:M	Yes [] N/A []
TS5	DTE 1.6TXS path data delay nanosecond resolution	171.6a.2	Path data delays reported with nanosecond or better resolution.	TS*DTEXS:M	Yes [] N/A []
TS6	DTE 1.6TXS path data delay sub-nanosecond resolution	171.6a.2	Path data delays reported with sub-nanosecond resolution.	TS*DTEXS:O	Yes [] No [] N/A []

Clause 172 was added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024:

172. Physical Coding Sublayer (PCS), type 800GBASE-R

172.1 Overview

172.1.3 Summary of functions

Change the text in 172.1.3 as follows:

The PCS service interface is the 800 Gb/s Media Independent Interface (800GMII), which is defined in Clause 170. The 800GMII provides a uniform interface to the Reconciliation Sublayer for all 800 Gb/s PHY implementations.

The 800GBASE-R PCS is based on the 400GBASE-R PCS defined in Clause 119.

The 800GBASE-R PCS provides all of the services required by the 800GMII, including the following:

- a) Encoding (decoding) of eight 800GMII data octets to (from) 66-bit blocks (64B/66B)
- b) Transcoding from (to) 66-bit blocks to (from) 257-bit blocks (256B/257B)
- c) Scrambling (descrambling) the 257-bit transcoded blocks
- d) Alignment marker insertion and removal
- e) Reed-Solomon encoding (decoding) the scrambled 257-bit blocks
- f) Distribution of Reed-Solomon encoded data to (from) PCS lanes
- g) Alignment marker lock, lane deskew and reordering of received PCS lanes
- h) Transferring encoded data to (from) the ~~PMA~~ sublayer below the PCS
- i) Compensation for any rate differences caused by the insertion or deletion of alignment markers or due to any rate difference between the 800GMII and ~~PMA~~ the sublayer below the PCS through the insertion or deletion of idle control characters

172.1.5 Inter-sublayer interfaces

Change the text in 172.1.5 as follows:

The upper interface of the PCS connects to the Reconciliation Sublayer through the 800GMII. The lower interface of the PCS connects to ~~the a PMA or a FEC sublayer~~ to support a PMD. The lower interface of the 800GBASE-R PCS has a nominal rate ~~at the PMA service interface~~ of 26.5625 Gb/s on each of 32 PCS lanes, which provides capacity for the MAC data rate of 800 Gb/s.

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience.

Change the title and text of 172.1.5.2 as follows:

172.1.5.2 Physical Medium Attachment (PMA) service interface~~Service interface below the PCS~~

The ~~PMA~~ service interface below the PCS supports the exchange of encoded data between the PCS and ~~PMA~~ the sublayer below the PCS. ~~The PMA service interface is defined in 173.3 and It is an instance of the inter-sublayer service interface definition defined in 169.3 and is described in an abstract manner which does not imply any particular implementation.~~

~~The PMA service interface for the PCS is described in an abstract manner and does not imply any particular implementation.~~

172.1.6 Functional block diagram

Replace Figure 172–2 with the following figure:

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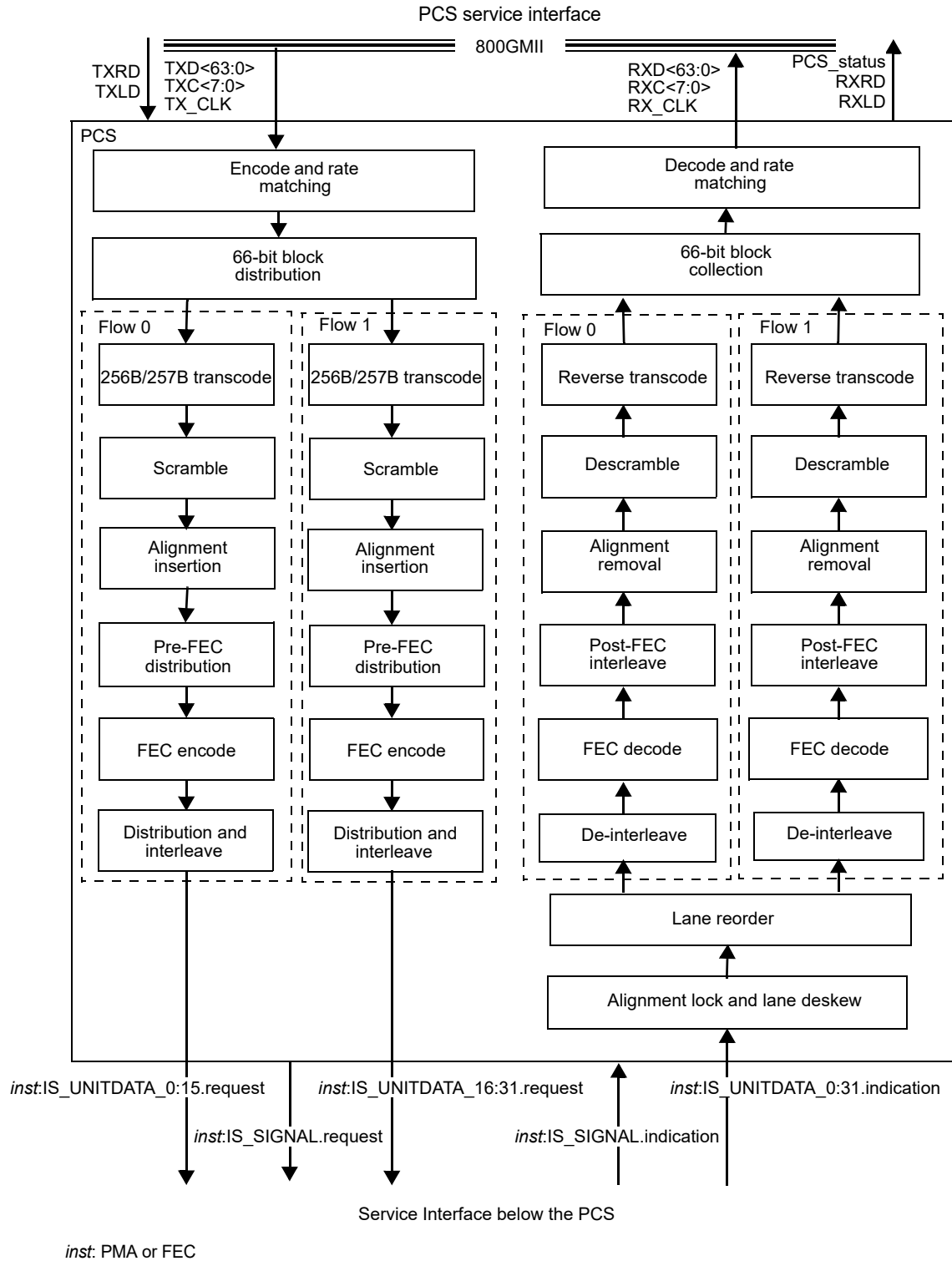


Figure 172–2—Functional block diagram

172.2 PCS functions

172.2.1 Overview

Change the text of 172.2.1 as follows:

The 800GBASE-R PCS is composed of the PCS Transmit and PCS Receive processes. The PCS shields the Reconciliation Sublayer (and MAC) from the specific nature of the underlying channel.

When communicating with the 800GMII, the PCS uses an eight octet-wide, synchronous data path, with packet delineation being provided by transmit control signals ($TXC_{<n>} = 1$) and receive control signals ($RXC_{<n>} = 1$). When communicating with the PMA sublayer below the PCS, the 800GBASE-R PCS uses 32 encoded bit streams also known as PCS lanes. Per direction (receive or transmit), the PCS lanes originate from a common clock but may vary in phase and skew dynamically. ~~The PMA operates independently of block and packet boundaries.~~ The PCS provides the functions necessary to map packets between the 800GMII ~~format~~ and the PMA service interface below the PCS format.

The PCS transmit function can operate in normal mode or test-pattern mode.

When the transmit function is in normal mode, the PCS Transmit process continuously generates 66-bit blocks based on the $TXD_{<63:0>}$ and $TXC_{<7:0>}$ signals on the 800GMII. This takes place in the encode and rate matching function shown in Figure 172–2. The 66-bit blocks are then distributed between two parallel transmit functions, referred to as flow 0 and flow 1. Within each flow, the 66-bit blocks are transcoded to 257-bit blocks and scrambled. Alignment markers are periodically inserted into the data stream of each flow following the requirements specified in 172.2.4.6. For each flow, the data stream is distributed to two 5140-bit blocks and then FEC encoded. For each flow, the two FEC codewords are then interleaved before data is distributed to individual PCS lanes.

Transmit data units are sent to the service interface below the PCS via the PMA:IS_UNITDATA_0:31.request ~~inst:IS_UNITDATA_0:31.request~~ primitive. The SIGNAL_OK parameter of the inst:IS_SIGNAL.request primitive is set to OK when the transmit data units are valid and is set to FAIL otherwise. The inst:IS_SIGNAL.request primitive is only required if the sublayer below the PCS is an SM-PMA (see Clause 176).

When the transmit function is in test-pattern mode, a test pattern is packed into the transmit data units that are sent to the ~~PMA service interface below the PCS~~ via the ~~PMA:IS_UNITDATA_0:31.request~~ inst:IS_UNITDATA_0:31.request primitive.

In the receive direction, the PCS Synchronization process continuously monitors PMA:IS_SIGNAL.indication(SIGNAL_OK) ~~inst:IS_SIGNAL.indication(SIGNAL_OK)~~. When SIGNAL_OK indicates OK, then the PCS synchronization process accepts data units via the PMA:IS_UNITDATA_0:31.indication ~~inst:IS_UNITDATA_0:31.indication~~ primitive. It attains alignment marker lock based on the common marker (CM) portion of the alignment markers that are periodically transmitted on every PCS lane and identifies individual PCS lanes using the unique marker (UM) portion of the alignment marker. The PCS lanes are then reordered and deskewed, and the align_status flag is set. Note that a particular transmit PCS lane can be received on any receive lane of the service interface due to the skew and multiplexing that occurs in the path.

The PCS Receive process separates the reordered PCS lanes into two sets of 16 PCS lanes belonging to each flow. Within a flow, the data from the 16 PCS lanes is de-interleaved, processed by the FEC decoder, and re-interleaved on a 10-bit basis to form a single data stream. The alignment markers are removed, the data is descrambled and reverse transcoded back to 66-bit blocks. A 66-bit block collection function merges the 66-bit blocks from the two flows into a single stream of blocks that are then 64B/66B decoded.

172.2.4 Transmit function

172.2.4.2 Rate matching

Change the text of 172.2.4.2 as follows:

The transmit PCS may remove idle control characters or sequence ordered sets to compensate for the insertion of alignment markers. The transmit PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the 800GMII and the PMA service interface below the PCS. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

172.2.4.10 Transmit bit ordering and distribution

Change the text of 172.2.4.10 as follows:

The transmit bit ordering and distribution is illustrated in Figure 172–4. The transmit bit ordering and distribution in “Flow 0 transmit function” and “Flow 1 transmit function” are identical to the 400GBASE-R transmit bit ordering and distribution in Figure 119–11 with the following exceptions:

- The “66-bit block distribution” of Figure 172–4 directly feeds into the “64B/66B to 256B/257B transcoder” in each flow’s transmit function shown in Figure 119–11, and the functional blocks above the “64B/66B to 256B/257B transcoder” in each flow are not used.
- For flow 1, the PMA service interface lane numbers 0:15 are replaced with 16:31, respectively.

Replace Figure 172–4 with the following figure:

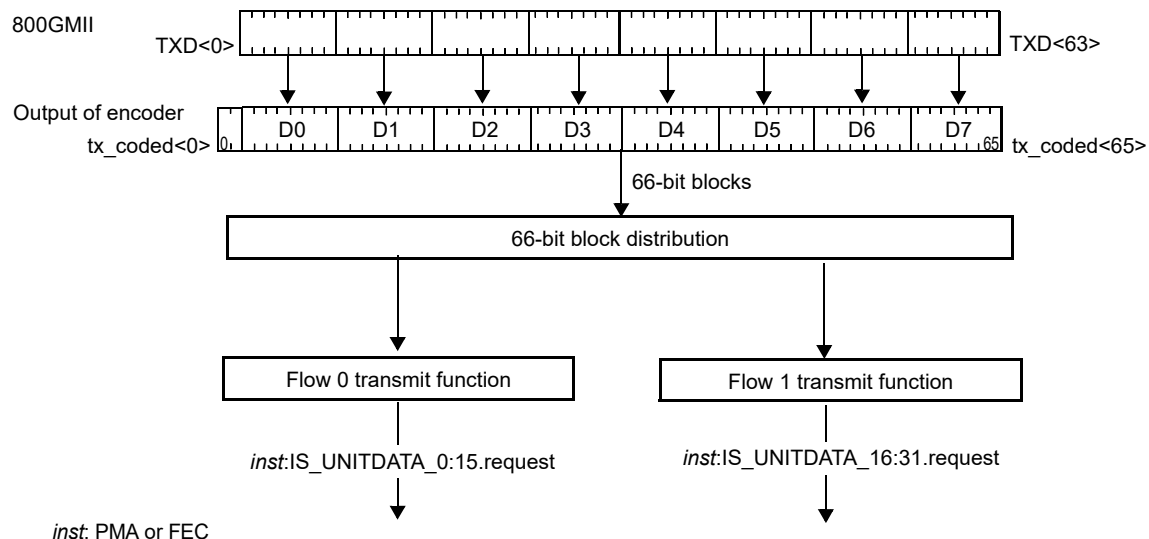


Figure 172–4—800GBASE-R PCS transmit bit ordering and distribution

172.2.5 Receive function

172.2.5.2 Lane reorder and de-interleave

Change the text of 172.2.5.2 as follows:

The PCS lanes might be received in any order on the ~~PMA~~-service interface lanes. The PCS receive function shall order the PCS lanes according to the PCS lane number. The PCS lane number is defined by the unique portion (UM₀ to UM₅) of the alignment marker that is mapped to each PCS lane (see 172.2.4.6).

After all 32 PCS lanes are aligned, deskewed, and reordered, they are separated into two sets of 16 PCS lanes belonging to each flow. PCS lanes 0 to 15 are sent to flow 0 and PCS lanes 16 to 31 are sent to flow 1. Within a flow, the data from the 16 PCS lanes is de-interleaved to reconstruct the original two streams of FEC codewords.

172.2.5.10 Rate matching

Change the text in 172.2.5.10 as follows:

The receive PCS may insert idle control characters to compensate for the deletion of alignment markers. The receive PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains ~~on the PMA between the service interface below the PCS~~ and the 800GMII. See 119.2.3.5 and 119.2.3.8 for the deletion and insertion rules.

NOTE—If no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 800GMII will be reduced by approximately 49 ppm.

172.6 Auto-Negotiation

Change the text in 172.6 as follows:

The following requirements apply to a PCS used with an 800GBASE-CR8, ~~800GBASE-CR4, or 800GBASE-KR8, or 800GBASE-KR4~~ PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN_LINK.indication(link_status) primitive (see 73.9). The parameter link_status shall take the value FAIL when PCS_status=false and the value OK when PCS_status=true. The primitive shall be generated when the value of link_status changes.

172.7 Protocol implementation conformance statement (PICS) proforma for Clause 172, Physical Coding Sublayer (PCS), type 800GBASE-R⁴

172.7.4 PICS proforma tables for Physical Coding Sublayer (PCS), type 800GBASE-R

172.7.4.7 Auto-Negotiation

Change the table in 172.7.4.7 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Support for use with an 800GBASE-CR8 ₂ or 800GBASE-CR4 ₂ 800GBASE-KR8 ₂ or PMD 800GBASE-KR4 PMD	172.6	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN1	AN_LINK.indication primitive	172.6	Support of the AN_LINK.indication(link_status) primitive	AN:M	Yes [] N/A []
AN2	link_status parameter	172.6	Takes the value OK or FAIL	AN:M	Yes [] N/A []
AN3	Generation of AN_LINK.indication primitive	172.6	Generated when the value of link_status changes	AN:M	Yes [] N/A []

⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Clause 173 was added to IEEE Std 802.3-2022 by IEEE Std 802.3df-2024:

173. Physical Medium Attachment (PMA) sublayer, type 800GBASE-R

173.1 Overview

173.1.1 Scope

Change 173.1.1 as follows:

~~This clause specifies the Physical Medium Attachment (PMA) sublayer for the family of 800 Gb/s Physical Layer implementations, known as 800GBASE-R. The PMA allows the PCS (specified in Clause 172) to connect in a media-independent way with a range of physical media, or for the DTE 800GXS to connect to the PHY 800GXS (specified in Clause 171). The 800GBASE-R PMA(s) can support any of the 800 Gb/s PMDs in Table 169-1.~~

This clause specifies forms of the Physical Medium Attachment (PMA) sublayer that uses bit-multiplexing for 800GBASE-R Physical Layer implementations.

When necessary to differentiate the bit-multiplexing PMA (BM-PMA) types defined in this clause from the symbol-multiplexing PMA (SM-PMA) types defined in Clause 176, the term BM-PMA is used. Within this clause the term PMA refers specifically to the BM-PMA.

Three types of the 800GBASE-R BM-PMA are defined: 32:8 PMA, 8:32 PMA, and 8:8 PMA.

Insert new subclause 173.1.1a as follows:

173.1.1a Introduction

The PMA allows the PCS (specified in Clause 172) to connect in a media-independent way with a range of physical media, or for the DTE 800GXS to connect to the PHY 800GXS (specified in Clause 171). The 800GBASE-R PMA supports any 8-lane PMD or 800GAUI-8 in Table 169-2 and Table 169-3.

173.4 PMA types

173.4.2 8:32 PMA

Change the last item in the dashed list in 173.4.2 as follows:

- The interface below the PMA (32 lanes) connects with a PHY 800GXS, 800GBASE-LR1 Inner FEC, or 800GBASE-ER1 FEC.

Replace Figure 173–3 with the following:

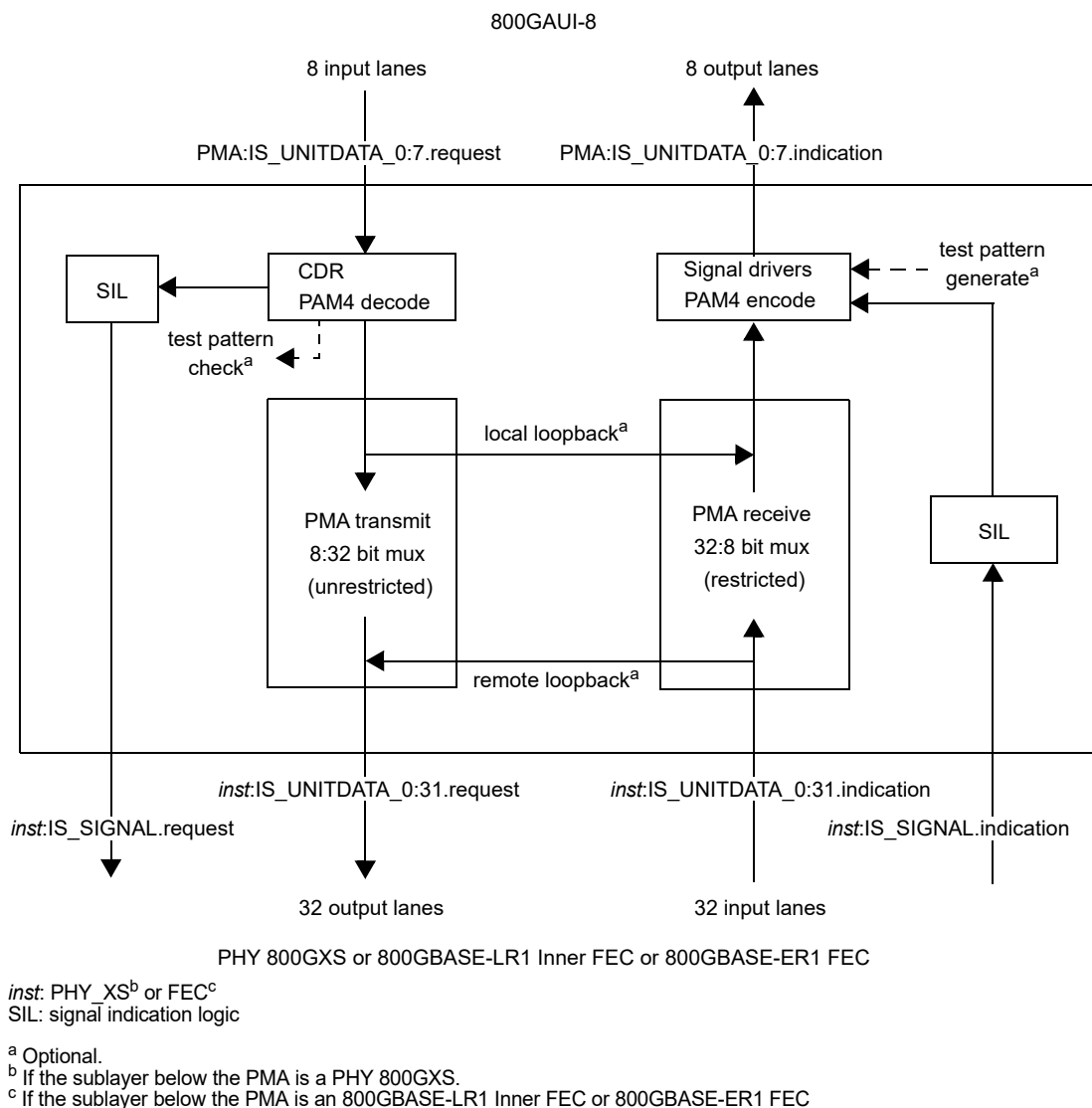


Figure 173–3—8:32 PMA functional block diagram

Insert new clauses Clause 174 through Clause 187 as follows:

174. Introduction to 1.6Tb/s networks

174.1 Overview

174.1.1 Scope

This clause describes the general requirements for 1.6 Terabit Ethernet.

1.6 Terabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 1.6 Tb/s, coupled with any IEEE 802.3 1.6TBASE Physical Layer implementation, and is defined for full duplex operation only.

174.1.2 Relationship of 1.6 Terabit Ethernet to the ISO OSI reference model

1.6 Terabit Ethernet couples the IEEE 802.3 MAC to a family of 1.6 Tb/s Physical Layer devices. The relationships among 1.6 Terabit Ethernet, the IEEE 802.3 MAC, and the ISO Open System Interconnection (OSI) reference model are shown in Figure 174–1.

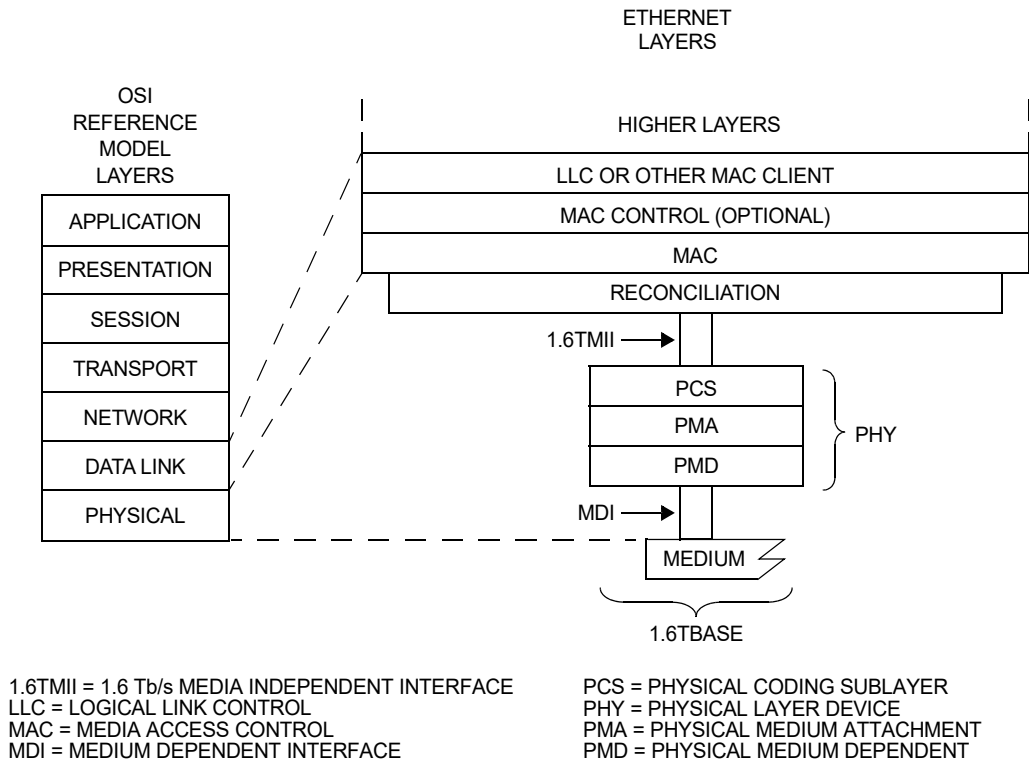


Figure 174–1—Architectural positioning of 1.6 Terabit Ethernet

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience, with the following exceptions:

- The management interface, when physically implemented as the MDIO/MDC (see Clause 45, 22.2.2.13, and 22.2.2.14)
- The PMA service interface, when physically implemented as 1.6TAUI-n (see 174.2.5)

— The MDI for a 1.6TBASE PMD (see 174.2.7)

174.1.3 Nomenclature

The nomenclature employed by the 1.6 Tb/s Physical Layer is explained as follows.

The alpha-numeric prefix 1.6TBASE in the port type (e.g., 1.6TBASE-R) represents a family of Physical Layer devices operating at a speed of 1.6 Tb/s.

The term 1.6TBASE-R represents a family of Physical Layer devices using the Physical Coding Sublayer (PCS) defined in Clause 175 for 1.6 Tb/s operation.

Physical Layer devices listed in Table 174–1 are defined for operation at 1.6 Tb/s.

Table 174–1—1.6 Tb/s PHYs

Name	Description
1.6TBASE-KR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight lanes of an electrical backplane (see Clause 178)
1.6TBASE-CR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight lanes of twinaxial copper cable (see 1.4.559 and Clause 179)
1.6TBASE-DR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight single-mode fibers with reach up to at least 500 m (see Clause 180)
1.6TBASE-DR8-2	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight single-mode fibers with reach up to at least 2 km (see Clause 182)

174.1.4 Physical Layer signaling systems

This standard specifies a family of Physical Layer implementations. Table 174–2 and Table 174–3 specify the correlation between PHY types and clauses. Implementations conforming to one or more PHY types meet the requirements of the corresponding clauses.

Table 174–2—PHY type and clause correlation (1.6TBASE-R optical)

PHY type	Clause ^a														
	Time Synchronization	90		170		171	175	176	177	120F	120G	176C	176D	180	182
		RS	1.6TMII	1.6TMII Extender	1.6TBASE-R PCS	1.6TBASE-R SM-PMA	1.6TBASE-R Inner FEC	1.6TAUI-16 C2C	1.6TAUI-16 C2M	1.6TAUI-8 C2C	1.6TAUI-8 C2M	1.6TBASE-DR8 PMD	1.6TBASE-DR8-2 PMD		
1.6TBASE-DR8	O	M	O	O	M	M	—	O	O	O	O	M	—		
1.6TBASE-DR8-2	O	M	O	O	M	M	M	O	O	O	O	—	M		

^a O = Optional, M = Mandatory.

Table 174–3—PHY type and clause correlation (1.6TBASE-R electrical)

PHY type	Clause ^a										
	Time Synchronization	90	170		171	175	176	120F	176C	178	179
			RS	1.6TMII							
1.6TBASE-KR8	O	M	O	O	M	M	O	O	M	—	
1.6TBASE-CR8	O	M	O	O	M	M	O	O	—	M	

^a O = Optional, M = Mandatory.

174.2 Summary of 1.6 Terabit Ethernet architecture

174.2.1 Reconciliation Sublayer (RS) and Media Independent Interface (MII)

The Media Independent Interface (MII) specified in Clause 170 provides a logical interconnection between the MAC sublayer and Physical Layer devices (PHYs). The MII is not intended to be physically instantiated, rather it logically connects layers within a device.

The 1.6TMII supports 1.6 Tb/s operation through its 64-bit-wide transmit and receive data paths. The Reconciliation Sublayer (RS) provides a mapping between the signals provided at the MII (1.6TMII) and the MAC/PLS service definition.

While the 1.6TMII is an optional interface, it is used extensively in this standard as a basis for functional specification and it provides a common service interface for the Physical Coding Sublayer (Clause 175).

174.2.2 1.6TMII Extender and 1.6TMII Extender Sublayer (1.6TXS)

The optional 1.6TMII Extender (Clause 171) may be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 1.6TMII.

The 1.6TMII Extender Sublayer (1.6TXS) is part of the 1.6TMII Extender. It is equivalent in function to the 1.6TBASE-R PCS in Clause 175 with the exceptions as defined in Clause 171.

174.2.3 Physical Coding Sublayer (PCS)

The PCS performs encoding of data from the 1.6TMII, applies FEC, distributes the data to multiple PCS lanes, and transfers the encoded data to the PMA.

The 1.6TBASE-R PCS is specified in Clause 175.

174.2.4 Physical Medium Attachment (PMA) sublayer

The PMA sublayer provides a medium-independent means to support the use of a range of physical media.

The 1.6TBASE-R SM-PMA is specified in Clause 176.

174.2.5 Attachment unit interface (1.6TAUI-n)

The 1.6 Tb/s Attachment Unit Interface (1.6TAUI-n) provides an electrical interface within a 1.6TBASE-R PHY or 1.6TMII Extender. The 1.6TAUI-n is defined for chip-to-chip (C2C) and chip-to-module (C2M) implementations.

The 1.6TAUI-n C2C is specified in Annex 120F and Annex 176C.

The 1.6TAUI-n C2M is specified in Annex 120G and Annex 176D.

The 1.6TAUI-n instantiations are described in 176B.7.

174.2.6 Inner FEC sublayer

The Inner FEC sublayer provides error correction, in addition to that provided by the 1.6TBASE-R PCS, for the PMD.

For 1.6TBASE-DR8-2 the 1.6TBASE-R Inner FEC is specified in Clause 177.

174.2.7 Physical Medium Dependent (PMD) sublayer

The PMD sublayer is responsible for interfacing to the transmission medium. The PMD is located just above the Medium Dependent Interface (MDI).

The MDI, logically subsumed within each PMD subclause, is the actual medium attachment for the various supported media.

1.6TBASE-R PMDs, including the corresponding MDI and media, are specified in Clause 178, Clause 179, Clause 180, and Clause 182.

174.2.8 Auto-Negotiation (AN)

Auto-Negotiation (AN) provides a device with the capability to detect the abilities (modes of operation) supported by the device at the other end of the link, determine common abilities, and configure for joint operation.

AN used by a 1.6 Tb/s backplane PHY (1.6TBASE-KR8) and the 1.6 Tb/s copper PHY (1.6TBASE-CR8) is specified in Clause 73.

174.2.9 Management interface (MDIO/MDC)

The optional MDIO/MDC management interface provides an interconnection between MDIO Manageable Devices (MMDs) and Station Management (STA) entities.

The MDIO/MDC management interface is specified in Clause 45.

174.2.10 Management

Managed objects, attributes, and actions are defined for all 1.6 Terabit Ethernet components. These items are specified in Clause 30.

174.2.11 Time Synchronization

A 1.6 Tb/s Physical Layer may optionally support time synchronization protocols that require knowledge of packet egress and ingress time. Ethernet support for time synchronization protocols is defined in Clause 90.

If time synchronization is supported, then the following capabilities are provided:

- The 1.6 Tb/s Reconciliation Sublayer (RS) provides a Time Synchronization Service Interface (TSSI) that connects to a TimeSync client (see 90.4).
- The path data delay values of each sublayer in a Physical Layer are reported as status variables, which may be mapped to MDIO status registers (see 90.6).

174.2.12 Inter-sublayer link training (ILT)

Inter-sublayer link training (ILT) facilitates the orderly start-up of an inter-sublayer link (ISL) and coordinates the start-up of a series of ISLs along a network path. For each ISL, ILT provides a mechanism for a receiver to control peer transmitter states (e.g., equalization, modulation, and precoding), to indicate the receiver state, and to coordinate the transition to DATA mode. ILT is specified in Annex 178B.

ILT is supported by the following PHY types:

- 1.6TBASE-KR8
- 1.6TBASE-CR8
- 1.6TBASE-DR8
- 1.6TBASE-DR8-2

174.2.13 FEC Degrade

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in 116.6.

FEC degrade detection and signaling specifications for the 1.6TXS and 1.6TBASE-R PCS are summarized in 171.6 and 175.3, respectively.

174.3 Service interface specification method and notation

The service interface specification for the 1.6TBASE-R Physical Layers is as per the definition in 1.2.2. Note that the 1.6TBASE-R inter-sublayer service interfaces use multiple scalar REQUEST and INDICATION primitives, to indicate the transfer of multiple independent streams of data units, as defined in 174.3.1 through 174.3.3.

174.3.1 Inter-sublayer service interface

The inter-sublayer service interface is described in an abstract manner and does not imply any particular implementation. The inter-sublayer service interface primitives are defined as follows:

IS_UNITDATA_*i*.request
IS_UNITDATA_*i*.indication
IS_SIGNAL.indication
IS_SIGNAL.request

The IS_UNITDATA_*i*.request (where *i* = 0 to *n*−1, and *n* is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units to a sublayer from the next higher (closer to the MAC) sublayer.

The IS_UNITDATA_*i*.indication (where *i* = 0 to *n*−1, and *n* is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next higher sublayer.

The IS_SIGNAL.indication primitive is used to define the transfer of signal status from a sublayer to the next higher sublayer.

The IS_SIGNAL.request primitive is used to define the transfer of signal status to a sublayer from the next higher sublayer.

174.3.2 Instances of the inter-sublayer service interface

The inter-sublayer service interface may be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

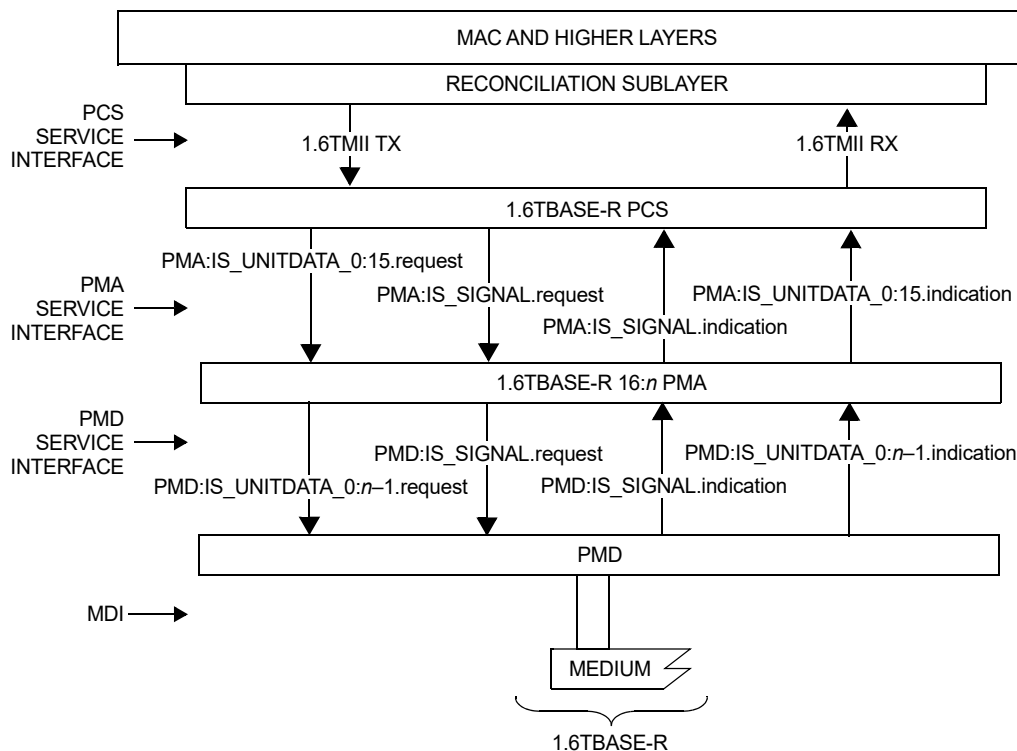
- PMA:—for primitives issued on the interface between the PMA and the sublayer above called the PMA service interface
- PHY_XS:—for primitives issued on the interface between the PHY 1.6TXS and the sublayer above called the PHY 1.6TXS service interface
- FEC:—for primitives issued on the interface between the Inner FEC and the sublayer above called the FEC service interface
- PMD:—for primitives issued on the interface between the PMD and the sublayer above called the PMD service interface

As an example, the primitives for the PMD service interface are identified as follows:

PMD:IS_UNITDATA_0.request
PMD:IS_UNITDATA_0.indication
PMD:IS_SIGNAL.request
PMD:IS_SIGNAL.indication

Examples of inter-sublayer service interfaces for 1.6TBASE-R with their corresponding instance names are illustrated in Figure 174–2 through Figure 174–4.

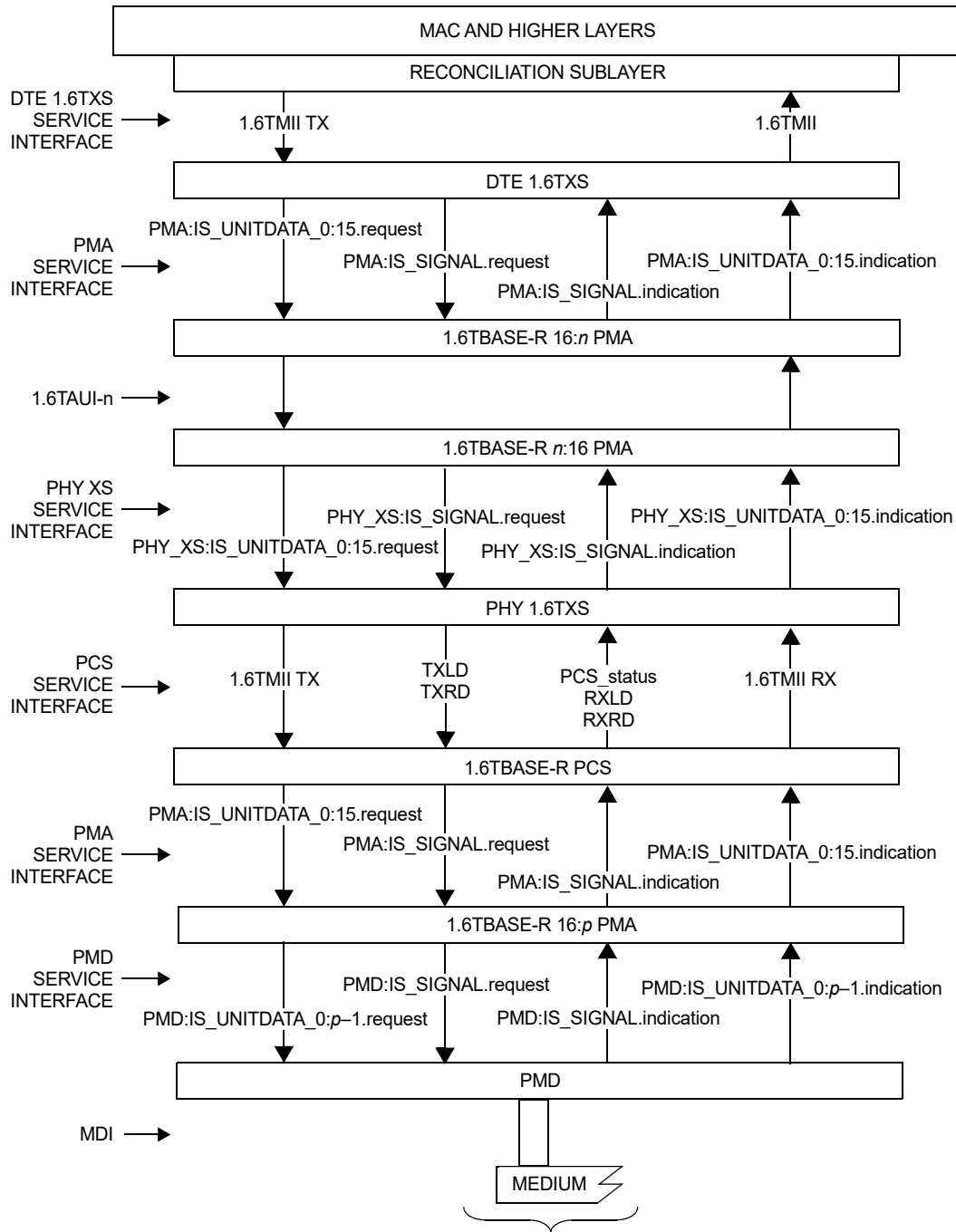
Primitives for other instances of inter-sublayer interfaces are represented in a similar manner.



1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
 n = NUMBER OF PARALLEL STREAMS

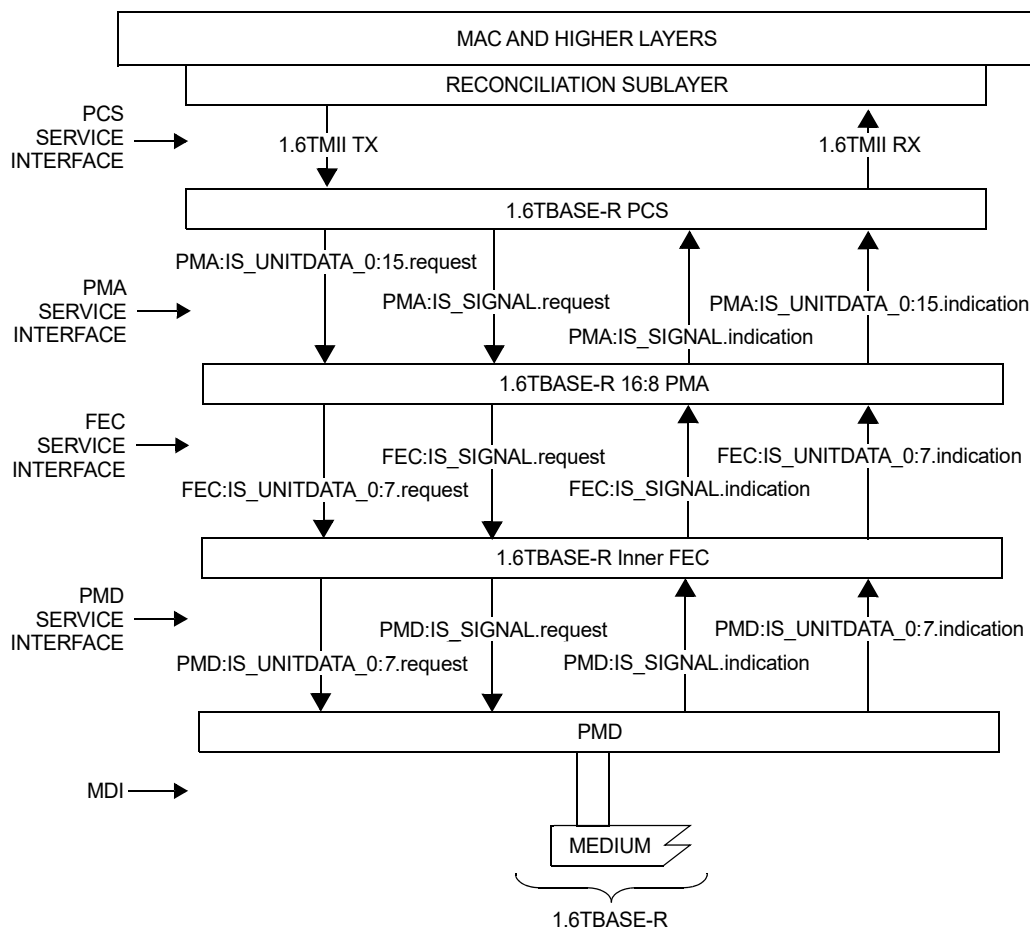
Figure 174–2—Inter-sublayer service interfaces for a 1.6TBASE-R PHY



1.6TAUI-n = 1.6 Tb/s ATTACHMENT UNIT INTERFACE
1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
1.6TXS = 1.6 Tb/s PHY EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

1.6TBASE-R
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
XS = EXTENDER SUBLAYER
n = NUMBER OF PARALLEL STREAMS
p = NUMBER OF PARALLEL STREAMS

Figure 174-3—Inter-sublayer service interfaces for a 1.6TBASE-R PHY and 1.6TMII Extender



1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 174-4—Inter-sublayer service interfaces for a 1.6TBASE-R PHY with 1.6TBASE-R Inner FEC

174.3.3 Semantics of inter-sublayer service interface primitives

The semantics of the inter-sublayer service interface primitives for the 1.6TBASE-R sublayers are described in 116.3.3.

174.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices. Table 174–4 contains the values of maximum sublayer delay (sum of transmit and receive delays at one end of the link). If a PHY contains an Auto-Negotiation sublayer, the delay of the Auto-Negotiation sublayer is included within the delay of the PMD and medium. The delay for a set of sublayers within the same package may be constrained by the sum of constraints for the set of the sublayers.

Table 174–4—Sublayer delay constraints

Sublayer	Maximum			Notes ^a
	(bit time) ^b	(pause_quantum) ^c	(ns)	
1.6T MAC, RS, and MAC Control	393 216	768	245.76	See 170.1.4.
1.6TBASE-R PCS or 1.6TXS ^d	640 000	1250	400	See 175.5.
1.6TBASE-R 8:16 or 16:8 PMA	24 576	48	15.36	See 176.8.
1.6TBASE-R 8:8 or 16:16 PMA	24 576	48	15.36	See 176.8.
1.6TBASE-R Inner FEC	138240	270	86.4	See 177.8.
1.6TAUI-4 C2C or C2M component	36 864	72	46.08	Includes allocation of 5 ns for one direction through AUI channel. See 176C.5 and 176D.5.
1.6TBASE-KR8 PMD	118 784	232	74.24	Includes allocation of 14 ns for one direction through the backplane medium. See 178.6.
1.6TBASE-CR8 PMD	118 784	232	74.24	Includes allocation of 14 ns for one direction through the cable medium. See 179.6.
1.6TBASE-DR8 PMD	118 784	232	74.24	Includes 2 m of fiber. See 180.4.1.
1.6TBASE-DR8-2 PMD	118 784	232	74.24	Includes 2 m of fiber. See 182.4.1.

^a Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^b For 1.6TBASE, 1 bit time is equal to 0.625 ps. (See 1.4.215 for the definition of bit time.)

^c For 1.6TBASE, 1 pause_quantum is equal to 320 ps. (See 1.4.459 for the definition of pause_quantum.)

^d If an implementation includes the 1.6TMII Extender, the delay associated with the 1.6TMII Extender includes two 1.6TXS sublayers.

The physical medium interconnecting two optical PHYs introduces additional delay in a link.

See 80.4 for the calculation of bit time per meter of fiber or electrical cable.

See 31B.3.7 for PAUSE reaction timing constraints for stations at operating speeds of 1.6 Tb/s.

174.5 Skew constraints

Skew (or relative delay) can be introduced between PCS lanes by both active and passive elements of a 1.6TBASE-R link. Skew is defined as the difference in time between the start of the alignment marker for the earliest and latest arriving PCS lanes. The PCS deskew function (see 175.2.5.1) compensates for all lane-to-lane Skew observed at the receiver. The Skew between the PCS lanes is kept within limits as shown in Table 174–5 so that the transmitted information on the PCS lanes can be reassembled by the receive PCS.

Skew Variation may be introduced due to variations in electrical, thermal or environmental characteristics. Skew Variation is defined as the change in Skew between any PCS lane and any other PCS lane over the entire time that the link is in operation. From the time the link is brought up, Skew Variation is limited so that each PCS lane always traverses the same lane between any pair of adjacent sublayers while the link remains in operation.

The maximum Skew and Skew Variation at physically instantiated interfaces is specified at Skew points SP1, SP2, and SP3 for the transmit direction and SP4, SP5, and SP6 for the receive direction as illustrated in Figure 174–5 for a PHY with one 1.6TAUI-n and Figure 174–6 for a PHY with two 1.6TAUI-n.

NOTE—No physically instantiated interfaces are specified at SP2 and SP5 (PMD service interface) for 1.6TBASE-R PHY types.

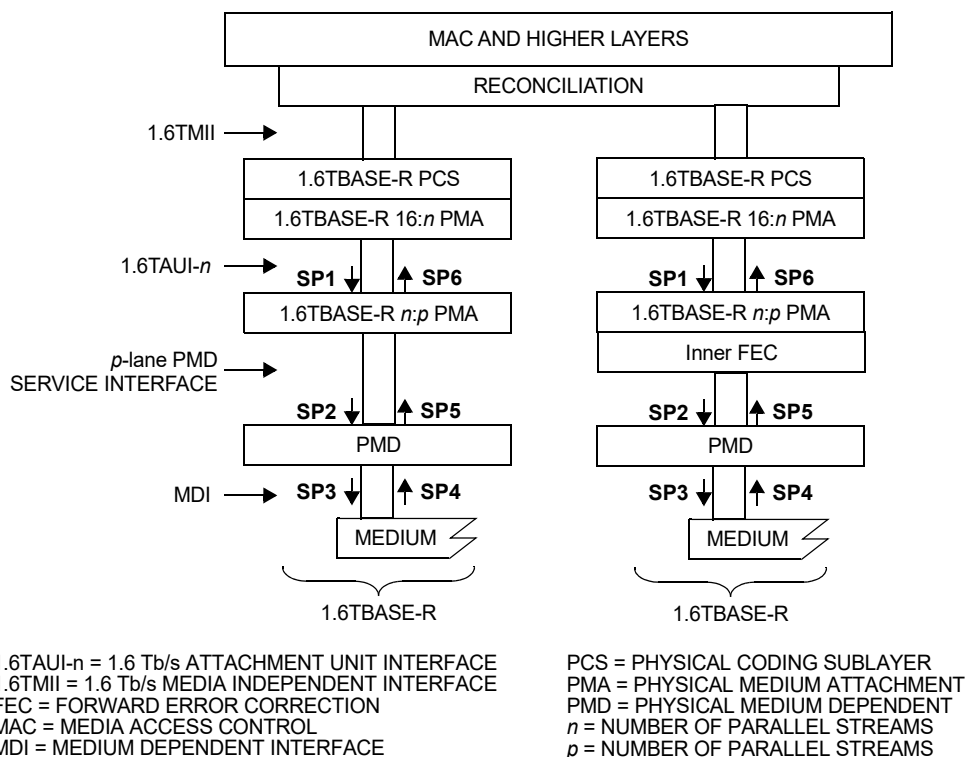
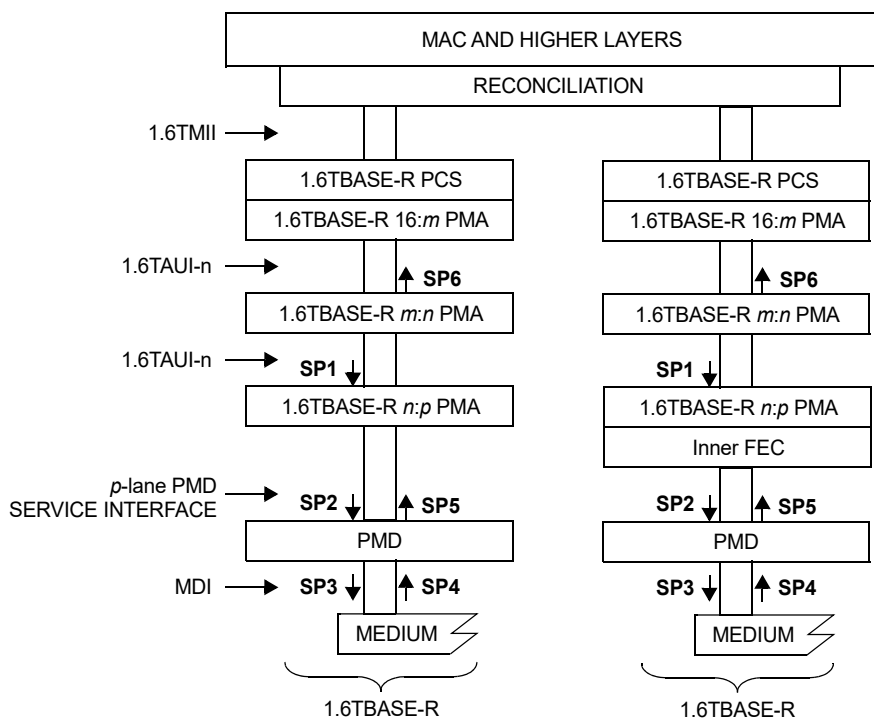


Figure 174–5—1.6TBASE-R Skew points for a PHY with one 1.6TAUI-n



1.6TAUI-n = 1.6 Tb/s ATTACHMENT UNIT INTERFACE
1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
 m = NUMBER OF PARALLEL STREAMS
 n = NUMBER OF PARALLEL STREAMS
 p = NUMBER OF PARALLEL STREAMS

Figure 174-6—1.6TBASE-R Skew points for a PHY with two 1.6TAUI-n

In the transmit direction, the Skew points are defined in the following locations:

- SP1 on the 1.6TAUI-n interface closest to the PMD at the input of the PMA
- SP2 on the PMD service interface, at the input of the PMD
- SP3 at the output of the PMD, at the MDI

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD
- SP5 on the PMD service interface, at the output of the PMD
- SP6 on the 1.6TAUI-n interface closest to the PCS or DTE 1.6TXS at the output of the PMA

The allowable limits for Skew are shown in Table 174-5 and the allowable limits for Skew Variation are shown in Table 174-6.

The Skew and Skew Variation requirements for the PCS, PMA and PMD sublayers are specified in the respective clauses as noted in Table 174-5 and Table 174-6.

Table 174–5—Summary of Skew constraints

Skew points	Maximum Skew		Notes ^a
	(ns) ^b	for 1.6TBASE-R PCS lane (bit) ^c	
SP1	16	≈ 1 700	See 176.9
SP2	N/A	N/A	
SP3	36	≈ 3 825	See 178.7, 179.7, 180.4.2, 182.4.2
SP4	116	≈ 12 325	See 178.7, 179.7, 180.4.2, 182.4.2
SP5	N/A	N/A	
SP6	145	≈ 15 406	See 176.9
At PCS receive	152	≈ 16 150	See 175.2.5.1

^a Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

^b The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^c The symbol ≈ indicates approximate equivalent of maximum Skew in bits based on 1 bit time equals 9.411765 ps at PCS lane signaling rate of 106.25 Gb/s.

Table 174–6—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation					Notes ^a
	(ns)	for 106.25 Gb/s PCS lane (bit) ^b	for 53.125 GBd PMD lane (UI) ^c	for 106.25 GBd PMD lane (UI) ^d	for 113.4375 GBd PMD lane (UI) ^e	
SP1	0.2	N/A	≈ 11	≈ 21	N/A	See 176.9
SP2	N/A	N/A	N/A	N/A	N/A	
SP3	0.6	N/A	N/A	≈ 64	≈ 68	See 178.7, 179.7, 180.4.2, 182.4.2
SP4	3.4	N/A	N/A	≈ 361	≈ 386	See 178.7, 179.7, 180.4.2, 182.4.2
SP5	N/A	N/A	N/A	N/A	N/A	
SP6	3.8	N/A	≈ 202	≈ 404	N/A	See 176.9
At PCS receive	4	≈ 425	N/A	N/A	N/A	See 175.2.5.1, 176.9

^a Should there be a discrepancy between this table and the Skew Variation requirements of the relevant sublayer clause, the sublayer clause prevails.

^b The symbol ≈ indicates approximate equivalent of maximum Skew Variation in bits based on 1 bit time equals 9.411765 ps at PCS lane signaling rate of 106.25 Gb/s.

^c The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^d The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 9.411765 ps at PMD lane signaling rate of 106.25 GBd.

^e The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 8.31543 ps at PMD lane signaling rate of 113.4375 GBd.

174.6 Protocol implementation conformance statement (PICS) proforma

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 175 through Clause 180, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. A PICS is included at the end of each clause as appropriate. Each of the 1.6 Terabit Ethernet PICS conforms to the notation and conventions used in 21.6.

175. Physical Coding Sublayer (PCS), type 1.6TBASE-R

175.1 Overview

175.1.1 Scope

This clause specifies the Physical Coding Sublayer (PCS) that is common to a family of Physical Layer implementations known as 1.6TBASE-R. The 1.6TBASE-R PCS is a sublayer of the 1.6 Tb/s PHYs listed in Table 174–1. The term 1.6TBASE-R is used when referring generally to Physical Layers using the PCS defined in this clause. Annex 175A provides examples of RS-FEC codewords constructed with the method specified in this clause.

The 1.6TBASE-R PCS borrows heavily from the 400GBASE-R PCS defined in Clause 119 and the 800GBASE-R PCS defined in Clause 172.

175.1.2 Relationship of the 1.6TBASE-R PCS to other standards

Figure 175–1 shows the relationship of the 1.6TBASE-R PCS, along with other Ethernet sublayers, to the ISO Open System Interconnection (OSI) reference model.

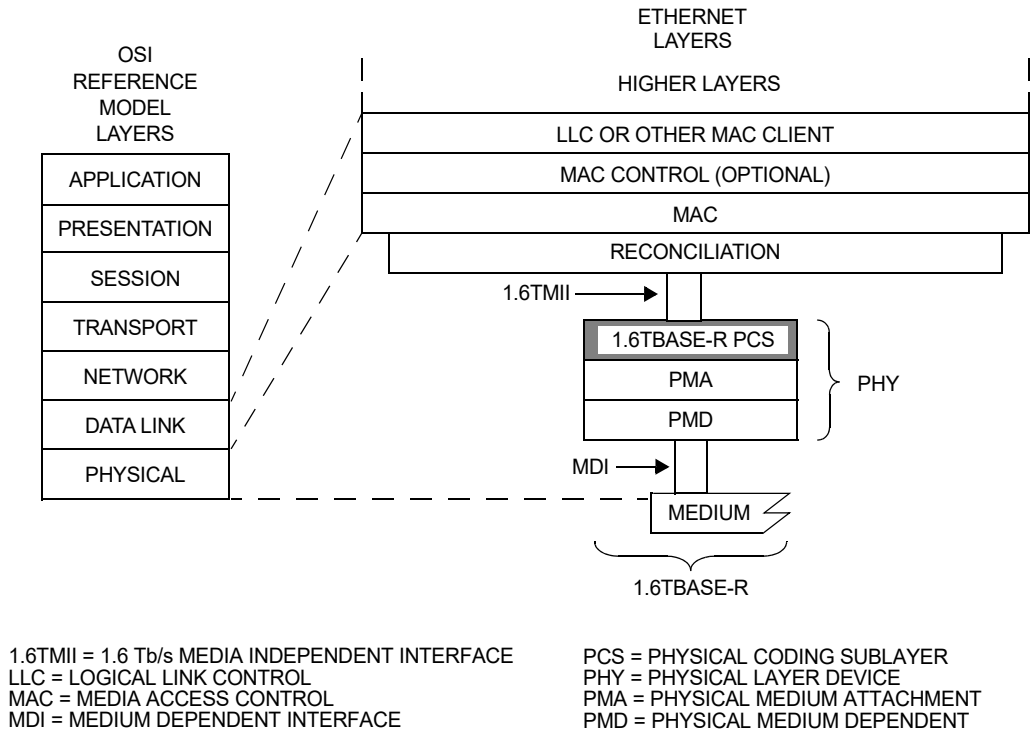


Figure 175–1—1.6TBASE-R PCS relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

175.1.3 Summary of functions

The 1.6TBASE-R PCS provides all services required by the MAC and RS, including the following:

- Encoding (decoding) of eight 1.6TMII data octets to (from) 66-bit blocks (64B/66B)
- Transcoding from (to) 66-bit blocks to (from) 257-bit blocks (256B/257B)
- Scrambling (descrambling) the 257-bit transcoded blocks
- Alignment marker (AM) insertion (removal)
- Reed-Solomon encoding (decoding) the scrambled 257-bit blocks
- FEC degrade detection and signaling
- Distribution of Reed-Solomon encoded data to (from) PCS lanes
- Alignment marker lock, lane deskew and reordering of received PCS lanes
- Transferring encoded data to (from) the sublayer below the PCS
- Compensation for any rate differences caused by the insertion or deletion of alignment markers or due to any rate difference between the 1.6TMII and the sublayer below the PCS through the insertion or deletion of idle control characters

175.1.4 Inter-sublayer interfaces

The upper interface of the PCS connects to the Reconciliation Sublayer or a PHY 1.6TXS, through the PCS service interface. The lower interface of the PCS connects to a PMA or Inner FEC through the service interface below the PCS as necessary to support a PMD. The lower interface of the 1.6TBASE-R PCS has a nominal rate of 106.25 Gb/s on each of 16 PCS lanes, which provides capacity for the MAC data rate of 1.6 Tb/s.

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience.

175.1.4.1 PCS service interface

The PCS service interface allows the 1.6TBASE-R PCS to transfer information to and from a PCS client as illustrated at the top of Figure 175–2.

When the client sublayer is the Reconciliation Sublayer, the PCS service interface is the 1.6 Tb/s Media Independent Interface (1.6TMII) defined in Clause 170.

When the client sublayer is the PHY 1.6TXS (see Figure 171–1), the PCS service interface is the 1.6TMII with additional signals TXRD, TXLD, RXRD, RXLD, and PCS_status. The TXRD and TXLD signals indicate the state of the remote degrade and local degrade, respectively, as detected by the PHY 1.6TXS in the transmit direction (see 171.3.3a). The RXRD signal indicates the state of the rx_rm_degraded variable (see 175.2.5.5) as detected by the PCS in the receive direction. The RXLD signal is the logical OR of the FEC_degraded_SER variable (see 175.2.5.3) and the rx_local_degraded variable (see 175.2.5.5) as detected by the PCS in the receive direction. The PCS_status signal indicates the state of the PCS_status variable (see 175.2.6.2.2).

175.1.4.2 Service interface below the PCS

The service interface below the PCS supports the exchange of encoded data between the PCS and the sublayer below the PCS. It is an instance of the inter-sublayer service interface defined in 174.3 and is described in an abstract manner which does not imply any particular implementation.

175.1.5 Functional block diagram

Figure 175–2 provides a functional block diagram of the 1.6TBASE-R PCS and its primary interfaces.

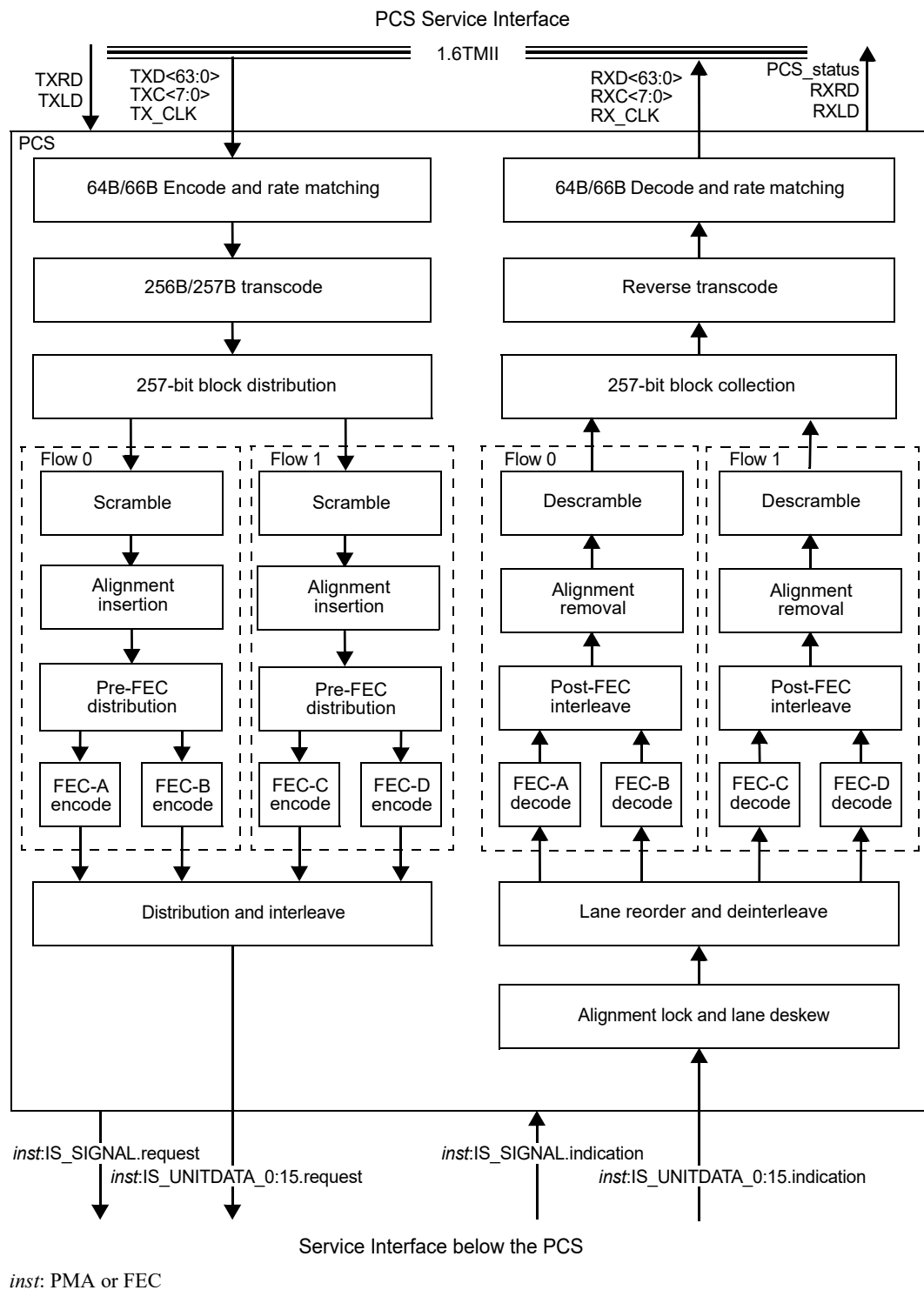


Figure 175–2—Functional block diagram

175.2 PCS functions

175.2.1 Overview

The 1.6TBASE-R PCS is composed of the PCS Transmit and PCS Receive processes. The PCS shields the Reconciliation Sublayer and MAC from the specific nature of the underlying channel.

When communicating with the 1.6TMII, the PCS uses an eight octet-wide, synchronous data path, with packet delineation being provided by transmit control signals ($\text{TXC}_{<n>} = 1$) and receive control signals ($\text{RXC}_{<n>} = 1$). When communicating with the PMA or inner FEC sublayer, the 1.6TBASE-R PCS uses 16 encoded bit streams also known as PCS lanes. Per direction (receive or transmit), the PCS lanes originate from a common clock but may vary in phase and skew dynamically. The PMA and inner FEC operate independently of packet boundaries. The PCS provides the functions necessary to map packets between the 1.6TMII format and the service interface below the PCS.

The PCS transmit function operates in normal mode or test-pattern mode.

When the transmit function is in normal mode, the PCS Transmit process continuously generates 66-bit blocks based on the $\text{TXD}_{<63:0>}$ and $\text{TXC}_{<7:0>}$ signals on the 1.6TMII. This takes place in the encode and rate matching function shown in Figure 175–2. The 66-bit blocks are transcoded to 257-bit blocks to reduce the line coding overhead. The 257-bit blocks are then distributed between two parallel transmit data flows, referred to as flow 0 and flow 1. Within each flow, the 257-bit blocks are first scrambled. Alignment markers are periodically inserted into the data stream of each flow following the requirements specified in 175.2.4.6. Within each flow, the data stream is distributed to two 5140-bit blocks and then FEC encoded. After FEC encoding, four codewords, one from each of the FEC encoders in each flow, are then interleaved and distributed to individual PCS lanes.

Transmit data units are sent to the service interface via the *inst:IS_UNITDATA_0:15.request* primitive. The *SIGNAL_OK* parameter of the *inst:IS_SIGNAL.request* primitive is set to OK when the transmit data units are valid and is set to FAIL otherwise.

When the transmit function is in test-pattern mode, a test pattern is packed into the transmit data units that are sent to the service interface below the PCS via the *inst:IS_UNITDATA_0:15.request* primitive.

In the receive direction, the PCS Synchronization process continuously monitors *inst:IS_SIGNAL.indication(SIGNAL_OK)*. When *SIGNAL_OK* indicates OK, the PCS synchronization process accepts data units via the *inst:IS_UNITDATA_0:15.indication* primitive. It attains alignment marker lock and deskews the PCS lanes based on the common marker (CM) portion of the alignment markers that are periodically transmitted on every PCS lane. Once all the PCS lanes are identified and re-ordered using the unique marker (UM) portion of the alignment markers, the *align_status* flag is set to true. Note that a particular transmit PCS lane might be received on any receive lane of the service interface due to the skew and multiplexing that occurs in the path.

The reordered PCS lanes are deinterleaved into two data flows, flow 0 and flow 1. Each data flow input consists of two FEC data streams. Within each flow, received data is processed by the FEC decoders, and re-interleaved on a 10-bit basis to form a data stream of 257-bit blocks. The alignment markers are removed, and the data is descrambled. After the descramble function, 257-bit blocks are collected from each flow and merged to form a single data flow. The 257-bit blocks are then reverse transcoded back to 66-bit blocks which are then 64B/66B decoded.

175.2.2 Use of 66-bit blocks

The PCS maps 1.6TMII signals into 66-bit blocks, and vice versa, using a 64B/66B coding scheme. The PCS functions ENCODE and DECODE generate, manipulate, and interpret 66-bit blocks as defined in 175.2.3.

175.2.3 64B/66B code

The 64B/66B code is identical to that specified in 119.2.3 with the following exceptions:

- 200GMII/400GMII is replaced with 1.6TMII.
- 200GBASE-R/400GBASE-R is replaced with 1.6TBASE-R.
- EEE and low power idle are not supported.

175.2.4 Transmit function

175.2.4.1 64B/66B encoder

The transmit PCS generates 66-bit blocks based on the TXD<63:0> and TXC<7:0> signals received on the 1.6TMII. Each 1.6TMII transfer is encoded into one 66-bit block. The contents of each 66-bit block are contained in a vector tx_coded<65:0> with tx_coded<1:0> containing the sync header and the remainder of the bits, the payload.

The transmit PCS may use either the state-diagram encoder defined by Figure 119–14 or the stateless encoder defined in 172.2.4.1.2.

175.2.4.2 Rate matching

The transmit PCS may remove idle control characters or sequence ordered sets to compensate for the insertion of alignment markers. The transmit PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the 1.6TMII and the service interface below the PCS. See 82.2.3.6 and 82.2.3.9 for the deletion and insertion rules.

175.2.4.3 64B/66B to 256B/257B transcoder

The 64B/66B to 256B/257B transcoder is identical to that specified in 119.2.4.2.

Four consecutive tx_coded<65:0> blocks are used to construct a 257-bit block, tx_xcoded<256:0>.

NOTE—The stream of 257-bit blocks generated by this process, together with the FEC_degraded_SER and rx_local_degraded variables should be used as the reference signal for mapping to OTN.

175.2.4.4 Block distribution

The 257-bit blocks, tx_xcoded<256:0>, from the transcoder are distributed to the two flows in an alternating fashion by the block distribution function such that one 257-bit block is sent to flow 0, the next 257-bit block is sent to flow 1, the following 257-bit block is sent to flow 0, and subsequent 257-bit blocks continue the distribution procedure across the two flows. This creates two streams of transcoded blocks, tx_xcoded_f0<256:0> to flow 0, and tx_xcoded_f1<256:0> to flow 1.

175.2.4.5 Scrambler

The transcoded 257-bit blocks, tx_xcoded_f0<256:0> in flow 0 and tx_xcoded_f1<256:0> in flow 1, are scrambled with a self-synchronizing scrambler within each flow to generate tx_scrambled_f0<256:0> and tx_scrambled_f1<256:0>, respectively. The scrambler is identical to the scrambler in 49.2.6 using the

polynomial in Equation (49–1). When reset is asserted, the two scramblers shall be initialized to non-zero values different from each other.

175.2.4.6 Alignment marker mapping and insertion

In order to support deskew and reordering of the individual PCS lanes at the receive PCS, alignment markers (AMs) corresponding to PCS lanes are periodically inserted after being processed by the alignment marker mapping function. The alignment marker mapping function compensates for the operation of the symbol distribution function and rearranges the alignment marker bits so that they appear on the PCS lanes intact and in the desired sequence. This preserves the properties of the alignment markers (e.g. DC balance, transition density) and provides a deterministic pattern for the purpose of synchronization.

An alignment marker group is composed of the alignment markers for all 16 PCS lanes plus an additional 133-bit pad and a 3-bit status field to yield the equivalent of eight 257-bit blocks. The alignment marker group is divided evenly between the two flows and is aligned to the beginning of four FEC messages across both flows. The alignment marker group interrupts any data transfer that is already in progress. The pad bits at the end of the alignment marker group shall be set to a free running PRBS9 pattern in each flow, defined by the polynomial $x^9 + x^5 + 1$. The initial value of the PRBS9 pattern generators may be any pattern other than all zeros. The 3-bit transmit alignment marker status field (tx_am_sf) carries local and remote FEC degrade status using an end-to-end process described in 116.6 and is appended to the padding in flow 1.

Idle control characters or sequence ordered sets are removed, if necessary, to accommodate the insertion of the alignment markers. The alignment marker group is not scrambled which allows the receiver to directly search for and find the individual alignment markers, deskew the PCS lanes, and reassemble the aggregate stream before FEC decoding is performed. The alignment markers are formed from a known pattern that is defined to be balanced and with many transitions and therefore scrambling is not necessary.

The format of each PCS lane's alignment marker is shown in Figure 175–3. There is a portion that is common across all alignment markers (designated as CM₀ to CM₅), a unique portion per PCS lane (designated as UM₀ to UM₅), and a unique pad per PCS lane (designated as UP₀ to UP₂). Common synchronization logic independent of the received PCS lane number may be used with the common portion of the alignment marker. The unique pad (UP₀ to UP₂) within the alignment markers and the PRBS9 pad at the end of the alignment maker group are ignored on receive.

The format shown in Figure 175–3 defines how the alignment markers appear on a given PCS lane. In the FEC codewords, they appear in a permuted format due to the codeword interleaving that occurs before FEC codewords are distributed to PCS lanes.

The 3-bit transmit alignment marker status field, tx_am_sf, allows the local PCS to communicate the status of the FEC degraded feature to the remote PCS and is assigned values as shown in Table 175–1.

Table 175–1—tx_am_sf field assignments

Bit	Field name	Assignment when the client sublayer is the RS	Assignment when the client sublayer is the PHY 1.6TXS
<2>	remote degraded	FEC_degraded_SER + rx_local_degraded	TXRD
<1>	local degraded	0	TXLD
<0>	<reserved for future use>	0	0

In Table 175–1, the PCS status variable FEC_degraded_SER is defined in 175.2.5.3. The rx_local_degraded variable is defined in 175.2.5.5. TXRD and TXLD are signals from the PHY 1.6TXS on the PCS service interface (see 175.1.4.1), and + denotes logical OR.

175.2.4.6.1 Alignment marker creation

The alignment marker mapping function creates a set of 16 alignment markers, and in combination with an additional 133-bit PRBS9 pad and 3-bit status field, the PCS generates an alignment marker group.

Let $am_x<119:0>$ be the alignment marker for PCS lane x , $x=0$ to 15, where bit 0 is the first bit transmitted.

$am_x<119:0>$ is constructed as shown in Figure 175–3 using the values in Table 175–2 for each PCS lane number x .

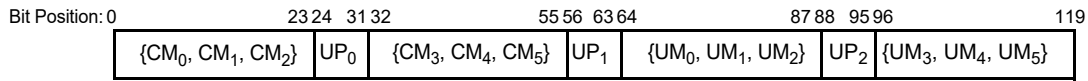


Figure 175–3—Alignment marker format

Table 175–2—1.6TBASE-R alignment marker encodings

PCS lane	Encoding ^a														
	CM0	CM1	CM2	UP0	CM3	CM4	CM5	UP1	UM0	UM1	UM2	UP2	UM3	UM4	UM5
0	0x9A	0x4A	0x26	0xB6	0x65	0xB5	0xD9	0xD9	0xFE	0x8E	0x0C	0x26	0x01	0x71	0xF3
1				0x04				0x67	0xA5	0x21	0x81	0x98	0x5A	0xDE	0x7E
2				0x46				0xFE	0xC1	0x0C	0xA9	0x01	0x3E	0xF3	0x56
3				0x5A				0x84	0x79	0x7F	0x2F	0x7B	0x86	0x80	0xD0
4				0xE1				0x19	0xD5	0xAE	0x0D	0xE6	0x2A	0x51	0xF2
5				0xF2				0x4E	0xED	0xB0	0x2E	0xB1	0x12	0x4F	0xD1
6				0x3D				0xEE	0xBD	0x63	0x5E	0x11	0x42	0x9C	0xA1
7				0x22				0x32	0x29	0x89	0xA4	0xCD	0xD6	0x76	0x5B
8				0x60				0x9F	0x1E	0x8C	0x8A	0x60	0xE1	0x73	0x75
9				0x6B				0xA2	0x8E	0x3B	0xC3	0x5D	0x71	0xC4	0x3C
10				0xFA				0x04	0x6A	0x14	0x27	0xFB	0x95	0xEB	0xD8
11				0x6C				0x71	0xDD	0x99	0xC7	0x8E	0x22	0x66	0x38
12				0x18				0x5B	0x5D	0x09	0x6A	0xA4	0xA2	0xF6	0x95
13				0x14				0xCC	0xCE	0x68	0x3C	0x33	0x31	0x97	0xC3
14				0xD0				0xB1	0x35	0x04	0x59	0x4E	0xCA	0xFB	0xA6
15				0xB4				0x56	0x59	0x45	0x86	0xA9	0xA6	0xBA	0x79

^a Each octet is transmitted LSB to MSB.

As an example, the variable `am_0` is sent as (left most bit sent first, showing the first 32 bits transmitted of `am_0`):

01011001 01010010 01100100 01101101

NOTE—A text file containing the alignment marker values shown in Table 175–2 is available at <https://standards.ieee.org/downloads/802.3>.

Editor's note (to be removed prior to publication): The following file will be placed in the 802.3 downloads folder referenced above when this amendment is published:
https://www.ieee802.org/3/dj/public/24_05/opsasnick_3dj_02_2405.txt

The alignment marker group is divided between the two flows and mapped into two variables, `am_mapped_f0` and `am_mapped_f1`. The variable `am_mapped_f0` is inserted in flow 0 which produces FEC codewords A and B, and `am_mapped_f1` is inserted in flow 1 which produces FEC codewords C and D. The 16 alignment markers, `am_x`, are mapped into these two variables using the following procedure to compensate for the pre-FEC and post-FEC symbol distribution so that the AMs appear on the PCS lanes in the correct sequence:

For all $k=0$ to 2

For all $j=0$ to 15

$\text{am_mapped_f0} \langle (320k+20j+9) : (320k+20j) \rangle = \text{am_}\{j\} \langle (40k+9) : (40k) \rangle$; CW A
 $\text{am_mapped_f0} \langle (320k+20j+19) : (320k+20j+10) \rangle = \text{am_}\{j\} \langle (40k+19) : (40k+10) \rangle$; CW B
 $\text{am_mapped_f1} \langle (320k+20j+9) : (320k+20j) \rangle = \text{am_}\{j\} \langle (40k+29) : (40k+20) \rangle$; CW C
 $\text{am_mapped_f1} \langle (320k+20j+19) : (320k+20j+10) \rangle = \text{am_}\{j\} \langle (40k+39) : (40k+30) \rangle$; CW D

The additional 133-bit pad is then appended to the variables `am_mapped_f0` and `am_mapped_f1`. In the following expressions, `PRBS9_f0` and `PRBS_f1` are outputs of the PRBS pattern generators in flow 0 and flow 1, respectively.

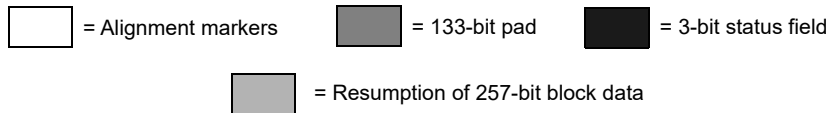
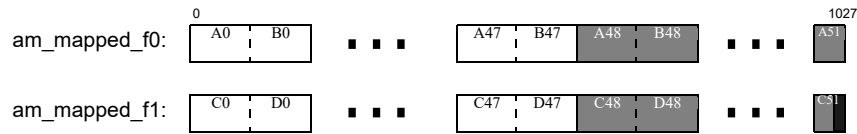
$\text{am_mapped_f0} \langle 1027:960 \rangle = \text{PRBS9_f0} \langle 67:0 \rangle$
 $\text{am_mapped_f1} \langle 1024:960 \rangle = \text{PRBS9_f1} \langle 64:0 \rangle$

The 3-bit transmit alignment marker status field is then appended to the variable `am_mapped_f1` as follows:

$\text{am_mapped_f1} \langle 1027:1025 \rangle = \text{tx_am_sf} \langle 2:0 \rangle$

The alignment marker mappings along with the padding and tx_am_sf fields are shown in Figure 175–4. Note that am_mapped_f0 contains 10-bit symbols of FEC codewords A and B, and am_mapped_f1 contains 10-bit symbols of FEC codewords C and D.

PCS lane	10-bit symbol index															
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	A0	B0	C0	D0	A16	B16	C16	D16	A32	B32	C32	D32	A48	B48	C48	D48
1	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
2	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
3	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
4	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
5	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
6	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
7	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
8	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
9	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
10	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
11	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
12	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
13	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
14	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
15	A15	B15	C15	D15	A31	B31	C31	D31	A47	B47	C47	D47	A	B	C	D



A = from FEC codeword A B = from FEC codeword B C = from FEC codeword C D = from FEC codeword D

Figure 175–4—1.6TBASE-R alignment marker mapping to PCS lanes

175.2.4.6.2 Alignment marker insertion

The alignment marker group shall be inserted so it appears in the output stream every 655360×257 -bit blocks (equivalent to 32768 codewords). The alignment marker group is divided evenly between the two flows such that am_mapped_f0<1027:0> and am_mapped_f1<1027:0> are inserted into flow 0 and flow 1 every 327680×257 -bit blocks.

In each flow, a 10280-bit block of data (equivalent to two codewords) is formed and represented by the variable tx_scrambled_am_f0<10279:0> in flow 0 and tx_scrambled_am_f1<10279:0> in flow 1. These variables are constructed in one of two ways, with or without alignment markers. With AMs, the block in each flow contains one-half of the alignment marker group followed by 36×257 -bit blocks of scrambled data. Without AMs, each block contains 40×257 -bit blocks of scrambled data.

Let the vector $\text{tx_scrambled_f0}_{i<256:0>}$ represent consecutive values of $\text{tx_scrambled_f0}_{<256:0>}$, and let $\text{tx_scrambled_f1}_{i<256:0>}$ represent consecutive values of $\text{tx_scrambled_f1}_{<256:0>}$.

For the 10280-bit blocks with an alignment marker group inserted:

$$\text{tx_scrambled_am_f0}_{<1027:0>} = \text{am_mapped_f0}_{<1027:0>}$$

$$\text{tx_scrambled_am_f1}_{<1027:0>} = \text{am_mapped_f1}_{<1027:0>}$$

For all $i=0$ to 35

$$\text{tx_scrambled_am_f0}_{<(257i+1284):(257i+1028)>} = \text{tx_scrambled_f0}_{i<256:0>}$$

$$\text{tx_scrambled_am_f1}_{<(257i+1284):(257i+1028)>} = \text{tx_scrambled_f1}_{i<256:0>}$$

For the 10280-bit blocks without an alignment marker group:

For all $i=0$ to 39

$$\text{tx_scrambled_am_f0}_{<(257i+256):(257i)>} = \text{tx_scrambled_f0}_{i<256:0>}$$

$$\text{tx_scrambled_am_f1}_{<(257i+256):(257i)>} = \text{tx_scrambled_f1}_{i<256:0>}$$

Alignment marker repetition rate within each flow is shown in Figure 175–5.

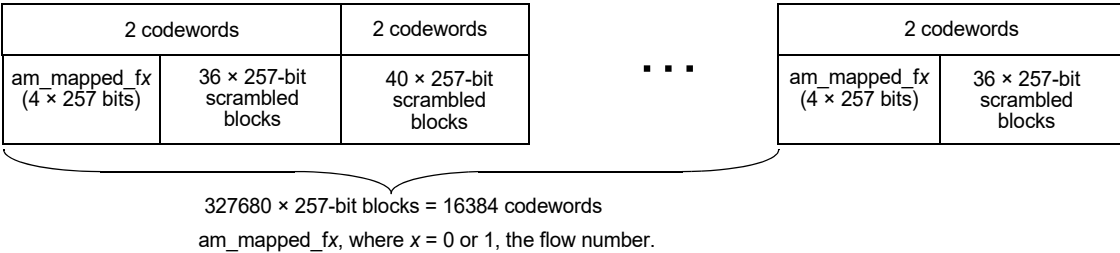


Figure 175–5—1.6TBASE-R alignment marker insertion period (per-flow)

The alignment marker insertion within each flow shall occur at the same point relative to the original stream of 257-bit blocks before block distribution. This requirement is illustrated in Figure 175–6 where the alignment marker group is inserted after block $k-1$ and prior block k .

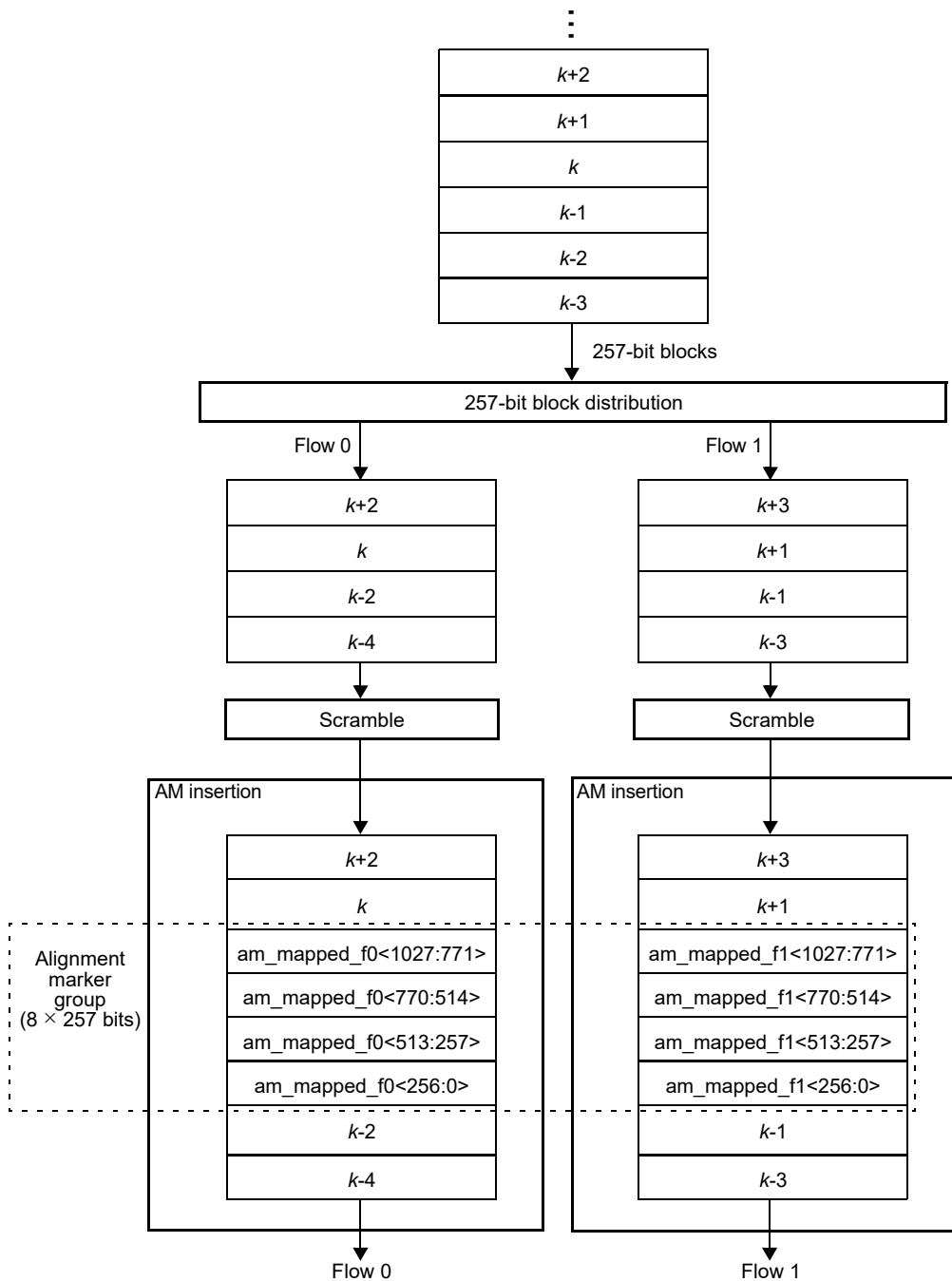


Figure 175–6—1.6TBASE-R PCS alignment marker insertion

175.2.4.7 Pre-FEC distribution

In order to improve error correction capability, four Reed-Solomon FEC codewords are interleaved when forming the PCS lanes. To enable this interleaving, a pre-FEC distribution function occurs in each flow. The pre-FEC distribution function receives a 10280-bit block and performs a 10-bit symbol round robin distribution to form two FEC messages. In flow 0, m_A and m_B are formed from $tx_scrambled_am_f0$. In flow 1, m_C and m_D are formed from $tx_scrambled_am_f1$. Each resulting FEC message consists of 514 10-bit symbols. The following procedure describes the 10-bit round robin distribution process.

For all $i=0$ to 513

$$\begin{aligned} m_A\langle(513-i)\rangle\langle 9:0\rangle &= tx_scrambled_am_f0\langle(20i+9) : (20i)\rangle \\ m_B\langle(513-i)\rangle\langle 9:0\rangle &= tx_scrambled_am_f0\langle(20i+19) : (20i+10)\rangle \\ m_C\langle(513-i)\rangle\langle 9:0\rangle &= tx_scrambled_am_f1\langle(20i+9) : (20i)\rangle \\ m_D\langle(513-i)\rangle\langle 9:0\rangle &= tx_scrambled_am_f1\langle(20i+19) : (20i+10)\rangle \end{aligned}$$

Where $m_X\langle 513\rangle$ is the first transmitted symbol and $m_X\langle 0\rangle$ is the last transmitted symbol of each message, and $bit\langle 0\rangle$ is the first transmitted bit and $bit\langle 9\rangle$ is the last transmitted bit of each symbol.

175.2.4.8 Reed-Solomon encoder

The Reed-Solomon encoder in each flow is identical to that specified in 119.2.4.6. In flow 0, two 5140-bit message blocks, m_A and m_B , are encoded using the RS(544,514) encoder into codeword A (c_A) and codeword B (c_B), respectively. In flow 1, two 5140-bit message blocks, m_C and m_D , are encoded using the RS(544,514) encoder into codeword C (c_C) and codeword D (c_D), respectively.

Examples of RS-FEC codewords are provided in Annex 175A.

175.2.4.9 Symbol distribution

After RS-FEC encoding, the four codewords across both flows are interleaved on a 10-bit symbol basis to form 16 PCS lanes as shown in Figure 175–7. Each PCS lane alternates 10-bit FEC symbols from the four codewords in the repeating pattern, {A, B, C, D, A, B, C, D, ...}.

Let $pcsl_i\langle 1359:0\rangle$ represent the 1360 bits transmitted on PCS lane i for each set of four codewords where $bit\langle 0\rangle$ is transmitted first on each PCS lane.

The interleaving of the four codewords follows this procedure:

For all $j=0$ to 33

For all $i=0$ to 15

$$\begin{aligned} pcsl_i\langle(40j+9) : (40j)\rangle &= c_A\langle 543-16j-i\rangle\langle 9:0\rangle \\ pcsl_i\langle(40j+19) : (40j+10)\rangle &= c_B\langle 543-16j-i\rangle\langle 9:0\rangle \\ pcsl_i\langle(40j+29) : (40j+20)\rangle &= c_C\langle 543-16j-i\rangle\langle 9:0\rangle \\ pcsl_i\langle(40j+39) : (40j+30)\rangle &= c_D\langle 543-16j-i\rangle\langle 9:0\rangle \end{aligned}$$

175.2.4.10 Transmit data

The 16 PCS lanes formed from the symbol distribution are transmitted on the service interface below the PCS. PCS lane i shall be transmitted on $inst:IS_UNITDATA_i.request$, for $i=0$ to 15.

The transmit data distribution and bit ordering are illustrated in Figure 175–7.

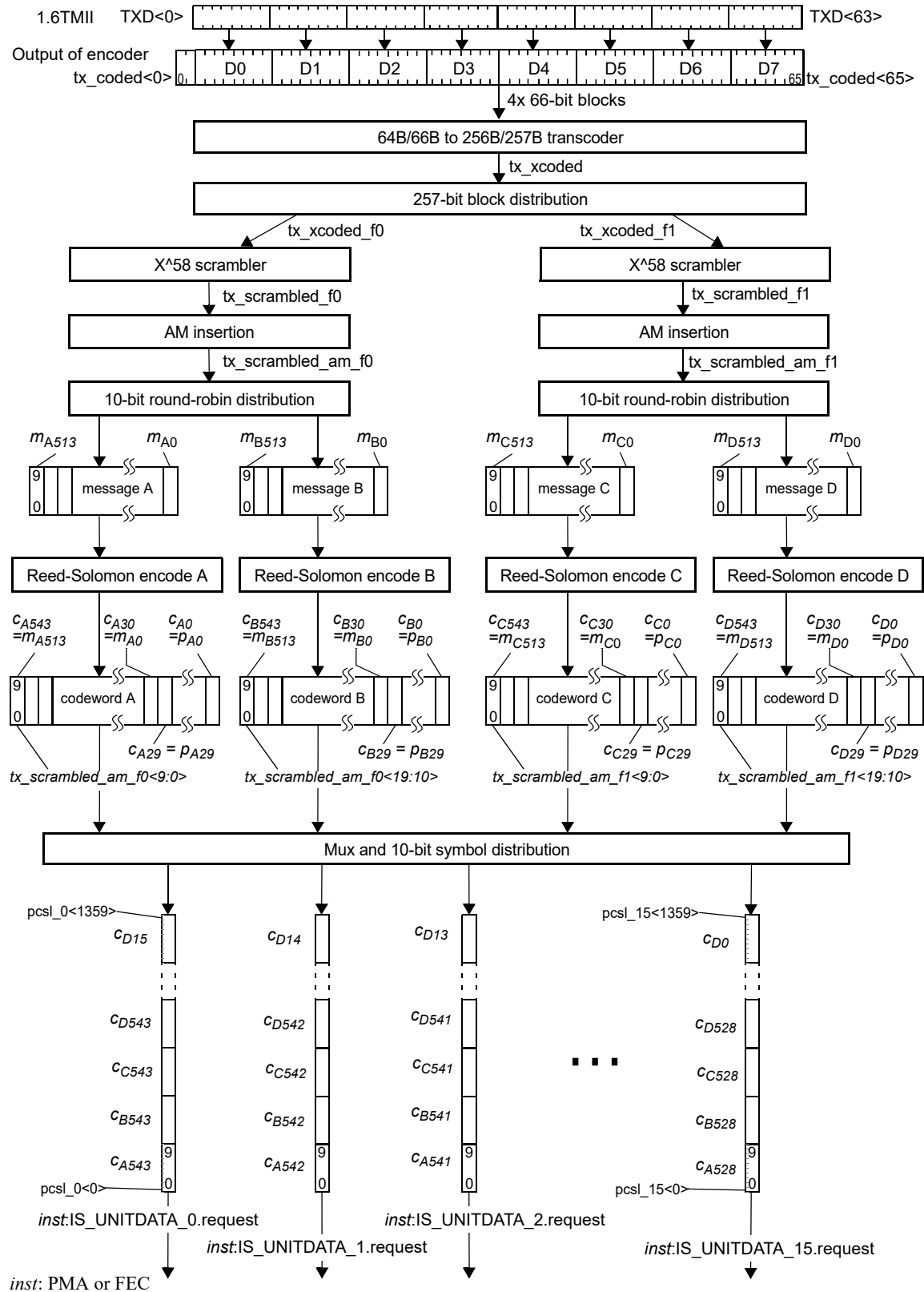


Figure 175-7—1.6TBASE-R PCS transmit data distribution and bit ordering

175.2.4.11 Test-pattern generator

The PCS shall have the ability to generate a scrambled idle test pattern which is suitable for receiver tests and for certain transmitter tests. When test pattern generation is enabled by the PCS control variable `tx_test_mode` (set to 1), the scrambled idle test pattern is generated by the PCS. The scrambled idle test pattern is the output of the PCS when the input to the PCS at the 1.6TmII is composed only of idle control characters.

175.2.5 Receive function

175.2.5.1 Alignment lock and deskew

The alignment lock and deskew is identical to that specified in 119.2.5.1 with $n = 16$ and the following exceptions:

- Figure 119–13 is replaced with Figure 175–8 and Figure 175–9.
- The maximum Skew is changed from 180 ns to 152 ns.

175.2.5.2 Lane reorder and deinterleave

The PCS lanes might be received in any order on the lanes of the service interface below the PCS. The PCS receive function shall order the PCS lanes according to the PCS lane number. The PCS lane number is defined by the unique portion (UM_0 to UM_5) of the alignment marker that is mapped to each PCS lane (see 175.2.4.6).

After all PCS lanes are aligned, deskewed, and reordered, they are deinterleaved to reconstruct the original four streams of FEC codewords.

175.2.5.3 Reed-Solomon decoder

The Reed-Solomon decoder is identical to that specified in 119.2.5.3, with the following exceptions:

- When `FEC_bypass_indication_enable` is asserted, the associated additional error monitoring counts the number of symbol errors detected in consecutive non-overlapping blocks of 8192 codewords across all four FEC decoders to determine whether to set `hi_ser`.
- When `FEC_degraded_SER_enable` is asserted, the associated additional error monitoring counts the number of symbol errors detected on all PCS lanes, where the least significant two bits of `FEC_degraded_SER_interval` are ignored (evaluated as ‘00’) to make the number of codewords a multiple of four.
- References to PCS management objects in 119.3 refer instead to 175.8.

The following counters shall be implemented:

FEC_corrected_cw_counter

A 32-bit counter that counts once for each corrected FEC codeword processed when `align_status` is true as set by the PCS synchronization process (see 175.2.6.3). A corrected FEC codeword is one that contains errors and was corrected by the Reed-Solomon decoder. Note that this counter counts all corrected codewords across all four FEC decoders.

FEC_uncorrected_cw_counter

A 32-bit counter that counts once for each uncorrected FEC codeword processed when `align_status` is true. An uncorrected FEC codeword is one that contains errors that were not corrected by the Reed-Solomon decoder. Note that this counter counts all uncorrected codewords across all four FEC decoders.

FEC_symbol_error_counter_{*i*}

A set of sixteen 32-bit counters where counter *i* counts once for each 10-bit symbol corrected on PCS lane *i* when align_status is true (*i*=0 to 15). Note that symbols from each PCS lane are distributed across all four FEC codeword decoders.

FEC_cw_counter

A 48-bit counter that counts all FEC codewords received by the PCS when align_status is true.

FEC_codeword_error_bin_{*i*}

A set of fifteen 32-bit counters where counter *i* counts once for each codeword received with exactly *i* correctable 10-bit symbols when align_status is true (*i*=1 to 15). For example, if a codeword has exactly 5 errored 10-bit symbols, then FEC_codeword_error_bin_5 is incremented. Note that these counters count all codewords with correctable errors across all four FEC decoders.

175.2.5.4 Post-FEC interleave

In each flow, after the Reed-Solomon decoders process the codewords, the two resulting message blocks are interleaved on a 10-bit basis to form rx_scrambled_am, corresponding to 40 transcoded blocks, in order to recreate the transmitted data stream.

175.2.5.5 Alignment marker removal

In each flow, every 327680 × 257-bit blocks (corresponding to 16 384 codewords), the first 1028 bits of an rx_scrambled_am block is the vector am_rx<1027:0>, where bit 0 is the first bit received. The specific codewords that include this vector are indicated by the alignment lock and deskew function. The 3-bit receive alignment marker status field (rx_am_sf) is assigned from the variable am_rx in flow 1 as follows:

rx_am_sf<2:0> = am_rx<1027:1025> (only in flow 1)

rx_am_sf<2:0> is used to generate the following PCS status variables:

rx_local_degraded

Boolean variable that is set to true when the receiver detects rx_am_sf<1> asserted for two consecutive alignment marker periods. It is deasserted when rx_am_sf<1> is deasserted for two consecutive alignment marker periods.

rx_rm_degraded

Boolean variable that is set to true when the receiver detects rx_am_sf<2> asserted for two consecutive alignment marker periods. It is deasserted when rx_am_sf<2> is deasserted for two consecutive alignment marker periods.

The vector am_rx is removed from rx_scrambled_am to create rx_scrambled prior to descrambling.

175.2.5.6 Descrambler

The descrambler shall process rx_scrambled<256:0> to reverse the effect of the scrambler using the polynomial given in Equation (49–1). It shall produce the same result as the implementation shown in Figure 49–10.

175.2.5.7 Block collection

The block collection reverses the block distribution performed in the transmitter (see 175.2.4.4) by combining the 257-bit blocks from the two flows in an alternating fashion to form a single stream of 257-bit blocks. The first 257-bit block after the alignment marker group from flow 0 shall be followed by the first 257-bit block after the alignment marker group from flow 1.

175.2.5.8 256B/257B to 64B/66B transcoder

The 256B/257B to 64B/66B transcoder in each flow is identical to that specified in 119.2.5.7.

175.2.5.9 64B/66B decoder

The receive PCS decodes 66-bit blocks to produce RXD<63:0> and RXC<7:0> for transmission to the 1.6TMII. One 1.6TMII transfer is decoded from each 66-bit block. The receive PCS may use either the state-diagram decoder defined by Figure 119–15 or the stateless decoder defined in 172.2.5.9.2.

175.2.5.10 Rate matching

The receive PCS may insert idle control characters to compensate for the deletion of alignment markers. The receive PCS may remove idle control characters or sequence ordered sets or may insert idle control characters to compensate for different clock domains on the service interface below the PCS and the 1.6TMII. See 82.2.3.6 and 82.2.3.9 for the deletion and insertion rules.

NOTE—If no idle control characters are inserted to compensate for the removal of alignment markers, the average data rate on the 1.6TMII will be reduced by approximately 12 ppm.

175.2.6 State diagrams

This subclause is composed of state diagrams, including the associated definitions of constants, variables, functions, and counters.

175.2.6.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

175.2.6.2 State diagram definitions

1.6TBASE-R PCS uses the following state diagrams defined in Clause 119:

- Figure 119–12, “Alignment marker lock state diagram”
- Figure 119–14, “Transmit state diagram”
- Figure 119–15, “Receive state diagram”

Any constants, variables, functions, or counters used in the above Clause 119 state diagrams which are not defined in the following subclauses shall use the values and definitions from 119.2.6.2.

175.2.6.2.1 Constants

The constants are the same as those specified in 119.2.6.2.1 with the following modification:

- 200GMII/400GMII is replaced with 1.6TMII.

175.2.6.2.2 Variables

align_status

A variable set by the PCS synchronization process to reflect the status of PCS lane-to-lane alignment. It is set to true when all lanes are synchronized and aligned and set to false when the PCS synchronization process is not complete.

all_locked	1
Boolean variable that is set to true when amps_lock<x> is true for all x and is set to false when amps_lock<x> is false for any x.	2
	3
amps_lock<x>	4
Boolean variable that is set to true when the receiver has detected the location of the alignment marker payload sequence for a given lane on the service interface below the PCS, where x=0 to 15 for 1.6TBASE-R.	5
	6
	7
	8
current_pcsI	9
Same behavior as 119.2.6.2.2 with the exception that the unique marker values are defined in Table 175–2.	10
	11
	12
cw_bad<z>	13
Boolean variable that is set to true if the Reed-Solomon decoder fails to correct the current FEC codeword z and is set to false otherwise. z = {A, B, C, or D}.	14
	15
deskew_done	16
Boolean variable that is set to true when pcs_enable_deskew is set to true and the deskew process is completed. Otherwise, this variable is set to false.	17
	18
	19
hi_ser	20
When FEC_bypass_indication_enable is set to one, this bit is set to one for a period of 60 ms to 75 ms if the number of RS-FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 175.2.5.3), and is set to zero otherwise. When FEC_bypass_indication_enable is set to zero, this bit is set to zero.	21
	22
	23
	24
pcs_alignment_valid	25
Boolean variable that is set to true if all PCS lanes are aligned. PCS lanes are considered to be aligned when amps_lock<x> is true for all x, each PCS lane is locked to a unique alignment marker payload sequence, and the PCS lanes are deskewed. Otherwise, this variable is set to false.	26
	27
	28
	29
pcs_enable_deskew	30
Boolean variable that enables and disables the PCS synchronization process. Received bits may be discarded whenever deskew is enabled. It is set to true when deskew is enabled and set to false when deskew is disabled.	31
	32
	33
pcs_lane_mapping<x>	34
A variable that holds the value of the PCS lane number received on the service interface below the PCS, inst:IS_UNITDATA_x.indication.	35
	36
	37
PCS_reset	38
Boolean variable that is true when set by a management entity and is false otherwise.	39
	40
PCS_status	41
Boolean variable that is true when align_status is true and is false otherwise.	42
	43
reset	44
Boolean variable that controls the resetting of the PCS sublayer. It is true whenever a reset is necessary including when PCS_reset is true and during power on.	45
	46
restart_lock	47
Boolean variable that is set by the PCS synchronization process to restart the alignment marker lock process on all PCS lanes. It is set to true in the DESKEW_FAIL state or if any restart_lock<z> bits are true. It is set to false upon entry into the LOSS_OF_ALIGNMENT state.	48
	49
	50
restart_lock<z>	51
Boolean variable that is set by the codeword monitor process to restart the alignment marker lock process on all PCS lanes. It is set to true after three consecutive uncorrected codewords are	52
	53
	54

received for FEC codeword z (in 3_BAD state) and set to false upon entry into the CW_WAIT state. $z = \{A, B, C, \text{ or } D\}$.

test_cw< z >

Boolean variable that is set to true when a new FEC codeword z is available for decoding and is set to false when the TEST_CW state is entered. $z = \{A, B, C, \text{ or } D\}$.

175.2.6.2.3 Functions

The functions are the same as those specified in 119.2.6.2.3 with the following modifications:

- 200GMII/400GMII is replaced with 1.6TMII.
- EEE and low power idle are not supported.

175.2.6.2.4 Counters

amp_counter

Counts the interval of 32768 FEC codewords containing alignment marker payload sequences for the 1.6TBASE-R PCS.

cw_bad_count< z >

Counts the number of consecutive uncorrected FEC codewords by FEC decoder z . This counter is set to zero when a FEC codeword z is received and cw_bad< z > is false. $z = \{A, B, C, \text{ or } D\}$.

175.2.6.3 State diagram figures

The 1.6TBASE-R PCS shall implement sixteen alignment marker lock processes as depicted in Figure 119–12 with these exceptions:

- The alignment marker values are used as specified in Table 175–2.
- amp_counter counts the interval of 32768 FEC codewords.

An alignment marker lock process operates independently on each input service lane. The alignment marker lock state diagram shown in Figure 119–12 determines when the PCS has obtained alignment marker lock to the received bit stream for a given lane of the service interface. Each alignment marker lock process looks for two valid alignment markers 1114112×10 -bit Reed-Solomon symbols apart to gain alignment marker lock. When the alignment marker lock process achieves lock for a lane, the variable amps_lock< x > is set to true ($x = 0$ to 15), and the PCS lane number received on that lane of the service interface is stored in the PCS status variable pcs_lane_mapping< x >. Once in lock, a lane goes out of alignment marker lock when restart_lock is signaled by the PCS synchronization process (when three uncorrectable codewords in a row are seen) or when the alignment marker lock process sees five alignment markers in a row that fail to match the expected pattern on a given lane.

The PCS shall run the synchronization process as depicted in Figure 175–8 and four independent codeword monitor processes as shown in Figure 175–9. The PCS synchronization process is responsible for determining if the PCS is capable of presenting coherent data to the 1.6TMII. The synchronization process ensures that all PCS lanes have alignment marker lock, are locked to different alignment markers, and that the skew is within the PCS deskew limit. When these conditions are met, the align_status variable is set to true. The synchronization process, along with the 16 alignment marker lock processes, are restarted if any of the four FEC decoder blocks process three consecutive uncorrectable codewords as indicated by the PCS codeword monitor processes.

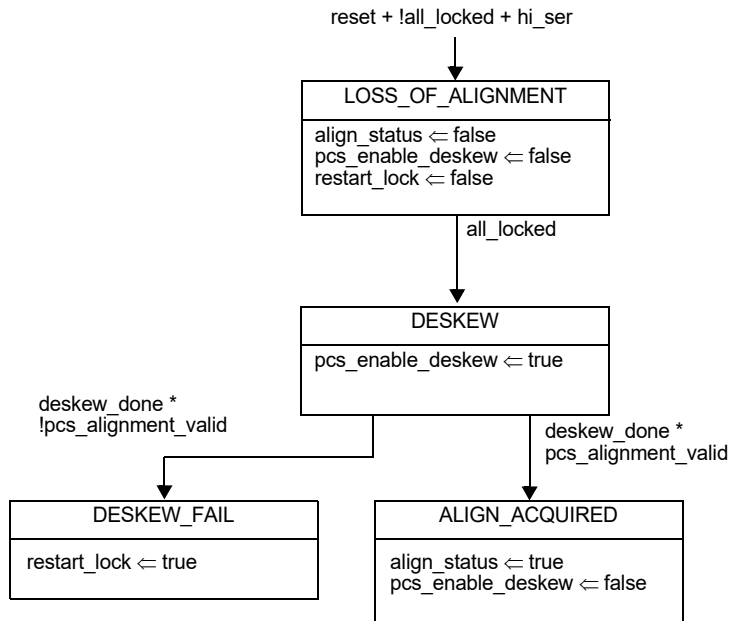


Figure 175–8—PCS synchronization state diagram

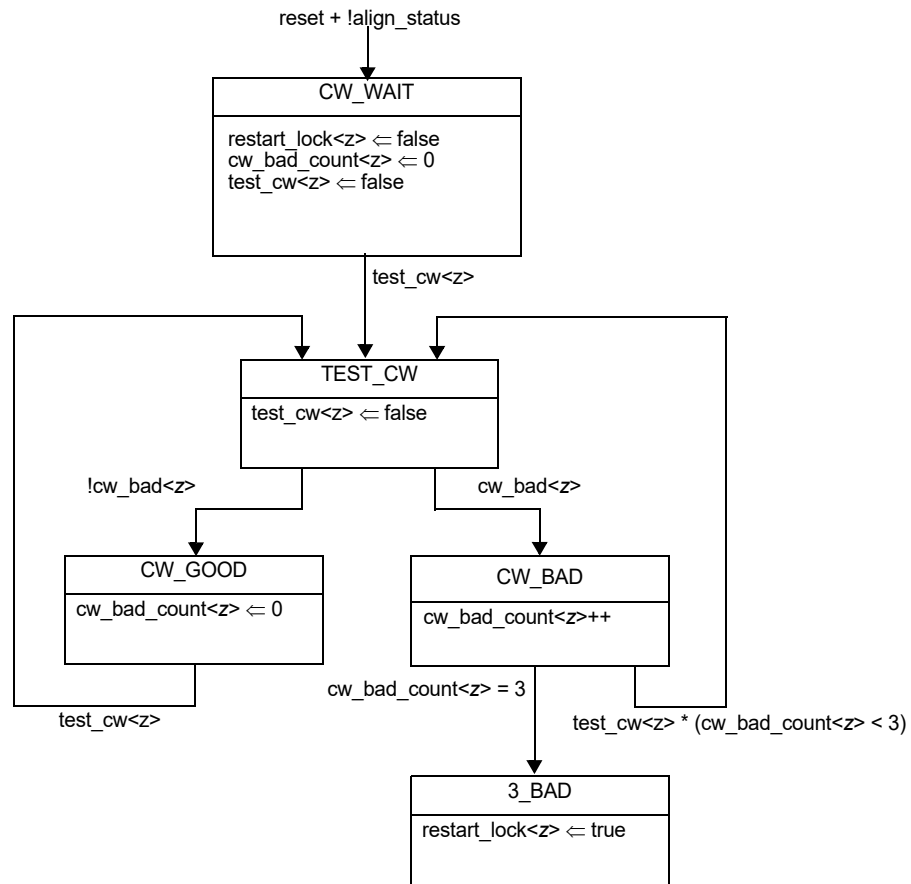


Figure 175-9—PCS codeword monitor state diagram

175.3 FEC Degrad

The FEC degrade feature provides the ability to detect degrade conditions at the RS-FEC decoder using FEC degrade detection and to propagate all detected FEC degrade conditions using FEC degrade signaling. The propagation of FEC degrade signals across PCS and XS sublayers is described in 116.6.

FEC degrade detection is specified in 175.2.5.3. FEC degrade detection is optional.

FEC degrade signaling using the PCS alignment marker status field is specified for the PCS in 175.2.4.6. FEC degrade signaling across the PCS service interface is specified in 175.1.4.1. FEC degrade signaling is required.

175.4 Loopback

Loopback is identical to that specified in 119.4.

The loopback function is enabled when the PCS control variable loopback is set to 1.

175.5 Delay constraints

The maximum delay contributed by the 1.6TBASE-R PCS (sum of transmit and receive delays at one end of the link) shall be no more than 640000 bit times, equivalent to 1250 pause_quanta or 400 ns. A description of overall system delay constraints and the definition of bit times and pause_quanta can be found in 174.4 and its references.

175.6 Path data delay for time synchronization

The PCS sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report PCS path data delays is indicated by the PCS_delay_ns_TX_ability, PCS_delay_subns_TX_ability, PCS_delay_ns_RX_ability, and PCS_delay_subns_RX_ability variables (see Table 175–4).

The transmit and receive path data delays are reported as if the DDMP (data delay measurement point) is at the start of the set of four interleaved RS-FEC codewords (see 90.7), and the PCS_timesync_multilane_ability variable is asserted.

Four separate delays are reported in the following eight path data delay status variables:

- PCS_delay_ns_TX_max and PCS_delay_subns_TX_max
- PCS_delay_ns_TX_min and PCS_delay_subns_TX_min
- PCS_delay_ns_RX_max and PCS_delay_subns_RX_max
- PCS_delay_ns_RX_min and PCS_delay_subns_RX_min

175.7 Auto-Negotiation

The following requirements apply to a PCS used with a 1.6TBASE-CR8 or 1.6TBASE-KR8 PMD where support for the auto-negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN_LINK.indication(link_status) primitive (see 73.9). The parameter link_status shall take the value FAIL when PCS_status=false and the value OK when PCS_status=true. The primitive shall be generated when the value of link_status changes.

The PCS status variable PCS_status is defined in 175.2.6.2.2.

175.8 Management variables

PCS control and status variables intended to be accessible via a management system are listed in Table 175–3 and Table 175–4.

If the MDIO Interface (see Clause 45) is implemented, PCS control and status variables shall be mapped to MDIO registers as shown in Table 175–3 and Table 175–4.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 175–3—PCS control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PCS_reset	175.2.6.2.2	3.0.15	45.2.3.1
loopback	175.4	3.0.14	45.2.3.1
tx_test_mode	175.2.4.11	3.42.3	45.2.3.19
FEC_bypass_indication_enable	175.2.5.3	3.800.1	45.2.3.60
FEC_degraded_SER_enable	175.2.5.3	3.800.2	45.2.3.60
FEC_degraded_SER_activate_threshold	175.2.5.3	3.806, 3.807	45.2.3.64
FEC_degraded_SER_deactivate_threshold	175.2.5.3	3.808, 3.809	45.2.3.65
FEC_degraded_SER_interval	175.2.5.3	3.810, 3.811	45.2.3.66

Table 175–4—PCS status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PCS_status	175.2.6.2.2	3.32.12	45.2.3.15.1
align_status	175.2.6.2.2 Figure 175–8	3.50.12	45.2.3.23.1
amps_lock<15:0>	175.2.6.2.2 Figure 119–12	3.53.7:0, 3.52.7:0	45.2.3.26, 45.2.3.25
FEC_cw_counter	175.2.5.3	3.300 to 3.302	45.2.3.48a
FEC_codeword_error_bin_<1:15>	175.2.5.3	3.340 to 3.369	45.2.3.48b
pcs_lane_mapping<0:15>	175.2.6.2.2 Figure 119–12	3.400 to 3.415	45.2.3.49
FEC_symbol_error_counter_<0:15>	175.2.5.3	3.600 to 3.631	45.2.3.58
FEC_bypass_indication_ability	119.2.5.3	3.801.1	45.2.3.61.6
hi_ser	175.2.5.3 175.2.6.2.2	3.801.2	45.2.3.61.5
FEC_degraded_SER_ability	119.2.5.3	3.801.3	45.2.3.61.4
FEC_degraded_SER	175.2.5.3	3.801.4	45.2.3.61.3
rx_rm_degraded	175.2.5.5	3.801.5	45.2.3.61.2
rx_local_degraded	175.2.5.5	3.801.6	45.2.3.61.1
FEC_corrected_cw_counter	175.2.5.3	3.802, 3.803	45.2.3.62
FEC_uncorrected_cw_counter	175.2.5.3	3.804, 3.805	45.2.3.63
PCS_delay_ns_TX_max	175.6	3.1801, 3.1802	45.2.3.68
PCS_delay_subns_TX_max	175.6	3.1809	45.2.3.68

Table 175–4—PCS status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PCS_delay_ns_TX_min	175.6	3.1803, 3.1804	45.2.3.68
PCS_delay_subns_TX_min	175.6	3.1810	45.2.3.68
PCS_delay_ns_RX_max	175.6	3.1805, 3.1806	45.2.3.69
PCS_delay_subns_RX_max	175.6	3.1811	45.2.3.69
PCS_delay_ns_RX_min	175.6	3.1807, 3.1808	45.2.3.69
PCS_delay_subns_RX_min	175.6	3.1812	45.2.3.69
PCS_timesync_multilane_ability	175.6	3.1800.11	45.2.3.67
PCS_delay_ns_RX_ability	175.6	3.1800.0	45.2.3.67
PCS_delay_ns_TX_ability	175.6	3.1800.1	45.2.3.67
PCS_delay_subns_RX_ability	175.6	3.1800.2	45.2.3.67
PCS_delay_subns_TX_ability	175.6	3.1800.3	45.2.3.67

175.9 Protocol implementation conformance statement (PICS) proforma for Clause 175, Physical Coding Sublayer (PCS), type 1.6TBASE-R⁵

175.9.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 175, Physical Coding Sublayer (PCS), type 1.6TBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

175.9.2 Identification

175.9.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

175.9.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 175, Physical Coding Sublayer (PCS), type 1.6TBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

⁵*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

175.9.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
MII	1.6TMII logical interface is supported	175.1.4.1		O	Yes [] No []
*MD	MDIO	175.8		O	Yes [] No []
*FDD	Support for FEC degrade detection	175.2.5.3		O	Yes [] No []

175.9.4 PICS proforma tables for Physical Coding Sublayer (PCS), type 1.6TBASE-R

175.9.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
*SE	Uses stateless encoder	175.2.4.1		O	Yes [] No []
TF1	Transmit 64B/66B encoder uses state diagram in Figure 119–14.	175.2.4.1		!SE:M	Yes [] N/A []
TF2	Transmit 64B/66B encoder uses stateless rules in 172.2.4.1.2.	175.2.4.1		SE:M	Yes [] N/A []
TF3	64B/66B to 256B/257B transcoder	175.2.4.3		M	Yes []
TF4	Scrambler	175.2.4.5		M	Yes []
TF5	Pad value	175.2.4.6		M	Yes []
TF6	Alignment marker insertion	175.2.4.6		M	Yes []
TF7	FEC degrade signaling	175.2.4.6		M	Yes []
TF8	Pre-FEC distribution	175.2.4.7		M	Yes []
TF9	Reed-Solomon encoder	175.2.4.8		M	Yes []
TF10	Symbol distribution	175.2.4.9		M	Yes []
TF11	Transmission bit ordering	175.2.4.10		M	Yes []
TF12	Scrambled idle transmit test-pattern generator	175.2.4.11		M	Yes []

175.9.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
*BI	Bypass indication supported	175.2.5.3		O	Yes [] No []
*SD	Uses stateless decoder	175.2.5.9		O	Yes [] No []
RF1	Alignment marker lock	175.2.5.1		M	Yes []
RF2	PCS lane number is captured to an MDIO register	175.2.6.3		MD:M	Yes [] N/A []
RF3	Deskew	175.2.5.1		M	Yes []
RF4	Lane reorder and de-interleave	175.2.5.2		M	Yes []
RF5	Reed-Solomon decoder corrects any combination of up to $t=15$ symbol errors in a codeword	175.2.5.3		M	Yes []
RF6	Reed-Solomon decoder is capable of indicating when a codeword was not corrected.	175.2.5.3		M	Yes []
RF7	Error monitoring and assertion of hi_ser, while error indication is bypassed	175.2.5.3		BI:M	Yes [] N/A []
RF8	Detect FEC degrade	175.2.5.3		FDD:M	Yes [] N/A []
RF9	FEC degrade signaling on PCS service interface	175.1.4.1		M	Yes []
RF10	Alignment marker removal	175.2.5.5		M	Yes []
RF11	Descrambler	175.2.5.6		M	Yes []
RF12	256B/257B to 64B/66B transcoder	175.2.5.8		M	Yes []
RF13	Receive 64B/66B decoder uses state diagram in Figure 119–15.	175.2.5.9		!SD:M	Yes [] N/A []
RF14	Receive 64B/66B decoder uses stateless rules in 172.2.5.9.2.	175.2.5.9		SD:M	Yes [] N/A []

175.9.4.3 64B/66B coding rules

Item	Feature	Subclause	Value/Comment	Status	Support
C1	Encoder (and ENCODE function) implements the code as specified	175.2.3, 175.2.6.2.3		M	Yes []
C2	Decoder (and DECODE function) implements the code as specified	175.2.3, 175.2.6.2.3		M	Yes []
C3	Only valid block types are transmitted	175.2.3		M	Yes []
C4	Invalid block types are treated as an error	175.2.3		M	Yes []
C5	Only valid control characters are transmitted	175.2.3		M	Yes []
C6	Invalid control characters are treated as an error	175.2.3		M	Yes []
C7	Idles do not interrupt data	175.2.3		M	Yes []
C8	IDLE control code insertion and deletion	175.2.3	Insertion or deletion in groups of 8 /I/s	M	Yes []
C9	Sequence ordered set deletion	175.2.3	Only one whole ordered set of two consecutive sequence ordered sets may be deleted	M	Yes []

175.9.4.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to PCS management objects is provided	175.8		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	175.8		MD:M	Yes [] N/A []

175.9.4.5 Loopback

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Supports loopback	175.4		M	Yes []
L2	When in loopback, transmits what it receives from the 1.6TMII	175.4		M	Yes []
L3	When in loopback, ignore all data presented by the sublayer below the PCS.	175.4		M	Yes []

175.9.4.6 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1	PCS delay constraint	175.5		M	Yes []

175.9.4.7 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	175.6		O	Yes [] No []
TS1	Path data delays	175.6	Path data delays reported as if DDMP is at start of a set of four interleaved RS-FEC codewords.	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	175.6	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	175.6	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

175.9.4.8 Auto-Negotiation

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Support for use with a 1.6TBASE-CR8 or 1.6TBASE-KR8 PMD	175.7	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN1	AN_LINK.indication primitive	175.7	Support of the AN_LINK.indication(link_status) primitive	AN:M	Yes [] N/A []
AN2	link_status parameter	175.7	Takes the value OK or FAIL.	AN:M	Yes [] N/A []
AN3	Generation of AN_LINK.indication primitive	175.7	Generated when the value of link_status changes	AN:M	Yes [] N/A []

176. Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R

176.1 Overview

176.1.1 Scope

This clause specifies the Physical Medium Attachment (PMA) sublayer that uses RS-FEC symbol multiplexing for 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Physical Layer implementations.

When necessary to differentiate the symbol-multiplexing PMA (SM-PMA) types defined in this clause from the bit-multiplexing PMA (BM-PMA) types defined in Clause 120 and Clause 173, the term SM-PMA is used. Within this clause, the term PMA refers specifically to the SM-PMA.

Three types of the 200GBASE-R SM-PMA are defined in this clause: 8:1 PMA, 1:8 PMA, and 1:1 PMA.

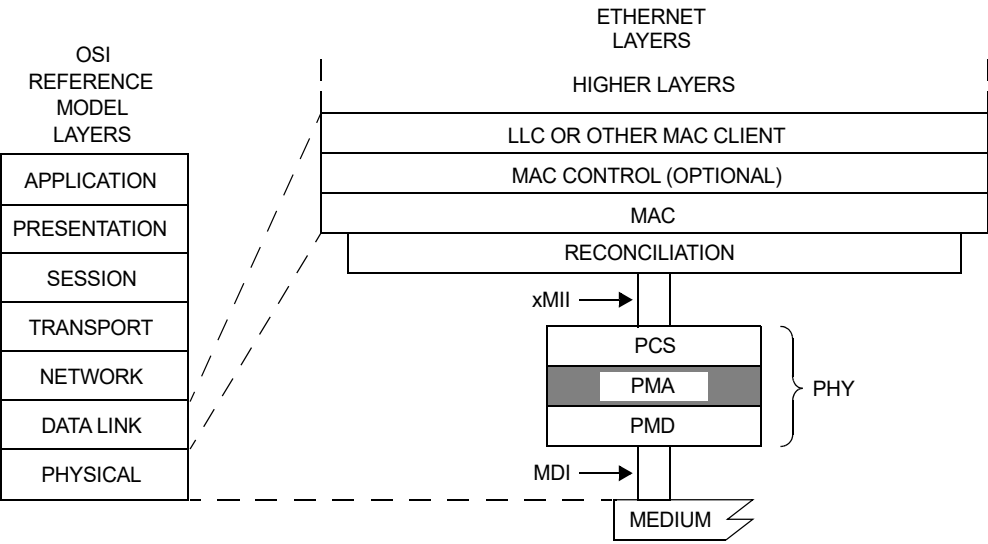
Three types of the 400GBASE-R SM-PMA are defined in this clause: 16:2 PMA, 2:16 PMA, and 2:2 PMA.

Three types of the 800GBASE-R SM-PMA are defined in this clause: 32:4 PMA, 4:32 PMA, and 4:4 PMA.

Four types of the 1.6TBASE-R SM-PMA are defined in this clause: 16:8 PMA, 8:16 PMA, 8:8 PMA, and 16:16 PMA.

176.1.2 Relationship of the PMA to the ISO OSI reference model

Figure 176–1 shows the relationship of the PMA sublayer, along with other Ethernet sublayers, to the ISO Open System Interconnection (OSI) reference model.



xMII = GENERIC MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 176–1—200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R SM-PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

176.1.3 Conventions

The following terms are defined for use within this clause as follows:

BM-PMA

This term refers to a 200GBASE-R, 400GBASE-R, or 800GBASE-R PMA which provides multiplexing by bits (rather than RS-FEC symbols) of PCS lanes as defined in Clause 120 for 200GBASE-R and 400GBASE-R, and in Clause 173 for 800GBASE-R.

m:n PMA

This term refers collectively to the 200GBASE-R 8:1 PMA, 400GBASE-R 16:2 PMA, 800GBASE-R 32:4 PMA, and 1.6TBASE-R 16:8 PMA.

n:m PMA

This term refers collectively to the 200GBASE-R 1:8 PMA, 400GBASE-R 2:16 PMA, 800GBASE-R 4:32 PMA, and 1.6TBASE-R 8:16 PMA.

n:n PMA

This term refers collectively to the 200GBASE-R 1:1 PMA, 400GBASE-R 2:2 PMA, 800GBASE-R 4:4 PMA, 1.6TBASE-R 8:8 PMA, and 1.6TBASE-R 16:16 PMA.

PMAL

This term refers to a PMA lane operating at 212.5 Gb/s.

SM-PMA

This term refers to a 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PMA which provides multiplexing by RS-FEC symbols (rather than bits) of PCS lanes as defined in this clause.

symbol-pair

A symbol-pair is defined as two adjacent RS-FEC symbols (for example, on a PCS lane) where the first symbol in the pair is from RS-FEC codeword A and the second symbol is from RS-FEC codeword B. Symbol-pairs are used in the 200GBASE-R, 400GBASE-R and 800GBASE-R symbol-multiplexing PMAs.

symbol-quartet

A symbol-quartet is defined as four adjacent RS-FEC symbols (for example, on a PCS lane) where the first symbol in the quartet is from RS-FEC codeword A, the second symbol is from RS-FEC codeword B, the third symbol is from RS-FEC codeword C, and the fourth symbol is from RS-FEC codeword D. Symbol-quartets are used in the 1.6TBASE-R symbol-multiplexing PMAs.

xAUI-n

This term refers collectively to 200GAUI-1, 400GAUI-2, 800GAUI-4, 1.6TAUI-8, and 1.6TAUI-16, where “n” is the number of physical lanes.

xBASE-R

This term refers collectively to 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R.

xMII

This term refers collectively to 200GMII, 400GMII, 800GMII, and 1.6TMII.

xXS

This term refers collectively to 200GXS, 400GXS, 800GXS, and 1.6TXS.

176.1.4 Summary of PMA functions

The following principal functions are provided (when required) by the PMA in both the transmit and receive directions:

- Adapting data from PCS lanes (PCSLs) to one or more PMALs
- Symbol-level multiplexing and demultiplexing
 - Alignment to RS-FEC symbols using the PCSL alignment markers
 - Delay alternating PCSLs by two RS-FEC codewords for 200GBASE-R and 400GBASE-R PMAs
 - Four-way RS-FEC codeword interleaving on all PMALs
- PAM4 encoding and decoding
- Optional local and remote loopbacks
- Test-pattern generation and detection

The m:n PMAs and n:m PMAs primarily provide RS-FEC symbol multiplexing and demultiplexing functions between PCSLs and PMALs. PCSL formats are defined by the xBASE-R PCS sublayer in Clause 119, Clause 172, and Clause 175.

An SM-PMA does not support bit multiplexing interfaces. To convert between bit multiplexing and symbol multiplexing interfaces, a BM-PMA and an SM-PMA are placed back-to-back with each PMA demultiplexing to PCSLs in between. For example, to convert from a 200GAUI-2 interface to a 200GAUI-1 interface, a 200GBASE-R 2:8 BM-PMA (see Clause 120) is placed next to a 200GBASE-R 8:1 SM-PMA.

See Annex 176B for a complete description of xBASE-R PMA instantiations using the SM-PMAs defined in this clause. PMA instantiations for xBASE-R PHYs are described in 176B.4 through 176B.7. The special cases that require back-to-back PMA sublayers are described in 176B.3.

The PMA transmit clock is passed from the interface above the PMA to the interface below in the transmit direction, and the PMA receive clock is passed from the interface below the PMA to the interface above in the receive direction.

176.1.5 PMA sublayer positioning

An implementation may use one or more PMA sublayers to adapt the number and rate of the PCS lanes to the number and rate of the PMD lanes. The number of PMA sublayers required depends on the partitioning of the functionality for a particular implementation. Several examples are illustrated in Annex 176B in Figure 176B–1, Figure 176B–2, Figure 176B–3 and Figure 176B–4.

Table 176–1 through Table 176–4 describe the possible PMA sublayer positioning.

Table 176–1—xBASE-R m:n PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
xBASE-R PCS, xBASE-R BM-PMA, or xAUI- <i>m</i> ^a	xAUI- <i>n</i> , xBASE-R Inner FEC, or xBASE-R <i>n</i> -lane PMD
DTE xXS	xAUI- <i>n</i>

^a 1.6TAUI-16 only.

Table 176–2—xBASE-R n:m PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
xAUI- <i>n</i>	PHY xXS, xBASE-R BM-PMA, or xAUI- <i>m</i> ^a
800GAUI-4	800GBASE-LR1 Inner FEC

^a 1.6TAUI-16 only.

Table 176–3—xBASE-R n:n PMA positioning^a

Sublayer or interface above PMA	Sublayer or interface below PMA
xAUI- <i>n</i>	xAUI- <i>n</i> , xBASE-R Inner FEC, or xBASE-R <i>n</i> -lane PMD

^a For all n:n PMAs with the exception of the 1.6TBASE-R 16:16 PMA.

Table 176–4—1.6TBASE-R 16:16 PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
1.6TBASE-R PCS, DTE 1.6TXS, or 1.6TAUI-16	1.6TAUI-16
1.6TAUI-16	PHY 1.6TXS

176.2 PMA service interface

The PMA service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R or 400GBASE-R, 169.3 for 800GBASE-R, or 174.3 for 1.6TBASE-R.

The PMA service interfaces are illustrated in Figure 176–2, Figure 176–12 and Figure 176–13.

The PMA service interface primitives (when applicable) are summarized as follows:

```

PMA:IS_UNITDATA_i.request(tx_symbol)
PMA:IS_SIGNAL.request(SIGNAL_OK)
PMA:IS_UNITDATA_i.indication(rx_symbol)
PMA:IS_SIGNAL.indication(SIGNAL_OK)

```

When the client sublayer is an xBASE-R PCS, DTE xXS or a BM-PMA, each instance of tx_symbol and rx_symbol takes on one of two values: 0 or 1. When the client sublayer is an xAUI-n, each instance of tx_symbol and rx_symbol takes on one of four values: 0, 1, 2, or 3 using PAM4 signaling.

In the transmit direction, the m:n PMAs receive m parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($m-1$).request. In the receive direction, the m:n PMAs send m parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($m-1$).indication. The nominal data rate per stream at the client interface for the 200GBASE-R 8:1, 400GBASE-R 16:2, and 800GBASE-R 32:4 PMAs is 26.5625 Gb/s. The nominal data rate per stream at the client interface for the 1.6TBASE-R 16:8 PMA is 106.25 Gb/s.

In the transmit direction, the n:m PMAs receive n parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($n-1$).request. In the receive direction, the n:m PMAs send n parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($n-1$).indication. The nominal data rate per stream at the client interface for all n:m PMAs is 212.5 Gb/s.

In the transmit direction, the n:n PMAs receive n parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($n-1$).request. In the receive direction, the n:n PMAs send n parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($n-1$).indication. The nominal data rate per stream at the client interface for the 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, and 1.6TBASE-R 8:8 PMAs is 212.5 Gb/s. The nominal data rate per stream at the client interface for the 1.6TBASE-R 16:16 PMA is 106.25 Gb/s.

The PMA receives signal status information from the sublayer above the PMA using the PMA:IS_SIGNAL.request primitive. The PMA provides signal status information to the client sublayer using the PMA:IS_SIGNAL.indication primitive. For the m:n and n:m PMAs the SIGNAL_OK parameter

at the client interface is set according to Table 176–5. For the $n:n$ PMAs, the SIGNAL_OK parameter at the client interface is set to the value of the received SIGNAL_OK parameter from the sublayer below the PMA (inst:IS_SIGNAL.indication(SIGNAL_OK)).

Table 176–5—PMA:IS_SIGNAL.indication(SIGNAL_OK) generation

<i>inst:IS_SIGNAL.indication</i> ^a SIGNAL_OK	<i>align_status_mux</i> ^b or <i>all_locked_demux</i> ^c	PMA:IS_SIGNAL.indication SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

^a From the sublayer below the PMA.

^b For $n:m$ PMAs (see 176.4.4.2.1).

^c For $m:n$ PMAs (see 176.4.4.2.1).

176.3 Service interface below PMA

The service interface below the PMA is an instance of the inter-sublayer service interface defined in 116.3 (for 200GBASE-R or 400GBASE-R), 169.3 (for 800GBASE-R), or 174.3 (for 1.6TBASE-R).

The service interface primitives (when applicable) are summarized as follows:

inst:IS_UNITDATA_i.request(tx_symbol)
inst:IS_SIGNAL.request(SIGNAL_OK)
inst:IS_UNITDATA_i.indication(rx_symbol)
inst:IS_SIGNAL.indication(SIGNAL_OK)

When the sublayer below the PMA is a PHY xXS, BM-PMA, or Inner FEC, then each instance of tx_symbol and rx_symbol takes on one of two (bit) values: 0 or 1. When the sublayer below the PMA is an xAUI-n or an xBASE-R PMD, then each instance of tx_symbol and rx_symbol takes on one of four values: 0, 1, 2, or 3 using PAM4 signaling.

In the transmit direction, the $m:n$ PMAs send n parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(n–1).request*. In the receive direction, the $m:n$ PMAs receive n parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(n–1).indication*. The nominal data rate per stream at the service interface below the $m:n$ PMAs is 212.5 Gb/s.

In the transmit direction, the $n:m$ PMAs send m parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(m–1).request*. In the receive direction, the $n:m$ PMAs receive m parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(m–1).indication*. The nominal data rate per stream at interface below the PMA for the 200GBASE-R 1:8, 400GBASE-R 2:16 and 800GBASE-R 4:32 PMAs is 26.5625 Gb/s. The nominal data rate per stream at the interface below the 1.6TBASE-R 8:16 PMA is 106.25 Gb/s.

In the transmit direction, the $n:n$ PMAs send n parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(n-1).request*. In the receive direction, the $n:n$ PMAs receive n parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(n-1).indication*. The nominal data rate per stream at the interface below the PMA for the 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4 and 1.6TBASE-R 8:8 PMAs is 212.5 Gb/s. The nominal data rate per stream at the interface below the PMA for the 1.6TBASE-R 16:16 PMA is 106.25 Gb/s.

The PMA receives signal status information from the sublayer below the PMA using the *inst:IS_SIGNAL.indication* primitive. The PMA provides signal status information to the sublayer below the PMA using the *inst:IS_SIGNAL.request* primitive. For $m:n$ and $n:m$ PMAs, the SIGNAL_OK parameter at the interface below the PMA is set according to Table 176–6. For the $n:n$ PMAs, the SIGNAL_OK parameter at the interface below the PMA is set to the value of the received SIGNAL_OK value.

Table 176–6—*inst:IS_SIGNAL.request*(SIGNAL_OK) generation

PMA:IS_SIGNAL.request ^a SIGNAL_OK	align_status_mux ^b or all_locked_demux ^c	<i>inst:IS_SIGNAL.request</i> ^d SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

^a From the sublayer above the PMA.

^b For $m:n$ PMAs (see 176.4.4.2.1).

^c For $n:m$ PMAs (see 176.4.4.2.1).

^d To the service interface below the PMA.

176.4 200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4 and 1.6TBASE-R 16:8 SM-PMAs

This subclause defines the $m:n$ PMAs using the variables m and n to denote the number of input and output lanes of each PMA, where:

- m is 8 and n is 1 for the 200GBASE-R 8:1 PMA
- m is 16 and n is 2 for the 400GBASE-R 16:2 PMA
- m is 32 and n is 4 for the 800GBASE-R 32:4 PMA
- m is 16 and n is 8 for the 1.6TBASE-R 16:8 PMA

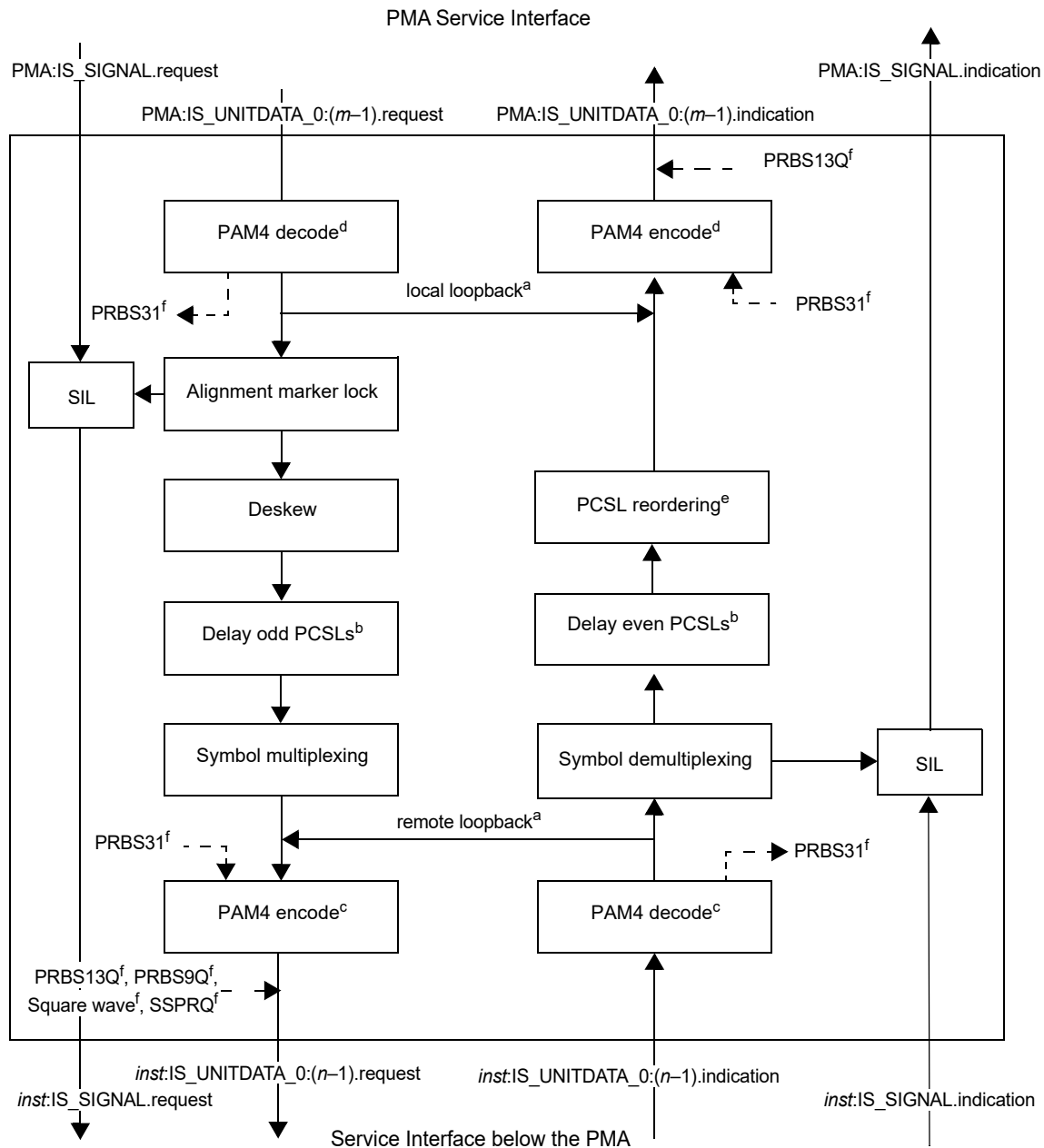
Note that m equals the number of PCSs and n equals the number of PMALs for each xBASE-R $m:n$ PMA.

176.4.1 Overview

In the transmit (multiplexing) direction, the $m:n$ PMAs multiplex RS-FEC symbols from m PCSs at the PMA service interface to n PMALs at the service interface below the PMA. In the receive (demultiplexing) direction, the $m:n$ PMAs demultiplex RS-FEC symbols from n PMALs at the service interface below the PMA to m PCSs toward the PMA service interface. The transmit and receive functions are described in 176.4.2 and 176.4.3, respectively. The sublayer or interfaces above and below the $m:n$ PMAs are shown in Table 176–1.

A functional block diagram of the m:n PMAs is shown in Figure 176–2.

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inst: PMA or PMD or FEC or AUI
SIL: Signal indication logic

^a Optional

^b If 200GBASE-R, 400GBASE-R or 800GBASE-R

^c If the sublayer below the PMA is an AUI or PMD

^d If the sublayer above the PMA is a 1.6TAUI-16

^e If the sublayer above the PMA is an 800GBASE-R BM-PMA

^f Required, optional, or not applicable per 176.7.4

m = 8 (200GBASE-R), 16 (400GBASE-R), 32 (800GBASE-R), or 16 (1.6TBASE-R)

n = 1 (200GBASE-R), 2 (400GBASE-R), 4 (800GBASE-R), or 8 (1.6TBASE-R)

**Figure 176–2—200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4,
1.6TBASE-R 16:8 PMAs functional block diagram**

176.4.2 Transmit function

The transmit function of the $m:n$ PMAs processes m input streams representing m PCS lanes to produce n output streams of symbol-multiplexed data running at 212.5 Gb/s.

As shown in Figure 176–2, the transmit function takes m input lanes from the top of the figure and processes the data through the functional blocks on the left-hand side to n output lanes at the bottom of the figure. PAM4 decoding is first provided at the inputs if necessary. The input PCSLs are then identified by searching for the PCS alignment markers utilizing the same alignment marker lock process as used in the corresponding PCS sublayer. This also identifies the RS-FEC symbol boundary and allows the PCSLs to then be deskewed and aligned to a multiple-symbol or codeword boundary which, in turn, allows the symbol multiplexing function to produce 212.5Gb/s output lanes with a round-robin pattern of RS-FEC symbols from four different FEC codewords for all MAC data rates supported by this process. At the output, PAM4 encoding is provided if necessary.

The following subclauses describe each of the transmit functional blocks in the transmit (multiplexing) data flow from the PMA service interface to the service interface below the PMA.

176.4.2.1 PAM4 decode

The transmit PAM4 decode is only required if there is a 1.6TAUI-16 above the PMA. The PAM4 encode and decode functions are described in 176.7.1.

176.4.2.2 Alignment marker lock

The alignment marker lock process for each input PCSL operates in the same manner as the xBASE-R PCS alignment marker lock process (see 119.2.5.1, 172.2.5.1, and 175.2.5.1).

Once alignment marker lock is achieved on all input lanes and each lane is locked to a unique alignment marker payload sequence, the PCSLs may be deskewed. Synchronization between the PCS lanes is performed by the PMA multiplex lane synchronization process as specified in the state diagram in Figure 176–10.

176.4.2.3 Deskew

The deskew process is performed to align the PCSLs to an appropriate boundary which enables the symbol multiplexing process to interleave RS-FEC symbols from four different FEC codewords in a round-robin pattern on each output lane. The deskew boundary is different for each $m:n$ PMA as described in 176.4.2.3.1, 176.4.2.3.2, and 176.4.2.3.3.

The deskew process is performed in the DESKEW state of the PMA multiplex lane synchronization process (see Figure 176–10). After the deskew process is complete, the RS-FEC symbol boundaries are known and appropriately aligned across all PCSLs.

176.4.2.3.1 4-codeword deskew

The 4-codeword deskew is used for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.

The 4-codeword deskew aligns PCSLs to a four codeword boundary relative to the start of the alignment marker block across all PCSLs. To achieve 4-codeword deskew, delay is added independently to the PCSL(s) that arrive earlier in time compared to the last arriving PCSL(s) until there is an integer multiple of four RS-FEC codewords between the start of the alignment markers on any two PCSLs.

After the 4-codeword deskew is complete, the remaining inter-lane skew between the alignment markers of the PCSLs is in multiples of four codewords. For the 200GBASE-R 8:1 PMA, it equals $N \times 272$ RS-FEC symbols, and for the 400GBASE-R 16:2 PMA, it equals $N \times 136$ RS-FEC symbols, where N is an integer.

- For 200GBASE-R: $N \times 4$ codewords $\times$ 544 symbols per codeword / 8 PCSLs = $N \times 272$ symbols.
- For 400GBASE-R: $N \times 4$ codewords $\times$ 544 symbols per codeword / 16 PCSLs = $N \times 136$ symbols.

NOTE— A specific case of 4-codeword deskew is the removal of all inter-lane skew between PCSL alignment markers. This corresponds to the case of $N = 0$ for all remaining inter-lane skew after deskew is complete.

176.4.2.3.2 20-bit deskew

The 20-bit deskew is used for the 800GBASE-R 32:4 PMA.

The 20-bit deskew aligns PCSLs to a 20-bit boundary (two RS-FEC symbols) relative to the start of the alignment marker block across all PCSLs.

To achieve the 20-bit deskew, delay is added independently to the PCSLs that arrive earlier in time compared to the last arriving PCSL(s), until there is an integer multiple of two RS-FEC symbols (20 bits) between the start of the alignment markers of any two PCSLs.

176.4.2.3.3 40-bit deskew

The 40-bit deskew is used for the 1.6TBASE-R 16:8 PMA.

The 40-bit deskew aligns PCSLs to a 40-bit boundary (4 RS-FEC symbols) relative to the start of the alignment marker block across all PCSLs. This delay aligns RS-FEC symbol-quartets across the PCS lanes.

To achieve the 40-bit deskew, delay is added independently to the PCSLs that arrive earlier in time compared to the last arriving PCSL(s), until there is an integer multiple of four RS-FEC symbols (40 bits) between the start of the alignment markers of any two PCSLs.

176.4.2.4 Delay odd PCSLs

The odd PCSLs are delayed by a specific number of RS-FEC symbols with respect to the even PCSLs to enable the multiplexing of RS-FEC symbols from four different codewords by the symbol-pair multiplexer.

This delay function is used by the 200GBASE-R 8:1, 400GBASE-R 16:2, and 800GBASE-R 32:4 PMAs which employ symbol-pair multiplexing, but not by the 1.6TBASE-R 16:8 PMA which employs symbol-quartet multiplexing.

Odd PCSLs are delayed by a single RS-FEC symbol as described in 176.4.2.4.1. In addition to the single RS-FEC symbol delay, the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs further delay the odd PCSLs by two codewords as described in 176.4.2.4.2.

This results in the following total delay for each odd PCSL:

- The 200GBASE-R 8:1 PMA delays each odd PCSL by 137 FEC symbols.
- The 400GBASE-R 16:2 PMA delays each odd PCSL by 69 FEC symbols.
- The 800GBASE-R 32:4 PMA delays each odd PCSL by one FEC symbol.
- The 1.6TBASE-R 16:8 PMA adds no delay to the odd PCS lanes.

176.4.2.4.1 Delay odd PCSLs by one symbol

This delay is performed by the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs.

These PMAs delay the odd PCSLs by one RS-FEC symbol (10 bits) to remove the checkerboard pattern created by the distribution of symbols to PCS lanes (see 119.2.4.7). This delay aligns RS-FEC symbol pairs across all PCS lanes. Aligning RS-FEC symbol-pairs eventually enables multiplexing on the symbol-pair boundaries to result in a round-robin pattern among symbols from four different codewords.

Figure 176–3 shows the result of the one symbol delay applied to the odd PCSLs for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.

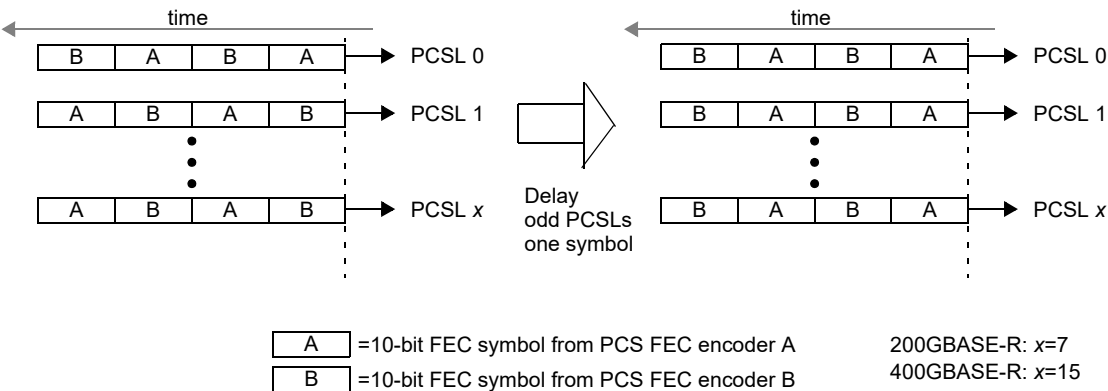


Figure 176–3—Delay odd PCSLs by one RS-FEC symbol
(200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs)

Figure 176–4 shows the result of the one symbol delay applied to the odd PCSLs for the 800GBASE-R 32:4 PMA.

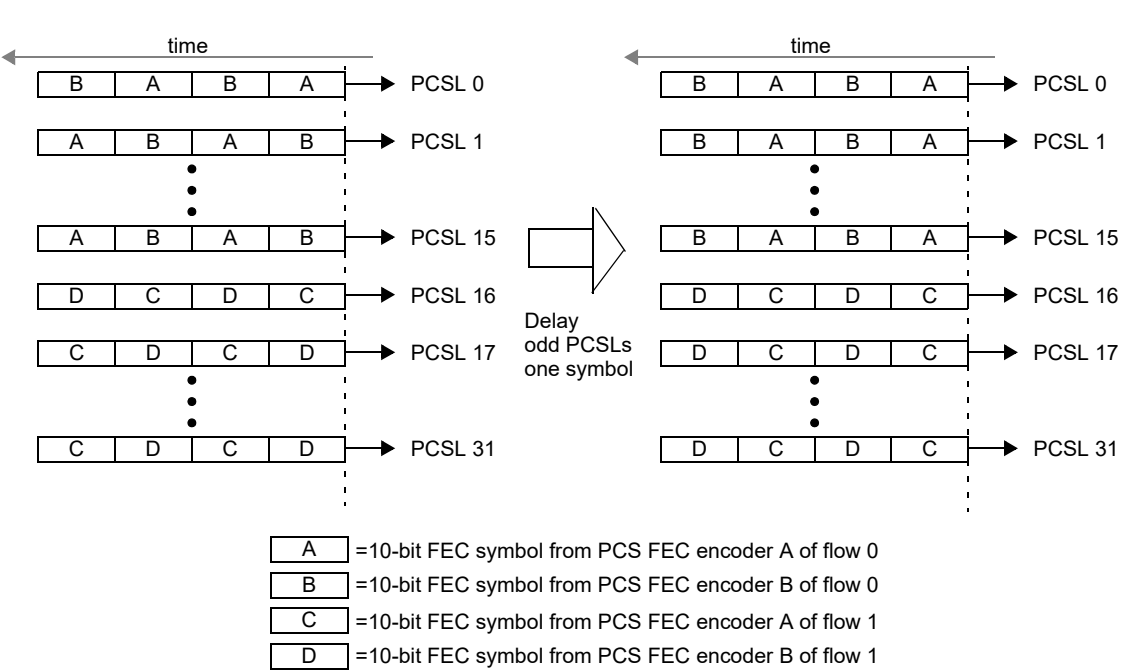


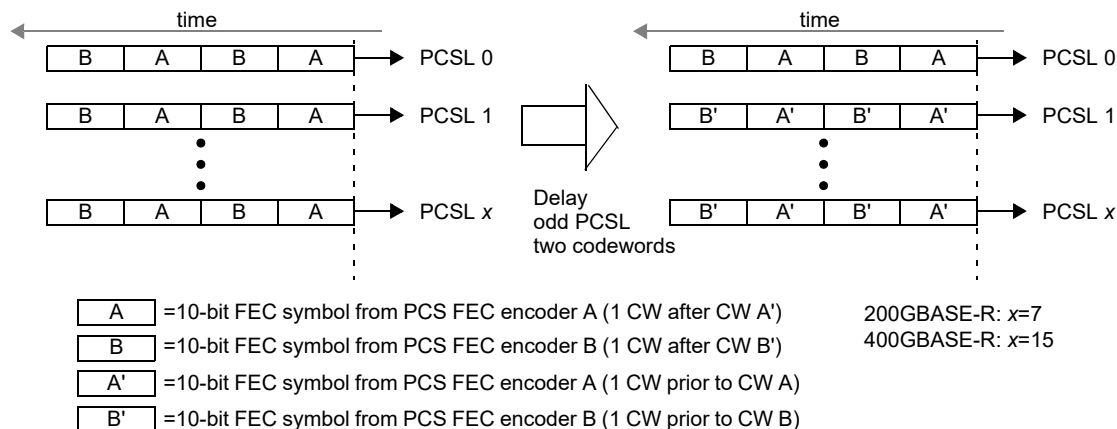
Figure 176–4—Delay Odd PCSLs by one RS-FEC symbol (800GBASE-R 32:4 PMA)

176.4.2.4.2 Delay odd PCSLs by two codewords

This delay is performed for the 200GBASE-R 8:1, 400GBASE-R 16:2 PMAs.

For the 200GBASE-R 8:1 PMA, the odd PCSLs are further delayed by 136 RS-FEC symbols (1360 bits), and for the 400GBASE-R 16:2 PMA, the odd PCSLs are further delayed by 68 RS-FEC symbols (680 bits) which is equivalent to adding a delay of two RS-FEC codewords to the odd PCSLs (2 codewords × 544 symbols per codeword / number of PCS lanes).

Figure 176–5 shows the result of the two codeword delay applied to the odd numbered PCSLs of the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.



**Figure 176-5—Delay odd PCSLs by two codewords
(200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs)**

176.4.2.5 Symbol multiplexing

The symbol multiplexing function multiplexes symbol-pairs or symbol-quartets. Symbol-pairs are multiplexed in the case of the 200GBASE-R, 400GBASE-R and 800GBASE-R PMAs. Symbol-quartets are multiplexed in the case of the 1.6TBASE-R PMAs. The symbol-pair and symbol-quartet multiplexing functions are defined below.

176.4.2.5.1 Symbol-pair multiplexing

A symbol-pair multiplexer is defined as having eight input lanes and one output lane. The symbol-pair multiplexer shall transmit symbol-pairs from the eight realigned input lanes (PCSLs) in a round-robin manner to the output lane.

For the 200GBASE-R 8:1 PMA and 400GBASE-R 16:2 PMA, the symbol-pair multiplexer input lanes shall consist of four even numbered and four odd numbered PCSLs, and the round-robin multiplexing process shall alternate between even and odd PCS lanes.

For the 800GBASE-R 32:4 PMA, the symbol-pair multiplexer input lanes shall consist of four PCSLs numbered between 0 to 15 (low PCSLs) and four PCSLs numbered between 16 to 31 (high PCSLs). The round-robin multiplexing process shall alternate between low PCSLs and high PCSLs.

Figure 176-6 illustrates an example of an 8:1 symbol-pair multiplexer function for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs starting with deskewed and delayed PCSLs as shown in Figure 176-5. Two such symbol-pair multiplexing processes are required for the 400GBASE-R 16:2 PMA to service all 16 input lanes. Different output lane orderings are allowed which conform to the above rules for PCSL input lane grouping and ordering.

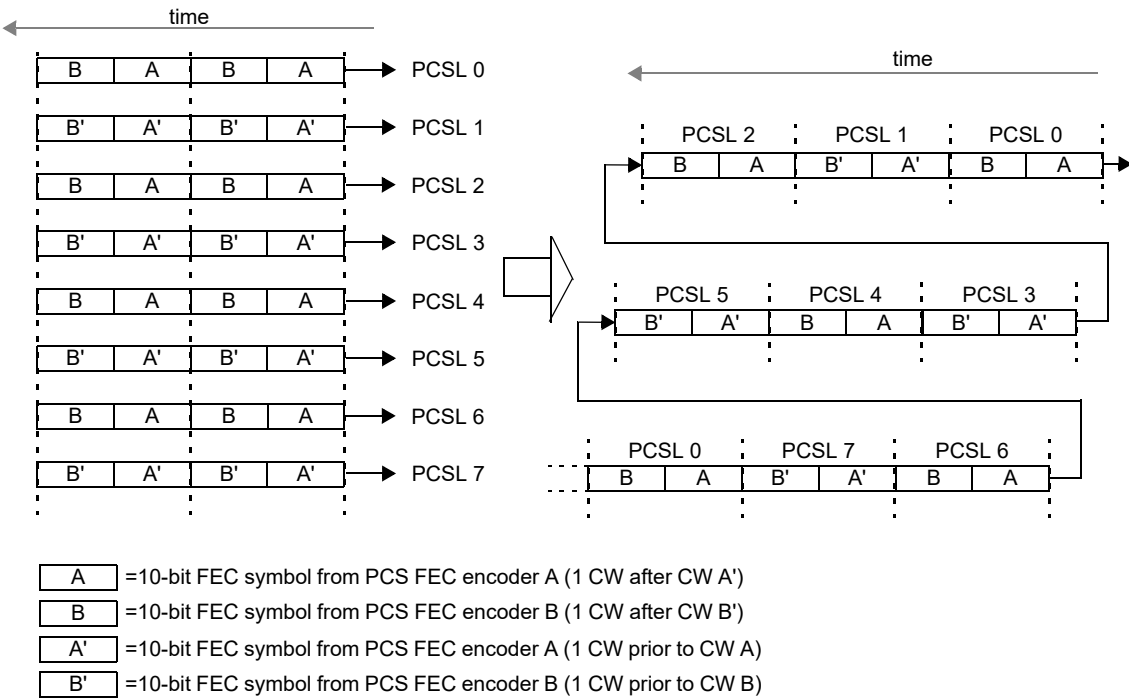


Figure 176-6—Example output lane of the 8:1 symbol-pair multiplexer used in a 200GBASE-R 8:1 PMA or 400GBASE-R 16:2 PMA

Figure 176-7 illustrates an example of an 8:1 symbol-pair multiplexing function for the 800GBASE-R 32:4 PMA starting with deskewed and delayed PCSLs as shown in Figure 176-4. Four similar symbol-pair multiplexing processes are required by 800GBASE-R 32:4 PMA to service all 32 input lanes.

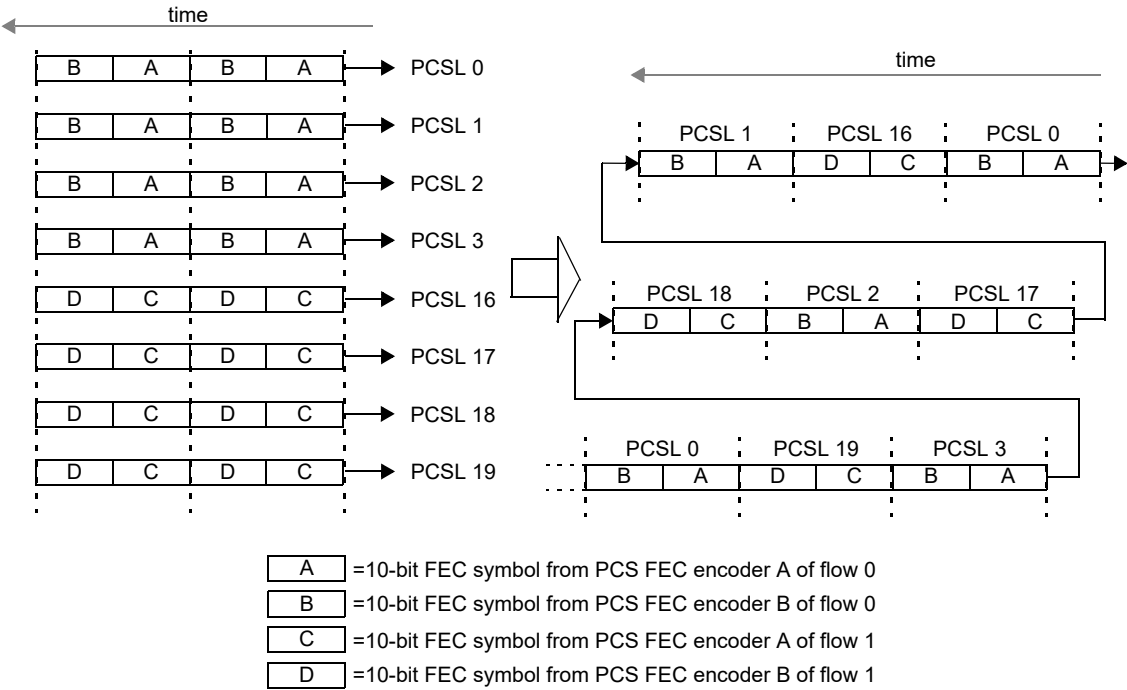


Figure 176-7—Example output lane of a 8:1 symbol-pair multiplexer used in an 800GBASE-R 32:4 PMA

176.4.2.5.2 Symbol-quartet multiplexing

A symbol-quartet multiplexer is defined as having two input lanes and one output lane. The symbol-quartet multiplexer shall transmit symbol-quartets by alternating them from the two realigned input lanes (PCSLs) to the output lane. A symbol-quartet multiplexer may use any two PCSLs as input lanes.

The 1.6TBASE-R 16:8 PMA uses eight symbol-quartet multiplexing processes to multiplex 16 input lanes to eight output lanes.

An example of the symbol-quartet multiplexing process is illustrated in Figure 176-8 starting with two input PCS lanes deskewed to a 40-bit boundary. Note that all 16 PCS lanes come with the same round-robin pattern of four RS-FEC symbols from four separate codewords.

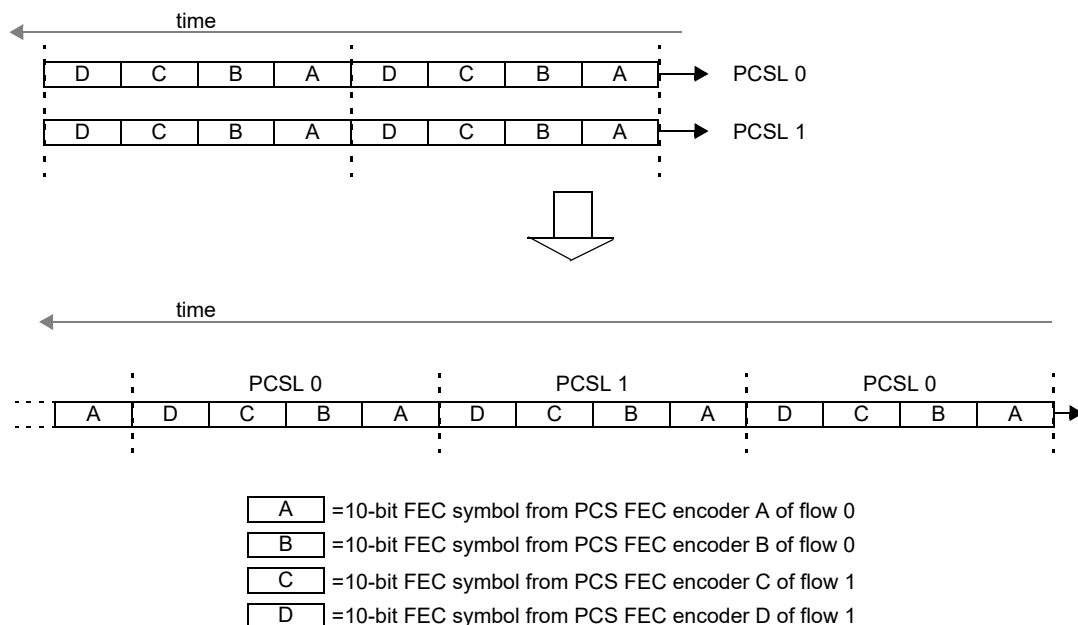


Figure 176-8—Example output lane of a symbol-quartet multiplexer used in a 1.6TBASE-R PMA

176.4.2.6 PAM4 encode

The transmit PAM4 encode process is only required if the sublayer below the m:n PMA is an AUI or PMD. The PAM4 encode function is defined in 176.7.1.

176.4.3 Receive function

The PMA receive function of the m:n PMAs processes n input streams of symbol-multiplexed data running at a 212.5 Gb/s to produce m output streams representing m PCS lanes.

As shown in Figure 176-2, the receive function takes n input lanes from the bottom of the figure and processes the data through the functional blocks on the right-hand side to m PCSL output lanes at the top of the figure. PAM4 decoding is provided at the input interface if necessary. The input streams are then demultiplexed at an arbitrary symbol-pair (or symbol-quartet) boundary to produce a set of 8 (or 2) candidate PCSLs from each input stream. These candidate PCSLs are searched for PCS alignment markers using an alignment marker lock process that observes only the symbol boundary given by the symbol demultiplexing function. If all 8 (or 2) PCS lanes successfully locate an alignment marker, then the symbol-pair (or symbol-quartet) boundary is confirmed using the `pmal_locked_demux<y>` indication back to the symbol demultiplexing process, where y is the input PMAL number. When all input PMALs are successfully demultiplexed, the full set of PCS lanes is allowed to proceed through the remaining receive functions. The even PCSLs are delayed to match the delay added to the odd PCSLs in the transmit flow, and are then reordered, if required, before being output on the PMA service interface. At the output, PAM4 encoding is provided if necessary.

The following subclauses describe each of the receive functional blocks in the receive (demultiplexing) data flow from the service interface below PMA service interface to the PMA service interface.

176.4.3.1 PAM4 decode

The receive PAM4 decode process is only required if the sublayer below the PMA is an AUI or PMD. The PAM4 encode and decode functions are defined in 176.7.1.

176.4.3.2 Symbol demultiplexing

There are two types of symbol demultiplexing, a symbol-pair demultiplexing function and a symbol-quartet demultiplexing function. For the 200GBASE-R, 400GBASE-R and 800GBASE-R PMAs, a 1:8 symbol-pair demultiplex is performed on each input lane to create eight separate PCSs. For the 1.6TBASE-R PMA, a 1:2 symbol-quartet demultiplex is performed on each input lane to create two separate PCSs from each input lane.

The symbol demultiplexing function locates the correct symbol demultiplex boundary and achieves symbol lock on each input PMAL. After all input PMALs from the service interface below the PMA have achieved symbol lock, the PMA then sends a full set of reconstructed PCSs to the next steps in the receive function flow.

176.4.3.2.1 Symbol-pair demultiplexing

Symbol-pair demultiplexing is performed for the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs.

A functional block diagram of a 1:8 symbol-pair demultiplexer, which is composed of a 20-bit demultiplexing function and an alignment marker lock function (see 176.4.3.2.3), is shown in Figure 176–9.

The 20-bit demultiplexer distributes 20 consecutive bits at a time from an input PMAL to each of eight PCS output lanes using a round-robin process. Alignment marker lock is only achieved on all eight PCS lanes if the 20-bit boundary chosen by the 20-bit demultiplexer aligns to the correct symbol-pair boundary. If alignment marker lock is not achieved on all eight PCS lanes within a time interval set by the symbol-pair demultiplexing process, the 20-bit demultiplexer shifts the 20-bit boundary by one bit (a one bit slip), and restarts the alignment lock process on all eight PCS output lanes. This process of a one-bit slip followed by alignment marker search continues until all eight PCS lanes have alignment marker lock using the 20-bit symbol-pair boundary selected by the 20-bit demultiplexer. When all eight PCS output lanes have alignment marker lock, the input PMAL has achieved symbol-pair lock and `pmal_locked_demux<y>` is set to true, where `y` is the input PMAL number.

The PMA shall implement one PMA demultiplex symbol lock process for each input PMAL as specified by the state diagram in Figure 176–11 to acquire symbol-pair lock and establish the symbol-pair boundary for each input PMAL.

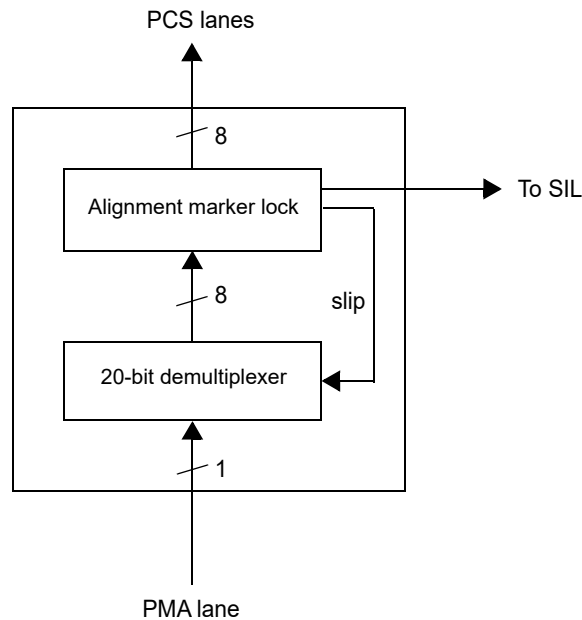


Figure 176–9—1:8 symbol-pair demultiplexer functional block diagram

176.4.3.2.2 Symbol-quartet demultiplexing

Symbol-quartet demultiplexing is performed in the 1.6TBASE-R 16:8 PMA.

Each 1:2 symbol-quartet demultiplexing process is similar to the process defined in 176.4.3.2.1 for the symbol-pair demultiplexing with the exception that it operates on 40-bit boundaries (four RS-FEC symbols) and distributes 40 consecutive bits at a time from an input lane to each of two PCS output lanes.

The 1.6TBASE-R PMA shall implement one PMA demultiplex symbol lock process for each input lane as specified by the state diagram in Figure 176–11 to establish the symbol-quartet boundary for each input lane.

176.4.3.2.3 Alignment marker lock

The alignment marker lock process for each PCSL after demultiplexing operates in the same manner as the xBASE-R PCS alignment marker lock process (see 119.2.5.1, 172.2.5.1, and 175.2.5.1) with the following exceptions:

- The SLIP function defined in 119.2.6.3.3 slips by 10 bits at a time for the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs. It slips by 40 bits at a time for the 1.6TBASE-R 16:8 PMA.
- The restart_lock variable in Figure 119–12 takes the value of restart_lock_demux<y> which is set in the state diagram shown in Figure 176–11.

176.4.3.3 Delay even PCS lanes

Delay is added to the even PCS lanes to match the delay added to the odd PCS lanes in the transmit function described in 176.4.2.4. This removes the artificial skew created by adding delay in the transmit (multiplexing) function.

- The 200GBASE-R 8:1 PMA adds delay equal to 137 FEC symbols to each even PCS lane.
- The 400GBASE-R 16:2 PMA adds delay equal to 69 FEC symbols to each even PCS lane.
- The 800GBASE-R 32:4 PMA adds delay equal to one FEC symbol to each even PCS lane.
- The 1.6TBASE-R 16:8 PMA adds no delay to the even PCS lanes.

176.4.3.4 PCS lane reordering

The PCS lane reordering function is required in the 800GBASE-R PMA when adjacent to an 800GBASE-R BM-PMA which requires its input PCS lanes to be identified by position on the PMA service interface.

PCS lane numbers are identified by the unique portion of the alignment markers, UM0 to UM5 (see 172.2.4.6). PCS lane reordering is performed such that PCS lane x is transmitted as PMA:IS_UNITDATA_x.indication of the PMA service interface, for all $x = 0$ to 31.

176.4.3.5 PAM4 encode

The receive PAM4 encode process is only required if there is a 1.6TAUI-16 above the PMA. The PAM4 encoding function is described in 176.7.1.

176.4.4 State diagrams

This subclause is composed of state diagrams, including the associated definitions of variables, functions, and counters.

176.4.4.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

176.4.4.2 State diagram definitions

The 200GBASE-R 8:1 PMA, 400GBASE-R 16:2 PMA, 800GBASE-R 32:4 PMA, and 1.6TBASE-R 16:8 PMA use the following state diagram from Clause 119:

- Figure 119–12, “Alignment marker lock state diagram”

Any constants, variables, functions, or counters used in the above Clause 119 state diagram which are not defined in the following subclauses shall use the values and definitions from 119.2.6.2 for the 200GBASE-R 8:1 PMA and 400GBASE-R 16:2 PMA, the values and definitions from 172.2.6.2 for the 800GBASE-R 32:4 PMA, and the values and definitions from 175.2.6.2 for the 1.6TBASE-R 16:8 PMA. The variable x in the definitions of 119.2.6.2 is used as an index for the PMA service interface lane number and shall use the values of $x = 0$ to 7 for the 200GBASE-R 8:1 PMA, $x = 0$ to 15 for the 400GBASE-R 16:2 PMA, $x = 0$ to 31 for the 800GBASE-R 32:4 PMA, and $x = 0$ to 15 for the 1.6TBASE-R 16:8 PMA.

176.4.4.2.1 Variables

The following variables are common to all PMA state machines:

PMA_reset

A Boolean variable that is true when set by a management entity and is false otherwise.

reset

A Boolean variable that controls the resetting of the PMA sublayer. It is true whenever a reset is necessary including when initiated by PMA_reset and during power on.

The following variables are used in the PMA multiplex synchronization state diagram (Figure 176–10) and the alignment marker lock state diagram (Figure 119–12) when used for the PMA multiplexing function (in the multiplexing direction).

<code>align_status_mux</code>	Boolean variable that is set to true when PCS lane synchronization is complete. It indicates that <code>all_locked_mux</code> is true and deskew is complete.
<code>all_locked_mux</code>	Boolean variable that takes the value of the variable <code>all_locked</code> as defined in 119.2.6.2.2.
<code>deskew_done_mux</code>	Boolean variable that is set to true when deskew is complete. Otherwise, it is set to false.
<code>deskew_enable_mux</code>	Boolean variable that is set to true in the DESKEW state (see Figure 176–10) to start the deskew process. Otherwise it is set to false.
<code>pcs_lanes_identified_mux</code>	Boolean variable that is set to true if each input lane is locked to a unique alignment marker sequence identified using the alignment markers in Table 119–1 for 200GBASE-R, Table 119–2 for 400GBASE-R, Table 172–2 and Table 172–3 for 800GBASE-R, or Table 175–2 for 1.6TBASE-R PMAs.
<code>restart_lock</code>	Boolean variable that is set to true to restart the alignment marker lock processes for each lane x as defined in Figure 119–12. For the multiplexing process, it is set to true when <code>restart_lock_mux</code> is true.
<code>restart_lock_mux</code>	Boolean variable that is set in the state diagram shown in Figure 176–10. The variable is set to true when the lane synchronization process fails to lock, and is set to false upon entering the LOSS_OF_ALIGNMENT state, causing the alignment marker lock process to restart on all input lanes.
<code>signal_ok_mux</code>	Boolean variable that is set based on the most recently received value of the SIGNAL_OK primitive on the service interface in the multiplexing direction. It is true if the value was OK and false otherwise.

The following variables are used in the PMA demultiplex symbol lock state diagram (Figure 176–11) and the alignment marker lock state diagram (Figure 119–12) when used for the PMA demultiplexing function (in the demultiplexing direction). The index variable n represents the number of input PMAs.

<code>all_locked_demux</code>	Boolean variable that is set to true when <code>pmal_locked_demux<y></code> is true for all y . For $y = 0$ to $(n-1)$.
<code>pcs_lanes_identified_demux</code>	Boolean variable that is set to true if all demultiplexed PCS lanes are locked to a unique alignment marker sequence identified using the alignment markers in Table 119–1 for 200GBASE-R, Table 119–2 for 400GBASE-R, Table 172–2 and Table 172–3 for 800GBASE-R, or Table 175–2 for 1.6TBASE-R PMAs.
<code>pmal_locked_demux<y></code>	Boolean variable that is set to true when <code>amps_lock<x></code> is true, as defined in 119.2.6.2.2, for all PCSs within the single input lane in the demultiplexing direction. For $y = 0$ to $(n-1)$.
<code>restart_lock</code>	Boolean variable that is set to true to restart the alignment marker lock processes for each lane x as

defined in Figure 119–12. For the demultiplexing process, restart_lock is replaced by the variable restart_lock_demux<y> which is applied to the alignment mark lock processes for all lanes x within a single input lane in the demultiplexing direction. There are eight such PCSLs multiplexed together within a single input lane for the 200G/400G/800GBASE-R PMAs, and two PCSLs multiplexed together within a single input lane for the 1.6TBASE-R PMAs.

restart_lock_demux<y>

Boolean variable that is set to true in the SYMBOL_LOCK_RESTART and SLIP_CONTROL states to restart the alignment marker lock processes for the PCSLs within a single input lane in the demultiplexing direction. For $y = 0$ to $(n-1)$.

signal_ok_demux

Boolean variable that is set based on the most recently received value of the SIGNAL_OK primitive of the service interface in the demultiplexing direction. It is true if the value was OK and false otherwise.

symbol_lock_counter_demux_done<y>

Boolean variable that indicates that the symbol_lock_counter_demux<y> has reached its terminal count. For $y = 0$ to $(n-1)$.

176.4.4.2.2 Functions

BIT_SLIP_DEMUX

This function shifts the symbol-pair or symbol-quartet boundary used by the symbol demultiplexing function by one bit. This causes the next candidate symbol-pair or symbol-quartet boundary to be tested by its associated alignment marker lock processes.

SLIP

The SLIP function used in the alignment marker lock process for the PMA multiplexing direction is the same as defined in 119.2.6.3.3.

The SLIP function used in the alignment marker lock process for the PMA demultiplexing direction is defined differently. It slips by ten bits at a time for the 200GBASE-R, 400GBASE-R, and 800GBASE-R PMAs. It slips by 40 bits at a time for the 1.6TBASE-R PMAs. These are the only valid starting locations for alignment markers at the symbol boundary set by the PMA demultiplex symbol lock process.

176.4.4.2.3 Counters

symbol_lock_counter_demux<y>

Counts three alignment marker intervals. The alignment marker interval is defined as 4096 FEC codewords for 200GBASE-R, as 8192 FEC codewords for 400GBASE-R, as 16384 FEC codewords for 800GBASE-R, and as 32768 FEC codewords for 1.6TBASE-R. There is one counter associated with each input lane, $y = 0$ to $(n-1)$.

176.4.4.3 State diagram figures

The PMA multiplexing function (in the multiplexing direction) and the PMA demultiplexing function (in the demultiplexing direction), shall each implement m alignment marker processes as depicted in Figure 119–12 where m is the number of input or output lanes (m equals the number of PCSLs).

The PMA multiplexing function shall implement a single instance of the PMA multiplex synchronization process as depicted in Figure 176–10. The synchronization process ensures that all PMA input lanes in the multiplexing direction have alignment marker lock, are locked to different alignment markers, and that deskew of all input lanes has been completed to the appropriate boundary for a specific PMA. When these conditions are met, the align_status_mux variable is set to true.

The PMA demultiplexing function shall implement n instances of the PMA demultiplex symbol lock process where n equals the number of input lanes in the demultiplexing direction. Each PMA demultiplex symbol lock process searches for the exact symbol boundary within one input data stream in order to correctly perform the symbol demultiplexing function. Each PMA demultiplex symbol lock process monitors the alignment marker lock processes for the PCSs demultiplexed from within its input lane. When all PCS lanes within its input PMAL have achieved alignment marker lock, the `pmal_locked_demux<y>` variable is set to true, where y is the input PMAL number. When `pmal_locked_demux<y>` is true for all y , where $y = 0$ to $(n-1)$, the `all_locked_demux` variable is set to true. When the `all_locked_demux` and `pcs_lanes_identified_demux` variables are both true, then the PMA demultiplexing function is able to present correct data on all m output lanes of the service interface. At any time, if `all_locked_demux` is true and `pcs_lanes_identified_demux` is false, then `restart_lock_demux<y>` is set to true for all y in order to restart the symbol lock process for all input lanes.

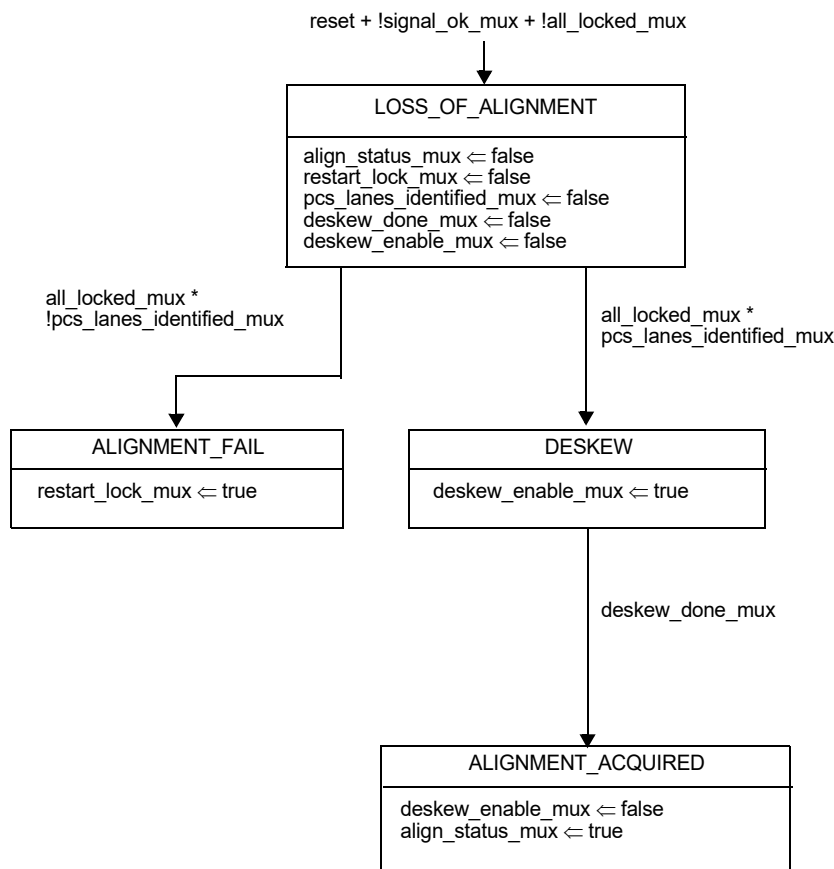


Figure 176–10—PMA multiplex synchronization state diagram

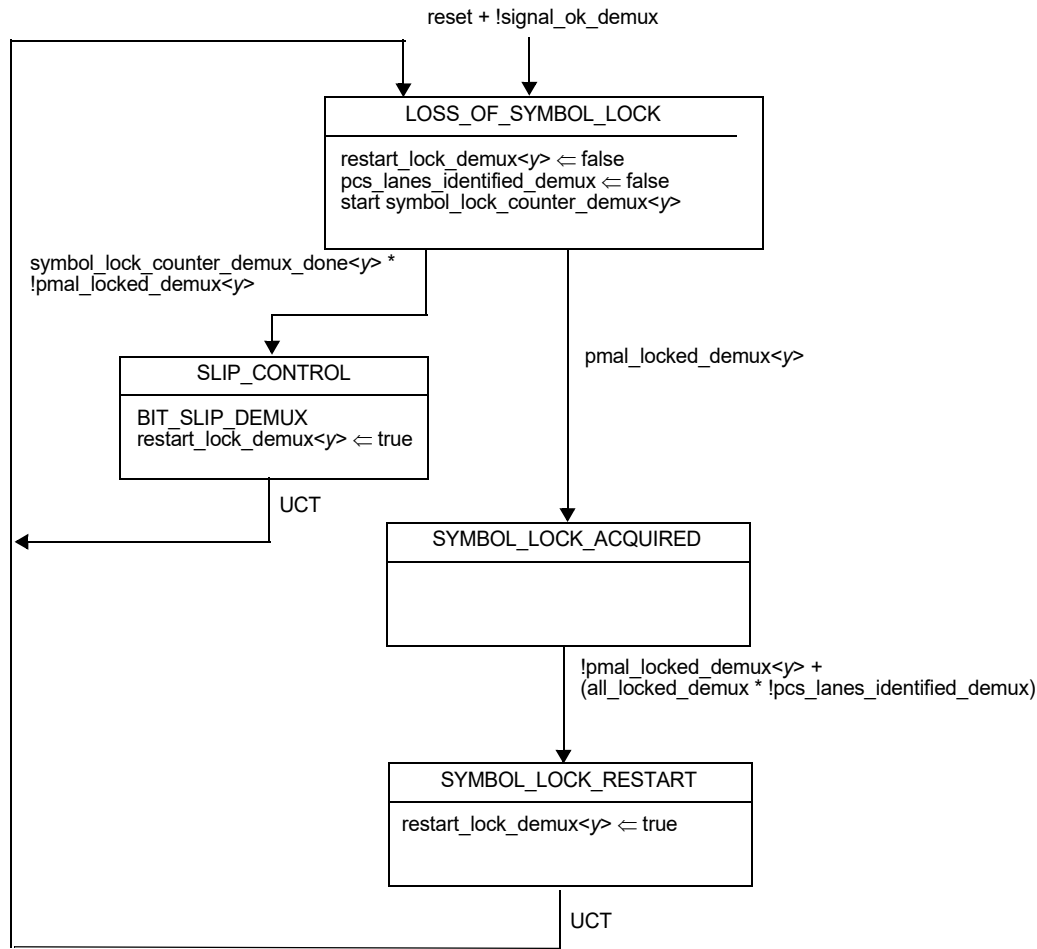


Figure 176–11—PMA demultiplex symbol lock state diagram

176.5 200GBASE-R 1:8, 400GBASE-R 2:16, 800GBASE-R 4:32, 1.6TBASE-R 8:16 SM-PMAs

This subclause defines the $n:m$ PMAs using the variables n and m to denote the number of input and output lanes of the PMA, where:

- n is 1 and m is 8 for the 200GBASE-R 1:8 PMA
- n is 2 and m is 16 for the 400GBASE-R 2:16 PMA
- n is 4 and m is 32 for the 800GBASE-R 4:32 PMA
- n is 8 and m is 16 for the 1.6TBASE-R 8:16 PMA

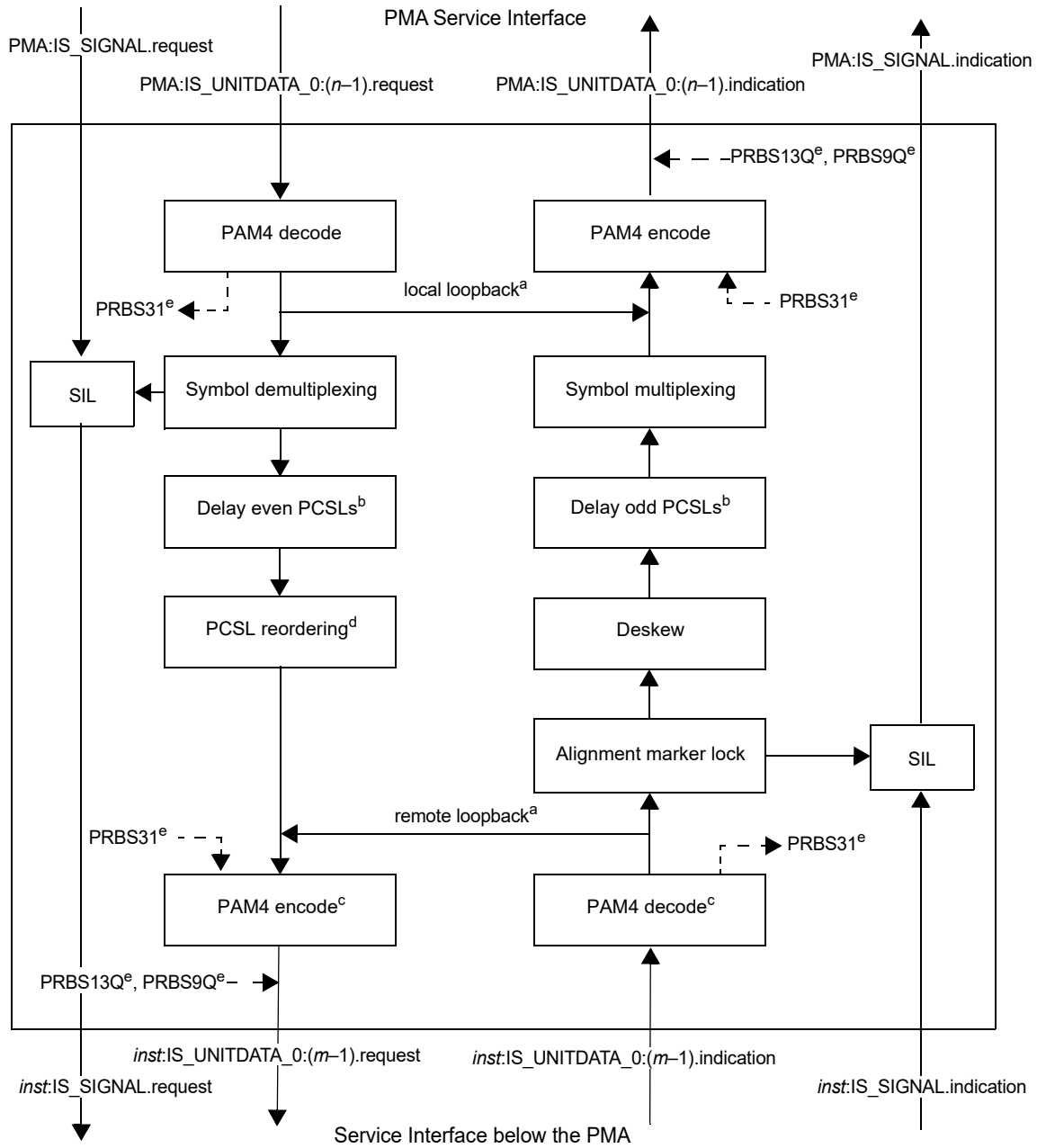
Note that m equals the number of PCSs and n equals the number of PMALs for each xBASE-R $n:m$ PMA.

176.5.1 Overview

In the transmit (demultiplexing) direction, the n : m PMAs demultiplex RS-FEC symbols from n PMALs at the PMA service interface to m PCSs at the service interface below the PMA. In the receive (multiplexing) direction, the n : m PMAs multiplex RS-FEC symbols from m PCSs at the service interface below the PMA to n PMALs at the PMA service interface. The transmit and receive functions are described in 176.5.2 and 176.5.3 respectively. The sublayer or interfaces above and below the n : m PMAs are shown in Table 176–2.

Figure 176–12 provides the functional block diagram of the n : m PMAs.

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inst: PMA or PHY XS or FEC or AUI
SIL: Signal indication logic

^a Optional

^b If 200GBASE-R, 400GBASE-R or 800GBASE-R

^c If the sublayer below the PMA is a 1.6T-AUI-16

^d If the sublayer below the PMA is an 800GBASE-R BM-PMA

^e Required, optional, or not applicable per 176.7.4

$n = 1$ (200GBASE-R), 2 (400GBASE-R), 4 (800GBASE-R), or 8 (1.6TBASE-R)

$m = 8$ (200GBASE-R), 16 (400GBASE-R), 32 (800GBASE-R), or 16 (1.6TBASE-R)

Figure 176-12—200GBASE-R 1:8, 400GBASE-R 2:16, 800GBASE-R 4:32, 1.6TBASE-R 8:16 PMAs functional block diagram

176.5.2 Transmit function

The transmit function of the $n:m$ PMAs is identical to the receive function of the $m:n$ PMAs (see 176.4.3).

176.5.3 Receive function

The receive function of the $n:m$ PMAs is identical to the transmit function of the $m:n$ PMA (see 176.4.2).

176.6 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, 1.6TBASE-R 8:8, 1.6TBASE 16:16 PMAs

This subclause defines the $n:n$ PMAs using the variable n to denote the number of input and output lanes of the PMA, where:

- n is 1 for the 200GBASE-R 1:1 PMA
- n is 2 for the 400GBASE-R 2:2 PMA
- n is 4 for the 800GBASE-R 4:4 PMA
- n is 8 for the 1.6TBASE-R 8:8 PMA
- n is 16 for the 1.6TBASE-R 16:16 PMA

176.6.1 Overview

In the transmit and receive directions, data received on each input lane is mapped to an output lane.

If the interface above or below the PMA is an AUI, PAM4 encoding and decoding is provided by the PMA at that interface.

If the interface above the PMA is an AUI and the interface below is an AUI or a PMD, the Gray mapped PAM4 symbol sequence on each output lane is identical to the Gray mapped PAM4 symbol sequence received on the corresponding input lane, in both transmit and receive directions.

The sublayer or interfaces above and below the $n:n$ PMAs are shown in Table 176–3 and Table 176–4.

Figure 176–13 provides the functional block diagram of the $n:n$ PMAs.

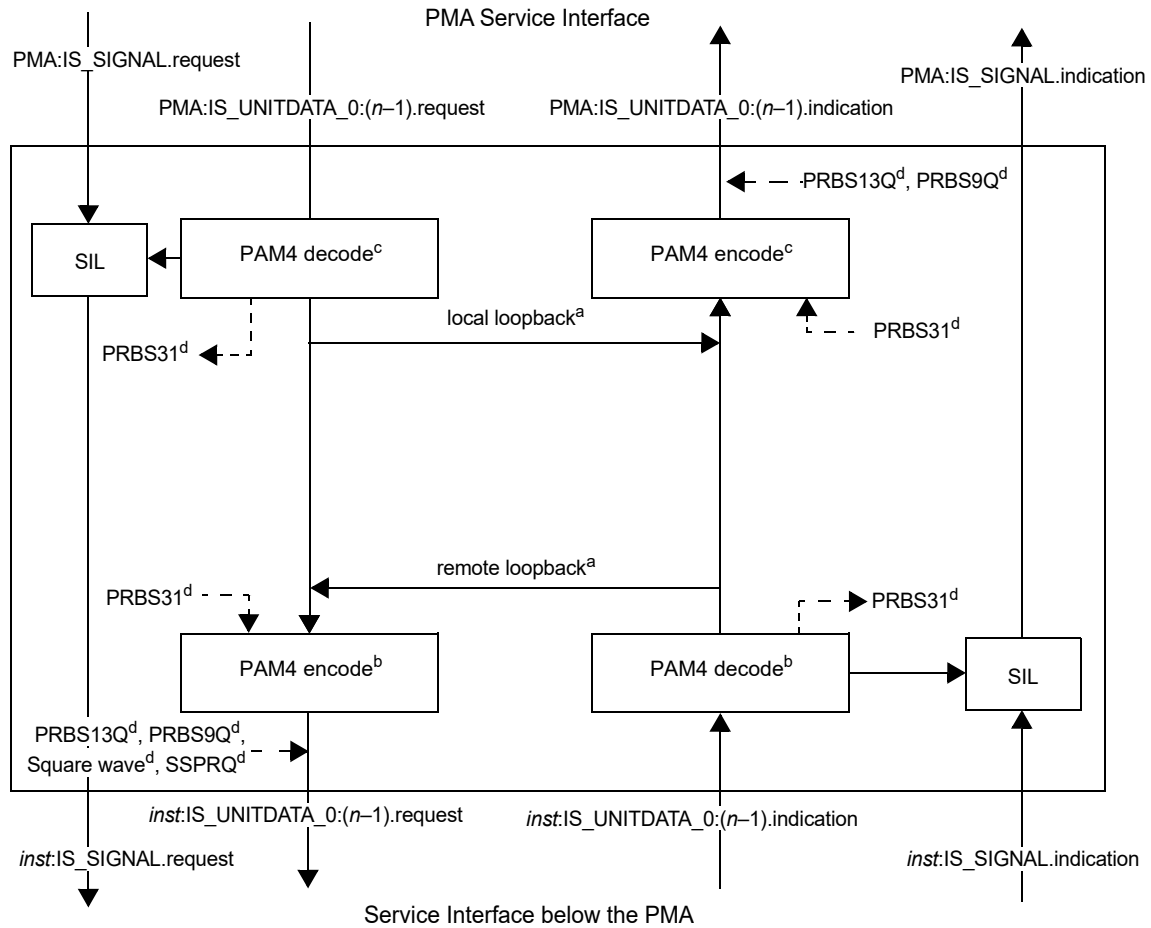


Figure 176–13—200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, 1.6TBASE-R 8:8, 1.6TBASE-R 16:16 PMAs functional block diagram

176.7 Common functions within the PMA

176.7.1 PAM4 encoding

The PAM4 encoding function includes Gray mapping as specified in 176.7.1.1 and $1/(1+D) \bmod 4$ precoding as specified in 176.7.1.2.

176.7.1.1 Gray mapping

The Gray mapping for PAM4 encoded lanes is identical to that specified in 120.5.7.1.

176.7.1.2 Precoding

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or are part of an xAUI-n C2C or C2M link.

The PMA shall provide $1/(1+D) \bmod 4$ precoding capability on each output lane and may optionally provide $1/(1+D) \bmod 4$ decoding capability on each input lane. Precoding is implemented as specified in 135.5.7.2.

The precoder is enabled independently on the Tx output, Rx input, Rx output, and Tx input on each lane. Precoding is enabled and disabled using variables `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei`, where *i* is in the range 0 to (n–1).

If the PMA is connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or is part of an xAUI-n C2C or C2M link, and training is enabled by the management variable `mr_training_enable` (see 178B.15), then `precoder_tx_out_enablei` and `precoder_rx_in_enablei` shall be set as determined by the inter-sublayer link training (ILT) function in the LINK_READY state on lane *i* (see Figure 178B–8). The method by which the ILT function affects these variables is implementation dependent.

If the PMA is connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or is part of an xAUI-n C2C or C2M link, and ILT is disabled by the management variable `mr_training_enable`, then `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei` are set as required by the implementation.

176.7.2 PMA local loopback mode (optional)

The PMA sublayer may provide a local loopback function. When local loopback mode enabled, each bit received from the PMA:IS_UNITDATA_i.request(tx_symbol) primitive is looped back toward the direction of the MAC using the PMA:IS_UNITDATA_i.indication(rx_symbol) primitive.

During local loopback, the PMA continues to propagate data in the Tx direction.

The ability to perform this function is indicated by the `local_loopback_ability` PMA status variable. A device is placed in local loopback mode when the `local_loopback_enable` PMA control variable is set to one, and removed from local loopback mode when this variable is set to zero.

176.7.3 PMA remote loopback mode (optional)

Remote loopback, if provided, should be implemented in a PMA sublayer close enough to the PMD to maintain the bit sequence on each individual PMD lane. When remote loopback mode is enabled, each bit received from the `inst:IS_UNIT_DATAi.indication(rx_symbol)` primitive is looped back toward the direction of the MDI using the `inst:IS_UNIT_DATAi.request(tx_symbol)` primitive.

During remote loopback, the PMA continues to propagate data in the Rx direction.

The ability to perform this function is indicated by the `remote_loopback_ability` PMA status variable. A device is placed in remote loopback mode when the `remote_loopback_enable` PMA control variable is set to one, and removed from remote loopback mode when this variable is set to zero.

176.7.4 PMA test patterns

Where the lanes of the PMA are connected to a physically instantiated interface xAUI-n, Inner FEC, or a PMD service interface, the PMA generates and detects test patterns. These test patterns are used to test

adjacent layer interfaces for an individual PMA sublayer or to perform testing between a physically instantiated interface of a PMA and external testing equipment.

The test patterns, described below, are applied on individual output lanes of the PMA.

176.7.4.1 PRBS31 test pattern

The PMA shall include a PRBS31 pattern generator and checker as specified in 120.5.11.1.1. The PRBS31 pattern generator and checker is required when the PMA is adjacent to an xAUI-n, an Inner FEC, or a PMD.

176.7.4.2 PRBS31Q test pattern

The PRBS31Q test pattern generation is provided by the PRBS31 generator (see 176.7.4.1), followed by Gray mapping, and precoding (if enabled) in the PAM4 encoder (see 176.7.1).

The PRBS31Q test pattern checking is provided by the PRBS31 checker (see 176.7.4.1), followed by inverse precoding (if enabled), and inverse Gray mapping in the PAM4 decoder (see 176.4.3.5).

176.7.4.3 PRBS13Q test pattern

The PMA may include a PRBS13Q pattern generator as specified in 120.5.11.2.1, when the PMA is adjacent to a xAUI-n or a PMD.

176.7.4.4 PRBS9Q test pattern

The PMA may include a PRBS9Q pattern generator as specified in 120.5.11.2.a, when the PMA is adjacent to an xAUI-n or a PMD defined in Clause 178 or Clause 179.

176.7.4.5 SSPRQ test pattern

The PMA may include a SSPRQ pattern generator as specified in 120.5.11.2.3, when the PMA is adjacent to a PMD defined in Clause 180 or Clause 181.

176.7.4.6 Square wave (quaternary) test pattern

The PMA may include a square wave (quaternary) test pattern generator as specified in 120.5.11.2.4, when the PMA is adjacent to a PMD defined in Clause 180 or Clause 181.

176.7.4.7 PMA block error detection and counters

The PRBS31 test pattern checker in each PMAL shall include block error detection and 17 related counters. Block error detection and behavior of the counters is defined in 174A.8.

The following counters are implemented per PMAL i , where $i = 0$ to $n-1$ and n is the number of PMALs:

test_block_error_bin_ i_k

A set of 16 48-bit counters where counter k counts once for each test block received with exactly k errored test symbols, $k = 0$ to 15.

test_block_error_bin_ i_{16p}

A 48-bit counter that counts once for each test block received with 16 or more errored test symbols.

176.8 Delay Constraints

There may be up to four instances of an SM-PMA within a Physical Layer, which is composed of an xBASE-R PHY and, optionally, an xMII Extender. The maximum delay (sum of transmit and receive delays) contributed by each instance of an xBASE-R PMA shall meet the values specified in Table 176–7. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4.

Table 176–7—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200GBASE-R 1:8 PMA or 8:1 PMA	13 312	26	66.56
200GBASE-R 1:1 PMA	3 072	6	15.36
400GBASE-R 2:16 PMA or 16:2 PMA	16 384	32	40.96
400GBASE-R 2:2 PMA	6 144	12	15.36
800GBASE-R 4:32 PMA or 32:4 PMA	12 288	24	15.36
800GBASE-R 4:4 PMA	12 288	24	15.36
1.6TBASE-R 8:16 PMA or 16:8 PMA	24 576	48	15.36
1.6TBASE-R 8:8 PMA or 16:16 PMA	24 576	48	15.36

176.9 Skew Constraints

The Skew (relative delay) between the PCSs is kept within limits so that the information on the lanes can be reassembled by the PCS.

Any PMA that combines PCSs from different input lanes onto the same output lane is required to tolerate Skew Variation between the input lanes without changing the PCSL positions on the output lane.

176.9.1 Skew constraints for 200GBASE-R and 400GBASE-R SM-PMAs

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 29 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 160 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.9.2 Skew constraints for 800GBASE-R SM-PMAs

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 as shown in Figure 169–4 and Figure 169–5.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 16 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 145 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.9.3 Skew constraints for 1.6TBASE-R SM-PMAs

Skew and Skew Variation are defined in 174.5 and specified at the points SP1 to SP6 as shown in Figure 174–5 and Figure 174–6.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 16 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 145 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.10 Path data delay for time synchronization

The SM-PMA sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report SM-PMA path data delays is indicated by the `PMA_delay_ns_TX_ability`, `PMA_delay_subns_TX_ability`, `PMA_delay_ns_RX_ability`, and `PMA_delay_subns_RX_ability` status variables (see Table 176–9).

The transmit and receive path data delays are reported as if the DDMP (data delay measurement point) occurs on an odd PCS lane (see 90.7).

Four separate delays are reported in the following eight path data delay status variables:

- `PMA_delay_ns_TX_max` and `PMA_delay_subns_TX_max`
- `PMA_delay_ns_TX_min` and `PMA_delay_subns_TX_min`
- `PMA_delay_ns_RX_max` and `PMA_delay_subns_RX_max`
- `PMA_delay_ns_RX_min` and `PMA_delay_subns_RX_min`

176.11 Management variables

PMA control and status variables intended to be accessible via a management system are listed in Table 176–8 and Table 176–9.

If the MDIO Interface (see Clause 45) is implemented, PMA control and status variables shall be mapped to MDIO registers as shown in Table 176–8 and Table 176–9. These tables provide the MDIO register and bit numbers for the PMA addressed as MMD 1. For implementations with multiple PMA sublayers, additional PMA sublayers use the corresponding register and bit numbers in MMDs 8, 9, 10, and 11 as necessary.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 176–8—PMA control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Remote_loopback_enable	176.7.3	1.0.1	45.2.1.1.4
Local_loopback_enable	176.7.2	1.0.0	45.2.1.1.5
precoder_tx_out_enable_<0:15>	176.7.1.2	1.600.0 to 1.600.15	45.2.1.139
precoder_rx_in_enable_<0:15>	176.7.1.2	1.601.0 to 1.601.15	45.2.1.140
precoder_rx_out_enable_<0:15>	176.7.1.2	1.602.0 to 1.602.15	45.2.1.141
precoder_tx_in_enable_<0:15>	176.7.1.2	1.603.0 to 1.603.15	45.2.1.142
PRBS31_pattern_enable	176.7.4.1	1.1501.7	45.2.1.170
SSPRQ_pattern_enable	176.7.4.5	1.1501.14	45.2.1.170
PRBS_Tx_gen_enable	176.7.4.2, 176.7.4.3	1.1501.3	45.2.1.170
PRBS_Tx_check_enable	176.7.4.2, 176.7.4.3	1.1501.2	45.2.1.170
PRBS_Rx_gen_enable	176.7.4.2, 176.7.4.3	1.1501.1	45.2.1.170
PRBS_Rx_check_enable	176.7.4.2, 176.7.4.3	1.1501.0	45.2.1.170
Square_wave_enable_<0:15>	176.7.4.6	1.1510.0 to 1.1510.15	45.2.1.171
PRBS9Q_enable_<0:7>	176.7.4.4	1.1511.0 to 1.1511.7	45.2.1.171a
PRBS13Q_enable_<0:7>	176.7.4.3	1.1512.0 to 1.1512.7	45.2.1.172
PMA_reset	176.4.4.2.1	1.0.15	45.2.1.1

Table 176–9—PMA status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
800G_Remote_loopback_ability	176.7.3	1.73.15	45.2.1.60b.100
Local_loopback_ability	176.7.2	1.8.0	45.2.1.7.15
precoder_rx_in_request_flag	176.7.1.2	1.604.0	45.2.1.143.2
precoder_tx_in_request_flag	176.7.1.2	1.604.1	45.2.1.143.1
precoder_rx_in_request_<0:7>	176.7.1.2	1.605.0 to 1.605.7	45.2.1.144
precoder_tx_in_request_<0:7>	176.7.1.2	1.606.0 to 1.606.7	45.2.1.145
Square_wave_ability	176.7.4.6	1.1500.12	45.2.1.169
PRBS13Q_gen_Tx_ability	176.7.4.3	1.1500.11	45.2.1.169
PRBS13Q_gen_Rx_ability	176.7.4.3	1.1500.10	45.2.1.169

Table 176–9—PMA status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
SSPRQ_gen_Tx_ability	176.7.4.5	1.1500.13	45.2.1.169
PRBS9Q_gen_Rx_ability	176.7.4.4	1.1500.14	45.2.1.169
PRBS9Q_gen_Tx_ability	176.7.4.4	1.1500.15	45.2.1.169
Ln<0:7>_PRBS_Tx_test_err_counter	176.7.4.2	1.1600 to 1.1607	45.2.1.173
Ln<0:7>_PRBS_Rx_test_err_counter	176.7.4.2	1.1700 to 1.1707	45.2.1.174
test_block_error_bin_0_<0:15>	176.7.4.7	1.2600 to 1.2647	45.2.1.264
test_block_error_bin_0_16p	176.7.4.7	1.2648 to 1.2650	45.2.1.264
test_block_error_bin_1_<0:15>	176.7.4.7	1.2651 to 1.2698	45.2.1.264
test_block_error_bin_1_16p	176.7.4.7	1.2699 to 1.2701	45.2.1.264
test_block_error_bin_2_<0:15>	176.7.4.7	1.2702 to 1.2749	45.2.1.264
test_block_error_bin_2_16p	176.7.4.7	1.2750 to 1.2752	45.2.1.264
test_block_error_bin_3_<0:15>	176.7.4.7	1.2753 to 1.2800	45.2.1.264
test_block_error_bin_3_16p	176.7.4.7	1.2801 to 1.2803	45.2.1.264
test_block_error_bin_4_<0:15>	176.7.4.7	1.2804 to 1.2851	45.2.1.264
test_block_error_bin_4_16p	176.7.4.7	1.2852 to 1.2854	45.2.1.264
test_block_error_bin_5_<0:15>	176.7.4.7	1.2855 to 1.2902	45.2.1.264
test_block_error_bin_5_16p	176.7.4.7	1.2903 to 1.2905	45.2.1.264
test_block_error_bin_6_<0:15>	176.7.4.7	1.2906 to 1.2953	45.2.1.264
test_block_error_bin_6_16p	176.7.4.7	1.2954 to 1.2956	45.2.1.264
test_block_error_bin_7_<0:15>	176.7.4.7	1.2957 to 1.3004	45.2.1.264
test_block_error_bin_7_16p	176.7.4.7	1.3005 to 1.3007	45.2.1.264
align_status_mux	176.4.4.2.1	1.1478.13	45.2.1.168d
all_locked_mux	176.4.4.2.1	1.1478.12	45.2.1.168d
deskew_done_mux	176.4.4.2.1	1.1478.11	45.2.1.168d
pcs_lanes_identified_mux	176.4.4.2.1	1.1478.10	45.2.1.168d
all_locked_demux	176.4.4.2.1	1.1478.9	45.2.1.168d
pcs_lanes_identified_demux	176.4.4.2.1	1.1478.8	45.2.1.168d
pmal_locked_demux<7:0>	176.4.4.2.1	1.1478.7:0	45.2.1.168d
PMA_delay_ns_TX_max	176.10	1.1801, 1.1802	45.2.1.176
PMA_delay_subns_TX_max	176.10	1.1809	45.2.1.176
PMA_delay_ns_TX_min	176.10	1.1803, 1.1804	45.2.1.176
PMA_delay_subns_TX_min	176.10	1.1810	45.2.1.176
PMA_delay_ns_RX_max	176.10	1.1805, 1.1806	45.2.1.177
PMA_delay_subns_RX_max	176.10	1.1811	45.2.1.177

Table 176–9—PMA status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMA_delay_ns_RX_min	176.10	1.1807, 1.1808	45.2.1.177
PMA_delay_subns_RX_min	176.10	1.1812	45.2.1.177
PMA_delay_ns_RX_ability	176.10	1.1800.0	45.2.1.175
PMA_delay_ns_TX_ability	176.10	1.1800.1	45.2.1.175
PMA_delay_subns_RX_ability	176.10	1.1800.2	45.2.1.175
PMA_delay_subns_TX_ability	176.10	1.1800.3	45.2.1.175

176.12 Protocol implementation conformance statement (PICS) proforma for Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R⁶

176.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

176.12.2 Identification

176.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

176.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

176.12.3 Major capabilities/options

Editor's note: This PICS will be implemented in Draft 2.1. Please do not submit comments against this subclause.

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177. Inner Forward Error Correction Sublayer (Inner FEC), type
200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R

177.1 Overview

177.1.1 Scope

This clause specifies the Inner Forward Error Correction (Inner FEC) sublayer for some 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Physical Layer implementations that use 200 Gb/s per lane signaling.

When necessary for disambiguation, to differentiate the Inner FEC defined in this clause from the 800GBASE-LR1 Inner FEC defined in Clause 184, the terms 200GBASE-R Inner FEC, 400GBASE-R Inner FEC, 800GBASE-R Inner FEC, and 1.6TBASE-R Inner FEC are used. Within this clause the term Inner FEC refers specifically to 200GBASE-R Inner FEC, 400GBASE-R Inner FEC, 800GBASE-R Inner FEC, and 1.6TBASE-R Inner FEC. Annex 177A provides test vectors constructed using the methods specified in this clause.

177.1.2 Relationship of the Inner FEC to other standards

Figure 177–1 shows the relationship of the Inner FEC sublayer, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model.

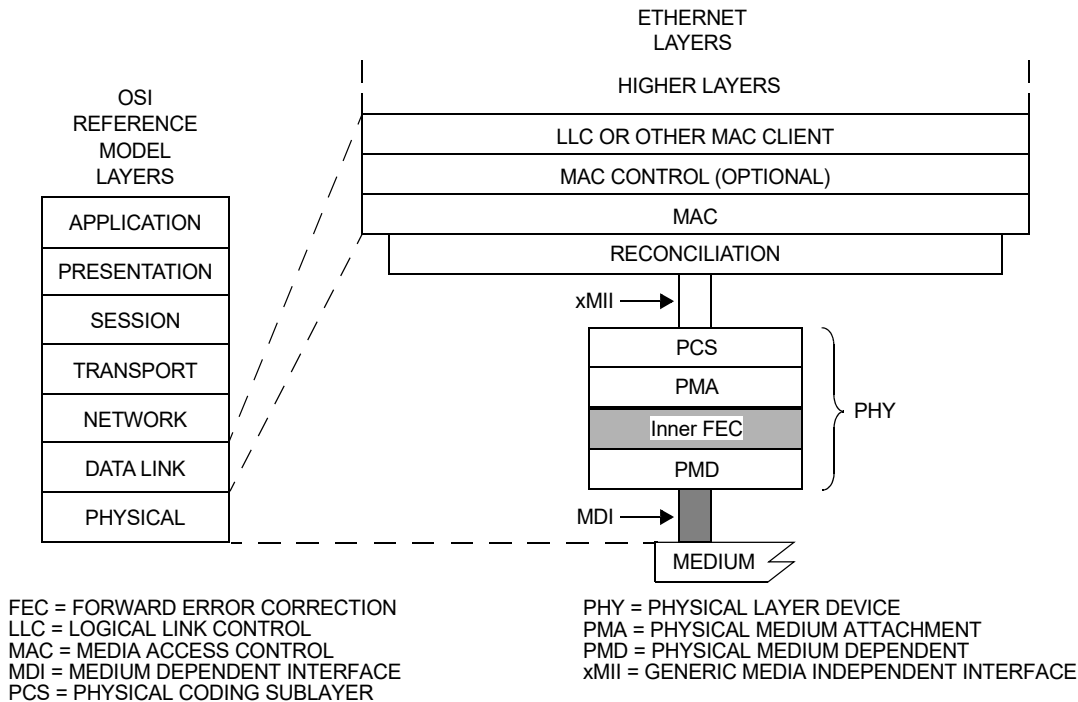


Figure 177–1—Relationship of Inner FEC to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

177.1.3 Summary of functions

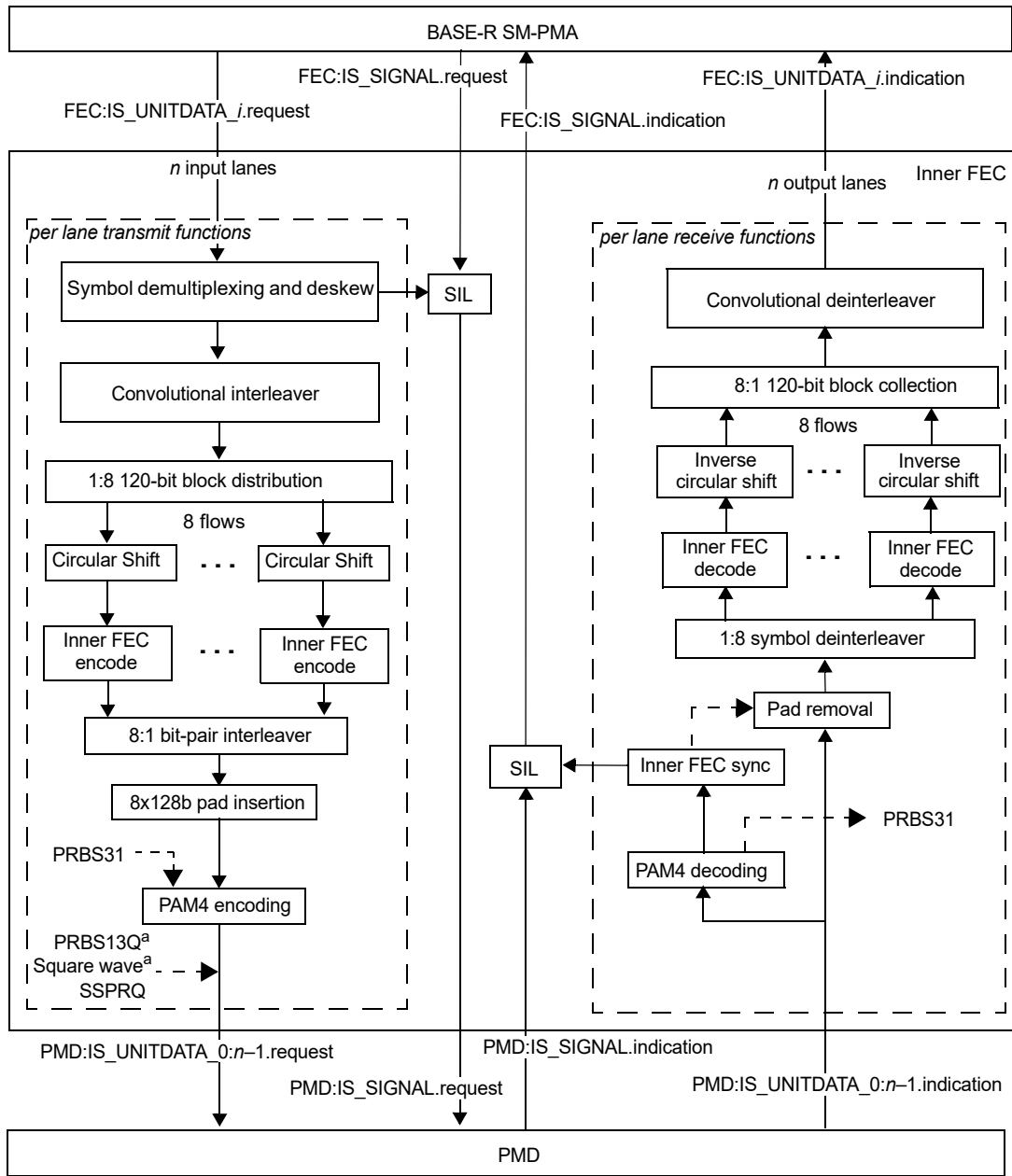
The Inner FEC provides a second layer of FEC protection without decoding the data encoded by the outer RS-FEC in the PCS.

The Inner FEC provides the following services:

- Interleaving (deinterleaving) the data from (to) the BASE-R SM-PMA using the convolutional interleaver (deinterleaver).
- Distributing (collecting) the convolutional interleaved data to (from) eight Inner FEC flows
- Performing circular shift and inverse circular shift on each Inner FEC flow
- Inner FEC encoding and decoding
- 8:1 interleaving (1:8 deinterleaving) the eight Inner FEC flows to (from) a single flow
- Pad insertion and removal
- PAM4 encoding and decoding

177.1.4 Functional block diagram

Figure 177–2 provides a functional block diagram of the Inner FEC. The number (n) of input and output lanes matches the number of Inner FEC instances, and this number is 1 for 200GBASE-R, 2 for 400GBASE-R, 4 for 800GBASE-R, and 8 for 1.6TBASE-R.



^a Optional

Figure 177-2—Functional block diagram

177.2 Inner FEC service interface

For the 200GBASE-R Inner FEC, the client sublayer is a 200GBASE-R 8:1 SM-PMA or a 200GBASE-R 1:1 SM-PMA. For the 400GBASE-R Inner FEC, the client sublayer is a 400GBASE-R 16:2 SM-PMA or a 400GBASE-R 2:2 SM-PMA. For the 800GBASE-R Inner FEC, the client sublayer is an 800GBASE-R 32:4 SM-PMA or an 800GBASE-R 4:4 SM-PMA. For the 1.6TBASE-R Inner FEC, the client sublayer is a 1.6TBASE-R 16:8 SM-PMA or a 1.6TBASE-R 8:8 SM-PMA.

The Inner FEC service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and 174.3 for 1.6TBASE-R.

The service interface primitives are summarized as follows:

FEC:IS_UNITDATA_*i*.request(tx_symbol)
FEC:IS_UNITDATA_*i*.indication(rx_symbol)
FEC:IS_SIGNAL.request(SIGNAL_OK)
FEC:IS_SIGNAL.indication(SIGNAL_OK)

The primitives are defined for $i = 0$ to $n - 1$. In the transmit direction, the Inner FEC sublayer receives n parallel bit streams, each at a nominal rate of 212.5 Gb/s, from the client sublayer. In the receive direction, the Inner FEC sublayer sends n parallel bit streams, each at a nominal rate of 212.5 Gb/s, to the client sublayer. The number of parallel streams, n , is 1 for 200GBASE-DR1-2, 2 for 400GBASE-DR2-2, 4 for 800GBASE-DR4-2, 800GBASE-FR4, and 800GBASE-LR4, and 8 for 1.6TBASE-DR8-2. Each instance of tx_symbol and rx_symbol takes on one of two values: 0 or 1.

The Inner FEC receives signal status information from the client sublayer (PMA) using the FEC:IS_SIGNAL.request(SIGNAL_OK) primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, and FAIL. When SIGNAL_OK is IN_PROGRESS or FAIL, the tx_symbol parameters are undefined.

The Inner FEC sublayer provides signal status information to the client sublayer (PMA) using the FEC:IS_SIGNAL.indication(SIGNAL_OK) primitive (see Figure 177–2). The SIGNAL_OK parameter takes one of four values according to Table 177–1. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding rx_symbol parameters on all lanes are unspecified.

Table 177–1—FEC:IS_SIGNAL.indication(SIGNAL_OK) generation

PMD:IS_SIGNAL.indication SIGNAL_OK value	all_synced (see 177.7.2.1)	FEC:IS_SIGNAL.indication SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

In addition to the primitives noted above, an associated clock is transferred from input to output along with the IS_UNITDATA primitives in the transmit and receive direction.

177.3 Service interface below the Inner FEC

The service interface below the Inner FEC is the PMD service interface defined in 182.3.

The PMD provides signal status information to the Inner FEC using the PMD:IS_SIGNAL.indication(SIGNAL_OK) primitive. The SIGNAL_OK parameter takes one of four values: OK, READY, IN_PROGRESS, and FAIL. When SIGNAL_OK is IN_PROGRESS or FAIL, the rx_symbol parameters are undefined.

The Inner FEC sublayer provides signal status information to the PMD using the PMD:IS_SIGNAL.request(SIGNAL_OK) primitive (see Figure 177–2). The SIGNAL_OK parameter takes

one of four values according to Table 177–2. When the value of SIGNAL_OK is READY, IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes are unspecified.

Table 177–2—PMD:IS_SIGNAL.request(SIGNAL_OK) generation

FEC:IS_SIGNAL.request SIGNAL_OK value	all_locked_demux (see 176.4.4.2.1)	PMD:IS_SIGNAL.request SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

NOTE—SIGNAL_OK = OK does not guarantee that the stream provided to the Inner FEC sublayer through PMD:IS_UNITDATA_i.indication is a valid signal.

177.4 Transmit function

The following processes are performed independently on each Inner FEC service interface input lane.

177.4.1 Symbol demultiplexing and deskew

The symbol demultiplexing and deskew function is illustrated in Figure 177–3.

The alignment marker lock process within the symbol demultiplexing function determines the RS-FEC symbol-pair, symbol-quartet and codeword boundaries, as described in 176.4.3.2. Alignment marker lock is required for 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Inner FEC. Note that for 200GBASE-R and 400GBASE-R, the data stream is not altered.

Deskew aligns the codeword boundaries on all PCS lanes within an Inner FEC lane. Deskew is required for 800GBASE-R and 1.6TBASE-R Inner FEC. After deskew, the PCS lanes are recombined by the symbol multiplexing function.

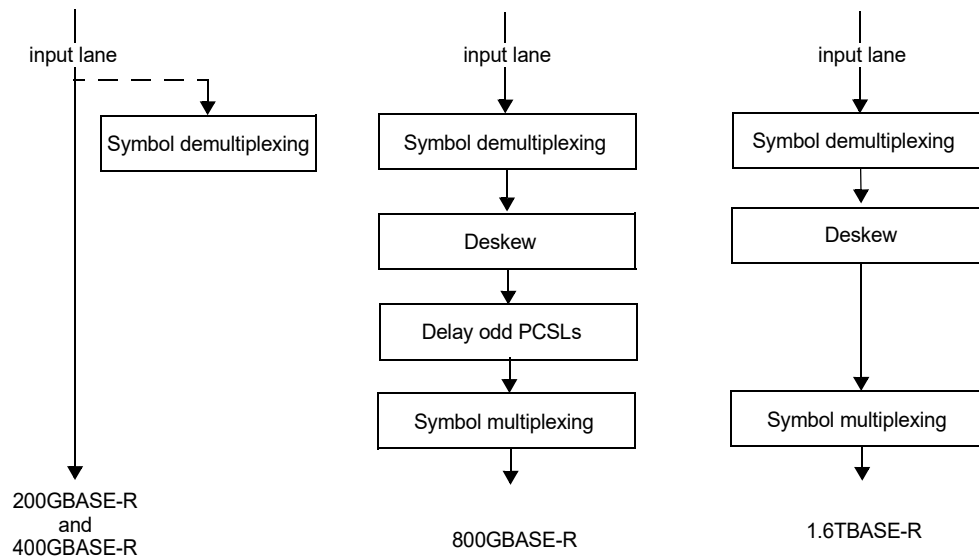


Figure 177-3—Symbol demultiplexing and deskew block diagrams

177.4.1.1 Symbol demultiplexing

The symbol demultiplexing function is performed as defined in 176.4.3.2, with the exception that it operates on the Inner FEC service interface input lanes.

177.4.1.2 Deskew

For 800GBASE-R PHYs, after alignment marker lock is achieved on each of the eight PCSs in an input stream, Skew between PCSs is removed as defined in 172.2.5.1, with the exception that the maximum Skew to be removed is the Skew at SP1 (see Table 169-5) plus the Skew added by the PMA above the Inner FEC.

For 1.6TBASE-R PHYs, after alignment marker lock is achieved on each of the two PCSs in an input stream, Skew between PCSs is removed as defined in 175.2.5.1, with the exception that the maximum Skew to be removed is the Skew at SP1 (see Table 174-5) plus the Skew added by the PMA above the Inner FEC.

NOTE— In 200GBASE-R and 400GBASE-R PHYs, deskew is not required because the SM-PMA above the Inner FEC already removes the Skew by aligning the PCS lanes within each PMA lane to a 4-codeword boundary.

177.4.1.3 Delay odd PCS lanes

Delay of odd PCS lanes is performed only by the 800GBASE-R PHYs. The odd PCSs are delayed by one RS-FEC symbol (10 bits), as illustrated in Figure 176-4, to remove the checkerboard pattern created by the distribution of RS-FEC symbols to PCS lanes.

177.4.1.4 Symbol multiplexing

For 800GBASE-R PHYs, after aligning the PCS lanes to the alignment markers and delaying the odd PCSs, eight PCSs are re-combined into a single data stream using the symbol-pair multiplexing function described in 176.4.2.5.1.

For 1.6TBASE-R PHYs, after aligning the PCSs to the alignment markers, two PCSs are re-combined into a single data stream using the symbol-quartet multiplexing function described in 176.4.2.5.2.

177.4.2 Convolutional interleaver

A convolutional interleaver shall be used on each deskewed Inner FEC lane to systematically distribute RS-FEC symbols from 12 different RS-FEC codewords into an Inner FEC codeword, in order to improve robustness of the system to bursts of errors. Using the RS-FEC symbol-quartet boundaries found by the symbol demultiplexing and deskew process (see 177.4.1) the convolutional interleaver creates groups of four RS-FEC symbols from four RS-FEC codewords. When using PRBS31 encoded by the Inner FEC test mode (see 177.6.1.1), the selection of the RS-FEC symbol-quartet boundary position is unspecified.

The convolutional interleaver contains three delay lines. The first line (Delay Line 0) delays the data by $2 \times Q \times 4$ RS-FEC symbols, the second line (Delay Line 1) by $1 \times Q \times 4$ RS-FEC symbols, and the last line (Delay Line 2) adds no delay. The values of Q are shown in table Table 177–3.

Table 177–3—Configuration of convolutional interleaver delay lines

Ethernet rate	Value of Q
200GBASE-R	192
400GBASE-R	96
800GBASE-R	48
1.6TBASE-R	24

The data from deskewed PMA lane is fed into each delay line one RS-FEC symbol-quartet at a time. The input data round-robins between the three delay lines in the order Delay Line 0, then Delay Line 1 and lastly Delay Line 2.

The output of the convolutional interleaver round-robins between the three delay lines, receiving one RS-FEC symbol-quartet from each at a time, in the order Delay Line 0, then Delay Line 1 and lastly Delay Line 2.

Figure 177–4 illustrates the delay lines of the convolutional interleaver using Q instances of a RS-FEC symbol-quartet sized base delay element labeled “D”. The input and output switches are always aligned to the same row.

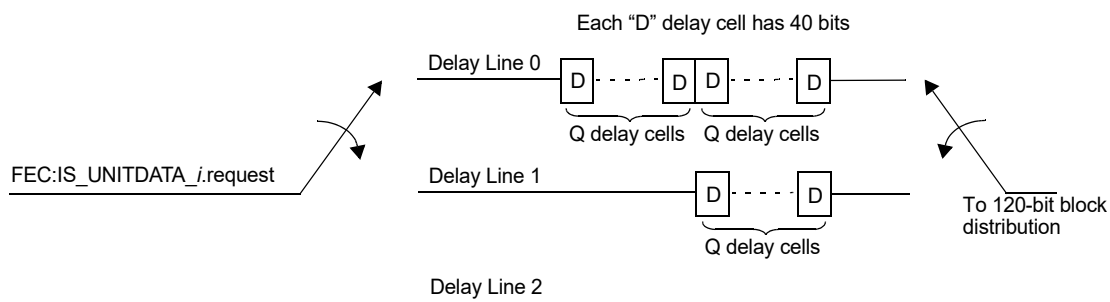


Figure 177–4—Convolutional interleaver

177.4.3 120-bit block distribution

The output of the convolutional interleaver is distributed to 8 Inner FEC flows, in blocks of 120 bits (12 RS-FEC symbols, or 3 RS-FEC symbol-quartets) in a round-robin fashion from flow 0 to 7. The first RS-FEC symbol-quartet within a 120-bit block is always from delay line 0. Within each RS-FEC symbol, bit 0 is transmitted first and bit 9 is transmitted last.

177.4.4 Circular shift

The circular shift function enhances the burst error tolerance of the Inner FEC. The 12 RS-FEC symbols within each 120-bit block are shuffled to maximize the distance between transmitted PAM4 symbols from the same RS-FEC codeword. The circular shift function, $CS(n)$, circularly shifts the 12 RS-FEC symbols by n RS-FEC symbols towards the last arrived RS-FEC symbol. The circular shift function applied to each Inner FEC flow is shown in Figure 177–5, where symbol A is the first arrived RS-FEC symbol in each 120-bit block from the convolutional interleaver.

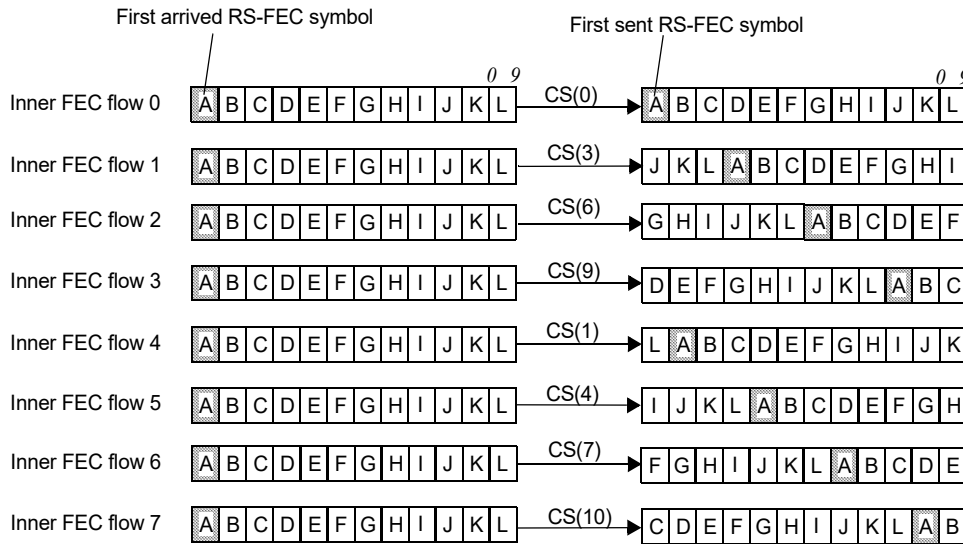


Figure 177–5—Circular shift

177.4.5 Inner FEC encode

In each of the 8 Inner FEC flows, the 120-bit output block of the circular shift, $m\langle 119:0 \rangle$, shall be encoded by a systematic binary(128,120) code, appending 8 parity bits at the end of each 120-bit block to produce a 128-bit Inner FEC codeword $c\langle 127:0 \rangle$. $m(119)$ is the first bit received from the circular shift and $c(127)$ is the first bit transmitted after Inner FEC encode.

The encode process is illustrated in Figure 177–6. First, each pair of bits to be modulated into a PAM4 symbol are XORed resulting in 60 bits, $m_xor\langle 59:0 \rangle$, from the 120-bit message.

For $k = 0:59$

$$m_xor(k) = m(2k) \text{ XOR } m(2k+1)$$

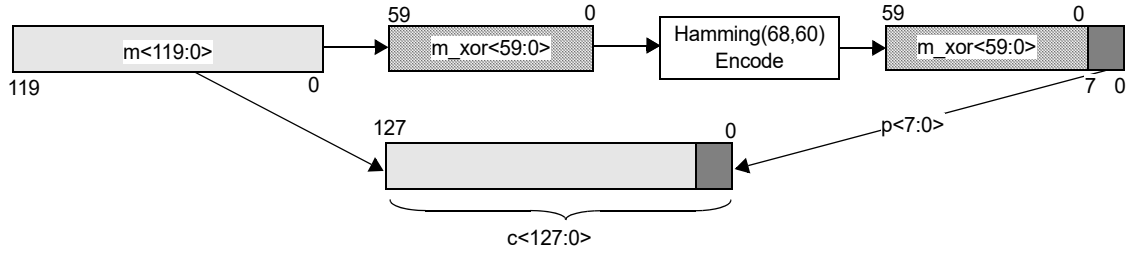


Figure 177-6—Inner FEC encode process

Next, a Hamming (68,60) encoder is used to generate 8 parity bits based on the 60 bits from the previous step, $m_xor<59:0>$.

The systematic Hamming code is most naturally defined in terms of its parity-check matrix. Consider the function g which maps an integer i , $0 \leq i \leq 67$, to a binary column vector as shown in Equation (177-1). The element α^i can be expressed as a linear combination of $\alpha^0, \alpha^1, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6$, as shown in Equation (177-2). ($s_{0,i}, s_{1,i}, s_{2,i}, s_{3,i}, s_{4,i}, s_{5,i}, s_{6,i}$) are the binary coefficients of the linear combination that creates the element α^i in the Galois Field $GF(2^7)$ with primitive polynomial $x^7 + x^3 + 1$.

The parity-check matrix H is then a 8×68 binary matrix as shown in Equation (177-3). To obtain the encoder matrix G , we calculate P and B using Equation (177-4) and Equation (177-5). The generator matrix of the Hamming(68,60) code is $G = [I_{60}; G_{60,8}]$, where I_{60} is a 60×60 identity matrix, and $G_{60,8} = P^T$ is a 60×8 matrix used to generate the 8 parity bits as given in Table 177-4. Within each row the MSB is on the left.

The eight parity bits are calculated as

$$p<7:0> = m_xor<59:0> \cdot G_{60,8}$$

The output codeword $c<127:0>$ is:

$$\begin{aligned} c<127:8> &= m<119:0> \\ c<7:0> &= p<7:0> \end{aligned}$$

$$g(i) = \begin{bmatrix} \alpha^i \\ 1 \end{bmatrix} = \begin{bmatrix} s_{0,i} \\ s_{1,i} \\ s_{2,i} \\ s_{3,i} \\ s_{4,i} \\ s_{5,i} \\ s_{6,i} \\ 1 \end{bmatrix} \quad (177-1)$$

$$\alpha^i = s_{0,i} \times \alpha^0 + s_{1,i} \times \alpha^1 + s_{2,i} \times \alpha^2 + s_{3,i} \times \alpha^3 + s_{4,i} \times \alpha^4 + s_{5,i} \times \alpha^5 + s_{6,i} \times \alpha^6 \quad (177-2)$$

$$H = [g(0), g(1), g(2), \dots, g(66), g(67)] \quad (177-3)$$

$$P = B \cdot [g(0), g(1), g(2), \dots, g(58), g(59)] \quad (177-4)$$

$$B = [g(60), g(61), \dots, g(66), g(67)]^{-1} \quad (177-5)$$

Table 177-4—Generation matrix for Hamming(68,60)^a

G(0:11)	G(12:23)	G(24:35)	G(36:47)	G(48:59)
1,0,0,1,0,1,0,0	1,0,1,0,1,0,0,0	0,0,1,1,0,1,1,1	1,1,1,0,0,0,0,0	0,1,1,1,1,0,0,1
0,1,0,0,1,0,1,0	0,1,0,1,0,1,0,0	1,1,0,0,0,0,1,0	0,1,1,1,0,0,0,0	1,1,1,0,0,1,0,1
0,0,1,0,0,1,0,1	0,0,1,0,1,0,1,0	0,1,1,0,0,0,0,1	0,0,1,1,1,0,0,0	1,0,1,0,1,0,1,1
1,1,0,0,1,0,1,1	0,0,0,1,0,1,0,1	1,1,1,0,1,0,0,1	0,0,0,1,1,1,0,0	1,0,0,0,1,1,0,0
1,0,1,1,1,1,0,0	1,1,0,1,0,0,1,1	1,0,1,0,1,1,0,1	0,0,0,0,1,1,1,0	0,1,0,0,0,1,1,0
0,1,0,1,1,1,1,0	1,0,1,1,0,0,0,0	1,0,0,0,1,1,1,1	0,0,0,0,0,1,1,1	0,0,1,0,0,0,1,1
0,0,1,0,1,1,1,1	0,1,0,1,1,0,0,0	1,0,0,1,1,1,1,0	1,1,0,1,1,0,1,0	1,1,0,0,1,0,0,0
1,1,0,0,1,1,1,0	0,0,1,0,1,1,0,0	0,1,0,0,1,1,1,1	0,1,1,0,1,1,0,1	0,1,1,0,0,1,0,0
0,1,1,0,0,1,1,1	0,0,0,1,0,1,1,0	1,1,1,1,1,1,1,0	1,1,1,0,1,1,1,1	0,0,1,1,0,0,1,0
1,1,1,0,1,0,1,0	0,0,0,0,1,0,1,1	0,1,1,1,1,1,1,1	1,0,1,0,1,1,1,0	0,0,0,1,1,0,0,1
0,1,1,1,0,1,0,1	1,1,0,1,1,1,0,0	1,1,1,0,0,1,1,0	0,1,0,1,0,1,1,1	1,1,0,1,0,1,0,1
1,1,1,0,0,0,1,1	0,1,1,0,1,1,1,0	0,1,1,1,0,0,1,1	1,1,1,1,0,0,1,0	1,0,1,1,0,0,1,1

^a Within each row, the MSB is on the left.

177.4.6 8:1 bit-pair interleaving

After Inner FEC encoding, the eight flows of Inner FEC codewords shall be multiplexed together to form a single output lane. The 8:1 multiplexing is performed from flow 0 to flow 7, in a round-robin fashion, on pairs of bits beginning with c<127:126>. Each pair of bits will later be modulated into a PAM4 symbol.

177.4.7 Pad insertion and format

A 1024-bit pad shall be inserted after every 8704 Inner FEC codewords (see Figure 177-7). The nominal rate for each Inner FEC lane after pad insertion is 226.875 Gb/s. The ratio between the rate before and after pad insertion is 1088/1089. The bit rate after pad insertion is 363/340 of the bit rate of the tx_symbol stream at the Inner FEC service interface.

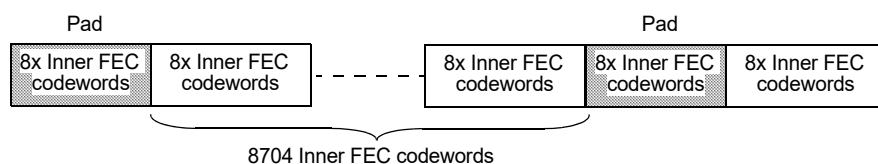


Figure 177-7—Pad insertion interval

The 1024-bit pad is equivalent to Inner FEC encoding (see 177.4.5) and 8:1 bit-pair interleaving (see 177.4.6) the $8 \times 120b$ data provided in Table 177-5, when Inner FEC flow i is fed pad_cw_ i ($i=0:7$) (see Figure 177-8).

Table 177–5—Pad format

	Pad frame alignment sequence $\text{pad_cw}_i<127:122>$, $\text{m}_i<119:114>$	Pad bits, $\text{pad_cw}_i<121:8>$, $\text{m}_i<113:0>$	Parity $\text{pad_cw}_i<7:0>$ $\text{p}_i<7:0>$
pad_cw_0	01 01 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_0<127:8>$
pad_cw_1	01 10 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_1<127:8>$
pad_cw_2	10 01 11	See 177.4.7.2	Calculated based on $\text{pad_cw}_2<127:8>$
pad_cw_3	01 00 01	See 177.4.7.2	Calculated based on $\text{pad_cw}_3<127:8>$
pad_cw_4	01 10 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_4<127:8>$
pad_cw_5	01 10 01	See 177.4.7.2	Calculated based on $\text{pad_cw}_5<127:8>$
pad_cw_6	00 01 10	See 177.4.7.2	Calculated based on $\text{pad_cw}_6<127:8>$
pad_cw_7	10 10 11	See 177.4.7.2	Calculated based on $\text{pad_cw}_7<127:8>$

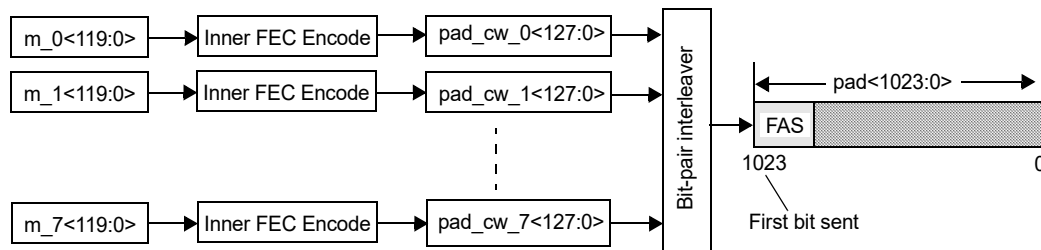


Figure 177–8—Pad codewords mapping

177.4.7.1 Pad frame alignment sequence

There is a fixed 48-bit Frame Alignment Sequence (FAS) pattern in each pad, in order to enable the receiver to locate and remove the pad. The FAS with the first bit transmitted on the leftmost is:
01011001 01010010 01100100 10100110 10101101 10011011.

177.4.7.2 Pad bits

Pad bits spanning across all 8 pad codewords, as shown in Table 177–5, are filled with 0 and then scrambled using a self-synchronizing PRBS13 scrambler using the same polynomial as Equation (94–3). The initial state of the scrambler is any non-zero state, and the scrambler state is retained from the previous pad.

177.4.7.3 Parity bits

The parity bits of a pad codeword are generated in the same way as for a payload codeword, as defined in 177.4.5.

177.4.8 PAM4 encoding

The PAM4 encoding function includes Gray mapping as specified in 177.4.8.1 and $1/(1+D) \bmod 4$ precoding as specified in 177.4.8.2.

177.4.8.1 Gray mapping

The Gray mapping for PAM4 encoded lanes is implemented as specified for output lanes in 120.5.7.1.

177.4.8.2 Precoding

The Inner FEC shall provide $1/(1+D) \bmod 4$ precoding capability on each transmit lane. Precoding is implemented as specified for output lanes in 135.5.7.2.

Precoding is enabled and disabled using variables `precoder_tx_out_enable_i` (where i is in the range 0 to 7). If training is enabled by the management variable `mr_training_enable` (see 178B.15), then `precoder_tx_out_enable_i` shall be set as determined by the inter-sublayer link training (ILT) function in the `PATH_READY` state on lane i (see Figure 178B–8). The method by which the ILT function affects this variable is implementation dependent. If ILT is disabled by the management variable `mr_training_enable`, `precoder_tx_out_enable_i` is set as required by the implementation.

177.5 Receive function

The following processes are performed independently on each PMD service interface input lane.

177.5.1 PAM4 decoding

The PAM4 decoding function includes inverse $1/(1+D) \bmod 4$ precoding as specified in 177.5.1.1 and inverse Gray mapping as specified in 177.5.1.2. The hard-decision PAM4 decoding function is depicted as a separate branch for synchronization and test pattern checkers in Figure 177–2, the soft-decision PAM4 decoding used for Inner FEC decoding process may be implemented anywhere in the receive function providing the net behavior is the same.

177.5.1.1 Inverse precoding

The Inner FEC may optionally provide inverse $1/(1+D) \bmod 4$ precoding capability on each receive lane.

If inverse precoding is implemented, it is enabled or disabled as determined by the implementation.

If inverse precoding is enabled, the Inner FEC receive function processes the detected data equivalent to the process specified for input lanes in 135.5.7.2.

If inter-sublayer link training (ILT) is enabled by the control variable `mr_training_enable` (see 178B.15), precoding of the received data is enabled at the peer interface transmitter as requested by the receiver using ILT. If ILT is disabled, then the precoding of data at the transmitter is controlled by a management entity.

NOTE— If inverse precoding operation is enabled as part of the PAM4 decoding, it also affects the decoding operation in 177.5.5, which does not use the output of the PAM4 decoding function.

177.5.1.2 Inverse Gray mapping

The inverse Gray mapping for PAM4 encoded lanes is identical to the process specified for input lanes in 120.5.7.1.

177.5.2 Inner FEC synchronization

On the receive side, the boundary of Inner FEC codewords shall be located before further processing is done. The eight codewords inserted as pad (see 177.4.7) are used to frame the data stream and are then removed before the received data is processed further.

Inner FEC synchronization may be performed on PAM4 decoded bit-pair stream using hard-decision decoding. 1:8 bit-pair deinterleaving (each pair of bits corresponding to a PAM4 symbol) is performed to eight Inner FEC flows. The initial position is not specified. Note that this may not be the correct position to perform the deinterleaving, as the first bit-pair after the deinterleaving position may not be on Inner FEC flow 0.

The Inner FEC achieves codeword synchronization as specified by the Inner FEC codeword synchronization state diagram shown in Figure 177–10. The synchronization process searches for a valid set of codewords in a window of 128-bit blocks, confirms the candidate location is valid for a second window of 128b-bit blocks and then monitors that the synchronization location continues to be valid during operation.

After all eight Inner FEC flows have successfully synchronized, Inner FEC flow 0 is identified based on the relative position of Inner FEC codeword boundaries in the synchronized Inner FEC flows. If the blind 1:8 bit-pair deinterleaving (each pair of bits corresponding to a PAM4 symbol) was performed on the correct location, the eight obtained Inner FEC codeword boundaries will be fully aligned, and the flow containing the first bit-pair (corresponding to one PAM4 symbol) after 1:8 bit-pair deinterleaving will be Inner FEC flow 0, as shown in Figure 177–9. Otherwise, there will be one bit-pair of skew between the eight Inner FEC flows, and Inner FEC flow 0 will be the one with a leading bit-pair (corresponding to one PAM4 symbol).

The Inner FEC codeword boundaries found by synchronization are used for deinterleaving of soft inputs into the Inner FEC decoding as described in 177.5.5.

177.5.3 Pad removal

Once Inner FEC flow 0 is identified, Inner FEC synchronization is achieved. Pad may then be searched based on the codeword boundaries and removed. The search of pad follows the same rules as AM common marker search defined in 119.2.6.2.2, that the 48-bit FAS is compared on a nibble-wise basis (12 comparisons). If nine or more nibbles in the candidate block match the FAS, the candidate block is considered a valid FAS. If two additional valid FASs (total of three consecutive FASs) are successfully located in the next two candidate positions each 8704 codewords away, FAS lock is achieved. The FAS may be optionally monitored, in which case if twelve consecutive FAS don't match the expected values, the FAS detect process may restart.

The state diagram of the FAS lock process is shown in Figure 177–11.

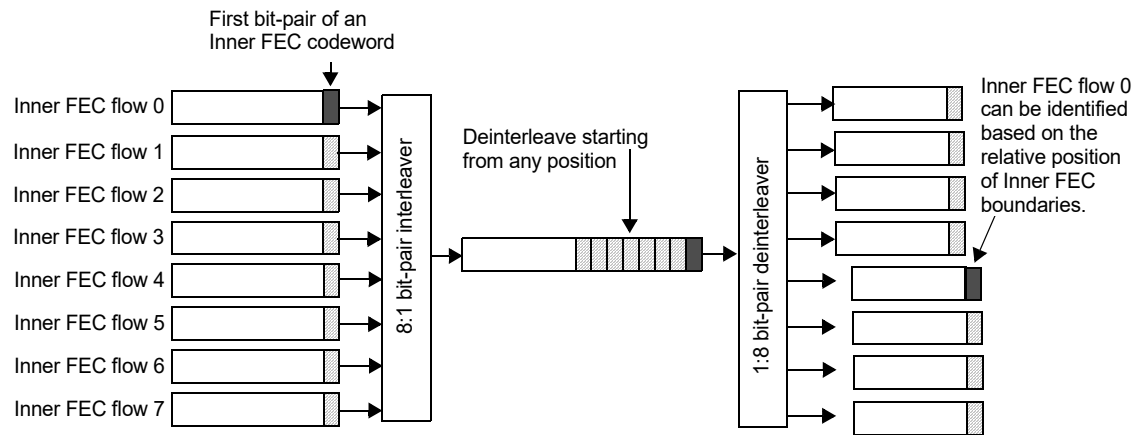


Figure 177–9—Identify Inner FEC flow 0

177.5.4 1:8 PAM4 symbol deinterleaving

The received data stream shall be deinterleaved in PAM4 symbols and distributed back to eight Inner FEC flows in a round-robin fashion. The first PAM4 symbol after the pad shall be sent to Inner FEC flow 0.

177.5.5 Inner FEC decode

The Inner FEC decoder is a soft-decision decoder that requires a higher resolution than four values for each received PAM4 symbol. The resolution is implementation specific and is beyond the scope of this standard. The decoder evaluates the incoming block of 64 rx_symbol inputs and determines the most likely payload value.

The decoder is expected to correct all codewords with one bit error. It may also be able to correct codewords with more than one bit error, with a chance of miscorrection.

The following counters shall be implemented per Inner FEC lane:

Inner_FEC_corrected_cw_counter

A 32-bit counter that counts once for each corrected Inner FEC codeword when Inner_FEC_sync_status is true. Mapping of the counters to management variables is specified in 177.10. A corrected Inner FEC codeword is a codeword that contained errors and was corrected or miscorrected. Note that when there is more than one bit error in a codeword there is a chance that the soft decision decoder could miscorrect the codeword. The Inner FEC decoder interprets miscorrected codewords as corrected codewords.

Inner_FEC_uncorrected_cw_counter

A 32-bit counter that counts once for each uncorrected Inner FEC codeword processed when Inner_FEC_sync_status is true. An uncorrected Inner FEC codeword is a codeword that contains errors that the decoder was unable to correct.

Inner_FEC_total_bits_counter

A 64-bit counter that counts once for each bit processed by the Inner FEC decoder. This may be used together with the Inner_FEC_corrected_bits_counter to roughly measure the pre-Inner-FEC BER.

Inner_FEC_corrected_bits_counter

A 64-bit counter that counts once for each bit modified by the Inner FEC decoder. This may be used together with the Inner_FEC_total_bits_counter to roughly measure the pre-Inner-FEC BER.

Inner_FEC_codeword_error_bin_k

A set of four 32-bit counters where counter k counts once for each codeword received with exactly k bits corrected (flipped) when fas_lock is true ($k = 0$ to 3). For example, if an Inner FEC codeword has exactly two bits corrected, then Inner_FEC_codeword_error_bin_2 is incremented. Error bin 3 increments when three or more bits are corrected in an Inner FEC codeword.

For each codeword processed, exactly one counter in Inner_FEC_codeword_error_bin_k or Inner_FEC_uncorrected_cw_counter is incremented.

177.5.6 Inverse circular shift

The inverse circular shift shall perform the inverse of the circular shift function (see 177.4.4) to restore the order of 12 RS-FEC symbols in each decoded codeword.

177.5.7 8:1 120-bit block multiplexer

After the inverse circular shift the eight Inner FEC flows shall be multiplexed into one stream on a 120-bit block (three RS-FEC symbol-quartet) basis in a round-robin fashion from Inner FEC flow 0 through 7 before being fed into the convolutional deinterleaver. This provides the inverse function of 177.4.3.

177.5.8 Convolutional deinterleaver

The convolutional deinterleaver shall perform the inverse function of the convolutional interleaver function (see 177.4.2) to restore the order of the data received to be compatible with the BASE-R SM-PMA.

177.6 Test patterns

177.6.1 Test pattern generation

The Inner FEC includes test pattern generators. All test pattern generators defined in this subclause are per Inner FEC lane. These test patterns are used to drive each transmit output lane towards the PMD service interface for PMD testing purposes.

Enabling any of the pattern generators on an Inner FEC lane only affects that lane. The behavior when more than one generator is enabled on the same lane is not specified.

177.6.1.1 PRBS31 encoded by Inner FEC test pattern

The Inner FEC shall support a test mode, where a PRBS31 test pattern generated from each transmit output lane i of the PMA above the Inner FEC (see 176.7.4.1) is processed by Inner FEC sublayer, with the exception that the alignment lock and deskew function on the Inner FEC transmit path (see 177.4.1) is disabled and bypassed.

This test mode is enabled on each FEC lane i using the PRBS31_encoded_by_Inner_FEC_ i management variable and enabling the PRBS31 generator on each transmit output lane i of the PMA above the Inner FEC.

177.6.1.2 PRBS31 test pattern

The Inner FEC shall include a PRBS31 test-pattern generator, as specified in 120.5.11.1.1, at the input to the PAM4 encoder.

177.6.1.3 PRBS13Q test pattern

The Inner FEC may include a PRBS13Q test-pattern generator, as specified in 120.5.11.2.1.

177.6.1.4 PRBS31Q test pattern

The Inner FEC shall include a PRBS31Q test pattern, provided by a PRBS31 generator (see 177.6.1.2) and the PAM4 encoder (see 177.4.8), on each transmit output lane towards the PMD service interface. The PAM4 encoder provides precoding (if enabled) and Gray mapping.

177.6.1.5 SSPRQ test pattern

The Inner FEC shall include a short stress pattern random quaternary (SSPRQ) test-pattern generator, as specified in 120.5.11.2.3.

177.6.1.6 Square wave (quaternary) test pattern

The Inner FEC may support a square wave (quaternary) test-pattern generator, as specified in 120.5.11.2.4.

177.6.2 Test pattern checking

The Inner FEC includes test pattern checkers. All test pattern checkers defined in this subclause are per Inner FEC lane. These test pattern checkers are used to check the signal on each receive input lane from the PMD service interface for PMD testing purposes.

177.6.2.1 PRBS31 test pattern checker

The Inner FEC shall include test pattern checkers for PRBS31, as specified in 120.5.11.1.1, at the output of the PAM4 decoder.

177.6.2.2 PRBS31Q test pattern checking

For PRBS31Q checking, the hard-decision PAM4 decoder (see 177.5.1) provides inverse precoding (if enabled) and inverse Gray mapping and the PRBS31 checker (see 177.6.2.1) checks for bit errors.

177.6.2.3 PRBS31 encoded by Inner FEC test pattern checker

The Inner FEC shall support a test mode, where a PRBS31 test pattern is encoded by the Inner FEC (see 176.7.4). For this test mode, the PRBS31 pattern is recovered by the Inner FEC receive function on each FEC lane i and the PRBS31 checkers in the PMA above the Inner FEC on each receive input lane i (see 176.7.4.1) are used to check the pattern.

177.7 State diagrams

This subclause is composed of state diagrams, including the associated definitions of constants, variables, functions, and counters.

177.7.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

177.7.2 State diagram definitions

177.7.2.1 Variables

all_synced

A Boolean variable that is set to true when sync_flow<x> is true for all eight flows and the Inner FEC flow 0 is identified, and is false otherwise.

fas_cnt_done

Boolean variable that indicates that fas_cnt has reached its terminal count.

fas_lock

A Boolean variable that is set to true when the receiver has detected the location of the frame alignment sequence within the pad codewords.

fas_valid

A Boolean variable that is set to true if the received 48-bit block is a valid frame alignment sequence payload. The frame alignment sequence payload is described in 177.4.7. The 48 bits are compared on a nibble-wise basis (12 comparisons). If nine or more nibbles in the candidate block match the corresponding nibbles in the sequence defined in 177.4.7, the candidate block is considered a valid frame alignment sequence payload.

FEC_reset

A Boolean variable that is true when set by a management entity and is false otherwise.

Inner_FEC_sync_status

A Boolean variable that is set to true when all_synced and fas_lock are both true and is set to false otherwise.

reset

A Boolean variable that controls the resetting of the Inner FEC sublayer. It is true whenever a reset is necessary including when initiated by FEC_reset and during power on.

restart_inner_fec_sync

A Boolean variable that is set by the Inner FEC synchronization process or the Inner FEC pad detection process.

signal_ok

A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive. It is true if the value is OK and false otherwise.

slip_done

A Boolean variable that is set to true when the SLIP requested by the Inner FEC synchronization state diagram has been completed indicating that the next candidate 128-bit block position can be tested.

sync_flow<x>

A Boolean variable that is set to true after the Inner FEC codeword boundary is found for an Inner FEC flow, where x = 0 to 7, and represents an Inner FEC flow ID before identifying the actual Inner FEC flow numbering.

test_cw
A Boolean variable that is set to true when a new Inner FEC codeword is available for decoding and is set to false when the CW_CHECK_N state is entered, where N = 1 to 3.

test_fas
A Boolean variable that is set to true when a candidate block position is available for testing and false when the FIND_1ST state is entered.

valid_cw
A Boolean variable that is set to true when the calculated syndrome of the Inner FEC codeword being tested is zero and is set to false otherwise.

177.7.2.2 Functions

SLIP
Causes the next candidate block position to be tested. The precise method for determining the next candidate block position is not specified and is implementation dependent. However, an implementation shall evaluate all possible block positions.

177.7.2.3 Counters

bad_cw_cnt
Counts the number of invalid Inner FEC codewords received within a period of 150 codewords. An invalid Inner FEC codeword is indicated when valid_cw is false.

bad_fas_cnt
Counts the number of consecutive FAS that don't match the expected values.

cw_cnt
Counts the number of Inner FEC codewords processed.

fas_cnt
Counts the interval of Inner FEC codewords between two adjacent pads as defined in 177.4.7.

valid_cw_cnt
Counts the number of valid Inner FEC codewords within a period of 50 codewords. A valid Inner FEC codeword is indicated when valid_cw is true.

177.7.3 State diagram figures

The Inner FEC sublayer shall implement eight self-synchronization processes, as shown in Figure 177–10, for each input lane in the receive direction. Each synchronization process operates independently on an Inner FEC flow to identify the boundaries of the Inner FEC codewords.

An Inner FEC Pad detection process as illustrated in the state diagram in Figure 177–10 shall be implemented for each input lane in the receive direction.

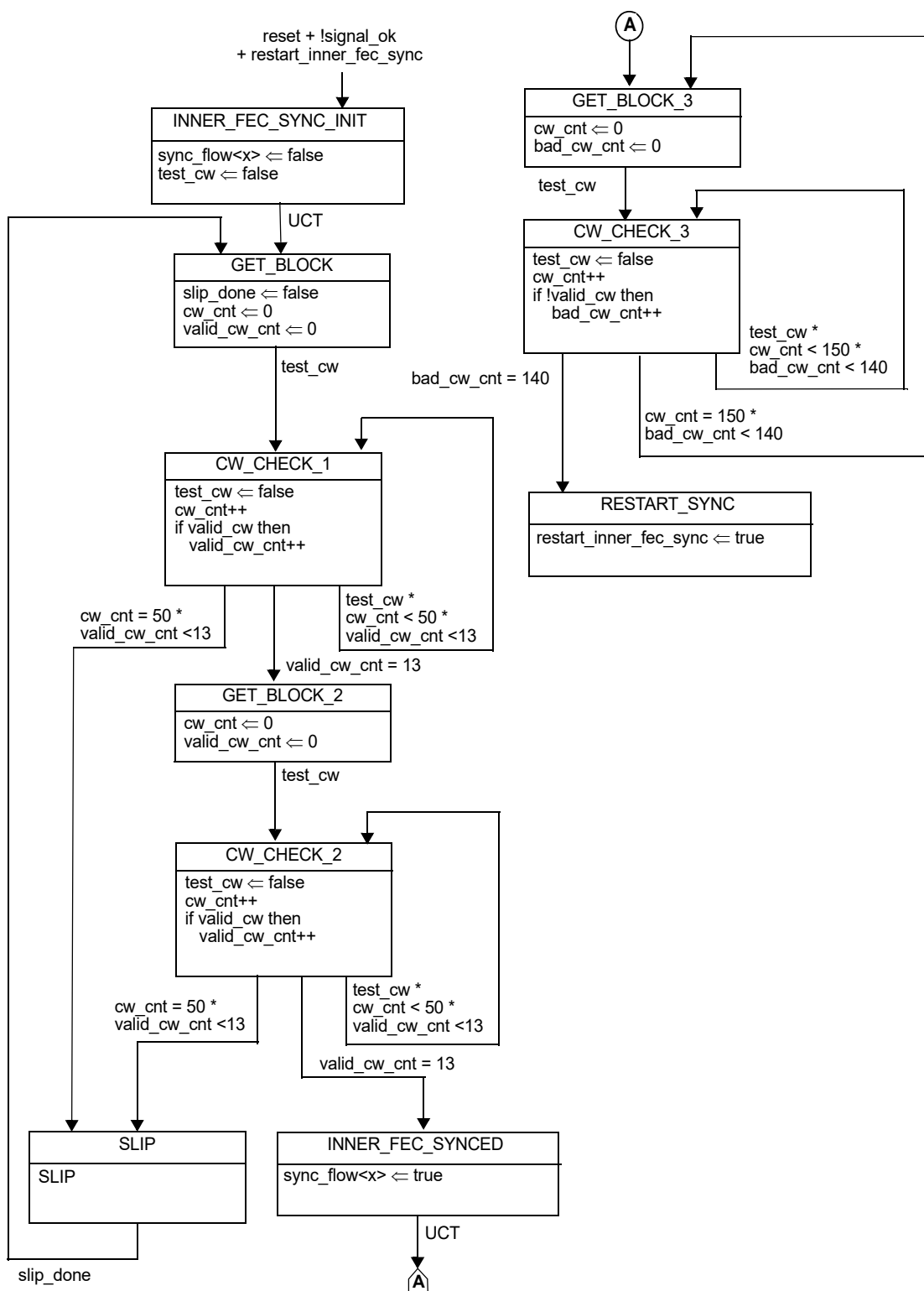


Figure 177–10—Inner FEC self-synchronization state diagram

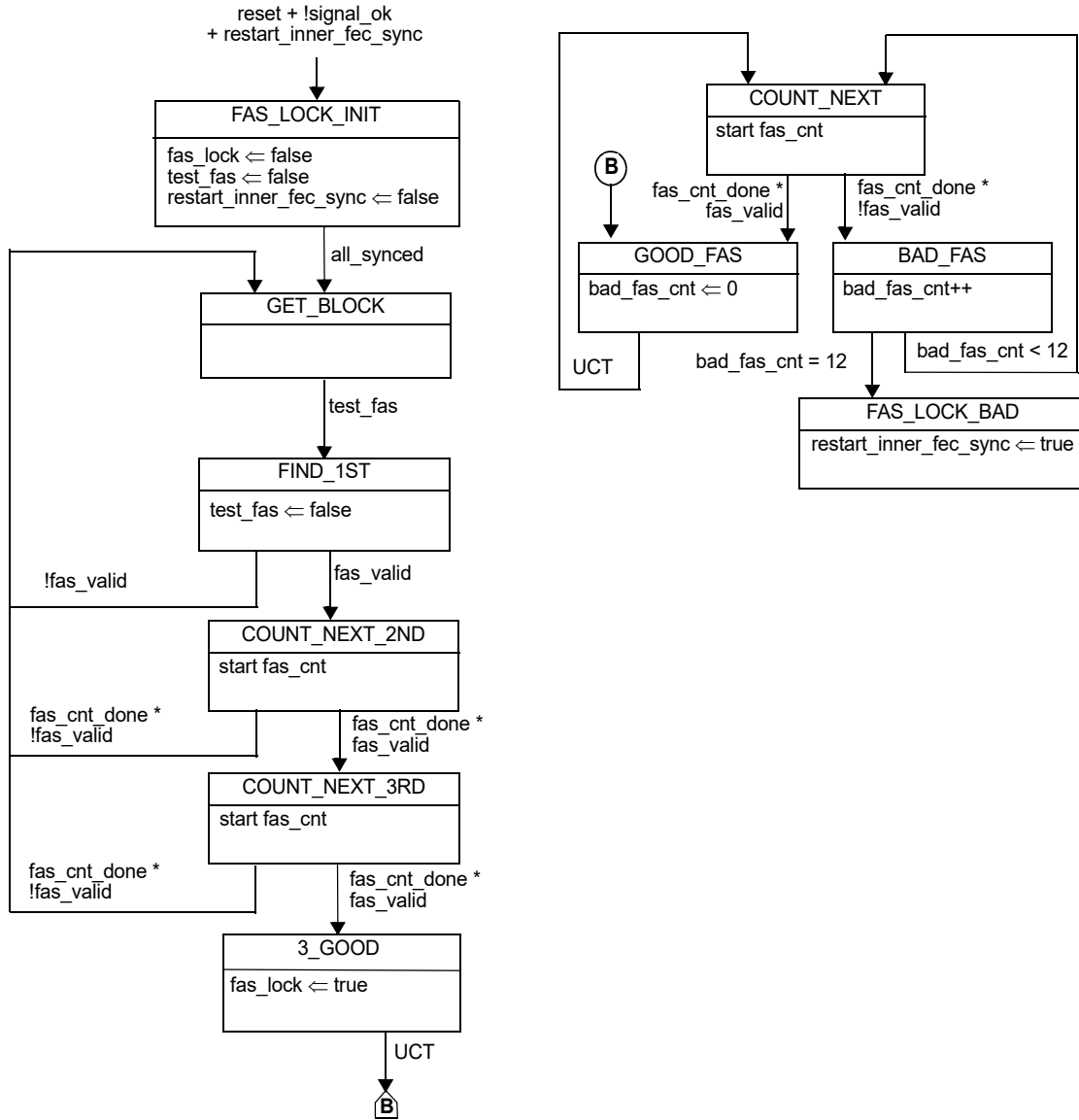


Figure 177-11—Inner FEC pad detection state diagram

177.8 Delay constraints

The maximum delay contributed by the Inner FEC (sum of transmit and receive delays at one end of the link) shall be no more than the values provided in Table 177–6. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4, 169.4, 174.4, and their references.

Table 177–6—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200GBASE-R Inner FEC	59 392	116	296.96
400GBASE-R Inner FEC	70 656	138	176.64
800GBASE-R Inner FEC	93 184	182	116.48
1.6TBASE-R Inner FEC	138 240	270	86.4

177.9 Path data delay for time synchronization

The Inner FEC sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report Inner FEC path data delays is indicated by the Inner_FEC_delay_ns_TX_ability, Inner_FEC_delay_subns_TX_ability, Inner_FEC_delay_ns_RX_ability, and Inner_FEC_delay_subns_RX_ability status variables (see Table 177–8).

The transmit and receive path data delays are reported as if the DDMP (data delay measurement point) occurs on the first symbol on FEC flow 0 after the 1024-bit pad insertion (see 177.4.7), corresponding to the longest delay for transmit and the shortest delay for receive (see 90.7).

Four separate delays are reported in the following eight path data delay status variables:

- Inner_FEC_delay_ns_TX_max, Inner_FEC_delay_subns_TX_max
- Inner_FEC_delay_ns_TX_min, Inner_FEC_delay_subns_TX_min
- Inner_FEC_delay_ns_RX_max, Inner_FEC_delay_subns_RX_max
- Inner_FEC_delay_ns_RX_min, Inner_FEC_delay_subns_RX_min

177.10 Management variables

Inner FEC control and status variables intended to be accessible via a management system are listed in Table 177–7 and Table 177–8.

If the MDIO Interface (see Clause 45) is implemented, Inner FEC control and status variables shall be mapped to MDIO registers as shown in Table 177–7 and Table 177–8.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 177–7—Inner FEC control variables and MDIO mapping

Control variable	Variable reference	MDIO Register/bit number	MDIO register/bit reference
FEC_reset	177.7.2.1	1.0.15	45.2.1.1
precoder_tx_out_enable_<0:7>	177.4.8.2	1.600.0:7	45.2.1.139
PRBS31_encoded_by_Inner_FEC<0:7>	177.6.1.1	1.2405.0:7	45.2.1.255
PRBS13Q_pattern_enable<0:7>	177.6.1.3	1.2401.0:7	45.2.1.251
PRBS31_pattern_enable<0:7>	177.6.1.2	1.2402.0:7	45.2.1.252
SSPRQ_pattern_enable<0:7>	177.6.1.5	1.2403.0:7	45.2.1.253
Square_wave_enable<0:7>	177.6.1.6	1.2404.0:7	45.2.1.254

Table 177–8—Inner FEC status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
pmal_locked_demux, Inner FEC lane 0 through 7	176.4.4.2.1	1.2410.8:15	45.2.1.256
Inner_FEC_sync_status, Inner FEC lane 0 through 7	177.7.2.1	1.2410.0:7	45.2.1.256
Inner_FEC_corrected_cw_counter (Inner FEC lane 0)	177.5.5	1.2412, 1.2413	45.2.1.258
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 0)	177.5.5	1.2414, 1.2415	45.2.1.259
Inner_FEC_total_bits_counter (Inner FEC lane 0)	177.5.5	1.2416, 1.2417, 1.2418, 1.2419	45.2.1.260
Inner_FEC_corrected_bits_counter (Inner FEC lane 0)	177.5.5	1.2420, 1.2421, 1.2422, 1.2423	45.2.1.261
Inner_FEC_codeword_error_bin_k (Inner FEC lane 0)	177.5.5	1.2424, 1.2425, 1.2426, 1.2427, 1.2428, 1.2429, 1.2430, 1.2431	45.2.1.262
Inner_FEC_corrected_cw_counter (Inner FEC lane 1)	177.5.5	1.2434, 1.2435	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 1)	177.5.5	1.2436, 1.2437	45.2.1.263

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_total_bits_counter (Inner FEC lane 1)	177.5.5	1.2438, 1.2439, 1.2440, 1.2441	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 1)	177.5.5	1.2442, 1.2443, 1.2444, 1.2445	45.2.1.263
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 1)	177.5.5	1.2446, 1.2447, 1.2448, 1.2449, 1.2450, 1.2451, 1.2452, 1.2453	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 2)	177.5.5	1.2456, 1.2457	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 2)	177.5.5	1.2458, 1.2459	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 2)	177.5.5	1.2460, 1.2461, 1.2462, 1.2463	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 2)	177.5.5	1.2464, 1.2465, 1.2466, 1.2467	45.2.1.263
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 2)	177.5.5	1.2468, 1.2469, 1.2470, 1.2471, 1.2472, 1.2473, 1.2474, 1.2475	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 3)	177.5.5	1.2478, 1.2479	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 3)	177.5.5	1.2480, 1.2481	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 3)	177.5.5	1.2482, 1.2483, 1.2484, 1.2485	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 3)	177.5.5	1.2486, 1.2487, 1.2488, 1.2489	45.2.1.263

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 3)	177.5.5	1.2490, 1.2491, 1.2492, 1.2493, 1.2494, 1.2495, 1.2496, 1.2497	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 4)	177.5.5	1.2500, 1.2501	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 4)	177.5.5	1.2502, 1.2503	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 4)	177.5.5	1.2504, 1.2505, 1.2506, 1.2507	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 4)	177.5.5	1.2508, 1.2509, 1.2510, 1.2511	45.2.1.263
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 4)	177.5.5	1.2512, 1.2513, 1.2514, 1.2515, 1.2516, 1.2517, 1.2518, 1.2519	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 5)	177.5.5	1.2522, 1.2523	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 5)	177.5.5	1.2524, 1.2525	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 5)	177.5.5	1.2526, 1.2527, 1.2528, 1.2529	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 5)	177.5.5	1.2530, 1.2531, 1.2532, 1.2533	45.2.1.263
Inner_FEC_codeword_error_bin_ <i>k</i> (Inner FEC lane 5)	177.5.5	1.2534, 1.2535, 1.2536, 1.2537, 1.2538, 1.2539, 1.2540, 1.2541	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 6)	177.5.5	1.2544, 1.2545	45.2.1.263

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 6)	177.5.5	1.2546, 1.2547	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 6)	177.5.5	1.2548, 1.2549, 1.2550, 1.2551	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 6)	177.5.5	1.2552, 1.2553, 1.2554, 1.2555	45.2.1.263
Inner_FEC_codeword_error_bin_k (Inner FEC lane 6)	177.5.5	1.2556, 1.2557, 1.2558, 1.2559, 1.2560, 1.2561, 1.2562, 1.2563	45.2.1.263
Inner_FEC_corrected_cw_counter (Inner FEC lane 7)	177.5.5	1.2566, 1.2567	45.2.1.263
Inner_FEC_uncorrected_cw_counter (Inner FEC lane 7)	177.5.5	1.2568, 1.2569	45.2.1.263
Inner_FEC_total_bits_counter (Inner FEC lane 7)	177.5.5	1.2570, 1.2571, 1.2572, 1.2573	45.2.1.263
Inner_FEC_corrected_bits_counter (Inner FEC lane 7)	177.5.5	1.2574, 1.2575, 1.2576, 1.2577	45.2.1.263
Inner_FEC_codeword_error_bin_k (Inner FEC lane 7)	177.5.5	1.2578, 1.2579, 1.2580, 1.2581, 1.2582, 1.2583, 1.2584, 1.2585	45.2.1.263
PRBS13Q_gen_Tx_ability	177.6.1.3	1.1500.11	45.2.1.169
Square_wave_ability	177.6.1.6	1.1500.12	45.2.1.169
Inner_FEC_delay_ns_TX_max	177.9	1.1813, 1.1814	45.2.1.177a
Inner_FEC_delay_subns_TX_max	177.9	1.1817	45.2.1.177a
Inner_FEC_delay_ns_TX_min	177.9	1.1815, 1.1816	45.2.1.177a
Inner_FEC_delay_subns_TX_min	177.9	1.1818	45.2.1.177a
Inner_FEC_delay_ns_RX_max	177.9	1.1819, 1.1820	45.2.1.177b
Inner_FEC_delay_subns_RX_max	177.9	1.1823	45.2.1.177b

Table 177–8—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_delay_ns_RX_min	177.9	1.1821, 1.1822	45.2.1.177b
Inner_FEC_delay_subns_RX_min	177.9	1.1824	45.2.1.177b
Inner_FEC_delay_ns_RX_ability	177.9	1.1800.4	45.2.1.175
Inner_FEC_delay_subns_RX_ability	177.9	1.1800.5	45.2.1.175
Inner_FEC_delay_ns_TX_ability	177.9	1.1800.6	45.2.1.175
Inner_FEC_delay_subns_TX_ability	177.9	1.1800.7	45.2.1.175

177.11 Protocol implementation conformance statement (PICS) proforma for Clause 177, Inner FEC sublayer, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R

177.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 177, Inner FEC, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

177.11.2 Identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

177.11.2.1 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 177, Inner FEC, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

177.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO	177.10		O	Yes [] No []

177.11.4 PICS proforma tables for Inner FEC sublayer, type 200GBASE-R, 400GBASE-R, 800GBASE-R and 1.6TBASE-R

177.11.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Convolutional interleaver	177.4.2		M	Yes []
TF2	120-bit block distribution	177.4.3		M	Yes []
TF3	Circular shift	177.4.4		M	Yes []
TF4	Inner FEC encode	177.4.5		M	Yes []
TF5	8:1 PAM4 interleaving	177.4.6		M	Yes []
TF6	Pad insertion and format	177.4.7		M	Yes []

177.11.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Inner FEC synchronization and pad removal	177.5.2		M	Yes []
RF2	1:8 PAM4 deinterleaving	177.5.4		M	Yes []
RF3	Inner FEC decode	177.5.5		M	Yes []
RF4	Inverse circular shift	177.5.6		M	Yes []
RF5	120-bit block multiplex	177.5.7		M	Yes []
RF6	Convolutional deinterleaver	177.5.8		M	Yes []

177.11.4.3 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	Inner FEC synchronization process	177.7.3		M	Yes []
SM2	Pad lock process	177.7.3		M	Yes []

177.11.4.4 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
TIM1		177.8		M	Yes []

177.11.4.5 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	177.9		O	Yes [] No []
TS1	Path data delays	177.9	Path data delays reported as if the DDMP is at the first symbol on FEC flow 0 after the 1024-bit pad insertion	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	177.9	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	177.9	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

178. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8

178.1 Overview

This clause specifies the 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 PMDs and the associated baseband media. The PMDs provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links on one, two, four, or eight lanes, respectively, over an electrical backplane. The specifications for the four PMDs are similar, except for the number of lanes and associated parameters, and the MDI.

This clause defines specifications for two classes of transmitters and two classes of receivers, identified by transmitter package class and receiver package class, respectively. The package is either class A or class B. Devices conform to electrical specifications of either class A or class B. The required characteristics of the electrical interconnect between two devices depend on the transmitter package class on one device and the receiver package class on the other device.

The associated Annex 178A provides measurement methods for electrical channels and interfaces.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate sublayers (as specified in Table 178–1 through Table 178–4) and to the medium through the appropriate MDI, as illustrated in Figure 178–1.

Table 178–1—Physical Layer clauses associated with the 200GBASE-KR1 PMD

Associated clause	200GBASE-KR1
73—AN	Required
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120F—200GAUI-2 C2C ^c	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
178B—ILT	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.4.1.

^c When implemented in a 200GBASE-KR1 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 200GBASE-KR1 PHY may include one instance of 200GAUI-n C2C as described in 176B.4.1.

Table 178–2—Physical Layer clauses associated with the 400GBASE-KR2 PMD

Associated clause	400GBASE-KR2
73—AN	Required
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
178B—ILT	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.5.1.

^c When implemented in a 400GBASE-KR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 400GBASE-KR2 PHY may include one instance of 400GAUI-n C2C as described in 176B.5.1.

Table 178–3—Physical Layer clauses associated with the 800GBASE-KR4 PMD

Associated clause	800GBASE-KR4
73—AN	Required
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
178B—ILT	Required

^a A 800GBASE-KR4 PHY may include one instance of 800GAUI-n C2C as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Table 178–4—Physical Layer clauses associated with the 1.6TBASE-KR8 PMD

Associated clause	1.6TBASE-KR8
73—AN	Required
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
170—1.6 Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII extender	Optional
172—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a
178B—ILT	Required

- ^a A 1.6TBASE-KR8 PHY may include one instance of 1.6TAUI-n C2C as described in 176B.7.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to the guidelines in 176B.7.1.

Figure 178–1 shows the relationship of the PMD and MDI, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model.

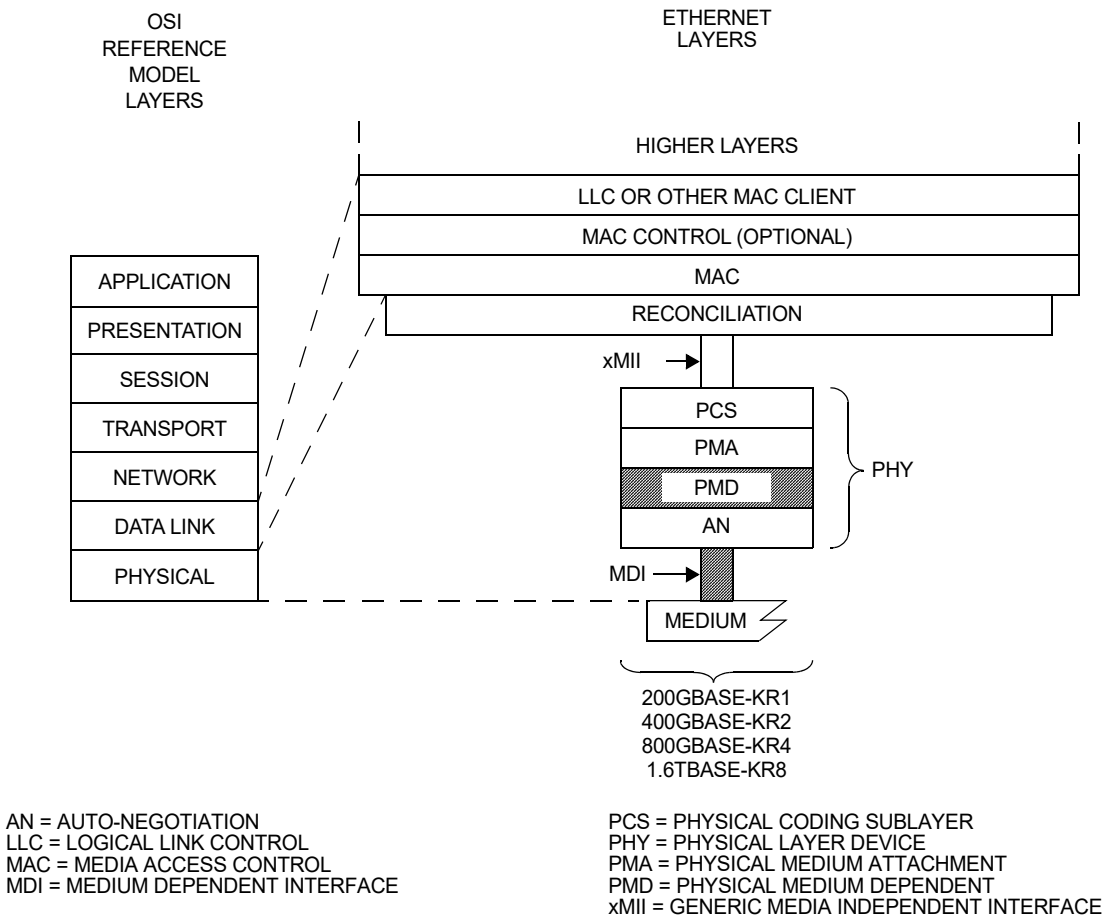


Figure 178–1—200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

178.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the PMD using the method described in 174A.8, with BER_{added} equal to 1.6×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 8×10^{-6} .

178.3 Conventions

This clause describes four PMDs, 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8, which have one, two, four, and eight lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 200GBASE-KR1, $n = 2$ for 400GBASE-KR2, $n = 4$ for 800GBASE-KR4, and $n = 8$ for 1.6TBASE-KR8.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 200GBASE-KR1 PMD, 400GBASE-KR2 PMD, 800GBASE-KR4 PMD, or 1.6TBASE-KR8 PMD.

Within this clause, the term FEC refers to the RS-FEC within the Clause 119 PCS for 200GBASE-KR1 and 400GBASE-KR2, the Clause 172 PCS for 800GBASE-KR4, and the Clause 175 PCS for 1.6TBASE-KR8.

178.4 Service interfaces

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-KR1 and 400GBASE-KR2, in 169.3 for 800GBASE-KR4, and in 174.3 for 1.6TBASE-KR8. The nominal signaling rate for each symbol stream in `PMD:IS_UNITDATA_i.request` and `PMD:IS_UNITDATA_i.indication`, where $i = 0$ to $n-1$, is 106.25 GBd.

The `SIGNAL_OK` parameter of the `PMD:IS_SIGNAL.indication` primitive corresponds to the variable `training_status` of the inter-sublayer training function, as defined in 178B.14.2.1. When `SIGNAL_OK` is either `IN_PROGRESS` or `FAIL`, the `rx_symbol` parameters of `PMD:IS_UNITDATA_i.indication` on all lanes are unspecified.

178.5 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive `AN_LINK.indication` defined in 73.9. The requirements are the same as those specified in 179.5.

178.6 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 178–5. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 178–5—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quantum) ^b	Maximum (ns)
200GBASE-KR1	14 848	29	74.24
400GBASE-KR2	29 696	58	74.24
800GBASE-KR4	59 392	116	74.24
1.6TBASE-KR8	118 784	232	74.24

^a One bit time is equal to 5 ps for 200GBASE-KR1, 2.5 ps for 400GBASE-KR2, 1.25 ps for 800GBASE-KR4, and 625 fs for 1.6TBASE-KR8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GBASE-KR1, 1.28 ns for 400GBASE-KR2, 640 ps for 800GBASE-KR4, and 256 ps for 1.6TBASE-KR8. See 31B.2 for the definition of pause_quantum.

Descriptions of overall system delay constraints can be found in 116.4 for 200GBASE-KR1 and 400GBASE-KR2, in 169.4 for 800GBASE-KR4, and in 174.4 for 1.6TBASE-KR8.

178.7 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

178.7.1 Skew Constraints for 200GBASE-KR1 and 400GBASE-KR2

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5. Skew Variation at SP2 to SP5 is specified only for 400GBASE-KR2. For 200GBASE-KR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

178.7.2 Skew Constraints for 800GBASE-KR4 and 1.6TBASE-KR8

Skew and Skew Variation are defined in 169.5 for 800GBASE-KR4 and in 174.5 for 1.6TBASE-KR8. They are specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5 for 800GBASE-KR4, and in Figure 174–5 and Figure 174–6 for 1.6TBASE-KR8.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5 and 174.5.

178.8 PMD functional specifications

178.8.1 Specified test points

The test points are illustrated in Figure 178–2, which shows one direction of a 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, or 1.6TBASE-KR8 link.

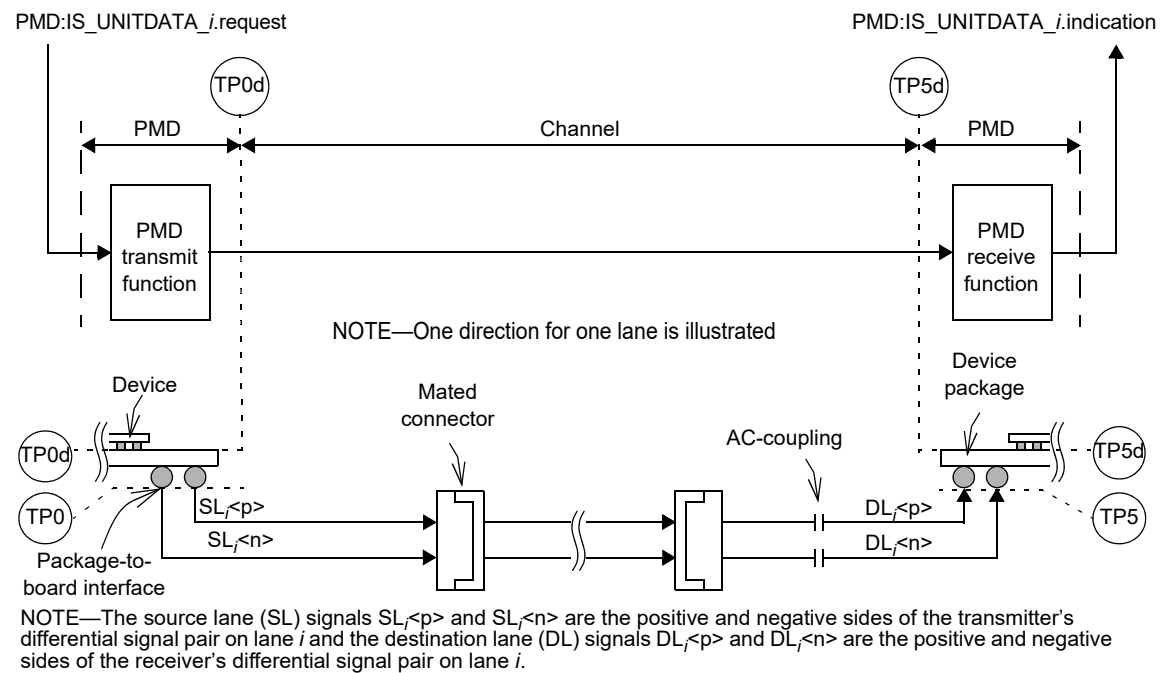


Figure 178–2—200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, or 1.6TBASE-KR8 link

178.8.2 PMD transmit function

The PMD transmit function specification is identical to that of 179.8.2 with the exception that electrical signals are delivered to the MDI according to the transmit electrical specifications in 178.9.2.

178.8.3 PMD receive function

The PMD receive function specification is identical to that of 179.8.3 with the exception that electrical signals are received from the MDI according to the receive electrical specifications in 178.9.3.

178.8.4 PMD global signal detect function

The PMD global signal detect function specification is identical to that of 179.8.4.

178.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function specification is identical to that of 179.8.5.

178.8.6 PMD global transmit disable function

The PMD global transmit disable function is optional. Its specification is identical to that of 179.8.6.

178.8.7 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function is optional. Its specification is identical to that of 179.8.7.

178.8.8 PMD fault functions

The specification of PMD fault functions is identical to that of 179.8.8.

178.8.9 Inter-sublayer link training (ILT) function

A PMD shall provide the inter-sublayer link training (ILT) function for a Type E1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), control the transmitter output on each lane of the MDI, indicate the receiver state, and coordinate the transition to DATA mode.

The PMD transmit function is configured in response to the requests in the incoming training frames as specified in 179.9.4.1.3 through 179.9.4.1.5.

The PMD transmit function supports the coefficient indexes `k_list` = {−3, −2 −1, 0, 1} and the initial conditions preset 1 through preset 6 and initialize (see Table 179–8).

178.8.10 PMD reset function

If the variable `PMD_reset` is asserted, the PMD shall be reset as defined in 45.2.1.1.1. PMD reset shall reset the ILT function associated with the PMD (see 178.8.9).

178.9 Electrical characteristics

178.9.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

178.9.2 Transmitter characteristics

The transmitter on each lane shall meet the specifications at TP0v given in Table 178–6 and detailed in the referenced subclauses.

Unless otherwise specified, transmitter signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP0v to the test equipment.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP0v where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

Table 178–6—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate (range)		106.25 $\pm$ 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.1		
Transmitter disabled		0.03	V
Transmitter enabled		1	V
DC common-mode voltage (range)	176D.8.1	0.2 to 1	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	179.9.4.1	0.03	V
Signal to AC common-mode noise ratio, $SCMR$ (min)	178.9.2.6	15	dB
Difference effective return loss, $dERL$ (min)	178.9.2.2	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	178.9.2.3	3.25	dB
Difference steady-state voltage, dv_f (range)	178.9.2.4	0 to 0.1	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	178.9.2.5	0	—
Level separation mismatch ratio, R_{LM} (min)	179.9.4.2	0.95	—
Transmitter waveform			
absolute value of step size for all taps (min)	179.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	179.9.4.1.4	0.025	—
value at minimum state for $c(-3)$ (max)	179.9.4.1.5	–0.06	—
value at maximum state for $c(-2)$ (min)	179.9.4.1.5	0.12	—
value at minimum state for $c(-1)$ (max)	179.9.4.1.5	–0.34	—
value at minimum state for $c(0)$ (max)	179.9.4.1.5	0.5	—
value at minimum state for $c(1)$ (max)	179.9.4.1.5	–0.2	—
Difference signal-to-noise-and-distortion ratio, $dSNDR$ (min)	178.9.2.7	0	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	179.9.4.3	28	dB
Output jitter (max)	179.9.4.6		
J_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
J_{4u03}			
Tx package Class A		0.118	UI
Tx package Class B		0.12	UI

178.9.2.1 Transmitter test fixture

Unless otherwise noted, measurements of the transmitter are made at the output of a test fixture (TP0v) as shown in Figure 178–3 and described in [Annex 163A](#). An example test fixture is described in [Annex 163B](#).

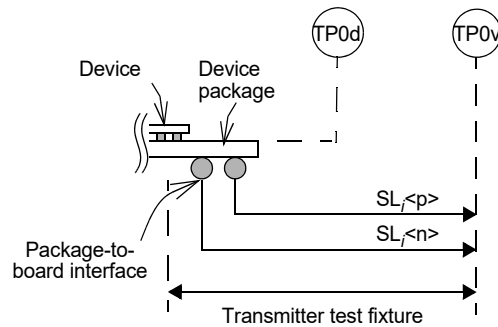


Figure 178-3—Transmitter test fixture and test points

178.9.2.1.1 Test fixture insertion loss

The insertion loss of the test fixture shall be between 3.4 dB and 4.4 dB at 53.125 GHz. The magnitude of the insertion loss deviation of the test fixture shall be less than or equal to 0.2 dB from 0.05 GHz to 67 GHz. Insertion loss deviation is calculated as specified in 93A.4, where T_t is 0.005 ns, and f_b and f_r values are taken from Table 178-12.

178.9.2.1.2 Test fixture effective return loss (ERL)

The ERL of the test fixture at TP0v is computed using the procedure in 93A.5 with the values in Table 178-7. Parameters that do not appear in Table 178-7 take values from Table 178-12.

Table 178-7—Test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

The ERL at TP0v shall be greater than or equal to 15 dB.

178.9.2.1.3 Test fixture common-mode to common-mode return loss

The common-mode to common-mode return loss of the test fixture shall be greater than or equal to 6 dB at all frequencies between 0.2 GHz and 67 GHz.

178.9.2.2 Transmitter difference ERL

The difference ERL of the transmitter at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 178–7 and Table 178–12, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v. The reference value, $ERL^{(ref)}$, is calculated based on the receiver package class to which the device adheres.

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) in Table 178–6

Table 178–8—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Tukey window flag	tw	1	—

178.9.2.3 Transmitter common-mode to common-mode return loss

The common-mode to common-mode return loss shall be greater than or equal to RL_{cc} (min) in Table 178–6 at all frequencies between 0.2 GHz and 60 GHz.

178.9.2.4 Difference steady-state voltage

The measured steady-state voltage of the transmitter at TP0v $v_f^{(meas)}$ and the difference steady-state voltage dv_f are computed using the procedure in 163A.3.2.1 with $N_v = 400$, $M = 32$, $D_p = 4$, and other parameter values specified in Table 178–12 and Table 178–13. The reference value $v_f^{(ref)}$ is calculated based on the receiver package class to which the device adheres.

The difference steady-state voltage at TP0v shall meet the specification dv_f (min) in Table 178–6.

178.9.2.5 Difference linear fit pulse peak ratio

The difference linear fit pulse peak ratio of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with parameters specified in Table 178–12.

The difference linear fit pulse peak ratio at TP0v shall meet the specification dR_{peak} (min) in Table 178–6. The reference value $R_{peak}^{(ref)}$ is calculated based on the transmitter package class to which the device adheres.

178.9.2.6 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is calculated using Equation (178–1).

$$SCMR = 20\log_{10}\left(\frac{v_{peak}}{VCM_{FB}}\right) \quad (178-1)$$

where

$SCMR$	is the signal to AC common-mode noise ratio in dB
v_{peak}	is defined in 179.9.4.1.2
VCM_{FB}	is the full-band peak-to-peak AC common-mode voltage defined by the method specified in 179.9.4.1 and measured with the transmitter equalization set to “no equalization”

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 178–6.

178.9.2.7 Difference signal-to-noise-and-distortion ratio

Difference signal-to-noise-and-distortion ratio (dSNDR) for transmitter output is defined in 179.9.4.5 with the following exceptions:

- Parameter values are taken from Table 178–13.
- In the calculation of the reference channel transfer function in 179.9.4.5.2, the transmitter S-parameter model, $S^{(t)}$, uses the parameters in Table 178–12 that correspond to the package class to which the device adheres and S is the measured S-parameter matrix of the test fixture described in 178.9.2.1.

178.9.3 Receiver characteristics

The receiver on each lane shall meet the specifications at TP5v given in Table 178–9 and detailed in the referenced subclauses.

Table 178–9—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate (range)	178.9.3.2	106.25 ± 50 ppm	GBd
Amplitude tolerance ^a	178.9.3.3	0.5	V
Interference tolerance	178.9.3.4	Table 178–10	—
Jitter tolerance	178.9.3.5	Table 179–17	—
Difference effective return loss, $dERL$ (min)	178.9.3.6	–3	dB
Differential-mode to common-mode return loss, RL_{cd}	178.9.3.7	Equation (178–4)	dB

^a Specified as the steady-state voltage $v_f^{(meas)}$ (as defined in 178.9.2.4) measured at the test transmitter’s output.

178.9.3.1 Receiver test fixture

Unless otherwise noted, measurements of the receiver are made at the input of a test fixture (TP5v) as shown in Figure 178–4.

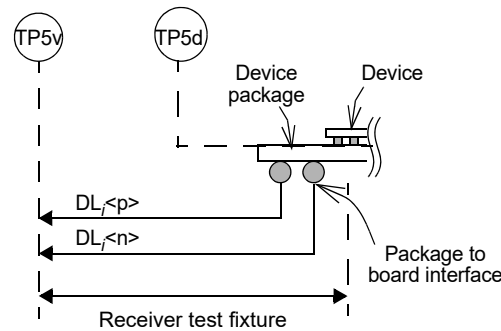


Figure 178–4—Receiver test fixture and test points

Receiver test fixture characteristics are the same as specified in 178.9.2.1.

178.9.3.2 Receiver signaling rate

A PHY shall comply with the receiver requirements of 178.9.3.4 and 178.9.3.5 for any signaling rate in the range specified in Table 178–9.

178.9.3.3 Receiver amplitude tolerance

When a PMD receiver is connected to a compliant transmitter that has the maximum allowed steady-state voltage (see Table 178–9), using a channel with the minimum insertion loss specified in 178.9.3.4, the PMD receiver operation shall enable a block error ratio as specified in 178.2.

The receiver is allowed to control the transmitter equalizer coefficients, using the ILT function (see 178.8.9) or an equivalent process, to meet these requirements.

178.9.3.4 Receiver interference tolerance

The receiver on each lane shall meet the expected block error ratio specified in 178.2 with channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 178–10.

178.9.3.4.1 Test setup

The interference tolerance test is performed with the setup shown in Figure 93C–2 or its equivalent. Characterization and calibration of the various elements in the test setup uses the test configurations in Figure 93C–3, Figure 93C–4, Figure 93C–5, and Figure 93C–6. In each figure TP0v replaces TP0a and TP5v replaces TP5a.

The transmitter meets the requirements stated in 178.9.2 with the added constraint that the differential peak-to-peak voltage (see 176D.8.1) is less than or equal to 0.8V for any transmitter equalizer setting. The ISI channel emulates the frequency dependent loss of the channel. The channel noise source emulates crosstalk, noise, and any other non-equalizable signal distortions that may be introduced by a transmitter or channel.

The ERL of the test setup in Figure 93C–4 measured at TP5 replica towards TPt meets the requirements in 178.10.3.

The channel noise source has an adjustable output amplitude that is set according to the calibration procedure in 178.9.3.4.2. The noise produced by the channel noise source is measured directly at the output of the noise source (see Figure 93C-6). The noise is Gaussian with a crest factor of at least 5. The noise spectral density, $NSD(f)$, is normalized and constrained according to the relations in Equation (93C-1), where f_b is the signaling rate and f_{NSDI} is 1 GHz. The average noise spectral density, $NSD_{average}$, is defined by Equation (93C-2).

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used to measure the output of the transmitter and the channel noise source.

178.9.3.4.2 Noise calibration

The output amplitude of channel noise source is set to yield a target test channel Channel Operating Margin (COM). The test channel COM is derived from the S-parameters for the test channel from TPt to TP5 replica $S^{(c)}$ measured using the test setup in Figure 93C-4. COM is calculated according to Annex 178A using the parameter values in Table 178-12 and Table 178-13 for the corresponding receiver class but with the following exceptions.

The transmitter model is determined in one of the following ways.

- a) If the transmitter is a device with known S-parameters and transition time T_r , these parameters are used instead of the transmitter device package model in 178A.1.4, and the $ILdd$ in Table 178-10 is the $ILdd$ of the concatenated S-parameters of the device and the channel between TPt and TP5 replica. T_r should be provided as the value at the input of the device S-parameters network, as defined in 120G.3.1.4 but with no observation filter.
- b) If a calibrated instrument-grade transmitter is used, the TP0 to TP0a trace in Figure 93C-2 and Figure 93C-3 and the TP0 to TP0a replica trace in Figure 93C-4 are omitted and the transmitter model $S^{(t)}$ is omitted from Equation (178A-2). In this case, the $ILdd$ in Table 178-10 is the $ILdd$ of the channel between TPt and TP5 replica. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A-10) uses the filter $H_i(f)$ defined in 178A.1.6.2, where T_r is the measured transmitter transition time (see 120G.3.1.4).

In addition, the following parameters are derived from measurements of the test transmitter collected using the test setup in Figure 93C-3.

- a) The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 85$.
- b) The parameter R_{LM} is set to the measured value of R_{LM} .
- c) The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{4u03} and J_{RMS} using Equation (178-2) and Equation (178-3) respectively.

The interference noise (see 178A.1.5) is derived from the S-parameters for the noise path from TPn to TP5 replica $S^{(n)}$ measured using the test setup shown in Figure 93C-5. For a given RMS noise amplitude σ_{ns} , the spectral density of the channel noise source $N_s(f)$ may be assumed to be $2\sigma_{ns}^2/f_b$ from DC to $f_b/2$ and 0 for frequencies greater than $f_b/2$. The spectral density measured at the output of the channel noise source, $NSD(f)$, may also be used as $N_s(f)$ for the calculation of COM and, in such cases, the value of σ_{ns}^2 is the integral of $NSD(f)$ over frequency.

COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 178-12 on the receiver side. The value of COM is taken as the lower of the two calculated values. The value of σ_{ns} that yields the target value of COM specified in Table 178-10 is determined (via iterative calculation or by other means). The amplitude of the channel noise source is then configured so that the measured RMS value at its output matches σ_{ns} .

$$A_{DD} = \frac{\frac{J4u_{03}}{2} + Q4d \sqrt{(Q4d^2 + 1) \times J_{RMS}^2 - \left(\frac{J4u_{03}}{2}\right)^2}}{Q4d^2 + 1} \quad (178-2)$$

$$\sigma_{RJ} = \frac{\frac{J4u_{03}}{2} - A_{DD}}{Q4d} \quad (178-3)$$

where

$$Q4d = 3.719$$

NOTE 1— $Q4d$ is an approximated solution of $Q(Q4d) = 1 \times 10^{-4}$, where the Q function is defined in Equation (95-1).

NOTE 2—Calculation of A_{DD} requires that $(Q4d^2 + 1) \times J_{RMS}^2 \geq \left(\frac{J4u_{03}}{2}\right)^2$. If this does not hold, a different transmitter should be used in the test setup.

178.9.3.4.3 Test procedure

The receiver of each lane is tested using the following procedure.

- 1) Calibrate the amplitude of the channel noise source using the procedure defined in 178.9.3.4.2.
- 2) Allow the receiver under test to configure the test transmitter equalizer using the inter-sublayer link training function (see 178.8.8) or any other method that yields the same result.
- 3) Configure the test transmitter to transmit either scrambled idle or PRBS31Q.
- 4) Verify that the block error ratio requirements are met using the test methods defined in 174A.8.1 or 174A.10.1.

Table 178-10—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
Block error ratio ^a	< 1.45×10^{-11}						—
Insertion loss, IL_{dd} , at 53.125 GHz ^b							dB
Receiver package class A	14.5	15.5	—	33.5	34.5	—	
Receiver package class B	14.5	15.5	—	30	31	—	
COM including effects of broadband noise ^c	—	—	3	—	—	3	dB

^aThe block error ratio (see 178.2) is measured instead of the FEC symbol error ratio in step 10) of the receiver interference tolerance method defined in 93C.2.

^b IL_{dd} measured between TPt and TP5 (see Figure 93C-4) plus IL_{dd} of the specific package used by the test transmitter. See 178.9.3.4.2.

^cThe COM value is the target for the receiver noise level calibration defined in 178.9.3.4.2. The channel noise voltage applied in 178.9.3.4.2 should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

178.9.3.5 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179-12. The test setup shown in Figure 93-12, or its equivalent, is used. TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A, Annex 93C, and Annex 120D. The test channel meets the insertion loss requirement for Test 2 in Table 178-10. The synthesizer frequency is set to the specified jitter frequency and

the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 178.9.3.4, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per the method in 178.9.3.4.2, is at least 3 dB.
- For the COM parameter calibration described in 93C.2 item 7), J4u is substituted by J4u₀₃, and both J_{RMS} and J4u₀₃ are measured with the jitter frequency and amplitude set according to Case F from Table 179–12.

The receiver under test shall meet the block error ratio in Table 178–10 for each case in Table 179–12.

178.9.3.6 Receiver difference ERL

The difference ERL of the receiver at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 178–8 and Table 178–12, and with the value of T_{fx} equal to twice the delay from TP5v to TP5. The reference value, $ERL^{(ref)}$, is calculated based on the receiver package class to which the device adheres.

The receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 178–9.

178.9.3.7 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (178–4). The differential-mode to common-mode return loss limit is illustrated by Figure 178–5.

$$RLcd(f) \geq \begin{cases} 25 - 20 \frac{f}{106.25} & 0.05 \leq f \leq 53.125 \\ 15 & 53.125 \leq f \leq 60 \end{cases} \quad (178-4)$$

where

$RLcd(f)$ is the differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

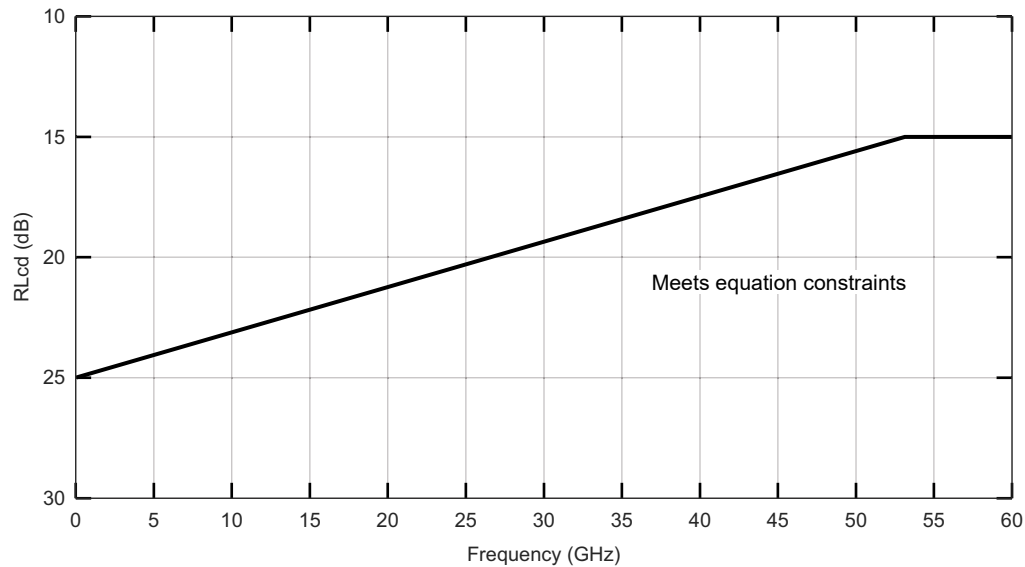


Figure 178–5—Receiver differential-mode to common-mode return loss limit

178.10 Channel characteristics

Table 178–11 provides a summary of the channel characteristics for 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 and references to the subclauses addressing each parameter.

Table 178–11—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	178.10.1	3	dB
Maximum insertion loss from Tp0d to Tp5d, IL_{dd} , at 53.125 GHz (recommended)	178.10.2	40	dB
Minimum channel ERL	178.10.3	11	dB
Differential-mode to common-mode return loss, RL_{cd}	178.10.4	Equation (178–5)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	178.10.5	Equation (178–6)	dB
Common-mode to differential-mode insertion loss, IL_{dc}	178.10.5	Equation (178–7)	dB
Maximum AC-coupling 3 dB corner frequency	178.10.6	100	kHz

178.10.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 178A.1 and the values in Table 178–12 and Table 178–13. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line stage 1 length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2.

The maximum likelihood sequence detection (MLSD) defined in 178A.1.10 is to be used for the calculation of COM.

NOTE—The parameters and values in Table 178–12 and Table 178–13 correspond to behavioral models of transmitters and receivers that are compliant to the PMD specifications in this clause. The purpose of these parameters and values is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementations. It is expected that the PMD specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation.

Table 178–12—Device termination and package model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1	$z_p^{(1)}$	33	mm
Transmission line 1 length, Test 2	$z_p^{(1)}$	12	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1, Tx / Rx	$z_p^{(1)}$	45 / 44	mm
Transmission line 1 length, Test 2, Tx / Rx	$z_p^{(1)}$	30 / 29	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Single-ended reference resistance	R_0	50	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package model parameters are based on the reference impedance R_0 . Channel scattering parameters can be measured with different reference impedance and renormalized to match the reference impedance before computation.			

Table 178–13—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—
Transmitter equalizer, coefficient –2	$c(-2)$		
Minimum value		0	
Maximum value		0.14	—
Step size		0.02	
Transmitter equalizer, coefficient –1	$c(-1)$		
Minimum value		–0.34	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, coefficient 0	$c(0)$		
Minimum value		0.54	—
Transmitter equalizer, coefficient 1	$c(1)$		
Minimum value		–0.2	
Maximum value		0	—
Step size		0.02	
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	10^{-8}	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI

Table 178–13—COM parameter values

Parameter	Symbol	Value	Units
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	6	—
Number of fixed-position taps ^a	N_{fix}	15	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index ^b	N_{max}	87	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	−0.7 −0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-4}	—
Include maximum likelihood sequence detection		yes	—

^a The number of fixed-position post-cursor taps is 8.

^b The latest post-cursor position for a floating tap is 80.

178.10.2 Channel insertion loss (recommended)

The recommended maximum channel insertion loss, $ILdd$, is 40 dB at 53.125 GHz.

178.10.3 Channel ERL

The ERL of the channel at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 178–13 and Table 178–14.

Channel ERL at TP0 and at TP5 shall be greater than or equal to 11 dB.

Table 178–14—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	7000	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

178.10.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (178–5).

$$RLcd(f) \geq \begin{cases} 22 - 10 \frac{f}{53.125} & 0.05 \leq f < 53.125 \\ 15 - 3 \frac{f}{53.125} & 53.125 \leq f \leq 60 \end{cases} \quad (178-5)$$

where

$RLcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The TP0 to TP5 channel differential-mode to common-mode return loss limit is illustrated in Figure 178–6.

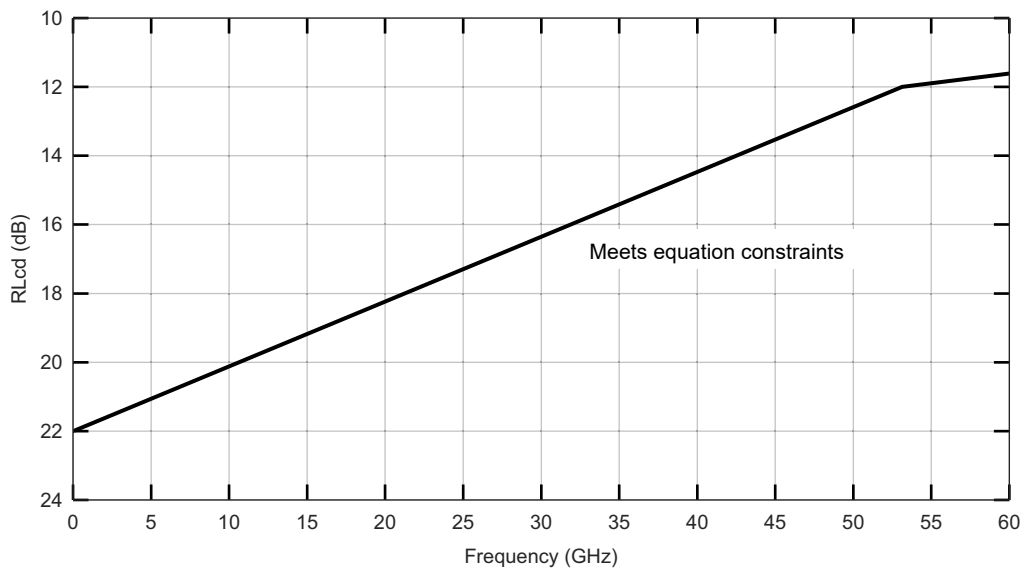


Figure 178–6— Channel differential-mode to common-mode return loss

178.10.5 Channel mode conversion insertion loss

The difference between the TP0 to TP5 channel differential-mode to common-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (178–6).

$$ILcd(f) - ILdd(f) \geq \Delta IL(f) \quad (178-6)$$

where

$ILcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode insertion loss at frequency f in dB

$ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB

$\Delta IL(f)$ is defined by Equation (178–8) and illustrated in Figure 178–7
 f is the frequency in GHz

The difference between the TP0 to TP5 channel common-mode to differential-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (178–7).

$$ILdc(f) - ILdd(f) \geq \Delta IL(f) \quad (178-7)$$

where

$ILdc(f)$ is the TP0 to TP5 channel common-mode to differential-mode insertion loss at frequency f in dB

$ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB

$\Delta IL(f)$ is defined by Equation (178–8) and illustrated in Figure 178–7

f is the frequency in GHz

$$\Delta IL(f) = \begin{cases} 10 & 0.05 \leq f < 26.5625 \\ 10 - 4 \frac{f - 26.5625}{26.5625} & 26.5625 \leq f \leq 60 \end{cases} \quad (178-8)$$

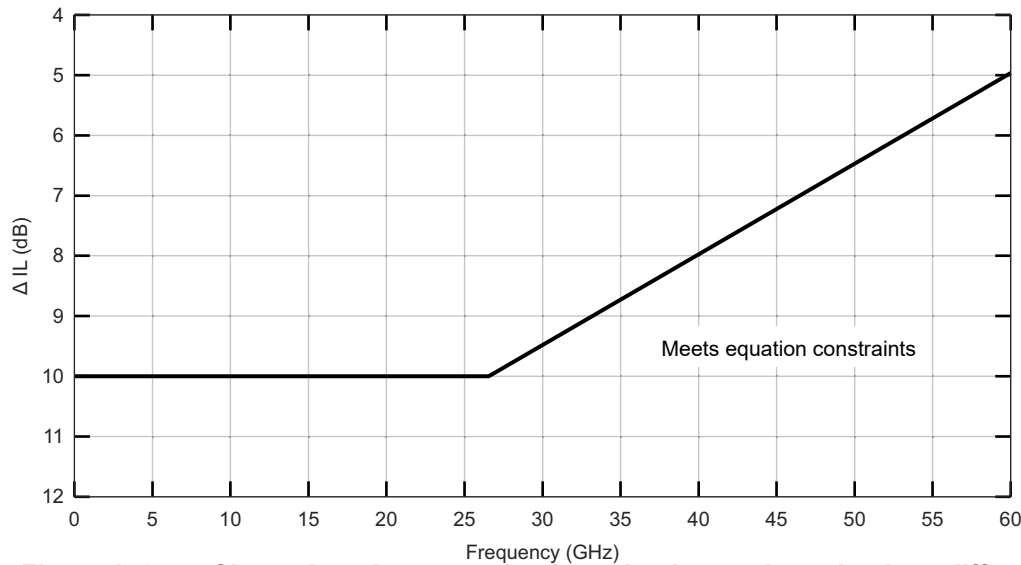


Figure 178–7—Channel mode conversion insertion loss to insertion loss difference

178.10.6 AC-coupling

AC-coupling shall be implemented within the channel (between TP0d and TP5d) using DC blocking capacitors. The low-frequency 3 dB cutoff of the channel shall be less than 100 kHz.

Systems with no AC-coupling within the channel are considered engineered links. It is the system integrator's responsibility to verify that the transmitter and the receiver are compatible with the common-mode voltage differences that may exist in this configuration.

178.11 MDI specifications

The MDI for the 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 178–2.

178.12 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 93.10.

178.13 Management variables

The PMD control and status variables are identical to those defined in 179.14.

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178.14 Protocol implementation conformance statement (PICS) proforma for Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8⁷

178.14.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

178.14.2 Identification

178.14.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

178.14.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 178, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

178.14.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*KR1	200GBASE-KR1 PMD	178.1	Can operate as a 200GBASE-KR1 PMD	O.1	Yes [] No []
*KR2	400GBASE-KR2 PMD	178.1	Can operate as a 400GBASE-KR2 PMD	O.1	Yes [] No []
*KR4	800GBASE-KR4 PMD	178.1	Can operate as a 800GBASE-KR4 PMD	O.1	Yes [] No []
*KR8	1.6TBASE-KR8 PMD	178.1	Can operate as a 1.6TBASE-KR8 PMD	O.1	Yes [] No []
PMA100	100GBASE-R PMA	178.1	Device implements Clause 135 PMA for 100GBASE-P	KR1:M	Yes [] N/A []
PMA200	200GBASE-P PMA	178.1	Device implements Clause 120 PMA for 200GBASE-R	KR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	178.1	Device implements Clause 120 PMA for 400GBASE-R	KR4:M	Yes [] N/A []
*AUIPMD	200GAUI-n C2C, 400GAUI-n C2C, or 800GAUI-n C2C	178.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	178.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!KR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	178.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	178.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC 100GBASE-P RS-FEC	178.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	KR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	178.1	Device implements Clause 82 100GBASE-R PCS	KR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	178.1	Device implements Clause 119 200GBASE-R PCS	KR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	178.1	Device implements Clause 119 400GBASE-R PCS	KR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	178.5	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	178.6	Conforms to constraints	M	Yes []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
SC	Skew and Skew Variation constraints	178.7	Conforms to constraints	M	Yes []
*MD	MDIO capability	178.13	Device implements Clause 45 MDIO	O	Yes [] No []
*CHNL	Channel	178.10	Item marked with CHNL are channel specifications, these items are the only ones applicable to channels and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	178.9.2		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

178.14.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8

178.14.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	178.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 178.9.2	M	Yes []
PF2	Transmit function in TRAINING mode	178.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 178.9.2	M	Yes []
PF4	Receive function	178.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	178.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	178.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	178.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	178.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
PF9	PMD transmit fault	178.8.8	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	178.8.8	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

178.14.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	178.8.9	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed_i at the start of the training pattern	M	Yes []
PC4	Control field structure	178.8.9	As shown in 179.8.9	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	136.8.11.4.1	When requested, set according to the request, with values per Table 179–8	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC9	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []
PC10	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []

178.14.4.3 Transmitter characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	178.9.2	Per Table 178–6	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD not in same package as PCS	178.9.2	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	TP0 to TP0v test fixture	178.9.2.1	Meets specifications	M	Yes []
TC4	Output voltage specifications	176D.8.1	Per Table 178–6	M	Yes []
TC5	Difference ERL at TP0v	178.9.2.2	Per Table 178–6	M	Yes []
TC6	Common-mode to common-mode return loss	93.8.1.4	Per Table 178–6	M	Yes []
TC7	Difference steady-state voltage	178.9.2.4	Per Table 178–6	M	Yes []
TC8	Difference linear fit pulse peak ratio	178.9.2.5	Per Table 178–6	M	Yes []
TC9	Level separation mismatch ratio	178.9.2	Per Table 178–6	M	Yes []
TC10	Absolute coefficient step size	179.9.4.1.4	Per Table 178–6	M	Yes []
TC11	Coefficient range	179.9.4.1.5	Per Table 178–6	M	Yes []
TC12	Signal-to-noise-and-distortion ratio	178.9.2	Per Table 178–6	M	Yes []
TC13	Output jitter	179.9.4.6	Per Table 178–6	M	Yes []
TC14	Transmitter signaling rate	178.9.2	Per Table 178–6	M	Yes []
TC15	Low-frequency peak-to-peak AC common-mode voltage	178.9.2	Per Table 178–6	M	Yes []
TC16	Signal to AC common-mode noise ratio	178.9.2	Per Table 178–6	M	Yes []
TC17	Signal-to-residual-intersymbol-interference ratio	178.9.2	Per Table 178–6	M	Yes []

178.14.4.4 Receiver characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	TP5 to TP5v test fixture	178.9.3.1	Meets specifications	M	Yes []
RC2	Difference ERL at TP5v	178.9.3.6	Greater than or equal to -3 dB	M	Yes []
RC3	Input differential-mode to common-mode return loss	178.9.3.7	Meets constraints of Equation (178–4)	M	Yes []
RC4	Interference tolerance	178.9.3.4	PHY meets FEC symbol error ratio requirement in all tests defined in Table 178–10	M	Yes []
RC5	Jitter tolerance	178.9.3.5	PHY meets FEC symbol error ratio requirement in all cases defined in Table 179–12	M	Yes []

178.14.4.5 Channel characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	ERL	178.10.3	Greater than or equal to 9.7 dB	CHNL:M	Yes [] N/A []
CC2	Channel Operating Margin (COM)	178.10	Greater than or equal to 3 dB	CHNL:M	Yes [] N/A []
CC3	AC-coupling	93.9.4	Between TP0d and TP5d, 3 dB cutoff frequency less than 100 kHz	CHNL:M	Yes [] N/A []
CC4	Channel differential-mode to common-mode return loss	178.10.4	Per Equation (178–5)	CHNL:M	Yes [] N/A []
CC5	Channel differential-mode to common-mode insertion loss	178.10.5	Per Equation (178–6)	CHNL:M	Yes [] N/A []
CC6	Channel common-mode to differential-mode insertion loss	178.10.5	Per Equation (178–7)	CHNL:M	Yes [] N/A []

178.14.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	178.12	Conform to applicable requirements of 93.10	M	Yes []

179. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8

179.1 Overview

This clause specifies the 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 PMDs and the associated baseband media. The PMDs provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links on one, two, four, or eight lanes, respectively, over twinaxial copper cable. The specifications for the four PMDs are similar, except for the number of lanes and associated parameters, and the MDI.

There are four associated annexes. Annex 179A provides information on parameters that might not be testable in an implemented system, since the test points they are associated with are typically inaccessible. Annex 179B specifies test fixtures. Annex 179C specifies MDIs. Annex 179D describes host and cable assembly types.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate sublayers (as specified in Table 179–1 through Table 179–4) and to the medium through the appropriate MDI, as illustrated in Figure 179–1.

PMDs defined in this clause conform to one of three host classes. The electrical specifications are separate for each host class.

Cable assemblies defined in this clause conform to one of four classes, which differ by the maximum insertion loss.

Operation of Ethernet links is provided for specific combinations of host classes and cable assembly classes, as listed in Table 179–15 and Table 179A–4.

Table 179–1—Physical Layer clauses associated with the 200GBASE-CR1 PMD

Associated clause	200GBASE-CR1
73—AN	Required
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
178B—ILT	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.4.1.

^c When implemented in a 200GBASE-CR1 PHY, the signaling rate range is constrained as specified to 120.1.4.

^d A 200GBASE-CR1 PHY may include one instance of 200GAUI-n C2C as described in 176B.4.1.

Table 179–2—Physical Layer clauses associated with the 400GBASE-CR2 PMD

Associated clause	400GBASE-CR2
73—AN	Required
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—400GBASE-R PCS	Optional
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d

Table 179–2—Physical Layer clauses associated with the 400GBASE-CR2 PMD (*continued*)

Associated clause	400GBASE-CR2
120D—400GAUI-8 C2C ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
178B—ILT	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.5.1.

^c When used within a 400GBASE-CR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d A 400GBASE-CR2 PHY may include one instance of 400GAUI-n C2C as described in 176B.5.1.

Table 179–3—Physical Layer clauses associated with the 800GBASE-CR4 PMD

Associated clause	800GBASE-CR4
73—AN	Required
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
178B—ILT	Required

^a A 800GBASE-CR4 PHY may include one instance of 800GAUI-n C2C as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Table 179–4—Physical Layer clauses associated with the 1.6TBASE-CR8 PMD

Associated clause	1.6TBASE-CR8
73—AN	Required
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
170—1.6 Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII extender	Optional
172—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a
178B—ILT	Required

^a A 1.6TBASE-CR8 PHY may include one instance of 1.6TAUI-n C2C as described in 176B.7.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to the guidelines in 176B.7.1.

Figure 179–1 shows the relationship of the PMD and MDI, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model.

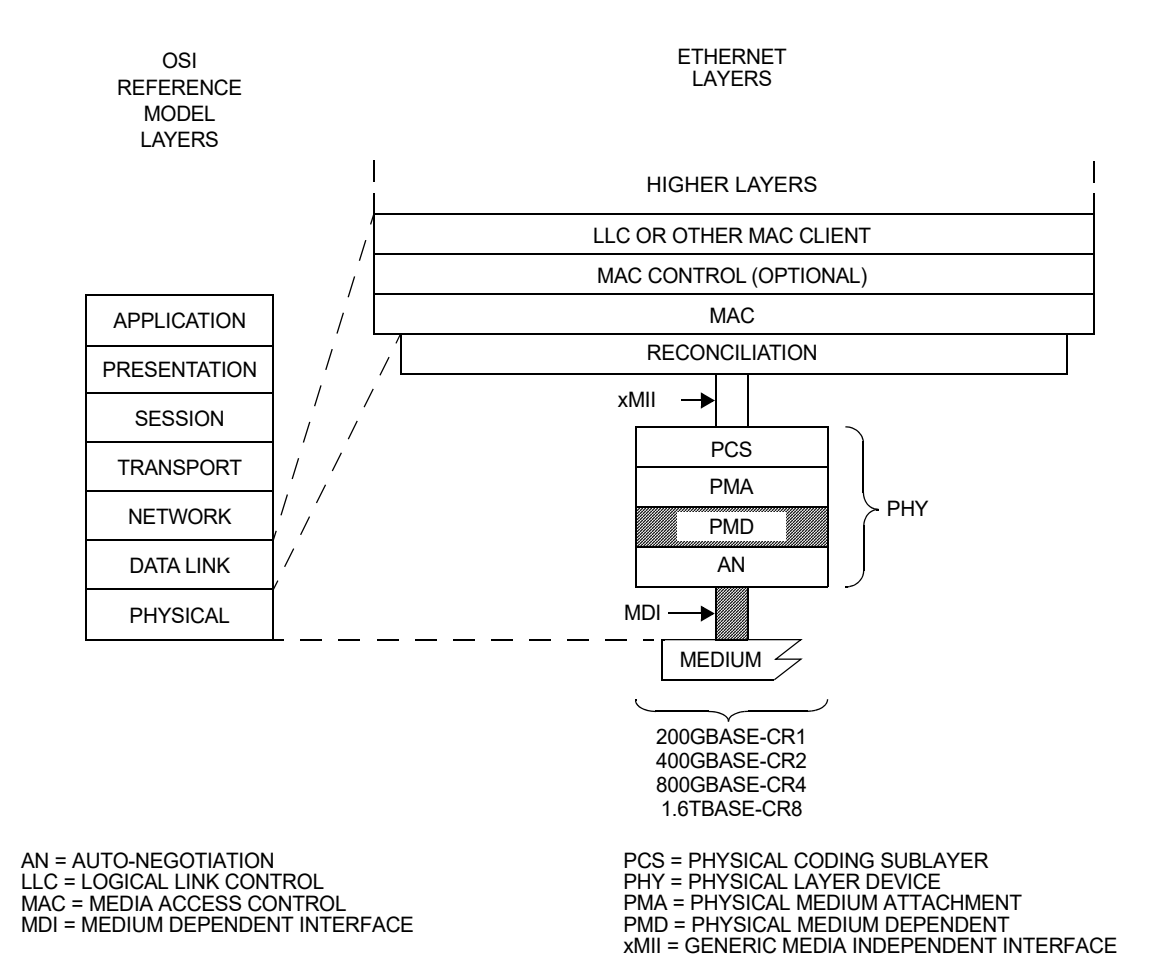


Figure 179–1—200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

179.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the PMD using the method described in 174A.8, with BER_{added} equal to 1.6×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 8×10^{-6} .

179.3 Conventions

This clause describes four PMDs, 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8, which have one, two, four, and eight lanes, respectively. For efficient description, the

parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n=1$ for 200GBASE-CR1, $n=2$ for 400GBASE-CR2, $n=4$ for 800GBASE-CR4, and $n=8$ for 1.6TBASE-CR8.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n-1$.

Within this clause, the unqualified term “PMD” refers to any of 200GBASE-CR1 PMD, 400GBASE-CR2 PMD, 800GBASE-CR4 PMD, or 1.6TBASE-CR8 PMD.

Within this clause, the term FEC refers to the Reed-Solomon code within the Clause 119 PCS for 200GBASE-CR1 and 400GBASE-CR2, the Clause 172 PCS for 800GBASE-CR4, and the Clause 175 PCS for 1.6TBASE-CR8.

179.4 Service interfaces

This subclause specifies the services provided by the PMD. The PMD service interface is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (PMA). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-CR1 and 400GBASE-CR2, in 169.3 for 800GBASE-CR4, and in 174.3 for 1.6TBASE-CR8. The nominal signaling rate for each symbol stream in PMD:IS_UNITDATA_ i .request and PMD:IS_UNITDATA_ i .indication, where $i = 0$ to $n-1$, is 106.25 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable training_status of the inter-sublayer training function, as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_ i .indication on all lanes are unspecified.

179.5 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive AN_LINK.indication defined in 73.9. (See 82.6, 119.6, 172.6, and 175.7.)

200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 and 1.6TBASE-CR8 PHYs may be extended using 200GAUI-n C2C, 400GAUI-n C2C, 800GAUI-n C2C, and 1.6TAUI-n C2C, respectively, as physical instantiations of the inter-sublayer service interface between devices. If such extension is used, then the AN_LINK(link_status).indication is relayed from the device with the PCS sublayer to the device with the AN sublayer by means at the discretion of the implementer. As examples, the implementer may employ use of pervasive management or employ a dedicated electrical signal to relay the state of link_status as indicated by the PCS sublayer on one device to the AN sublayer on the other device.

179.6 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 179–5. It is assumed that the one-way delay through the medium is no more than 14 ns.

Descriptions of overall system delay constraints can be found in 116.4 for 200GBASE-CR1 and 400GBASE-CR2, in 169.4 for 800GBASE-CR4, and in 174.4 for 1.6TBASE-CR8.

Table 179–5—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GBASE-CR1	14 848	29	74.24
400GBASE-CR2	29 696	58	74.24
800GBASE-CR4	59 392	116	74.24
1.6TBASE-CR8	118 784	232	74.24

^a One bit time is equal to 5 ps for 200GBASE-CR1, 2.5 ps for 400GBASE-CR2, 1.25 ps for 800GBASE-CR4, and 625 fs for 1.6TBASE-CR8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GBASE-CR1, 1.28 ns for 400GBASE-CR2, 640 ps for 800GBASE-CR4, and 256 ps for 1.6TBASE-CR8. See 31B.2 for the definition of pause_quanta.

179.7 Skew constraints

The Skew (relative delay) between the PMD lanes is kept within limits so that the information on the PCS lanes can be reassembled by the PCS. The Skew Variation is also limited to ensure that a given PCS lane always traverses the same PMD lane.

179.7.1 Skew Constraints for 200GBASE-CR1 and 400GBASE-CR2

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5. Skew Variation at SP2 to SP5 is specified only for 400GBASE-CR2. For 200GBASE-CR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

179.7.2 Skew Constraints for 800GBASE-CR4 and 1.6TBASE-CR8

Skew and Skew Variation are defined in 169.5 for 800GBASE-CR4 and in 174.5 for 1.6TBASE-CR8. They are specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5 for 800GBASE-CR4, and in Figure 174–5 and Figure 174–6 for 1.6TBASE-CR8.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5 and 174.5.

179.8 PMD functional specifications

179.8.1 Specified test points

For purposes of system conformance, the PMD sublayer is standardized at the test points described in Table 179–6.

The test points are illustrated in Figure 179–2, which shows one direction of a 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8 link. Each differential pair in the cable is separately shielded. The signal shields are connected to ground contacts in the MDI plug connectors on both ends of the cable assembly, creating a common ground reference for both PMDs.

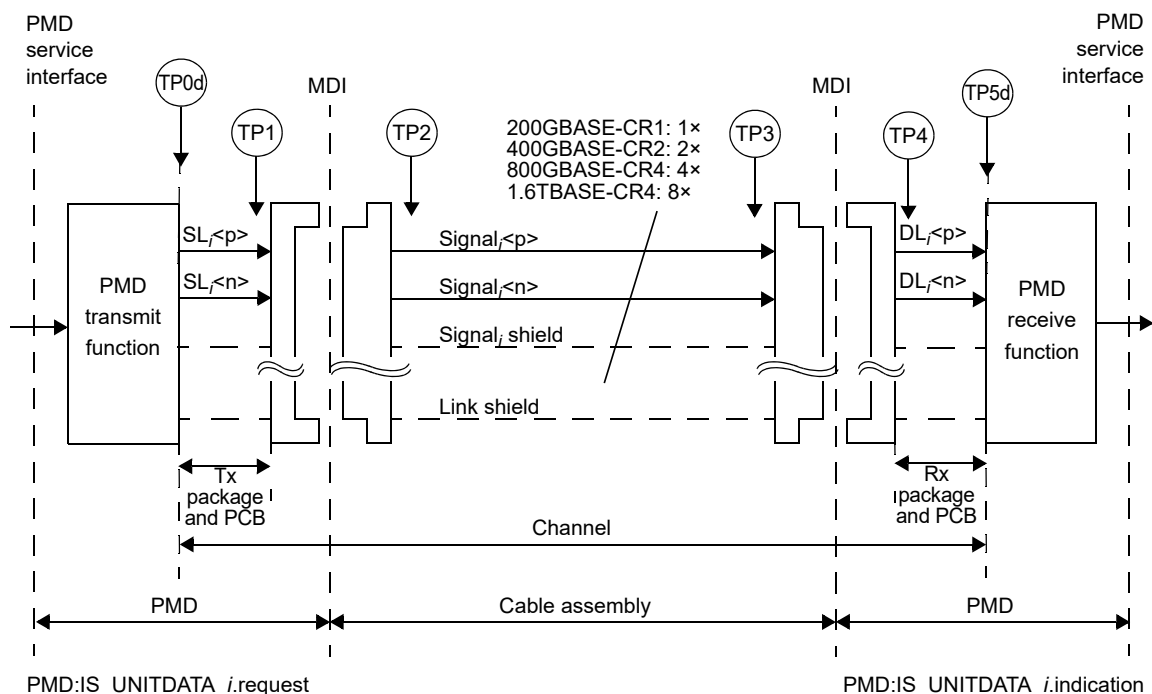
Table 179–6—Test points

Test point	Description
TP0d	Represents the transmitter output at the die bump. This point is not intended to be accessible for measurements.
TP1	The cable assembly input (corresponding to MDI signals $SL_i<p>$ and $SL_i<n>$) on a cable assembly test fixture (see 179B.3), used in cable assembly specifications.
TP2	The host output (corresponding to MDI signals $SL_i<p>$ and $SL_i<n>$) on a TP2 or TP3 test fixture (see 179B.2). Host output measurements and tests defined in 179.9.4 are made at this point.
TP3	The host input (corresponding to MDI signals $DL_i<p>$ and $DL_i<n>$) on a TP2 or TP3 test fixture (see 179B.2). Host input measurements and tests defined in 179.9.5 are made at this point.
TP4	The cable assembly output (corresponding to MDI signals $DL_i<p>$ and $DL_i<n>$) on a cable assembly test fixture (see 179B.3), used in cable assembly specifications.
TP5d	Represents the receiver input at the die bump. This point is not intended to be accessible for measurements.
NOTE 1—The channel between TP0d to TP5d includes the transmitter and receiver packages and PCBs and the cable assembly. NOTE 2—Cable assembly measurements for the specifications in 179.11 are made between TP1 and TP4 as illustrated in Figure 179A–1. These measurements include a mated connector pair at each end of the cable assembly. NOTE 3—A mated connector pair is included in transmitter specifications at TP2 and in receiver specifications at TP3. The recommended maximum insertion loss from TP0d to TP2 or from TP3 to TP5d, excluding the test fixtures, is provided in 179A.4. NOTE 4—Test fixture connector labeling may differ from the MDI signal names.	

Annex 179A provides information on parameters associated with test points TP0d and TP5d, which are typically inaccessible in an implemented system.

179.8.2 PMD transmit function

The PMD transmit function has two operating modes: DATA and TRAINING. The operating mode is controlled by the ILT function (see 179.8.9).



NOTE—The source lane (SL) signals SL_i<p> and SL_i<n> are the positive and negative sides of the transmitter's differential signal pair on lane *i* and the destination lane (DL) signals DL_i<p> and DL_i<n> are the positive and negative sides of the receiver's differential signal pair on lane *i*.

Figure 179-2—200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 or 1.6TBASE-CR8 link diagram

When operating in DATA mode, the PMD transmit function shall convert the tx_symbol parameters of the PMD service interface message PMD:IS_UNITDATA_i.request on each lane into a stream of PAM4 symbols for transmission as electrical signals on the corresponding contacts of the MDI (see 179.12). The PAM4 symbol values 0, 1, 2, and 3 correspond to the tx_symbol values zero, one, two, and three, respectively. When operating in TRAINING mode, the PAM4 symbol stream on each lane is taken from the output of the training pattern generator in the PMD control function (see Figure 178B-4).

Each lane has two output signals, SL_i<p> and SL_i<n>, which create a differential signal SL_i<p> – SL_i<n> and a common-mode signal (SL_i<p> + SL_i<n>) / 2. The differential and common-mode signals on each lane comply with the transmit electrical specifications in 179.9.4.

The polarity of the PMD output on each of the lanes is either normal, where the highest differential output voltage corresponds to the PAM4 symbol 3 and the lowest differential output voltage corresponds to the PAM4 symbol 0, or inverted, where the highest differential output voltage corresponds to the PAM4 symbol 0 and the lowest differential output voltage corresponds to the PAM4 symbol 3. If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 179.8.3). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

179.8.3 PMD receive function

The PMD receive function converts the differential electrical signal on each of the lanes on the MDI (DL_i<p> – DL_i<n>) into a symbol stream for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_i.indication(rx_symbol).

The polarity of the PMD input on each of the lanes is either normal, where the highest differential input voltage at the MDI input corresponds to rx_symbol = three and the lowest differential input voltage corresponds to rx_symbol = zero, or inverted, where the highest differential input voltage at the MDI corresponds to rx_symbol = zero and the lowest differential input voltage corresponds to rx_symbol = three. If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.10). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

The PMD receive function also recovers an appropriate clock from the input signal on each lane for timing of the delivery of PMD:IS_UNITDATA_*i*.indication messages to the PMD client.

179.8.4 PMD global signal detect function

The PMD global signal detect function is used by the PMD to indicate the successful completion of the start-up protocol by the inter-sublayer training (ILT) function (see 179.8.9). The variable Global_PMD_signal_detect is set to the value of remote_rts in the ILT function (see 178B.14.2.1).

179.8.5 PMD lane-by-lane signal detect function

The variable PMD_signal_detect_*i* is the logical AND of the variables local_rx_ready and remote_rx_ready in lane *i* of the PMD control function. These variables are set independently on each lane.

179.8.6 PMD global transmit disable function

The PMD global transmit disable function is optional. If this function is supported, it shall meet the following requirements.

Setting the Global_PMD_transmit_disable variable to 1 has the same effect as if tx_disable (see 179.8.7) is set to 1 on all lanes (*i* = 0 to *n*–1). Global_PMD_transmit_disable does not overwrite the tx_disable or PMD_transmit_disable_*i* variables.

179.8.7 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function enables selective disabling of the transmitters in each lane. This function is used by the PMD control function. It may optionally be provided through control variables PMD_transmit_disable_*i*.

If either PMD_transmit_disable_*i* or the variable tx_disable in lane *i* of the PMD control function (see 178B.14.2.1) are set to one, the transmitter output on lane *i* of the PMD shall not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 179–7.

179.8.8 PMD fault functions

PMD transmit fault indication is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable PMD_transmit_fault to one. The assertion of Global_PMD_transmit_disable is not considered a transmit fault. However, if PMD_transmit_fault is set to one, then Global_PMD_transmit_disable should also be set to one.

PMD receive fault indication is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable PMD_receive_fault to one.

The variable PMD_fault is the logical-OR of PMD_receive_fault, PMD_transmit_fault, and any other implementation-specific fault indications.

179.8.9 Inter-sublayer link training (ILT) function

A PMD shall provide the inter-sublayer link training (ILT) function for a Type E1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), control the transmitter output on each lane of the MDI, indicate the receiver state, and coordinate the transition to DATA mode.

The PMD transmit function is configured in response to the requests in the incoming training frames as specified in 179.9.4.1.3 through 179.9.4.1.5.

The PMD transmit function supports the coefficient indexes $k_list = \{-3, -2, -1, 0, 1\}$ and the initial conditions preset 1 through preset 6 and initialize (see Table 179–8).

179.8.10 PMD reset function

If the variable `PMD_reset` is asserted, the PMD shall be reset as defined in 45.2.1.1.1. PMD reset shall reset the ILT function associated with the PMD (see 179.8.9).

179.9 PMD electrical characteristics

179.9.1 AC-coupling

Interoperability between PMD components operating from different supply voltages is facilitated by AC-coupling within the cable assembly as specified in 179.11.

179.9.2 Signal paths

The MDI transmit and receive paths are point-to-point connections. Each path corresponds to one MDI lane and is composed of two complementary signals, which form a balanced differential pair.

For 200GBASE-CR1, there is one differential lane in each direction for a total of two pairs, or four connections. For 400GBASE-CR2, there are two differential lanes in each direction for a total of four pairs, or eight connections. For 800GBASE-CR4, there are four differential lanes in each direction for a total of eight pairs, or 16 connections. For 1.6TBASE-CR8, there are eight differential lanes in each direction for a total of 16 pairs, or 32 connections.

179.9.3 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

179.9.4 Transmitter characteristics

The transmitter on each lane shall meet the specifications at TP2 given in Table 179–7 and detailed in the referenced subclauses.

Unless specified otherwise, transmitter signal measurements are made for each lane separately using a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz, with AC-coupled connection from TP2 to the test equipment.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP2 where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

Recommended transmitter characteristics at TP0d are provided in 179A.2.

Table 179–7—Summary of transmitter specifications at TP2

Parameter	Subclause reference	Value	Units
Signaling rate (range)		106.25 ± 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.1	0.03	V
Transmitter disabled		1	V
Transmitter enabled			
DC common-mode voltage (max)	176D.8.1	1	V
AC common-mode peak-to-peak voltage (max)	176D.8.1	0.03	V
Low-frequency, $V_{CM_{LF}}$		0.08	V
Full-band, $V_{CM_{FB}}$			
Effective return loss, ERL (min)	179.9.4.7	7.3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	179.9.4.8	See Equation (179–19)	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	179.9.4.9	See Equation (179–20)	dB
Transmitter steady-state voltage, v_f (range)	179.9.4.1.2	0.4 to 0.5	V
Linear fit pulse peak ratio, R_{peak} (min)	179.9.4.1.2		
Host class HL		0.456	—
Host class HN		0.345	—
Host class HH		0.234	—
Level separation mismatch ratio R_{LM} (min)	179.9.4.2	0.95	—
Transmitter waveform			
absolute value of step size for all taps (min)	179.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	179.9.4.1.4	0.025	—
value at minimum state for $c(-3)$ (max)	179.9.4.1.5	–0.06	—
value at maximum state for $c(-2)$ (min)	179.9.4.1.5	0.12	—
value at minimum state for $c(-1)$ (max)	179.9.4.1.5	–0.34	—
value at minimum state for $c(0)$ (max)	179.9.4.1.5	0.5	—
value at minimum state for $c(1)$ (max)	179.9.4.1.5	–0.2	—
Difference signal-to-noise-and-distortion ratio, dSNDR (min)	179.9.4.5	0	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	179.9.4.3	26.7	dB
Output jitter (max)	179.9.4.6		
J_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
$J4u_{03}$			
Host class HL		0.120	UI
Host class HN		0.124	UI
Host class HH		0.128	UI

179.9.4.1 Transmitter output waveform

The PMD transmit function includes programmable equalization to compensate for the frequency-dependent loss of the channel and facilitate data recovery at the receiver. The functional model for the transmit equalizer on each lane is the five-tap transversal filter shown in Figure 179–3.

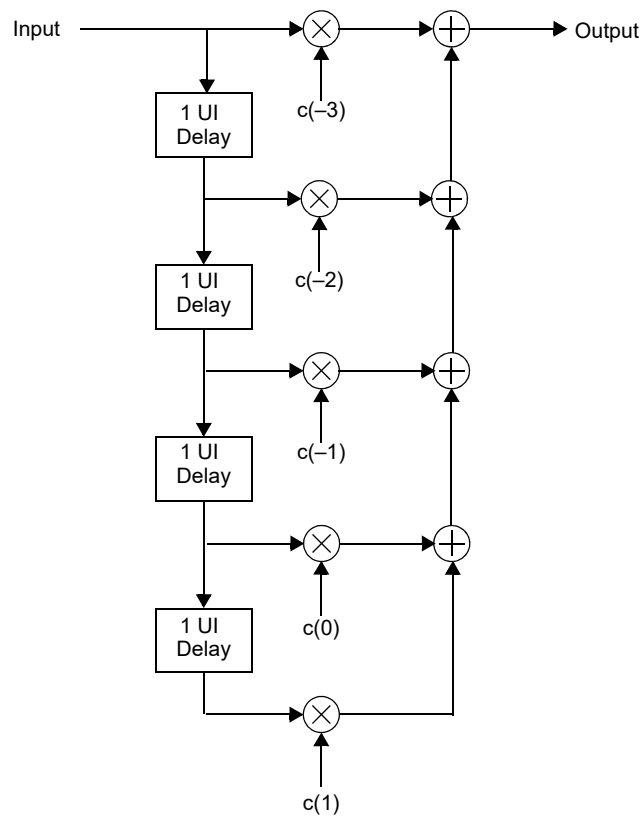


Figure 179-3—Transmit equalizer functional model

The state of the transmit equalizer and hence the transmitted output waveform may be manipulated via the ILT function (see 179.8.9) or via the PMD management interface. The transmit function responds to a set of commands issued by the peer interface receive function and conveyed by a back-channel communications path. This command set includes instructions to increment, decrement, hold, or set to zero a selected coefficient $c(k)$, where $k = -3$ to 1. In addition, it includes commands to set all coefficients to one of six initial conditions.

In response, the transmit function relays status information to the peer interface receive function. The status messages indicate the selected coefficient index k , the status of the coefficient $c(k)$ (updated, not updated, or at limit), and the status of the initial condition (updated or not updated).

179.9.4.1.1 Linear fit to the measured waveform

The following procedure is used to determine the linear fit pulse response, linear fit error, and normalized transmitter coefficient values.

Set the transmitter under test to transmit the PRBS13Q test pattern (defined in 120.5.11.2.1). For each configuration of the transmit equalizer, capture at least one complete cycle of the test pattern at TP2, as specified in 85.8.3.3.4. The reference clock recovery unit (CRU) used in the measurement acts as a high-pass jitter filter with a corner frequency of 4 MHz and a slope of 20 dB/decade.

In the following calculations, M should be an integer greater than or equal to 32. Interpolation of the captured waveform may be used to achieve this.

Compute the linear fit pulse response $p(k)$ and linear fit error $e(k)$, $k=1$ to $M \times N_p$, from the captured waveform, as specified in 85.8.3.3.5, with $N_p = 400$ and $D_p = 4$, where the aligned symbols $x(n)$ are assigned normalized amplitudes -1 , $-ES$, ES , and 1 to represent the PAM4 symbol values 0, 1, 2, and 3 respectively. ES is defined as $(|ES1| + |ES2|)/2$ where $ES1$ and $ES2$ are calculated according to 120D.3.1.2.

Define $r(k)$ to be the linear fit pulse response when transmit equalizer coefficients have been set to the “preset 1” values. The normalized coefficients for any configuration of the transmit equalizer are computed using the following method.

For each value of m in the range $-M/2$ to $M/2 - 1$ when M is even and $-(M-1)/2$ to $(M-1)/2$ when M is odd:

- a) Define an $M \times N_p$ -by-5 matrix R_m . The elements of R_m are assigned values per Equation (179-1) where $i = -3$ to 1 and $j = 1$ to $M \times N_p$.

$$R_m(j, i+4) = \begin{cases} r(m+j-M \times i) & \text{if } 1 \leq m+j-M \times i \leq M \times N_p \\ 0 & \text{otherwise} \end{cases} \quad (179-1)$$

- b) The normalized coefficients of the transmit equalizer are computed using Equation (179-2).

$$\begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} = (R_m^T R_m)^{-1} R_m^T \begin{bmatrix} p(1) \\ \dots \\ p(M \times N_p) \end{bmatrix} \quad (179-2)$$

- c) The linear fit pulse response is reconstructed from the matrix R_m and the normalized coefficients using Equation (179-3).

$$\begin{bmatrix} p_m(1) \\ \dots \\ p_m(M \times N_p) \end{bmatrix} = R_m \begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} \quad (179-3)$$

- d) $\varepsilon^2(m)$ is computed using Equation (179-4).

$$\varepsilon^2(m) = \sum_{k=1}^{M \times N_p} (p(k) - p_m(k))^2 \quad (179-4)$$

The normalized transmit equalizer coefficients $c(i)$ for a given linear fit pulse $p(k)$ are the values $c_m(i)$ for the value of m that minimizes $\varepsilon^2(m)$.

179.9.4.1.2 Steady-state voltage and linear fit pulse peak ratio

The linear fit pulse peak v_{peak} and steady-state voltage v_f are defined using the linear fit pulse response $p(1)$ through $p(M \times N_p)$ with $N_p = 400$, measured with transmit equalizer set to preset 1 (no equalization). The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 179.9.4.1.1.

v_{peak} is defined as the maximum value of $p(k)$.

v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_v)$ divided by M .

The linear fit pulse peak ratio R_{peak} is defined by Equation (179–5).

$$R_{peak} = \frac{v_{peak}}{v_f} \quad (179-5)$$

179.9.4.1.3 Coefficient initialization

When the Coefficient update state diagram (Figure 178B–10) is in either the OUT_OF_SYNC state or the NEW_IC state, the coefficients of the transmit equalizer shall be configured to values within the ranges specified in Table 179–8 (according to the value of the variable `ic_req`, which is defined in 178B.14.3.1). These requirements apply upon the assertion of initial condition status of “Updated”.

Table 179–8—Coefficient initial conditions

ic_req	$c(-3)$	$c(-2)$	$c(-1)$	$c(0)$	$c(1)$
initialize	0	0	0	1	0
preset 1 ^a	0	0	0	1	0
preset 2	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.5 ± 0.025	0 ± 0.025
preset 3	0 ± 0.025	0 ± 0.025	–0.075 ± 0.025	0.75 ± 0.025	0 ± 0.025
preset 4	0 ± 0.025	0.05 ± 0.025	–0.2 ± 0.025	0.75 ± 0.025	0 ± 0.025
preset 5	–0.025 ± 0.025	0.075 ± 0.025	–0.25 ± 0.025	0.65 ± 0.025	0 ± 0.025
preset 6	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.75 ± 0.025	0 ± 0.025

^a Preset 1 is the reference for the calculation of the normalized coefficients of the transmit equalizer (see 179.9.4.1.1). As a result, the normalized coefficients for preset 1 and initialize do not include any tolerances.

179.9.4.1.4 Coefficient step size

The change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “increment” shall be between 0.005 and 0.025, and the change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “decrement” shall be between –0.025 and –0.005. It is recommended that the same step size is used for all coefficients.

The coefficients other than $c(\text{coef_sel})$ are not expected to change. The absolute change in any coefficient other than $c(\text{coef_sel})$ shall be less than 0.005.

The change in the normalized transmit equalizer coefficient is defined to be the difference in the value measured prior to the assertion of the “increment” or “decrement” request (i.e., `coef_req` is “Hold”) and the value upon the assertion of a coefficient status of “updated”.

179.9.4.1.5 Coefficient range

When sufficient “increment” or “decrement” requests have been received for a given coefficient, the coefficient reaches a lower or upper bound based on the range of that coefficient or the combination of coefficients.

The sum of the absolute values of $c(-3)$, $c(-2)$, and $c(-1)$, $c(0)$, and $c(1)$ shall be less than or equal to 1.

With $c(-3)$, $c(-2)$, and $c(-1)$ set to zero and both $c(0)$ and $c(1)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(1)$ shall be less than or equal to -0.2 .

With $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$ set to zero and having received sufficient “decrement” requests so that it is at its minimum value, $c(0)$ shall be less than or equal to 0.5 .

With $c(-3)$, $c(-2)$, and $c(1)$ set to zero and both $c(-1)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-1)$ shall be less than or equal to -0.34 .

With $c(-3)$, $c(-1)$, and $c(1)$ set to zero, $c(0)$ having received sufficient “decrement” requests so that it is at its minimum value, and $c(-2)$ having received sufficient “increment” requests so that it is at its maximum value, $c(-2)$ shall be greater than or equal to 0.12 .

With $c(-2)$, $c(-1)$, and $c(1)$ set to zero and both $c(-3)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-3)$ shall be less than or equal to -0.06 .

NOTE—Any of the coefficients $c(-3)$, $c(-2)$, $c(-1)$, or $c(1)$ may be set to zero by asserting a coefficient request of “no equalization” for that coefficient, using the control function specified in 179.8.9, or by implementation specific means.

179.9.4.2 Transmitter linearity

Transmitter linearity is defined using the method in 120D.3.1.2.

The transmitter linearity shall meet the requirement when the transmitter equalization is set to each of the initial conditions defined in Table 179–8.

179.9.4.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is determined using the method in 120D.3.1.7 with the following exceptions:

- The linear fit pulse response $p(k)$ is determined using the linear fit procedure in 179.9.4.1.1.
- The continuous time filter parameters are provided in Table 179–18.
- For calculation of SNR_{ISI} using Equation (120D–8) a value of 20 is used for N_b .
- A time offset is added to t_p whose value is swept from -0.5 UI to 0.5 UI when calculating $ISI_{cursors}$. SNR_{ISI} is defined as the lowest value found across the time offset sweep.
- The transmitter equalizer setting is chosen, within the required coefficient ranges (see Table 179.9.4.1.5), to result in the highest SNR_{ISI} value.

179.9.4.4 Host channel insertion loss

The recommended insertion loss for host channels is provided in 179A.4

179.9.4.5 Difference signal-to-noise-and-distortion ratio

Difference signal-to-noise-and-distortion ratio for a specific transmitter equalization preset i , $d\text{SNDR}(i)$, is calculated using Equation (179–6). The transmitter dSNDR is defined as the minimum value of $d\text{SNDR}(i)$ across the presets defined in Table 179–8.

$$d\text{SNDR}(i) = \text{SNDR}_i^{(meas)} - \text{SNDR}^{(ref)}(\mathbf{c}_i) \quad (179-6)$$

where

- $\text{SNDR}_i^{(meas)}$ is the measured SNDR of the transmitter, as defined in 179.9.4.5.1, measured with all lanes set to equalization preset i (see Table 179–8)
- $\mathbf{c}_i$ is the vector of coefficients corresponding to preset i in Table 179–8 (using the center of the range of each coefficient)
- $\text{SNDR}^{(ref)}(\mathbf{c}_i)$ is the reference SNDR for the coefficient vector $\mathbf{c}_i$, defined in 179.9.4.5.3 using values in Table 179–18.

The method for calculating $d\text{SNDR}(i)$ is illustrated in Figure 179–4.

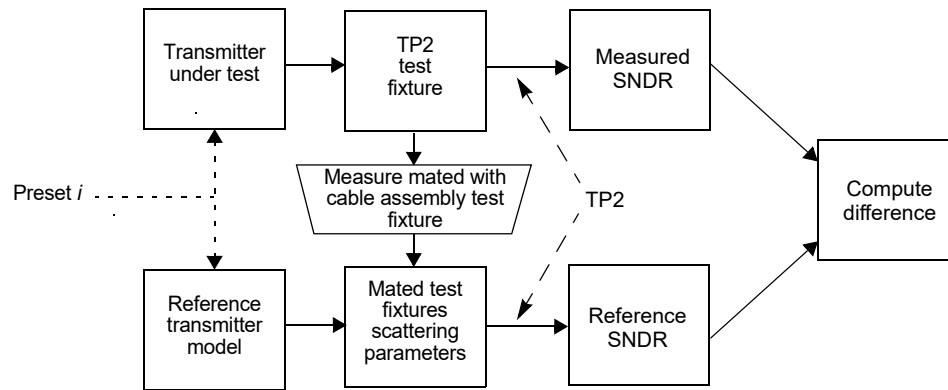


Figure 179–4—Calculation method for transmitter ΔSNDR

179.9.4.5.1 Measured SNDR

$\text{SNDR}^{(meas)}$ of a specific lane, at a specific equalization setting, is calculated by the following procedure.

Set the transmitter on the lane under test to transmit PRBS13Q and the transmitters on all other lanes to transmitting either PRBS1Q or PCS data.

Capture the output of the lane under test and compute the linear fit to the captured waveform and the linear fit pulse response, $p(k)$, and the linear fit error, $e(k)$, according to 179.9.4.1.1. The standard deviation of $e(k)$ is denoted by σ_e .

Measure the RMS deviation from the mean voltage at a fixed low-slope point in runs of at least 6 consecutive identical PAM4 symbols. PRBS13Q includes such a run for each of the PAM4 levels. The average of the four measurements is denoted by σ_n .

The signal power P_{signal} is defined by Equation (179–7) and Equation (179–8).

$$m_0 = 1 + (k_{peak} - 1) \bmod M \quad (179-7)$$

$$P_{Signal} = \sum_{i=0}^{N_p-1} p(M \times i + m_0)^2 \quad (179-8)$$

where

k_{peak} is the value of k that maximizes $p(k)$
 M and N_p are defined in 179.9.4.1.1.

$SNDR^{(meas)}$ is defined by Equation (179-9).

$$SNDR^{(meas)} = 10 \log_{10} \left(\frac{P_{Signal}}{\sigma_e^2 + \sigma_n^2} \right) \quad (179-9)$$

179.9.4.5.2 Reference channel transfer function

The reference channel transfer function $H_{21}^{(ref)}(f)$ is calculated using Equation (178A-3), where $S^{(k)}$ is defined by Equation (178A-2) with the following terms:

- The transmitter S-parameter model $S^{(t)}$ is calculated using the method described in 178A.1.4, using the parameters in Table 179-16 and Table 179-17 for the transmitter host class that the transmitter adheres to.
- S is the measured S-parameter matrix of the TP2 test fixture used in the test mated with a cable assembly test fixture such that the mated test fixtures comply with the requirements in 179B.4.
- The receiver S-parameter model $S^{(r)}$ is omitted.

179.9.4.5.3 Reference SNDR

$SNDR^{(ref)}(c)$ for a specific lane, where c is a vector of transmit equalizer coefficients, is calculated by Equation (179-10).

$$SNDR^{(ref)}(c) = 10 \log_{10} \left(\frac{\sigma_{ts}^2}{\sigma_{in}^2} \right) \quad (179-10)$$

where

σ_{ts} is calculated by Equation (179-11) through Equation (179-13).
 σ_{in} is calculated by Equation (179-14) and Equation (179-15).

In Equation (179-11) through Equation (179-15) and the referenced equations in Annex 178A, the parameters f_b , T_r , and SNR_{TX} take the values in Table 179-18.

$$H_{noffe}(f) = H_t(f) H_{21}^{(ref)}(f) H_{BT}(f) \quad (179-11)$$

where

$H_t(f)$ is the input rise time transfer function defined in 178A.1.6.2.
 $H_{21}^{(ref)}(f)$ is the reference channel transfer function defined in 179.9.4.5.2.
 $H_{BT}(f)$ is the fourth-order Bessel-Thomson low-pass transfer function defined by Equation (52-2) and Equation (52-3) with the exception that $f_r = 60$ GHz.

$$h_{txffe}(t) = \sum_{n=-3}^1 c(n)h_{noffe}(t - nT_b) \quad (179-12)$$

where

$c(n)$ is n^{th} element of the vector $\mathbf{c}$.
 $h_{noffe}(t)$ is defined by Equation (178A–19) with the $H_{noffe}(f)$ calculated by Equation (179–11).
 T_b is the unit interval $1/f_b$.

$$\sigma_{ts}^2 = \sum_{n=-D_p}^{N_p-D_p-1} h_{txffe}^2(t_{max} + nT_b) \quad (179-13)$$

where t_{max} is the value of t that maximizes $h_{txffe}(t)$.

$$h_{tn}(n) = h_{noffe}(t_{max} + nT_b) \quad (179-14)$$

$$\sigma_{tn}^2 = f_b \int_{-\pi}^{\pi} S_{tn}(\theta) d\theta \quad (179-15)$$

where $S_{tn}(\theta)$ is calculated from $h_{tn}(n)$ using Equation (178A–18).

179.9.4.6 Output jitter

Output jitter is characterized by three parameters: J_{RMS} , J_{4u03} , and EOJ_{03} . These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the host channel. The equalizer setting is chosen to minimize any or all of the jitter parameters.

Jitter is measured separately on each lane. The lane under test transmits either PRBS13Q or PRBS9Q (defined in 120.5.11.2.a). Lanes not under test transmit either PRBS31Q or scrambled idle. For J_{4u03} and EOJ_{03} measurements, the transmitter output is disabled.

NOTE—Outputs of lanes not under test are disabled in order to minimize the effect of crosstalk on the measurement. Other circuitry in lanes not under test should be kept active during the measurement.

Measurement is performed with a clock recovery unit (CRU) with a corner frequency of 4 MHz and a slope of 20 dB/decade.

179.9.4.6.1 J_{RMS}

Calculation of the jitter parameter J_{RMS} uses the timing of multiple transitions in the pattern to estimate the jitter caused by clock phase noise while minimizing the effect of additive noise, using the method described in this subclause.

Select a set A of transitions from the test pattern used in the test. The size of this set should be large enough to enable calculation of J_{RMS} (as defined below) with sufficient accuracy. The set A should include multiple transitions between different PAM4 levels. The location of the specific transitions in the test pattern and the threshold levels at which the transition times are extracted are chosen from the measured signal to minimize J_{RMS} .

For each specific transition $i \in A$:

- Obtain a set $A_i = \{t_i(1), t_i(2), \dots\}$ of transition times modulo the period of the pattern. The size of this set should be large enough to enable estimating the mean and standard deviation with sufficient accuracy.
- estimate the mean μ_i and standard deviation σ_i from the measured dataset A_i .
- Measure the slope of the signal at time μ_i , denoted s_i .

From the collections of σ_i and s_i , calculate the best-fit coefficients for the polynomial in $1/s_i^2$ defined by Equation (179–16):

$$\sigma_i^2 = a + b \frac{1}{s_i^2} + c \frac{1}{s_i^4} \quad (179-16)$$

The free coefficient a is the extrapolation of the measured data to an infinite slope. J_{RMS} is defined as the square root of a .

179.9.4.6.2 J4u₀₃

Calculation of the jitter parameter J4u₀₃ uses the timing of two specific transitions, R03 (from the symbol 0 to the symbol 3) and F30 (from the symbol 3 to the symbol 0). The location of the specific transitions in the test pattern and the threshold levels at which the transition times are extracted are chosen from the measured signal to minimize J4u₀₃. The transitions and thresholds in Table 120D–4 (for PRBS13Q) and Table 162–13 (for PRBS9Q) are examples that may be chosen in many practical cases.

For each of the two transitions ($i=1$ to 2):

- Obtain a set $A_i = \{t_i(1), t_i(2), \dots\}$ of transition times modulo the period of the pattern. The sizes of each set should be large enough to enable estimation of Jnu_{03} (as defined below) with sufficient accuracy.
- From the set A_i , create an estimated probability distribution $f_i(t)$.
- Denote by Jnu_i the time interval that includes all but 10^{-n} of $f_i(t)$, between the $5 \times 10^{-(n+1)}$ and the $1 - 5 \times 10^{-(n+1)}$ quantiles of $f_i(t)$

Jnu_{03} is defined by Equation (179–17).

$$Jnu_{03} = \sqrt{\frac{Jnu_1^2 + Jnu_2^2}{2}} \quad (179-17)$$

J4u₀₃ is defined as Jnu_{03} with $n=4$.

179.9.4.6.3 EOJ₀₃

EOJ₀₃ is calculated from measurements using the method specified in 162.9.4.7 for even-odd jitter. Only the transitions R03 and F30 defined in 179.9.4.6.2 are used. The measured values for these two transitions are denoted EOJ₁ and EOJ₂.

EOJ₀₃ is defined by Equation (179–18).

$$EOJ_{03} = \sqrt{\frac{EOJ_1^2 + EOJ_2^2}{2}} \quad (179-18)$$

179.9.4.7 Transmitter effective return loss (ERL)

The ERL of the transmitter at TP2 is defined by the procedure in 93A.5 using the values in Table 179–9 and Table 179–16, and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the test connector and the host-facing connection, excluding the mating interface discontinuity. T_{fx} is provided by the test fixture provider.

Table 179–9—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	1600	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

Transmitter ERL at TP2 shall meet the requirement in Table 179–7.

179.9.4.8 Common-mode to common-mode return loss

The transmitter common-mode to common-mode return loss shall meet Equation (179–19) as illustrated in Figure 179–5.

$$RL_{cc}(f) \geq \begin{cases} 2 & 0.05 \leq f < 4 \\ 2 + \frac{2}{40}(f - 4) & 4 \leq f < 44 \\ 4 + \frac{2}{16}(44 - f) & 44 \leq f \leq 60 \end{cases} \quad (179-19)$$

where

$RL_{cc}(f)$ is the transmitter common-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

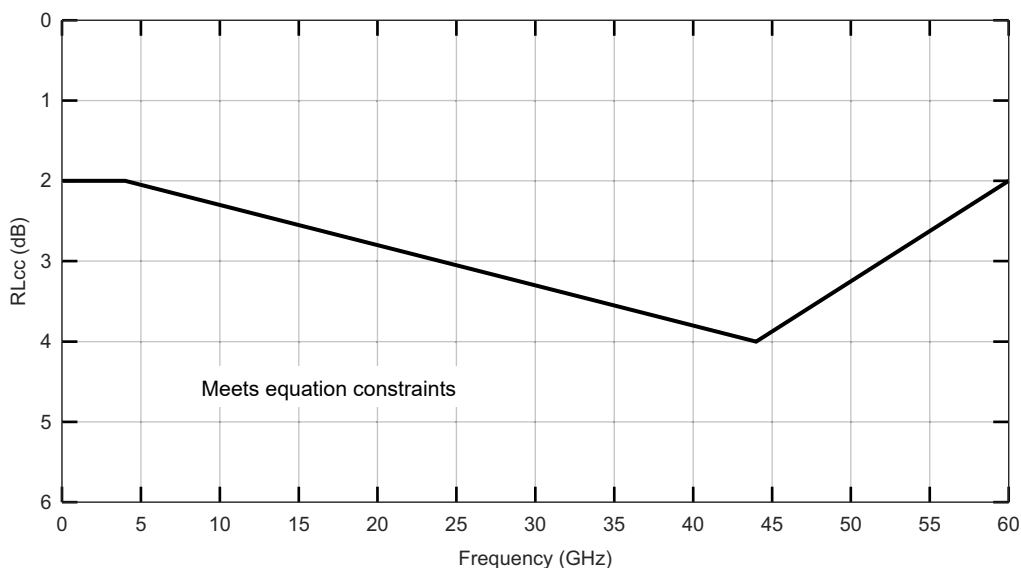


Figure 179-5—Transmitter common-mode to common-mode return loss

179.9.4.9 Common-mode to differential-mode return loss

The transmitter common-mode to differential-mode return loss shall meet Equation (179-20) as illustrated in Figure 179-6.

$$RL_{dc}(f) \geq \begin{cases} 28 - 15 \frac{f}{12.89} & 0.05 \leq f < 12.89 \\ 13 - 5 \frac{f - 12.89}{35 - 12.89} & 12.89 \leq f < 35 \\ 8 & 35 \leq f \leq 60 \end{cases} \quad (179-20)$$

where

$RL_{dc}(f)$ is the transmitter common-mode to differential-mode return loss at frequency f in dB

f is the frequency in GHz

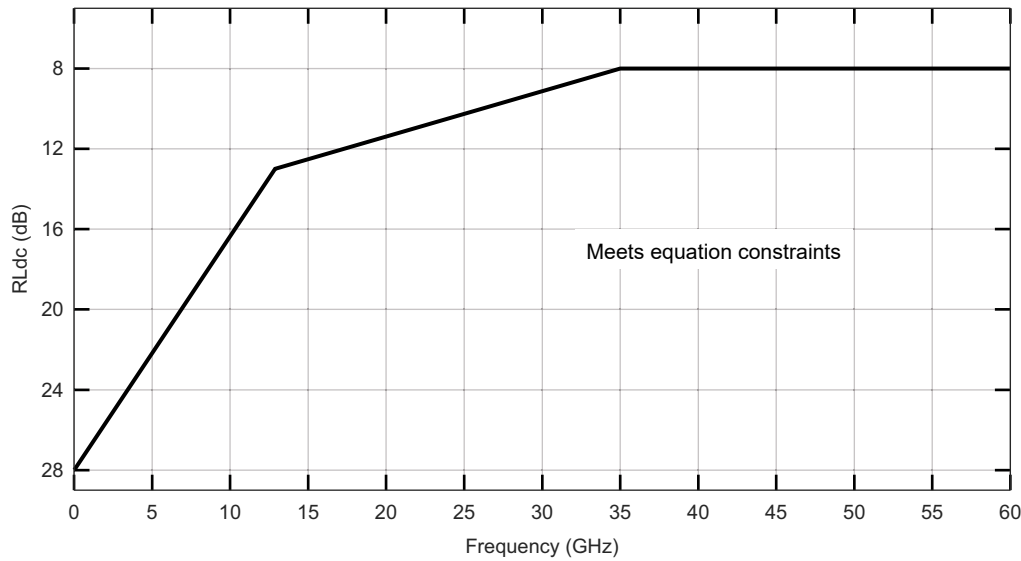


Figure 179-6—Transmitter common-mode to differential-mode return loss

179.9.5 Receiver characteristics

The receiver on each lane shall meet the specifications given in Table 179-10 and detailed in the referenced subclauses. Unless specified otherwise, all receiver measurements are made for each lane separately, at TP3, utilizing the test fixtures specified in Annex 179B.

Recommended receiver characteristics at TP5d are provided in Annex 179A.3.

Table 179-10—Summary of receiver specifications at TP3

Parameter	Subclause reference	Value	Units
Signaling rate (range)	179.9.5.1	106.25 ± 50 ppm	GBd
Amplitude tolerance ^a	179.9.5.2	0.5	V
Interference tolerance	179.9.5.3	Table 179-11	—
Jitter tolerance	179.9.5.4	Table 179-12	—
Effective return loss, ERL (min)	179.9.5.5	7.3	dB
Differential-mode to common-mode return loss, RL_{cd} (min)	179.9.5.6	Equation (179-27)	dB

^a Specified as the steady-state voltage (as defined in 179.9.4.1.2) measured at the test transmitter's output.

179.9.5.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 179.9.5.2, 179.9.5.3, and 179.9.5.4 for any signaling rate in the range specified in Table 179-10.

179.9.5.2 Receiver amplitude tolerance

When a PMD receiver is connected to a compliant transmitter that has a steady-state voltage (see 179.9.4.1.2) equal to the Amplitude tolerance value in Table 179–10, using a compliant cable assembly with the minimum insertion loss specified in 179.11.2, the PMD receiver operation shall enable a block error ratio as specified in 179.2.

The receiver is allowed to control the transmitter equalizer coefficients, using the ILT function (see 179.8.9) or an equivalent process, to meet these requirements.

179.9.5.3 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 179.9.5.3.1 through 179.9.5.3.5. Receiver interference tolerance test requirements are specified in Table 179–11.

Two tests are defined. Test 1 includes a low-loss channel. Test 2 includes a high-loss channel. The cable assembly used in the test channel (see 110.8.4.2.2) shall meet the cable assembly Channel Operating Margin (COM) as specified in 179.11.7.

Table 179–11—Interference tolerance test parameters

Parameter	Test 1 (low loss)		Test 2 (high loss)		Units
	Min	Max	Min	Max	
Test pattern	PRBS31Q				
Block error ratio required ^a	$< 1.45 \times 10^{-11}$				
Test channel insertion loss at 53.125 GHz ^b					
Host class HL	15.5	16.5	36.5	37.5	dB
Host class HN	15.5	16.5	31.5	32.5	dB
Host class HH	15.5	16.5	26.5	27.5	dB
Cable assembly insertion loss at 53.125 GHz					
Host class HL	15.5	16.5	33.5	34.5	dB
Host class HN	15.5	16.5	28.5	29.5	dB
Host class HH	15.5	16.5	23.5	24.5	dB
COM ^c	3		3		dB

^a See 179.2 for definition of block error ratio.

^b Insertion loss between the two test references (see Figure 110–3b).

^c COM is calculated as defined in 179.9.5.3.3. Meeting the test requirements with a lower value of COM demonstrates margin to the specification but is not required for compliance.

179.9.5.3.1 Test setup

The interference tolerance test is performed with the setup similar to the one described in 110.8.4.2.1. The test setup includes noise injection for all lanes of the pattern generator.

179.9.5.3.2 Test channel

The test channel is the same as the one defined in 110.8.4.2.2, except that the cable assembly meets the requirements of 179.11, the test channel loss meets the requirements of Table 179–11, and the cable assembly test fixture meets the requirements of 179B.3.

179.9.5.3.3 Test channel calibration

The scattering parameters of the test channel are measured at the test references as illustrated in Figure 110–3b using the cable assembly test fixtures specified in 179B.3.

The insertion loss at 53.125 GHz of the signal path between the test references in Figure 110–3b is within the limits in Table 179–11.

The COM is calculated using the method and parameters of 179.11.7 with the following considerations:

- The test channel is measured between the Tx and Rx test references shown in Figure 110–3b. COM is calculated using the receiver host channel, package, and device termination models in Table 179–16 corresponding to the host class of the receiver under test.
- The transition time T_r is measured at the Tx test reference with the transmit equalizer set to preset 1 (i.e., equalization off with maximum amplitude), using the method in 120E.3.1.5, with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz.
- EOJ₀₃, J4u₀₃, and J_{RMS} are measured at the Tx test reference with pattern generator noise injection disabled (see 179.9.5.3.4). The measured values comply with the specification in Table 179–7. In the calculation of COM, A_{DD} and σ_{RJ} are calculated from the measured values of J4u₀₃ and J_{RMS} using Equation (179–21) and Equation (179–22), replacing the values in Table 179–18. It is recommended to adjust the pattern generator jitter such that J4u₀₃ and J_{RMS} are as close as practical to their limits in Table 179–7.
- In the calculation of the channel S-parameters in 178A.1.2, the transmitter S-parameter model $S^{(t)}$ is omitted. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A–10) uses a rise time filter $H_t(f)$ (see 178A.1.6.2) calculated with T_r measured in step b.
- In the calculation of COM, SNR_{TX} is set to the SNDR value with $N_p=85$ measured at the Tx test reference with pattern generator noise disabled.
- The interference noise (see 178A.1.7.5) is derived from the S-parameters of the test channel. For a given noise RMS noise voltage σ_{ns} measured at the Tx test reference, the spectral density of the noise source $N_s(f)$ used for calculation of COM may be calculated using Equation (179–23). Alternatively, $N_s(f)$ may be obtained by measurement of the noise spectral density at the output of the noise source.
- The value of σ_{ns} that yields the target value of COM specified in Table 179–11 is determined (e.g., via iterative calculation of COM). The noise level at the Tx test reference is then configured so that the measured RMS voltage matches σ_{ns} .

$$A_{DD} = \frac{\frac{J4u_{03}}{2} + Q4d \sqrt{(Q4d^2 + 1) \times J_{RMS}^2 - \left(\frac{J4u_{03}}{2}\right)^2}}{Q4d^2 + 1} \quad (179-21)$$

$$\sigma_{RJ} = \frac{\frac{J4u_{03}}{2} - A_{DD}}{Q4d} \quad (179-22)$$

where $Q4d = 3.719$.

NOTE 1— $Q4d$ is an approximated solution of $Q(Q4d) = 10^{-4}$, where the Q function is defined in Equation (95–1).

NOTE 2—The calculation of A_{DD} can, under certain conditions, result in a negative argument of the square root. If this condition occurs, the recommended solution is to increase sinusoidal jitter or bounded uncorrelated jitter such that the argument is positive.

$$N_s(f) = \begin{cases} \left(\frac{2\sigma_{ns}^2}{\beta \cdot f_b} \right) \frac{(f/f_{hp})^2}{1 + (f/f_{hp})^2} & 0 \leq f \leq \frac{f_b}{2} \\ 0 & \text{otherwise} \end{cases} \quad (179-23)$$

where

$$\begin{aligned} f_b & \text{ is defined in Table 179-18} \\ f_{hp} & \text{ is 6 GHz} \\ \beta & \text{ is } 1 - (2f_{hp}/f_b) \operatorname{atan}\left(\frac{f_b}{2f_{hp}}\right). \end{aligned}$$

179.9.5.3.4 Pattern generator and noise injection

The pattern generator transmits data to all lanes of the device under test. At the start of transmitter training, the pattern generator output amplitude on all lanes shall be 0.8 V peak-to-peak differential when measured on an alternating zero-three sequence. The output amplitude, measured on an alternating zero-three sequence, is not permitted to exceed 0.8 V peak-to-peak differential during transmitter training. The output waveform and the ERL of the pattern generator comply with 179.9.4.

Noise is added to the signal before the Tx test reference, with noise level set according to step g) in 179.9.5.3.3.

The noise required for each lane is calibrated. The noise may be added either to one lane at a time or using multiple noise sources to all lanes at the same time.

To avoid excessive low-frequency weighting to the receiver input noise, the noise added to the signal is bounded by the normalized spectral density mask defined in Equation (179-24), Equation (179-25), and Equation (179-26) with $f_1 = 8$ GHz and $f_2 = 5$ GHz, and illustrated in Figure 179-7.

$$NSD_{avg} = \frac{1}{(f_b/2 - f_1)} \int_{f_1}^{f_b/2} NSD(f) df \quad (179-24)$$

$$10 \log_{10} \left(\frac{NSD(f)}{NSD_{avg}} \right) \geq -3 + 3.6(f/f_b) \quad f_1 \leq f \leq f_b/2 \quad (179-25)$$

$$10 \log_{10} \left(\frac{NSD(f)}{NSD_{avg}} \right) \leq \begin{cases} -12 + 15f/f_2 & 0 \leq f < f_2 \\ 3 & f_2 \leq f \leq f_b/2 \end{cases} \quad (179-26)$$

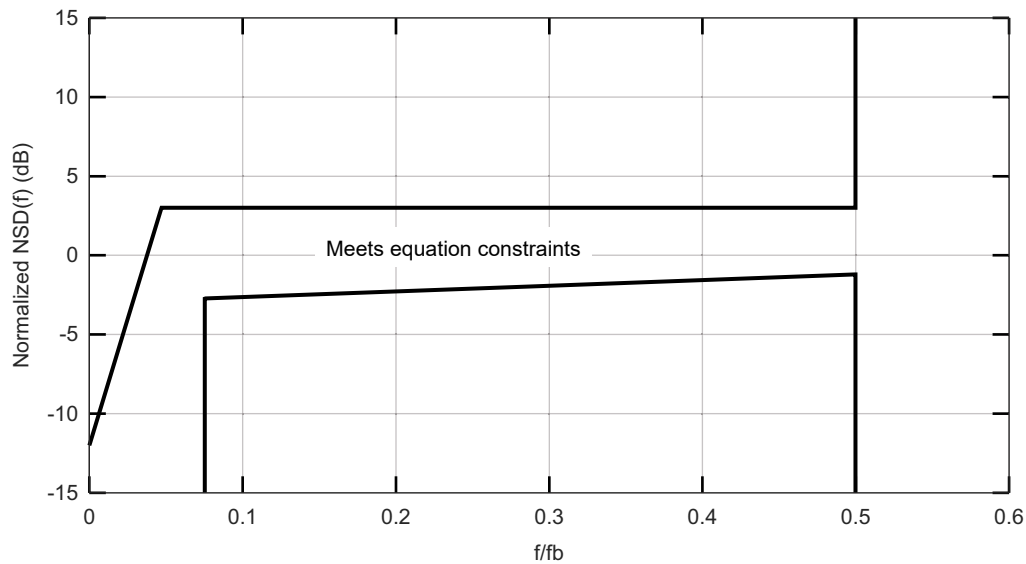


Figure 179-7—NSD(f) constraints

179.9.5.3.5 Test procedure

The pattern generator is first configured to transmit the training pattern defined in 179.8.9. During this initialization period, the device under test (DUT) configures the pattern generator transmit equalizer to the coefficient settings it would select using the start-up protocol described in 179.8.9 and the receiver is tuned using its optimization method. The coefficient settings may be communicated via the start-up protocol or by other means.

After the pattern generator equalizer has been configured and the receiver tuned, the pattern generator is set to generate the test pattern specified in Table 179-11. During the test, the transmitters in the device under test transmit the same pattern type specified for the test, with equalization turned off (preset 1 condition).

A PHY shall meet the block error ratio requirement in all tests defined in Table 179-11 with broadband noise added to all lanes (see 179.9.5.3.4).

Block error ratio is defined in 174A.8.

179.9.5.4 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 179.9.5.4.1 and 179.9.5.4.2.

179.9.5.4.1 Test setup

Jitter tolerance is measured with a channel meeting the insertion loss of Test 2 as specified in Table 179-11. The pattern generator includes sinusoidal jitter injection.

179.9.5.4.2 Test procedure

The jitter tolerance test procedure is similar to that of 179.9.5.3, with the exception that no noise is injected (i.e., step g in 179.9.5.3.3 is not performed). Instead, jitter with the specified frequency and amplitude is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency in Table 179–12 at the Tx test reference (see Figure 110–3a). The test channel COM, calculated per 179.9.5.3.3 with the jitter-stressed transmitter output, shall not be lower than the value in Table 179–11.

A PHY shall meet the block error ratio requirement defined in Table 179–11 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179–12 with jitter added to all lanes.

Table 179–12—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Case G	Units
Jitter frequency	0.04	0.1333	0.4	1.333	4	12	40	MHz
Jitter amplitude (peak-to-peak)	5	1.5	0.5	0.15	0.05	0.05	0.05	UI

NOTE—The A_{DD} (Equation (179–21)) and σ_{RJ} (Equation (179–22)) calculated from transmitter measurements in this test may be higher than the values in Table 179–18. A suitable channel should be chosen in order to meet the COM requirement with these values.

179.9.5.5 Receiver ERL

The ERL of the receiver at TP3 is computed using the procedure in 93A.5 with the values in Table 179–9 and Table 179–16, and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the test connector and the host-facing connection, excluding the mating interface discontinuity. T_{fx} is provided by the test fixture provider.

Receiver ERL at TP3 shall meet the requirement specified in Table 179–10.

179.9.5.6 Receiver differential-mode to common-mode return loss

The receiver differential-mode to common-mode return loss shall meet Equation (179–27) as illustrated in Figure 179–8.

$$RLcd(f) \geq \begin{cases} 28 - 15 \frac{f}{12.89} & 0.01 \leq f < 12.89 \\ 13 - 5 \frac{f - 12.89}{35 - 12.89} & 12.89 \leq f < 35 \\ 8 & 35 \leq f \leq 60 \end{cases} \quad (179-27)$$

where

$RLcd(f)$ is the receiver differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

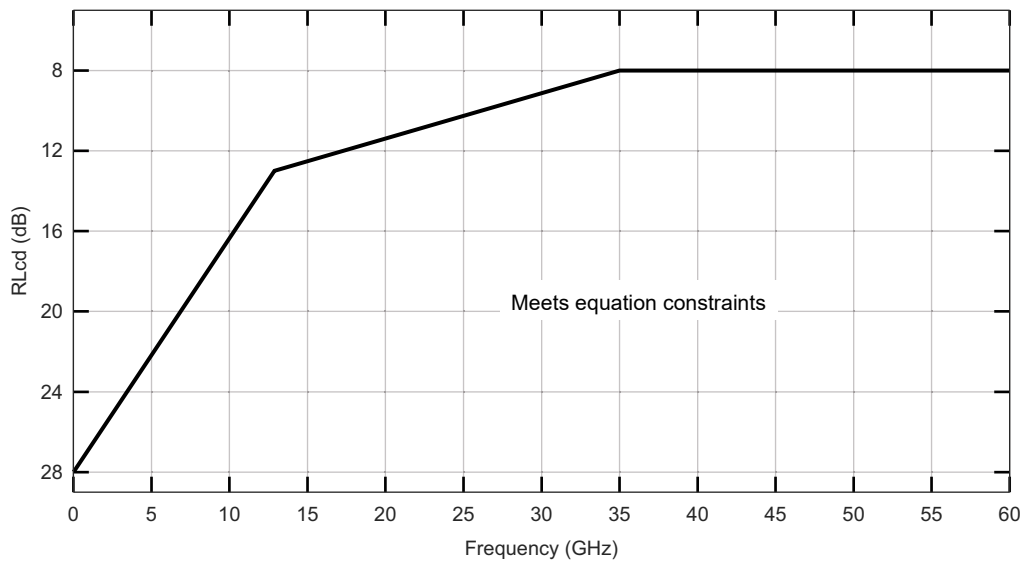


Figure 179-8—Receiver differential-mode to common-mode return loss

179.10 Channel (TP0d-TP5d) characteristics

The channel is defined between TP0d and TP5d to include the transmitter and receiver differential controlled impedance package and PCB and the cable assembly as illustrated in Figure 179-2. The channel insertion loss, return loss, COM, and the host channel parameters are provided informatively in 179A.4 through 179A.7.

Channel definitions apply for links between two PHYs of the same type: 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8.

179.11 Cable assembly characteristics

Cable assemblies defined in this subclause contain insulated conductors terminated in a connector at each end for use as link segments between MDIs. Cable assemblies are primarily intended as point-to-point links between 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8 PHYs using controlled impedance cables.

Four cable assembly types are specified:

- 200GBASE-CR1: Cable assembly that supports single-lane links between two 200GBASE-CR1 PHYs.
- 400GBASE-CR2: Cable assembly that supports two-lane links between two 400GBASE-CR2 PHYs.
- 800GBASE-CR4: Cable assembly that supports four-lane links between two 800GBASE-CR4 PHYs.
- 1.6TBASE-CR8: Cable assembly that supports eight-lane links between two 1.6TBASE-CR8 PHYs.

The specifications for the four cable assembly types are identical except the number of lanes.

For each of the cable assembly types, four cable assembly classes are defined, labeled CA-A, CA-B, CA-C, and CA-D. The cable assembly classes differ by the maximum insertion loss (see 179.11.2).

The cable assembly types are described in Annex 179D. The MDIs are defined in Annex 179C.

Within the cable assembly, all lanes shall be AC-coupled. It is recommended that AC-coupling is within the plug connectors. The low-frequency 3 dB cutoff of the AC-coupling shall be less than 100 kHz. If AC-coupling is implemented using coupling capacitors, it is recommended that the value of the capacitors be less than or equal to 100 nF, to limit the inrush charge.

NOTE—There are various methods for AC-coupling in actual implementations.

All cable assembly measurements are to be made between TP1 and TP4 with cable assembly test fixtures as specified in Annex 179B. These cable assembly specifications are based upon twinaxial cable characteristics, but other cable types are acceptable if the specifications of this subclause are met.

Table 179–13 provides a summary of the cable assembly characteristics and references to the subclauses addressing each parameter.

Table 179–13—Cable assembly characteristics summary

Description	Reference	Value	Unit
Insertion loss at 53.125 GHz, IL_{dd} (max)	179.11.2		
CA-A		19	dB
CA-B		24	dB
CA-C		29	dB
CA-D		34	dB
Insertion loss at 53.125 GHz, IL_{dd} (min)	179.11.2	16	dB
Minimum cable assembly ERL ^a	179.11.3	8.25	dB
Differential-mode to common-mode return loss, RL_{cd}	179.11.4	Equation (179–27)	dB
Mode conversion insertion loss	179.11.5	Equation (179–28) and Equation (179–29)	dB
Common-mode to common-mode return loss, RL_{cc}	179.11.6	Equation (179–19)	dB
Minimum COM	179.11.7	3	dB
NOTE—The expected cable assembly reach is 0.5 m for CA-A, 1 m for CA-B, 1.5 m for CA-C, and 2 m for CA-D. Compliant cable assemblies may be longer. The length of a cable assembly does not imply compliance to specifications.			

^a Cable assemblies with a COM greater than 4 dB are not required to meet minimum ERL.

179.11.1 Reference impedance

The reference differential impedance for cable assembly specifications is 100 Ω . The reference common-mode impedance for cable assembly specifications is 25 Ω .

179.11.2 Cable assembly insertion loss

The measured insertion loss at 53.125 GHz of a cable assembly shall be less than or equal to IL_{dd} (max) in Table 179–13 for the cable assembly class the cable adheres to.

The measured insertion loss at 53.125 GHz of a cable assembly shall be greater than or equal to IL_{dd} (min) in Table 179–13.

179.11.3 Cable assembly ERL

The cable assembly ERL at TP1 and at TP4 is defined by the procedure in 93A.5 with the values in Table 179–14 and Table 179–16, and with T_{fx} equal to twice the test fixture delay. The test fixture delay is defined as the propagation delay between the test connector and the cable-facing connection, excluding the mating interface discontinuity. T_{fx} is provided by the test fixture provider.

The values of cable assembly ERL at TP1 and at TP4 shall meet the requirement specified in Table 179–13 for cable assemblies that have a COM less than 4 dB.

Table 179–14—Cable assembly ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	9000	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

179.11.4 Differential-mode to common-mode return loss

The cable assembly differential-mode to common-mode return loss shall meet Equation (179–27) as illustrated in Figure 179–8.

179.11.5 Mode conversion insertion loss

The cable assembly differential-mode to common-mode insertion loss and common-mode to differential-mode insertion loss are specified relative to the insertion loss.

The difference between the cable assembly differential-mode to common-mode insertion loss and the cable assembly insertion loss shall meet Equation (179–28).

$$IL_{cd}(f) - IL_{dd}(f) \geq \Delta IL(f) \quad (179-28)$$

where

$IL_{cd}(f)$	is the cable assembly differential-mode to common-mode insertion loss at frequency f in dB
$IL_{dd}(f)$	is the cable assembly insertion loss at frequency f in dB
$\Delta IL(f)$	is defined by Equation (179–30) and illustrated in Figure 179–9
f	is the frequency in GHz

The difference between the cable assembly common-mode to differential-mode insertion loss and the cable assembly insertion loss shall meet Equation (179–29).

$$ILdc(f) - ILdd(f) \geq \Delta IL(f) \quad (179-29)$$

where

$ILdc(f)$ is the cable assembly common-mode to differential-mode insertion loss at frequency f in dB

$ILdd(f)$ is the cable assembly insertion loss at frequency f in dB

$\Delta IL(f)$ is defined by Equation (179-30) and illustrated in Figure 179-9

f is the frequency in GHz

$$\Delta IL(f) = \begin{cases} 10 & 0.05 \leq f < 26.5625 \\ 10 - 4 \frac{f - 26.5625}{26.5625} & 26.5625 \leq f \leq 60 \end{cases} \quad (179-30)$$

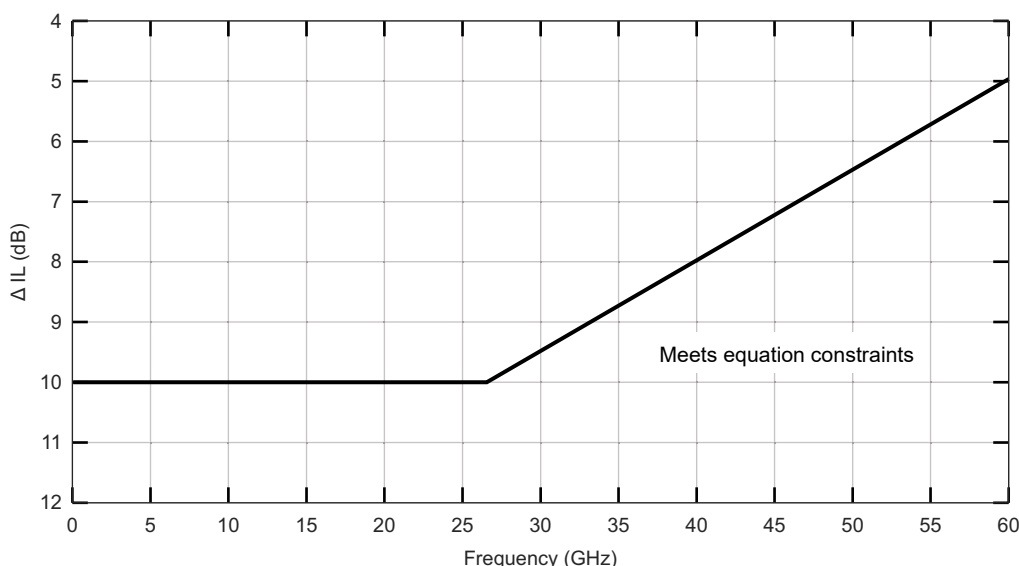


Figure 179-9—Cable assembly mode conversion insertion loss to insertion loss difference

179.11.6 Common-mode to common-mode return loss

The cable assembly common-mode to common-mode return loss shall meet Equation (179-19) as illustrated in Figure 179-5.

179.11.7 Cable assembly Channel Operating Margin

The cable assembly Channel Operating Margin (COM) for each lane is derived from measurements of the cable assembly signal, near-end crosstalk, and far-end crosstalk paths. COM is computed using the procedure in 178A.1 and parameter values provided in 179.11.7.1. The signal paths for calculation of COM are defined in 179.11.7.2.

Calculation of COM is done using the maximum likelihood sequence detection (MLSD) defined in 178A.1.10.

For each cable assembly class, the valid combinations of transmitter host class and receiver host class are listed in Table 179–15. For each valid combination., COM for any lane within the cable assembly shall be greater than or equal to the minimum COM specified in Table 179–13.

Table 179–15—Cable assembly class and host class valid combinations

Cable assembly class	Host classes, transmitter side	Host classes, receiver side	Number of combinations
CA-A	HN or HL	HL, HN, or HH	6
	HH	HL or HN	2
CA-B	HL	HL, HN, or HH	3
	HN	HL or HN	2
	HH	HL	1
CA-C	HL	HL or HN	2
	HN	HL	1
CA-D	HL	HL	1

179.11.7.1 COM parameters

The parameters in Table 179–16 and Table 179–17 represent a partial host channel that includes a combination of a packaged device and a PCB, using the S-parameter model defined in 178A.1.4 and 178A.1.5. This model is not a host channel specification.

The parameters in Table 179–18 correspond to behavioral models of transmitters and receivers that, with suitable host channel models, are compliant to the PMD specifications in this clause. The purpose of these parameters is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementations.

It is expected that the PMD specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation and host channel construction.

Table 179–16—Host model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF

Table 179–16—Host model parameters (*continued*)

Parameter	Symbol	Value	Units
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	See Table 179–17	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	See Table 179–17	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Partial host channel model			
Single-ended board capacitance at the package-to-board interface	$C_0^{(h)}$	0	nF
Transmission line parameter γ_0	$\gamma_0^{(h)}$	0	1/mm
Transmission line parameter a_1	$a_1^{(h)}$	5.95×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	$a_2^{(h)}$	2.6×10^{-5}	ns/mm
Transmission line parameter τ	$\tau^{(h)}$	5.79×10^{-3}	ns/mm
Transmission line characteristic impedance	$Z_c^{(h)}$	92.5	Ω
Transmission line length	$z_p^{(h)}$	See Table 179–17	mm
Single-ended capacitance at the model-to-measurement interface	C_1	0	nF
Single-ended reference resistance	R_0	50	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package and partial host channel model parameters are based on the reference impedance R_0 . Channel scattering parameters measured with different reference impedance need to be renormalized to match the reference impedance before computation.			

Table 179–17—Partial host channel model parameters per Host class

Parameter	Host class			Units
	HL	HN	HH	
Package class	A	B	B	—
Package transmission line 1 length, $z_p^{(1)}$	8	15	45	mm
Partial host PCB transmission line length, $z_p^{(h)}$	9	70	60	mm

NOTE—For each host class, the sum of the differential insertion loss (IL_{dd}) at 53.125 GHz of the partial host channel (excluding the device termination) and the reference mated test fixtures (see Equation (179B–5) and Figure 179A–1) is equal to the recommended maximum host channel insertion loss in 179A.4 for that host class.

Table 179–18—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—
Transmitter equalizer, coefficient –2	$c(-2)$	0	—
Minimum value		0.14	
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient –1	$c(-1)$	–0.34	—
Minimum value		0	
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient 0	$c(0)$	0.54	—
Minimum value			
Transmitter equalizer, coefficient 1	$c(1)$	–0.2	—
Minimum value		0	
Maximum value		0.02	
Step size			

Table 179–18—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	10^{-8}	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	6	—
Number of fixed-position taps	N_{fix}	15 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	87 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$	$w_{max}(j)$		
$1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$		0.7	—
$N_{fix} < j$		0.05	—
Normalized lower limit on feed-forward coefficient $w(j)$	$w_{min}(j)$		
$1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$		–0.7	—
$N_{fix} < j$		–0.05	—
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-4}	—
Include maximum likelihood sequence detection		yes	—

^a There are 8 fixed-position post-cursor taps.

^b The latest post-cursor position for a floating tap is 80.

179.11.7.2 Channel signal and crosstalk path calculations

The channel paths between TP0d and TP5d used for calculation of the cable assembly COM consist of measured cable assembly signal and crosstalk paths, a reference transmitter model, and a reference receiver model. Reference transmitter and receiver models are defined in 178A.1.4 and 178A.1.5, respectively, using the parameters in Table 179–16 and Table 179–17, which define a model for each of the three host classes HL, HN, and HH.

179.11.7.2.1 Channel signal path

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–15), the S-parameters of the channel signal path from TP0d to TP5d are calculated using Equation (178A–2), where

- S is the measured cable assembly signal differential-mode scattering matrix (TP1 to TP4).
- The transmitter and receiver S-parameter models $S^{(t)}$ and $S^{(r)}$ are calculated using the method described in 178A.1.4 and 178A.1.5, respectively, using the parameters in Table 179–16 and Table 179–17.
- The parameter $z_p^{(h)}$ for the transmitter is taken from the signal path column and the host class of the transmitter in Table 179–17.
- The parameter $z_p^{(h)}$ for the receiver is taken from the signal path column and the host class of the receiver in Table 179–17.

179.11.7.2.2 Channel crosstalk paths

The MDI is a significant contributor to crosstalk and is included in and characterized by the cable assembly crosstalk measurements. Crosstalk includes a near-end path where the aggressor is the PMD transmitter, and in some cases, additional near-end and far-end crosstalk paths where the aggressors are other PMD transmitters that are connected to the same cable assembly.

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–15), the S-parameters of each channel near-end crosstalk path from TP0d to TP5d are calculated using Equation (178A–2), as specified in 179.11.7.2.1, with the following exceptions:

- S is the measured cable assembly near-end crosstalk differential-mode scattering matrix (TP1 to TP4).
- The parameter $z_p^{(h)}$ for the transmitter is taken from the signal path column and the host class of the receiver, instead of the transmitter, in Table 179–17.

For a valid combination of a cable assembly class, transmitter host class, and receiver host class (see Table 179–15), the S-parameters of each channel far-end crosstalk path from TP0d to TP5d are calculated using Equation (178A–2), as specified in 179.11.7.2.1, with the following exceptions:

- S is the measured cable assembly far-end crosstalk differential-mode scattering matrix (TP1 to TP4).

Cable assemblies have several types, as described in Annex 179D. The number of near-end and far-end crosstalk paths per cable assembly type is given in Table 179–19.

Table 179–19—Number of crosstalk paths used in calculation of COM

Victim (one end)	NEXT	FEXT (other end)			
		SFP224	SFP-DD224	QSFP224	QSFP-DD1600 or OSFP1600
SFP224	1	0	1	3	7
SFP-DD224	2	1	1	3	7
QSFP224	4	3	3	3	7
QSFP-DD1600 or OSFP1600	8	7	7	7	7

179.12 MDI specifications

The MDI couples the PMD (specified in 179.8 and 179.9) to the cable assembly (specified in 179.11).

Annex 179C specifies the MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

- 200GBASE-CR1 has five specified MDI connectors: SFP224, SFP-DD224, QSFP224, QSFP-DD1600, and OSFP1600.
- 400GBASE-CR2 has four specified MDI connectors: SFP-DD224, QSFP224, QSFP-DD1600, and OSFP1600.
- 800GBASE-CR4 has three specified MDI connectors: QSFP224, QSFP-DD1600, and OSFP1600.
- 1.6TBASE-CR8 has two specified MDI connectors: QSFP-DD1600 and OSFP1600.

179.13 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 14.7.

179.14 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 179–20 and Table 179–21. Additional variables associated with the ILT function are listed in Table 178B–6 and Table 178B–7.

If the MDIO Interface (see Clause 45) is implemented, PCS control and status variables shall be mapped to MDIO registers as shown in Table 179–20, Table 179–21, Table 178B–6, and Table 178B–7.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 179–20—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	179.8.10	1.0.15	45.2.1.1
Global_PMD_transmit_disable	179.8.6	1.9.0	45.2.1.8
PMD_transmit_disable_7, PMD_transmit_disable_6, PMD_transmit_disable_5, PMD_transmit_disable_4 ^a	179.8.7	1.9.8:5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	179.8.7	1.9.4:3	45.2.1.8
PMD_transmit_disable_1 ^c	179.8.7	1.9.2	45.2.1.8
PMD_transmit_disable_0	179.8.7	1.9.1	45.2.1.8

^a Available only in 1.6TBASE-CR8.

^b Available only in 800GBASE-CR4 and 1.6TBASE-CR8.

^c Available only in 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

Table 179–21—PMD status variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	179.8.8	1.1.7	45.2.1.2
PMD_transmit_fault	179.8.8	1.8.11	45.2.1.7
PMD_receive_fault	179.8.8	1.8.10	45.2.1.7
PMD_signal_detect_7, PMD_signal_detect_6, PMD_signal_detect_5, PMD_signal_detect_4 ^a	179.8.5	1.10.8:5	45.2.1.9
PMD_signal_detect_3, PMD_signal_detect_2 ^b	179.8.5	1.10.4:3	45.2.1.9
PMD_signal_detect_1 ^c	179.8.5	1.10.2	45.2.1.9
PMD_signal_detect_0	179.8.5	1.10.1	45.2.1.9
Global_PMD_signal_detect	179.8.4	1.10.0	45.2.1.9

^a Available only in 1.6TBASE-CR8.

^b Available only in 800GBASE-CR4 and 1.6TBASE-CR8.

^c Available only in 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8.

179.15 Protocol implementation conformance statement (PICS) proforma for Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8⁸

179.15.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

179.15.2 Identification

179.15.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

179.15.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 179, Physical Medium Dependent (PMD) sublayer and baseband medium, type 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁸Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

179.15.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CR1	100GBASE-CR1 PMD	162.1	Can operate as a 100GBASE-CR1 PMD	O.1	Yes [] No []
*CR2	200GBASE-CR2 PMD	162.1	Can operate as a 200GBASE-CR2 PMD	O.1	Yes [] No []
*CR4	400GBASE-CR4 PMD	162.1	Can operate as a 400GBASE-CR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-P PMA	162.1	Device implements Clause 135 PMA for 100GBASE-P	CR1:M	Yes [] N/A []
PMA200	200GBASE-R PMA	162.1	Device implements Clause 120 PMA for 200GBASE-R	CR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	162.1	Device implements Clause 120 PMA for 400GBASE-R	CR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	162.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	162.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!CR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	162.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	162.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC	162.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	CR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	162.1	Device implements Clause 82 100GBASE-R PCS	CR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	162.1	Device implements Clause 119 200GBASE-R PCS	CR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	162.1	Device implements Clause 119 400GBASE-R PCS	CR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	162.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	162.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	162.6	Conforms to constraints	M	Yes []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO capability	162.7	Device implements Clause 45 MDIO with the variable mapping in 162.7.	O	Yes [] No []
*CBL	Cable assembly	162.11	Item marked with CBL are cable assembly specifications, these items are the only ones applicable to cable assemblies and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	162.9.4		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

179.15.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

179.15.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

179.15.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	A136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed_ <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	162.9.4.1.3	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []
PC9	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC10	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []

179.15.4.3 Transmitter specifications

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, PMD in same package as PCS	162.9.4	Per Table 179–7	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, PMD is not in same package as PCS	162.9.4	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	Output voltage specifications	93.8.1.3	Per Table 179–7	M	Yes []
TC4	ERL	162.9.4.8	Per Table 179–7	M	Yes []
TC5	Common-mode to differential-mode return loss	162.9.4.10	Per Table 179–7	M	Yes []
TC6	Common-mode to common-mode return loss	162.9.4.9	Per Table 179–7	M	Yes []
TC7	Steady-state voltage, v_f	162.9.4.1.2	Per Table 179–7	M	Yes []
TC8	Linear fit pulse peak (min)	162.9.4.1.2	Per Table 179–7	M	Yes []
TC9	Level separation mismatch ratio	162.9.4	Per Table 179–7	M	Yes []
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 179–7	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Absolute change in coefficients other than $c(\text{coef_sel})$	162.9.4.1.4	Per Table 179–7	M	Yes []
TC12	Coefficient range	162.9.4.1.5	Per Table 179–7	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120D.3.1.6	Per Table 179–7	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	162.9.4	Per Table 179–7	M	Yes []
TC15	Output jitter	162.9.4.7	Per Table 179–7	M	Yes []
TC16	Low-frequency AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 179–7	M	Yes []
TC17	Full-band AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 179–7	M	Yes []

179.15.4.4 Receiver specifications

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Signaling rate range	162.9.5.1	Per Table 162–15	M	Yes []
RC2	Differential-mode to common-mode return loss at TP3	162.9.5.6	Per Table 162–15	M	Yes []
RC3	Input amplitude tolerance	162.9.5.2	Meets FEC symbol error ratio requirement under described conditions	M	Yes []
RC4	Interference tolerance	162.9.5.3	PHY meets FEC symbol error ratio requirement in all tests defined in Table 162–16 with the specified test setup and procedure	M	Yes []
RC5	Jitter tolerance	162.9.5.4	PHY meets FEC symbol error ratio requirement with the specified test setup and procedure	M	Yes []
RC6	ERL	162.9.5.5	Per Table 162–15	M	Yes []

179.15.4.5 Cable assembly specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CA1	Differential reference impedance	162.11.1	100 Ω	CBL:M	Yes [] N/A []
CA2	Minimum insertion loss	162.11.2	Per Equation (162–19)	CBL:M	Yes [] N/A []
CA3	Maximum insertion loss at 53.125 GHz	162.11.2	19.75 dB	CBL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CA4	ERL	162.11.3	Greater than or equal to 8.25 dB for cable assemblies that have a COM less than 4 dB	CBL:M	Yes [] N/A []
CA5	Differential-mode to common-mode input and output return loss	162.11.4	Per Equation (162–20)	CBL:M	Yes [] N/A []
CA6	Differential-mode to common-mode insertion loss	162.11.5	Per Equation (162–21)	CBL:M	Yes [] N/A []
CA7	Common-mode to common-mode return loss	162.11.6	Per Equation (162–22)	CBL:M	Yes [] N/A []
CA8	Cable assembly Channel Operating Margin (COM)	162.11.7	Greater than or equal to 3 dB for both Test 1 and Test 2, for any lane within the cable assembly	CBL:M	Yes [] N/A []
CA9	AC-coupling	162.11	3 dB cutoff frequency less than 100 kHz	CBL:M	Yes [] N/A []

179.15.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Environmental specifications	162.13	Conform to applicable requirements of 14.7	M	Yes []

180. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

180.1 Overview

This clause specifies the 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMDs together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. These PMDs types provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links over one, two, four, or eight pairs, respectively, of single-mode fiber, with a reach of up to at least 500 m.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 180–1 through Table 180–4, to the medium through the medium dependent interface (MDI).

Table 180–1—Physical Layer clauses associated with the 200GBASE-DR1 PMD

Associated clause	200GBASE-DR1
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120C—200GAUI-8 C2M ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120E—200GAUI-4 C2M ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
120G—200GAUI-2 C2M	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
176D—200GAUI-1 C2M	Optional ^d
178B—ILT	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.4.1.

^c When implemented in a 200GBASE-DR1 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 200GAUI-n may be instantiated within a 200GBASE-DR1 PHY as described in 176B.4.1.

Table 180–2—Physical Layer clauses associated with the 400GBASE-DR2 PMD

Associated clause	400GBASE-DR2
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120C—400GAUI-16 C2M ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120E—400GAUI-8 C2M ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
120G—400GAUI-4 C2M	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
176D—400GAUI-2 C2M	Optional ^d
178B—ILT	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.5.1.

^c When implemented in a 400GBASE-DR2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 400GAUI-n may be instantiated within a 400GBASE-DR2 PHY as described in 176B.5.1.

Table 180–3—Physical Layer clauses associated with the 800GBASE-DR4 PMD

Associated clause	800GBASE-DR4
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
178B—ILT	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-DR4 PHY as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Table 180–4—Physical Layer clauses associated with the 1.6TBASE-DR8 PMD

Associated clause	1.6TBASE-DR8
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
120G—1.6TAUI-16 C2M	Optional ^a
170—1.6Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII Extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a
176D—1.6TAUI-8 C2M	Optional ^a
178B—ILT	Required

^a One or two 1.6TAUI-n may be instantiated within a 1.6TBASE-DR8 PHY as described in 176B.7.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to the guidelines in 176B.7.1.

Figure 180–1 shows the relationship of the PMD and MDI (shown shaded) with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 200 Gb/s Ethernet and 400 Gb/s Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2. 1.6 Tb/s Ethernet is introduced in Clause 174 and the purpose of each PHY sublayer is summarized in 174.2.

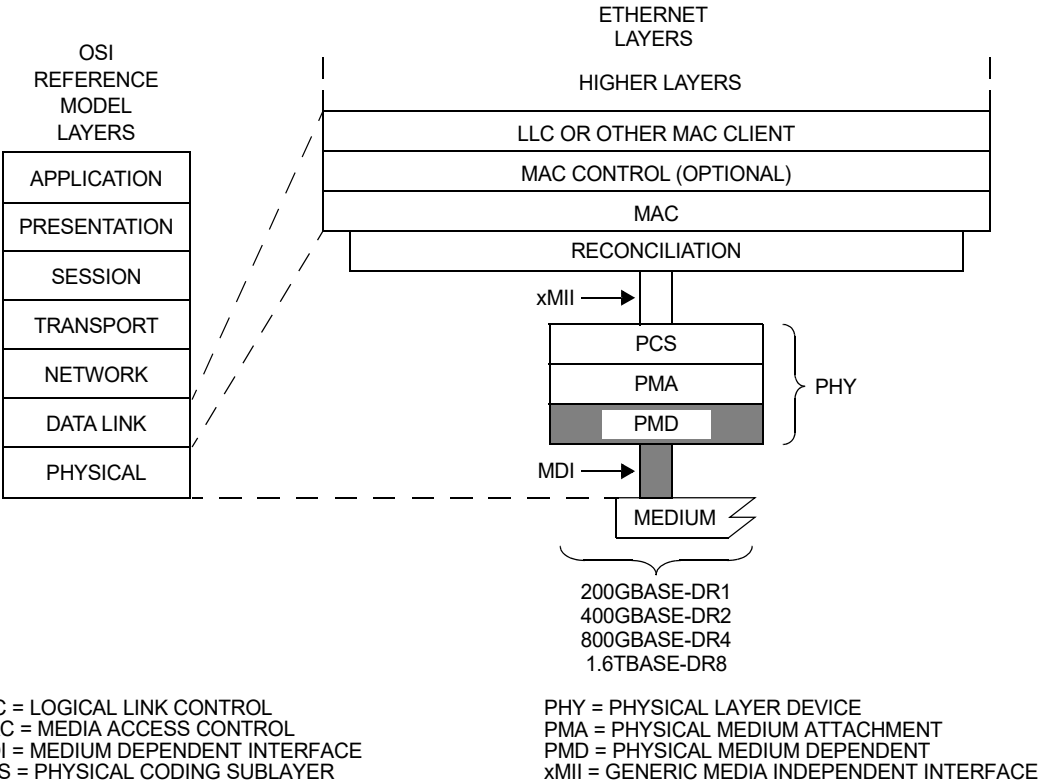


Figure 180-1—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

180.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the PMD using the method described in 174A.8, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 3.2×10^{-5} .

180.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the PMD. The service interface for these PMDs are described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and 174.3 for 1.6TBASE-R. For PMD:IS_UNITDATA_1.request in the transmit direction and PMD:IS_UNITDATA_1.indication in the

receive direction, where $i = 0$ to $n-1$, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 106.25 GBd. The number of parallel streams, n , is 1 for 200GBASE-DR1, 2 for 400GBASE-DR2, 4 for 800GBASE-DR4, and 8 for 1.6TBASE-DR8.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable training_status of the inter-sublayer training (ILT) function (see 180.5.12), as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication on all lanes are unspecified.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.request provides the status from ISLs above the PMD.

180.4 Delay and Skew

180.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-DR1 PMD including 2 m of fiber in one direction shall be no more than 14 848 bit times (29 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR2 PMD including 2 m of fiber in one direction shall be no more than 29 696 bit times (58 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-DR4 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 1.6TBASE-DR8 PMD including 2 m of fiber in one direction shall be no more than 118 784 bit times (232 pause_quanta or 74.24 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 for 200GBASE-DR1 and 400GBASE-DR2, in 169.4 for 800GBASE-DR4, and in 174.4 for 1.6TBASE-DR8.

180.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS.

180.4.2.1 Skew constraints for 200GBASE-DR1 and 400GBASE-DR2

Skew and Skew Variation for 200GBASE-DR1 and 400GBASE-DR2 are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–4 and Figure 116–5. For 200GBASE-DR1, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 200GBASE-DR1 and 400GBASE-DR2 see 116.5.

180.4.2.2 Skew constraints for 800GBASE-DR4 and 1.6TBASE-DR8

Skew and Skew Variation for 800GBASE-DR4 are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

Skew and Skew Variation for 1.6TBASE-DR8 are defined in 174.5 and specified at the points SP1 to SP6 shown in Figure 174–5 and Figure 174–6.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 800GBASE-DR4 see 169.5 and for 1.6TBASE-DR8 see 174.5.

180.5 PMD functional specifications

The 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

180.5.1 Block diagram

The block diagram for 800GBASE-DR4 is shown in Figure 180–2.

The block diagram for 200GBASE-DR1 is equivalent to Figure 180–2, but for one lane per direction, instead of four.

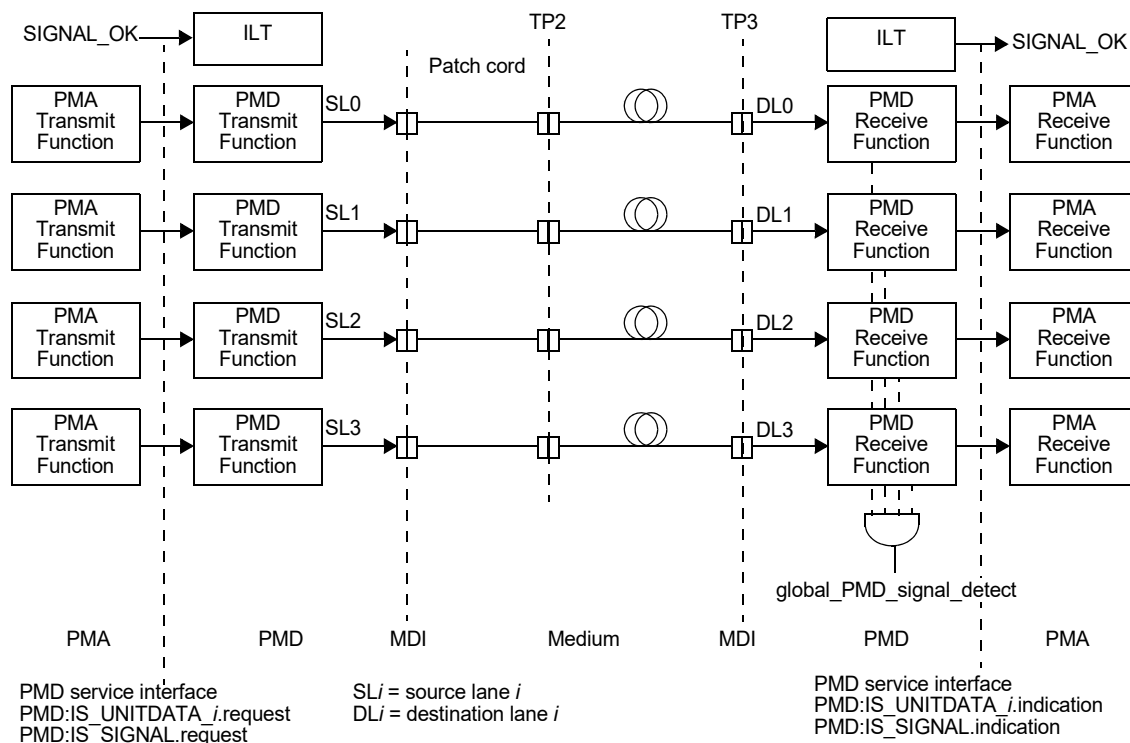
The block diagram for 400GBASE-DR2 is equivalent to Figure 180–2, but for two lanes per direction, instead of four.

The block diagram for 1.6TBASE-DR8 is equivalent to Figure 180–2, but for eight lanes per direction, instead of four.

The ILT function indicated in Figure 180–2 is defined in Annex 178B.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 180.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3)

at the MDI (see 180.8.3). Unless specified otherwise, all receiver measurements and tests defined in 180.9 are made at TP3.



NOTE: For clarity, only one direction of transmission is shown

Figure 180–2—Block diagram for 800GBASE-DR4 transmit/receive paths

180.5.2 PMD transmit function

The PMD Transmit function shall convert the *n* symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA₀.request to PMD:IS_UNITDATA_{*n*–1}.request into *n* separate optical signals. The *n* optical signals shall then be delivered to the MDI, which contains *n* parallel light paths for transmit, according to the transmit optical specifications in this clause. The highest optical power level in each signal shall correspond to tx_symbol = three and the lowest shall correspond to tx_symbol = zero.

180.5.3 PMD receive function

The PMD Receive function shall convert the *n* parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA₀.indication to PMD:IS_UNITDATA_{*n*–1}.indication, all according to the receive optical specifications in this clause. The higher optical power level in each signal shall correspond to rx_symbol = three and the lowest shall correspond to rx_symbol = zero.

180.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all *n* lanes. For the 200GBASE-DR1 PMD, the variable Global_PMD_signal_detect is set to

PMD_signal_detect_0 (see 180.5.5). For the 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMDs, the variable Global_PMD_signal_detect is set to the logical-and of PMD_signal_detect_i from all of the PMD lanes.

180.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable PMD_signal_detect_i to either 0 or 1 according to the criteria in Table 180–5.

Table 180–5—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_i
Average optical power at TP3 ≤ -15 dBm	0
[The input signal is a compliant 200GBASE-R, 400GBASE-R, 800GBASE-R or 1.6TBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 180–8]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_i variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

180.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

180.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 180–7.

If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

180.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitters in each lane to be selectively disabled.

When a `PMD_transmit_disable_i` variable (where i represents the lane number in the range $0:n-1$) is set to one, this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 180–7.

If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disable_i` variable to one, turning off the optical transmitter in each lane.

180.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set `PMD_fault` variable to one.

180.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the `PMD_transmit_fault` variable to one.

180.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the `PMD_receive_fault` variable to one.

180.5.12 Inter-sublayer link training (ILT) function

A PMD shall provide the ILT function for a Type O1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

180.6 Lane assignments

In order to support the ILT function as described in 180.5.12, optical lane assignments (within a group of transmit or receive lanes) for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 are specified. The positioning and ordering of transmit and receive lanes at the MDI is specified in 180.9.

Each fiber between the transmitter of one PHY and the receiver of another PHY shall connect to the same signal lane number. For example, a fiber connects SL1 at the transmitting end to DL1 at the receiving end.

180.7 PMD to MDI optical specifications

The operating range for the 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 PMD is defined in Table 180–6. A 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers according to the specifications defined in Table 180–11. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 400GBASE-DR4 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Table 180–6—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 operating range

PMD type	Required operating range
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8	2 m to 500 m

180.7.1 Transmitter optical specifications

A transmitter shall meet the specifications defined in Table 180–7 per the definitions in 180.9.

Table 180–7—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 transmit characteristics

Description	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Signaling rate, each lane (range)	106.25 ± 50 ppm		GBd
Modulation format	PAM4		—
Lane wavelength (range)	1304.5 to 1317.5		nm
Side-mode suppression ratio (SMSR), each lane (min)	30		dB
Average launch power, each lane (max)	4		dBm
Average launch power, each lane ^a (min)	−3.3 ^b		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.2		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) for max(TECQ, TDECQ) < 0.9 dB for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB	−0.3 −1.2 + max(TECQ, TDECQ)		dBm dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4		dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	3.4		dB
TDECQ − TECQ , each lane (max)	2.5		dB
Transmitter overshoot and undershoot, each lane (max)	22		%
Transmitter power excursion, each lane (max)	2.3		dBm
Extinction ratio, each lane (min)	3.5		dB

Table 180–7—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 transmit characteristics (*continued*)

Description	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Transmitter transition time, each lane (max)	8		ps
Average launch power of OFF transmitter, each lane (max)	–15		dBm
RIN _{xx} OMA, each lane (max) xx = 15.5 xx = 21.4	–139 —	— –139	dB/Hz dB/Hz
Optical return loss tolerance, each lane (max)	15.5	21.4	dB
Transmitter reflectance ^c , each lane (max)	–26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of –3.3 dBm corresponds to an OMA of –0.3 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 180–7 vary with max(TECQ, TDECQ). The relationships are illustrated in Figure 180–3 along with the values for OMA_{outer} each lane (max).

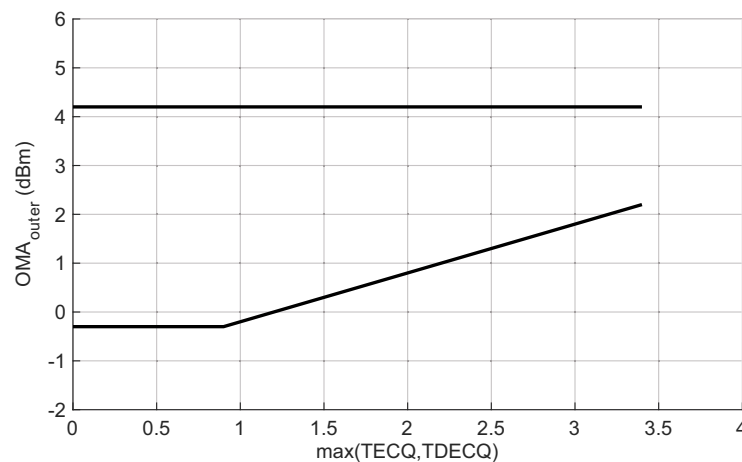


Figure 180–3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max(TECQ, TDECQ)

180.7.2 Receive optical specifications

A receiver shall meet the specifications defined in Table 180–8 per the definitions in 180.9.

Table 180–8—200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 receive characteristics

Description	Value	Unit
Signaling rate, each lane (range)	106.25 ± 50 ppm	GBd
Modulation format	PAM4	—
Lane wavelengths (range)	1304.5 to 1317.5	nm
Damage threshold ^a , each lane	5	dBm
Average receive power, each lane (max)	4	dBm
Average receive power, each lane ^b (min)	−6.3	dBm
Receive power (OMA_{outer}), each lane (max)	4.2	dBm
Receiver reflectance (max)	−26	dB
Receiver sensitivity (OMA_{outer}), each lane ^c (max) for $TECQ < 0.9$ dB for $0.9 \text{ dB} \leq TECQ \leq SECQ$	−3.4 −4.3 + TECQ	dBm
Stressed receiver sensitivity (OMA_{outer}), each lane ^c (max)	−0.9	dBm
Conditions of stressed receiver sensitivity test ^d :		
Stressed eye closure for PAM4 (SECQ), lane under test	3.4	dB
OMA_{outer} of each aggressor lane ^e	4.2	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 180.8) for the block error ratio specified in 180.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^e No aggressors needed for 200GBASE-DR1 in a single lane device.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 180–7 vary depending on the value of TECQ for the test source used for measurement. Figure 180–4 illustrates this.

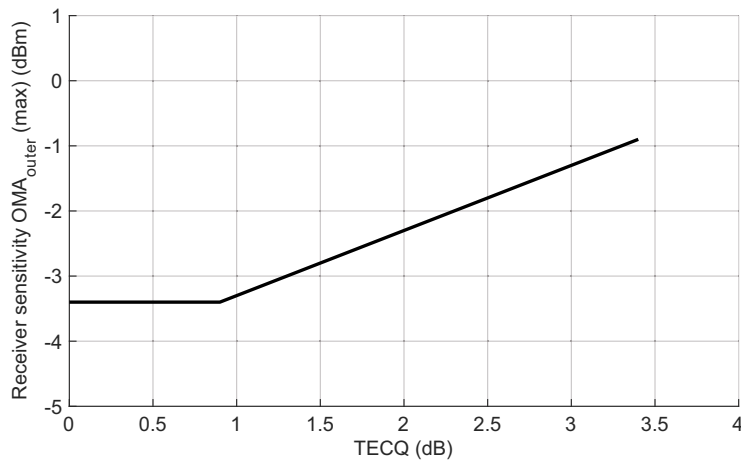


Figure 180-4—Receiver sensitivity (OMA_{outer}), each lane (max)

180.7.3 Illustrative link power budget

An illustrative power budget and penalties for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 channels are shown in Table 180-9.

Table 180-9—Illustrative link power budget

Parameter	Value	Unit
Power budget (for max TDECQ)	6.5	dB
Operating distance	500	m
Channel insertion loss ^{a, b}	3	dB
Maximum discrete reflectance	−35	dB
Allocation for penalties ^c (for max TDECQ)	3.5	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 180-6 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 180.8.2.

^b The channel insertion loss of 3 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.1 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.4 dB depending on the discrete reflectances in the link as shown in Table 180-12.

^c Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. In this illustrative link power budget the value shown includes an allocation of 0.1 dB for MPI and DGD penalties. For cases with a channel insertion loss less than 3 dB, as shown in Table 180-12, the allocation for penalties should be “6.5 – channel insertion loss”.

The values of transmitter $\text{OMA}_{\text{outer}}$ (max), transmitter $\text{OMA}_{\text{outer}}$ (min) versus $\max(\text{TECQ}, \text{TDECQ})$, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) (max) versus TECQ are illustrated in Figure 180–5.

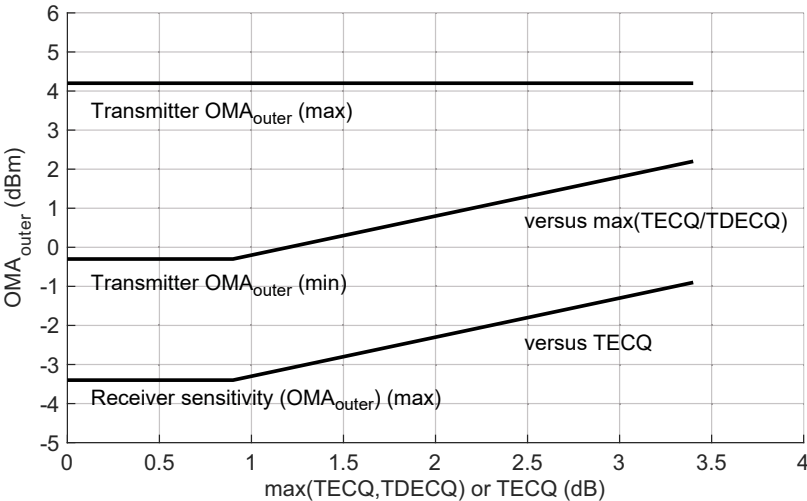


Figure 180–5—Transmitter $\text{OMA}_{\text{outer}}$ each lane versus $\max(\text{TECQ}, \text{TDECQ})$ and receiver sensitivity ($\text{OMA}_{\text{outer}}$) each lane versus TECQ

180.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 180–6.

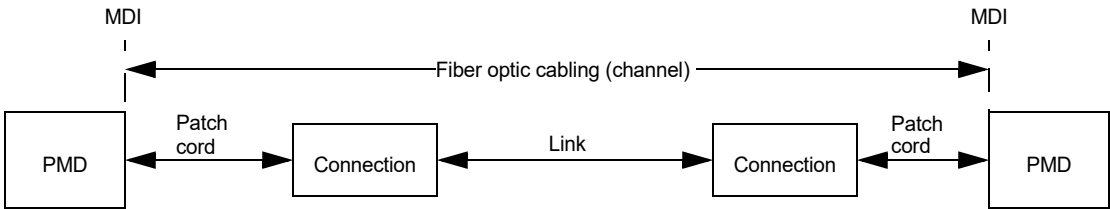


Figure 180–6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 180–10. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. The maximum value of channel insertion loss is dependent on the number and maximum value of the discrete reflectances within the channel as given in Table 180–12. Discrete reflectances below –55 dB may be ignored when determining the supported channel insertion loss.

Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 series one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber

that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 180–10—Optical channel characteristics

Description	200GBASE-DR1	400GBASE-DR2 800GBASE-DR4 1.6TBASE-DR8	Unit
Operating distance (max)	500		m
Channel insertion loss ^{a,b} (max)	See Table 180–12		dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	0.65		ps/nm
Negative dispersion ^b (min)	–0.85		ps/nm
DGD_max ^c	2.24		ps
Optical return loss (min)	27	37	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

180.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 180–11 where they differ.

Table 180–11—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3.

180.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 180–6, consists of a mated pair of optical connectors.

The maximum link distance is based on an allocation of 0.25 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 180–12 being allocated to connection and splice loss. For example, with no discrete reflections greater than –55 dB, a total connection and splice loss allocation of 2.75 dB supports five connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 180–10 are met.

Table 180–12—Maximum channel insertion loss versus number of discrete reflectances

Maximum channel insertion loss (dB)		Number of discrete reflectances > –55 dB and ≤ –45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectances > –45 dB and ≤ –35 dB	0	3	3	3	3	3	3	3	3	3
	1	3	3	3	3	3	3	3	3	3
	2	3	3	3	2.9	2.9	2.9	2.9	2.9	2.9
	3	2.9	2.9	2.9	2.9	2.9	2.8	2.8	2.8	— ^a
	4	2.8	2.8	2.8	2.8	2.7	2.7	2.7	— ^a	— ^a
	5	2.8	2.8	2.7	2.7	2.7	2.6	— ^a	— ^a	— ^a
	6	2.6	2.6	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

180.8.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 180–6). The PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 180A–1. Example constructions of the MDI include the following:

- PMD with a connectorized fiber pigtail plugged into an adapter
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 180.5.1, not at the MDI.

Annex 180A specifies the details of the MDIs for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8.

For 200GBASE-DR1, besides the option to connect to a single fiber MDI, there are two additional specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 400GBASE-DR2, and 800GBASE-DR4 there are two specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 1.6TBASE-DR8 there is one specified MDI optical receptacle, a single-row 16 fiber interface.

180.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

180.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 180–14 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 180–14 may be used to perform that test. The test patterns used in this clause are shown in Table 180–13.

Table 180–13—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.5
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle	119.2.4.9, 172.2.4.11, 175.2.4.11
6	SSPRQ	120.5.11.2.4
7	Valid 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R signal	119.2.4, 172.2.4, 175.2.4

Table 180–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6 or 7	180.9.2
Side mode suppression ratio	3, 5, 6 or 7	180.9.2
Average optical power	3, 5, 6 or 7	180.9.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	180.9.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	180.9.5
Transmitter eye closure for PAM4 (TECQ)	6	180.9.6
Transmitter overshoot and undershoot	6	180.9.7
Transmitter power excursion	6	180.9.8
Extinction ratio	4 or 6	180.9.9
Transmitter transition time	Square wave or 6	180.9.10

Table 180–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
RIN _{xx} OMA	4 or 6	180.9.11
Receiver sensitivity	3 or 5	180.9.12
Stressed receiver conformance test signal calibration	6	180.9.13
Stressed receiver sensitivity	3 or 5	180.9.13

180.9.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 180–7 if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 180–14.

180.9.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 180–7 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 180–14.

180.9.4 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limit given in Table 180–7. The OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 180–14 as the difference between the average optical launch power level P₃, measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P₀, measured over the central 2 UI of a run of 6 zeros, as shown in Figure 180–7. OMA_{outer} is measured using waveforms captured at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

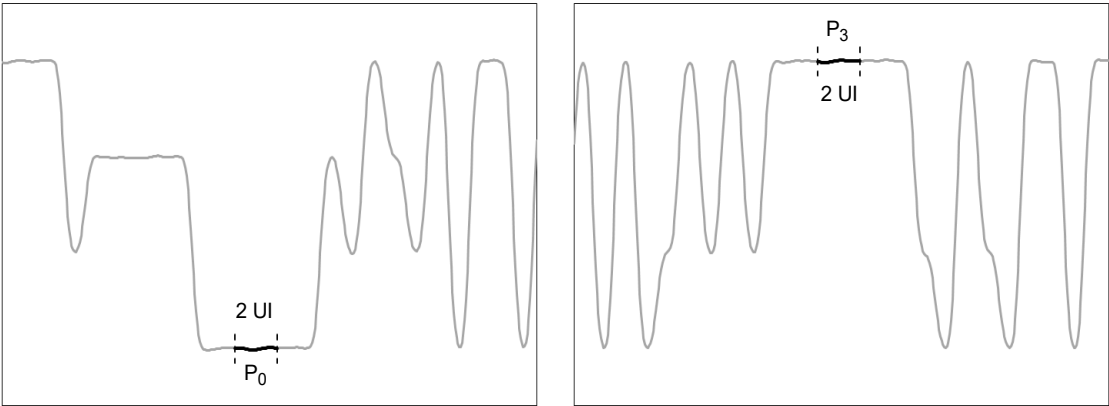


Figure 180–7—Example power levels P₀ and P₃ from PRBS13Q test pattern

180.9.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limits given in Table 180–7 if measured using the methods specified in 121.8.5.1, 121.8.5.3, and 180.9.5.1, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 180–7 and uses the test pattern specified for TDECQ in Table 180–14.
- TDECQ is defined with all receive lanes in operation using test pattern 3 or 5 (see Table 180–13). The received test patterns shall be asynchronous to the pattern used to test the transmitter, and shall have power levels as specified in Table 180–8 for the aggressor lanes in the stressed receiver sensitivity test.
- For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 180–1 through Table 180–4), the clock source for the SSPRQ test pattern is derived from the clock recovered from the xAUI-n input signal. The pattern of the xAUI-n input signal may be PRBS31Q, scrambled idle, or a valid xBASE-R signal.
- The reference receiver, composed of the combination of the O/E converter and the oscilloscope, has a 3 dB bandwidth of approximately 53.125 GHz with a fourth-order Bessel-Thomson response to at least 1.3×106.25 GHz, and at frequencies above 1.3×106.25 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The reference equalizer is a 15-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period, with equalizer coefficient constraints as shown in Table 180–15. The reference equalizer may be implemented in the oscilloscope, but it is not considered part of the reference receiver.
- The target PAM4 symbol error ratio is 4.56×10^{-4} and the related Q_t value (see 121.8.5.3) is 3.428.
- The normalized noise power density spectrum $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 53.125 GHz.
- The optical return loss is as given in Table 180–7.
- The lowest measured TDECQ values are achieved with the equalizer optimization method described in 121.8.5. Alternative optimization methods such as minimum mean squared error (MMSE) may be used to determine equalizer tap weights to reduce test time, and are expected to report equal or higher values of TDECQ. These alternative methods should not be used for receiver sensitivity and stressed receiver sensitivity calibration.

Table 180–15—Reference equalizer tap coefficients

Parameter	Symbol	Value	
		Minimum	Maximum
Feed-forward equalizer (FFE) length	N_b	15	
Number of equalizer pre-cursor taps	—	0	3
Main tap coefficient limit	$c(0)$	0.9	2.5
Normalized equalizer coefficient limits: ^a	$c(i)$		
$i = -3$		−0.15	0.1
$i = -2$		−0.1	0.25
$i = -1$		−0.5	0.1
$i = 1$		−0.6	0.2
$i = 2$		−0.2	0.3
$i = 3$		−0.15	0.15
$i = 4$		−0.15	0.15
$i = 5$		−0.15	0.15
$i = 6$		−0.15	0.15
$i \geq 7$		−0.1	0.1
Equalizer gain ^b	—	1	

^a The main tap is marked by $i = 0$. The minimum and maximum values are relative to this tap's coefficient.

^b The sum of the equalizer coefficients.

180.9.5.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 180–16.

Table 180–16—Transmitter compliance channel specifications

PMD type	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
	Minimum	Maximum			
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8	$0.0463 \times (\lambda - 1311) - 0.55$	$0.0443 \times (\lambda - 1311) + 0.37$	Minimum	See Table 180–7	0.5

^a Dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 500 m distance. The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024).

^b The value “Minimum” implies that the test channel insertion loss should be sufficiently low that it does not significantly stress the tester receiver.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 180–16 for the wavelength of

the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 180–16. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 180–16.

180.9.6 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter's eye closure at TP2. The TECQ shall be within the limit given in Table 180–7 if measured using a test pattern specified for TECQ in Table 180–14. TECQ is measured using the methods specified for TDECQ in 180.9.5, except that the test fiber is not used.

180.9.7 Transmitter overshoot and undershoot

The overshoot and undershoot of each lane shall be within the limit given in Table 180–7 if measured using a test pattern specified for overshoot and undershoot in Table 180–14.

Overshoot and undershoot are measured using the waveforms captured for the TDECQ test (see 180.9.5) and the waveform captured for the TECQ test (see 180.9.6), at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

Overshoot and undershoot are defined in 140.7.7.

180.9.8 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 180–7 if measured using a test pattern specified for transmitter power excursion in Table 180–14.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 180.9.6) at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

Transmitter power excursion is defined in 140.7.8.

180.9.9 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 180–7 if measured using a test pattern specified for extinction ratio in Table 180–14. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 180–7. The extinction ratio is measured using waveforms captured at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

180.9.10 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 180–7 if measured using a test pattern specified for transmitter transition time in Table 180–14.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} , or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling

edges respectively. The transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 180.9.4), with the exception that the square wave test pattern may be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

180.9.11 Relative intensity noise ($RIN_{xx}OMA$)

$RIN_{xx}OMA$ of each lane, with “xx” referring to the value for optical return loss tolerance in Table 180–7, shall be within the limit given in Table 180–7 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 180.9.4, but with applied xx dB optical return loss and the reference receiver specified for TDECQ measurement in 180.9.5. The noise is measured at the output of the reference receiver defined in 180.9.5, before the reference equalizer.

$RIN_{xx}OMA$ is calculated as defined in Equation (180–1):

$$RIN_{xx}OMA = 10\log_{10}\left[\frac{(N_3 + N_0)^2}{OMA_{outer}^2 B}\right] \quad (180-1)$$

where:

- $RIN_{xx}OMA$ = relative intensity noise referred to optical modulation amplitude, measured with xx dB optical return loss,
- N_3 is the RMS deviation from the mean of the optical power at the P_3 level,
- N_0 is the RMS deviation from the mean of the optical power at the P_0 level,
- N_0 and N_3 are to be measured on a region in a place in the pattern that is selected to minimize the measurement error,
- B = Low-pass bandwidth of filter of the reference receiver (Hz).

180.9.12 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 180–8 if measured using a test pattern for receiver sensitivity in Table 180–14. The conformance test signal at TP3 meets the requirements for a transmitter followed by an attenuator.

The TECQ of the conformance test signal is measured according to 180.9.5, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 180–8.

180.9.13 Stressed receiver sensitivity

Stressed receiver sensitivity of each lane shall be within the limit given in Table 180–8 if measured using the method defined in 121.8.10 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 180.9.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 180–7.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $RIN_{xx}OMA$ of the SRS test source should be no greater than the value specified in Table 180–7.

- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 180–7 using test patterns specified in Table 180–14.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ OMA_{outer} of each aggressor lane” are as given in Table 180–8.

180.10 Safety, installation, environment, and labeling

180.10.1 General safety

All equipment subject to this clause shall conform to IEC 62368-1.

180.10.2 Laser safety

200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.⁹

180.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

180.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 PMD over the life of the product while the product operates within the manufacturer’s range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

180.10.5 Electromagnetic emission

A system integrating a 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, or 1.6TBASE-DR8 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

⁹A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

180.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

180.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-DR4).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 180.10.2.

180.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 180–17 and Table 180–18. Additional variables associated with the ILT function are listed in Table 178B–6 and Table 178B–7.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 180–17 and Table 180–18.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 180–17—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	180.5.6	1.0.15	45.2.1.1
PMD_global_transmit_disable	180.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_7 to PMD_transmit_disable_4 ^a	180.5.8	1.9.8 to 1.9.5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	180.5.8	1.9.4, 1.9.3	45.2.1.8
PMD_transmit_disable_1 ^c	180.5.8	1.9.2	45.2.1.8
PMD_transmit_disable_0	180.5.8	1.9.1	45.2.1.8

^a Available only in 1.6TBASE-DR8

^b Available only in 800GBASE-DR4 and 1.6TBASE-DR8

^c Available only in 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

Table 180–18—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	180.5.9	1.1.7	45.2.1.2

Table 180–18—PMD status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_transmit_fault	180.5.10	1.8.11	45.2.1.7
PMD_receive_fault	180.5.11	1.8.10	45.2.1.7
PMD_global_signal_detect	180.5.4	1.10.0	45.2.1.9
PMD_signal_detect_7 to PMD_signal_detect_4 ^a	180.5.5	1.10.8 to 1.10.5	45.2.1.9
PMD_signal_detect_3, PMD_signal_detect_2 ^b	180.5.5	1.10.4, 1.10.3	45.2.1.9
PMD_signal_detect_1 ^c	180.5.5	1.10.2	45.2.1.9
PMD_signal_detect_0	180.5.5	1.10.1	45.2.1.9

^a Available only in 1.6TBASE-DR8

^b Available only in 800GBASE-DR4 and 1.6TBASE-DR8

^c Available only in 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8

180.12 Protocol implementation conformance statement (PICS) proforma for Clause 180, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4¹⁰

180.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 180, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

Editor's note: This PICS will be implemented in Draft 2.1. Please do not submit comments against this subclause.

¹⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

181. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500

181.1 Overview

This clause specifies the 800GBASE-FR4-500 PMD together with the single-mode fiber medium. The optical signals generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMD provide point-to-point 800 Gb/s Ethernet links over four wavelength division multiplexing (WDM) lanes on single-mode fiber, with a reach of up to at least 500 m.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 181–1 to the medium through the medium dependent interface (MDI) and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent.

Table 181–1—Physical Layer clauses associated with the 800GBASE-FR4-500 PMDs

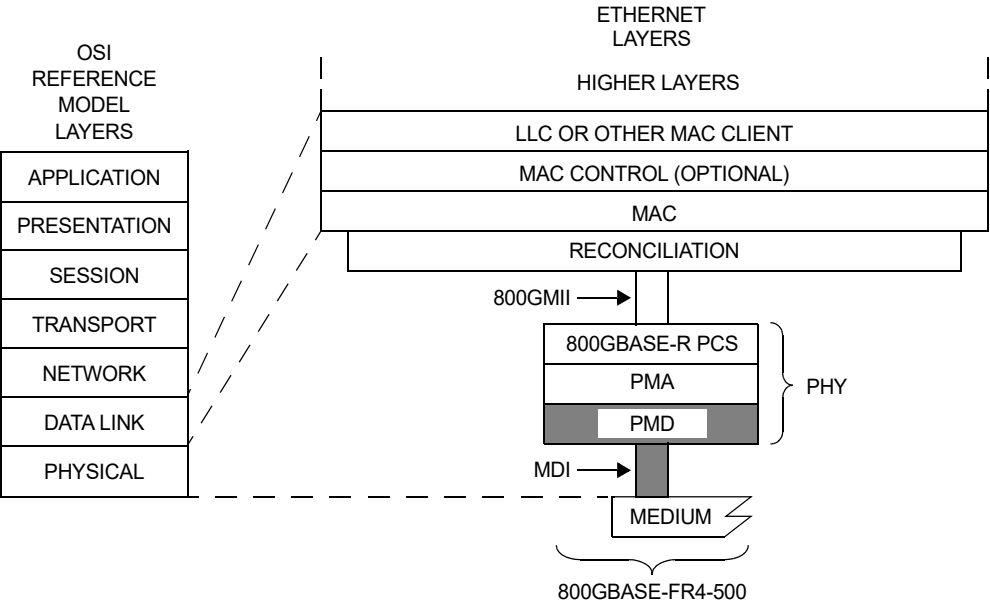
Associated clause	800GBASE-FR4-500
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
178B—ILT	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-FR4-500 PHY as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Figure 181–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
FR4-500 = PMD FOR SINGLE-MODE FIBER — 500 m

Figure 181–1—800GBASE-FR4-500 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

181.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the PMD using the method described in 174A.8, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 3.2×10^{-5} .

181.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 169.3. For PMD:IS_UNITDATA_i.request in the transmit direction and PMD:IS_UNITDATA_i.indication in the receive direction, where $i = 0$ to 3, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 106.25 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable training_status of the inter-sublayer training (ILT) function (see 181.5.12), as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication on all lanes are unspecified.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.request provides the status from ISLs above the PMD.

181.4 Delay and Skew

181.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-FR4-500 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

181.4.2 Skew constraints

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5. The measurements of Skew and Skew Variation are defined in 86.8.3.1.

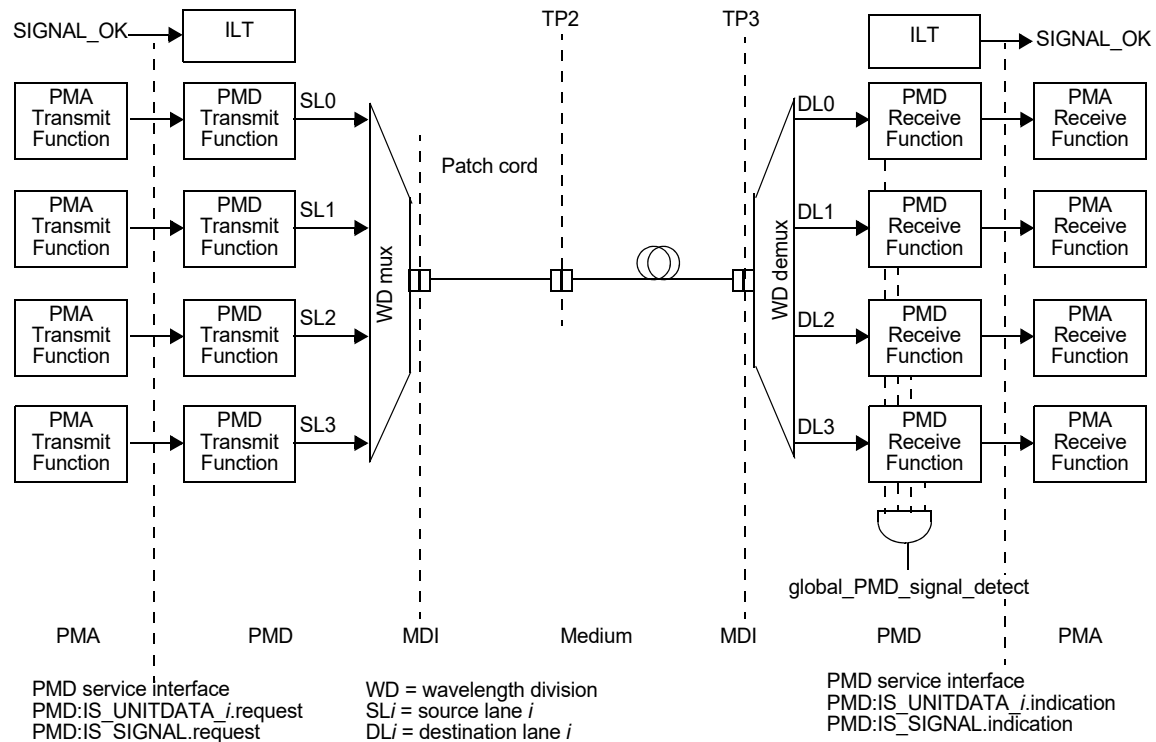
181.5 PMD functional specifications

The 800GBASE-FR4-500 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

181.5.1 Block diagram

The block diagram is shown in Figure 181–2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 181.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see Clause 181.8.3). Unless specified otherwise, all receiver measurements and tests defined in 181.9 are made at TP3.

The ILT function indicated in Figure 181–2 is defined in Annex 178B.



NOTE: For clarity, only one direction of transmission is shown

Figure 181–2—Block diagram for 800GBASE-FR4-500 transmit/receive paths

181.5.2 PMD transmit function

The PMD Transmit function shall convert the four symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA₀.request to PMD:IS_UNITDATA₃.request into four separate optical signals. The four optical signals shall then be wavelength division multiplexed and delivered to the MDI, all according to the transmit optical specifications in this clause. The four optical power levels in the signal in order from lowest to highest shall correspond to tx_symbol values zero, one, two, and three, respectively.

181.5.3 PMD receive function

The PMD Receive function shall demultiplex the composite optical signal received from the MDI into four separate optical signals. The four optical signals shall then be converted into four symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA₀.indication to PMD:IS_UNITDATA₃.indication, all according to the receive optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to rx_symbol values zero, one, two, and three, respectively.

181.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all 4 lanes and it is set to the logical-and of PMD_signal_detect_i from all of the PMD lanes.

181.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to 3, a receiver shall set the variable PMD_signal_detect_ i to either 0 or 1 according to the criteria in Table 181–2

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_ i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_ i variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 181–2—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_ i
For any lane; Average optical power at TP3 ≤ -16 dBm	0
For all lanes; [The input signal is a compliant 800GBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 181–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

181.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

181.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 181–5.

If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

181.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitters in each lane to be selectively disabled.

When a PMD_transmit_disable_ i variable (where i represents the lane number in the range 0:3) is set to one, this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 181–5.

If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_i variable to one, turning off the optical transmitter in each lane.

181.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

181.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

181.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

181.5.12 Inter-sublayer link training (ILT) function

A PMD shall provide the ILT function for a Type O1 interface, specified in Annex 178B. When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

181.6 Wavelength-division-multiplexed lane assignments

The wavelength range for each lane of the 800GBASE-FR4-500 is defined in Table 181–3. The 800GBASE-FR4-500 center wavelengths are members of the CWDM wavelength grid defined in ITU-T G.694.2 and are spaced at 20 nm.

Table 181–3—800GBASE-FR4-500 wavelength-division-multiplexed lane assignments

Lane	Center wavelength	Wavelength range
0	1271 nm	1264.5 to 1277.5 nm
1	1291 nm	1284.5 to 1297.5 nm
2	1311 nm	1304.5 to 1317.5 nm
3	1331 nm	1324.5 to 1337.5 nm

NOTE—Each functional lane, denoted SLi at the transmitter and DLi at the receiver, is mapped to a specific wavelength to support ILT operation.

181.7 PMD to MDI optical specifications

The operating range for the PMD is defined in Table 181–4. A 800GBASE-FR4-500 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-FR4-500 PMD operating at 600 m meets the operating range requirement of 2 m to 500 m).

Table 181–4—800GBASE-FR4-500 operating range

PMD type	Required operating range
800GBASE-FR4-500	2 m to 500 m

181.7.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 181–5 per the definitions in 181.9.

Table 181–5—800GBASE-FR4-500 transmit characteristics

Description	800GBASE-FR4-500	Unit
Signaling rate, each lane (range)	106.25 ± 50 ppm	GBd
Modulation format	PAM4	—
Lane wavelengths (range)	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	nm
Side-mode suppression ratio (SMSR), each lane (min)	30	dB
Total average launch power (max)	10.9	dBm
Average launch power, each lane (max)	4.9	dBm
Average launch power, each lane ^a (min)	−2.2 ^b	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.8	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min) for TDECQ < 0.9 dB for $0.9 \text{ dB} \leq \max(\text{TECQ}, \text{TDECQ}) \leq 3.4 \text{ dB}$	0.8 −0.1 + max(TECQ, TDECQ)	dBm dBm
Difference in launch power between any two lanes (OMA _{outer}) (max)	4	dB
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	3.4	dB
$ \text{TDECQ} - \text{TECQ} $, each lane (max)	2.5	dB
Transmitter overshoot and undershoot, each lane (max)	22	%
Transmitter power excursion, each lane (max)	2.9	dBm
Extinction ratio, each lane (min)	3.5	dB
Transmitter transition time, each lane (max)	8	ps
Average launch power of OFF transmitter, each lane (max)	−15	dBm
RIN _{xx} OMA, each lane (max)	−139	dB/Hz
Optical return loss tolerance (max)	17.1	dB
Transmitter reflectance ^c (max)	−26	dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of −2.2 dBm corresponds to an OMA of 0.8 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 181–5 vary with max(TECQ, TDECQ). The relationships are illustrated in Figure 181–3 along with the values for OMA_{outer} each lane (max).

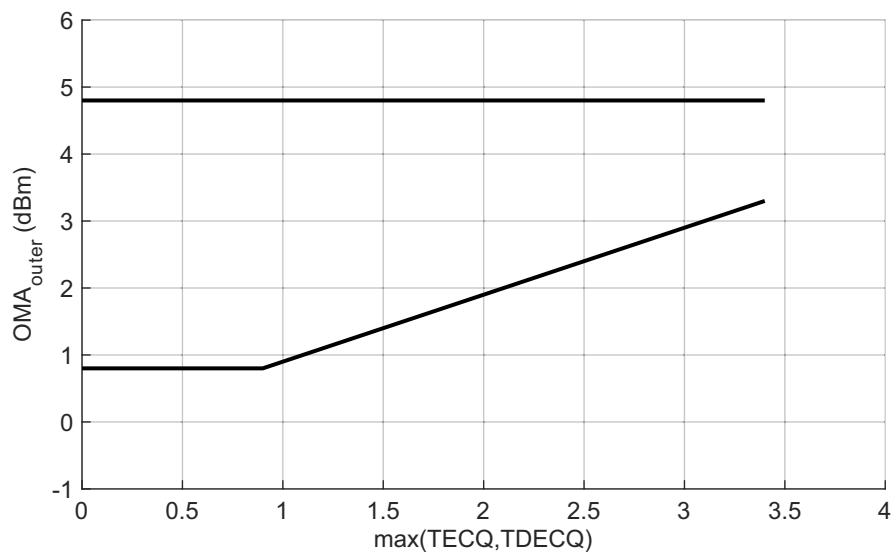


Figure 181-3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max(TECQ, TDECQ)

181.7.2 800GBASE-FR4-500 receive optical specifications

The receiver shall meet the specifications in Table 181-6 per the definitions in 181.9.

Table 181–6—800GBASE-FR4-500 receive characteristics

Description	800GBASE-FR4-500	Unit
Signaling rate, each lane (range)	106.25 ± 50 ppm	GBd
Modulation format	PAM4	—
Lane wavelengths (range)	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	nm
Damage threshold ^a , each lane	5.9	dBm
Average receive power, each lane (max)	4.9	dBm
Average receive power, each lane ^b (min)	−5.7	dBm
Receive power (OMA _{outer}), each lane (max)	4.8	dBm
Difference in receive power between any two lanes (OMA _{outer}) (max)	4.1	dB
Receiver reflectance (max)	−26	dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max) for TECQ < 0.9 dB for 0.9 dB ≤ TECQ ≤ SECQ	−3.2 −4.1 + TECQ	dBm dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	−0.7	dBm
Conditions of stressed receiver sensitivity test: ^d		
Stressed eye closure for PAM4 (SECQ), lane under test	3.4	dB
OMA _{outer} of each aggressor lane	1.3	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 181.9.13) for the block error ratio specified in 181.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 181–6 vary depending on the value of TECQ for the test source used for measurement. Figure 181–4 illustrates this.

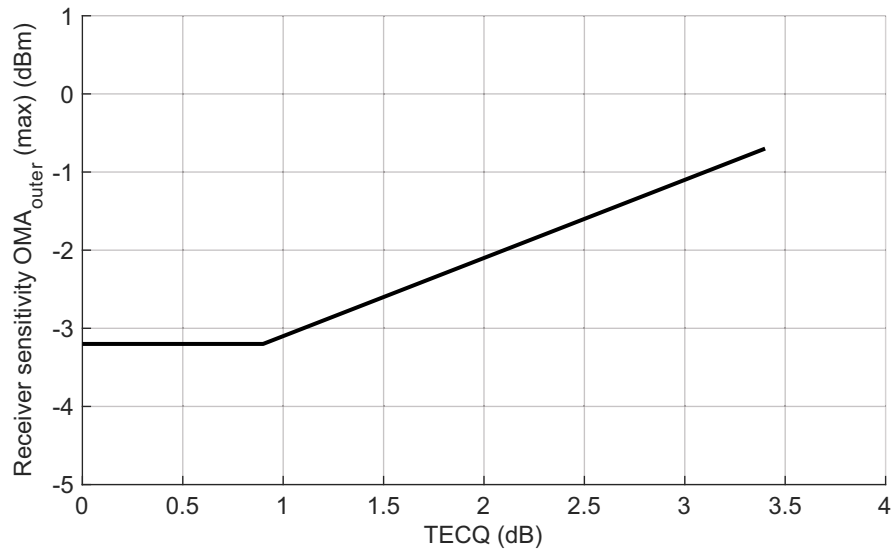


Figure 181-4—Receiver sensitivity (OMA_{outer}), each lane (max)

181.7.3 800GBASE-FR4-500 illustrative link power budget

Illustrative power budgets and penalties for the channels are shown in Table 181-7.

Table 181-7—800GBASE-FR4-500 illustrative link power budgets

Parameter	800GBASE-FR4-500	Unit
Power budget (for maximum TDECQ)	7.4	dB
Operating distance	500	m
Channel insertion loss ^a	3.5	dB
Maximum discrete reflectance ^b	−35 ^c	dB
Allocation for penalties ^d (for maximum TDECQ)	3.9	dB
Additional insertion loss allowed ^e	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 181-4 and fiber attenuation of 0.5 dB/km plus an allocation for connection and splice loss given in 181.8.2.1.

^b See 181.8.2.2 for details and specifications as a function of the number of discrete reflectances within the channel.

^c Maximum value for each discrete reflectance with 4 discrete reflectances above −55 dB within the channel.

^d Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. This value includes an allocation of 0.5 dB for MPI and DGD penalties.

^e The additional insertion loss may be allocated for cable, connectors or splices, provided the fiber optic cabling (channel) characteristics in 181.8 are met.

The values of transmitter OMA_{outer} (max), transmitter OMA_{outer} (min) versus max(TECQ, TDECQ), and receiver sensitivity (OMA_{outer}) (max) versus TECQ are illustrated in Figure 181-5.

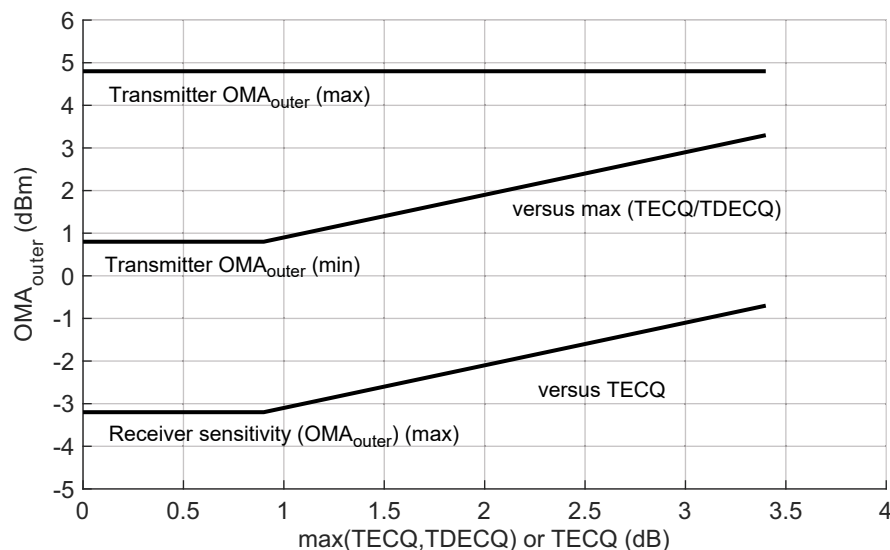


Figure 181-5—Transmitter OMA_{outer} each lane versus $\max(TECQ, TDECQ)$ and receiver sensitivity (OMA_{outer}) each lane versus TECQ

181.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 181-6.

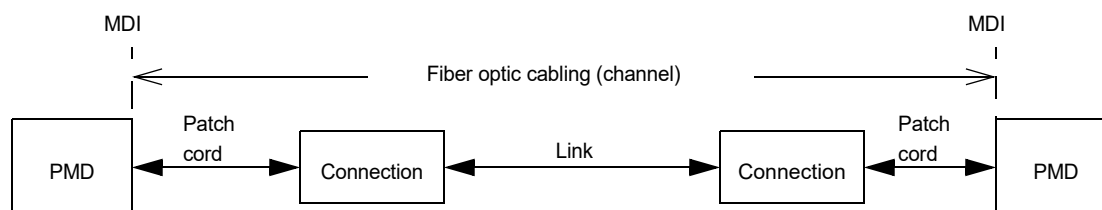


Figure 181-6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 181-8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. The channel insertion loss is given in Table 181-8. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 181–8—Optical channel characteristics

Description	800GBASE-FR4-500	Unit
Operating distance (max)	500	m
Channel insertion loss ^{a, b} (max)	3.5	dB
Channel insertion loss (min)	0	dB
Positive dispersion ^b (max)	1.5	ps/nm
Negative dispersion ^b (min)	–2.82	ps/nm
DGD_max ^c	2.24	ps
Optical return loss (min)	27	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1264.5 nm to 1337.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

181.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 181–9 where they differ.

Table 181–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3.

181.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 181–6, consists of a mated pair of optical connectors.

181.8.2.1 Connection insertion loss

The maximum link distance for 800GBASE-FR4-500 is based on an allocation of 3.25 dB total connection and splice loss. Connections with different loss characteristics may be used provided the requirements of Table 181–9 are met.

181.8.2.2 Maximum discrete reflectance

The maximum value for each discrete reflectance shall be less than or equal to the value shown in Table 181–10 corresponding to the number of discrete reflectances above –55 dB within the channel. For numbers of discrete reflectances in between two numbers shown in the table, the lower of the two corresponding maximum discrete reflectance values applies.

Table 181–10—Maximum value of each discrete reflectance for 800GBASE-FR4-500

Number of discrete reflectances above –55 dB	Maximum value for each discrete reflectance	Unit
1	–25	dB
2	–31	dB
4	–35	dB
6	–38	dB
8	–40	dB
10	–41	dB

181.8.3 Medium Dependent Interface (MDI)

The 800GBASE-FR4-500 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 181–6). Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 181.5.1, not at the MDI.

181.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

181.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 181–12 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 181–12 may be used to perform that test. The test patterns used in this clause are shown in Table 181–11.

Table 181–11—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.5
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle	172.2.4.11
6	SSPRQ	120.5.11.2.4
7	Valid 800GBASE-R signal	172.2.4

Table 181–12—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6 or 7	181.9.2
Side mode suppression ratio	3, 5, 6 or 7	181.9.2
Average optical power	3, 5, 6 or 7	181.9.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	181.9.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	181.9.5
Transmitter eye closure for PAM4 (TECQ)	6	181.9.6
Transmitter overshoot and undershoot	6	181.9.7
Transmitter power excursion	6	181.9.8
Extinction ratio	4 or 6	181.9.9
Transmitter transition time	Square wave or 6	181.9.10
RIN _{17.1} OMA	4 or 6	181.9.11
Receiver sensitivity	3 or 5	181.9.12
Stressed receiver conformance test signal calibration	6	181.9.13
Stressed receiver sensitivity	3 or 5	181.9.13

181.9.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 181–5 if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 181–12.

181.9.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 181–5 if measured using the methods given in IEC 61280-1-1, with the sum of the optical power from all of the lanes not under test below –30 dBm, per the test setup in Figure 53–6.

181.9.4 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 181–5. OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 181–12 as the difference between the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 181–7. OMA_{outer} is measured using waveforms captured at the output of the reference receiver defined in 181.9.5, before the reference equalizer.

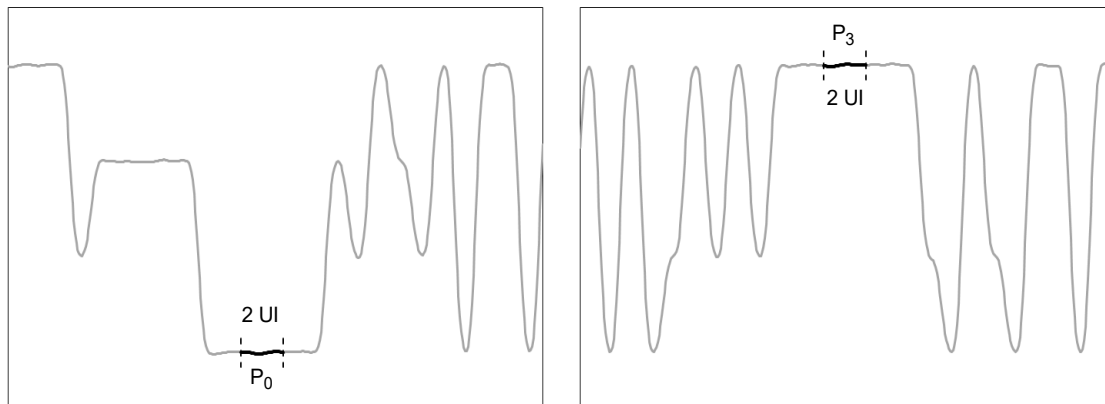


Figure 181–7—Example power levels P_0 and P_3 from PRBS13Q test pattern

181.9.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limits given in Table 181–5 if measured using the methods specified in 121.8.5.1, 121.8.5.3, and 181.9.5.1, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 181–5 and uses the test pattern specified for TDECQ in Table 181–12.
- TDECQ is defined with all receive lanes in operation using pattern 3, or 5 (see Table 181–11). The received test patterns shall be asynchronous to the pattern used to test the transmitter, and shall have power levels as specified in Table 181–6 for the aggressor lanes in the stressed receiver sensitivity test.
- For those cases where there is an 800GAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 181–1), the clock source for the SSPRQ test pattern is derived from the clock recovered

- from the 800GAUI-n input signal. The pattern of the 800GAUI-n input signal may be PRBS31Q, scrambled idle, or a valid 800GBASE-R signal.
- The reference receiver, composed of the combination of the O/E converter and the oscilloscope, has a 3 dB bandwidth of approximately 53.125 GHz with a fourth-order Bessel-Thomson response to at least 1.3×106.25 GHz, and at frequencies above 1.3×106.25 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
 - The reference equalizer is a 15-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period, with equalizer coefficient constraints as shown in Table 181–13. The reference equalizer may be implemented in the oscilloscope, but it is not considered part of the reference receiver.
 - The target PAM4 symbol error ratio is 4.56×10^{-4} and the related Q_t value (see 121.8.5.3) is 3.428.
 - The normalized noise power density spectrum $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 53.125 GHz.
 - The lowest measured TDECQ values are achieved with the equalizer optimization method described in 121.8.5. Alternative optimization methods such as minimum mean squared error (MMSE) may be used to determine equalizer tap weights to reduce test time, and are expected to report equal or higher values of TDECQ. These alternative methods should not be used for receiver sensitivity and stressed receiver sensitivity calibration.

Table 181–13—Reference equalizer tap coefficients

Parameter	Symbol	Value	
		Minimum	Maximum
Feed-forward equalizer (FFE) length	N_b	15	
Number of equalizer pre-cursor taps	—	0	3
Main tap coefficient limit	$c(0)$	0.9	2.5
Normalized equalizer coefficient limits: ^a	$c(i)$		
$i = -3$		–0.15	0.1
$i = -2$		–0.1	0.25
$i = -1$		–0.5	0.1
$i = 1$		–0.6	0.2
$i = 2$		–0.2	0.3
$i = 3$		–0.15	0.15
$i = 4$		–0.15	0.15
$i = 5$		–0.15	0.15
$i = 6$		–0.15	0.15
$i \geq 7$		–0.1	0.1
Equalizer gain ^b	—	1	

^a The main tap is marked by $i = 0$. The minimum and maximum values are relative to this tap's coefficient.

^b The sum of the equalizer coefficients.

181.9.5.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 181–14.

Table 181–14—Transmitter compliance channel specifications

Lane	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
	Minimum	Maximum			
0	$0.0508 \times (\lambda - 1271) - 2.49$	$0.0468 \times (\lambda - 1271) - 1.45$	Minimum	17.1	0.5
1	$0.0485 \times (\lambda - 1291) - 1.5$	$0.0458 \times (\lambda - 1291) - 0.53$			
2	$0.0463 \times (\lambda - 1311) - 0.55$	$0.0443 \times (\lambda - 1311) + 0.37$			
3	$0.044 \times (\lambda - 1331) + 0.35$	$0.0423 \times (\lambda - 1331) + 1.23$			

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 500 m distance.

^b The value “Minimum” implies that the test channel insertion loss should be sufficiently low that it does not significantly stress the tester receiver.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A 800GBASE-FR4-500 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 181–14 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 181–14. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 181–14.

181.9.6 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter’s eye closure at TP2. The TECQ of each lane shall be within the limit given in Table 181–5 if measured using a test pattern specified for TECQ in Table 181–12. The TECQ of each lane is measured using the methods specified for TDECQ in 181.9.5, except that the test fiber is not used.

181.9.7 Transmitter overshoot and undershoot

The transmitter overshoot and undershoot of each lane shall be within the limit given in Table 181–5 if measured using a test pattern specified for transmitter overshoot and undershoot in Table 181–12.

Overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 181.9.5) and the waveform captured for the TECQ test (see 181.9.6), at the output of the reference receiver defined in 181.9.5, before the reference equalizer.

Overshoot and undershoot are calculated using the methods in 140.7.7.

181.9.8 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 181–5 if measured using a test pattern specified for transmitter power excursion in Table 181–12.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 181.9.6), at the output of the reference receiver defined in 181.9.5, before the reference equalizer.

Transmitter power excursion is defined in 140.7.8.

181.9.9 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 181–5 if measured using a test pattern specified for extinction ratio in Table 181–12. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 181–7. The extinction ratio is measured using waveforms captured at the output of the reference receiver defined in 181.9.5, before the reference equalizer.

181.9.10 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 181–5 if measured using a test pattern specified for transmitter transition time in Table 181–12.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. The transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 181.9.5, before the reference equalizer.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 181.9.4), with the exception that the square wave test pattern may be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

181.9.11 Relative intensity noise (RIN_{xxOMA})

RIN_{xxOMA} of each lane, with “xx” referring to 17.1, which is the value for optical return loss tolerance in Table 181–5, shall be within the limit given in Table 181–5 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 181.9.4, but with applied xx dB optical return loss and the reference receiver specified for TDECQ measurement in 181.9.5. RIN_{xxOMA} is measured using the methods specified in 180.9.11.

181.9.12 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 181–6 if measured using a test pattern specified for receiver sensitivity in Table 181–12.

The conformance test signal applied at TP3 meets the requirements for a 800GBASE-FR4-500 transmitter followed by an attenuator. An optical demultiplexer may be used to separate the lane having the wavelength for the lane under test as specified in Table 181–6 for calibrating the TECQ.

The TECQ of the conformance test signal is measured according to 181.9.6, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 181–6.

181.9.13 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limit given in Table 181–6 if measured using the method defined in 122.8.10, with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 181.9.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 181–5.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $RIN_{xx}OMA$ of the SRS test source (where x is the value for optical return loss tolerance from Table 181–5) shall be no greater than the value specified in Table 181–5.
- An example stressed receiver conformance test setup is shown in Figure 122–8; however, alternative test setups that generate equivalent stress conditions may be used.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 181–5.
- The values of transmitter overshoot and undershoot and transmitter power excursion of the stressed receiver conformance test signal are within the limits specified in Table 181–5.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ OMA_{outer} of each aggressor lane” are as given in Table 181–6.

181.10 Safety, installation, environment, and labeling

181.10.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

181.10.2 Laser safety

Optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product’s laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹¹

181.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

¹¹A host system that fails to meet the manufacturer’s requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

181.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-FR4-500 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

181.10.5 Electromagnetic emission

A system integrating a 800GBASE-FR4-500 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

181.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

181.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-FR4-500).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 181.10.2.

181.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 181–15 and Table 181–16. Additional variables associated with the ILT function are listed in Table 178B–6 and Table 178B–7.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 181–15 and Table 181–16.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 181–15—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	181.5.6	1.0.15	45.2.1.1

Table 181–15—PMD control variables and MDIO mapping (*continued*)

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_global_transmit_disable	181.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_3 to PMD_transmit_disable_0	181.5.8	1.9.4 to 1.9.1	45.2.1.8

Table 181–16—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	181.5.9	1.1.7	45.2.1.2
PMD_transmit_fault	181.5.10	1.8.11	45.2.1.7
PMD_receive_fault	181.5.11	1.8.10	45.2.1.7
PMD_global_signal_detect	181.5.4	1.10.0	45.2.1.9
PMD_signal_detect_3 to PMD_signal_detect_0	181.5.5	1.10.4 to 1.10.1	45.2.1.9

181.12 Protocol implementation conformance statement (PICS) proforma for Clause 181, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500¹²

181.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 181, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4-500, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

Editor's note: This PICS will be implemented in Draft 2.1. Please do not submit comments against this subclause.

¹²*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

182. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

182.1 Overview

This clause specifies the 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. These PMDs types provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links over one, two, four, or eight pairs, respectively, of single-mode fiber, with a reach of up to at least 2 km.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 182–1 through Table 182–4, to the medium through the medium dependent interface (MDI).

Table 182–1—Physical Layer clauses associated with the 200GBASE-DR1-2 PMD

Associated clause	200GBASE-DR1-2
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120C—200GAUI-8 C2M ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120E—200GAUI-4 C2M ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
120G—200GAUI-2 C2M	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
176D—200GAUI-1 C2M	Optional ^d
177—200GBASE-R Inner FEC	Required
178B—ILT	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.4.1.

^c When implemented in a 200GBASE-DR1-2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 200GAUI-n may be instantiated within a 200GBASE-DR1-2 PHY as described in 176B.4.1.

Table 182–2—Physical Layer clauses associated with the 400GBASE-DR2-2 PMD

Associated clause	400GBASE-DR2-2
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120C—400GAUI-16 C2M ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120E—400GAUI-8 C2M ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
120G—400GAUI-4 C2M	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
176D—400GAUI-2 C2M	Optional ^d
177—400GBASE-R Inner FEC	Required
178B—ILT	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.5.1.

^c When implemented in a 400GBASE-DR2-2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 400GAUI-n may be instantiated within a 400GBASE-DR2-2 PHY as described in 176B.5.1.

Table 182–3—Physical Layer clauses associated with the 800GBASE-DR4-2 PMD

Associated clause	800GBASE-DR4-2
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
177—800GBASE-R Inner FEC	Required
178B—ILT	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-DR4-2 PHY as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Table 182–4—Physical Layer clauses associated with the 1.6TBASE-DR8-2 PMD

Associated clause	1.6TBASE-DR8-2
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
120G—1.6TAUI-16 C2M	Optional ^a
170—1.6Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII Extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a

Table 182–4—Physical Layer clauses associated with the 1.6TBASE-DR8-2 PMD

Associated clause	1.6TBASE-DR8-2
176D—1.6TAUI-8 C2M	Optional ^a
177—1.6TBASE-R Inner FEC	Required
178B—ILT	Required

^a One or two 1.6TAUI-n may be instantiated within a 1.6TBASE-DR8-2 PHY as described in 176B.7.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to the guidelines in 176B.7.1.

Figure 182–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 200 Gb/s Ethernet and 400 Gb/s Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 116.2. 1.6 Tb/s Ethernet is introduced in Clause 174 and the purpose of each PHY sublayer is summarized in 174.2.

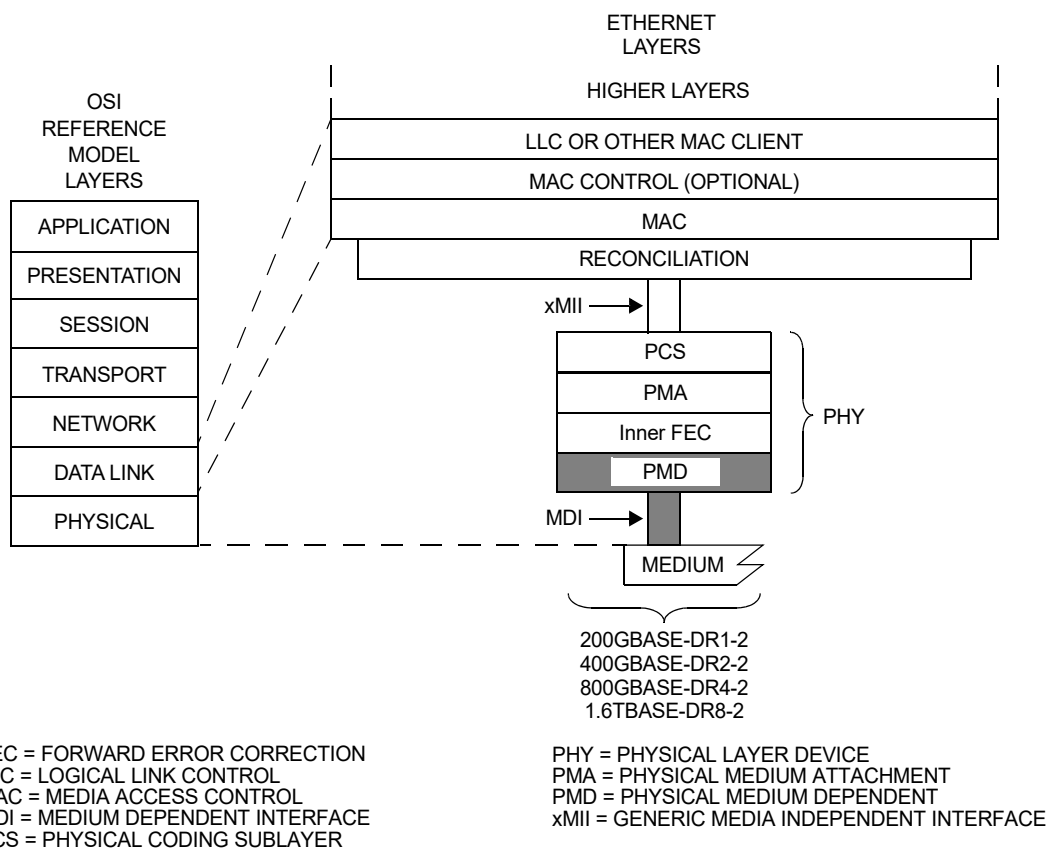


Figure 182–1—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

182.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA nearest to the PMD using the method described in 174A.8, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 3.2×10^{-5} .

182.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and 174.3 for 1.6TBASE-R, except that the rx_symbol parameter is redefined.

For PMD:IS_UNITDATA_i.request in the transmit direction and PMD:IS_UNITDATA_i.indication in the receive direction, where $i = 0$ to $n-1$, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 113.4375 GBd. The number of parallel streams, n , is 1 for 200GBASE-DR1-2, 2 for 400GBASE-DR2-2, 4 for 800GBASE-DR4-2, and 8 for 1.6TBASE-DR8-2. Each of the tx_symbol parameters takes on one of four (PAM4) values: 0, 1, 2, or 3. Each of the rx_symbol parameters takes a set of four or more implementation-specific values to support the Inner FEC soft-decision decoder (see 177.5.4).

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable training_status of the inter-sublayer training (ILT) function (see 182.5.12), as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication on all lanes are unspecified.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.request provides the status from ISLs above the PMD.

182.4 Delay and Skew

182.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-DR1-2 PMD including 2 m of fiber in one direction shall be no more than 14 848 bit times (29 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR2-2 PMD including 2 m of fiber in one direction shall be no more than 29 696 bit times (58 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-DR4-2 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 1.6TBASE-DR8-2 PMD including 2 m of fiber in one direction shall be no more than 118 784 bit times (232 pause_quanta or 74.24 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 for 200GBASE-DR1-2 and 400GBASE-DR2-2, in 169.4 for 800GBASE-DR4-2, and in 174.4 for 1.6TBASE-DR8-2.

182.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS.

182.4.2.1 Skew constraints for 200GBASE-DR1-2 and 400GBASE-DR2-2

Skew and Skew Variation for 200GBASE-DR1-2 and 400GBASE-DR2-2 are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–4 and Figure 116–5. For 200GBASE-DR1-2, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 200GBASE-DR1-2 and 400GBASE-DR2-2 see 116.5.

182.4.2.2 Skew constraints for 800GBASE-DR4-2 and 1.6TBASE-DR8-2

Skew and Skew Variation for 800GBASE-DR4-2 are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

Skew and Skew Variation for 1.6TBASE-DR8-2 are defined in 174.5 and specified at the points SP1 to SP6 shown in Figure 174–5 and Figure 174–6.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 800GBASE-DR4-2 see 169.5 and for 1.6TBASE-DR8-2 see 174.5.

182.5 PMD functional specifications

The 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

182.5.1 Block diagram

The block diagram for 800GBASE-DR4-2 is shown in Figure 182–2.

The block diagram for 200GBASE-DR1-2 is equivalent to Figure 182–2, but for one lane per direction, instead of four.

The block diagram for 400GBASE-DR2-2 is equivalent to Figure 182–2, but for two lanes per direction, instead of four.

The block diagram for 1.6TBASE-DR8-2 is equivalent to Figure 182–2, but for eight lanes per direction, instead of four.

The ILT function indicated in Figure 182–2 is defined in Annex 178B.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 182.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 182.8.3). Unless specified otherwise, all receiver measurements and tests defined in 182.9 are made at TP3.

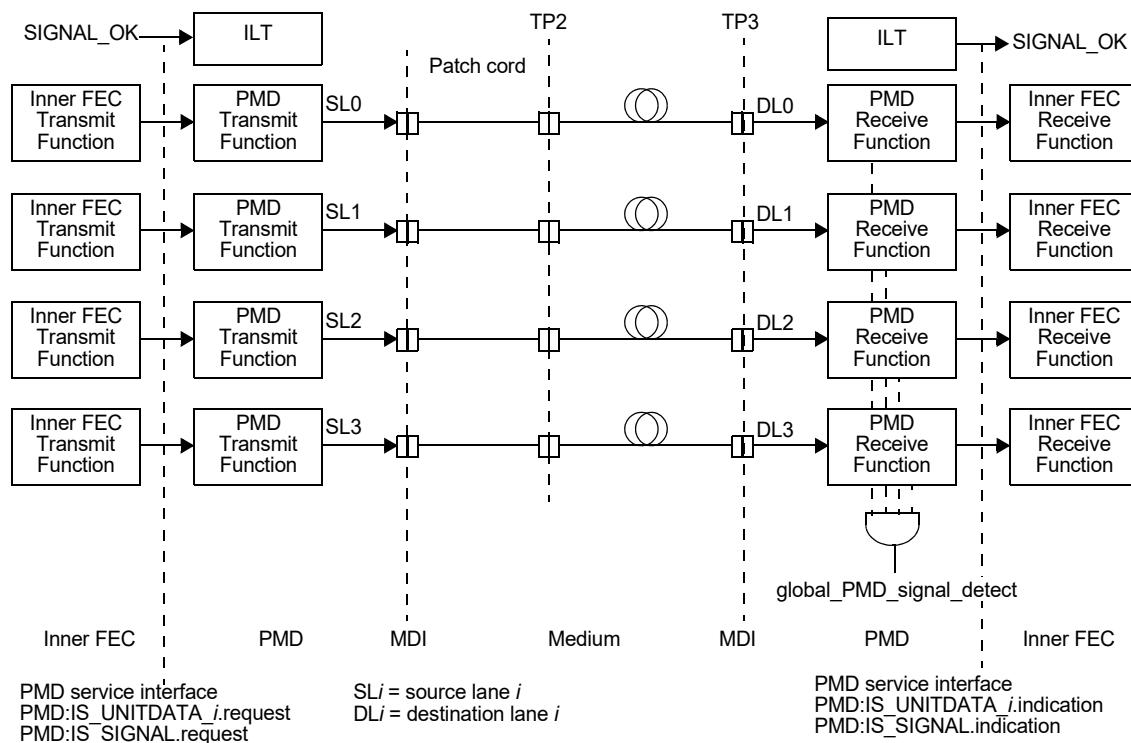


Figure 182–2—Block diagram for 800GBASE-DR4-2 transmit/receive paths

182.5.2 PMD transmit function

The PMD Transmit function shall convert the n symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA₀.request to PMD:IS_UNITDATA _{$n-1$} .request into n separate optical signals. The n optical signals shall then be delivered to the MDI, which contains n parallel light paths for transmit, according to the transmit optical specifications in this clause. The highest optical power level in each signal shall correspond to tx_symbol = three and the lowest shall correspond to tx_symbol = zero.

182.5.3 PMD receive function

The PMD Receive function shall convert the n parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_ $n-1$.indication, all according to the receive optical specifications in this clause. The higher optical power level in each signal shall correspond to rx_symbol = three and the lowest shall correspond to rx_symbol = zero.

182.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all n lanes. For the 200GBASE-DR1-2 PMD, the variable Global_PMD_signal_detect is set to PMD_signal_detect_0 (see 182.5.5). For the 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs, the variable Global_PMD_signal_detect is set to the logical-and of PMD_signal_detect_ i from all of the PMD lanes.

182.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable PMD_signal_detect_ i to either 0 or 1 according to the criteria in Table 182–5.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_ i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_ i variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 182–5—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_ i
Average optical power at TP3 ≤ -15 dBm	0
[The input signal is a compliant 200GBASE-R, 400GBASE-R, 800GBASE-R or 1.6TBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 182–8]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

182.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

182.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 182–7.

If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

182.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitters in each lane to be selectively disabled.

When a PMD_transmit_disable_*i* variable (where *i* represents the lane number in the range 0:*n*–1) is set to one, this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 182–7.

If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_*i* variable to one, turning off the optical transmitter in each lane.

182.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

182.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

182.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

182.5.12 Inter-sublayer link training (ILT) function

A PMD shall provide the ILT function for a Type O1 interface, specified in Annex 178B. When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

182.6 Lane assignments

In order to support the ILT function as described in 182.5.12, optical lane assignments (within a group of transmit or receive lanes) for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 are specified. The positioning and ordering of transmit and receive lanes at the MDI is specified in 182.9.

Each fiber between the transmitter of one PHY and the receiver of another PHY shall connect to the same signal lane number. For example, a fiber connects SL1 at the transmitting end to DL1 at the receiving end.

182.7 PMD to MDI optical specifications

The operating range for the 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs is defined in Table 182–6. A 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers according to the specifications defined in Table 182–11. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-DR4-2 PMD operating at 2.5 km meets the operating range requirement of 2 m to 2 km).

Table 182–6—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 operating range

PMD type	Required operating range
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2	2 m to 2 km

182.7.1 Transmitter optical specifications

A transmitter shall meet the specifications defined in Table 182–7 per the definitions in 182.9.

Table 182–7—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 transmit characteristics

Description	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Signaling rate, each lane (range)	113.4375 ± 50 ppm		GBd
Modulation format	PAM4		—
Lane wavelength (range)	1304.5 to 1317.5		nm
Side-mode suppression ratio (SMSR), each lane (min)	30		dB
Average launch power, each lane (max)	4		dBm
Average launch power, each lane ^a (min)	−3.3 ^b		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.2		dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min)	−0.3		dBm
for max(TECQ, TDECQ) < 0.9 dB	−1.2 + max(TECQ, TDECQ)		dBm
for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB			
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4		dB

Table 182–7—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 transmit characteristics (*continued*)

Description	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Transmitter eye closure for PAM4 (TECQ), each lane (max)	3.4		dB
TDECQ – TECQ , each lane (max)	2.5		dB
Transmitter overshoot and undershoot, each lane (max)	22		%
Transmitter power excursion, each lane (max)	2.3		dBm
Extinction ratio, each lane (min)	3.5		dB
Transmitter transition time, each lane (max)	8		ps
Average launch power of OFF transmitter, each lane (max)	–15		dBm
RIN _{xx} OMA, each lane (max) xx = 17.1 xx = 21.4	–139 —	— –139	dB/Hz dB/Hz
Optical return loss tolerance, each lane (max)	17.1	21.4	dB
Transmitter reflectance ^c , each lane (max)	–26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of –3.3 dBm corresponds to an OMA of –0.3 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 182–7 vary with max (TECQ, TDECQ). The relationships are illustrated in Figure 182–3 along with the values for OMA_{outer} each lane (max).

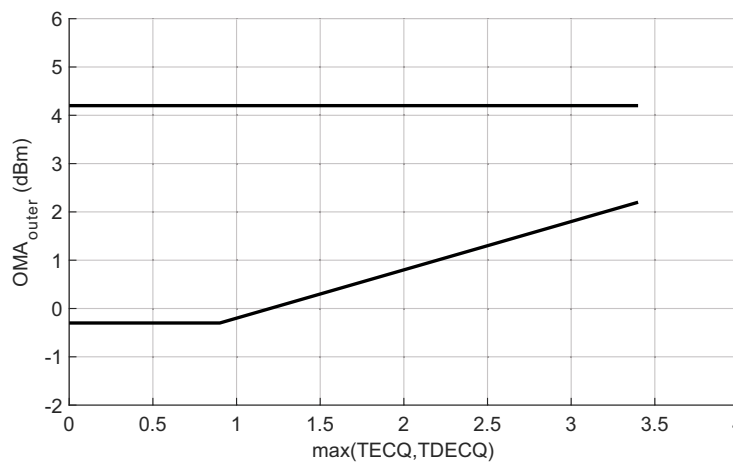


Figure 182–3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max (TECQ, TDECQ)

182.7.2 Receive optical specifications

A receiver shall meet the specifications defined in Table 182–8 per the definitions in 182.9.

Table 182–8—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 receive characteristics

Description	Value	Unit
Signaling rate, each lane (range)	113.4375 ± 50 ppm	GBd
Modulation format	PAM4	—
Lane wavelengths (range)	1304.5 to 1317.5	nm
Damage threshold ^a , each lane	5	dBm
Average receive power, each lane (max)	4	dBm
Average receive power, each lane ^b (min)	−7.3	dBm
Receive power (OMA_{outer}), each lane (max)	4.2	dBm
Receiver reflectance (max)	−26	dB
Receiver sensitivity (OMA_{outer}), each lane ^c (max) for $TECQ < 0.9$ dB for $0.9 \text{ dB} \leq TECQ \leq SECQ$	−4.7 −5.6 + TECQ	dBm
Stressed receiver sensitivity (OMA_{outer}), each lane ^c (max)	−2.2	dBm
Conditions of stressed receiver sensitivity test ^d :		
Stressed eye closure for PAM4 (SECQ), lane under test	3.4	dB
OMA_{outer} of each aggressor lane ^e	4.2	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 182.8) for the block error ratio specified in 182.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^e No aggressors needed for 200GBASE-DR1-2 in a single lane device.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 182–8 vary depending on the value of TECQ for the test source used for measurement. Figure 182–4 illustrates this.

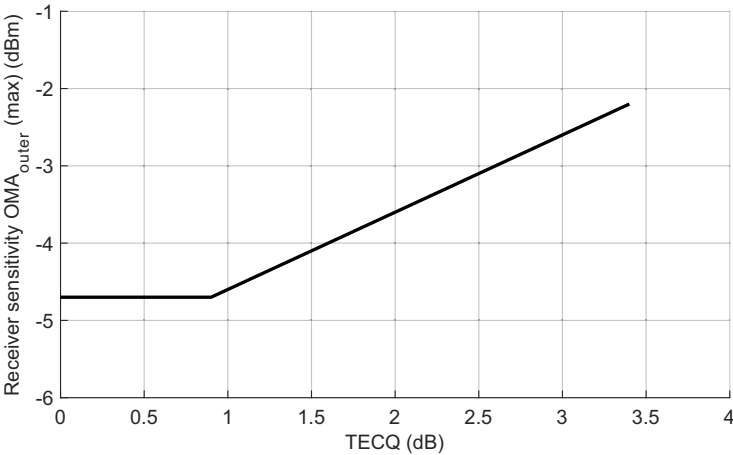


Figure 182-4—Receiver sensitivity (OMA_{outer}), each lane (max)

182.7.3 Illustrative link power budget

An illustrative power budget and penalties for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 channels are shown in Table 182-9.

Table 182-9—Illustrative link power budget

Parameter	Value	Unit
Power budget (for max TDECQ)	7.8	dB
Operating distance	2000	m
Channel insertion loss ^a	4	dB
Maximum discrete reflectance	−35	dB
Allocation for penalties ^b (for max TDECQ)	3.8	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 182-6 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 182.8.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. This value includes an allocation of 0.4 dB for MPI and DGD penalties.

The values of transmitter OMA_{outer} (max), transmitter OMA_{outer} (min) versus max(TECQ, TDECQ), and receiver sensitivity (OMA_{outer}) (max) versus TECQ are illustrated in Figure 182-5.

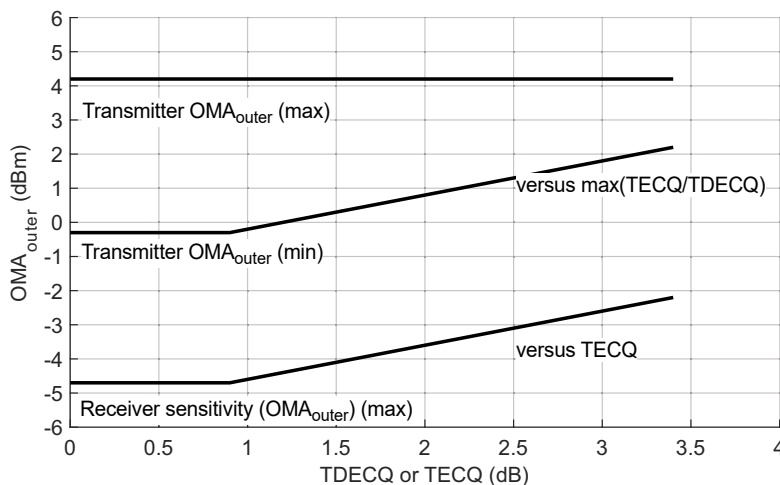


Figure 182-5—Transmitter OMA_{outer} each lane versus $\max(TECQ, TDECQ)$ and receiver sensitivity (OMA_{outer}) each lane versus TECQ

182.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 182-6.

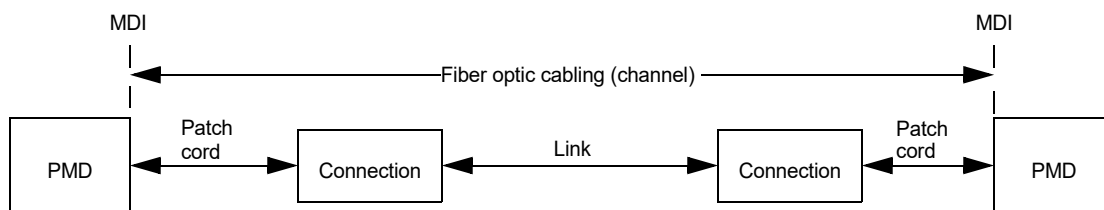


Figure 182-6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 182-10. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 series one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 182–10—Optical channel characteristics

Description	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Operating distance (max)	2000		m
Channel insertion loss ^{a,b} (max)	4		dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	2.62		ps/nm
Negative dispersion ^b (min)	−3.41		ps/nm
DGD_max ^c	2.3		ps
Optical return loss (min)	25	37	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

182.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 182–11 where they differ.

Table 182–11—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3.

182.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 182–6, consists of a mated pair of optical connectors.

182.8.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 3 dB total connection and splice loss. For example, this allocation supports six connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 182–10 are met.

182.8.2.2 Maximum discrete reflectance

The maximum value for each discrete reflectance shall be less than or equal to the value shown in Table 182–12 corresponding to the number of discrete reflectances above –55 dB within the channel. For numbers of discrete reflectances in between two numbers shown in the table, the lower of the two corresponding maximum discrete reflectance values applies.

Table 182–12—Maximum value of each discrete reflectance

Number of discrete reflectances above –55 dB	Maximum value for each discrete reflectance for 200GBASE-DR1-2	Maximum value for each discrete reflectance for 400GBASE-DR2-2, 800GBASE-DR4-2, 1.6TBASE-DR8-2
1	–25 dB	–37 dB
2	–31 dB	–42 dB
4	–35 dB	–45 dB
6	–38 dB	–47 dB
8	–40 dB	–48 dB
10	–41 dB	–49 dB

182.8.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 182–6). The PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 180A–1. Example constructions of the MDI include the following:

- PMD with a connectorized fiber pigtail plugged into an adapter
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 182.5.1, not at the MDI.

Annex 180A specifies the details of the MDIs for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2.

For 200GBASE-DR1-2, besides the option to connect to a single fiber MDI, there are two additional specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 400GBASE-DR2-2, and 800GBASE-DR4-2 there are two specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 1.6TBASE-DR8-2 there is one specified MDI optical receptacle, a single-row 16 fiber interface.

182.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

182.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 182–14 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 182–14 may be used to perform that test. The test patterns used in this clause are shown in Table 182–13.

Table 182–13—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	177.6.1.6
3	PRBS31Q	177.6.1.4
4	PRBS13Q	177.6.1.3
5	Scrambled idle test pattern encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	119.2.4.9, 172.2.4.11, 175.2.4.11, 177.4
6	SSPRQ	177.6.1.5
7	Valid 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R signal encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	119.2.4, 172.2.4, 175.2.4, 177.4
8	PRBS31 pattern encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	177.6.1.1

Table 182–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6, 7 or 8	182.9.2
Side mode suppression ratio	3, 5, 6, 7 or 8	182.9.2
Average optical power	3, 5, 6, 7 or 8	182.9.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	182.9.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	182.9.5
Transmitter eye closure for PAM4 (TECQ)	6	182.9.6
Transmitter overshoot and undershoot	6	182.9.7
Transmitter power excursion	6	182.9.8
Extinction ratio	4 or 6	182.9.9
Transmitter transition time	Square wave or 6	182.9.10
RIN _{xx} OMA	4 or 6	182.9.11
Receiver sensitivity	5 or 8	182.9.12
Stressed receiver conformance test signal calibration	6	182.9.13
Stressed receiver sensitivity	5 or 8	182.9.13

182.9.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 182–7 if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 182–14.

182.9.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 182–7 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 182–14.

182.9.4 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 182–7. The OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 182–14 as the difference between the average optical launch power level P₃, measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P₀, measured over the central 2 UI of a run of 6 zeros, as shown in Figure 182–7. OMA_{outer} is measured using waveforms captured at the output of the reference receiver defined in 182.9.5, before the reference equalizer.

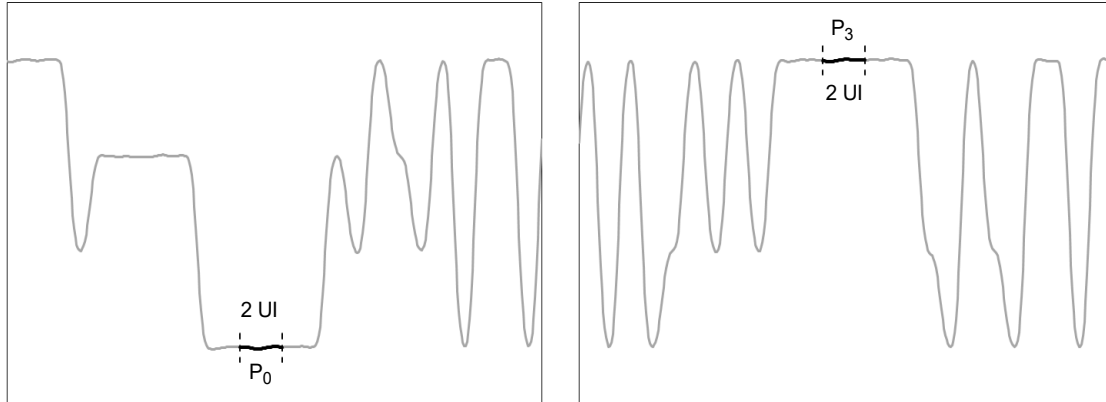


Figure 182-7—Example power levels P_0 and P_3 from PRBS13Q test pattern

182.9.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limit given in Table 182-7 if measured using the methods specified in 121.8.5.1, 121.8.5.3, and 182.9.5.1, with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 182-7 and uses the test pattern specified for TDECQ in Table 182-14.
- TDECQ is defined with all receive lanes in operation using pattern 5, 7, or 8 (see Table 182-13). The received test patterns shall be asynchronous to the pattern used to test the transmitter, and shall have power levels as specified in Table 182-8 for the aggressor lanes in the stressed receiver sensitivity test.
- For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 182-1 through Table 182-4), the clock source for the SSPRQ test pattern is derived from the clock recovered from the xAUI-n input signal. The pattern of the xAUI-n input signal may be PRBS31Q, scrambled idle, or a valid xBASE-R signal.
- The reference receiver, composed of the combination of the O/E converter and the oscilloscope, has a 3 dB bandwidth of approximately 56.71875 GHz with a fourth-order Bessel-Thomson response to at least 1.3×113.4375 GHz, and at frequencies above 1.3×113.4375 GHz, the response should not exceed -20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The reference equalizer is a 15-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period, with equalizer coefficient constraints as shown in Table 182-15. The reference equalizer may be implemented in the oscilloscope, but it is not considered part of the reference receiver.
- The target PAM4 symbol error ratio of 9.6×10^{-3} and the related Q_t value (see 121.8.5.3) is 2.489.
- The normalized noise power density spectrum $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 56.71875 GHz.
- The clock recovery unit (CRU) has a corner frequency of 4.27 MHz.
- The optical return loss is as given in Table 182-7.
- The lowest measured TDECQ values are achieved with the equalizer optimization method described in 121.8.5. Alternative optimization methods such as minimum mean squared error (MMSE) may be used to determine equalizer tap weights to reduce test time, and are expected to report equal or

higher values of TDECQ. These alternative methods should not be used for receiver sensitivity and stressed receiver sensitivity calibration.

Table 182–15—Reference equalizer tap coefficients

Parameter	Symbol	Value	
		Minimum	Maximum
Feed-forward equalizer (FFE) length	N_b	15	
Number of equalizer pre-cursor taps	—	0	3
Main tap coefficient limit	$c(0)$	0.9	2.5
Normalized equalizer coefficient limits: ^a	$c(i)$		
$i = -3$		−0.15	0.1
$i = -2$		−0.1	0.25
$i = -1$		−0.5	0.1
$i = 1$		−0.6	0.2
$i = 2$		−0.2	0.3
$i = 3$		−0.15	0.15
$i = 4$		−0.15	0.15
$i = 5$		−0.15	0.15
$i = 6$		−0.15	0.15
$i \geq 7$		−0.1	0.1
Equalizer gain ^b	—	1	

^a The main tap is marked by $i = 0$. The minimum and maximum values are relative to this tap's coefficient

^b The sum of the equalizer coefficients.

182.9.5.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 182–16.

Table 182–16—Transmitter compliance channel specifications

PMD type	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
	Minimum	Maximum			
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2	$0.1850 \times (\lambda - 1311) - 2.22$	$0.1770 \times (\lambda - 1311) + 1.47$	Minimum	See Table 182–7	0.8

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 2 km distance. The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024).

^b The value “Minimum” implies that the test channel insertion loss should be sufficiently low that it does not significantly stress the tester receiver.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 182–16 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 182–16. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 182–16.

182.9.6 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter’s eye closure at TP2. The TECQ shall be within the limit given in Table 182–7 if measured using a test pattern specified for TECQ in Table 182–14. TECQ is measured using the methods specified for TDECQ in 182.9.5, except that the test fiber is not used.

182.9.7 Transmitter overshoot and undershoot

The overshoot and undershoot of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for overshoot and undershoot in Table 182–14.

Overshoot and undershoot are measured using the waveforms captured for the TDECQ test (see 182.9.5) and the waveform captured for the TECQ test (see 182.9.6), at the output of the reference receiver defined in 182.9.5, before the reference equalizer.

Overshoot and undershoot are defined in 140.7.7.

182.9.8 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for transmitter power excursion in Table 182–14.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 182.9.6), at the output of the reference receiver defined in 182.9.5, before the reference equalizer.

Transmitter power excursion is defined in 140.7.8.

182.9.9 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for extinction ratio in Table 182–14. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 182–7. The extinction ratio is measured using waveforms captured at the output of the reference receiver defined in 182.9.5, before the reference equalizer

182.9.10 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for transmitter transition time in Table 182–14.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} , or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. The transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 182.9.5, before the reference equalizer

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 182.9.4), with the exception that the square wave test pattern may be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

182.9.11 Relative intensity noise ($RIN_{xx}OMA$)

$RIN_{xx}OMA$ of each lane, with “xx” referring to the value for optical return loss tolerance in Table 182–7, shall be within the limit given in Table 182–7 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 182.9.4, but with applied xx dB optical return loss and the reference receiver specified for TDECQ measurement in 182.9.5. $RIN_{xx}OMA$ is measured using the methods specified in 180.9.11.

182.9.12 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 182–8 if measured using a test pattern for receiver sensitivity in Table 182–14. The conformance test signal at TP3 meets the requirements for a transmitter followed by an attenuator.

The TECQ of the conformance test signal is measured according to 182.9.5, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 182–8.

182.9.13 Stressed receiver sensitivity

Stressed receiver sensitivity of each lane shall be within the limit given in Table 182–8 if measured using the method defined in 121.8.10 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 182.9.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 182–7.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $RIN_{xx}OMA$ of the SRS test source should be no greater than the value specified in Table 182–7.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 182–7 using test patterns specified in Table 182–14.
- The stressed receiver conformance input signal meets the sinusoidal jitter specified in 182.9.13.1.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ OMA_{outer} of each aggressor lane” are as given in Table 182–8.

182.9.13.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 182–17 and is illustrated in Figure 182–8.

Table 182–17—Applied sinusoidal jitter

Frequency range	Sinusoidal jitter, peak-to-peak (UI)
$f < 42.7 \text{ kHz}$	Not specified
$42.7 \text{ kHz} < f \leq 4.27 \text{ MHz}$	$2.13 \times 10^5 \text{ Hz}/f$
$4.27 \text{ MHz} < f < 10 \text{ LB}^a$	0.05

^a *LB* = loop bandwidth; upper frequency bound for added sine jitter should be at least 10 times the loop bandwidth of the receiver being tested.

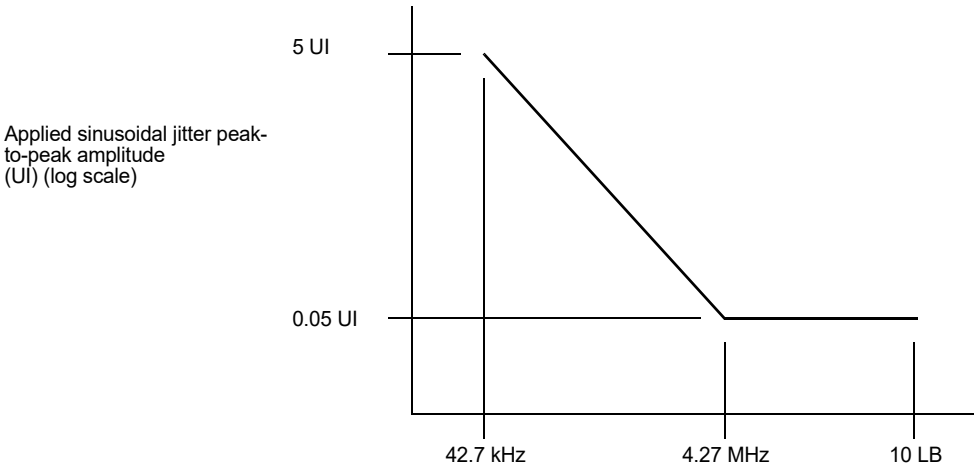


Figure 182–8—Illustration of the mask of the sinusoidal component of jitter tolerance

182.10 Safety, installation, environment, and labeling

182.10.1 General safety

All equipment subject to this clause shall conform to IEC 63268-1.

182.10.2 Laser safety

200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹³

182.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

182.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

182.10.5 Electromagnetic emission

A system integrating a 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

182.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

182.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-DR4-2).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 182.10.2.

182.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 182–18 and Table 182–19. Additional variables associated with the ILT function are listed in Table 178B–6 and Table 178B–7.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 182–18 and Table 182–19.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

¹³A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

Table 182–18—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	182.5.6	1.0.15	45.2.1.1
PMD_global_transmit_disable	182.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_7 to PMD_transmit_disable_4 ^a	182.5.8	1.9.8 to 1.9.5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	182.5.8	1.9.4, 1.9.3	45.2.1.8
PMD_transmit_disable_1 ^c	182.5.8	1.9.2	45.2.1.8
PMD_transmit_disable_0	182.5.8	1.9.1	45.2.1.8

^aAvailable only in 1.6TBASE-DR8-2

^bAvailable only in 800GBASE-DR4-2 and 1.6TBASE-DR8-2

^cAvailable only in 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

Table 182–19—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	182.5.9	1.1.7	45.2.1.2
PMD_transmit_fault	182.5.9	1.8.11	45.2.1.7
PMD_receive_fault	182.5.11	1.8.10	45.2.1.7
PMD_global_signal_detect	182.5.4	1.10.0	45.2.1.9
PMD_signal_detect_7 to PMD_signal_detect_4 ^a	182.5.5	1.10.8 to 1.10.5	45.2.1.9
PMD_signal_detect_3, PMD_signal_detect_2 ^b	182.5.5	1.10.4, 1.10.3	45.2.1.9
PMD_signal_detect_1 ^c	182.5.5	1.10.2	45.2.1.9
PMD_signal_detect_0	182.5.5	1.10.1	45.2.1.9

^aAvailable only in 1.6TBASE-DR8-2

^bAvailable only in 800GBASE-DR4-2 and 1.6TBASE-DR8-2

^cAvailable only in 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

182.12 Protocol implementation conformance statement (PICS) proforma for Clause 182, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4¹⁴

182.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 182, Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

Editor's note: This PICS will be implemented in Draft 2.1. Please do not submit comments against this subclause.

¹⁴*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

183. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4

183.1 Overview

This clause specifies the 800GBASE-FR4 and 800GBASE-LR4 PMDs together with the single-mode fiber medium. The optical signals generated by these two PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMDs provide point-to-point 800 Gb/s Ethernet links over four wavelength division multiplexing (WDM) lanes on single-mode fiber, with a reach of up to at least 2 km or 10 km, respectively.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 183–1 to the medium through the MDI.

Table 183–1—Physical Layer clauses associated with the 800GBASE-FR4 and 800GBASE-LR4 PMDs

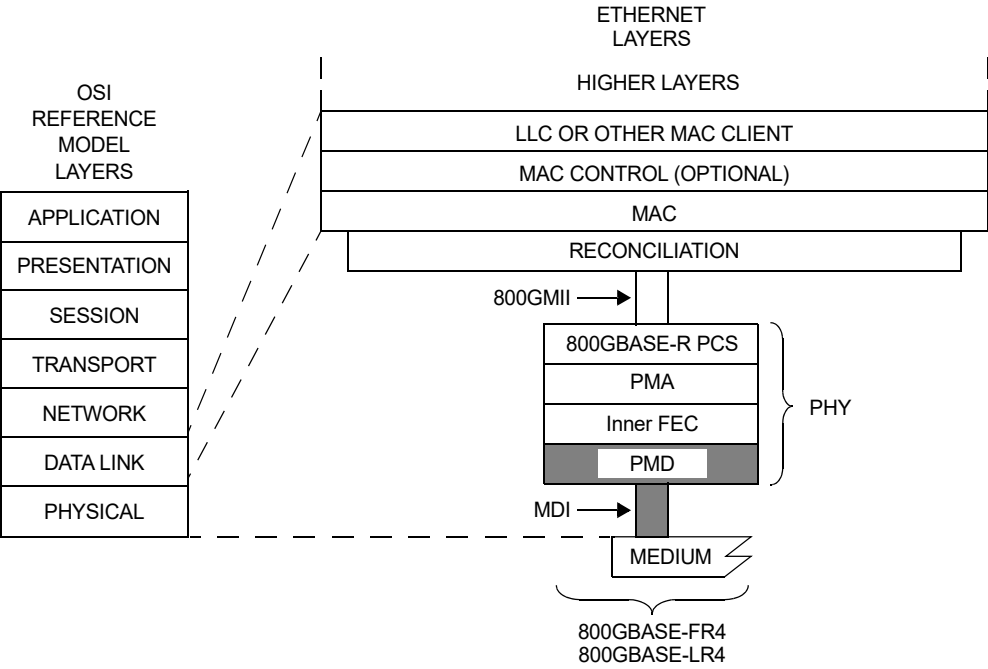
Associated clause	800GBASE-FR4 800GBASE-LR4
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
177—800GBASE-R Inner FEC	Required
178B—ILT	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-FR4 PHY or 800GBASE-LR4 PHY as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or two 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Figure 183–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
FR4 = PMD FOR SINGLE-MODE FIBER — 2 km
LR4 = PMD FOR SINGLE-MODE FIBER — 10 km

Figure 183–1—800GBASE-FR4 and 800GBASE-LR4 PMDs relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

183.2 Error ratio allocation

A complete PHY shall meet the specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA nearest to the PMD using the method described in 174A.8, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 3.2×10^{-5} .

183.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client. The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 169.3. For PMD:IS_UNITDATA_i.request in the transmit direction and PMD:IS_UNITDATA_i.indication in the

receive direction, where $i = 0$ to $n-1$, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 113.4375 GBd. The number of parallel streams, n , is 4. Each instance of tx_symbol and rx_symbol takes on one of four (PAM4) values: 0, 1, 2, or 3. In order to support the Inner FEC soft-decision decoder (see 177.5.5), rx_symbol may take on an implementation-dependent set of values beyond the scope of this standard.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable training_status of the inter-sublayer training (ILT) function (see 183.5.12), as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication on all lanes are unspecified.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.request provides the status from ISLs above the PMD.

183.4 Delay and Skew

183.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-FR4 or 800GBASE-LR4 PMD including 2 m of fiber in one direction shall be no more than 59392 bit times (96 pause_quanta or 74.24 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

183.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS. Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169-4 and Figure 169-5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5. The measurements of Skew and Skew Variation are defined in 87.8.2.

183.5 PMD functional specifications

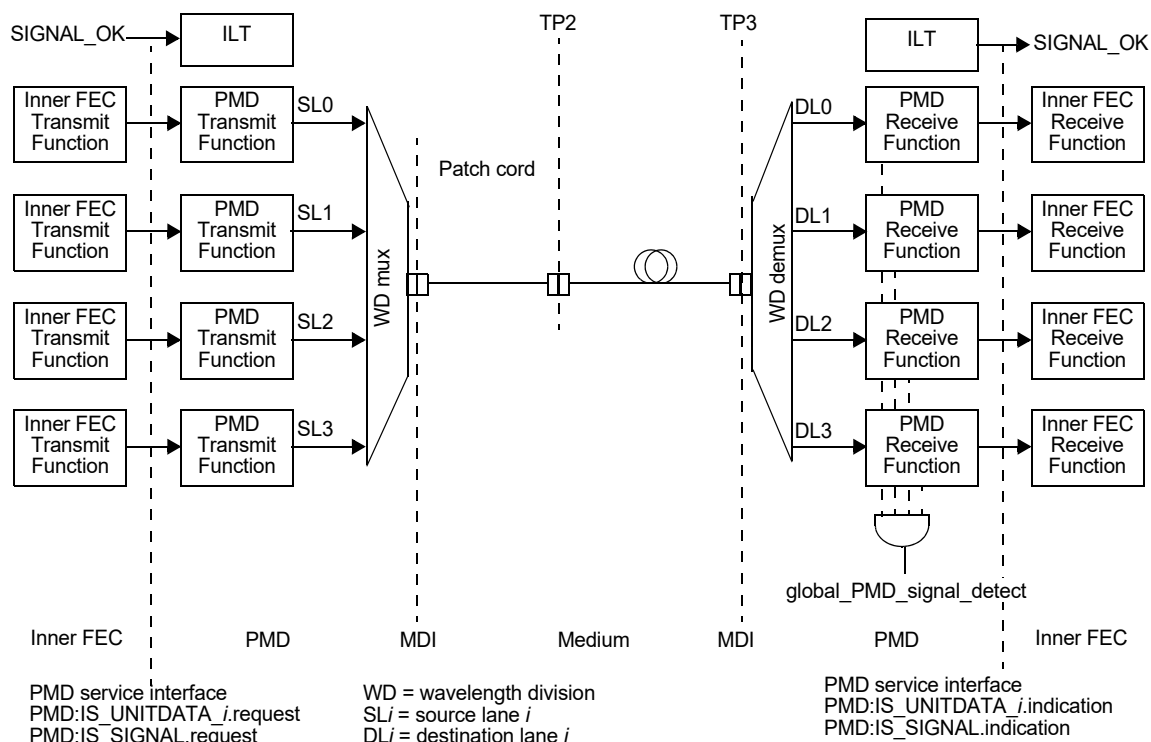
The 800GBASE-FR4 and 800GBASE-LR4 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

183.5.1 Block diagram

The PMD block is shown in Figure 183-2. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end

of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 183.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 183.8.3). Unless specified otherwise, all receiver measurements and tests defined in 183.9 are made at TP3.

The ILT function indicated in Figure 183–2 is defined in Annex 178B.



NOTE: For clarity, only one direction of transmission is shown

Figure 183–2—Block diagram for 800GBASE-FR4 and 800GBASE-LR4 transmit/receive paths

183.5.2 PMD transmit function

The PMD Transmit function shall convert the four symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_0.request to PMD:IS_UNITDATA_3.request into four separate optical signals. The four optical signals shall then be wavelength division multiplexed and delivered to the MDI, all according to the transmit optical specifications in this clause. The four optical power levels in the signal in order from lowest to highest shall correspond to tx_symbol values zero, one, two, and three, respectively.

183.5.3 PMD receive function

The PMD Receive function shall demultiplex the composite optical signal received from the MDI into four separate optical signals. The four optical signals shall then be converted into four symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_3.indication, all according to the receive optical specifications in this clause. The

four optical power levels in each signal in order from lowest to highest shall correspond to rx_symbol values zero, one, two, and three, respectively.

183.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all n lanes and it is set to the logical-and of PMD_signal_detect_ i from all of the PMD lanes.

183.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable PMD_signal_detect_ i to either 0 or 1 according to the criteria in Table 183–2

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the PMD_signal_detect_ i variable.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect_ i variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 183–2—Signal detection criteria for each lane

Receive conditions	PMD_signal_detect_ i
For any lane; Average optical power at TP3 ≤ -16 dBm	0
For all lanes; [The input signal is a compliant 800GBASE-R signal input] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 183–7]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

183.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

183.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 183–6.

183.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitters in each lane to be selectively disabled.

When a `PMD_transmit_disable_i` variable (where i represents the lane number in the range 0:3) is set to one, this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 183–6.

If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disable_i` variable to one, turning off the optical transmitter in each lane.

183.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set `PMD_fault` to one.

183.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the `PMD_transmit_fault` variable to one.

183.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the `PMD_receive_fault` variable to one.

183.5.12 Inter-sublayer link training (ILT) function

A PMD shall provide the ILT function for a Type O1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function for a Type O1 interface is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

183.6 Wavelength-division-multiplexed lane assignments

The wavelength range for each lane of the 800GBASE-FR4 is defined in Table 183–3. The 800GBASE-FR4 center wavelengths are members of the CWDM wavelength grid defined in ITU-T G.694.2 and are spaced at 20 nm.

The wavelength range for each lane of the 800GBASE-LR4 is defined in Table 183–4. The 800GBASE-LR4 center frequencies are members of the frequency grid for 100 GHz spacing defined in ITU-T G.694.1 and are spaced at 800 GHz.

Table 183–3—800GBASE-FR4 wavelength-division-multiplexed lane assignments

Lane	Center wavelength	Wavelength range
0	1271 nm	1264.5 to 1277.5 nm
1	1291 nm	1284.5 to 1297.5 nm
2	1311 nm	1304.5 to 1317.5 nm
3	1331 nm	1324.5 to 1337.5 nm

Table 183–4—800GBASE-LR4 wavelength-division-multiplexed lane assignments

Lane	Center frequency	Center wavelength	Wavelength range
0	231.4 THz	1295.56 nm	1294.53 to 1296.59 nm
1	230.6 THz	1300.05 nm	1299.02 to 1301.09 nm
2	229.8 THz	1304.58 nm	1303.54 to 1305.63 nm
3	229 THz	1309.14 nm	1308.09 to 1310.19 nm

NOTE—Each functional lane, denoted SL*i* at the transmitter and DL*i* at the receiver, is mapped to a specific wavelength to support ILT operation.

183.7 PMD to MDI optical specifications

The operating ranges for the PMDs are defined in Table 183–5. A 800GBASE-FR4 or 800GBASE-LR4 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-FR4 PMD operating at 2.5 km meets the operating range requirement of 2 m to 2 km).

Table 183–5—800GBASE-FR4 and 800GBASE-LR4 operating ranges

PMD type	Required operating range
800GBASE-FR4	2 m to 2 km
800GBASE-LR4	2 m to 10 km

183.7.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 183–6 per the definitions in 183.9.

Table 183–6—800GBASE-FR4 and 800GBASE-LR4 transmit characteristics

Description	800GBASE-FR4	800GBASE-LR4	Unit
Signaling rate, each lane (range)	113.4375 ± 50 ppm		GBd
Modulation format	PAM4		—
Lane wavelengths (range)	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	1294.53 to 1296.59 1299.02 to 1301.09 1303.54 to 1305.63 1308.09 to 1310.19	nm
Side-mode suppression ratio (SMSR), each lane (min)	30		dB
Total average launch power (max)	10.9	11.5	dBm
Average launch power, each lane (max)	4.9	5.5	dBm
Average launch power, each lane ^a (min)	−2.2 ^b	−1.1 ^c	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (max)	4.8	5.7	dBm
Outer Optical Modulation Amplitude (OMA _{outer}), each lane (min)	0.8 Equation (183–1) — —	— — 1.9 Equation (183–2)	dBm
for max(TECQ, TDECQ) < 0.9 dB			dBm
for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB			dBm
for max(TECQ, TDECQ) < 1.4 dB			dBm
for 1.4 dB ≤ max(TECQ, TDECQ) ≤ 3.9 dB			dBm
Difference in launch power between any two lanes (OMA _{outer}) (max)	4	3	dB
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	3.4	3.9	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	3.4	3.2	dB
TDECQ – TECQ , each lane (max)	2.5	2.5	dB
Transmitter overshoot and undershoot, each lane (max)	22		%
Transmitter power excursion, each lane (max)	2.9	3.8	dBm
Extinction ratio, each lane (min)	3.5		dB
Transmitter transition time, each lane (max)	8	13	ps
Average launch power of OFF transmitter, each lane (max)	−16	−16	dBm
RIN _{xx} OMA, each lane (max)	−139		dB/Hz
Optical return loss tolerance, each lane (max)	17.1	15.6	dB
Transmitter reflectance ^d , each lane (max)	−26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b Average launch power of −2.2 dBm corresponds to an OMA of 0.8 dBm with an infinite extinction ratio.

^c Average launch power of −1.1 dBm corresponds to an OMA of 1.9 dBm with an infinite extinction ratio.

^d Transmitter reflectance is defined looking into the transmitter.

$$-0.1 + \max(\text{TECQ}, \text{TDECQ}) \tag{183-1}$$

$$0.5 + \max(\text{TECQ}, \text{TDECQ}) \tag{183-2}$$

The values for OMA_{outer} each lane (min) in Table 183–6 vary with max(TECQ, TDECQ). The relationships are illustrated in Figure 183–3 along with the values for OMA_{outer} each lane (max).

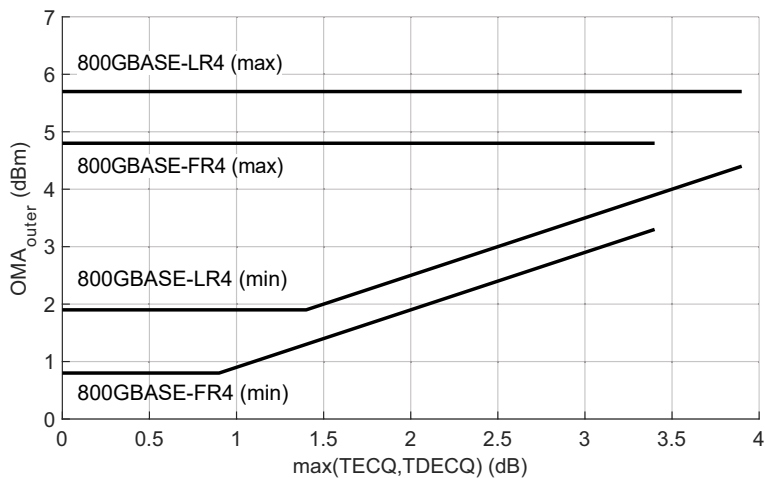


Figure 183–3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max(TECQ, TDECQ) for 800GBASE-FR4 and 800GBASE-LR4

183.7.2 800GBASE-FR4 and 800GBASE-LR4 receive optical specifications

The receiver shall meet the specifications in Table 183–7 per the definitions in 183.9.

Table 183–7—800GBASE-FR4 and 800GBASE-LR4 receive characteristics

Description	800GBASE-FR4	800GBASE-LR4	Unit
Signaling rate, each lane (range)	113.4375 ± 50 ppm		GBd
Modulation format	PAM4		—
Lane wavelengths (range)	1264.5 to 1277.5 1284.5 to 1297.5 1304.5 to 1317.5 1324.5 to 1337.5	1294.53 to 1296.59 1299.02 to 1301.09 1303.54 to 1305.63 1308.09 to 1310.19	nm
Damage threshold ^a , each lane	5.9	6.5	dBm
Average receive power, each lane (max)	4.9	5.5	dBm
Average receive power, each lane ^b (min)	−6.2	−7.4	dBm
Receive power (OMA _{outer}), each lane (max)	4.8	5.7	dBm
Difference in receive power between any two lanes (OMA _{outer}) (max)	4.1	3.3	dB
Receiver reflectance (max)	−26		dB
Receiver sensitivity (OMA _{outer}), each lane ^c (max)			
for TECQ < 0.9 dB	−3.7	−	dBm
for 0.9 dB ≤ TECQ ≤ SECQ	−4.6 + TECQ	−	dBm
for TECQ < 1.4 dB	−	−5.5	dBm
for 1.4 dB ≤ TECQ ≤ SECQ	−	−6.9 + TECQ	dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	−1.2	−3	dBm
Conditions of stressed receiver sensitivity test: ^d			
Stressed eye closure for PAM4 (SECQ), lane under test	3.4	3.9	dB
OMA _{outer} of each aggressor lane	0.8	0.3	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 183.9.13) for the block error ratio specified in 183.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 183–7 vary depending on the value of TECQ for the test source used for measurement. Figure 183–4 illustrates this.

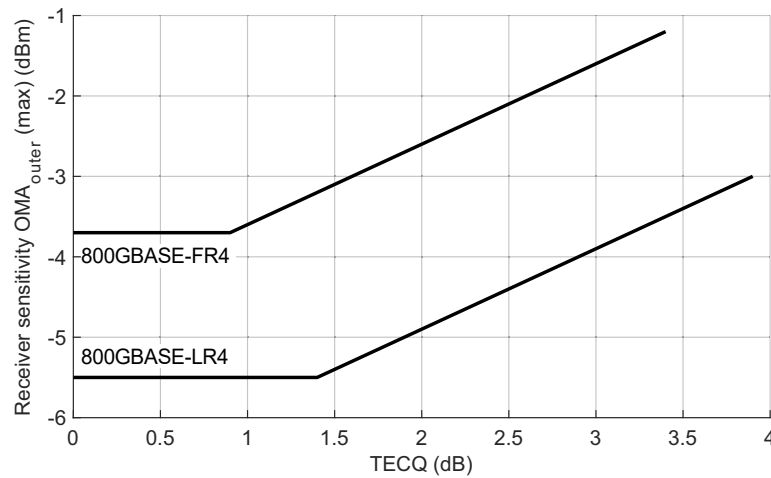


Figure 183-4—Receiver sensitivity (OMA_{outer}), each lane (max) for 800GBASE-FR4 and 800GBASE-LR4

183.7.3 800GBASE-FR4 and 800GBASE-LR4 illustrative link power budgets

Illustrative power budgets and penalties for the channels are shown in Table 183-8.

Table 183-8—800GBASE-FR4 and 800GBASE-LR4 illustrative link power budgets

Parameter	800GBASE-FR4	800GBASE-LR4	Unit
Power budget (for maximum TDECQ)	7.9	11.3	dB
Operating distance	2	10	km
Channel insertion loss ^a	4	6.3	dB
Maximum discrete reflectance ^b	-35 ^c	-35 ^d	dB
Allocation for penalties ^e (for maximum TDECQ)	3.9 ^f	5 ^g	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 183-5 and fiber attenuation of 0.5 dB/km plus an allocation for connection and splice loss given in 183.8.2.1.

^b See 183.8.2.2 for details and specifications as a function of the number of discrete reflectances within the channel.

^c Maximum value for each discrete reflectance with 4 discrete reflectances above -55 dB within the channel.

^d Maximum value for each discrete reflectance with 6 discrete reflectances above -55 dB within the channel.

^e Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

^f This value includes an allocation of 0.5 dB for MPI and DGD penalties.

^g This value includes an allocation of 0.7 dB for DGD penalties, and 0.4dB for MPI penalties.

The values of transmitter $\text{OMA}_{\text{outer}}$ (max), transmitter $\text{OMA}_{\text{outer}}$ (min) versus $\max(\text{TECQ}, \text{TDECQ})$, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) (max) versus TECQ are illustrated in Figure 183–5 for 800GBASE-FR4 and in Figure 183–6 for 800GBASE-LR4.

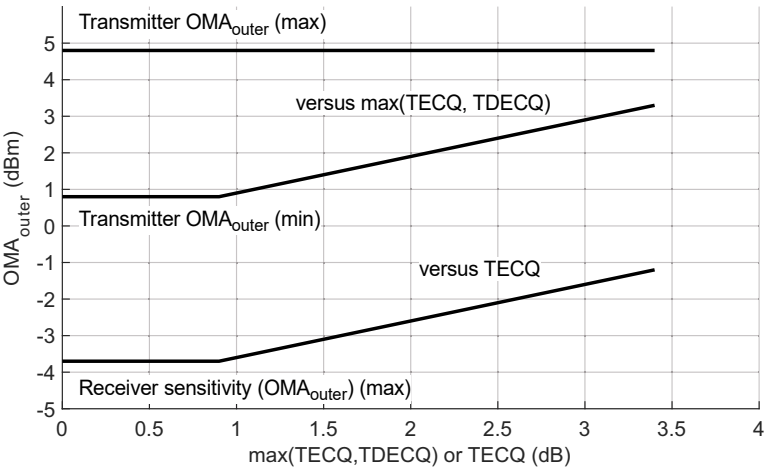


Figure 183–5—Transmitter $\text{OMA}_{\text{outer}}$ each lane versus $\max(\text{TECQ}, \text{TDECQ})$ and receiver sensitivity ($\text{OMA}_{\text{outer}}$) each lane versus TECQ for 800GBASE-FR4

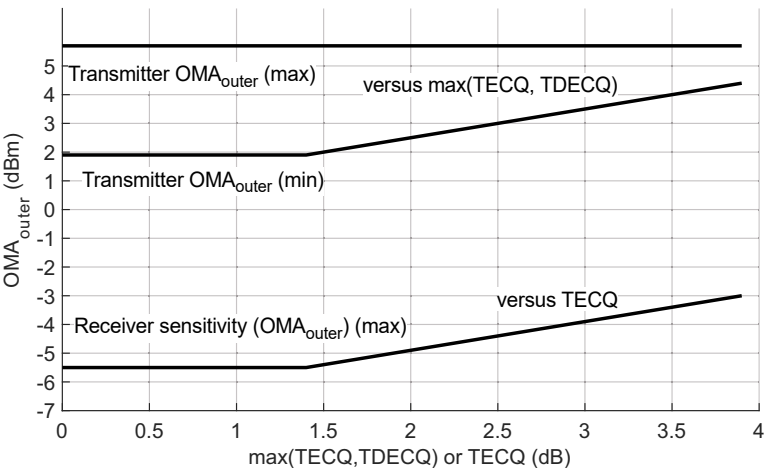


Figure 183–6—Transmitter $\text{OMA}_{\text{outer}}$ each lane versus $\max(\text{TECQ}, \text{TDECQ})$ and receiver sensitivity ($\text{OMA}_{\text{outer}}$) each lane versus TECQ for 800GBASE-LR4

183.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 183–7.

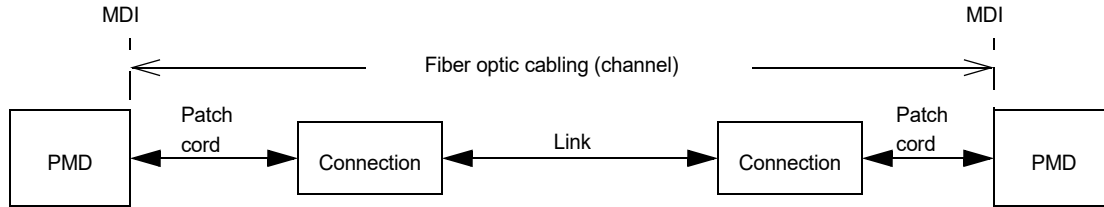


Figure 183–7—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 183–9. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 183–9—Optical channel characteristics

Description	800GBASE-FR4	800GBASE-LR4	Unit
Operating distance (max)	2	10	km
Channel insertion loss ^{a, b} (max)	4	6.3	dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	6.02	2.8	ps/nm
Negative dispersion ^b (min)	–11.26	–24.6	ps/nm
DGD_max ^c	2.3	4	ps
Optical return loss (min)	25	22	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b Over the wavelength range 1264.5 nm to 1337.5 nm for 800GBASE-FR4, and 1294.53 nm to 1310.19 nm for 800GBASE-LR4. The dispersion specifications are based on the statistical link design methodology documented in ITU-T REC G.652, Appendix I.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

183.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 183–10 where they differ.

Table 183–10—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in ANSI/TIA 568-C.3.

183.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 183–7, consists of a mated pair of optical connectors.

183.8.2.1 Connection insertion loss

The maximum link distance for 800GBASE-LR4 is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB. The maximum link distance for 800GBASE-FR4 is based on an allocation of 3 dB total connection and splice loss. Connections with different loss characteristics may be used provided the requirements of Table 183–9 are met.

183.8.2.2 Maximum discrete reflectance

The maximum value for each discrete reflectance shall be less than or equal to the value shown in Table 183–11 corresponding to the number of discrete reflectances above –55 dB within the channel. For numbers of discrete reflectances in between two numbers shown in the table, the lower of the two corresponding maximum discrete reflectance values applies.

Table 183–11—Maximum value of each discrete reflectance

Number of discrete reflectances above –55 dB	Maximum value for each discrete reflectance		Unit
	800GBASE-FR4	800GBASE-LR4	
1	–25	–22	dB
2	–31	–29	dB
4	–35	–33	dB
6	–38	–35	dB
8	–40	–37	dB
10	–41	–39	dB

183.8.3 Medium Dependent Interface (MDI)

The 800GBASE-FR4 or 800GBASE-LR4 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 183–7). Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 183.5.1, not at the MDI.

183.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

183.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 183–13 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 183–13 may be used to perform that test. The test patterns used in this clause are shown in Table 183–12.

Table 183–12—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	177.6.1.6
3	PRBS31Q	177.6.1.4
4	PRBS13Q	177.6.1.3
5	Scrambled idle test pattern encoded by the 800GBASE-R Inner FEC	175.2.4.11, 177.4
6	SSPRQ	177.6.1.5
7	Valid 800GBASE-R signal encoded by the 800GBASE-R Inner FEC	175.2.4, 177.4
8	PRBS31 pattern encoded by 800GBASE-R Inner FEC	177.6.1.1

Table 183–13—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	Square wave, 3, 4, 5, 6, 7 or 8	183.9.2
Side mode suppression ratio	3, 5, 6, 7 or 8	183.9.2
Average optical power	3, 5, 6, 7 or 8	183.9.3
Outer Optical Modulation Amplitude (OMA _{outer})	4 or 6	183.9.4
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	183.9.5
Transmitter eye closure for PAM4 (TECQ)	6	183.9.6
Overshoot and undershoot	6	183.9.7
Transmitter power excursion	6	183.9.8
Extinction ratio	4 or 6	183.9.9
Transmitter transition time	Square wave or 6	183.9.10
RIN _{xx} OMA	4 or 6	183.9.11
Receiver sensitivity	5 or 8	183.9.12
Stressed receiver conformance test signal calibration	6	183.9.13
Stressed receiver sensitivity	5 or 8	183.9.13

183.9.2 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 183–6 if measured per IEC 61280-1-3. The lane under test is modulated using the test pattern defined in Table 183–13.

183.9.3 Average optical power

The average optical power of each lane shall be within the limits given in Table 183–6 if measured using the methods given in IEC 61280-1-1, with the sum of the optical power from all of the lanes not under test below –30 dBm, per the test setup in Figure 53–6.

183.9.4 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 183–6. OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 183–13 as the difference between the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 183–8. OMA_{outer} is measured using waveforms captured at the output of the reference receiver defined in 183.9.5, before the reference equalizer.

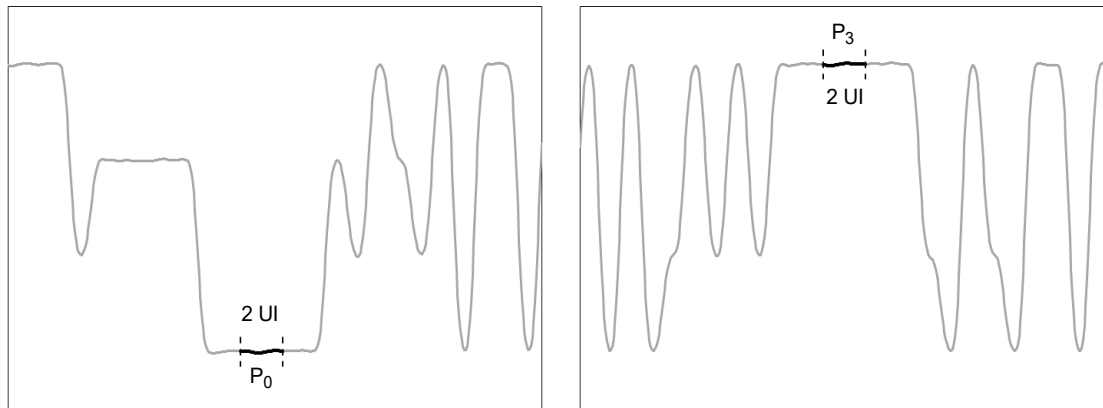


Figure 183–8—Example power levels P_0 and P_3 from PRBS13Q test pattern

183.9.5 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limit given in Table 183–6 if measured using the test setup specified in 122.8.5.1, with the optical channel specified in 183.9.5.1, the measurement method in 121.8.5.3 with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 183–6 and uses a test pattern specified for TDECQ in Table 183–13.
- TDECQ is defined with all receive lanes in operation using test pattern 3 or 5 (see Table 183–12). The received test patterns shall be asynchronous to the pattern used to test the transmitter, and shall have power levels as specified in Table 183–7 for the aggressor lanes in the stressed receiver sensitivity test.

- For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 183–1), the clock source for the SSPRQ test pattern is derived from the clock recovered from the 800GAUI-n input signal. The pattern of the 800GAUI-n input signal may be PRBS31Q, scrambled idle, or a valid 800GBASE-R signal.
- The reference receiver, composed of the combination of O/E converter and the oscilloscope has a 3 dB bandwidth of approximately 56.71875 GHz with a fourth-order Bessel-Thomson response to at least 1.3×113.4375 GHz and at frequencies above 1.3×113.4375 GHz the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The reference equalizer is a 15-tap, T-spaced, feed-forward equalizer (FFE), where T is the symbol period, with equalizer coefficient constraints as shown in Table 183–14. The reference equalizer may be implemented in the oscilloscope, but it is not considered part of the reference receiver.
- The target PAM4 symbol error ratio is 9.6×10^{-3} and the related Q_1 value (see 121.8.5.3) is 2.489.
- The normalized noise power density spectrum, $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 56.71875 GHz.
- The clock recovery unit (CRU) has a corner frequency of 4.27 MHz.
- The optical return loss is as given in Table 183–6.
- The lowest measured TDECQ values are achieved with the equalizer optimization method described in 121.8.5. Alternative optimization methods such as minimum mean squared error (MMSE) may be used to determine equalizer tap weights to reduce test time, and are expected to report equal or higher values of TDECQ. These alternative methods should not be used for receiver sensitivity and stressed receiver sensitivity calibration.

Table 183–14—Reference equalizer tap coefficients

Parameter	Symbol	Value	
		Minimum	Maximum
Feed-forward equalizer (FFE) length	N_b	15	
Number of equalizer pre-cursor taps	—	0	3
Main tap coefficient limit	$c(0)$	0.9	2.5
Normalized equalizer coefficient limits: ^a	$c(i)$		
$i = -3$		–0.15	0.1
$i = -2$		–0.1	0.25
$i = -1$		–0.5	0.1
$i = 1$		–0.6	0.2
$i = 2$		–0.2	0.3
$i = 3$		–0.15	0.15
$i = 4$		–0.15	0.15
$i = 5$		–0.15	0.15
$i = 6$		–0.15	0.15
$i \geq 7$		–0.1	0.1
Equalizer gain ^b	—	1	

^aThe main tap is marked by $i = 0$. The minimum and maximum values are relative to this tap's coefficient.

^bThe sum of the equalizer coefficients.

183.9.5.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 183–15.

Table 183–15—Transmitter compliance channel specifications

PMD type	Lane	Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
		Minimum	Maximum			
800GBASE-FR4	0	$0.203 \times (\lambda - 1271) - 9.96$	$0.187 \times (\lambda - 1271) - 5.81$	Minimum	17.1	0.8
	1	$0.194 \times (\lambda - 1291) - 5.99$	$0.183 \times (\lambda - 1291) - 2.12$			
	2	$0.185 \times (\lambda - 1311) - 2.22$	$0.177 \times (\lambda - 1311) + 1.47$			
	3	$0.176 \times (\lambda - 1331) + 1.38$	$0.169 \times (\lambda - 1331) + 4.92$			
800GBASE-LR4	All	$0.225 \times \lambda \times [1 - (1321.1 / \lambda)^4]$	$0.2175 \times \lambda \times [1 - (1307 / \lambda)^4]$	Minimum	15.6	0.8

^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 2 km for 800GBASE-FR4 and 10 km for 800GBASE-LR4. The dispersion specifications are based on the statistical link design methodology documented in ITU-T REC G.652, Appendix I.

^b The value “Minimum” implies that the test channel insertion loss should be sufficiently low that it does not significantly stress the tester receiver.

^c The back-reflection is applied at TP2.

^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution due to chromatic dispersion.

A 800GBASE-FR4 or 800GBASE-LR4 transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 183–15 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 183–15. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 183–15.

183.9.6 Transmitter eye closure for PAM4 (TECQ)

The transmitter eye closure for PAM4 (TECQ) is a measure of the optical transmitter’s eye closure at TP2. The TECQ of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for TECQ in Table 183–13. The TECQ of each lane is measured using the methods specified for TDECQ in 183.9.5, except that the test fiber is not used.

183.9.7 Transmitter overshoot and undershoot

The overshoot and undershoot of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for overshoot and undershoot in Table 183–13.

Overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 183.9.5) and the waveform captured for the TECQ test (see 183.9.6), at the output of the reference receiver defined in 183.9.5, before the reference equalizer.

Overshoot and undershoot are calculated using the methods in 140.7.7.

183.9.8 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for transmitter power excursion in Table 183–13.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 183.9.6), at the output of the reference receiver defined in 183.9.5, before the reference equalizer.

Transmitter power excursion is defined in 140.7.8.

183.9.9 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for extinction ratio in Table 183–13. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power level P_3 , measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P_0 , measured over the central 2 UI of a run of 6 zeros, as shown in Figure 183–8. Extinction ratio is measured using waveforms captured at the output of the reference receiver defined in 183.9.5, before the reference equalizer.

183.9.10 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 183–6 if measured using a test pattern specified for transmitter transition time in Table 183–13.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. Transmitter transition time is measured using waveforms captured at the output of the reference receiver defined in 183.9.5, before the reference equalizer.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 151.8.4), with the exception that the square wave test pattern can be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

183.9.11 Relative intensity noise ($RIN_{xx}OMA$)

$RIN_{xx}OMA$, with “xx” referring to the value for optical return loss tolerance in Table 183–6, shall be within the limit given in Table 183–6 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 183.9.4, but with applied xx dB optical reflection and the reference receiver specified for TDECQ measurement in 183.9.5. The noise is measured at the output of the reference receiver defined in 183.9.5, before the reference equalizer. $RIN_{xx}OMA$ is measured using the methods specified in 180.9.11.

183.9.12 Receiver sensitivity

The receiver sensitivity ($\text{OMA}_{\text{outer}}$) of each lane shall be within the limits given in Table 183–7 if measured using a test pattern specified for receiver sensitivity in Table 183–13.

The conformance test signal applied at TP3 meets the requirements for a 800GBASE-FR4 or 800GBASE-LR4 transmitter followed by an attenuator. An optical demultiplexer may be used to separate the lane having the wavelength for the lane under test as specified in Table 183–7 for calibrating the TECQ.

The TECQ of the conformance test signal is measured according to 183.9.6, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity ($\text{OMA}_{\text{outer}}$) as specified in Table 183–7.

183.9.13 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limit given in Table 183–7 if measured using the method defined in 122.8.10, with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 183.9.5, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 183–6.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $\text{RIN}_{\text{xxOMA}}$ of the SRS test source (where x is the value for optical return loss tolerance from Table 183–6) shall be no greater than the value specified in Table 183–6.
- An example stressed receiver conformance test setup is shown in Figure 122–8; however, alternative test setups that generate equivalent stress conditions may be used.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 183–6.
- The values of transmitter overshoot and undershoot and transmitter power excursion of the stressed receiver conformance test signal are within the limits specified in Table 183–6.
- The stressed receiver conformance input signal meets the sinusoidal jitter specified in 183.9.13.1.
- The required values of the “Stressed receiver sensitivity ($\text{OMA}_{\text{outer}}$), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “ $\text{OMA}_{\text{outer}}$ of each aggressor lane” are as given in Table 183–7.

183.9.13.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 182–17 and is illustrated in Figure 182–8.

183.10 Safety, installation, environment, and labeling

183.10.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

183.10.2 Laser safety

Optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹⁵

183.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

183.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-FR4 or 800GBASE-LR4 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause are met.

183.10.5 Electromagnetic emission

A system integrating a 800GBASE-FR4 or 800GBASE-LR4 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

183.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

183.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-FR4).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 183.10.2.

183.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 183–16 and Table 183–17. Additional variables associated with the ILT function are listed in Table 178B–6 and Table 178B–7.

¹⁵A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 183–16 and Table 183–17.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 183–16—MDIO/PMD control variable mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	183.5.6	1.0.15	45.2.1.242.1
PMD_global_transmit_disable	183.5.7	1.9.0	45.2.1.8
PMD_transmit_disable _{n-1} to PMD_transmit_disable ₀	183.5.7	1.9.4 to 1.9.1	45.2.1.8

Table 183–17—MDIO/PMD status variable mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	183.5.9	1.1.7	45.2.1.2.3
PMD_transmit_fault	183.5.10	1.8.11	45.2.1.7
PMD_receive_fault	183.5.11	1.8.10	45.2.1.7
PMD_global_signal_detect	183.5.4	1.10.0	45.2.1.9
PMD_signal_detect _{n-1} to PMD_signal_detect ₀	183.5.5	1.10.4 to 1.10.1	45.2.1.9

183.12 Protocol implementation conformance statement (PICS) proforma for Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4¹⁶

183.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

183.12.2 Identification

183.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

183.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-202x, Clause 183, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-FR4 and 800GBASE-LR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-202x.)	

Date of Statement	
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¹⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Editor's note: This PICS will be implemented in Draft 2.1. Please do not submit comments against this subclause.

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184. Inner Forward Error Correction (Inner FEC), type 800GBASE-LR1

184.1 Overview

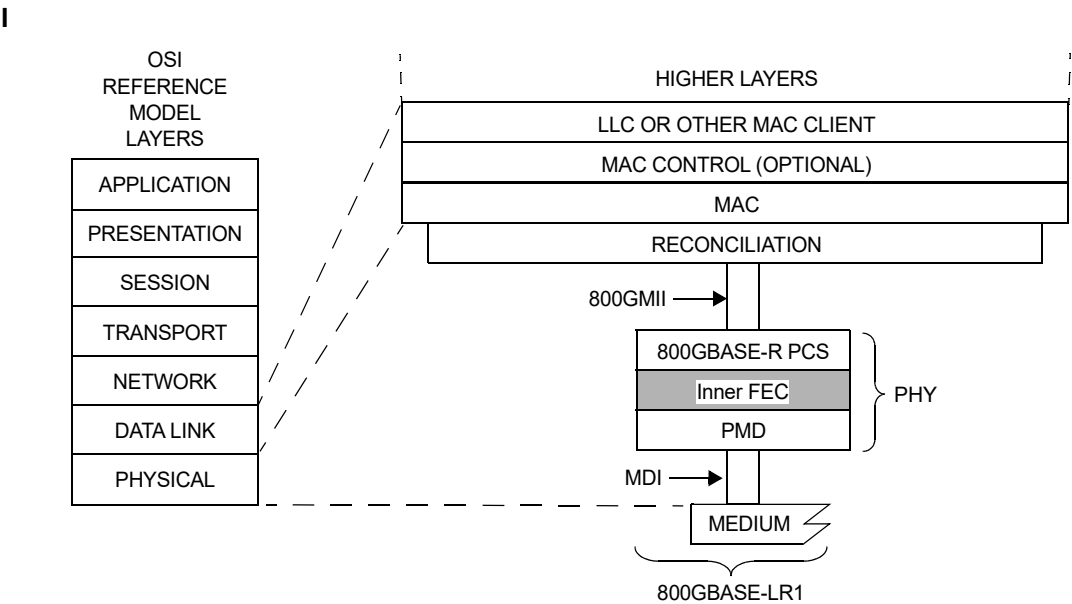
184.1.1 Scope

This clause specifies the Inner FEC sublayer for the Physical Layer implementations known as 800GBASE-LR1. This Inner FEC sublayer includes functionality often associated with the PMA sublayer.

When necessary for disambiguation, to differentiate the Inner FEC defined in this clause from the 800GBASE-R Inner FEC defined in Clause 177, the term 800GBASE-LR1 Inner FEC is used. Within this clause the term Inner FEC refers specifically the 800GBASE-LR1 Inner FEC.

184.1.2 Position of the Inner FEC to the ISO OSI reference model

Figure 184–1 depicts the relationship of the Inner FEC sublayer (shown shaded), along with other Ethernet sublayers, to the ISO Open System Interconnection (OSI) reference model.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 184–1—Inner FEC relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

184.1.3 Summary of functions

The Inner FEC provides a second layer of FEC protection without decoding the data encoded by the outer RS-FEC in the PCS.

The Inner FEC provides the following services:

- Alignment lock, deskew, and reordering of 32 received PCS lanes
- Permutation of symbols across the 32 PCS lanes to create 32 Inner FEC flows
- Per flow convolutional interleaving (deinterleaving)
- Per flow BCH(126,110) encoding (decoding), and circular shift (inverse circular shift)
- DSP frame generation (transmit) and synchronization and termination (receive)
- Mapping (demapping) to (from) DP-16QAM symbols
- In the transmit direction, creates analog signals to drive the optical modulator in the PMD
- In the receive direction, processes the analog signals from the optical demodulator of the PMD

184.2 Inner FEC overview

A functional block diagram of the Inner FEC is shown in Figure 184–2.

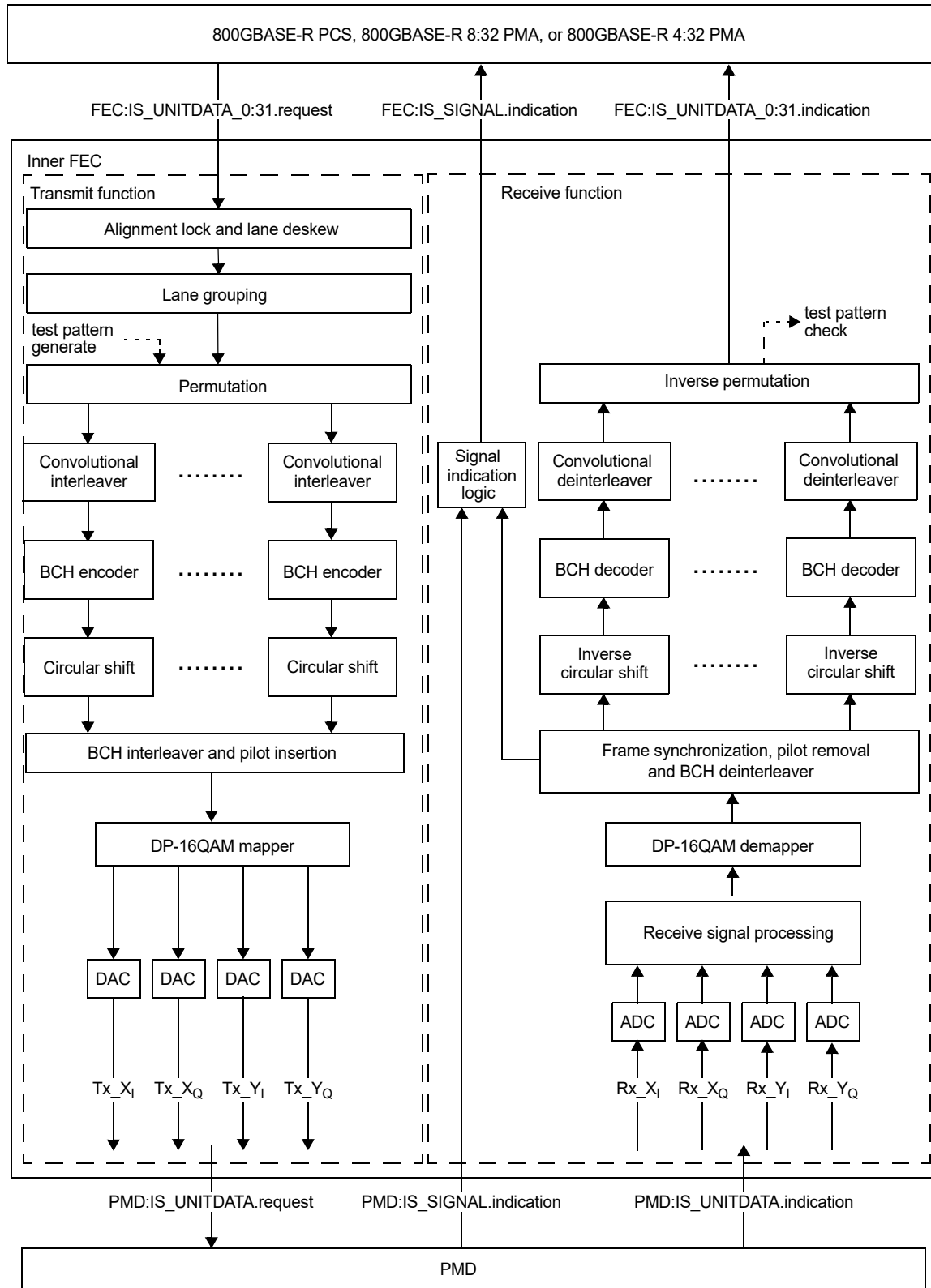


Figure 184-2—Inner FEC functional block diagram

The Inner FEC is composed of the Inner FEC transmit and the Inner FEC receive functions. Most of the encoding and decoding is done per Inner FEC flow.

The transmit functions receive 32 lanes of continuous RS(544,514) encoded 10-bit RS-FEC symbols. If necessary, the lanes are reordered and deskewed. The RS-FEC symbols are then rearranged across the 32 lanes by a permutation function to create 32 Inner FEC flows. The RS-FEC symbols in each of the 32 flows are interleaved by a convolutional interleaver and then encoded with a BCH(126,110) FEC. A circular shift is performed before interleaving the BCH(126,110) encoded flows and generating the DSP frames for each polarization. The DSP frames are then mapped into 16QAM symbols, converted into four analog signals, and sent to the PMD via the PMD:IS_UNITDATA.request primitive.

In the receive direction, the Inner FEC synchronization process continually monitors the PMD:IS_SIGNAL.indication primitive. When the SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive is OK, it converts the four analog received signals into four-level signals and demaps the 16QAM symbols to recover the DSP frames. The synchronization process synchronizes on the received DSP frames, and recovers the information required to decode the bits mapped on each DP-16QAM symbol. It reverses the circular shift and BCH interleaving and performs decoding and error correction. Convolutional interleaving and permutation are undone to restore the original lanes order. The 32 lanes are forwarded to the client sublayer via the FEC:IS_UNITDATA.indication primitive along with the SIGNAL_OK parameter of the FEC:IS_SIGNAL.indication primitive.

184.3 Inner FEC service interface

The client sublayer above the Inner FEC (see Figure 184–2) is a 800GBASE-R PCS, 800GBASE-R 8:32 PMA, or 800GBASE-R 4:32 PMA.

The service interface primitives are summarized as follows:

```
FEC:IS_UNITDATA_i.request(tx_symbol)
FEC:IS_UNITDATA_i.indication(rx_symbol)
FEC:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to 31. In the transmit direction, the Inner FEC receives 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, from the client sublayer above. In the receive direction, the Inner FEC sends 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, to the client sublayer.

FEC:IS_UNITDATA_i.request is the same as PMA:IS_UNITDATA_i.request for the PMA 32:8 defined in 173.2.

FEC:IS_UNITDATA_i.indication is the same as PMA:IS_UNITDATA_i.indication for the PMA 32:8 defined in 173.2.

The Inner FEC provides signal status information to the client sublayer above using the FEC:IS_SIGNAL.indication(SIGNAL_OK) primitive (see Figure 184–2). The SIGNAL_OK parameter is set to OK when alignment_status (see 184.7.2.2) is true and FAIL when alignment_status is false. When SIGNAL_OK is FAIL, the tx_symbol parameters of the FEC:IS_UNITDATA_i.indication primitives are undefined.

184.4 Transmit function

184.4.1 Alignment lock and deskew

The alignment lock and deskew function, when implemented, shall be identical to the processes specified in 172.2.5.1 except that only a deskew to 20-bit (two RS-FEC symbols) boundaries across all PCS lanes is required.

184.4.2 Lane grouping

The 32 Inner FEC sublayer interface bit streams might be received in any order. The bit stream number is defined by the unique portion (UM0 to UM5) of the alignment marker that is mapped to each PCS lane (see 172.2.4.6) carried by each bit stream.

The vector $pcsla[q, i]$ represents the Inner FEC flows at the output of the lane grouping function, aligned to 10-bit RS-FEC symbols, where the index i represents the sequence of 10-bit RS-FEC symbols and q the Inner FEC flow number.

The lane grouping function shall organize the 32 bit streams into two groups according to the PCS lane number so that all bit streams with PCS lane numbers 0 to 15 are assigned to one group and all bit streams with PCS lane numbers 16 to 31 to another group. One group is then assigned to $pcsla[q, i]$ for $q = 0$ to 15 and the other is assigned to $pcsla[q, i]$ for $q = 16$ to 31.

184.4.3 Permutation

The lane permutation function shall rearrange the RS-FEC symbols of the $pcsla[q, i]$ Inner FEC flows to create 32 output flows such that each group of four consecutive symbols on each output flow contains one symbol from each of the four RS-FEC encoders in the 800GBASE-R PCS. Figure 184–3 shows an example of the lane permutation function operation.

The permutation function maps the RS-FEC symbols received on 32 $pcsla$ flows to 32 output permo flows.

- Define $permo[q, i]$ to be the 10-bit symbol in output flow q at time i at the output of the permutation function.

The permutation function is defined by the following pseudocode:

```

For each  $i$ 
  For each  $q = 0$  to 31
     $permo[q, i] = pcsla[(q + 16 \times \text{floor}(i/2)) \bmod 32, i]$ 
  End for
End for

```

Where $\text{floor}(a)$ denotes the greatest integer less than or equal to the real number a .

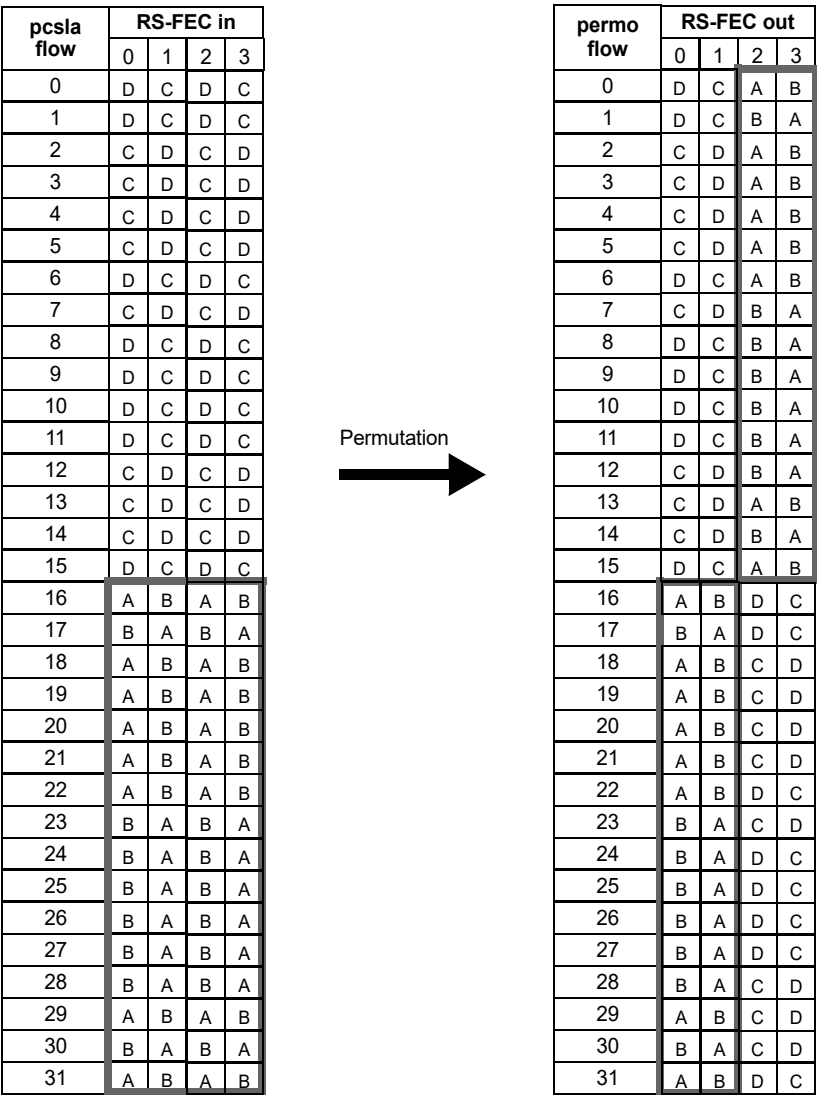


Figure 184–3—Example of the lane permutation function

184.4.4 Convolutional interleaver

The convolutional interleaver shall spread out the transmission order of consecutive units of 40 bits (four RS-FEC symbols) from the permo flows, such that the 110-bit input to the BCH encoder function contains no more than one RS-FEC symbol from any RS codeword. The convolutional interleaving function consists of three parallel delay lines (numbered 0 to 2), as illustrated in Figure 184–4. Each delay operator “D” represents a storage element of 40 bits. From one delay line to the next higher delay line, six delay operators are added. The bit ordering at the input of a delay line is preserved at the output.

The following list describes the convolutional interleaver process:

- a) The input and output switches are always aligned to the same row b , where $b = 0$ to 2
- b) A block of 40 bits is read from row b .
- c) The contents of row b are shifted to the right by 40 bits.
- d) A block of 40 bits is written to row b .
- e) The position of the switches is updated to $(b + 1) \bmod 3$

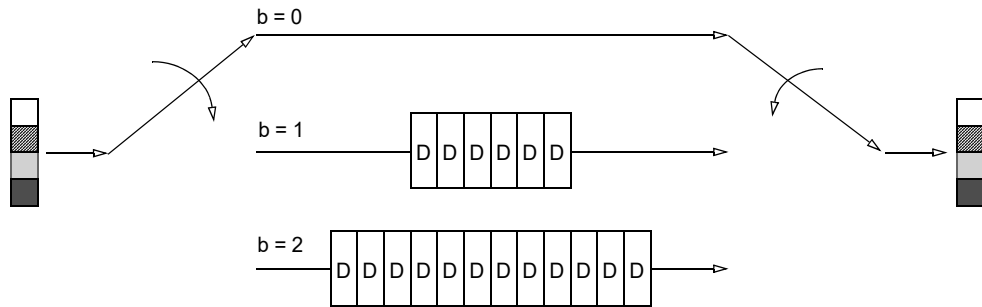


Figure 184-4—Convolutional interleaver

Initialization of the convolutional interleaver switches (to their topmost positions) is defined to occur at the start of every transmit DSP frame (see 184.4.8).

The following is performed individually on each of the q flows of permo, operating on 40-bit symbols (consisting of four RS-FEC symbols) j :

For each j
 $\text{convio}[j] = \text{permo}[j - 18 \times j \bmod 3]$
End for

NOTE — $\text{convio}[j]$ is undefined when the index to permo is negative.

184.4.5 BCH encoder

The BCH encoder works in conjunction with the outer RS(544,514) FEC to provide a high performance FEC for 800GBASE-LR1.

The BCH code generator polynomial shall be derived starting with a BCH(127,113) code that is shortened to a BCH(124,110) code and then extended to the final BCH(126,110) code by adding even and odd parity check bits.

The code is defined as follows:

- Define c as a binary vector of length 126, and $c(x)$ a polynomial of degree 125 with the coefficients defined by c (where the bit 0 of c represents the coefficient of power 125)
- Then c is a codeword of the BCH(126,110) code if $c(x)$ is divisible (modulo 2) by the defined binary generator polynomial $g(x)$

BCH (126,110) code operates on 11 RS-FEC symbols provided by the convolutional interleaver

- Denote the index of the BCH codeword as u and the bit index within the BCH codeword as v where $v = 0$ to 125.

The encoding of each BCH codeword u on flow q ($q = 0$ to 31) of convio is defined as follows:

- A message polynomial $m(x)$ of degree 109 is defined where the coefficient of the x^{109-v} term is the BCH encoder convio[110 $\times$ u + v] for $v = 0$ to 109
- A generator polynomial $g(x)$ of degree 16 is defined as shown in Equation (184–1)
- A parity polynomial $p(x)$ of degree 15 shown in Equation (184–2), is defined as the remainder from the division (modulo 2) of $m(x) \times x^{16}$ by the generator polynomial shown in Equation (184–1).

$$g(x) = x^{16} + x^{14} + x^{11} + x^{10} + x^9 + x^7 + x^5 + x^3 + x + 1 \quad (184-1)$$

$$p(x) = p_{15} \times x^{15} + p_{14} \times x^{14} + \dots + p_1 \times x + p_0 \quad (184-2)$$

Define parity[15:0] to be the coefficients of the computed parity polynomial where parity[15] corresponds to p_{15} in Equation (184–2), parity[14] to p_{14} and so on.

The bit mapping between each convio input flow and encodeo output flow is:

```

For each  $u$ 
    encodeo[126 $u$ :126 $u$ +109] = convio[110 $u$ :110 $u$ +109]
    encodeo[126 $u$ +110:126 $u$ +125] = parity[15:0]
End for

```

184.4.6 Circular shift

The circular shift function shall be applied to each flow. It rearranges the 110 payload bits of each BCH FEC codeword to further increase robustness to burst errors. Consider each 126-bit BCH FEC codeword as a vector of bits, encodeo[j], and apply the following process:

```

For  $j = 0$  to 109
    circo[ $j$ ] = encodeo[( $j - 20q$ ) mod 110]
End for

For  $j = 110$  to 125
    circo[ $j$ ] = encodeo[ $j$ ]
End for

```

Where q corresponds to the flow number (0 to 31).

184.4.7 BCH interleaver

The BCH interleaver shall work in conjunction with the circular shift to improve robustness to burst errors and unequal bit error rates of the bits extracted from the DP-16QAM symbols, due to constellation position and different PDL noise on different polarizations.

The BCH interleaver spreads the bits of each BCH codeword evenly across all dimensions of the DP-QAM16 symbols.

The BCH interleaver works on a block of 32 synchronized BCH codewords generated on the 32 circo flows.

- Denote the index of the 126-bit codeword block aligned across all 32 circo flows as k .
- Denote the index of the BCH interleaver output intero flows as q where $q = 0$ to 3.
- Denote $2l + j$ where $l = 0$ to 503 and $j = 0$ to 1, as an index of the bits within block k of output bits formed from input k block bits.

The mapping between each circo input flows and intero output flow is:

```

For each  $k$ 
  For  $q = 0$  to 3
    For  $l = 0$  to 503
      For  $j = 0$  to 1
        intero[ $q, 2l + j$ ] =
          circo[ $4 \times \text{floor}(l/63) + (2l + \text{floor}(l/2) \bmod 2 + q) \bmod 4, 2 \times l \bmod 63 + (l + j) \bmod 2$ ]
      End for
    End for
  End for
End for

```

184.4.8 DSP frame

The DSP frame is defined as a set of $96 \times 64 = 6144$ 4-bit blocks. 4-bit pilot symbols (PS) are inserted every 64 4-bit blocks (one 4-bit PS, 63 4-bit payload blocks). The details of the DSP frame are shown in Figure 184–5.

Two streams of DSP frames, one for each polarization, are generated by the inner FEC. The 4-bit blocks for DSP frame_0 are constructed from two consecutive bits from intero[0] (to be mapped to X_I) and two consecutive bits from intero[1] (to be mapped to X_Q), and the 4-bit blocks for the DSP frame_1 are constructed from two consecutive bits from intero[2] (to be mapped to Y_I) and two consecutive bits from intero[3] (to be mapped to Y_Q). See 184.4.11.1 for the mapping function.

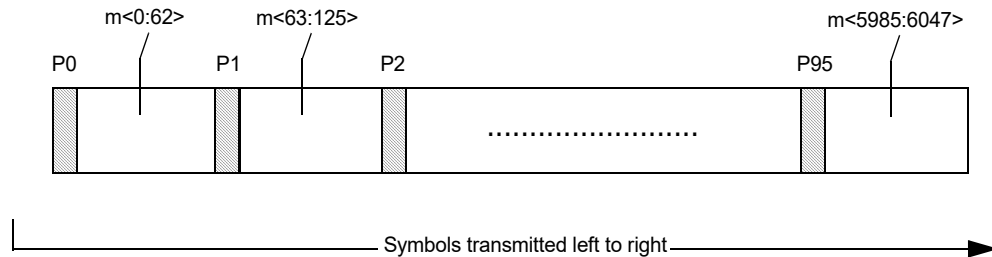


Figure 184–5—DSP frame

184.4.9 Pilot sequence

The pilot sequence shall be a fixed PRBS9 sequence with different seed values for DSP frame_0 and for DSP frame_1. The generator and seed values for each pilot sequence are shown in Figure 184–6 and listed in Table 184–1. The seeds are selected so that the pilot sequences are DC balanced.

For both DSP frame_0 and DSP frame_1, the generator is initialized using the seed at the start of every DSP frame. The generator produces a sequence of 192 bits. A zero bit is inserted after each bit to generate the bit-pairs that form the pilot symbols, which use the outer points of the 16QAM constellation. The bit-pairs that compose the pilot sequence are shown in Table 184–2. They are selected such that they will produce symbols that use the outer 16QAM constellation points, as shown in Figure 184–7.

The bit-pairs for the X polarization are distributed in a round-robin manner to outputs 0 and 1. The bit-pairs for the Y polarization are distributed in a round-robin manner to outputs 2 and 3.

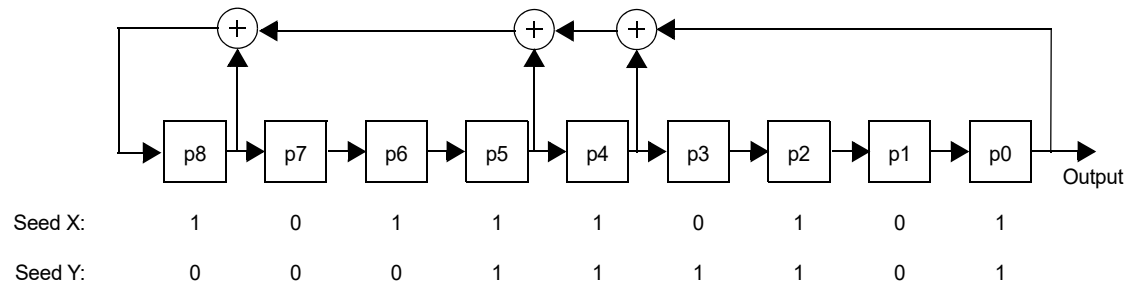


Figure 184-6—Pilot seed and sequence

Table 184-1—Pilot sequence generator polynomial and seed values

Generator polynomial	Seed for X-polarization	Seed for Y-polarization
$x^9 + x^8 + x^5 + x^4 + 1$	0x175	0x03D

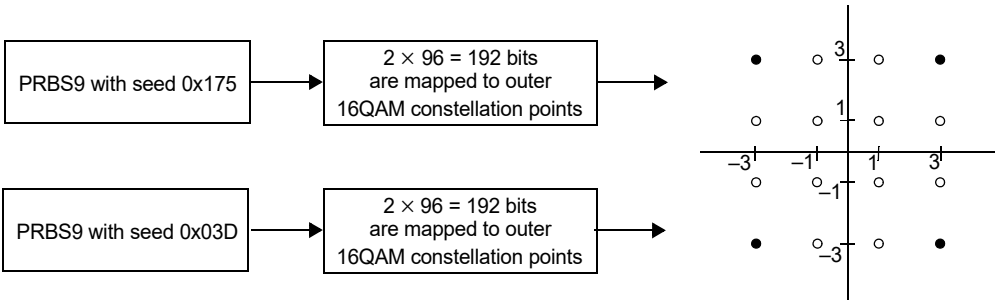


Figure 184-7—Pilot sequence mapping

Table 184-2—Pilot sequence

	Output					Output					Output					Output			
Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3
0	10	00	10	00	24	00	10	00	00	48	10	10	00	10	72	00	00	00	10
1	10	00	10	10	25	00	00	00	10	49	00	00	10	00	73	10	10	00	10
2	10	10	10	10	26	10	10	10	00	50	10	00	10	10	74	00	10	00	00
3	10	00	00	00	27	10	10	00	00	51	00	00	10	10	75	10	00	10	00
4	10	00	00	10	28	10	10	10	00	52	10	10	00	00	76	00	10	00	00

Table 184–2—Pilot sequence (*continued*)

Output					Output					Output					Output				
Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3	Index	0	1	2	3
5	00	00	00	10	29	00	00	00	00	53	00	00	10	00	77	10	10	00	10
6	10	10	00	00	30	00	10	00	00	54	10	00	10	00	78	00	10	00	00
7	00	10	00	10	31	10	00	00	00	55	10	10	00	00	79	00	10	00	00
8	00	10	00	00	32	10	10	00	10	56	00	00	00	10	80	00	10	10	10
9	00	10	10	00	33	00	10	10	10	57	00	00	10	00	81	10	10	10	10
10	00	00	00	00	34	00	00	10	00	58	10	00	00	00	82	10	00	10	10
11	00	10	10	00	35	00	00	00	00	59	00	10	00	10	83	10	10	10	10
12	10	10	10	10	36	10	00	00	10	60	10	00	10	10	84	00	10	10	00
13	00	10	00	00	37	10	10	00	10	61	00	10	00	00	85	10	00	10	00
14	00	00	10	10	38	10	00	00	00	62	10	10	00	00	86	00	00	10	10
15	10	00	00	10	39	10	10	10	00	63	10	00	10	10	87	00	00	10	10
16	00	10	00	10	40	10	10	10	10	64	10	00	00	10	88	10	10	10	00
17	10	00	10	00	41	00	10	10	00	65	00	10	10	10	89	00	10	00	10
18	10	00	10	10	42	00	10	00	00	66	10	10	00	10	90	00	00	10	10
19	00	00	00	10	43	00	00	00	00	67	00	00	10	00	91	10	00	00	00
20	10	00	10	10	44	00	00	00	10	68	00	10	00	10	92	10	00	10	00
21	10	00	10	10	45	00	00	00	10	69	00	00	00	00	93	00	10	10	10
22	10	10	10	00	46	10	10	10	00	70	10	10	10	00	94	10	00	10	10
23	00	10	10	10	47	00	00	10	00	71	10	10	00	10	95	00	00	10	10

184.4.10 DSP frame generation

The BCH interleaver and pilot insert function shall map the bits received on intero flows to dspfo.

- Denote the index of the intero flows and the dspfo flows as q , where $q = 0$ to 3,
- Denote u as the index of the BCH codeword at intero and pilot inserter function aligned across the four intero flows, and aligned to the boundaries of the BCH encode function
- Denote the bit index within a DSP frame as v , where $v = 0$ to 127
- Denote the pilot sequence as $ps[q,k]$ for flow $dspfo[q]$ where $q = 0$ to 3, and PS symbol index k where $k = 0$ to 95 (see Table 184–2)

The mapping between the ps and intero input flows, and the dspfo output flows is:

For every u

For $q = 0$ to 3

If $v = 0$, $dspfo[q, 128u + v] = ps[q, u \bmod 96]$

If $v = 1$, $dspfo[q, 128u + v] = 0$

For $v = 2$ to 127

```

        dspfo[q, 128u + v] = intero[q, 126u + v - 2]
    End for
End for
End for

```

184.4.11 DP-16QAM mapper

The DP-16QAM mapper shall convert every 2-bit pairs from the four dspfo flows to four flows of four-level signals. These four-level signals are used by the 800GBASE-LR1 PMD to generate a single optical DP-16QAM signal with orthogonal polarizations (see 185.5.2).

Table 184–3 shows the resulting 2-bit pair to four-level signal mapping.

Table 184–3—2-bit pair to four-level signal mapping

2-bit pair	Level
00	–3
01	–1
10	3
11	1

184.4.11.1 Symbol mapping to analog signals

The four signals from the DP-16QAM mapper shall be converted to four analog signals, two for each polarization: X_I , X_Q , Y_I , and Y_Q . The analog signals are passed to the 800GBASE-LR1 PMD sublayer using the tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq parameters of the PMD:IS_UNITDATA.request primitive.

184.5 Receive function

184.5.1 Receive signal sampling

The PMD:IS_UNITDATA.indication rx_signal_xi, rx_signal_xq, rx_signal_yi, and rx_signal_yq parameters are mapped to four analog signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q.

Each Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q signal shall be sampled at a sampling rate equal to or greater than the 800GBASE-LR1 signaling rate. The details of the sampling, including any quantization and the chosen sampling rate, are implementation specific.

The signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q each carry a combination of the transmitting Inner FEC Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

The resulting signals are forwarded to the receive signal processing function.

184.5.2 Receive signal processing

The four analog signals Rx_{X_I} , Rx_{X_Q} , Rx_{Y_I} , and Rx_{Y_Q} shall be processed and converted into two streams of 16QAM symbols representing the X and Y polarization of the DP-16QAM symbols.

The DP-16QAM symbols should be sampled with more than two bits per dimension, in order to enable the BCH decoder to correct errors and recover the bits from the symbols based on the mappings in Table 184–3.

Functions within the coherent receiver include, but are not limited to:

- Retiming and resampling to one sample per symbol after applying an antialiasing filter with a –3 dB bandwidth of 0.5 times the signaling rate to the incoming samples
- Equalization of signal distortion caused by chromatic dispersion, group delay, and polarization mode dispersion in the channel
- Clock phase recovery
- Frequency offset estimation and removal assuming a constant frequency offset over a given block size N which is implementation specific
- Carrier phase recovery
- Identification of the X and Y polarizations using the pilot sequences
- Identify the I and Q component of each polarization
- IQ offset evaluation and compensation

The details of the receive signal processing are implementation specific.

184.5.3 DP-16QAM demapper

The DP-16QAM demapper function shall map the four bit streams from the receive signal processing to 2-bit pairs along with any additional information required for BCH decoding. The precision of this information is implementation dependent.

The demapper implements the inverse of the DP-16QAM mapper function.

184.5.4 DSP frame synchronization and pilot removal

DSP synchronization shall comply with the DSP lock state diagram shown in Figure 184–9.

The pilot sequence bits are used to synchronize to the each of the two DSP frames, to identify the I and Q component of each polarization, to identify DSP frame_0 and DSP frame_1, and to remove the skew between polarizations. These symbols shall be removed prior to the BCH deinterleaver function.

The DSP frame synchronization process is restarted when synchronization is lost, the received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive changes to FAIL, or when a reset of the Inner FEC is required (see Figure 184–9).

184.5.5 BCH deinterleaver

The BCH deinterleaver shall perform the inverse of the BCH interleaver function (see 184.4.7) to map the information received on its four input flows to the correct location within its 32 output flows.

184.5.6 Inverse circular shift

The inverse circular shift shall perform the inverse of the circular shift function (see 184.4.6) to map the information received on its 32 input flows to the correct location within its 32 output flows. There are 32 inverse circular shift functions.

184.5.7 BCH decoder

The BCH decoder is a soft decision decoder that requires a higher resolution than two bits for each of the signals Rx_X_I, Rx_X_Q, Rx_Y_I, and Rx_Y_Q. The decoder evaluates the incoming codeword and determines the most likely message value.

The BCH decoder shall extract the information bits from the codeword, correct them as necessary, and discard the parity symbols. The BCH decoder correction capability is implementation specific. If the BCH decoder is not able to correct a codeword the parity bits are discarded and the information bits are passed on without correction.

There are 32 BCH decoder functions one for each flow.

The following counters shall be implemented to aid a network operator in determining the link quality.

Inner_FEC_corrected_cw_counter

A 32-bit counter that counts once for each corrected FEC codeword processed by any of the 32 BCH decoders when alignment_status is true. A corrected Inner FEC codeword is a codeword that contained errors and was corrected or miscorrected. Note that when there is more than one bit error in a codeword there is a chance that the soft decision decoder could miscorrect the codeword. The Inner FEC decoder interprets miscorrected codewords as corrected codewords.

Inner_FEC_uncorrected_cw_counter

A 32-bit counter that counts once for each uncorrected FEC codeword processed by any of the 32 BCH decoders when alignment_status is true. An uncorrected FEC codeword is a codeword that contains errors that the decoder was unable to correct.

Inner_FEC_total_bits_counter

A 64-bit counter that counts once for each bit processed by any of the 32 BCH decoders when alignment_status is true. This may be used together with the FEC_corrected_bits_counter to calculate pre-FEC BER.

Inner_FEC_corrected_bits_counter

A 64-bit counter that counts once for each bit corrected by any of the 32 BCH decoders when alignment_status is true. This may be used together with the FEC_total_bits_counter to calculate pre-FEC BER.

Inner_FEC_codeword_error_bin_k

A set of $k+1$ 32-bit counters where $k = 0$ to 4. While alignment_status is true, for each Inner FEC codeword received with exactly k corrected (flipped) bits, Inner_FEC_codeword_error_bin_k is incremented. For example, if an Inner FEC codeword has exactly two bits corrected, then Inner_FEC_codeword_error_bin_2 is incremented. Note that bin 4 is for 4 or more bits corrected in an Inner FEC codeword.

For each codeword processed, exactly one counter in Inner_FEC_codeword_error_bin_k or Inner_FEC_uncorrected_cw_counter is incremented.

184.5.8 Convolutional deinterleaver

The convolutional deinterleaver shall perform the inverse of the convolutional interleaver function (see 184.4.4) to map the information received on its 32 input flows to the correct location within its 32 output flows. There are 32 convolutional deinterleaver functions.

The convolutional deinterleaving function consists of three parallel delay lines (numbered 0 to 2), as illustrated in Figure 184–8. Each delay operator “D” represents a storage element of 40 bits. From one delay line to the next higher delay line, six delay operators are added. The bit ordering at the input of a delay line is preserved at the output.

The following list describes the convolutional interleaver process:

- The input and output switches are always aligned to the same row b , where $b = 0$ to 2
- A block of 40 bits is read from row b .
- The contents of row b are shifted to the right by 40 bits.
- A block of 40 bits is written to row b .
- The position of the switches is updated to $(b + 1) \bmod 3$

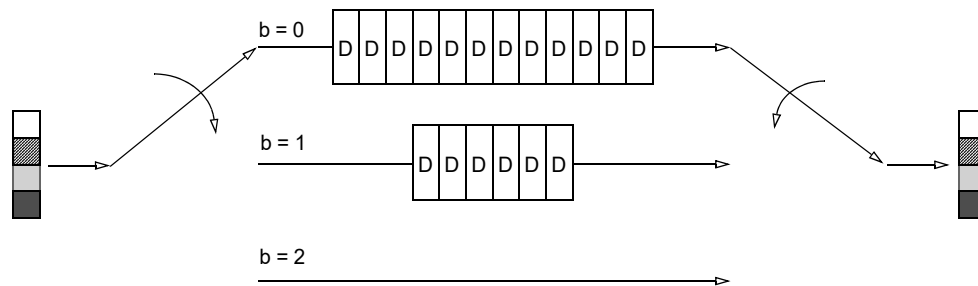


Figure 184–8—Convolutional deinterleaver

Initialization of the convolutional deinterleaver switches (to their topmost positions) is defined to occur at the start of every received DSP frame (see 184.4.8).

The following is performed individually on each of the 32 flows:

- Denote the input and output of the convolutional de-interleaver as $\text{input}[j]$ and $\text{output}[j]$, where the index j identifies 40-bit symbols.

For each j
 $\text{output}[j + 18 \times (2 - j \bmod 3)] = \text{input}[j]$
 End for

$\text{Output}[j]$ is undefined when the index is negative.

184.5.9 Inverse permutation

The inverse permutation function shall map the bits received from the convolutional deinterleaver on input_p , where $p = 0$ to 31, to 32 PCS lanes, and provide this data through the $\text{FEC:IS_UNITDATA}_i.\text{indication}$ primitives.

The mapping between the bit sequences on input_p flows and the PCS lanes is the inverse of the permutation mapping function (see 184.4.3).

184.6 Test pattern

184.6.1 Test pattern generator

The Inner FEC shall include a PRBS31 test pattern generator (see Figure 184–2). In test mode, 10-bit blocks from the PRBS31 generator are round-robin distributed to the 32 pcsla Inner FEC flows.

The PRBS31 generator implements the PRBS31 pattern defined by Equation (49–2) and shown in Figure 49–9.

The test pattern generator is enabled by the tx_generator_enable control variable.

184.6.2 Test pattern checker

The Inner FEC shall include a PRBS31 test pattern checker over the inverse permutation output PCS lanes (see 184.4.3). To recover the PRBS31 test pattern, 10-bit blocks are round-robin collected from the 32 inverse permutation function output flows.

The test pattern checker is enabled by the rx_checker_enable control variable.

The PRBS31 test pattern checker shall include block error detection and 17 related counters. Block error detection and behavior of the counters is defined in 174A.8.

The following counters shall be implemented:

test_block_error_bin_0_k

A set of 16 48-bit counters where counter *k* counts once for each test block received with exactly *k* errored test symbols, *k* = 0 to 15.

test_block_error_bin_0_16p

A 48-bit counter that counts once for each test block received with 16 or more errored test symbols.

184.7 State diagrams

This subclause is composed of detailed state diagram processes, including the associated definition of constants, variables, functions and counters.

184.7.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

184.7.2 State diagram definitions

184.7.2.1 Constants

M

The number of consecutive PS symbols that fail to match the expected value for a given polarization stream required to exit DSP frame lock (see 184.5.4). M = 8.

N
The number of consecutive PS symbols matching the expected value for a given polarization stream required to enter DSP frame lock (see 184.5.4). N = 12.

184.7.2.2 Variables

alignment_status
A Boolean variable set by the deskew process to reflect the status of the X polarization symbol stream to Y polarization symbol stream alignment. Set to true when the polarization symbol streams are synchronized and aligned and set to false otherwise.

alignment_valid
A Boolean variable that is set to true if the polarization symbol streams are aligned. Polarization symbol streams are considered to be aligned when dsp_lock<x> is true for both x, each polarization symbol stream has a unique identifier (dsp_ps_id<0> is different from dsp_ps_id<1>), and the polarization symbol streams are deskewed. Otherwise, this variable is set to false.

all_locked
A Boolean variable that is set to true when dsp_lock<x> is true for both x and is set to false when dsp_lock<x> is false for either x.

current_ps_id
A variable that holds the polarization symbol stream identifier corresponding to the current PS that is recognized on a given polarization symbol stream.
Values:
0: Value of the PS for the given polarization symbol stream maps to Table 184–2 sequence X_I, X_Q.
1: Value of the PS for the given polarization symbol stream maps to Table 184–2 sequence Y_I, Y_Q.

dsp_ps_id<x>
A variable that holds the value of the polarization symbol stream identifier received on the 800GBASE-LR1 PMD physical signal x, where x = 0 or x = 1.

dsp_lock<x>
A Boolean variable that is set to true when the receiver has detected the location of the PS for a given polarization symbol stream on the 800GBASE-LR1 PMD service interface, where x = 0:1.

enable_deskew
A Boolean variable that set to true when deskew is enabled and set to false when deskew is disabled. Received symbols may be discarded whenever deskew is enabled.

FEC_reset
A Boolean variable that is true when set by a management entity and is false otherwise.

reset
A Boolean variable that controls the resetting of the Inner FEC. It is true whenever a reset is necessary including when initiated by FEC_reset and during power on.

restart_lock
A Boolean variable that is set by the DSP frame lock process to reset the synchronization process on each polarization symbol stream. It is set to true when M PS symbols in a row fail to match (M_BAD state) on a given polarization symbol stream.

signal_ok
A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive. It is true if the value was OK and false if the value was FAIL.

sym_counter_done
A Boolean variable that indicates that sym_counter has reached its terminal count.

`sym_slip_done`
A Boolean variable that is set to true when the SLIP requested by the DSP frame lock state diagram has been completed indicating that the next candidate PS position is available for testing.

`sym_valid`
A Boolean variable that is set to true if the received symbol is a valid PS symbol according to the state of the pilot sequences generator (see 184.4.9) for the value of the `current_ps_id` variable.

`test_ps`
A Boolean variable that is set to true when a candidate PS symbol position is available for testing and false when the `FIND_1ST` state is entered.

184.7.2.3 Functions

`FIRST_COMPARE`
This function initializes the PS generator (see 184.4.9) and compares the value of the first PS symbol to the current symbol to determine if a PS symbol has been detected and returns the result of the comparison using the variable `sym_valid`.

`POL_IDENTIFY`
This function compares the value of the second PS symbol of each polarization sequence, to the current symbol to determine if a PS symbol has been detected and returns the result of the comparison using the variable `sym_valid` and the identified polarization using the variable `current_ps_id`.

`SYM_COMPARE`
This function compares the value of the expected PS symbol according to the state of the PS generator (see 184.4.9) and the value of the `current_ps_id` variable, to the current symbol to determine if a PS symbol has been detected and returns the result of the comparison using the variable `sym_valid`.

`SYM_SLIP`
Causes the next candidate PS symbol position to be tested. The precise method for determining the next candidate PS symbol position is not specified and is implementation dependent. However, an implementation shall evaluate all possible symbol positions.

184.7.2.4 Counters

`ps_bad_count`
Counts the number of consecutive PS symbols that don't match the expected value for a given polarization stream.

`ps_good_count`
Counts the number of consecutive PS symbols that match the expected value for a given polarization stream.

`sym_counter`
This counter counts the interval of 64 symbols between the position of one PS symbol and the expected position of the next PS symbol on a polarization symbol stream. This counter starts with a value of zero when `sym_counter` is asserted, is incremented for each symbol on that polarization symbol stream, and reaches a terminal value of 64 at the expected position of the next PS symbol on that polarization symbol stream.

184.7.3 State diagram figures

The Inner FEC shall implement two DSP frame lock processes as shown in Figure 184–9 to identify the boundaries of the DSP frames. The synchronization process operates independently on each polarization.

The state diagram determines when the Inner FEC has detected the location of the PS in the signal received from the PMD service interface.

The Inner FEC shall also implement the deskew process as shown in Figure 184–10.

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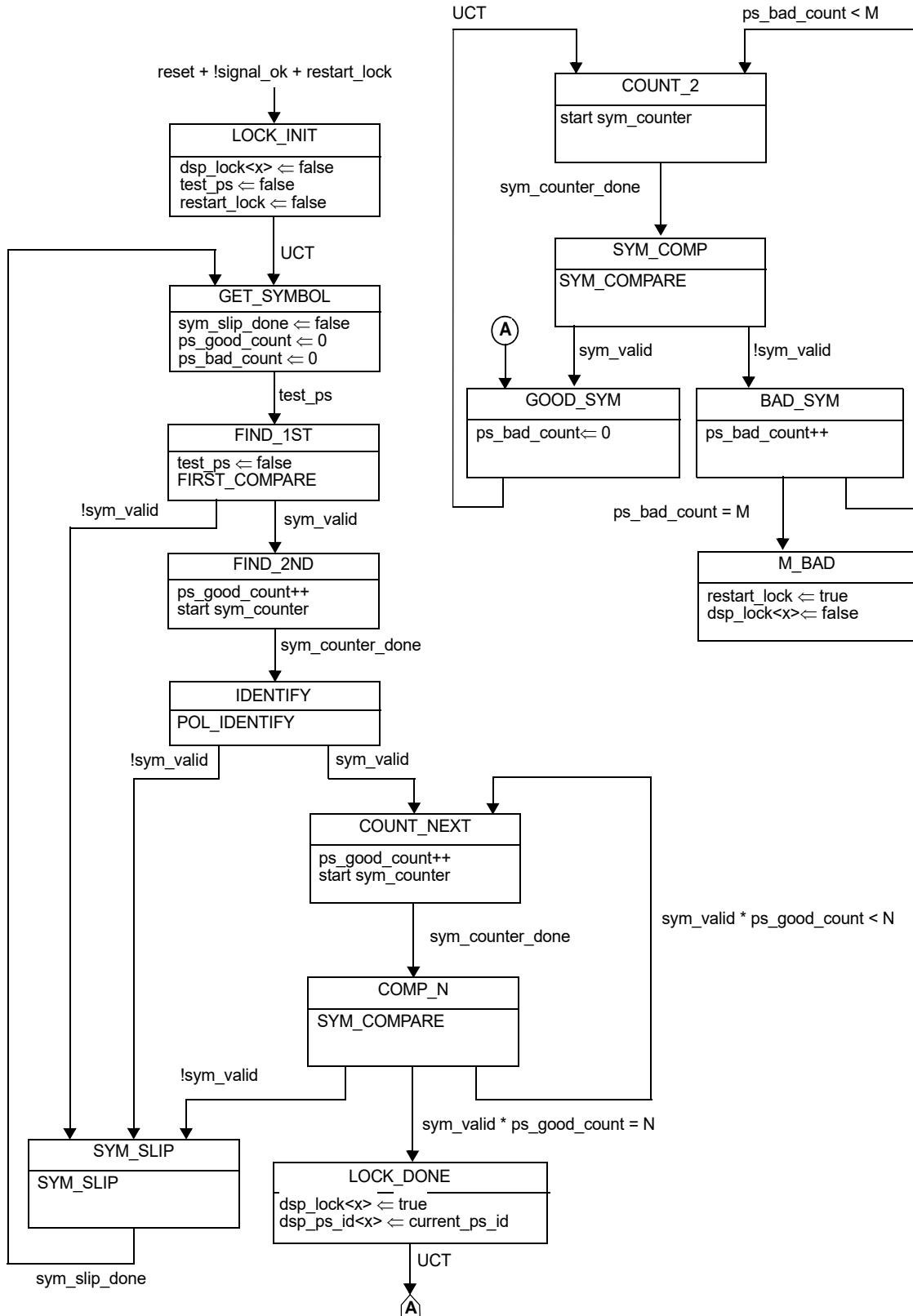


Figure 184–9—DSP lock state diagram

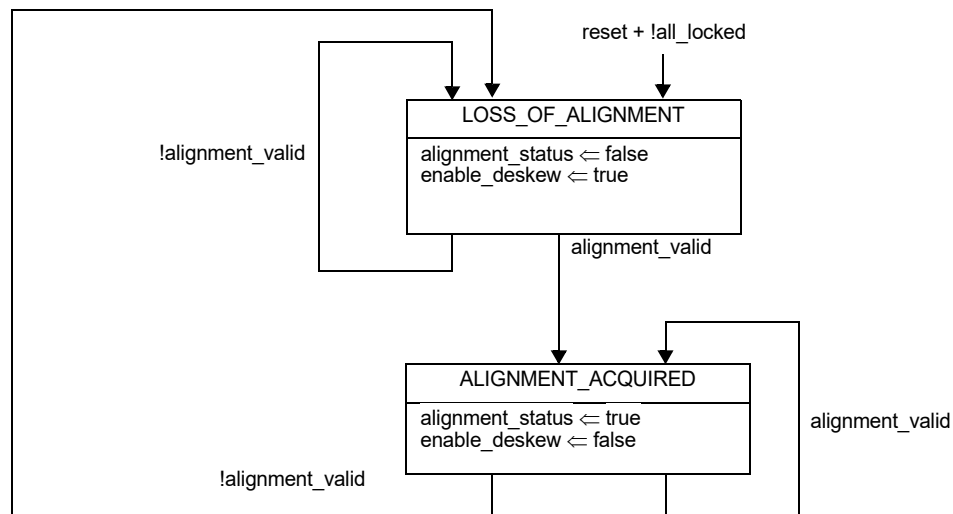


Figure 184–10—Deskew state diagram

184.8 Delay constraints

The maximum delay contributed by the Inner FEC (sum of transmit and receive delays at one end of the link) shall be no more than 131 072 bit times (256 pause_quanta or 163.84 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

184.9 Path data delay for time synchronization

The Inner FEC sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report Inner FEC path data delays is indicated by the Inner_FEC_delay_ns_TX_ability, Inner_FEC_delay_subns_TX_ability, Inner_FEC_delay_ns_RX_ability, and Inner_FEC_delay_subns_RX_ability status variables (see Table 184–5).

The transmit and receive path data delay values are reported as if the DDMP (data delay measurement point) occurs on dspfo[3,1894] (see 184.4.10), corresponding to the longest delay for transmit and the shortest delay for receive (see 90.7).

Four separate delays are reported in the following eight path data delay status variables (see Table 184–5):

- Inner_FEC_delay_ns_TX_max, Inner_FEC_delay_subns_TX_max
- Inner_FEC_delay_ns_TX_min, Inner_FEC_delay_subns_TX_min
- Inner_FEC_delay_ns_RX_max, Inner_FEC_delay_subns_RX_max
- Inner_FEC_delay_ns_RX_min, Inner_FEC_delay_subns_RX_min

184.10 Management variables

Inner FEC control and status variables intended to be accessible via a management system are listed in Table 184–4 and Table 184–5.

If the MDIO interface (see Clause 45) is implemented, Inner FEC control and status variables shall be mapped to the MDIO registers as shown in Table 184–4 and Table 184–5.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 184–4—Inner FEC control variables and MDIO mapping

Control variable	Variable reference	MDIO Register/bit number	MDIO register/bit reference
FEC_reset	184.7.2.2	1.0.15	45.2.1.1

Table 184–5—Inner FEC status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
dsp_lock<0>	184.7.2.2	1.2411.0	45.2.1.257
dsp_lock<1>	184.7.2.2	1.2411.1	45.2.1.257
alignment_valid	184.7.2.2	1.2410.0	45.2.1.256
tx_generator_enable	184.4.3	1.1501.3	45.2.1.170
rx_checker_enable	184.6.1	1.1501.0	45.2.1.170
Inner_FEC_corrected_cw_counter	184.5.7	1.2402, 1.2403	45.2.1.258
Inner_FEC_uncorrected_cw_counter	184.5.7	1.2404, 1.2405	45.2.1.259
Inner_FEC_total_bits_counter	184.5.7	1.2406, 1.2407, 1.2408, 1.2409	45.2.1.260
Inner_FEC_corrected_bits_counter	184.5.7	1.2410, 1.2411, 1.2412, 1.2413	45.2.1.260
Inner_FEC_codeword_error_bin_0	184.5.7	1.2424.15:0 1.2425.15:0	45.2.1.262
Inner_FEC_codeword_error_bin_1	184.5.7	1.2426.15:0 1.2427.15:0	45.2.1.262
Inner_FEC_codeword_error_bin_2	184.5.7	1.2428.15:0 1.2429.15:0	45.2.1.262

Table 184–5—Inner FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Inner_FEC_codeword_error_bin_3	184.5.7	1.2430.15:0 1.2431.15:0	45.2.1.262
Inner_FEC_codeword_error_bin_4	184.5.7	1.2432.15:0 1.2433.15:0	45.2.1.262
test_block_error_bin_0_<0:15>	184.6.2	1.2600 to 1.2647	45.2.1.264
test_block_error_bin_0_16p	184.6.2	1.2648 to 1.2650	45.2.1.264
Inner_FEC_delay_ns_TX_max	184.9	1.1813, 1.1814	45.2.1.177a
Inner_FEC_delay_subns_TX_max	184.9	1.1817	45.2.1.177a
Inner_FEC_delay_ns_TX_min	184.9	1.1815, 1.1816	45.2.1.177a
Inner_FEC_delay_subns_TX_min	184.9	1.1818	45.2.1.177a
Inner_FEC_delay_ns_RX_max	184.9	1.1819, 1.1820	45.2.1.177b
Inner_FEC_delay_subns_RX_max	184.9	1.1823	45.2.1.177b
Inner_FEC_delay_ns_RX_min	184.9	1.1821, 1.1822	45.2.1.177b
Inner_FEC_delay_subns_RX_min	184.9	1.1824	45.2.1.177b
Inner_FEC_delay_ns_RX_ability	184.9	1.1800.4	45.2.1.175
Inner_FEC_delay_subns_RX_ability	184.9	1.1800.5	45.2.1.175
Inner_FEC_delay_ns_TX_ability	184.9	1.1800.6	45.2.1.175
Inner_FEC_delay_subns_TX_ability	184.9	1.1800.7	45.2.1.175

184.11 Protocol implementation conformance statement (PICS) proforma for Clause 184, Inner FEC type 800GBASE-LR1¹⁷

184.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 184, Inner FEC sublayer type 800GBASE-LR1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

184.11.2 Identification

184.11.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

184.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 184, Inner FEC sublayer type 800GBASE-LR1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

¹⁷*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

184.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO	Clause 45	Registers and interface supported	O	Yes [] No []
DC	Delay constraints	184.8	Device conforms to delay constraints	M	Yes []

184.11.4 PICS proforma tables for Inner FEC Sublayer type 800GBASE-LR1

184.11.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Align lock and deskew	184.4.1		O	Yes []
TF2	Lane grouping	184.4.2		O	Yes []
TF3	Test pattern generator	184.6.1		M	Yes []
TF4	Permutation	184.4.3		M	Yes []
TF5	Convolutional interleaver	184.4.4		M	Yes []
TF6	BCH encoder	184.4.5		M	Yes []
TF7	Circular shift	184.4.6		M	Yes []
TF8	BCH interleaver	184.4.7		M	Yes []
TF9	Pilot sequence	184.4.9		M	Yes []
TF10	DSP frame generation	184.4.10		M	Yes []
TF11	DP-16QAM mapper	184.4.11		M	Yes []
TF12	Symbol mapping to analog signals	184.4.11.1		M	Yes []

184.11.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Receiver signal processing	184.5.2		M	Yes []
RF2	DP-16QAM demapper	184.5.3		M	Yes []
RF3	DSP frame synchronization	184.5.4		M	Yes []
RF4	Pilot symbols removal	184.5.4		M	Yes []
RF5	BCH deinterleaver	184.5.5		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
RF6	Inverse circular shift	184.5.6		M	Yes []
RF7	BCH decoder	184.5.7		M	Yes []
RF8	Convolutional deinterleaver	184.5.8		M	Yes []
RF9	Inverse permutation	184.5.9		M	Yes []
RF10	Test pattern checker	184.6.2		M	Yes []

184.11.4.3 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to Inner FEC management object is provided	184.10		O	Yes [] No []
M2	Mapping of MDIO control bits and MDIO status bits	184.10	Table 184–4 and Table 184–5	MD:M	Yes [] N/A []

184.11.4.4 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	DSP frame lock process	184.7.3	Implements two DSP frame lock processes as depicted in Figure 184–9	M	Yes []
SM2	Deskew process	184.7.3	Meets the requirements of Figure 184–10	M	Yes []

184.11.4.5 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	184.9		O	Yes [] No []
TS1	Path data delays	184.9	The transmit and receive path data delay values are reported as if the DDMP (data delay measurement point) occurs on dspfo[3,1894]	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	184.9	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	184.9	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

185. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1

185.1 Overview

This clause specifies the 800GBASE-LR1 PMD together with the associated single-mode fiber medium. The optical signal generated by this PMD type is modulated using a dual polarization 16-state quadrature amplitude modulation (DP-16QAM) format suitable for reception by a coherent optical receiver. The PMD provides point-to-point 800 Gb/s Ethernet links over one pair of single-mode fiber, with a reach of up to at least 10 km.

When forming a complete Physical Layer, a PMD shall be connected to the 800GBASE-LR1 Inner FEC as shown in Table 185–1.

Table 185–1—Physical Layer clauses associated with the 800GBASE-LR1 PMD

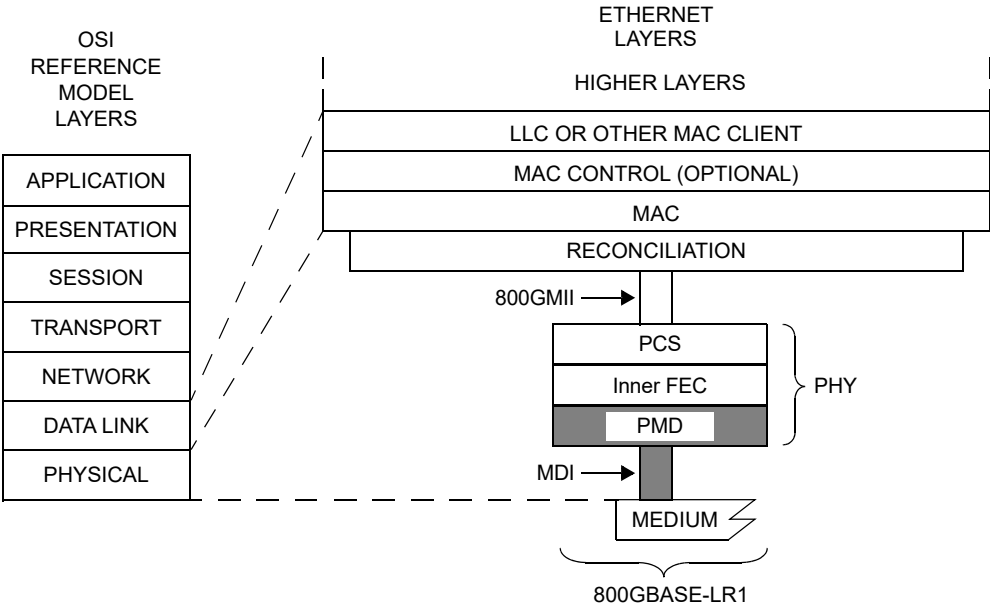
Associated clause	800GBASE-LR1
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Conditional ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
184—800GBASE-LR1 Inner FEC	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-LR1 PHY as described in 176B.6.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or two 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Figure 185–1 shows the relationship of the PMD and MDI (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMD = PHYSICAL MEDIUM DEPENDENT
LR1 = PMD FOR SINGLE MODE FIBER — 10 km

Figure 185–1—800GBASE-LR1 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

185.2 Error ratio allocation

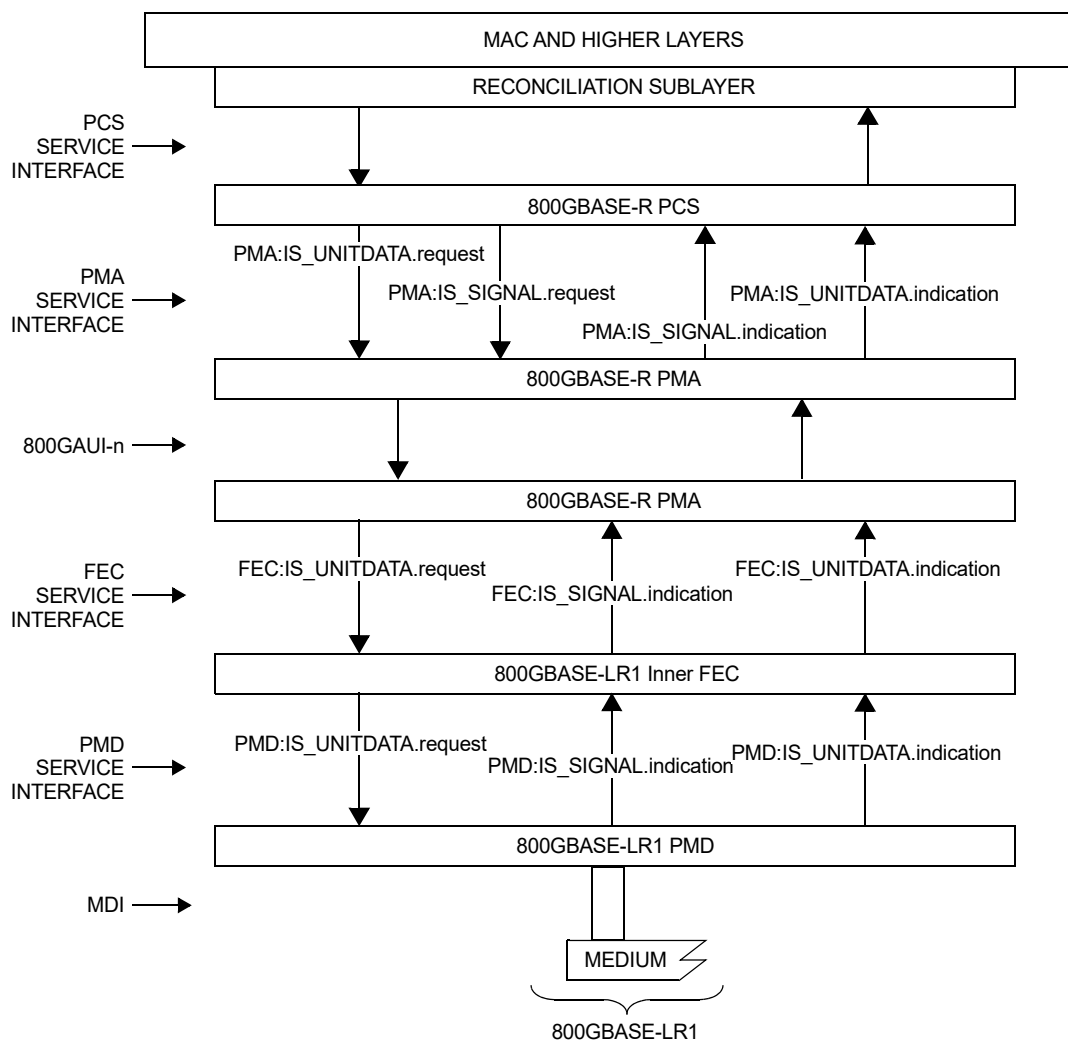
A complete PHY is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the Inner FEC using the method described in 174A.9, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PCS using the method described in 174A.10, with BER_{added} equal to 3.2×10^{-5} .

185.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 800GBASE-LR1 PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the Inner FEC sublayer that resides just above the PMD, and the PMD sublayer. The PMD translates the encoded data to and from signals suitable for the specified medium.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 185-2—800GBASE-LR1 inter-sublayer service interfaces including 800GMII Extender

185.3.1 800GBASE-LR1 PMD service interface

The 800GBASE-LR1 PMD service interface uses the following primitives to exchange signals and status information:

PMD:IS_UNITDATA.request
PMD:IS_UNITDATA.indication
PMD:IS_SIGNAL.indication

185.3.1.1 PMD:IS_UNITDATA.request

The PMD:IS_UNITDATA.request primitive defines the transfer of four analog signals from the 800GBASE-LR1 Inner FEC to the 800GBASE-LR1 PMD (see 184.4.11.1) that are inputs to the DP-16QAM modulator to generate a single optical signal with orthogonal polarizations (see 185.5.2).

185.3.1.1.1 Semantics of the primitive

PMD:IS_UNITDATA.request(tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq)

The PMD_UNITDATA.request primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the tx_signal_xi, tx_signal_xq, tx_signal_yi, and tx_signal_yq parameters.

185.3.1.1.2 When generated

The 800GBASE-LR1 Inner FEC generates PMD:IS_UNITDATA.request continuously.

185.3.1.1.3 Effect of receipt

The effect of receipt of the PMD:IS_UNITDATA.request by the 800GBASE-LR1 PMD is defined in 185.5.2.

185.3.1.2 PMD:IS_UNITDATA.indication

The PMD:IS_UNITDATA.indication primitive defines the transfer of four analog signals from the 800GBASE-LR1 PMD to the 800GBASE-LR1 Inner FEC that are the outputs of the coherent DP-16QAM receiver (see 185.5.3.)

185.3.1.2.1 Semantics of the primitive

PMD:IS_UNITDATA.indication(rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq)

The PMD:UNITDATA.indication primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq parameters, respectively.

185.3.1.2.2 When generated

The 800GBASE-LR1 PMD generates PMD:IS_UNITDATA.indication continuously.

185.3.1.2.3 Effect of receipt

The 800GBASE-LR1 Inner FEC processes the four analog signals as defined in 184.5.

185.3.1.3 PMD:IS_SIGNAL.indication(SIGNAL_OK)

The PMD:IS_SIGNAL.indication primitive is generated by the 800GBASE-LR1 PMD to indicate the status of the receive function via the SIGNAL_OK parameter.

185.3.1.3.1 Semantics of the service primitive

PMD:IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter corresponds to the variable Global_PMD_signal_detect as defined in 185.5.4. When Global_PMD_signal_detect is 1, SIGNAL_OK shall be assigned the value OK. When Global_PMD_signal_detect is 0, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_i.indication parameter is undefined.

NOTE—SIGNAL_OK = OK does not guarantee that the rx_signal parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_OK = OK indication and still not meet the frame loss ratio defined in 185.2.

185.3.1.3.2 When generated

The 800GBASE-LR1 PMD generates the PMD:IS_SIGNAL.indication primitive whenever there is a change in the value of the SIGNAL_OK parameter.

185.3.1.3.3 Effect of receipt

The effect of receipt by the Inner FEC is defined in 184.5.1.

185.4 Delay constraints

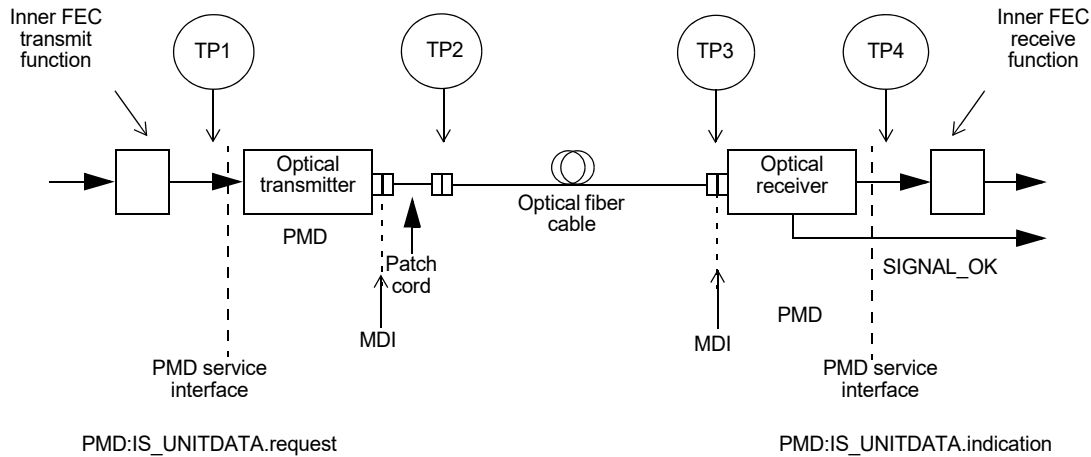
The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-LR1 PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

185.5 PMD functional specifications

The 800GBASE-LR1 PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

185.5.1 Block diagram

A block diagram for the PMD transmit/receive paths is shown in Figure 185–3 and a block diagram of the PMD is shown in Figure 185–4. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 185.8 and 185.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 185.7). Unless specified otherwise, all receiver measurements and tests defined in 185.8 are made at TP3.



For clarity, only one direction of transmission is shown

Figure 185-3—Block diagram for 800GBASE-LR1 transmit/receive paths

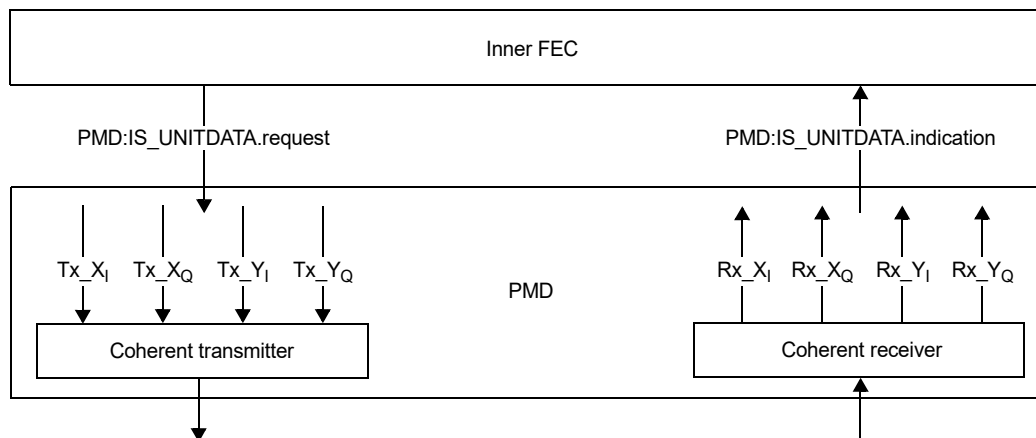


Figure 185-4—Block diagram of 800GBASE-LR1 PMD

185.5.2 PMD transmit function

The PMD transmit function shall convert the four analog streams (Tx_XI, Tx_XQ, Tx_YI, Tx_YQ) from the Inner FEC passed across the PMD service interface via the PMD:IS_UNITDATA.request primitive (see 185.3.1.1) into a single optical signal with orthogonal polarizations modulated using 16QAM and deliver them to the MDI, all according to the transmit optical specifications in this clause.

The analog signals use any of the mappings in Table 185–2.

Table 185–2—Allowed analog signals to symbol mappings

Mapping	Tx_X _I	Tx_X _Q	Tx_Y _I	Tx_Y _Q
1	X _I	X _Q	Y _I	Y _Q
2	X _Q	X _I	Y _Q	Y _I
3	X _I	X _Q	Y _Q	Y _I
4	X _Q	X _I	Y _I	Y _Q
5	Y _I	Y _Q	X _I	X _Q
6	Y _Q	Y _I	X _Q	X _I
7	Y _Q	Y _I	X _I	X _Q
8	Y _I	Y _Q	X _Q	X _I

185.5.3 PMD receive function

The PMD receive function shall convert the composite optical signal received from the MDI into four analog streams (Rx_X_I, Rx_X_Q, Rx_Y_I, Rx_Y_Q) passed across the PMD service interface to the Inner FEC via the PMD:IS_UNITDATA.indication primitive (see 185.3.1.2), all according to the receive optical specifications in this clause.

The four analog streams carry a combination of the transmitting Inner FEC Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

185.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of an optical signal. For the 800GBASE-LR1 PMD, the variable Global_PMD_signal_detect is set to PMD_signal_detect (see 185.5.5).

The state of the Global_PMD_signal_detect variable is conveyed to PMD client sublayers via the PMD service interface (see 185.3).

185.5.5 PMD signal detect function

The PMD signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input. A receiver shall set the variable PMD_signal_detect to either 0 or 1 according to the criteria in Table 185–6.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of PMD_signal_detect.

Implementations need to provide adequate margin between the input optical power level at which the PMD_signal_detect is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 185–3—Signal detection criteria

Receive conditions	PMD_signal_detect
Average optical power at TP3 ≤ -19 dBm	0
[The input signal is a compliant 800GBASE-LR1 signal] ^a AND [Optical power at TP3 $\geq$ average receive power (min) in Table 185–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

185.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

185.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows the optical transmitter to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off the optical transmitter so that it meets the requirements of the average launch power of the OFF transmitter in Table 185–5.

If a PMD fault is detected (see 185.5.8), then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter.

185.5.8 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault variable to one.

185.5.9 PMD transmit fault function (optional)

If the PMD has detected a local fault on the transmit path, the PMD shall set the PMD_transmit_fault variable to one.

185.5.10 PMD receive fault function (optional)

If the PMD has detected a local fault on the receive path, the PMD shall set the PMD_receive_fault variable to one.

185.6 PMD to MDI optical specifications

The operating range for the 800GBASE-LR1 PMD is defined in Table 185–4. A 800GBASE-LR1 compliant PMD operates over single mode fibers meeting the specifications in Table 185–9. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-LR1 PMD that could operate over 10 km would meet the operating range requirement of 2 m to 10 km).

Table 185–4—800GBASE-LR1 operating range

PMD type	Required operating range
800GBASE-LR1	2 m to 10 km

185.6.1 800GBASE-LR1 transmitter optical specifications

The 800GBASE-LR1 transmitter shall meet the specifications defined in Table 185–5 per the definitions in 185.8.

Table 185–5—800GBASE-LR1 transmit characteristics

Description	Value	Unit
Signaling rate (range)	123.6364 ± 50 ppm	GBd
Modulation format	DP-16QAM	—
Average launch power (max)	–6	dBm
Average launch power (min) for $ETCC \leq 1$ dB for $1 < ETCC \leq 3.4$ dB	–11.2 –12.2 + ETCC	dBm
Carrier frequency (range)	228.675 ± 20 GHz	THz
Power difference between X and Y polarizations (max)	1.5	dB
Skew between X and Y polarizations (max)	5	ps
Extended transmit constellation closure (ETCC) (max)	3.4	dB
Instantaneous I-Q offset per polarization (max)	–20	dB
Mean I-Q offset per polarization (max)	–26	dB
I-Q amplitude imbalance (mean)	1	dB
I-Q phase error magnitude (max)	5	deg
I-Q quadrature skew (max)	0.75	ps
Transmitter OSNR in a 12.5 GHz resolution bandwidth (min)	40	dB
Average launch power of OFF transmitter (max)	–20	dBm
Transmitter reflectance ^a (max)	–20	dB
RIN average (max)	–145	dB/Hz
RIN peak (max)	–140	dB/Hz
Laser linewidth (max)	1	MHz
Tx laser frequency slew rate: pre acquisition (max)	10	GHz/s
Tx laser frequency slew rate: post acquisition (max)	1	GHz/s

Table 185–5—800GBASE-LR1 transmit characteristics (*continued*)

Description	Value	Unit
Laser relative frequency tracking accuracy	± 0.9	GHz
Tx clock phase noise: total integrated random jitter (max)	0.015	UI _{RMS}
Tx clock phase noise: total periodic jitter (max)	0.03	UI _{pp}

^a Transmitter reflectance is defined looking into the transmitter.

185.6.2 800GBASE-LR1 receive optical specifications

The 800GBASE-LR1 receiver shall meet the specifications defined in Table 185–6 per the definitions in 185.8.

Table 185–6—800GBASE-LR1 receive characteristics

Description	Value	Unit
Signaling rate (range)	123.6364 ± 50 ppm	GBd
Modulation format	DP-16QAM	—
Carrier frequency (range)	228.675 ± 20 GHz	THz
Damage threshold ^a	–2	dBm
Average receive power tolerance (max)	–6	dBm
Average receive power tolerance (min) for $ETCC \leq 1$ dB for $1 < ETCC \leq 3.4$ dB	–17.5 –18.5 + ETCC	dBm
Receiver sensitivity (max) ^b for $ETCC \leq 1$ dB for $1 < ETCC \leq 3.4$ dB	–18 –19 + ETCC	dBm
Receiver reflectance (max)	20	dB
Frequency offset between received carrier and local oscillator (max)	40	GHz
Polarization dependent loss (max)	2	dB
Polarization rotation speed (max)	50	krad/s

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Receiver sensitivity is optional.

185.6.3 Illustrative link power budget

An illustrative power budget and penalties for 800GBASE-LR1 channels are shown in Table 185–7.

Table 185–7—800GBASE-LR1 illustrative link power budget

Parameter	Value	Unit
Power budget	6.8	dB
Operating distance	10	km
Channel insertion loss ^a	6.3	dB
Maximum discrete reflectance	–27	dB
Allocation for penalties ^b	0.5	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 185–4 and cabled optical fiber attenuation of 0.43 dB/km at 1311 nm plus an allocation for connection and splice loss given in 185.7.2.1.

^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

185.7 Optical channel characteristics

The fiber optic cabling model is shown in Figure 185–5.

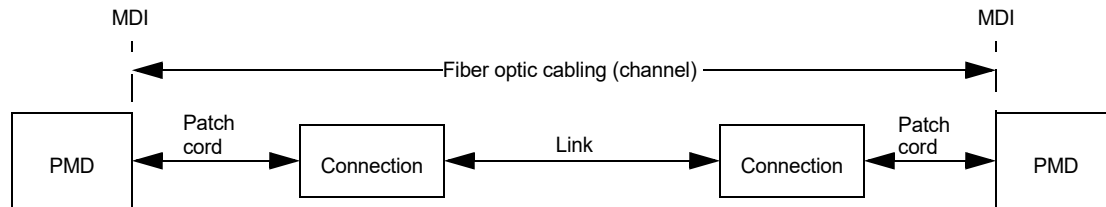


Figure 185–5—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 185–8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 185–8—Optical channel characteristics

Description	800GBASE-LR1	Unit
Operating distance (max)	10	km
Channel insertion loss ^{a,b} (max)	6.3	dB
Channel insertion loss (min)	0	dB
Positive dispersion (max)	10.1	ps/nm
Negative dispersion (min)	–12.3	ps/nm
Differential group delay (max) ^c	5	ps
Optical return loss (min)	24	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b At 1311 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_{max} is the maximum differential group delay that the system is required to tolerate.

185.7.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T G.652.B (dispersion unshifted), type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 185–9 where they differ.

Table 185–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.43 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² × km)

^a The 0.43 dB/km attenuation is provided for Outside Plant cable as defined in ITU-T G.695 Appendix I.

185.7.2 Optical fiber connection

An optical fiber connection, as shown in Figure 185–5, consists of a mated pair of optical connectors.

185.7.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 185–8 are met.

185.7.3 Medium Dependent Interface (MDI)

The 800GBASE-LR1 PMD is coupled to the fiber optic cabling via one fiber per direction at the MDI, being the interface between the PMD and the medium. Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 185.5.1, not at the MDI.

185.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

185.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 185–11 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 185–11 may be used to perform that test. The test patterns used in this clause are shown in Table 185–10.

Table 185–10—Test patterns

Pattern	Pattern description	Defined in
5	Scrambled idle test pattern encoded by the 800GBASE-LR1 Inner FEC	172.2.4.11, 184.4
7	Valid 800GBASE-R signal encoded by the 800GBASE-LR1 Inner FEC	172.2.4, 184.4
8	PRBS31 pattern encoded by 800GBASE-LR1 Inner FEC	184.4.3

Table 185–11—Mapping of parameters to test-patterns and related subclauses

Parameter	Pattern	Related subclause
Average channel output power	5 or 7	185.8.2
Carrier frequency (range)	5 or 7	185.8.3
Power difference between X and Y polarizations	5	185.8.4
Skew between X and Y polarizations	5	185.8.5
ETCC	5	185.8.6
Instantaneous I-Q offset per polarization	5	185.8.7
Mean I-Q offset per polarization	5	185.8.8
I-Q amplitude imbalance	5	185.8.9
I-Q phase error magnitude	5	185.8.10
I-Q quadrature skew	5	185.8.11
Transmitter in-band OSNR	5	185.8.12
Tx clock phase noise: total integrated random jitter	5	185.8.13
Tx clock phase noise: total periodic jitter	5	185.8.14
Average receive power tolerance	5, 7 or 8	185.8.15
Receiver sensitivity	5, 7 or 8	185.8.16

185.8.2 Average channel output power

The average channel output power shall be within the limits given in Table 185–5 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 185–11, per the test setup in Figure 53–6.

185.8.3 Carrier frequency (range)

The carrier frequency (range) shall be within the range given in Table 185–5 if measured per IEC 61280-1-3. The transmitter is modulated using the test pattern defined in Table 185–11.

185.8.4 Power difference between X and Y polarizations

The power difference between X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 185–5.

185.8.5 Skew between X and Y polarizations

The skew between the X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 185–5.

185.8.6 Extended transmitter constellation closure (ETCC)

ETCC is the metric used to define the quality of a 800 Gb/s DP-16QAM transmitter. The method and ETCC calculation are defined in 185.9.

The ETCC shall be within the limit given in Table 185–5 if measured using the methods specified in 185.9.

185.8.7 Instantaneous I-Q offset per polarization

The instantaneous I-Q offset per polarization shall be within the limit given in Table 185–5. The instantaneous I-Q offset per polarization is the maximum value per polarization and is calculated as $I\text{-Q offset} = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$ with a measurement interval of 1 μs .

185.8.8 Mean I-Q offset per polarization

The mean I-Q offset per polarization shall be within the limit given in Table 185–5. The mean I-Q offset per polarization is the mean value per polarization and is calculated as $I\text{-Q offset (mean)} = 10\log_{10}[(I_{\text{mean}}^2 + Q_{\text{mean}}^2)/P_{\text{signal}}]$.

185.8.9 I-Q amplitude imbalance

The I-Q amplitude imbalance is the amplitude difference in dB of the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 185–5.

185.8.10 I-Q phase error magnitude

The I-Q phase error magnitude is the phase difference of the in-phase component I and quadrature component Q of the signal, measured relative to local oscillator and shall be within the limit given in Table 185–5.

185.8.11 I-Q quadrature skew

The I-Q quadrature skew is the relative skew between the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 185–5.

185.8.12 Transmitter in-band OSNR

The transmitter in-band OSNR shall be within the limit given in Table 185–5. The transmitter in-band OSNR is defined as the ratio of the total signal power in transmitted channel to the highest noise power density (in a 12.5 GHz resolution bandwidth) in the range of the central frequency.

185.8.13 Tx clock phase noise: total integrated random jitter

The Tx clock phase noise: total integrated random jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213b, shall be within the limits given in Table 185–5.

185.8.14 Tx clock phase noise: total periodic jitter

The Tx clock phase noise: total periodic jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213c, shall be within the limits given in Table 185–5.

185.8.15 Average receive power tolerance

The average receive power tolerance defines the range of average receiver input power at TP3 over which the block error ratio requirement in 185.2 is met. Average receiver power tolerance is measured using the patterns listed in Table 185–11.

The conformance test signal meets the requirements for an 800GBASE-LR1 transmitter followed by a channel with the impairments: polarization dependent loss and polarization rotation speed as specified in Table 185–6, and differential group delay and dispersion as specified in Table 185–8.

The ETCC of the transmitter is measured according to 185.9. The ETCC is then used to calculate the minimum average receive power tolerance specified in Table 185–6.

The average receive power tolerance shall meet the limits given in Table 185–6.

185.8.16 Receiver sensitivity

Receiver sensitivity is an optional parameter defined as the lowest average receiver input power at TP3 with no link impairments at which the block error ratio requirement in 185.2 is met. Receiver sensitivity is measured using the patterns listed in Table 185–11.

The conformance test signal meets the requirements for an 800GBASE-LR1 transmitter followed by an optical attenuator.

The ETCC of the transmitter is measured according to 185.9. The ETCC is then used to calculate the maximum receiver sensitivity specified in Table 185–6.

185.9 Extended transmitter constellation closure (ETCC)

The ETCC is computed using the test setup and calculation defined in Annex 185A and the parameter values listed in Table 185–12 and Table 185–13.

185.9.1 Coherent detector front-end

Parameters for the coherent detector front-end shall meet the values listed in Table 185–12.

Table 185–12—Coherent detector front-end parameters

Description	Value	Unit
3 dB bandwidth (min)	65	GHz
Effective number of bits (<i>ENOB</i>)	5	bits
Oversampling ratio	1.25	—
Local oscillator linewidth (max)	100	kHz
BER_{ref}	1.1×10^{-2}	—

185.9.2 Calibrated front-end sampling

The post calibration residuals for the coherent detector front-end shall meet the values listed in Table 185–13.

Table 185–13—Coherent front-end post calibration residuals

Description	Value	Unit
X-Y gain error (max)	0.2	dB
Skew between X-Y polarizations (max)	0.5	ps
I-Q phase error for X (max)	2	degree
I-Q gain error for X (max)	0.2	dB
I-Q skew for X (max)	0.5	ps
I-Q phase error for Y (max)	2	degree
I-Q gain error for Y (max)	0.2	dB
I-Q skew for Y (max)	0.5	ps
Carrier frequency offset (max)	0.1	GHz

185.10 Safety, installation, environment, and labeling

185.10.1 General safety

All equipment subject to this clause shall conform to the general safety requirements in [J.2](#).

185.10.2 Laser safety

800GBASE-LR1 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹⁸

185.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

¹⁸A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

185.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-LR1 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

185.10.5 Electromagnetic emission

A system integrating a 800GBASE-LR1 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

185.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

185.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-LR1).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 185.10.2. Protocol implementation conformance statement (PICS) proforma for Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1.¹⁹

185.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 185–14 and Table 185–15.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 185–14 and Table 185–15.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

¹⁹*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Table 185–14—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Reset	185.5.6	1.0.15	45.2.1.1.1
Global PMD transmit disable	185.5.7	1.9.0	45.2.1.8.7

Table 185–15—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Fault	185.5.8	1.1.7	45.2.1.2.3
Transmit fault	185.5.9	1.8.11	45.2.1.7.4
Receive fault	185.5.10	1.8.10	45.2.1.7.5
Global PMD signal detect	185.5.5	1.10.0	45.2.1.9.7

185.12 Protocol implementation conformance statement (PICS) proforma for Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1²⁰

185.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

185.12.2 Identification

185.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

185.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 185, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

²⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

185.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
LR1	800GBASE-LR1	185.6	Device supports requirements for 800GBASE-LR1 PHY	O.1	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	185.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	185.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	185.4	Device conforms to delay constraints	M	Yes []
MD	MDIO capability	185.11	Registers and interface supported	O	Yes [] No []

185.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-LR1

185.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-LR1 PCS and PMA	185.1		M	Yes []
F2	Integration with management functions	185.1		O	Yes [] No []
F3	Error ratio allocation	185.2	Meets error ratio allocation specified in 185.2	M	Yes []
F4	Transmit function	185.5.2	Conveys analog data from PMD service interface to MDI	M	Yes []
F5	Receive function	185.5.3	Conveys analog data from PMD service interface to MDI	M	Yes []
F6	Global signal detect	185.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication (SIGNAL_DETECT)	O	Yes [] No []
F7	PMD reset function	185.5.5	Resets the PMD sublayer	O	Yes [] No []

185.12.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	185.11		MD:M	Yes [] N/A []
M2	PMD global transmit disable function	185.5.7	Disables the optical transmitter with the PMD_global_transmit_disable variable	MD:O	Yes [] No [] N/A []
M3	PMD fault function	185.5.8	Sets PMD_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
M4	PMD transmit fault function	185.5.9	Sets PMD_transmit_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
M5	PMD receive fault function	185.5.10	Sets PMD_receive_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []

185.12.4.3 PMD to MDI optical specifications for 800GBASE-LR1

Item	Feature	Subclause	Value/Comment	Status	Support
LR1	Transmitter meets specifications in Table 185–6	185.6.1	Per definitions in 185.8	M	Yes []
LR2	Receiver meets specifications in Table 185–7	185.6.2	Per definitions in 185.8	M	Yes []

185.12.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	185.8	2 m to 5 m in length	M	Yes []
OM2	Average channel output power	185.8.2	Per IEC 61280-1-1	M	Yes []
OM3	Carrier frequency (range)	185.8.3		M	Yes []
OM4	Power difference between X and Y polarizations	185.8.4		M	Yes []
OM5	Skew between X and Y polarizations	185.8.5		M	Yes []
OM6	ETCC	185.8.6		M	Yes []
OM7	Instantaneous I-Q offset per polarization	185.8.7		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM8	Mean I-Q offset per polarization	185.8.8		M	Yes []
OM9	I-Q amplitude imbalance	185.8.9		M	Yes []
OM10	I-Q phase error magnitude	185.8.10		M	Yes []
OM11	I-Q quadrature skew	185.8.11		M	Yes []
OM12	Transmitter in-band OSNR	185.8.12		M	Yes []
OM13	Tx clock phase noise: total integrated random jitter	185.8.13		M	Yes []
OM14	Tx clock phase noise: total periodic jitter (max)	185.8.14		M	Yes []
OM15	Average receive power	185.8.15		M	Yes []

185.12.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	185.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety	185.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Temperature, humidity, and handling	185.10.6	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

185.12.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	185.7.1	Meets the requirements in Table 185–9	M	Yes []
OC2	MDI requirements	185.7.3	Meets IEC 61753-1 and IEC 61753-021-02	INS:M	Yes [] N/A []

186. Forward Error Correction (FEC) sublayer and Physical Medium Attachment (PMA), type 800GBASE-ER1

186.1 Overview

186.1.1 Scope

This clause specifies the Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer for the Physical Layer implementations known as 800GBASE-ER1-20 and 800GBASE-ER1. The 800GBASE-ER1-20 and 800GBASE-ER1-PHYs (see 187.1) use the 800GBASE-R PCS defined in Clause 172 and the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA defined in this clause.

186.1.2 Relationship of 800GBASE-ER1 FEC and 800GBASE-ER1 PMA to other standards

Figure 186–1 shows the relationship of the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.

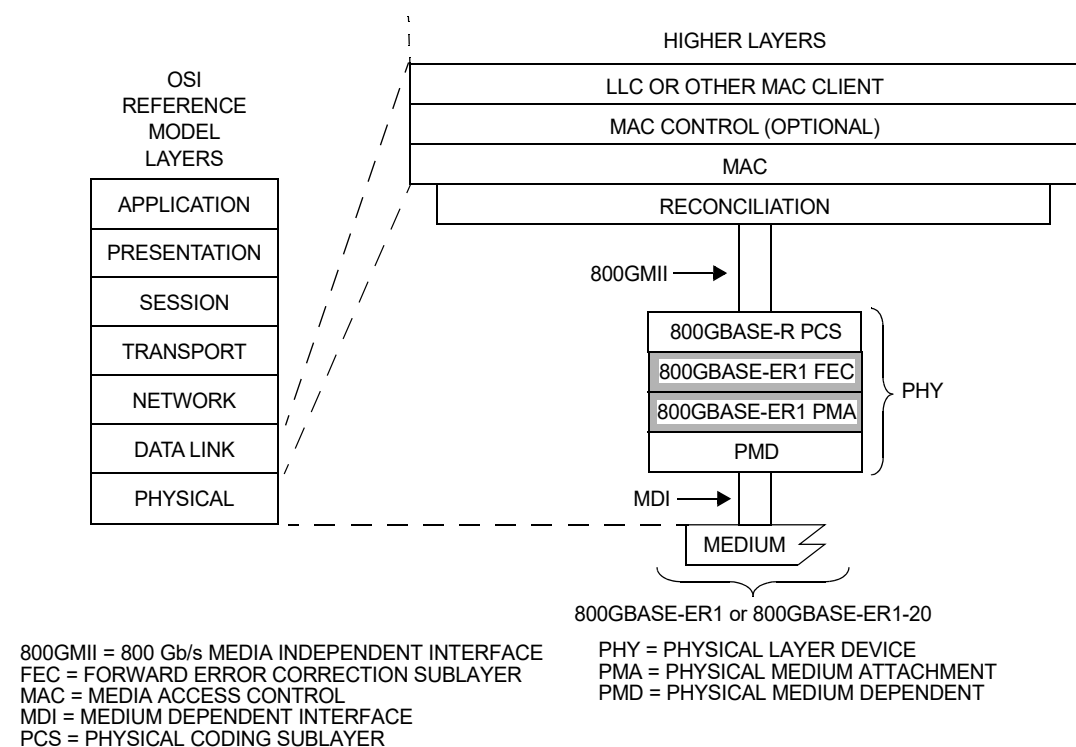


Figure 186–1—800GBASE-ER1 FEC and 800GBASE-ER1 PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

186.1.3 Summary of operation

In the transmit direction, the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers together provide RS-FEC decoding, mapping, ER1 FEC encoding, and generation of dual polarization, 16-state quadrature amplitude modulation (DP-16QAM) symbols at the 800GBASE-ER1-20 or 800GBASE-ER1 PMD service

interface. In the receive direction, the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers together decode DP-16QAM symbols from the 800GBASE-ER1-20 or 800GBASE-ER1 PMD service interface, perform ER1 FEC error detection, correction, demapping and decoding, and RS-FEC encoding at the 800GBASE-ER1 FEC service interface. A high-level block diagram of the 800GBASE-ER1 FEC and the 800GBASE-ER1 PMA is shown in Figure 186–2.

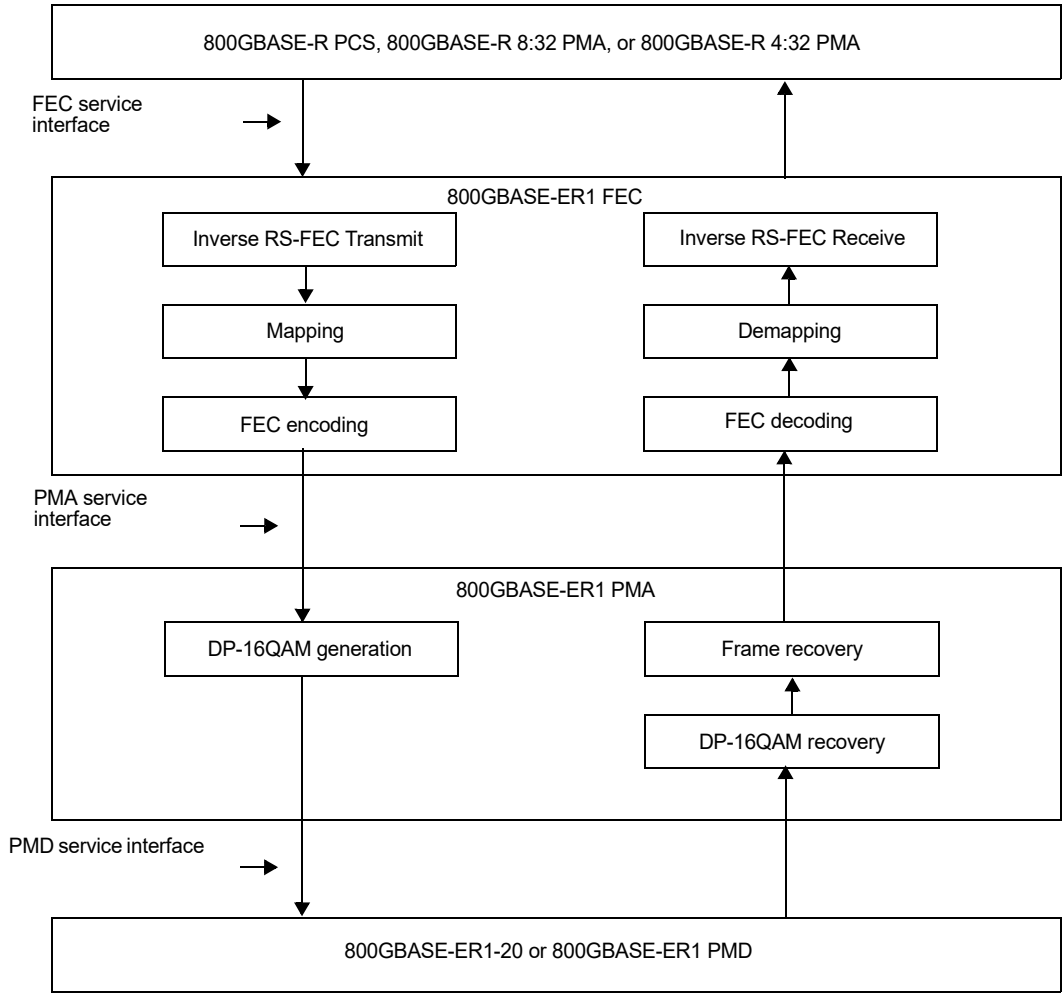


Figure 186–2—800GBASE-ER1 FEC and 800GBASE-ER1 PMA high level block diagram

186.2 Forward Error Correction sublayer (FEC), type 800GBASE-ER1

186.2.1 800GBASE-ER1 FEC overview

A functional block diagram of the 800GBASE-ER1 FEC sublayer is shown in Figure 186–3.

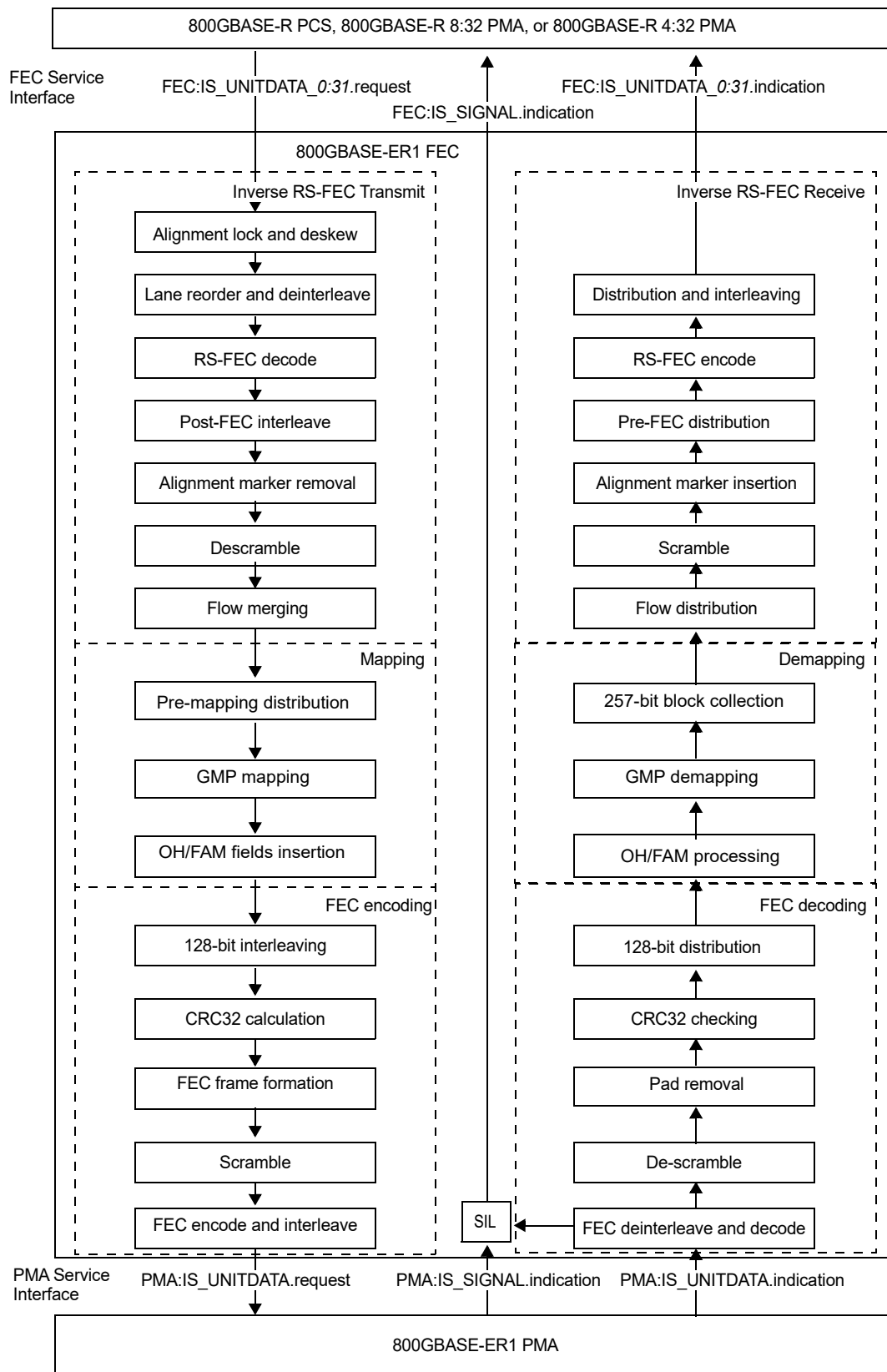


Figure 186-3—800GBASE-ER1 FEC functional block diagram

The dashed lines in Figure 186–3 relate to the high-level functions shown in Figure 186–2.

The 800GBASE-ER1 FEC sublayer is composed of the 800GBASE-ER1 FEC transmit and 800GBASE-ER1 FEC receive functions. The 800GBASE-ER1 FEC decodes the RS-FEC that the 800GBASE-R PCS uses and adds a stronger FEC to mitigate transmission errors over the medium.

The 800GBASE-ER1 FEC communicates with the client sublayer using 32 PCS lanes operating at a rate of 26.5625 Gb/s. For communication with the 800GBASE-ER1 PMA in the transmit direction, the 800GBASE-ER1 FEC provides ER1 FEC codewords (see 186.2.3.10) to the 800GBASE-ER1 PMA. For communication with the 800GBASE-ER1 PMA in the receive direction, the 800GBASE-ER1 FEC receives ER1 FEC codewords (see 186.3.2.2.1) from the 800GBASE-ER1 PMA.

The 800GBASE-ER1 FEC transmit function operates in normal mode or test-pattern mode.

When the transmit function is in normal mode, it receives 32 PCS lanes in the form of two flows with 16 lanes each. It aligns and reorders the 32 lanes to recover the two flows. For each flow, it decodes the RS(544,514) FEC, removes alignment markers, descrambles the blocks, and interleaves the lanes to form a serial stream of 257-bit blocks. The 257-bit blocks from the two flows are then merged on a 257-bit block basis to form a single stream of 257-bit blocks. Those blocks are distributed to 8 lanes, each of which is mapped into a tributary slot within the 800GBASE-ER1 frame (see 186.2.3.3). Frame alignment and other overhead elements are added to the 800GBASE-ER1 frames. The 8 lanes are interleaved and encoded with an extended BCH(256,239) code. The ER1 FEC codewords are sent to the 800GBASE-ER1 PMA via the PMA:IS_UNITDATA.request primitive as a stream of symbols.

When the transmit function is in test-pattern mode, the 800GBASE-ER1 FEC output is generated as specified in 186.2.3.12.

In the receive direction, the 800GBASE-ER1 FEC synchronization process continually monitors the PMA:IS_SIGNAL.indication primitive. When SIGNAL_OK parameter of the PMA:IS_SIGNAL.indication primitive is OK, the 800GBASE-ER1 FEC synchronization process accepts a stream of m -bit digitized DP-16QAM symbols via the PMA:IS_UNITDATA.indication primitive and forms a stream of ER1 FEC codewords. The ER1 FEC decoder generates error-corrected codewords from the incoming data stream on the 800GBASE-ER1 PMA service interface, which are then descrambled. The pad bits are removed and the CRC checking is performed before the 257-bit blocks are distributed to eight lanes. The 800GBASE-ER1 FEC then removes the frame alignment and overhead elements and passes the data to the demapper (see 186.2.4.7). The 8 lanes are then interleaved to a serial stream of 257-bit blocks and then distributed to 2 flows. In each flow, scrambling, alignment marker insertion, and RS-FEC encoding are performed, and finally distribution to 32 PCS lanes.

186.2.2 800GBASE-ER1 FEC service interface

The client sublayer above the 800GBASE-ER1 FEC (see Figure 186–2) is either the 800GBASE-R PCS, 800GBASE-R 8:32 PMA, or 800GBASE-R 4:32 PMA.

The service interface primitives are summarized as follows:

```
FEC:IS_UNITDATA_i.request(tx_symbol)
FEC:IS_UNITDATA_i.indication(rx_symbol)
FEC:IS_SIGNAL.indication(SIGNAL_OK)
```

The primitives are defined for $i = 0$ to 31. In the transmit direction, the 800GBASE-ER1 FEC receives 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, from the client sublayer above. In the receive direction, the 800GBASE-ER1 FEC sends 32 parallel bit streams, each at a nominal signaling rate of 26.5625 Gb/s, to the client sublayer.

The 800GBASE-ER1 FEC provides signal status information to the client sublayer above using the FEC:IS_SIGNAL.indication(SIGNAL_OK) primitive (see Figure 186–3). The SIGNAL_OK parameter is set to OK when fec_all_mfas_locked (see 186.4.2.1) is true and is set to FAIL when fec_all_mfas_locked is false. When SIGNAL_OK is FAIL, the rx_symbol parameters of the FEC:IS_UNITDATA_i.indication primitives are undefined.

186.2.3 Transmit function

186.2.3.1 Inverse RS-FEC Transmit

The inverse RS-FEC Transmit function decodes the RS(544,514) FEC that was added by the 800GBASE-R PCS and creates a single stream of 257-bit blocks to be mapped into the 800GBASE-ER1 frame.

186.2.3.1.1 Alignment lock and deskew

The Alignment lock and deskew function is identical to that specified in 172.2.5.1.

186.2.3.1.2 Lane reordering and deinterleaving

The Lane reordering and deinterleave function is identical to that specified in 172.2.5.2.

186.2.3.1.3 Reed-Solomon decoder

The Reed-Solomon decoder is identical to that specified in 172.2.5.3.

186.2.3.1.4 Post-FEC interleave

The Post-FEC interleave function is identical to that specified in 172.2.5.4.

186.2.3.1.5 Alignment marker removal

The alignment marker removal function is identical to that specified in 172.2.5.5.

The alignment marker removal function may provide the ability to record the location of the removed alignment markers using the OH field as described in 186.2.3.5.10. The presence of this option is indicated by the assertion of the FEC_alignment_marker_location_ability status variable (set to 1). When the option is provided, it is enabled by the assertion of the FEC_alignment_marker_location_enable control variable (set to 1).

The am_sf<2:0> bits are extracted from the alignment markers in the two flows, logically ORed together, and encoded into the 800GBASE-ER1 tributary frame as specified in 186.2.3.5.11.

186.2.3.1.6 Descrambler

The Descrambler function is identical to that specified in 172.2.5.6.

186.2.3.1.7 Flow merge

Following descrambling, the two flows of 257-bit blocks shall be merged in a round-robin fashion into a single stream of 257-bit blocks, starting with a block from flow 0.

186.2.3.2 Pre-mapping distribution

The stream of 257-bit blocks shall be distributed to 8 lanes in a round-robin manner. The GMP mapper (see 186.2.3.3), FAM and pad insertion (see 186.2.3.4), and OH insertion (see 186.2.3.5) functions are performed individually on each lane. The 8 lanes are then interleaved as specified in 186.2.3.6.

186.2.3.3 GMP mapper

GMP is a generic mechanism that uses a sigma/delta distribution algorithm to accommodate an arbitrary bit rate difference between a payload and the space in which it is carried. The principles of the GMP mapper used in the 800GBASE-ER1 FEC are described in ITU-T G.709 Annex D and ITU-T G.709.1 Annex B. Details of the overhead encoding for 800GBASE-ER1 are described in 186.2.3.5.

The GMP mapper shall insert the serialized stream of 257-bit blocks corresponding to each lane into the payload area of a 800GBASE-ER1 tributary frame. The 800GBASE-ER1 tributary frame is a structure with 5140 columns and 128 rows that contains 1280 bits of overhead followed by a payload area of 656 640 bits. The payload area is organized into a pad of 5 bits followed by 2555 blocks of 257 bits, as illustrated in Figure 186–4, with transmission order from top row to bottom row and from left to right within each row.

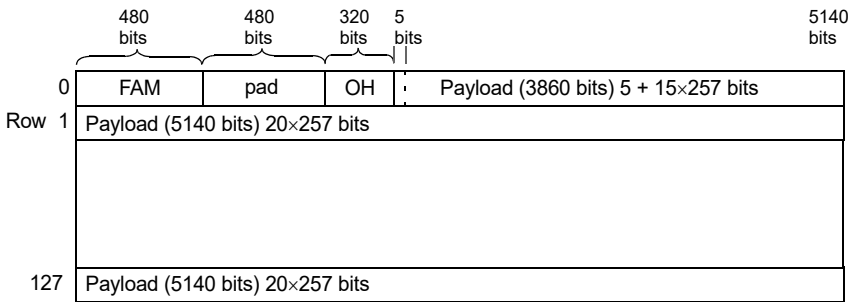


Figure 186–4—800GBASE-ER1 tributary frame structure

The information in the 800GBASE-ER1 tributary frame includes the following:

- 1) The first 480 bits of the 800GBASE-ER1 tributary frame carry the Frame Alignment Mechanism (FAM) field, defined in 186.2.3.4.1.
- 2) The next 480 bits are the pad field, defined in 186.2.3.4.2.
- 3) The next 320 bits carry the OH field, as described in 186.2.3.5.
- 4) When the transmit function is in normal mode, a 5-bit pad of all zeros is added after the OH field (i.e., as the first 5 bits of the payload area). This aligns the encoded MAC frames to 257-bit boundaries. The remaining bits, from bit 1286 of the first row to the end of the frame, are the payload area that consists of 2555 blocks of 257 bits.
- 5) When the transmit function is in test-pattern mode, the entire payload area is filled with the test pattern (i.e., there is no 5-bit pad following the OH field).

The eight 800GBASE-ER1 tributary frames share a common clock and the start of the frames are aligned.

For each of the eight 800GBASE-ER1 tributary frames, the payload area of a four frame multi-frame (identified by the two low-order bits of the multi-frame alignment signal, see 186.2.3.5.1) is divided into 10 220 GMP words of 257 bits. Each 257-bit GMP word is filled with either data bits (i.e., the logically

serialized stream of 257-bit blocks produced according to 186.2.3.2) or stuff bits, which are transmitted as zero and ignored on receipt.

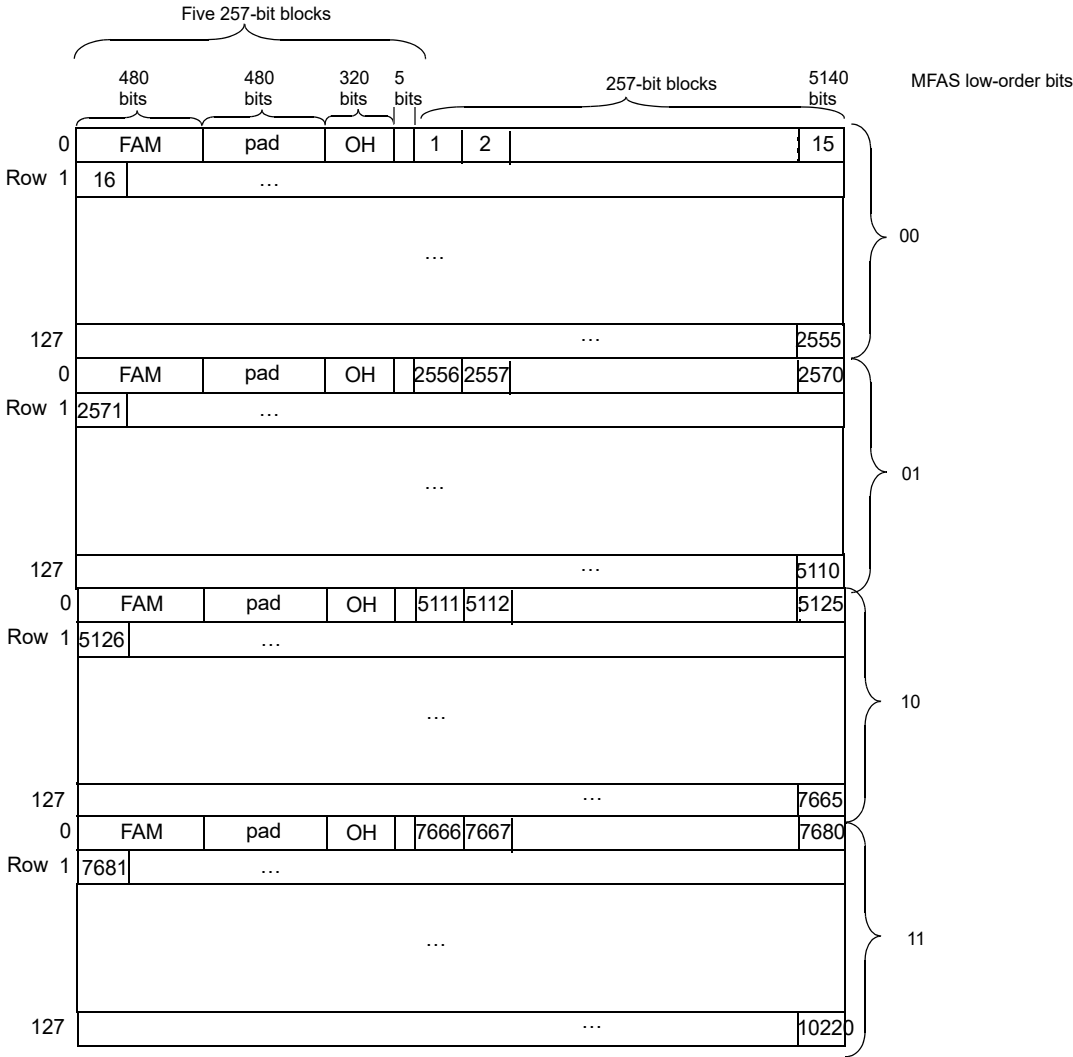


Figure 186–5—800GBASE-ER1 four-frame multi-frame for GMP mapping

The clocks for the encoded 800GBASE-R PCS data and the 800GBASE-ER1 tributary frame may be independent. The nominal rate required for the 257-bit encoded data in each lane is 100.385723 Gb/s $\pm$ 50 ppm (for an aggregate rate of 803.085785 Gb/s $\pm$ 50 ppm). The payload area of the 800GBASE-ER1 tributary frame has a capacity of 100.622438 Gb/s $\pm$ 20 ppm (for an aggregate capacity of 804.979507 Gb/s $\pm$ 20 ppm). The average number of GMP words carrying client data per multi-frame ranges from approximately 10 215.195 to approximately 10 216.625. Since the number of GMP words carrying client data in the multi-frame is always an integer, a given multi-frame will have between 10 215 and 10 217 GMP words that carry data. While the GMP mechanism is generic, the particular clock rates and tolerances for this application result in only three cases, allowing the positions of data and stuff GMP words to be pre-computed. The values in Table 186–1 shows the possible locations of GMP stuff words for each possible number of GMP words that are filled with client data.

Table 186–1—GMP stuff word locations in 800GBASE-ER1 multi-frame

Filled GMP words	Stuff GMP word locations (starting from 1)
10 217	1, 3407, 6814
10 216	1, 2556, 5111, 7666
10 215	1, 2045, 4089, 6133, 8177

186.2.3.4 Frame Alignment Mechanism (FAM) and pad fields

This clause specifies the FAM and pad fields of the 800GBASE-ER1 tributary frame.

186.2.3.4.1 FAM field

The FAM field is used to provide frame delineation for the 800GBASE-ER1 tributary frame. It is inserted before ER1 FEC encoding and removed after ER1 FEC decoding (see Figure 186–3). The content of the FAM field is specified in clause 9.1 of Recommendation ITU-T G.709.6, and consists of 16 bytes of 0x09, 16 bytes of 0xD7, and 28 bytes of 0x00. All bytes of this field are transmitted MSB first. The FAM field is protected by the ER1 FEC. FAM field alignment is processed post-ER1 FEC decode to locate the row corresponding to the start of the 800GBASE-ER1 tributary frame within the payload area of the ER1 FEC frame.

NOTE - this field is named “AM” in Recommendation ITU-T G.709.1 and in OIF-800ZR-01.0.

186.2.3.4.2 Pad

Pad is a 1920-bit all-zeros field. It is ignored upon receipt.

NOTE - this field is named “EOH” in Recommendation ITU-T G.709.6.

186.2.3.5 OH field

The 800GBASE-ER1 tributary frame overhead is present in each of the eight 800GBASE-ER1 tributary frames and is based on the FlexO overhead described in clause 9.2 of Recommendation ITU-T G.709.1. The overhead uses an eight-frame multi-frame, as shown in Figure 186–6 and described in 186.2.3.5.1 through 186.2.3.5.11. The frame overhead is based on the frame defined in subclause 4.3.3 of OIF-800ZR-01.0, which is itself a subset of what is defined in Recommendation ITU-T G.709.1. The 800GBASE-ER1 tributary frame adds additional overhead beyond what is specified in OIF-800ZR-01.0 and Recommendation ITU-T G.709.1 (see 186.2.3.5.10). The bit locations that are not labeled are unused, and are transmitted as zero and ignored by the receiver.

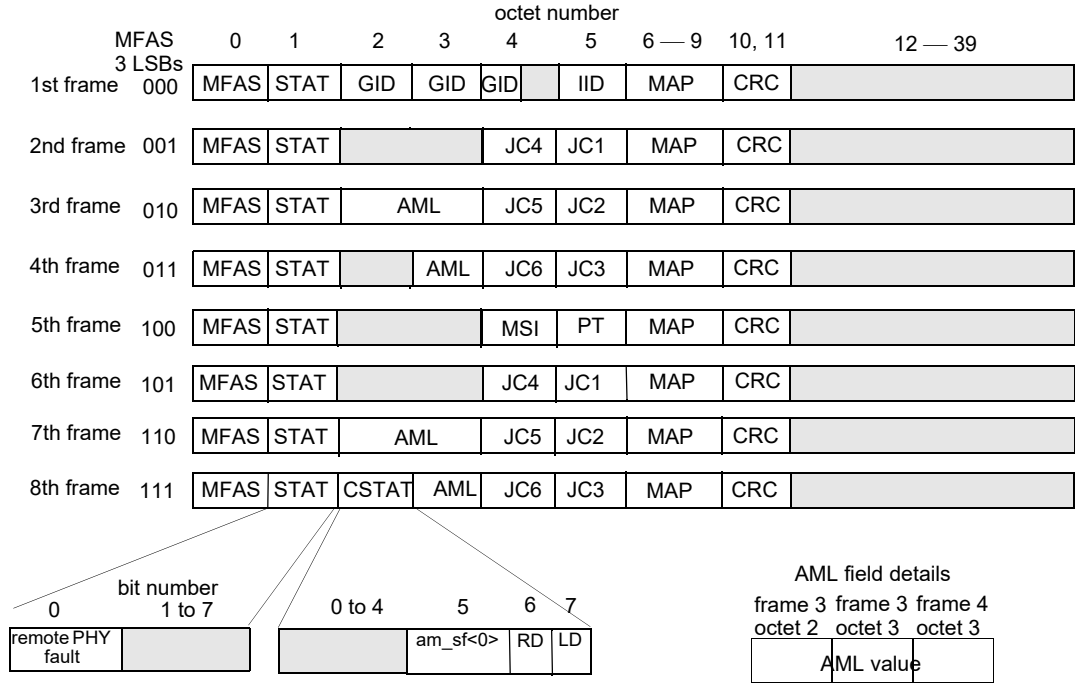


Figure 186–6—Contents of the OH field in eight frame multi-frame

186.2.3.5.1 Multi-frame alignment signal (MFAS)

The multi-frame alignment signal (MFAS) in the first octet of the overhead. As specified in clause 9.2.1 of Recommendation ITU-T G.709.1, it is an auto-wrapping 8-bit counter that is incremented in each 800GBASE-ER1 tributary frame. In the 800GBASE-ER1 FEC sublayer, two multi-frames are used to enable sharing of overhead bytes by information elements that do not need to be repeated every frame (GID, IID, PT, MSI, JC1-JC6, AML) or concatenation of bytes across multiple frames to form a larger field (MAP, AML). The Group Identification (see 186.2.3.5.3), Instance Identification (see 186.2.3.5.4), MAP (see 186.2.3.5.5), Payload Type (see 186.2.3.5.7), and Multiplex Structure Identifier (see 186.2.3.5.8) use an eight-frame multi-frame. The GMP overhead (see 186.2.3.5.9) and Alignment Marker Location overhead (see 186.2.3.5.10) uses a four-frame multi-frame.

186.2.3.5.2 Status octet (STAT)

STAT is present in every frame and provides status information for the 800GBASE-ER1-20 or 800GBASE-ER1 PHY.

The remote PHY fault indication bit is used to signal a remote 800GBASE-ER1 fault. It is set to one by the 800GBASE-ER1 FEC transmit function to indicate that the local 800GBASE-ER1 FEC receive function has not detected the location of the FAM field among the stream of 257-bit blocks delivered by the ER1 FEC decoder; otherwise it is set to zero. The remote PHY fault indication bit is in bit 0 of STAT.

186.2.3.5.3 Group ID

The Group ID is a 20-bit field that is specified in clause 9.2.2.1 of Recommendation ITU-T G.709.1. It is present in the first frame of the 8-frame multi-frame. The value is set to 1.

186.2.3.5.4 Instance ID (IID)

The Instance ID is an 8-bit field that is specified in clause 9.2.2.2 of Recommendation ITU-T G.709.1. It is present in the first frame of the 8-frame multi-frame. It is used for the purpose of identifying each 100G frame structure within the group. In the context of the 800GBASE-ER1 FEC sublayer, there are always eight 100G frame structures (i.e., eight 800GBASE-ER1 tributary frames), and the values 1 through 8 are used to identify them.

186.2.3.5.5 MAP

The MAP is a 256-bit field (spread across bytes 6 through 9 of the eight frame multi-frame) that is specified in clause 9.2.2.3 of Recommendation ITU-T G.709.1. It is used to indicate the IIDs that are part of the group. Since the IID values are fixed (see 186.2.3.5.4), the value of this field is also fixed. Byte 6 of the first frame has the value 0x7F, byte 8 of the first frame has the value 0x80, and all other bytes have the value 0x00.

186.2.3.5.6 CRC

The CRC is specified in clause 9.2.5 of Recommendation ITU-T G.709.1. It protects the integrity of bytes 2 through 10 of the overhead field.

186.2.3.5.7 Payload Type (PT)

The payload type is specified in clause 9.2.4.2 of Recommendation ITU-T G.709.1. When the 800GBASE-ER1 FEC sublayer is in normal mode, the value 0x42 is used. When the 800GBASE-ER1 FEC sublayer is in test-pattern mode, the value 0x41 is used.

186.2.3.5.8 Multiplex Structure Identifier (MSI)

The multiplex structure identifier is specified in clause 10.2.3 of Recommendation ITU-T G.709.1. When the 800GBASE-ER1 FEC is in normal mode, the value 0x80 is used. When the 800GBASE-ER1 FEC is in test-pattern mode, the value is 0x00.

186.2.3.5.9 GMP justification control overhead (JC1–JC6)

The justification control information is specified in clause 10.2.3 of Recommendation ITU-T G.709.1 and is carried in octets 4 and 5 of the second, third, and fourth frames of the four-frame multi-frame, as shown in Figure 186–6.

When the 800GBASE-ER1 FEC sublayer is in normal mode, the transmit path GMP mapper generates justification control octets (JC1–JC6) which are then used by the receive path GMP demapper to recover the 257-bit data blocks and re-time them to the same clock as the remote 800GBASE-R transmit clock. JC1–JC6 are carried once per 4-frame 800GBASE-ER1 tributary frame multi-frame. Because each 800GBASE-ER1 tributary is created from a common clock, the GMP value of the JC1–JC6 overhead is the same in all eight of the 800GBASE-ER1 tributary frames.

There are two parameters that are encoded into the overhead: $C_m(t)$ indicates the number of 257-bit GMP data words that will be transmitted during the next 800GBASE-ER1 FEC multi-frame, while $\Sigma C_{nD}(t)$ nominally indicates the running remainder. The long-term average value of $C_m(t) + \Sigma C_{nD}(t)$ represents the incoming serial stream rate as the number of information octets arriving at the GMP encoder per 800GBASE-ER1 tributary frame multi-frame. $C_m(t)$ is encoded in bits C1–C14 of JC1 and JC2, with the MSB in C1. $\Sigma C_{nD}(t)$ is encoded in bits D1–D7 of JC4 and JC5 with the MSB in D1. Refer to Annex B of Recommendation ITU-T G.709.1 for additional information on the encoding of JC1–JC6.

When the 800GBASE-ER1 FEC sublayer is in test-pattern mode, the justification control overhead bytes are transmitted as all zeros (there is no rate adaptation when in test-pattern mode; the test pattern is generated using the clock for the 800GBASE-ER1 tributary frame).

186.2.3.5.10 Alignment marker location control overhead (AML)

The three bytes of the AML field are used to encode information about the location of 800GBASE-R PCS alignment markers that were removed by the Inverse RS-FEC Tx function (see 186.2.3.1) within the stream of 257-bit blocks that are mapped into the 800GBASE-ER1 tributary multi-frame payload area, such that the 800GBASE-R PCS alignment markers can be re-inserted in the same location by 800GBASE-ER1 FEC receive function. The 3 bytes of the AML field (row 3, octets 2 and 3, and row 4, octet 3) in each multi-frame form a single 24-bit field, as shown in Figure 186–6.

When the alignment marker location feature is supported as indicated by the FEC status variable FEC_alignment_marker_location_ability (set to 1) and enabled by the FEC control variable FEC_alignment_marker_location_enable (set to 1), the 800GBASE-ER1 FEC maintains a counter of 257-bit blocks between consecutive sets of 800GBASE-R PCS alignment markers, with the first 257-bit block after the alignment markers in flow 0 being block 1. The AML field is encoded with the value of the counter for the first non-stuff 257-bit block that is mapped into the payload area of the 800GBASE-ER1 tributary multi-frame. For example, if the first 257-bit block in an 800GBASE-ER1 tributary frame is the ninth 257-bit block after 800GBASE-R PCS alignment markers were removed, the AML field would have the value 9, indicating that the 800GBASE-R PCS alignment markers were removed nine 257-bit blocks prior to that block. This is illustrated in Figure 186–7.

The upper part of the figure shows the stream of 257-bit blocks to be mapped into the 800GBASE-ER1 tributary multi-frame, along with the location of the alignment markers that were removed. The bottom part of the figure shows the 800GBASE-ER1 tributary multi-frame.

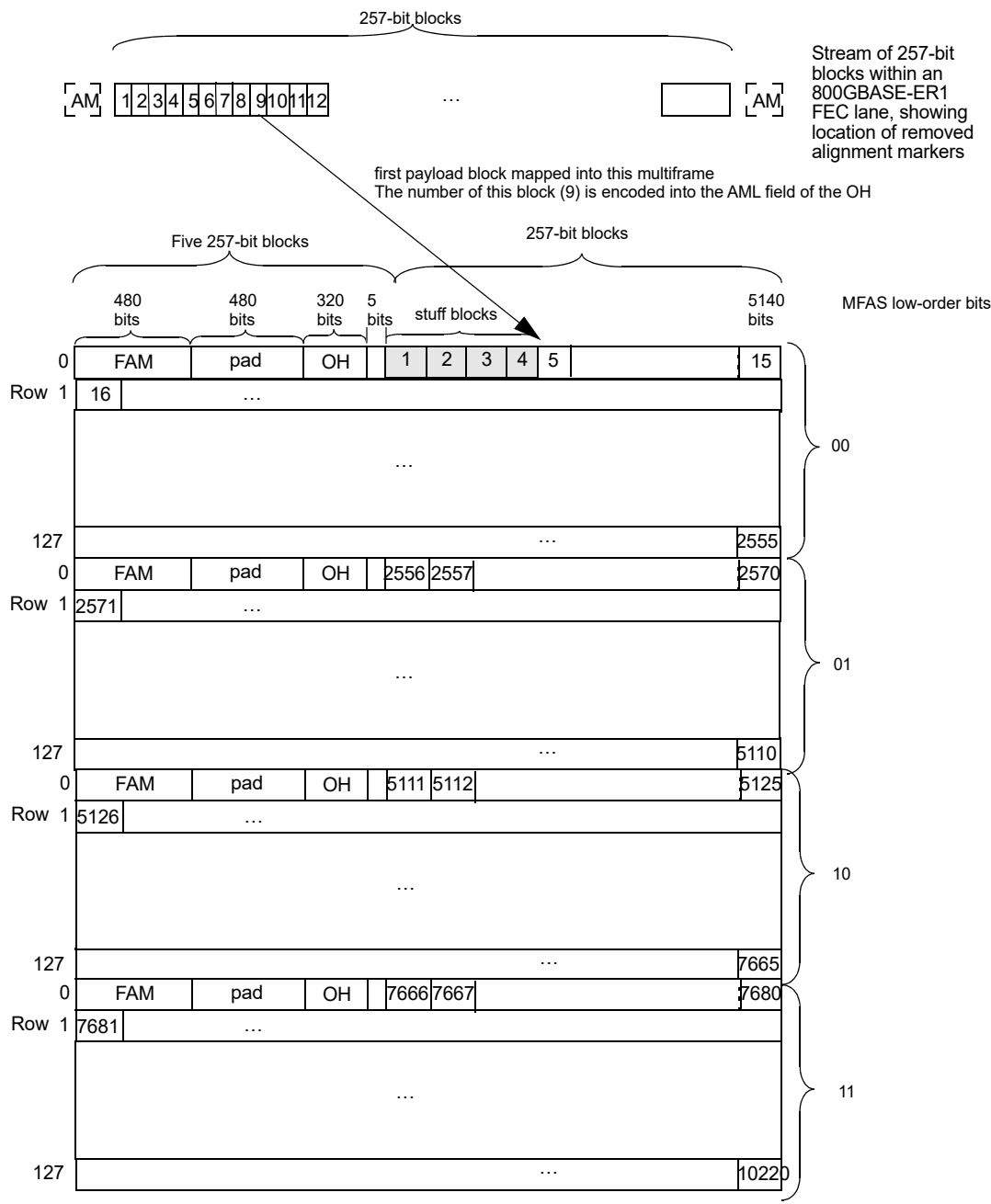


Figure 186-7— Example of AML field usage

If the alignment marker location feature is not supported or not enabled, the value of the AML field is set to all zeros.

186.2.3.5.11 Client status (STAT)

The client status information is present in frame 8 of the 8-frame multi-frame and is used to convey the information encoded in `am_sf<2:0>` across the 800GBASE-ER1 or 800GBASE-ER1-20 link according to the following rules:

- The $am_sf<2:0>$ bits from the two flows are logically ORed together
- The resulting $am_sf<0>$ signal is placed in CSTAT<6>
- The resulting $am_sf<1>$ signal is logically ORed with IFEC_degraded_SER and placed in CSTAT<8>
- The resulting $am_sf<2>$ signal is placed in CSTAT<7>

186.2.3.6 128-bit interleaving

The eight 800GBASE-ER1 tributary frames are interleaved on a 128-bit basis in a round-robin manner to form the 800GBASE-ER1 frame. This produces a structure with 128 rows of 41 120 bits. This structure may alternatively be represented as a frame with 512 rows of 10 280 bits. Figure 186–8 shows both of these representations of the 800GBASE-ER1 frame.

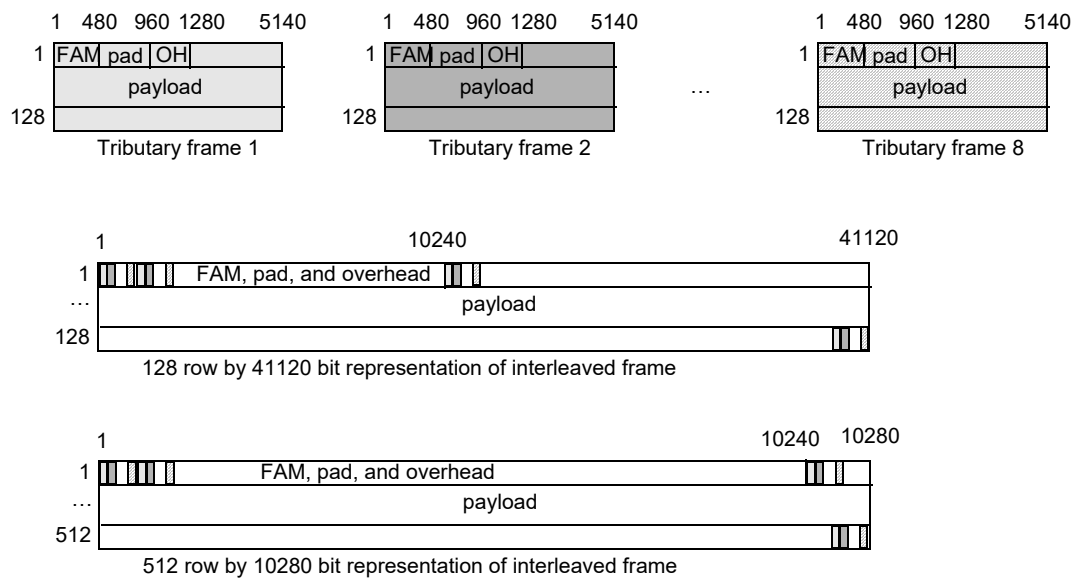


Figure 186–8—128-bit block interleaving of tributary frames

186.2.3.7 CRC32 insertion

A 32-bit cyclic redundancy code is calculated over each group of 41 120 bits (i.e., one row of the 128-row representation or four rows of the 512-row representation of the 800GBASE-ER1 frame) as specified in clause 12.4.1 of Recommendation ITU-T G.709.6. The CRC is inserted following the 41 120 bits.

186.2.3.8 ER1 FEC frame formation

The ER1 FEC frame is formed from 116 rows of the 512-row representation of the 800GBASE-ER1 frame (1 192 480 bits). Each group of four rows is extended with the CRC32 (see 186.2.3.7). The 29th group of four rows is further extended with a 64-bit pad. The FEC frame consists of 1 193 472 bits. The location of the 800GBASE-ER1 frames within the ER1 FEC frame will vary in a pattern that repeats every 128 ER1 FEC frames. Figure 186–9 illustrates an example of the ER1 FEC frame relative to an 800GBASE-ER1 frame.

NOTE - The ER1 FEC frame in this standard corresponds to the OBF84 structure defined in Recommendation ITU-T G.709.6.

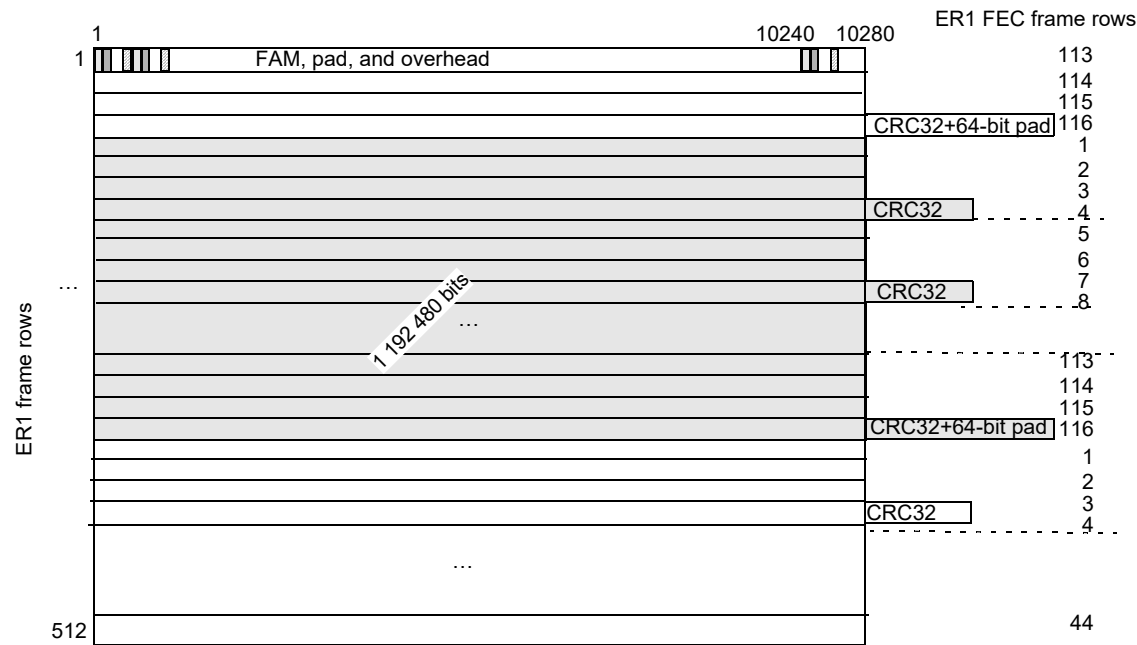


Figure 186–9—ER1 FEC frame formation

186.2.3.9 Scrambler

A frame-synchronous scrambler as specified in clause 12.4.3 of Recommendation ITU-T G.709.6 is applied to the ER1 FEC frame.

186.2.3.10 ER1 FEC encoder and interleaving

The scrambled ER1 FEC frame is encoded as described in clauses 12.4.4 and 12.4.5 of Recommendation ITU-T G.709.6, for the case of FlexO-8e-DO with parameters $k = 8$ and $j = 4$.

The output of the ER1 FEC encoders is interleaved as described in clause 12.4.6 of Recommendation ITU-T G.709.6.

186.2.3.11 Transmit bit ordering and distribution

The transmit bit ordering and distribution for the 800GBASE-ER1 FEC shall be as illustrated in Figure 186–10.

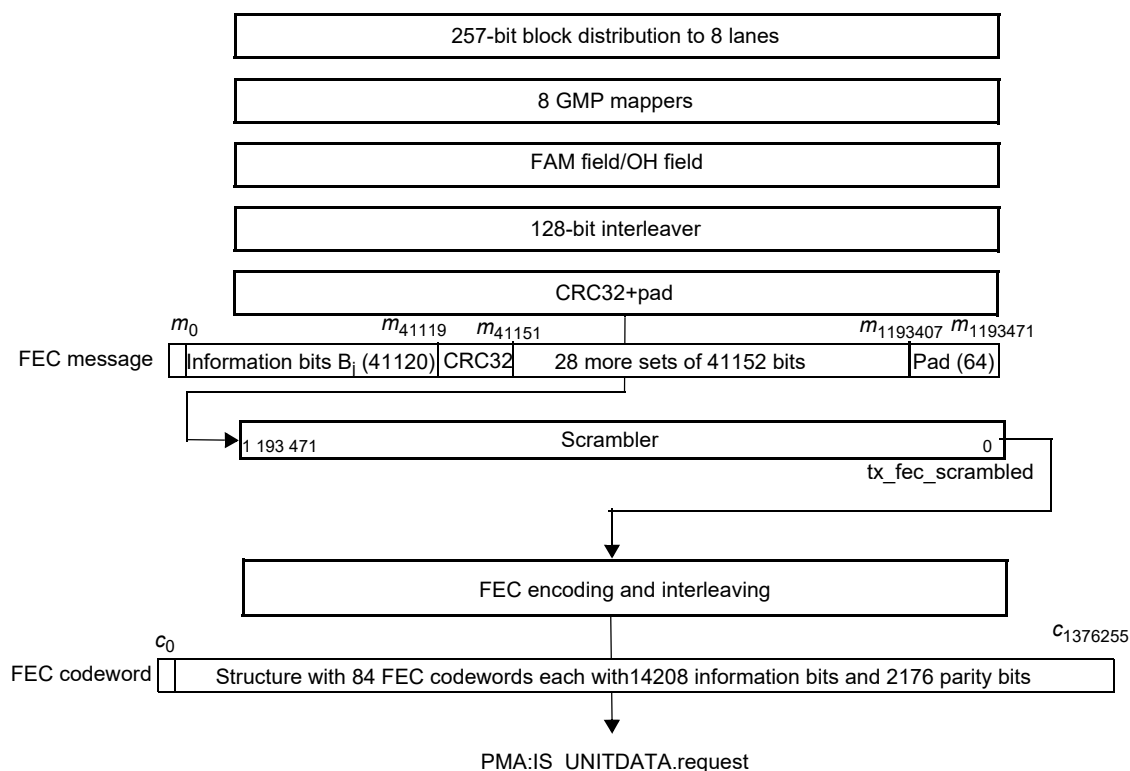


Figure 186-10—800GBASE-ER1 FEC transmit bit ordering and distribution

186.2.3.12 Test-pattern generator

The 800GBASE-ER1 FEC shall have the ability to generate a PRBS31 test pattern, with initial state of all ones. When test pattern generation is enabled by the FEC control variable `tx_test_mode` (set to 1), the test pattern is generated by the 800GBASE-ER1 FEC into each of the eight 800GBASE-ER1 tributary frames, filling the entire payload area as shown in Figure 186-11. The PRBS31 test pattern is as specified in 49.2.8.

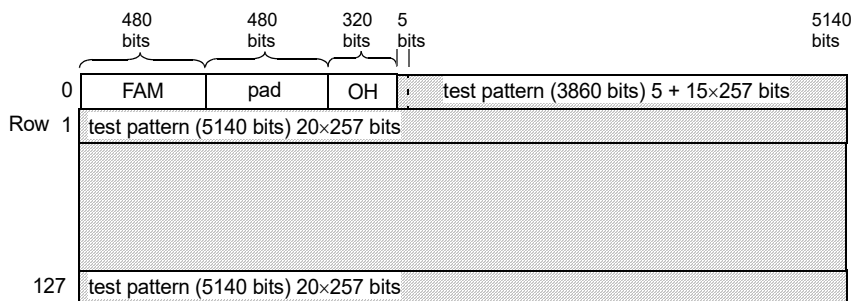


Figure 186-11—Location of test pattern in 800GBASE-ER1 tributary frame

186.2.4 Receive function

186.2.4.1 ER1 FEC deinterleaving and decoder

The interleaved bits of the ER1 FEC block are deinterleaved by reversing the interleaving process described in 186.2.3.10.

The ER1 FEC decoder shall extract blocks of 1 193 472 bits from an incoming ER1 FEC codeword of 1 376 256 bits. The incoming FEC codeword is formed from a digitized representation of 344 064 DP-16QAM symbols. The incoming DP-16QAM symbols are digitized to an m -bit resolution by the 800GBASE-ER1 PMA sublayer receive direction (see 186.3.4.1) and provided to the 800GBASE-ER1 FEC receive direction by the PMA_IS_UNITDATA.indication primitive in 186.3.2.

The ER1 FEC decoder is a soft decision decoder and so requires a higher resolution than 2 bits / 4 levels for each of the signals X_I , X_Q , Y_I , and Y_Q . The decoder evaluates the incoming codeword and determines the most likely message value.

The decoded blocks of 1 193 472 bits are passed to the descrambler.

The following counters shall be implemented:

FEC_corrected_cw_counter

The FEC_corrected_cw_counter is a 32-bit counter that counts once for each corrected FEC codeword processed when pma_alignment_valid is true. A corrected FEC codeword is a codeword that contained errors and was corrected (including FEC codewords that were mis-corrected).

FEC_uncorrected_cw_counter

The FEC_uncorrected_cw_counter is a 32-bit counter that counts once for each uncorrected FEC codeword processed when pma_alignment_valid is true. An uncorrected FEC codeword is a codeword that contains errors that were not corrected

FEC_total_bits_counter

The FEC_total_bits_counter is a 64-bit counter that counts once for each bit processed by the FEC decoder. This may be used together with the FEC_corrected_bits_counter as a measure of pre-FEC BER.

FEC_corrected_bits_counter

The FEC_corrected_bits_counter is a 64-bit counter that counts once for each bit corrected by the FEC decoder. This may be used together with the FEC_total_bits_counter to calculate pre-FEC BER.

186.2.4.2 Descrambler

The descrambler shall process sequences of 1 193 472 bits to reverse the effects of the scrambler using the polynomial given in 186.2.3.9.

186.2.4.3 Remove pad

The descrambled 1 193 472 bits are organized into the ER1 FEC frame described in 186.2.3.8, and the 64 pad bits are removed. The remaining 1 193 408 bits are the 116 rows of the 800GBASE-ER1 frame extended with a CRC32 every four rows. (see Figure 186–9).

186.2.4.4 CRC32 check and error marking

The CRC32 for each set of 4 rows is checked and removed. When errors are detected, the receiver determines which 257-bit blocks within the 4 rows protected by that CRC32 (41 120 bits) carry payload

information and replaces only those 257-bit blocks with transcoded error control blocks (EBLOCK_R), leaving any other 257-bit block unmodified. The details of how to determine which bits correspond to 257-bit blocks carrying payload information or overhead information from the 800GBASE-ER1 frame are beyond the scope of this standard.

The 800GBASE-ER1 FEC may provide the ability to signal the presence of a degraded received signal. The presence of this option is indicated by the assertion of the FEC_degraded_SER_ability variable (see 186.7). When the option is provided it is enabled by the assertion of the FEC_degraded_SER_enable variable (see 186.7).

When FEC_degraded_SER_enable is asserted, the 800GBASE-ER1 FEC counts the number of bit errors detected by CRC32 check. When the number of bit errors exceeds the threshold set in FEC_degraded_SER_activate_threshold (see 186.7), the FEC_degraded_SER bit (see 186.7) is set. At the end of each interval, if the number of bit errors detected is less than FEC_degraded_SER_deactivate_threshold, the FEC_degraded_SER bit is cleared. If either FEC_degraded_SER_ability or FEC_degraded_SER_enable is deasserted then the FEC_degraded_SER bit is cleared.

186.2.4.5 128-bit distribution

The set of 1 192 480 bits is already row-aligned with respect to the 800GBASE-ER1 frame. Groups of 128 bits are distributed to 8 lanes in a round-robin manner. The functions of FAM and OH field detection and removal and GMP demapping are performed on each of the 8 lanes.

186.2.4.6 FAM and OH fields detection and removal

The 800GBASE-ER1 frame floats within the ER1 FEC frame. The FAM field is the first 480 bits of each of the eight 800GBASE-ER1 tributary frames, and repeats every 1 320 980 bits. This field shall be aligned and locked using the state diagram in Figure 186–18.

Once FAM field lock has been acquired, the MFAS, status, and JC1–JC6 information is extracted from the OH field for use by the GMP demapper and for error signaling.

186.2.4.6.1 MFAS alignment

The purpose of MFAS for the 800GBASE-ER1 receiver is to identify which 800GBASE-ER1 tributary frames contain the PT, MSI, and JC1–JC6 octets which are needed for the GMP demapping process and the AML octets which are needed for the optional alignment marker location feature. Alignment to the four-frame 800GBASE-ER1 multi-frame is achieved via the two LSBs of the MFAS. Alignment to the eight-frame 800GBASE-ER1 multi-frame is achieved via the three LSBs of the MFAS.

The overhead bytes over the four 800GBASE-ER1 tributary frame multi-frame are shown in Figure 186–6.

186.2.4.6.2 Link status monitoring

The STAT octet present in every 800GBASE-ER1 tributary frame shall be monitored. It includes the remote PHY fault indication.

186.2.4.6.3 Payload type and multiplex structure identifier

The payload type (PT) is extracted from the fifth frame of the eight-frame multi-frame. The value indicates what signal is present in the payload area (800GE or test pattern). When test pattern generation is disabled by the FEC control variable tx_test_mode (set to 0), the value is expected to be 0x42. When test pattern generation is enabled by the FEC control variable tx_test_mode (set to 1), the value is expected to be 0x41.

The multiplex structure identifier (MSI) is extracted from the fifth frame of the eight-frame multi-frame. When test pattern generation is disabled by the FEC control variable `tx_test_mode` (set to 0), the value is expected to be 0x80. When test pattern generation is enabled by the FEC control variable `tx_test_mode` (set to 1), the value is expected to be 0x00.

186.2.4.6.4 GMP justification control overhead

The JC1-JC6 octets are found in the second, third, and fourth frames of the four-frame multi-frame. They are extracted and verified as described in Recommendation ITU-T G.709.1 Annex B, and used to control the operation of the GMP demapper (see 186.2.4.7).

186.2.4.6.5 Alignment Marker Location control overhead (AML)

When the alignment marker location feature is supported as indicated by the FEC status variable `FEC_alignment_marker_location_ability` (set to 1) and enabled by the FEC control variable `FEC_alignment_marker_location_enable` (set to 1), the contents of the AML field are extracted from the third and fourth frames of the four-frame multi-frame. A non-zero value indicates the offset between the first 257-bit block that is demapped from the frame and the previous set of 800GBASE-R PCS alignment markers. Since the number of 257-bit blocks between alignment markers is known, the AML information can be used to determine when the next set of 800GBASE-R PCS alignment markers will need to be inserted by the inverse RS-FEC function.

The value 0x000000 indicates that the transmit function did not encode any information about the location of alignment markers in this frame.

When alignment market location is not supported or not enabled, the AML field is ignored.

186.2.4.6.6 Client status (CSTAT)

The client status information is found in byte 2 of the eighth frame of the eight-frame multi-frame. The information is extracted and used to populate the `am_sf<2:0>` bits of the alignment markers that are subsequently inserted according to the following rules:

- The value in `CSTAT<6>` is replicated to `am_sf<0>` of the two flows
- The value in `CSTAT<7>` is replicated to `am_sf<2>` of the two flows
- The value in `CSTAT<8>` is logically ORed with `FEC_degraded_SER` and replicated to `am_sf<1>` of the two flows

186.2.4.6.7 Other overhead

The other overheads (GID, IID, MAP) do not need to be processed. They contain fixed values and may be ignored by the receiver.

186.2.4.7 GMP demapping

The GMP demapper shall decode the JC1-JC6 octets according to the procedures described in Recommendation ITU-T G.709.1 Annex B, recover the parameters $C_m(t)$ and $\Sigma C_{nD}(t)$, and use them to recover the 257-bit data blocks that were inserted into the frame by the GMP mapper and the signal stream rate.

If the payload type or multiplex structure identification overhead (see 186.2.4.6.3) did not match the expected value, the 257-bit data blocks extracted from the GMP frame are replaced with LF ordered sets.

In addition to being protected by the CRC32, the JC1-JC2 field information is also protected by limits on how the JC1-JC2 fields might change in successive multi-frames (see Table 186–1) and the coding technique for indicating these changes, which combine with the CRC8 in JC3 and the CRC4 in JC6 to provide error correction capability for bit and burst errors impacting JC1-JC6.

186.2.4.8 257-bit block collection

The 257-bit blocks from each of the eight lanes are interleaved to form a single stream of 257-bit blocks.

186.2.4.9 Inverse RS-FEC Receive

The inverse RS-FEC Receive function distributes the stream of 257-bit blocks to two flows and adds RS(544,514) FEC and alignment markers used by the 800GBASE-R PCS.

186.2.4.9.1 Flow distribution

The stream of 257-bit blocks is distributed to two flows on a round-robin basis. The first 257-bit block of each 800GBASE-ER1 tributary multi-frame is distributed to flow 0.

186.2.4.9.2 Scrambler

The Scrambler function shall be as specified in 172.2.4.5.

186.2.4.9.3 Alignment marker mapping and insertion

The Alignment marker mapping and insertion function shall be as specified in 172.2.4.6, with the following exceptions:

- The bits tx_am_sf<2:0> are assigned as described in 186.2.4.6.6.
- The location of alignment markers is determined by counting 257-bit blocks between alignment markers. If the alignment marker location feature is supported (FEC_alignment_marker_location_ability is set to 1) and is enabled by the FEC control variable FEC_alignment_marker_location_enable (set to 1), the location of the alignment markers is synchronized to information received in the AML field. The AMs are inserted at their original position (matching the position from before AMs were removed by far-end transmit function) as indicated by the RAML value. When an unexpected RAML value arrives, the previous position of the AM is maintained until 8 consecutive unexpected RAML values are received, after which the AM position is updated to the new position indicated by the RAML

186.2.4.9.4 Pre-FEC distribution

The Pre-FEC distribution function shall be as specified in 172.2.4.7.

186.2.4.9.5 Reed-Solomon encoder

The Reed-Solomon encoder function shall be as specified in 172.2.4.8.

186.2.4.9.6 Symbol distribution

The Symbol distribution function shall be as specified in 172.2.4.9.

186.3 Physical Medium Attachment (PMA) sublayer, type 800GBASE-ER1

186.3.1 Overview

186.3.1.1 Scope

This subclause specifies the Physical Medium Attachment (PMA) sublayer for the 800GBASE-ER1 and 800GBASE-ER1-20 PHYs. The 800GBASE-ER1 PMA allows the 800GBASE-ER1 FEC (see 186.2) to connect in a media independent way to the 800GBASE-ER1 or 800GBASE-ER1-20 PMD (specified in Clause 187).

The interfaces of the 800GBASE-ER1 PMA are defined in an abstract manner and do not imply any particular implementation.

186.3.1.2 Position of the 800GBASE-ER1 PMA in the 800GBASE-ER1 sublayers

Figure 186–1 shows the relationship of the 800GBASE-ER1 PMA sublayer (shown shaded) with other sublayers to the ISO Open System Interconnection (OSI) reference model.

186.3.1.3 Summary of functions

The purpose of the 800GBASE-ER1 PMA is to adapt the 800GBASE-ER1 FEC layer digital signals to and from the four analog signals comprising each DP-16QAM symbol, and to recover timing information from the received signal. The following is a summary of the principal functions of the 800GBASE-ER1 PMA in both the transmit and receive directions:

- Gray mapping and distribution of symbols for transmission to the two polarizations, X and Y.
- Insertion of the pilot sequence (PS) field into the stream of symbols on each polarization.
- Insertion of the frame alignment word (FAW), training sequence (TS), and reserved fields into the stream of symbols on each polarization.
- Encoding and digital-to-analog conversion of the in-phase and quadrature components of the X and Y symbol streams.
- Sampling of the incoming signals from the 800GBASE-ER1 PMD coherent receiver.
- Signal processing of the incoming samples including: equalization of the incoming samples for the effects of chromatic dispersion and other fixed impairments caused by the link and by the receive portion of the 800GBASE-ER1 PMD; equalization of the incoming samples for dynamic impairments including both state of polarization and polarization mode dispersion; carrier phase and frequency recovery; and symbol timing recovery and retiming of the samples to the signaling rate.
- Synchronization to the FAW followed by: I-Q offset evaluation and compensation; and polarization offset evaluation and compensation.
- Removal of the PS field from the received stream of symbols on each polarization.
- Removal of the FAW, TS, and reserved fields from the received stream of symbols on each polarization.
- Polarization combining.
- Providing signal status information in the receive direction.

A functional diagram of the 800GBASE-ER1 PMA sublayer is shown in Figure 186–12. PMA implementations may differ from the structure shown in the functional diagram as long as the observable behavior of the 800GBASE-ER1 PHY is compliant.

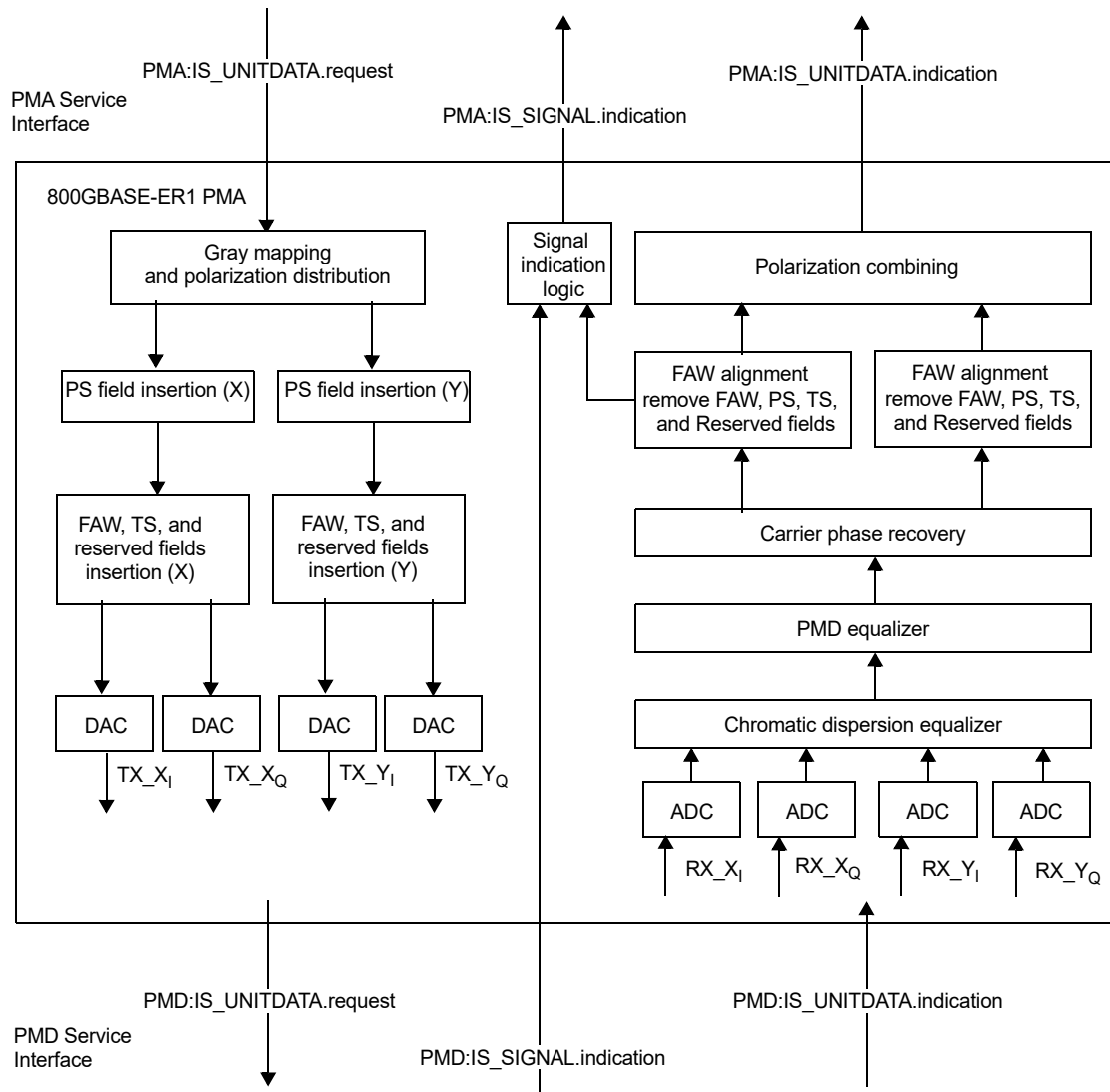


Figure 186-12—800GBASE-ER1 PMA functional block diagram

186.3.2 800GBASE-ER1 PMA service interface

The 800GBASE-ER1 PMA service interface uses the following primitives to exchange codewords and status information across the service interfaces:

PMA:IS_UNITDATA.request
PMA:IS_UNITDATA.indication
PMA:IS_SIGNAL.indication

186.3.2.1 PMA:IS_UNITDATA.request

The PMA:IS_UNITDATA.request primitive defines the transfer of a 1 376 256-bit ER1 FEC codeword from the 800GBASE-ER1 FEC to the 800GBASE-ER1 PMA via the tx_codeword parameter.

186.3.2.1.1 Semantics of the primitive

PMA:IS_UNITDATA.request(tx_codeword)

During transmission, the PMA_UNITDATA.request primitive conveys the FEC codeword in the tx_codeword parameter.

186.3.2.1.2 When generated

The 800GBASE-ER1 FEC generates a PMA_UNITDATA.request message for each 1 376 256-bit ER1 FEC codeword generated by the FEC encoder (see 186.2.3.10).

186.3.2.1.3 Effect of receipt

The 800GBASE-ER1 PMA passes each ER1 FEC codeword to the Gray mapping and polarization distribution function (see 186.3.3.1).

186.3.2.2 PMA:IS_UNITDATA.indication

The PMA:IS_UNITDATA.indication primitive defines the transfer of a ER1 FEC codeword from the 800GBASE-ER1 PMA to the 800GBASE-ER1 FEC via the rx_codeword parameter.

186.3.2.2.1 Semantics of the primitive

PMA_IS_UNITDATA.indication(rx_codeword)

The PMA_UNITDATA.indication primitive simultaneously conveys the best estimate of the in-phase and quadrature-phase of the X and Y polarization components of a DP-16QAM symbol spanning an ER1 FEC codeword.

The rx_codeword parameter conveys the four components of 344064 DP-16QAM symbols in $1\,376\,256 \times m$ bits as follows, where m is the implementation-dependent resolution of each component of the DP-16QAM symbol in bits and k is the DP-16QAM index with a range of 0 to 344064.

The DP-16AM symbols are mapped to rx_codeword as follows:

rx_codeword $\langle (k \times 4 + 1) \times m - 1 : k \times 4 \times m \rangle$ are the in-phase component of the X polarization (XI)
rx_codeword $\langle (k \times 4 + 2) \times m - 1 : (k \times 4 + 1) \times m \rangle$ are the quadrature component of the X polarization (XQ)
rx_codeword $\langle (k \times 4 + 3) \times m - 1 : (k \times 4 + 2) \times m \rangle$ are the in-phase component of the Y polarization (YI)
rx_codeword $\langle (k \times 4 + 4) \times m - 1 : (k \times 4 + 3) \times m \rangle$ are the quadrature component of the X polarization (YQ)

186.3.2.2.2 When generated

The 800GBASE-ER1 PMA generates a PMA:IS_UNITDATA.indicate message for each ER1 FEC codeword accumulated by the polarization combining and symbol deinterleaving function (see 186.3.4.4).

186.3.2.2.3 Effect of receipt

Upon receipt of each PMA:IS_UNITDATA.indication message, the 800GBASE-ER1 FEC passes each rx_codeword to the ER1 FEC decoder function (see 186.2.4.1).

186.3.2.3 PMA:IS_SIGNAL.indication

The PMA:IS_SIGNAL.indication primitive defines the transfer of the status of the 800GBASE-ER1 PMA receive function from the 800GBASE-ER1 PMA to the 800GBASE-ER1 FEC via the SIGNAL_OK parameter.

186.3.2.3.1 Semantics of the service primitive

PMA:IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter takes one of two values of the form:

- OK The 800GBASE-ER1 PMA receive function is not detecting a fault as defined in 186.3.4.5.
- FAIL The 800GBASE-ER1 PMA receive function is detecting a fault as defined in 186.3.4.5.

186.3.2.3.2 When generated

The 800GBASE-ER1 PMA generates a PMA:IS_SIGNAL.indication message whenever there is change in the value of the SIGNAL_OK parameter (see 186.3.4.5).

186.3.2.3.3 Effect of receipt

The effect of the SIGNAL_OK parameter when received by the 800GBASE-ER1 FEC is defined in 186.2.2.

186.3.3 800GBASE-ER1 PMA transmit function

The input of the 800GBASE-ER1 PMA transmit function receives 1 376 256-bit ER1 FEC codewords at 1/172 032 the DP-16QAM signaling rate from the 800GBASE-ER1 FEC sublayer. The output signals between the 800GBASE-ER1 PMA transmit function and 800GBASE-ER1 or 800GBASE-ER1-20 PMD carry analog signals representing the components of DP-16QAM symbols (namely X_I , X_Q , Y_I , and Y_Q).

186.3.3.1 Gray mapping and polarization distribution

The 800GBASE-ER1 PMA transmit function shall map sets of eight bits to DP-16QAM codewords, with a pair of bits mapped to each of the four components of the DP-16QAM symbol using Gray coding. Each ER1 FEC codeword $[c_0, c_1, \dots, c_{1376255}]$ passed across the PMA service interface from the ER1 FEC encoder in the tx_codeword parameter of the PMA:IS_UNITDATA.request primitive is mapped to 172 032 DP-16QAM symbols $[S_0, S_1, \dots, S_{172031}]$ where, for $i = 0$ to 172 031:

- (c_{8i}, c_{8i+2}) maps to the in-phase (I) component of the X-polarization of S_i
- (c_{8i+4}, c_{8i+6}) maps to the quadrature-phase (Q) component of the X-polarization of S_i
- (c_{8i+1}, c_{8i+3}) maps to the in-phase (I) component of the Y-polarization of S_i
- (c_{8i+5}, c_{8i+7}) maps to the quadrature-phase (Q) component of the Y-polarization of S_i

The complete mapping between the bits of ER1 FEC codewords and Gray-coded DP-16QAM symbols is detailed in Table 186–2

Table 186–2—In-phase (I) and quadrature-phase (Q) symbol values

$X: (c_{8i}, c_{8i+2}, c_{8i+4}, c_{8i+6})$ $Y: (c_{8i+1}, c_{8i+3}, c_{8i+5}, c_{8i+7})$	X_I Y_I	X_Q Y_Q
(0,0,0,0)	–3	–3
(0,0,0,1)	–3	–1
(0,0,1,0)	–3	3
(0,0,1,1)	–3	1
(0,1,0,0)	–1	–3
(0,1,0,1)	–1	–1
(0,1,1,0)	–1	3
(0,1,1,1)	–1	1
(1,0,0,0)	3	–3
(1,0,0,1)	3	–1
(1,0,1,0)	3	3
(1,0,1,1)	3	1
(1,1,0,0)	1	–3
(1,1,0,1)	1	–1
(1,1,1,0)	1	3
(1,1,1,1)	1	1

The 175 104 DP-16QAM symbols for the X and Y polarizations are distributed to two streams of 175 104 16QAM symbols for further processing.

186.3.3.2 800GBASE-ER1 PMA frame and multi-frame assembly

The stream of 16QAM symbols for each polarization is assembled into the 800GBASE-ER1 PMA frame and multi-frame. The frame structure consists of 7296 symbols that are organized into 114 rows of 64 symbols each. The multi-frame consists of 24 frames, for a total of 175 104 symbols per multi-frame. The details of the frame and multi-frame structures are defined in clause 11 and 12.1 of [ITU-T G.709.6], respectively, with the parameter $k = 8$.

Each 800GBASE-ER1 PMA frame contains 114 rows of 64 symbols. The first symbol of each row is a pilot symbol [P0, P1, ..., P113]. Each 800GBASE-ER1 PMA frame begins with an 11-symbol TS field carrying a training sequence (ts<0:10>). The first symbol of the training sequence is the pilot symbol P0.

The first 800GBASE-ER1 PMA frame in the multi-frame includes a 22-symbol multi-frame alignment field (mfas<0:21>), a 74-symbol reserved field (rsvd<0:73>), and 7076 payload symbols (S<0:7075>). All other frames in the multi-frame have 7172 payload symbols (63 in each row, following the pilot symbol).

Each 800GBASE-ER1 PMA multi-frame shall be serialized into a stream of 16QAM symbols for transmission. Relative to Figure 186–13, the 800GBASE-ER1 PMA frames shall be transmitted from top to bottom, and the symbols of each 800GBASE-ER1 PMA frame shall be transmitted from left to right. The assembly of symbols into 800GBASE-ER1 PMA frames and multi-frames is continuous.

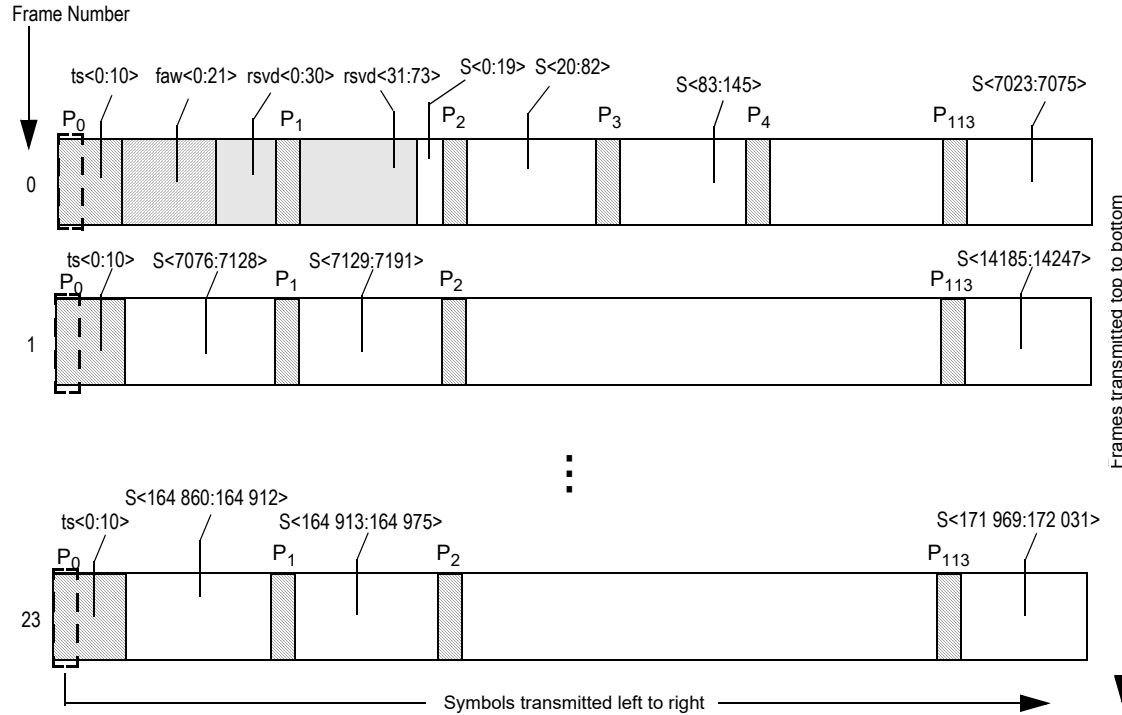


Figure 186–13—800GBASE-ER1 PMA Multi-frame and frame formats

186.3.3.3 Pilot sequence

The pilot sequence is a fixed PRBS10 sequence mapped to 16QAM symbols with different seed values for X and Y polarizations, as described in Annex A.1.2 of [ITU-T G.709.6]. The generator for each pilot sequence is shown in Figure 186–14, with the polynomial and seed values also indicated in Table 186–3. The seeds are selected so that the pilot and training sequences, when combined, are DC balanced. The seeds are also selected so that the first symbol in the training sequence is also the first symbol in the pilot sequence.

The generator is initialized using the seed at the start of every 800GBASE-ER1 PMA frame, so that the same 114 symbols, $[P_0, \dots, P_{113}]$ are inserted into every 800GBASE-ER1 PMA frame PS field of the same polarization. For each polarization X and Y, the generator produces 228 bits PRBS[227:0] that are mapped to 114 16QAM symbols, $[P_0, \dots, P_{113}]$ where for $i = 0$ to 113:

- PRBS[2*i*] maps to the in-phase (I) component of the 16QAM symbol $[P_i]$ for the respective polarization
- PRBS[2*i*+1] maps to the quadrature-phase (Q) component of the 16QAM symbol $[P_i]$ for the respective polarization

and where,

- 0 maps to -3 for the respective 16QAM symbol component
- 1 maps to $+3$ for the respective 16QAM symbol component

The complete pilot sequence is shown in Table 186–4.

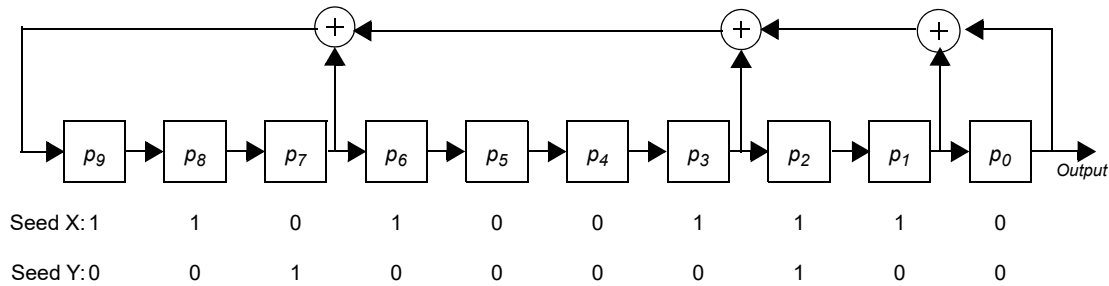


Figure 186–14—Pilot seed and sequence

Table 186–3—Pilot sequence generator polynomial and seed values

Generator polynomial	Seed for X-polarization	Seed for Y-polarization
$x^{10} + x^7 + x^3 + x + 1$	0x34E	0x084

Table 186–4—Pilot sequence

Index	X _I	X _Q	Y _I	Y _Q	Index	X _I	X _Q	Y _I	Y _Q	Index	X _I	X _Q	Y _I	Y _Q	Index	X _I	X _Q	Y _I	Y _Q
0	–3	3	–3	–3	29	3	–3	–3	3	58	3	–3	–3	3	87	–3	3	–3	–3
1	3	3	3	–3	30	3	3	3	3	59	3	–3	–3	3	88	3	3	3	3
2	–3	–3	–3	–3	31	3	–3	3	3	60	–3	–3	–3	–3	89	3	–3	3	3
3	3	–3	–3	3	32	–3	3	–3	–3	61	3	3	–3	–3	90	3	–3	3	–3
4	3	3	–3	–3	33	–3	–3	–3	3	62	3	–3	3	3	91	–3	–3	3	3
5	–3	3	3	3	34	3	–3	–3	3	63	–3	–3	–3	–3	92	–3	3	3	3
6	3	–3	3	3	35	–3	3	–3	–3	64	3	3	–3	3	93	–3	3	–3	–3
7	3	3	–3	3	36	–3	–3	3	3	65	3	3	–3	3	94	–3	3	3	3
8	–3	–3	–3	–3	37	–3	–3	3	3	66	3	3	–3	3	95	–3	–3	–3	3
9	–3	–3	–3	–3	38	–3	3	3	–3	67	–3	–3	3	3	96	–3	–3	–3	–3
10	3	3	3	–3	39	–3	–3	–3	3	68	–3	–3	–3	3	97	–3	3	3	–3
11	–3	–3	3	–3	40	–3	3	3	–3	69	3	3	3	3	98	–3	3	3	–3
12	3	3	3	–3	41	–3	–3	–3	–3	70	3	3	3	3	99	–3	3	3	–3
13	–3	–3	–3	–3	42	–3	–3	–3	–3	71	3	–3	3	3	100	3	–3	3	3
14	–3	3	–3	–3	43	3	3	–3	3	72	3	–3	–3	3	101	–3	3	–3	3
15	–3	–3	3	–3	44	3	3	3	–3	73	3	3	3	3	102	–3	3	–3	–3
16	–3	3	3	–3	45	–3	–3	–3	3	74	3	–3	–3	–3	103	3	–3	–3	–3
17	–3	3	3	3	46	3	3	–3	–3	75	3	–3	–3	–3	104	–3	3	–3	3

Table 186–4—Pilot sequence

18	−3	−3	−3	−3	47	3	3	−3	3	76	3	3	3	−3	105	−3	−3	3	3
19	−3	3	3	−3	48	−3	3	3	−3	77	−3	3	−3	3	106	3	−3	−3	−3
20	3	−3	3	3	49	3	−3	3	3	78	3	3	−3	3	107	3	3	3	−3
21	3	−3	−3	−3	50	−3	3	3	−3	79	−3	3	3	3	108	−3	3	3	−3
22	3	−3	3	−3	51	3	−3	−3	−3	80	3	3	3	−3	109	−3	−3	−3	3
23	3	−3	−3	3	52	−3	3	3	3	81	−3	3	3	3	110	3	−3	−3	−3
24	3	−3	−3	3	53	3	−3	−3	3	82	3	−3	−3	3	111	−3	−3	3	−3
25	3	3	3	−3	54	3	−3	3	−3	83	−3	3	3	3	112	3	3	3	−3
26	3	3	3	−3	55	3	3	−3	−3	84	−3	−3	3	−3	113	3	3	−3	3
27	−3	3	−3	3	56	−3	−3	3	3	85	−3	−3	3	−3					
28	−3	−3	−3	−3	57	−3	3	3	3	86	3	−3	3	3					

186.3.3.4 Training sequence

An 11-symbol training sequence as defined in Annex A.1.1 of [ITU-T G.709.6] is added at the beginning (transmitted first) of each 800GBASE-ER1 PMA multi-frame. The training sequences for X and Y polarizations are different in order to aid in identification by receivers. The first symbol of the training sequence is processed as part of the pilot sequence. It is made up of the outer four points of the 16QAM constellation symbol values and are listed in Table 186–5.

Table 186–5—Training sequences

Index	X _I	X _Q	Y _I	Y _Q
0	−3	3	−3	−3
1	3	3	−3	−3
2	−3	3	3	−3
3	3	3	−3	3
4	−3	−3	−3	3
5	3	3	3	3
6	−3	−3	−3	−3
7	−3	−3	−3	3
8	3	3	3	−3
9	3	−3	3	3
10	3	−3	3	−3

186.3.3.5 Multi-frame alignment word (FAW) and Reserved

The 22-symbol multi-frame alignment word defined in Annex A.1.3 of [ITU-T G.709.6] is inserted in the first frame of each multi-frame. The values for each polarization are listed in terms of constellation points in Table 186–6. It is made up of the outer four points of the 16QAM constellation symbol values and designed for DC balance. The receive function aligns to the FAW field in order to identify the 800GBASE-ER1 PMA multi-frame boundaries and for skew and skew variation removal between polarizations.

Following the FAW, a reserved field of 74 symbols is inserted. The value of the reserved symbols should be randomized, but is not otherwise specified.

Table 186–6—Multi-frame alignment word symbols

Index	X _I	X _Q	Y _I	Y _Q
0	3	−3	3	3
1	3	3	−3	3
2	3	3	−3	−3
3	3	3	−3	3
4	3	−3	3	−3
5	3	−3	3	3
6	−3	−3	3	−3
7	3	3	3	−3
8	−3	−3	−3	−3
9	−3	3	3	−3
10	−3	3	3	3
11	3	−3	−3	3
12	−3	−3	−3	3
13	−3	−3	3	3
14	−3	3	−3	−3
15	3	3	3	3
16	−3	−3	−3	−3
17	3	−3	−3	3
18	−3	3	3	−3
19	3	3	−3	−3
20	−3	−3	3	−3
21	−3	3	−3	3

186.3.3.6 Analog signal generation

The two streams of symbols, one for each polarization, shall be converted to four analog signals, two for each polarization: X_I , X_Q , Y_I , and Y_Q . The analog signals are passed to the 800GBASE-ER1 or 800GBASE-ER1-20 PMD sublayer using the tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq parameters of the PMD:IS_UNITDATA.request primitive.

186.3.3.7 Transmit bit ordering and distribution

The transmit bit ordering and distribution for the 800GBASE-ER1 PMA are illustrated in Figure 186–15.

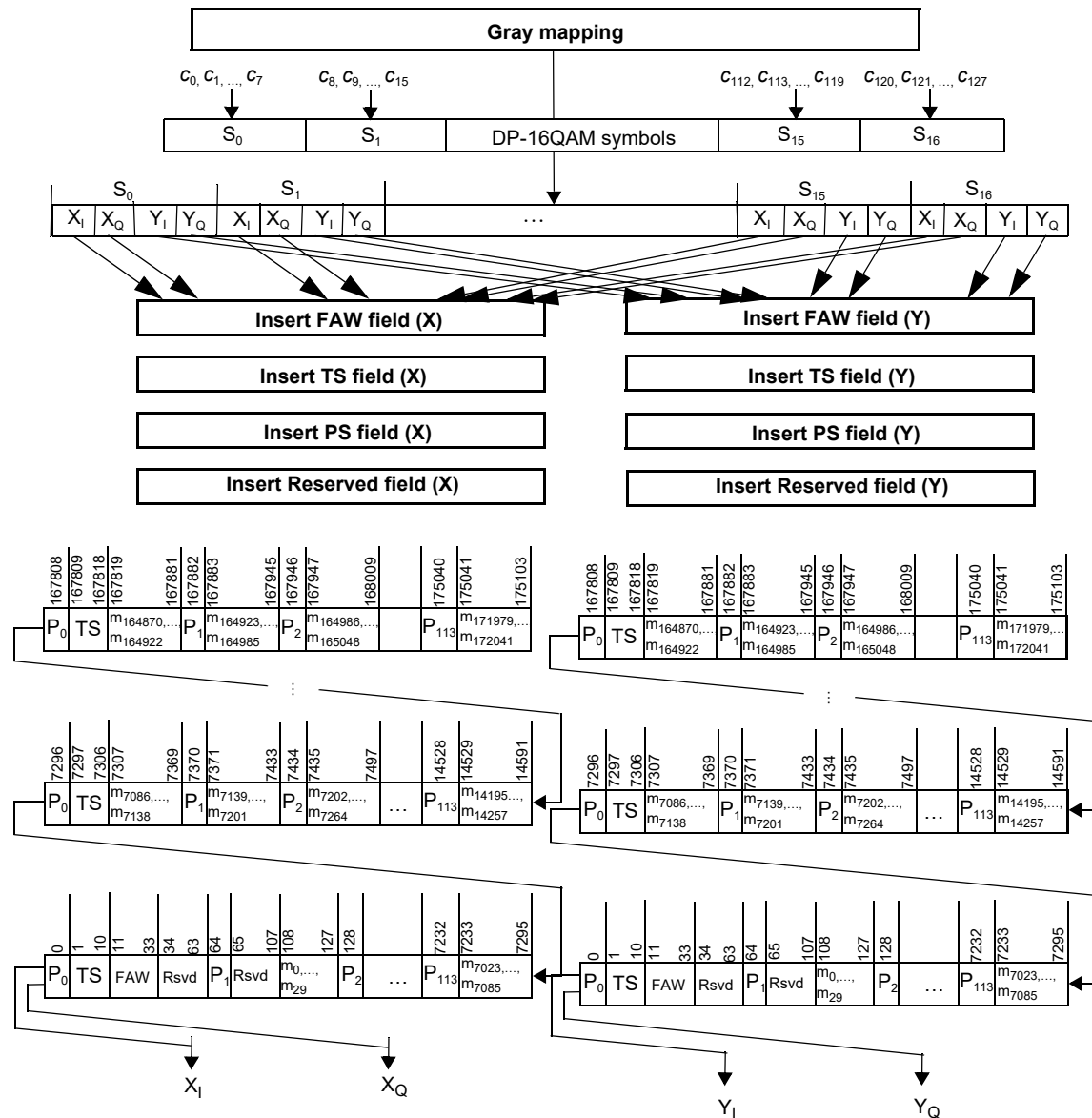


Figure 186–15—800GBASE-ER1 PMA transmit bit ordering and distribution

186.3.4 800GBASE-ER1 PMA receive function

The input of the 800GBASE-ER1 PMA receive function receives analog signals representing the components of DP-16QAM symbols (namely X_I , X_Q , Y_I , and Y_Q) from the 800GBASE-ER1 or 800GBASE-ER1-20 PMD.

The output of the 800GBASE-ER1 PMA receive function provides $1\ 376\ 256 \times m$ bits representing the ER1 FEC decoder input at $1/172\ 032$ the DP-16QAM signaling rate to the 800GBASE-ER1 FEC.

186.3.4.1 Receive signal sampling

Four analog signals RX_X_I , RX_X_Q , RX_Y_I , and RX_Y_Q are received from 800GBASE-ER1 or 800GBASE-ER1-20 PMD using the `rx_signal_xi`, `rx_signal_xq`, `rx_signal_yi`, `rx_signal_yq` parameters of the `PMD:IS_UNITDATA.indication` primitive. Each parameter of the `PMD:IS_UNITDATA.indication` primitive shall be sampled at a sampling rate equal to or larger than the 800GBASE-ER1 signaling rate. The resulting samples are forwarded to the receive signal processing block.

The details of the sampling, including any quantization and the chosen sampling rate are implementation specific.

186.3.4.2 Receive signal processing

The four incoming streams of samples shall be processed and converted into two streams of 16QAM symbols representing the X and Y polarizations of the DP-16QAM symbols. The 16QAM symbols should be sampled with more than two bits per dimension, in order to enable the ER1 FEC decoder to correct errors and recover the bits from the symbols based on the mapping in Table 186–2.

Functions within the signal processing block include, but are not limited to:

- Re-timing and re-sampling to one sample per symbol after applying an anti-aliasing filter with a –3 dB bandwidth of 0.5 times the signaling rate to the incoming samples.
- Equalization of signal distortion caused by chromatic dispersion, group delay and polarization mode dispersion in the channel.
- Clock phase recovery.
- Frequency offset estimation and removal assuming a constant frequency offset over a given block size N which is implementation specific.
- Carrier phase recovery.
- Identification of the X and Y polarizations and I and Q component of each polarization.
- I-Q offset evaluation and compensation.

The details of the receive signal processing are implementation specific. A PMD in combination with the PMA and ER1 FEC is required to meet the CRC error ratio specifications in 187.2.

186.3.4.3 FAW, TS, Reserved, and PS fields removal

The 800GBASE-ER1 PMA receive path attains lock to the 22-symbol FAW field that is transmitted on each of the two transmission polarizations on the in-phase and quadrature-phase signals. The FAW field appears in the first frame of each 24-frame multi-frame as shown in Figure 186–13. After FAW fields are found on each of the receive polarization, the transmit X and Y polarization symbol streams are identified and deskewed.

After the X and Y polarization symbol streams are identified and aligned, the FAW, TS, Reserved, and PS fields shall be removed leaving the remaining 172 032 message symbols for further processing.

186.3.4.4 Polarization combining and symbol deinterleaving

The message symbols from the X and Y polarization streams shall be combined to form ER1 FEC codewords comprising 1 376 256 symbols encoded as shown in Table 186–2 but at a higher resolution in order to aid the ER1 FEC error detection and correction process.

The 1 376 256 encoded DP-16QAM symbols are transferred to the 800GBASE-ER1 FEC sublayer as *m*-bit DP-16QAM symbols using PMA:IS_UNITDATA.indication messages.

186.3.4.5 Signal indication logic

The 800GBASE-ER1 PMA receive signal status is conveyed using the SIGNAL_OK parameter via the PMA:IS_SIGNAL.indication primitive (see 186.3.2.3.1).

SIGNAL_OK is set to OK if all of the following conditions are met:

- the SIGNAL_OK parameters in PMD:IS_SIGNAL.indication primitive is OK
- data is being successfully processed by all 800GBASE-ER1 PMA receive signal processing functions

Otherwise, the SIGNAL_OK parameter is set to the value FAIL.

186.4 State diagrams

This subclause is composed of state diagrams, including the associated definitions of constants, variables, functions, and counters.

186.4.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

186.4.2 State diagram definitions

186.4.2.1 Variables

block_rx

Boolean variable that is set to true when the next non-stuff 257b block is demapped by the GMP demapper function.

current_fam

A variable that holds the alignment pattern corresponding to the current FAM overhead that is recognized on a given 800GBASE-ER1 FEC sublayer tributary lane. It is compared to the variable first_fam to confirm the location of the FAM field has been detected.

current_mfas

A variable that holds the MFAS pattern corresponding to the current MFAS overhead that is recognized on a given 800GBASE-ER1 FEC sublayer tributary lane. It is compared to the variable first_mfas to confirm multi-frame alignment.

current_pma_pss	1
A variable that holds the 800GBASE-ER1 PMA polarization symbol stream identifier corresponding to the current frame alignment word sequence that is recognized on a given polarization symbol stream of the 800GBASE-ER1 PMD service interface. It is compared to the variable first_pma_pss to confirm that the location of the frame alignment word sequence has been detected.	2 3 4 5 6
Values:	7
See first_pma_pss.	8 9
fam_counter_done	10
A Boolean variable that indicates that amp_counter has reached its terminal count.	11 12
fam_match	13
A Boolean variable holds the output of the function FAM_COMPARE.	14 15
fam_slip_done	16
A Boolean variable that is set to true when the FAM_SLIP requested by the FAM field lock state diagram has been completed and the next candidate 480-bit block position is available to be tested.	17 18
fam_valid	19
A Boolean variable that is set to true if the first 256 bits of the received 480-bit block are a valid 800GBASE-ER1 FEC tributary frame alignment mechanism sequence as described in 186.2.3.4.1. The sequence is considered to be valid if all bits match the known bits of the pattern described in 186.2.3.4.1.	20 21 22 23 24
fam_lock<x>	25
A Boolean variable that is set to true when the receiver has detected the location of the FAM field among the stream of 257-bit blocks on an 800GBASE-ER1 FEC sublayer tributary lane, where $x = 0$ to 7.	26 27 28
faw_counter_done	29
A Boolean variable that indicates that faw_counter has reached its terminal count.	30 31
faw_match	32
A Boolean variable that holds the output of the function FAW_COMPARE.	33 34
faw_slip_done	35
A Boolean variable that is set to true when the SLIP requested by the FAW field lock state diagram has been completed indicating that the next candidate 22-symbol block position is available for testing.	36 37 38
faw_valid	39
A Boolean variable that is set to true if, after removing the PS, the candidate 22 16QAM symbol block from the received polarization symbol stream is a valid frame alignment word sequence. The candidate 22 16QAM symbol block is compared to the frame alignment word sequences defined in Table 186–6, considering all permitted 800GBASE-ER1 PMD service interface physical signal mappings. The candidate 22 16QAM symbol block is considered to be a valid frame alignment word sequence if at least 18 symbols match the 22 known symbols of the frame alignment word sequence described in 186.3.3.5.	40 41 42 43 44 45 46
faws_lock<x>	47
A Boolean variable that is set to true when the receiver has detected the location of the FAW field for a given polarization symbol stream on the 800GBASE-ER1 PMD service interface, where $x = 0:1$.	48 49 50 51
first_fam	52
A variable that holds the alignment pattern corresponding to the first FAM overhead that is	53 54

recognized on a given 800GBASE-ER1 FEC tributary lane. It is compared to the next received FEC frame alignment mechanism pattern that is tested.	1
	2
first_mfas	3
A variable that holds the MFAS pattern corresponding to the first MFAS overhead that is recognized on a given 800GBASE-ER1 FEC sublayer tributary lane. It is compared to the next received MFAS pattern that is tested.	4
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first_pma_pss	8
A variable that holds the 800GBASE-ER1 PMA receive polarization symbol stream identifier corresponding to the first frame alignment word sequence that is recognized on a given physical signal of the 800GBASE-ER1 PMD service interface. It is compared to the 800GBASE-ER1 PMA receive polarization symbol stream identifier corresponding to the next frame alignment word sequence that is tested.	9
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Values:	14
	15
X: Value of the frame alignment word sequence for the given 800GBASE-ER1 PMA receive polarization symbol stream maps to Table 186–6 sequence X_I , X_Q .	16
	17
Y: Value of the frame alignment word sequence for the given 800GBASE-ER1 PMA receive polarization symbol stream maps to Table 186–6 sequence Y_I , Y_Q .	18
	19
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frame_counter_done	21
A Boolean variable that indicates that frame_counter has reached its terminal count.	22
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mfas_lock<x>	24
A Boolean variable that is set to true when the receiver has detected a valid MFAS sequence on an 800GBASE-ER1 FEC sublayer tributary lane, where $x = 0$ to 7.	25
	26
mfas_valid	27
A Boolean variable that is set to true if the value of the MFAS overhead described in 186.2.3.5.1 is valid. The overhead is considered to be valid if the value increments in each frame, modulo 256.	28
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fec_all_locked	31
A Boolean variable that is set to true when fam_lock<x> is true for all lanes and is set to false when fam_lock<x> is false for any x .	32
	33
fec_all_mfas_locked	34
A Boolean variable that is set to true when mfas_lock<x> is true for all lanes and is set to false when mfas_lock<x> is false for any x .	35
	36
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fec_restart_lock	38
A Boolean variable that is set by the FAM field lock process to reset the synchronization process. It is set to true when 5 consecutive FEC frame alignment mechanism patterns fail to match (5_BAD state) on a given 800GBASE-ER1 tributary lane.	39
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fec_mfas_restart_lock	42
A Boolean variable that is set by the MFAS field lock process to reset the synchronization process. It is set to true when 5 consecutive MFAS values do not match the expected value (5_BAD state) on a given 800GBASE-ER1 FEC sublayer tributary lane.	43
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FEC_reset	46
Boolean variable that is true when set by a management entity and is false otherwise.	47
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fec_signal_ok	49
A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the PMA_IS_SIGNAL.indication primitive. It is true if the value was OK and false if the value was FAIL.	50
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pma_align_status	1
A Boolean variable set by the 800GBASE-ER1 PMA deskew process to reflect the status of the	2
800GBASE-ER1 PMA X polarization symbol stream to Y polarization symbol stream alignment.	3
Set to true when the polarization symbol streams are synchronized and aligned and set to false	4
when the 800GBASE-ER1 PMA synchronization process is not complete.	5
	6
pma_alignment_valid	7
A Boolean variable that is set to true if the 800GBASE-ER1 PMA polarization symbol streams are	8
aligned. 800GBASE-ER1 PMA polarization symbol streams are considered to be aligned when	9
faws_lock<x> is true for both x, each 800GBASE-ER1 PMA polarization symbol stream has a	10
unique identifier, and the 800GBASE-ER1 PMA polarization symbol streams are deskewed.	11
Otherwise, this variable is set to false.	12
	13
pma_all_locked	14
A Boolean variable that is set to true when faws_lock<x> is true for both x and is set to false when	15
faws_lock<x> is false for either x.	16
	17
pma_enable_deskew	18
A Boolean variable that set to true when deskew is enabled and set to false when deskew is	19
disabled. Received symbols may be discarded whenever deskew is enabled.	20
	21
PMA_reset	22
Boolean variable that is true when set by a management entity and is false otherwise.	23
	24
pma_restart_lock	25
A Boolean variable that is set by the FAW field lock process to reset the synchronization process	26
on 800GBASE-ER1 PMA polarization symbol streams. It is set to true when 15 consecutive frame	27
alignment word sequences to match (15_BAD state) on a given 800GBASE-ER1 PMA	28
polarization symbol stream.	29
	30
pma_signal_ok	31
A Boolean variable that is set based on the most recently received SIGNAL_OK parameter of the	32
PMD_IS_SIGNAL.indication primitive. It is true if the value was OK and false if the value was	33
FAIL.	34
	35
pma_pss_mapping<x>	36
A variable that holds the value of the pma_pss received on the 800GBASE-ER1 PMD physical	37
signal x.	38
	39
raml_align	40
Boolean variable that is set to true if the raml_counter needs to be aligned to a new value	41
	42
raml_max_count	43
Indicates the number of 257-bit blocks between alignment markers.	44
	45
raml_valid	46
Boolean variable that indicates if the received information in the AML field is valid. It is set by the	47
RAML_CHECK function.	48
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reset_fec	50
Boolean variable that controls the resetting of the 800GBASE-ER1 FEC sublayer. It is true	51
whenever a reset is necessary including when FEC_reset is true and during power on.	52
	53
reset_pma	54
Boolean variable that controls the resetting of the PMA sublayer. It is true whenever a reset is	
necessary including when PMA_reset is true and during power on.	
sof_raml	
Indicates the value of raml_counter at the start of an 800GBASE-ER1 FEC sublayer multi-frame.	

test_fam
A Boolean variable that is set to true when a candidate FAM field position is available for testing and false when the FIND_1ST state is entered.

test_faw
A Boolean variable that is set to true when a candidate FAW field position is available for testing and false when the FIND_1ST state is entered.

rx_local_degraded
A Boolean variable that is asserted true when the receiver detects the value 1 in the local degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames. It is deasserted when local degrade is deasserted for two consecutive 800GBASE-ER1 FEC frame periods.

rx_rm_degraded
A Boolean variable that is asserted true when the receiver detects the value 1 in the remote degrade bit of the STAT octet of two consecutive 800GBASE-ER1 FEC frames. It is deasserted when the value 0 is detected in the remote degrade bit for two consecutive 800GBASE-ER FEC sublayer frames.

186.4.2.2 Functions

FAM_COMPARE
This function compares the values of first_fam and current_fam to determine if a valid alignment mechanism pattern has been detected and returns the result of the comparison using the variable fam_match. If current_fam and first_fam found a match, fam_match is set to true.

FAM_SLIP
Causes the next candidate FAM field position to be tested. The precise method for determining the next candidate FAM field position is not specified and is implementation dependent. However, an implementation shall ensure that all possible FAM field positions are evaluated.

FAW_COMPARE
This function compares the values of first_pma_pss and current_pma_pss to determine if a valid frame alignment word sequence has been detected and returns the result of the comparison using the variable faw_match. If current_pma_pss and first_pma_pss indicate the same 800GBASE-ER1 PMA polarization symbol stream identifier, faw_match is set to true

FAW_SLIP
Causes the next candidate FAW field position to be tested. The precise method for determining the next candidate FAW field position is not specified and is implementation dependent. However, an implementation shall ensure that all possible FAW field positions are evaluated.

MFAS_COMPARE
This function compares the values of first_mfas and current_mfas to determine if a valid multi-frame alignment sequence has been detected and returns the result of the comparison using the variable mfas_valid. If current_mfas is one more than first_mfas, modulo 256, mfas_valid is set to true.

RAML_ALIGN
Updates the value of raml_counter based on the received AML field after compensating for the number of payload 257-bit blocks processed in the 800GBASE-ER1 FEC sublayer multi-frame

RAML_CHECK
Compares the value of sof_raml with the value of the received AML field. If they match, raml_valid is set to true, otherwise raml_valid is set to false.

186.4.2.3 Counters

fam_bad_count

Counts the number of consecutive 800GBASE-ER1 FEC sublayer frame alignment mechanism patterns that don't match the expected value.

fam_counter

Counts the interval of 2560 257-bit blocks containing 800GBASE-ER1 FEC sublayer frame alignment mechanism patterns.

faws_bad_count

Counts the number of consecutive frame alignment word sequences that don't match the expected value for a given 800GBASE-ER1 PMA polarization stream.

faw_counter

This counter counts the interval of 175 104 symbols between the starting position of one FAW field and the expected starting position of the next FAW field on a 800GBASE-ER1 PMA polarization symbol stream. This counter starts with a value of zero when start_faw_counter is asserted (at the position of the first symbol of a matched frame alignment word sequence on a 800GBASE-ER1 PMA polarization symbol stream), is incremented for each symbol on that 800GBASE-ER1 PMA polarization symbol stream, and reaches a terminal value of 175 104 at the expected position of the next FAW field on that 800GBASE-ER1 PMA polarization symbol stream.

frame_counter

Counts the interval of 82 240 bytes between 800GBASE-ER1 multi-frame alignment overhead.

mfas_bad_count

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frame alignment overhead sequences that don't match the expected value.

raml_bad_cnt

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frames where the received AML value doesn't match the expected value

raml_counter

Counts the number of 257-bit blocks since the last alignment marker location

zero_aml_cnt

Counts the number of consecutive 800GBASE-ER1 FEC sublayer multi-frames where the received AML value is 0.

186.4.3 State diagram figures

The 800GBASE-ER1 PMA shall implement two FAW field lock processes as shown in Figure 186–16 to identify the boundaries of 800GBASE-ER1 PMA multi-frames by observing the stream of symbols on four physical signals. The synchronization process operates independently on each polarization. The state diagram determines when the 800GBASE-ER1 PMA has detected the location of the frame alignment word sequence in the received symbol stream for the 800GBASE-ER1 PMD service interface.

The 800GBASE-ER1 PMA shall also implement the deskew process as shown in Figure 186–17.

The 800GBASE-ER1 FEC shall implement the FAM field lock process as shown in Figure 186–18 to identify the FAM field at the start of each of the eight tributary lanes distributed from the 800GBASE-ER1 FEC sublayer tributary frame by observing data from the ER1 FEC decoder output after removal of CRC32 and pad fields and distribution to the eight tributary lanes.

The 800GBASE-ER1 FEC shall implement the multi-frame alignment process as shown in Figure 186–19 on each of the eight tributary lanes to identify the overhead multi-frame by observing the pattern in the MFAS field.

If the alignment marker location feature is enabled, the 800GBASE-ER1 FEC shall implement the alignment marker location synchronization process shown in Figure 186–20. The AMs are inserted at their original position (matching the position from before AMs were removed by far-end transmit function) as indicated by the RAML value. When an unexpected RAML value arrives, the previous position of the AM is maintained until 8 consecutive unexpected RAML values are received, after which the AM position is updated to the new position indicated by the RAML.

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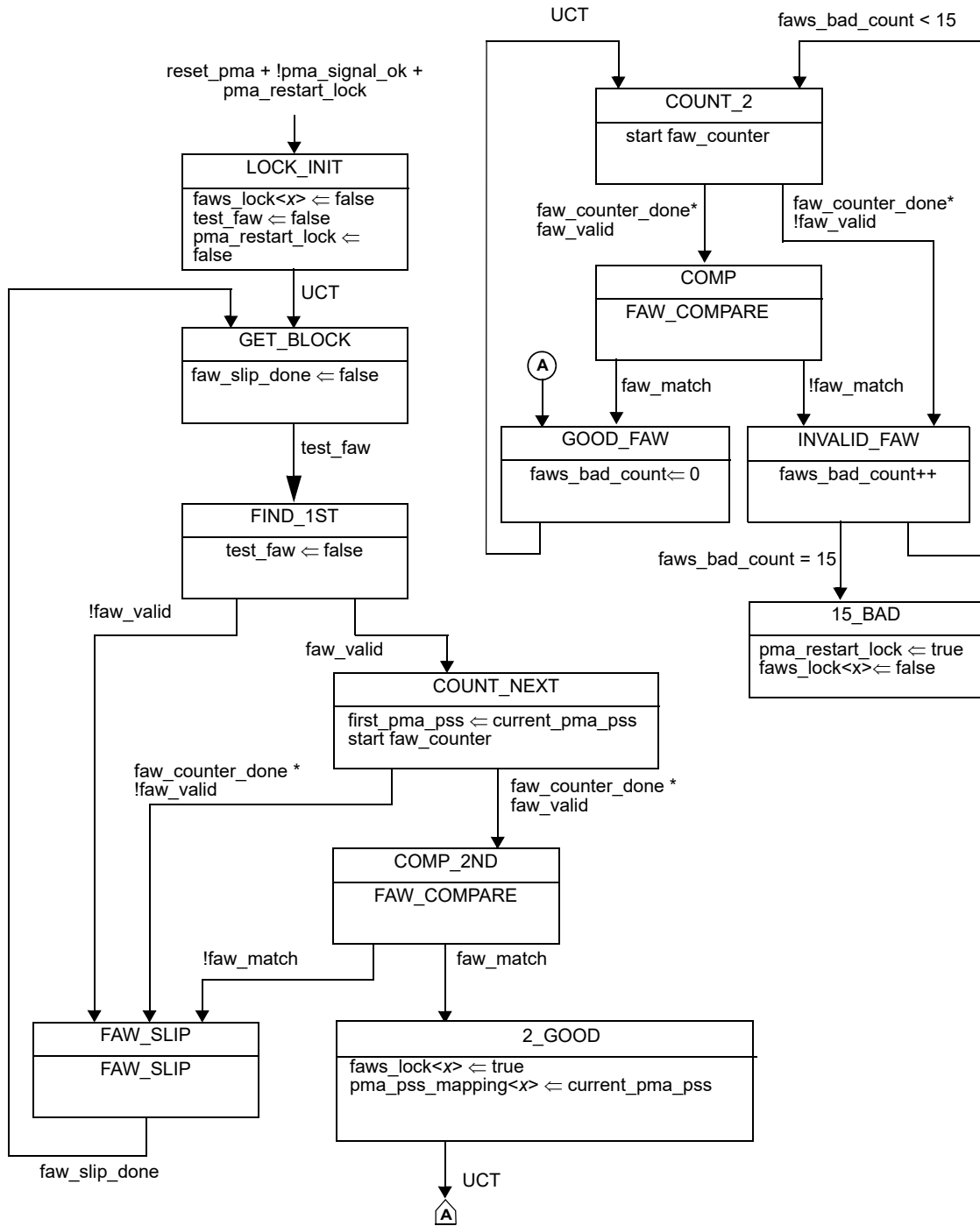


Figure 186-16—800GBASE-ER1 PMA FAW field lock state diagram

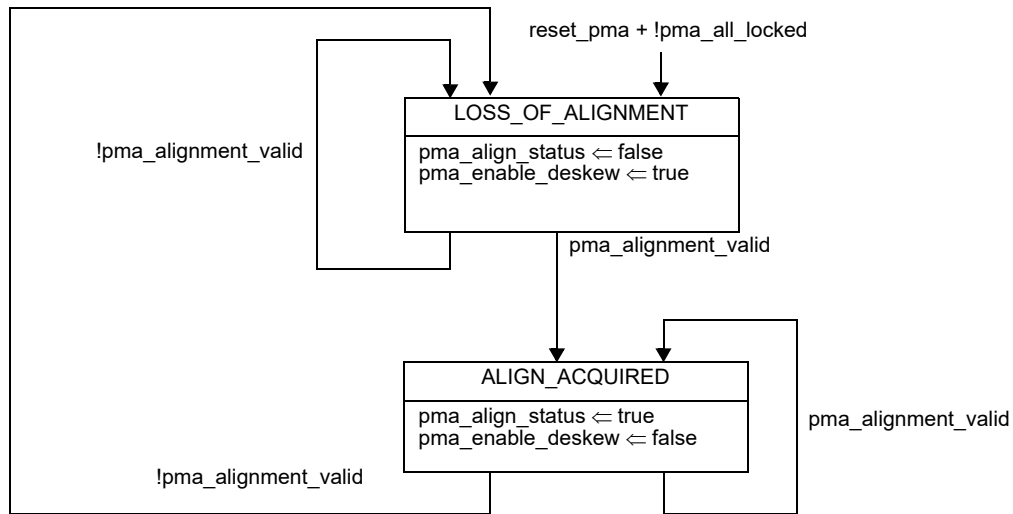


Figure 186–17—800GBASE-ER1 PMA deskew state diagram

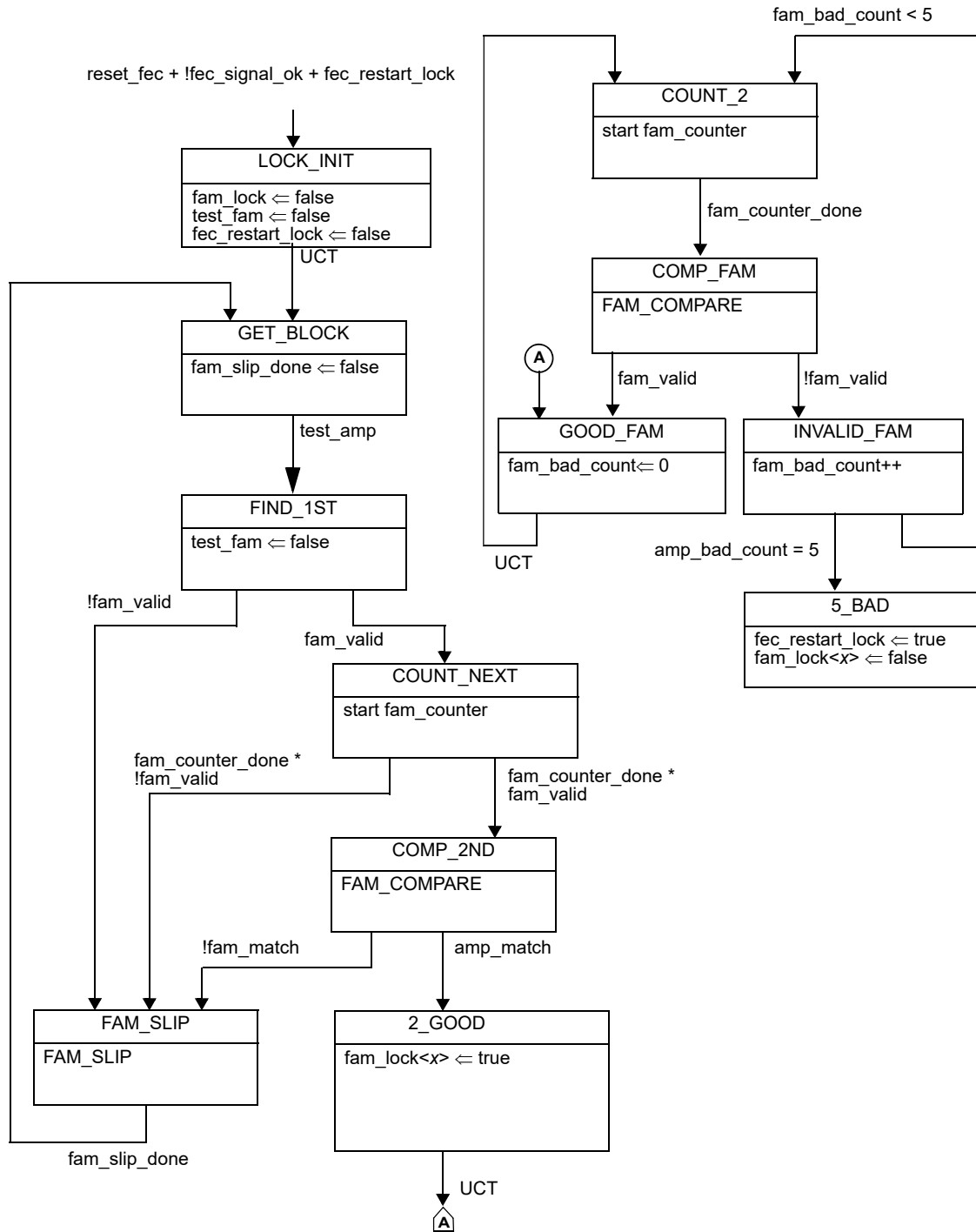


Figure 186–18—800GBASE-ER1 FEC FAM field lock state diagram

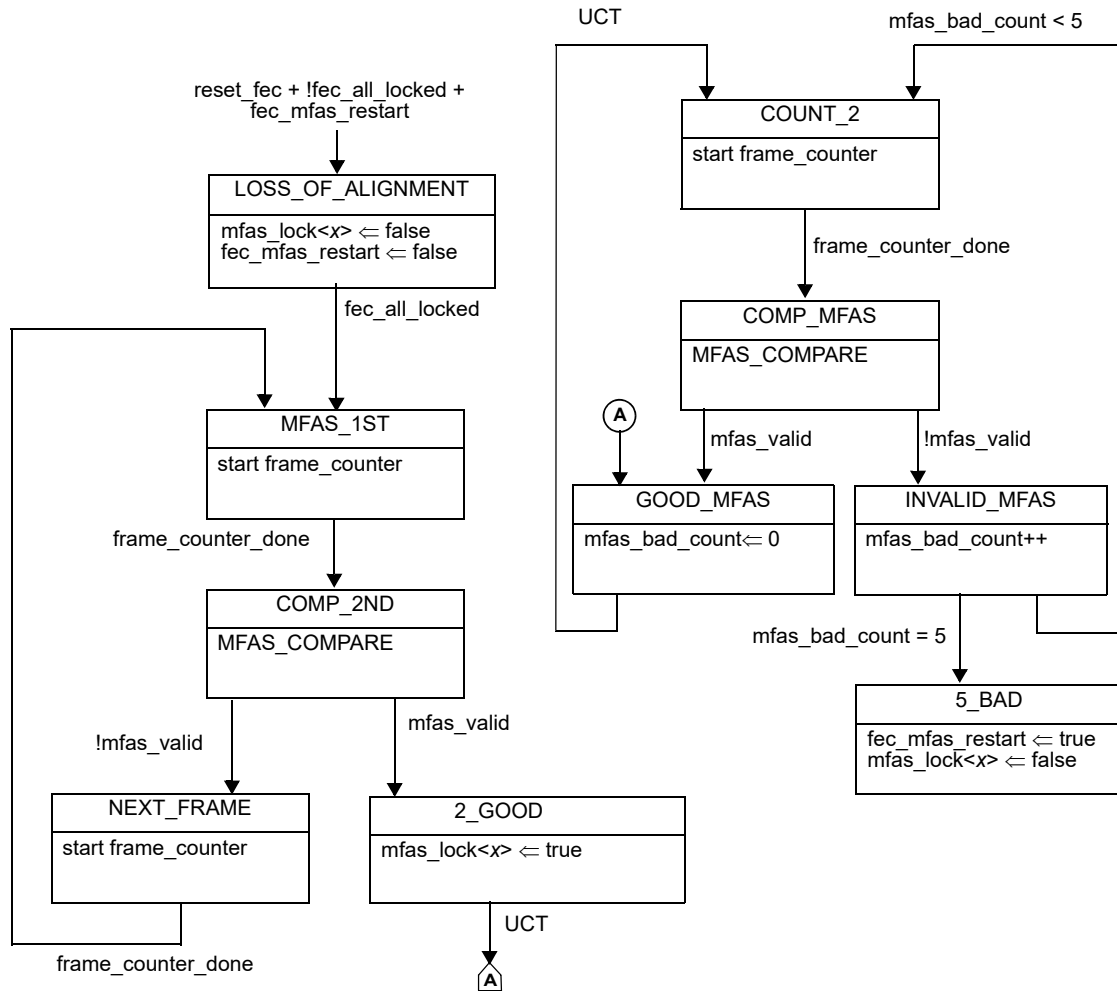


Figure 186-19—800GBASE-ER1 FEC multi-frame alignment state diagram

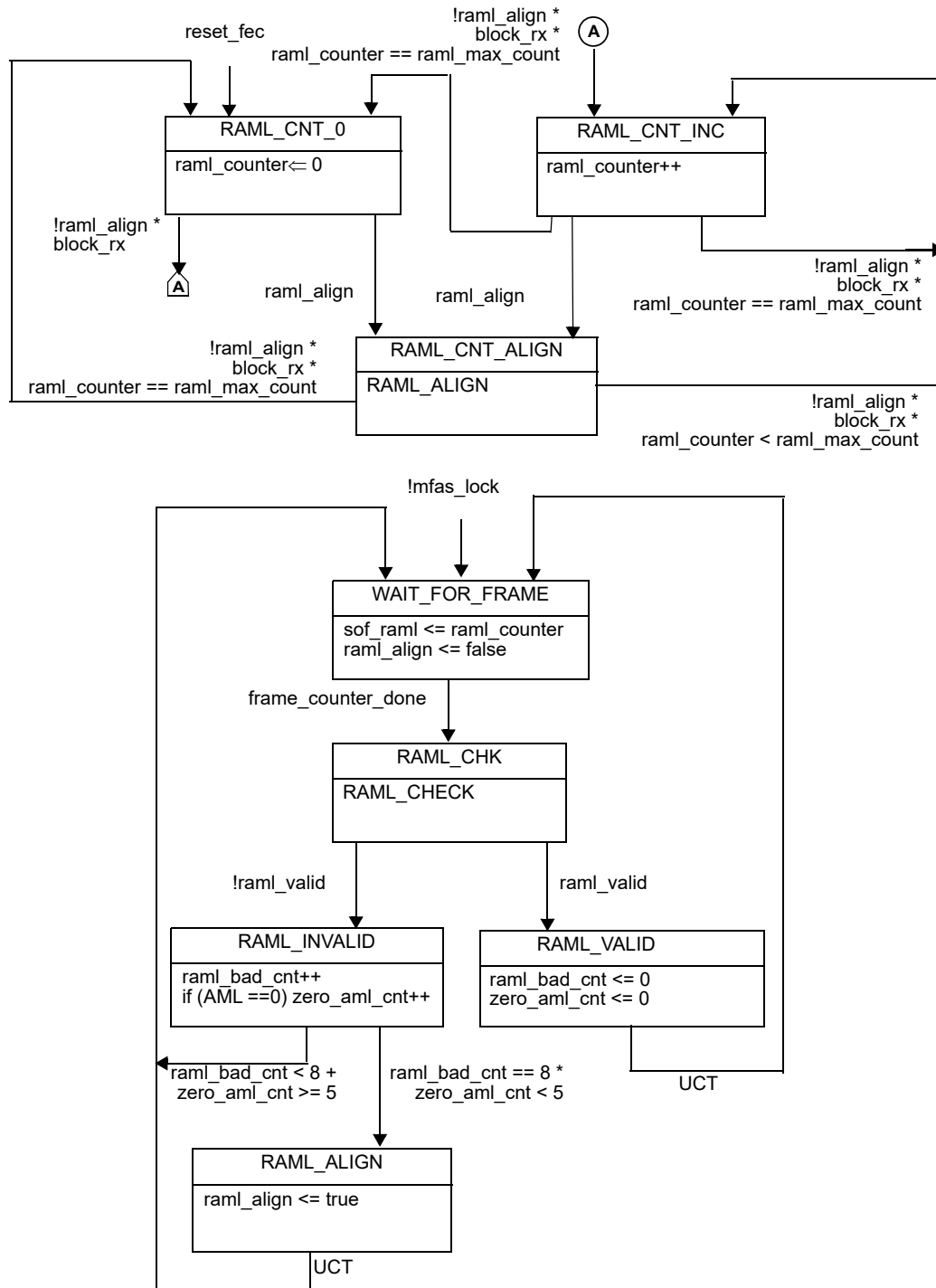


Figure 186-20—800GBASE-ER1 FEC Alignment marker location state diagram

186.5 Delay constraints

The maximum delay contributed by the 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers (sum of transmit and receive delays at one end of the link) shall be no more than 4 000 256 bit times (7813 pause_quanta or 5000.32 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

186.6 Path data delay for time synchronization

The 800GBASE-ER1 FEC and 800GBASE-ER1 PMA sublayers optionally provide path data delay information for time synchronization (see Clause 90).

186.6.1 800GBASE-ER1 FEC path data delay for time synchronization

The ability to report 800GBASE-ER1 FEC path data delays is indicated by the FEC_delay_ns_TX_ability, FEC_delay_subns_TX_ability, FEC_delay_ns_RX_ability, and FEC_delay_subns_RX_ability status variables (see Table 186–8).

The transmit and receive path data delays are reported as if the data delay measurement point (DDMP) occurs on the start of the first non-fixed-stuff 257-bit GMP word of the tributary 0 multi-frame, where the start of the 800GBASE-ER1 tributary frame is also the start of a FEC frame, taking into account the maximum (transmit) and minimum (receive) data delay through the GMP mechanism. This corresponds to the longest delay through the transmit ER1 FEC and shortest delay through the receive ER1 FEC (see 90.7).

Four separate delays are reported in the following eight path data delay status variables (see Table 186–8):

- FEC_delay_ns_TX_max, FEC_delay_subns_TX_max
- FEC_delay_ns_TX_min, FEC_delay_subns_TX_min
- FEC_delay_ns_RX_max, FEC_delay_subns_RX_max
- FEC_delay_ns_RX_min, FEC_delay_subns_RX_min

186.6.2 PMA path data delay for time synchronization

The ability to report PMA path data delays is indicated by the PMA_delay_ns_TX_ability, PMA_delay_subns_TX_ability, PMA_delay_ns_RX_ability, and PMA_delay_subns_RX_ability status variables (see Table 186–10).

The transmit and receive path data delays are reported as if the DDMP occurs on the first data symbol of the PMA frame S<0>, corresponding to the maximum delay for transmit and minimum delay for receive (see 90.7).

Four separate delays are reported in the following eight path data delay status variables (see Table 186–10):

- PMA_delay_ns_TX_max, PMA_delay_subns_TX_max
- PMA_delay_ns_TX_min, PMA_delay_subns_TX_min
- PMA_delay_ns_RX_max, PMA_delay_subns_RX_max
- PMA_delay_ns_RX_min, PMA_delay_subns_RX_min

186.7 Management variables

186.7.1 800GBASE-ER1 FEC management

FEC control and status variables intended to be accessible via a management system are listed in Table 186–7 and Table 186–8.

If the MDIO Interfaces (see Clause 45) is implemented, FEC control and status variables shall be mapped to MDIO registers as shown in Table 186–7 and Table 186–8.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 186–7—800GBASE-ER1 FEC control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
FEC_reset	186.4.2.1	1.0.15	45.2.1.1
FEC_tx_test_mode	186.2.3.12	1.2402.0	45.2.1.252
FEC_alignment_marker_location_enable	186.2.3.5.10	1.1825.14	45.2.1.177c
IFEC_bypass_indication_enable	186.2.3.1.3	1.2200.1	45.2.1.216
IFEC_degraded_SER_enable	186.2.3.1.3	1.2200.4	45.2.1.116
IFEC_degraded_SER_activate_threshold	186.2.3.1.3	1.650, 1.651	45.2.1.146
IFEC_degraded_SER_deactivate_threshold	186.2.3.1.3	1.652, 1.653	45.2.1.147
IFEC_degraded_SER_interval	186.2.3.1.3	1.654, 1.655	45.2.1.148

Table 186–8—800GBASE-ER1 FEC status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
FEC_erlfec_corrected_cw_counter	186.2.4.1	1.2412 to 1.2413	45.2.1.258
FEC_erlfec_uncorrected_cw_counter	186.2.4.1	1.2414 to 1.2415	45.2.1.259
FEC_erlfec_total_bits_counter	186.2.4.1	1.2416 to 1.2419	45.2.1.260
FEC_erlfec_corrected_bits_counter	186.2.4.1	1.2420 to 1.2423	45.2.1.261
FEC_alignment_marker_location_ability	186.2.3.5.10	1.1800.8	45.2.1.175
FEC_delay_ns_TX_max	186.6.1	1.1813, 1.1814	45.2.1.177a

Table 186–8—800GBASE-ER1 FEC status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
FEC_delay_subns_TX_max	186.6.1	1.1817	45.2.1.177a
FEC_delay_ns_TX_min	186.6.1	1.1815, 1.1816	45.2.1.177a
FEC_delay_subns_TX_min	186.6.1	1.1818	45.2.1.177a
FEC_delay_ns_RX_max	186.6.1	1.1819, 1.1820	45.2.1.177b
FEC_delay_subns_RX_max	186.6.1	1.1823	45.2.1.177b
FEC_delay_ns_RX_min	186.6.1	1.1821, 1.1822	45.2.1.177b
FEC_delay_subns_RX_min	186.6.1	1.1824	45.2.1.177b
FEC_delay_ns_RX_ability	186.6.1	1.1800.4	45.2.1.175
FEC_delay_ns_TX_ability	186.6.1	1.1800.5	45.2.1.175
FEC_delay_subns_RX_ability	186.6.1	1.1800.6	45.2.1.175
FEC_delay_subns_TX_ability	186.6.1	1.1800.7	45.2.1.175
IFEC_align_status	186.2.3.1.1	1.2201.14	45.2.1.217
IFEC_amps_lock<0:31>	186.2.3.1.1	1.2201.8 to 1.2201.11	45.2.1.217
IFEC_cw_counter	186.2.3.1.3	1.2207, 1.2208, 1.2209	45.2.1.220a
IFEC_codeword_error_bin<1:15>	186.2.3.1.3	1.3134 to 1.3163	45.2.1.268
IFEC_symbol_error_counter<0:31>	186.2.3.1.3	1.2212 to 1.2241 1.3170 to 1.3199	45.2.1.222
IFEC_pcs_lane_mapping<0:31>	186.2.3.1.3	1.3100 to 1.3131	45.2.1.265, 45.2.1.266
IFEC_bypass_indication_ability	186.2.3.1.3	1.2201.1	45.2.1.217
IFEC_degraded_SER_ability	186.2.3.1.3	1.2201.3	45.2.1.217
IFEC_degraded_SER	186.2.3.1.3	1.2201.2	45.2.1.217
IFEC_rx_rm_degraded	186.2.3.1.3	1.2201.4	45.2.1.217
IFEC_rx_local_degraded	186.2.3.1.3	1.2201.5	45.2.1.217
IFEC_corrected_cw_counter	186.2.3.1.3	1.2202, 1.2203	45.2.1.218
IFEC_uncorrected_cw_counter	186.2.3.1.3	1.2204, 1.2205	45.2.1.219

186.7.2 800GBASE-ER1 PMA management

PMA control and status variables intended to be accessible via a management system are listed in Table 186–9 and Table 186–10.

If the MDIO Interfaces (see Clause 45) is implemented, PMA control and status variables shall be mapped to MDIO registers as shown in Table 186–9 and Table 186–10.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 186–9—PMA control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMA_reset	186.4.2.1	1.0.15	45.2.1.1

Table 186–10—PMA status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMA_delay_ns_TX_max	186.6.2	1.1801, 1.1802	45.2.1.176
PMA_delay_subns_TX_max	186.6.2	1.1809	45.2.1.176
PMA_delay_ns_TX_min	186.6.2	1.1803, 1.1804	45.2.1.176
PMA_delay_subns_TX_min	186.6.2	1.1810	45.2.1.176
PMA_delay_ns_RX_max	186.6.2	1.1805, 1.1806	45.2.1.177
PMA_delay_subns_RX_max	186.6.2	1.1811	45.2.1.177
PMA_delay_ns_RX_min	186.6.2	1.1807, 1.1808	45.2.1.177
PMA_delay_subns_RX_min	186.6.2	1.1812	45.2.1.177
PMA_delay_ns_RX_ability	186.6.2	1.1800.0	45.2.1.175
PMA_delay_ns_TX_ability	186.6.2	1.1800.1	45.2.1.175
PMA_delay_subns_RX_ability	186.6.2	1.1800.2	45.2.1.175
PMA_delay_subns_TX_ability	186.6.2	1.1800.3	45.2.1.175

186.8 Protocol implementation conformance statement (PICS) proforma for Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1

186.8.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

186.8.2 Identification

186.8.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

186.8.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 186, Forward Error Correction sublayer (FEC) and Physical Medium Attachment (PMA) sublayer type 800GBASE-ER1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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186.8.3 PICS proforma tables for Forward Error Correction sublayer (FEC), type 800GBASE-ER1

186.8.3.1 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MD	MDIO	186.7.1		O	Yes [] No []
DC	Delay constraints	186.5		M	Yes []

186.8.3.2 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Inverse RS-FEC	186.2.3.1		M	Yes []
TF2	257-bit block distribution	186.2.3.2		M	Yes []
TF3	GMP mapper	186.2.3.3		M	Yes []
TF4	FAM and pad insertion	186.2.3.4		M	Yes []
TF5	General overhead insertion	186.2.3.5.1 - 186.2.3.5.1 1		M	Yes []
TF6	Alignment marker location feature	186.2.3.5.1 0		O	Yes [] No []
TF7	128-bit interleaving	186.2.3.6		M	Yes []
TF8	CRC32	186.2.3.7		M	Yes []
TF9	ER1 FEC frame	186.2.3.8		M	Yes []
TF10	Scrambler	186.2.3.9		M	Yes []
TF11	ER1 FEC encoding and interleaving	186.2.3.10		M	Yes []
TF12	Transmission bit ordering	186.2.3.11		M	Yes []
TF13	Test pattern generator	186.2.3.12		M	Yes []

186.8.3.3 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	ER1 FEC deinterleave and decode	186.2.4.1		M	Yes []
RF2	Descrambler	186.2.4.2		M	Yes []
RF3	Pad removal	186.2.4.3		M	Yes []
RF4	CRC32 check and error marking	186.2.4.4		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
RF5	128-bit distribution	186.2.4.5		M	Yes []
RF6	FAM and general OH fields	186.2.4.6.1 - 186.2.4.6.7		M	Yes []
RF7	GMP demapper	186.2.4.7		M	Yes []
RF8	257-bit collection	186.2.4.8		M	Yes []
RF9	Inverse RS-FEC	186.2.4.9		M	Yes []
RF10	Alignment marker location feature	186.2.4.9.3		O	Yes [] No []

186.8.3.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to FEC management objects is provided	186.7.1		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	186.7.1		MD:M	Yes []

186.8.3.5 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	186.6.1		O	Yes [] No []
TS1	Path data delays	186.6.1	Path data delays reported using DDMP as defined in 186.6.1	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	186.6.1	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	186.6.1	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

186.8.4 PICS proforma tables for Physical Media Attachment (PMA) sublayer, type 800GBASE-ER1

186.8.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Gray mapping and polarization distribution	186.3.3.1		M	Yes []
TF2	PMA frame and multiframe	186.3.3.2		M	Yes []
TF3	Pilot sequence	186.3.3.3		M	Yes []
TF4	Training sequence	186.3.3.4		M	Yes []
TF5	Multi-frame alignment word and Reserved field	186.3.3.5		M	Yes []
TF6	16QAM encode	186.3.3.6		M	Yes []
TF8	Transmit bit ordering	186.3.3.7		M	Yes []

186.8.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Receive signal sampling	186.3.4.1		M	Yes []
RF2	Receive signal processing	186.3.4.2		M	Yes []
RF3	FAW, TS, Reserved, and PS removal	186.3.4.3		M	Yes []
RF4	Polarization combining and symbol deinterleaving	186.3.4.4		M	Yes []
RF5	Signal indication logic	186.3.4.5		M	Yes []

186.8.4.3 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Equivalent access to PMA management objects is provided	186.7.2		!MD:O	Yes [] No [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	186.7.2		M	Yes []

186.8.4.4 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	186.6.2		O	Yes [] No []
TS1	Path data delays	186.6.2	Path data delays reported using DDMP as defined in 186.6.2	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	186.6.2	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	186.6.2	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

187. Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1

187.1 Overview

This clause specifies the 800GBASE-ER1-20 and 800GBASE-ER1 PMDs together with the associated single-mode fiber medium. The optical signals generated by these PMD types are modulated using a dual polarization 16-state quadrature amplitude modulation (DP-16QAM) format suitable for reception by a coherent optical receiver.

When forming a complete Physical Layer, a PMD shall be connected to the 800GBASE-ER1 PMA as shown in Table 187–1.

The 800GBASE-ER1 Physical Layer may optionally include an 800GMII Extender (see Clause 171).

Table 187–1—Physical Layer clauses associated with the 800GBASE-ER1-20 and 800GBASE-ER1 PMD

Associated clause	800GBASE-ER1-20 and 800GBASE-ER1
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional ^c
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^d
176—800GBASE-R SM-PMA	Conditional ^d
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
186—800GBASE-ER1 FEC	Required
186—800GBASE-ER1 PMA	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-ER1 or 800GBASE-ER1-20 PHY as described in 176B.6.1, except that if an 800GMII Extender is included in the Physical Layer implementation then no 800GAUI-n is permitted in the PHY.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If one or more 800GAUI-n are implemented in the PHY, then an 800GMII Extender is not permitted in the Physical Layer implementation.

^d If one or two 800GAUI-n are implemented in a PHY, 800GBASE-R BM-PMA or SM-PMA sublayers are required according to the guidelines in 176B.6.1.

Figure 187–1 shows the relationship of the PMD and MDI (shown shaded), along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. 800 Gigabit Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2.

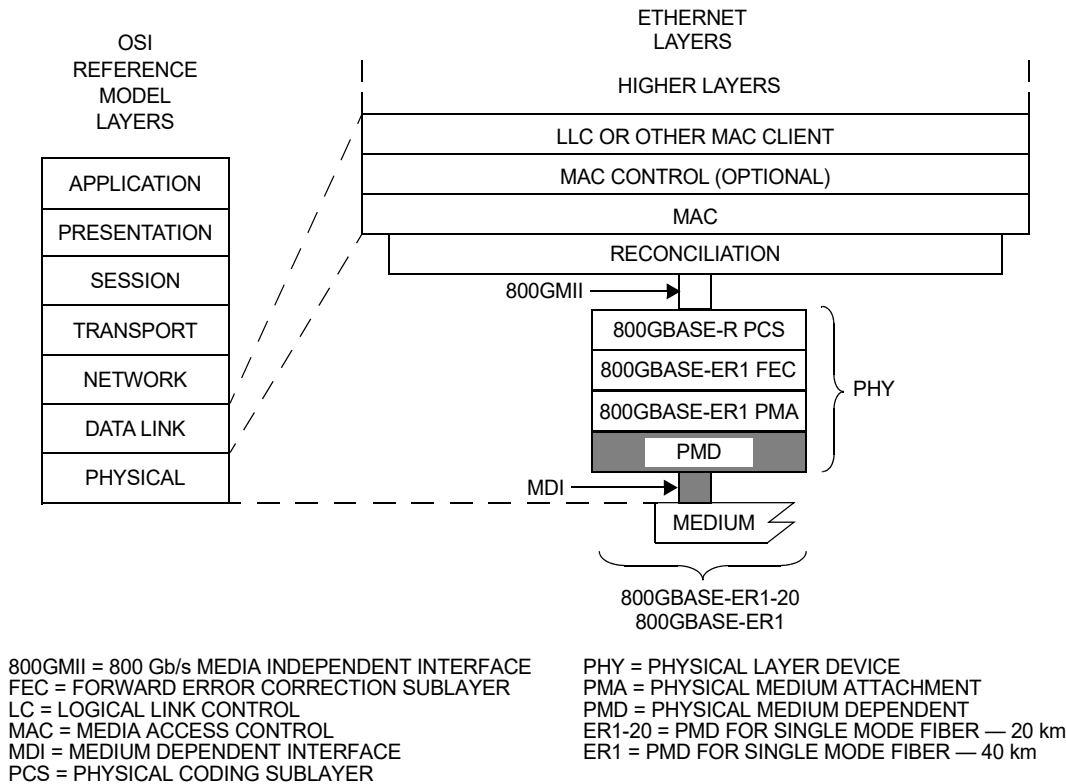


Figure 187–1—800GBASE-ER1-20 and 800GBASE-ER1 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

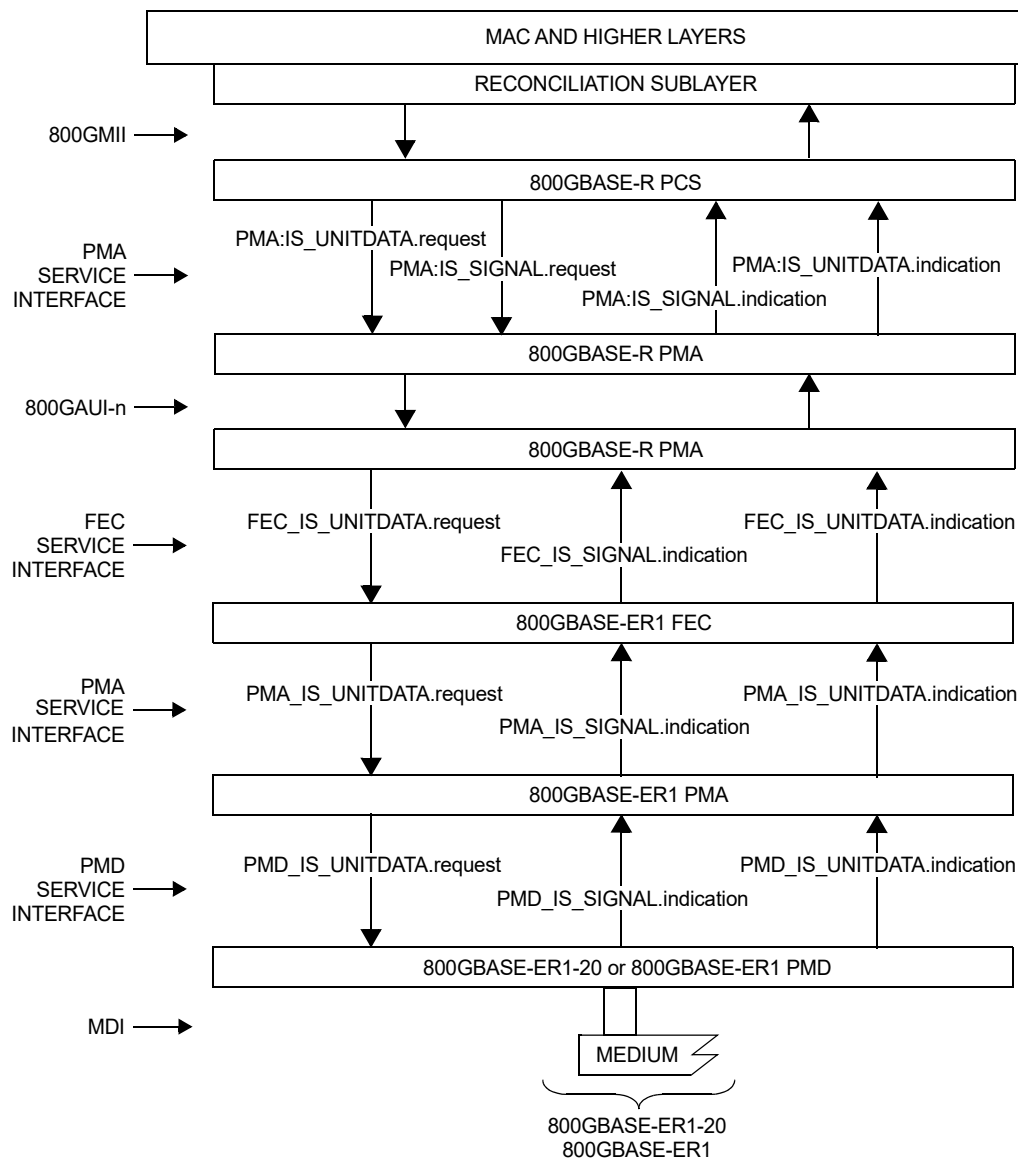
187.2 Error ratio allocation

With a compliant input signal, a PHY receiver is expected to meet the frame loss ratio specifications in 174A.5.

With a compliant input signal, a PMD receiver is expected to meet the CRC error ratio of 5.903×10^{-11} (see 174A.6), measured at the FEC decoder output.

187.3 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD sublayer. The PMD translates the encoded data to and from signals suitable for the specified medium.



800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
FEC = FORWARD ERROR CORRECTION
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 187-2—800GBASE-ER1 inter-sublayer service interfaces

187.3.1 PMD service interface

The PMD service interface uses the following primitives to exchange signals and status information:

PMD_IS_UNITDATA.request
PMD_IS_UNITDATA.indication
PMD_IS_SIGNAL.indication

187.3.1.1 PMD_IS_UNITDATA.request

The PMD_IS_UNITDATA.request primitive defines the transfer of four analog signals from the 800GBASE-ER1 PMA to the PMD (see 186.3.3.6) that are inputs to the DP-16QAM modulator to generate a single optical signal with orthogonal polarizations (see 187.5.2).

187.3.1.1.1 Semantics of the primitive

PMD_IS_UNITDATA.request(tx_signal_xi, tx_signal_xq, tx_signal_yi, tx_signal_yq)

The PMD_IS_UNITDATA.request primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the tx_signal_xi, tx_signal_xq, tx_signal_yi, and tx_signal_yq parameters.

187.3.1.1.2 When generated

The 800GBASE-ER1 PMA generates PMA_UNITDATA.request continuously.

187.3.1.1.3 Effect of receipt

The effect of receipt of the PMA_UNITDATA.request by the PMD is defined in 187.5.2.

187.3.1.2 PMD_IS_UNITDATA.indication

The PMD_IS_UNITDATA.indication primitive defines the transfer of four analog signals from the PMD to the 800GBASE-ER1 PMA that are the outputs of the coherent DP-16QAM receiver (see 187.5.3).

187.3.1.2.1 Semantics of the primitive

PMD_IS_UNITDATA.indication(rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq)

The PMD_IS_UNITDATA.indication primitive conveys four analog signals, representing the in-phase (I) and quadrature (Q) components for each of the polarizations (X and Y), via the rx_signal_xi, rx_signal_xq, rx_signal_yi, rx_signal_yq parameters, respectively.

187.3.1.2.2 When generated

The PMD generates PMA_UNITDATA.indication continuously.

187.3.1.2.3 Effect of receipt

The 800GBASE-ER1 PMA processes the four analog signals as defined in 186.3.4.

187.3.1.3 PMD_IS_SIGNAL.indication(SIGNAL_OK)

The PMD_IS_SIGNAL.indication primitive is generated by the PMD to indicate the status of the receive function via the SIGNAL_OK parameter.

187.3.1.3.1 Semantics of the service primitive

PMD_IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter corresponds to the variable Global_PMD_signal_detect as defined in 187.5.4. When Global_PMD_signal_detect is 1, SIGNAL_OK shall be assigned the value OK. When

Global_PMD_signal_detect is 0, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_i.indication parameter is undefined.

NOTE—SIGNAL_OK = OK does not guarantee that the rx_signal parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_OK = OK indication and still not meet the frame loss ratio defined in 187.2.

187.3.1.3.2 When generated

The PMD generates the PMD_IS_SIGNAL.indication primitive whenever there is a change in the value of the SIGNAL_OK parameter.

187.3.1.3.3 Effect of receipt

The effect of receipt by the PMA is defined in 186.3.4.1.

187.4 Delay constraints

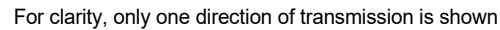
The sum of the transmit and receive delays at one end of the link contributed by the PMD including 2 m of fiber in one direction shall be no more than 16 384 bit times (32 pause_quanta or 20.48 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 169.4 and its references.

187.5 PMD functional specifications

The PMD performs the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

187.5.1 Block diagram

A block diagram for the transmit/receive paths is shown in Figure 187–3 and a block diagram of the PMD is shown in Figure 187–4. For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 187.8 and 187.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 187.7). Unless specified otherwise, all receiver measurements and tests defined in 187.8 are made at TP3.



The diagram illustrates the PMD interface between the PMA and PMD layers. The PMA layer is at the top, and the PMD layer is below it. The PMD layer is divided into two main functional blocks: a Coherent transmitter on the left and a Coherent receiver on the right. The Coherent transmitter receives four input signals from the PMA: Tx_X_I , Tx_X_Q , Tx_Y_I , and Tx_Y_Q . These signals are processed by the Coherent transmitter, which then outputs a signal to the PMA. The Coherent receiver receives four input signals from the PMA: Rx_X_I , Rx_X_Q , Rx_Y_I , and Rx_Y_Q . These signals are processed by the Coherent receiver, which then outputs a signal to the PMA. The PMA layer is labeled "PMA" and the PMD layer is labeled "PMD".

187.5.2 PMD transmit function

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The analog signals use any of the mappings in Table 187–2.

Table 187–2—Allowed analog signal to modulator input mappings

Mapping	Tx_X _I	Tx_X _Q	Tx_Y _I	Tx_Y _Q
1	X _I	X _Q	Y _I	Y _Q
2	X _Q	X _I	Y _Q	Y _I
3	X _I	X _Q	Y _Q	Y _I
4	X _Q	X _I	Y _I	Y _Q
5	Y _I	Y _Q	X _I	X _Q
6	Y _Q	Y _I	X _Q	X _I
7	Y _Q	Y _I	X _I	X _Q
8	Y _I	Y _Q	X _Q	X _I

187.5.3 PMD receive function

The PMD receive function shall convert the composite optical signal received from the MDI into four analog streams (Rx_X_I, Rx_X_Q, Rx_Y_I, Rx_Y_Q) passed across the PMD service interface to the PMA via the PMD_IS_UNITDATA.indication primitive (see 187.3.1.2), all according to the receive optical specifications in this clause.

The four analog streams carry a combination of the transmitting PMA Tx_X_I, Tx_X_Q, Tx_Y_I, and Tx_Y_Q signals used by the transmitting PMD to generate the DP-16QAM symbols.

187.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of an optical signal. For the PMD, the Global_PMD_signal_detect is set to PMD_signal_detect (see 187.5.5).

The state of Global_PMD_signal_detect is conveyed to PMD client sublayers via the PMD service interface (see 187.3).

187.5.5 PMD signal detect function

The PMD signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input. A receiver shall set the variable PMD_signal_detect to either 0 or 1 according to the criteria in Table 187–6.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of PMD_signal_detect.

Implementations need to provide adequate margin between the input optical power level at which PMD_signal_detect is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 187–3—Signal detection criteria

Receive conditions	PMD_signal_detect
Average optical power at TP3 ≤ -19.5 dBm	0
[The input signal is a compliant 800GBASE-ER1-20 or 800GBASE-ER1 signal] ^a AND [Optical power at TP3 $\geq$ average receive power (min) in Table 187–6]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

187.5.6 PMD reset function

If the variable PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

187.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows the optical transmitter to be disabled.

When the PMD_global_transmit_disable variable is set to one, this function shall turn off the optical transmitter so that it meets the requirements of the average launch power of the OFF transmitter in Table 187–5.

If a PMD fault is detected (see 187.5.8), then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter.

187.5.8 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault variable to one.

187.5.9 PMD transmit fault function (optional)

If the PMD has detected a local fault on the transmit path, the PMD shall set the PMD_transmit_fault variable to one.

187.5.10 PMD receive fault function (optional)

If the PMD has detected a local fault on the receive path, the PMD shall set the PMD_receive_fault variable to one.

187.6 PMD to MDI optical specifications

The operating range for the PMD is defined in Table 187–4. A compliant PMD operates over single mode fibers meeting the specifications in Table 187–9. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-ER1 PMD that could operate over 40 km would meet the operating range requirement of 2 m to 40 km).

Table 187–4—Operating range

PMD type	Required operating range
800GBASE-ER1-20	2 m to 20 km
800GBASE-ER1	2 m to 40 km

187.6.1 Transmitter optical specifications

The transmitter shall meet the specifications defined in Table 187–5 per the definitions in 187.8.

Table 187–5—800GBASE-ER1-20 and 800GBASE-ER1 transmit characteristics

Description	800GBASE-ER1-20	800GBASE-ER1	Unit
Signaling rate (range)	118.203351 ± 20 ppm		GBd
Modulation format	DP-16QAM		—
Average launch power (max)	−7	−1	dBm
Average launch power (min)	−11	−5	dBm
Carrier frequency (range)	193.7 ± 1.8 GHz		THz
Power difference between X and Y polarizations (max)	1		dB
Skew between X and Y polarizations (max)	5		ps
ETCC	2.5		dB
Instantaneous I-Q offset per polarization (max)	−20		dB
Mean I-Q offset per polarization (max)	−26		dB
I-Q amplitude imbalance (mean)	1		dB
I-Q phase error magnitude (max)	5		deg
I-Q quadrature skew (max)	0.75		ps
Transmitter OSNR in a 12.5 GHz bandwidth (min)	35		dB
Average launch power of OFF transmitter (max)	−20		dBm
Transmitter reflectance ^a (max)	−20		dB
RIN average (max)	−145		dB/Hz
RIN peak (max)	−140		dB/Hz
Laser linewidth (max)	1		MHz

Table 187–5—800GBASE-ER1-20 and 800GBASE-ER1 transmit characteristics (continued)

Description	800GBASE-ER1-20	800GBASE-ER1	Unit
Tx clock phase noise: phase noise mask frequency (max)			
1×10^4	–100		dBc/Hz
4×10^5	–132		dBc/Hz
1×10^6	–136		dBc/Hz
$\leq 1 \times 10^7$	–146		dBc/Hz
Tx clock phase noise: total integrated random jitter (max)	0.015		UI _{RMS}
Tx clock phase noise: total periodic jitter (max)	0.03		UI _{pp}

^a Transmitter reflectance is defined looking into the transmitter.

187.6.2 Receive optical specifications

The receiver shall meet the specifications defined in Table 187–6 per the definitions in 187.8.

Table 187–6—800GBASE-ER1-20 and 800GBASE-ER1 receive characteristics

Description	800GBASE-ER1-20	800GBASE-ER1	Unit
Signaling rate (range)	118.203351 ± 20 ppm		GBd
Modulation format	DP-16QAM		—
Carrier frequency (range)	193.7 ± 1.8 GHz		THz
Damage threshold ^a	2		dBm
Average receive power tolerance (max)	–1		dBm
Average receive power tolerance (min)	–18		dBm
Receiver sensitivity (max) ^b	–18.5	–19	dBm
Receiver reflectance (max)	20		dB
Frequency offset between received carrier and local oscillator (max)	± 1.8		GHz
Polarization dependent loss (max)	1.5		dB
Polarization rotation speed (max)	50		krad/s

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Receiver sensitivity is optional.

187.6.3 Illustrative link power budget

An illustrative power budget and penalties for 800GBASE-ER1-20 and 800GBASE-ER1 channels are shown in Table 187–7.

Table 187–7—800GBASE-ER1-20 and 800GBASE-ER1 illustrative link power budget

Parameter	800GBASE-ER1-20	800GBASE-ER1	Unit
Power budget	7.5	13	dB
Operating distance	20	40	km
Channel insertion loss ^a	7	12	dB
Maximum discrete reflectance	–27		dB
Allocation for penalties ^b	0.5	1	dB
Additional insertion loss allowed	0	1	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 187–4 and cabled optical fiber attenuation of 0.25 dB/km at 1550 nm plus an allocation for connection and splice loss given in 187.7.2.1.
^b Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

187.7 Optical channel characteristics

The fiber optic cabling model is shown in Figure 187–5.

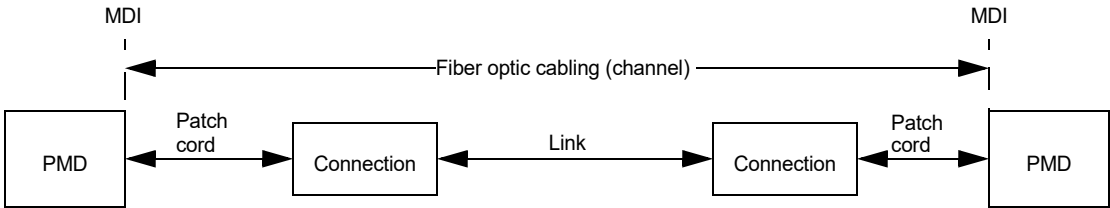


Figure 187–5—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 187–8. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4-2 one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 187–8—Optical channel characteristics

Description	800GBASE-ER1-20	800GBASE-ER1	Unit
Operating distance (max)	20	40	km
Channel insertion loss ^{a,b} (max)	7	12	dB
Channel insertion loss (min)	0		dB
Chromatic dispersion (max)	400	800	ps/nm
Chromatic dispersion (min)	0		ps/nm
Differential group delay (max) ^c	7	10	ps
Optical return loss (min)	24		dB

^a These channel insertion loss values include cable, connectors, and splices.

^b At 1550 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

187.7.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables containing ITU-T G.652.B (dispersion unshifted), type G.652.D (low water peak, dispersion unshifted), or type G.657.A1, or type G.657.A2 (bend insensitive) fibers, or the requirements in Table 187–9 where they differ.

Table 187–9—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1550	nm
Cabled optical fiber attenuation (max)	0.25 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² × km)

^a The 0.25 dB/km attenuation provided for outside plant cable has been conservatively derived using the link attenuation model described in ITU-T G.652 Appendix I using attenuation coefficient values from ITU-T G.695 Appendix I or equivalent.

187.7.2 Optical fiber connection

An optical fiber connection, as shown in Figure 187–5, consists of a mated pair of optical connectors.

187.7.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 2 dB total connection and splice loss. For example, this allocation supports four connections with an average insertion loss per connection of 0.5 dB.

Connections with different loss characteristics may be used provided the requirements of Table 187–8 are met.

187.7.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling via one fiber per direction at the MDI, being the interface between the PMD and the medium. Examples of an MDI include the following:

- Connectorized fiber pigtail
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 187.5.1, not at the MDI.

187.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

187.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 187–11 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 187–11 may be used to perform that test. The test patterns used in this clause are shown in Table 187–10.

Table 187–10—Test patterns

Pattern	Pattern description	Defined in
5	PRBS31 encoded by the 800GBASE-ER1 FEC sublayer	186.2.3.12
7	Valid 800GBASE-R signal encoded by the 800GBASE-ER1 FEC sublayer	172.2.4, 186.2.3.12

Table 187–11—Mapping of parameters to test-patterns and related subclauses

Parameter	Pattern	Related subclause
Average channel output power	5 or 7	187.8.2
Carrier frequency (range)	5 or 7	187.8.3
Power difference between X and Y polarizations	5	187.8.4
Skew between X and Y polarizations	5	187.8.5
ETCC	5	187.8.6
Instantaneous I-Q offset per polarization	5	187.8.7

Table 187–11—Mapping of parameters to test-patterns and related subclauses (*continued*)

Parameter	Pattern	Related subclause
Mean I-Q offset per polarization	5	187.8.8
I-Q amplitude imbalance	5	187.8.9
I-Q phase error magnitude	5	187.8.10
I-Q quadrature skew	5	187.8.11
Transmitter in-band OSNR	5	187.8.12
Tx clock phase noise: phase noise mask frequency	5	187.8.13
Tx clock phase noise: total integrated random jitter	5	187.8.14
Tx clock phase noise: total periodic jitter	5	187.8.15
Average receive power tolerance	5 or 7	187.8.16
Receiver sensitivity	5 or 7	187.8.17

187.8.2 Average channel output power

The average channel output power shall be within the limits given in Table 187–5 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using the test pattern defined in Table 187–11, per the test setup in Figure 53–6.

187.8.3 Carrier frequency (range)

The carrier frequency (range) shall be within the range given in Table 187–5 if measured per IEC 61280-1-3. The transmitter is modulated using the test pattern defined in Table 187–11.

187.8.4 Power difference between X and Y polarizations

The power difference between X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 187–5.

187.8.5 Skew between X and Y polarizations

The skew between the X and Y polarizations, as defined in Recommendation ITU-T G.698.2, shall be within the limit given in Table 187–5.

187.8.6 Extended transmitter constellation closure (ETCC)

ETCC is the metric used to define the quality of a 800 Gb/s DP-16QAM transmitter. The method and ETCC calculation are defined in 187.9.

The ETCC shall be within the limit given in Table 187–5 if measured using the methods specified in 187.9.

187.8.7 Instantaneous I-Q offset per polarization

The instantaneous I-Q offset per polarization shall be within the limit given in Table 187–5. The instantaneous I-Q offset per polarization is the maximum value per polarization and is calculated as $I\text{-}Q\text{ offset} = 10\log_{10}[(I\text{mean}^2 + Q\text{mean}^2)/P_{\text{signal}}]$ with a measurement interval of 1 μs .

187.8.8 Mean I-Q offset per polarization

The mean I-Q offset per polarization shall be within the limit given in Table 187–5. The mean I-Q offset per polarization is the mean value per polarization and is calculated as $I\text{-}Q\text{ offset (mean)} = 10\log_{10}[(I\text{mean}^2 + Q\text{mean}^2)/P\text{signal}]$.

187.8.9 I-Q amplitude imbalance

The I-Q amplitude imbalance is the amplitude difference in dB of the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 187–5.

187.8.10 I-Q phase error magnitude

The I-Q phase error magnitude is the phase difference of the in-phase component I and quadrature component Q of the signal, measured relative to local oscillator and shall be within the limit given in Table 187–5.

187.8.11 I-Q quadrature skew

The I-Q quadrature skew is the relative skew between the in-phase component I and quadrature component Q of the signal and shall be within the limit given in Table 187–5.

187.8.12 Transmitter in-band OSNR

The transmitter in-band OSNR shall be within the limit given in Table 187–5. The transmitter in-band OSNR is defined as the ratio of the total signal power in transmitted channel to the highest noise power density (in a 12.5 GHz resolution bandwidth) in the range of the central frequency.

187.8.13 Tx clock phase noise: phase noise mask frequency

The Tx clock phase noise: phase noise mask frequency, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213a, shall be within the limits given in Table 187–5.

187.8.14 Tx clock phase noise: total integrated random jitter

The Tx clock phase noise: total integrated random jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213b, shall be within the limits given in Table 187–5.

187.8.15 Tx clock phase noise: total periodic jitter

The Tx clock phase noise: total periodic jitter, as defined in OIF-800ZR-01.0, Implementation Agreement for 800ZR Coherent Interfaces, section 6.1.213c, shall be within the limits given in Table 187–5.

187.8.16 Average receive power tolerance

The average receive power tolerance defines the range of average receiver input power at TP3 over which the CRC ratio requirement in 187.2 is met. Average receiver power tolerance is measured using the patterns listed in Table 187–10.

The conformance test signal meets the requirements for an 800GBASE-ER1 transmitter, with ETCC equal to the maximum value specified in Table 187–5, followed by a channel with the following impairments: polarization dependent loss and polarization rotation speed as specified in Table 187–6, and differential group delay and dispersion as specified in Table 187–8.

The average receive power tolerance shall meet the limits given in Table 187–6.

187.8.17 Receiver sensitivity

Receiver sensitivity is an optional parameter defined as follows: the lowest average receiver input power at TP3 without link impairments at which the CRC error ratio requirement in 187.2 is met. Receiver sensitivity is measured using the patterns listed in Table 187–10.

The conformance test signal meets the requirements for an 800GBASE-ER1 transmitter with ETCC equal to the maximum value specified in Table 187–5, followed by an optical attenuator.

187.9 Extended transmitter constellation closure (ETCC)

The ETCC is computed using the test setup and calculation defined in Annex 185A and the parameter values listed in Table 187–12 and Table 187–13.

187.9.1 Coherent detector front-end

Parameters for the coherent detector front-end shall meet the values listed in Table 187–12.

Table 187–12—Coherent detector front-end parameters

Description	Value	Unit
3 dB bandwidth (min)	65	GHz
Effective number of bits (<i>ENOB</i>)	5	bits
Oversampling ratio	1.25	—
Local oscillator linewidth (max)	100	kHz
BER_{ref}	2×10^{-2}	—

187.9.2 Calibrated front-end sampling

The post calibration residuals for the coherent detector front-end shall meet the values listed in Table 187–13.

Table 187–13—Coherent front-end post calibration residuals

Description	Value	Unit
X-Y gain error (max)	0.2	dB
Skew between X-Y polarizations (max)	0.5	ps
I-Q phase error for X (max)	2	degree
I-Q gain error for X (max)	0.2	dB

Table 187–13—Coherent front-end post calibration residuals (*continued*)

Description	Value	Unit
I-Q skew for X (max)	0.5	ps
I-Q phase error for Y (max)	2	degree
I-Q gain error for Y (max)	0.2	dB
I-Q skew for Y (max)	0.5	ps
Carrier frequency offset (max)	0.1	GHz

187.10 Safety, installation, environment, and labeling

187.10.1 General safety

All equipment subject to this clause shall conform to the general safety requirements in [J.2](#).

187.10.2 Laser safety

The optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.²¹

187.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

187.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 800GBASE-ER1-20 or 800GBASE-ER1 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

²¹A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

187.10.5 Electromagnetic emission

A system integrating a 800GBASE-ER1-20 or 800GBASE-ER1 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

187.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

187.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-ER1).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 187.10.2.

187.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 187–14 and Table 187–15.

If the MDIO Interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 187–14 and Table 187–15.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 187–14—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Reset	187.5.6	1.0.15	45.2.1.1.1
Global PMD transmit disable	187.5.7	1.9.0	45.2.1.8.7

Table 187–15—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Fault	187.5.8	1.1.7	45.2.1.2.3
Transmit fault	187.5.9	1.8.11	45.2.1.7.4
Receive fault	187.5.10	1.8.10	45.2.1.7.5
Global PMD signal detect	187.5.5	1.10.0	45.2.1.9.7

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187.12 Protocol implementation conformance statement (PICS) proforma for Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1²²

187.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

187.12.2 Identification

187.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

187.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 187, Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1-20 and 800GBASE-ER1
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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²²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

187.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
ER1	800GBASE-ER1	187.6	Device supports requirements for 800GBASE-ER1 PHY	O.1	Yes [] No []
TP1	Reference point TP1 exposed and available for testing	187.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
TP4	Reference point TP4 exposed and available for testing	187.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [] No []
DC	Delay constraints	187.4	Device conforms to delay constraints	M	Yes []
MD	MDIO capability	187.11	Registers and interface supported	O	Yes [] No []

187.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 800GBASE-ER1

187.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 800GBASE-ER1 PCS and PMA	187.1		M	Yes []
F2	Integration with management functions	187.1		O	Yes [] No []
F3	Error ratio allocation	187.2	Meets error ratio allocation specified in 187.2	M	Yes []
F4	Transmit function	187.5.2	Conveys analog data from PMD service interface to MDI	M	Yes []
F5	Receive function	187.5.3	Conveys analog data from PMD service interface to MDI	M	Yes []
F6	Global signal detect	187.5.4	Report to the PMD service interface the message PMD_IS_SIGNAL.indication (SIGNAL_DETECT)	O	Yes [] No []
F7	PMD reset function	187.5.6	Resets the PMD sublayer	O	Yes [] No []

187.12.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	187.11		MD:M	Yes [] N/A []
M2	PMD global transmit disable function	187.5.7	Disables the optical transmitter with the PMD_global_transmit_disable variable	MD:O	Yes [] No [] N/A []
M3	PMD fault function	187.5.8	Sets PMD_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
M4	PMD transmit fault function	187.5.9	Sets PMD_transmit_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []
M5	PMD receive fault function	187.5.10	Sets PMD_receive_fault to one if a local fault is detected	MD:O	Yes [] No [] N/A []

187.12.4.3 PMD to MDI optical specifications for 800GBASE-ER1

Item	Feature	Subclause	Value/Comment	Status	Support
ER1	Transmitter meets specifications in Table 187–5	187.6.1	Per definitions in 187.8	M	Yes []
ER2	Receiver meets specifications in Table 187–6	187.6.2	Per definitions in 187.8	M	Yes []

187.12.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	187.8	2 m to 5 m in length	M	Yes []
OM2	Average channel output power	187.8.2	Per IEC 61280-1-1	M	Yes []
OM3	Carrier frequency (range)	187.8.3		M	Yes []
OM4	Power difference between X and Y polarizations	187.8.4		M	Yes []
OM5	Skew between X and Y polarizations	187.8.5		M	Yes []
OM6	ETCC	187.8.6		M	Yes []
OM7	Instantaneous I-Q offset per polarization	187.8.7		M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM8	Mean I-Q offset per polarization	187.8.8		M	Yes []
OM9	I-Q amplitude imbalance	187.8.9		M	Yes []
OM10	I-Q phase error magnitude	187.8.10		M	Yes []
OM11	I-Q quadrature skew	187.8.11		M	Yes []
OM12	Transmitter in-band (OSNR)	187.8.12		M	Yes []
OM13	Tx clock phase noise: phase noise mask frequency	187.8.13		M	Yes []
OM14	Tx clock phase noise: total integrated random jitter	187.8.14		M	Yes []
OM15	Tx clock phase noise: total periodic jitter (max)	187.8.15		M	Yes []
OM16	Average receive power	187.8.16		M	Yes []

187.12.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	187.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety	187.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Temperature, humidity, and handling	187.10.6	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

187.12.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	187.7.1	Meets the requirements in Table 187-9	M	Yes []
OC2	MDI requirements	187.7.3	Meets IEC 61753-1 and IEC 61753-021-02	INS:M	Yes [] N/A []

NOTE—This annex is numbered in correspondence to its associated clause; i.e., Annex 4A corresponds to Clause 4.

Annex 4A

(normative)

Simplified full duplex media access control

4A.4 Specific implementations

4A.4.2 MAC parameters

Change Note 4 in 4A.4.2 (as amended by IEEE Std 802.3df-2024) as follows:

NOTE 4—For 40 Gb/s, 50 Gb/s, 100 Gb/s, 200 Gb/s, 400 Gb/s, ~~and~~ 800 Gb/s, and 1.6 Tb/s operation, the received interpacket gap (the spacing between two packets, from the last bit of the FCS field of the first packet to the first bit of the Preamble of the second packet) can have a minimum value of 8 BT (bit times), as measured at the XLGMII, 50GMII, CGMII, 200GMII, 400GMII, ~~or~~ 800GMII, or 1.6TMII receive signals at the DTE due to clock tolerance and lane alignment requirements.

Annex 31B

(normative)

MAC Control PAUSE operation

31B.3 Detailed specification of PAUSE operation

31B.3.7 Timing considerations for PAUSE operation

Insert the following paragraph into 31B.3.7 (as amended by IEEE Std 802.3df-2024) after the “At operating speeds of 800 Gb/s, ...” paragraph:

At operating speeds of 1.6 Tb/s, a station shall not begin to transmit a (new) frame more than 3620 pause_quanta after the reception of a valid PAUSE frame that contains a non-zero value of pause_time, as measured at the MDI.

Insert the following calculation for 1.6 Tb/s operation into 31B.3.7 (as amended by IEEE Std 802.3df-2024) after the 800 Gb/s calculation listed with the last paragraph (“The PAUSE response time ...”):

1.6 Tb/s – max_ overrun = 231680 + frame_length.

31B.4 Protocol implementation conformance statement (PICS) proforma for MAC Control PAUSE operation

31B.4.3 Major capabilities/options

Change the table in 31B.4.3 (as amended by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*MIIp	At operating speeds of 800 Gb/s.	31B.3.7	N/A	O	Yes [] No []
*MIIq	At operating speeds of 1.6 Tb/s.	31B.3.7	N/A	O	Yes [] No []

31B.4.6 PAUSE command MAC timing considerations

Change the table in 31B.4.6 (as amended by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
TIM17	Measurement point for station at 800 Gb/s	31B.3.7	Delay at MDI ≤ 1810 pause_quanta ^a	MIIp: M	Yes [] N/A []
<u>TIM18</u>	<u>Measurement point for station at 1.6 Tb/s</u>	<u>31B.3.7</u>	<u>Delay at MDI ≤ 3620 pause_quanta^a</u>	<u>MIIq: M</u>	<u>Yes []</u> <u>N/A []</u>

Annex 73A

(normative)

Next page message code field definitions

Change Table 73A–1 as follows:

Table 73A–1—Message code field values

Message code	M 10	M 9	M 8	M 7	M 6	M 5	M 4	M 3	M 2	M 1	M 0	Message code description
1	0	0	0	0	0	0	0	0	0	0	1	Null Message
2	0	0	0	0	0	0	0	0	0	1	0	Extended FEC and Technology Ability Message
5	0	0	0	0	0	0	0	0	1	0	1	Organizationally Unique Identifier Tagged Message
6	0	0	0	0	0	0	0	0	1	1	0	AN device Identifier Tag Code
10	0	0	0	0	0	0	0	1	0	1	0	EEE Technology Message Code. EEE capability is advertised using unformatted message code field in the Message Next Page (see 73A.4).

Insert 73A.1a after 73A.1 as follows:

73A.1a Message code 2—Extended FEC and Technology Ability Message code

The Extended FEC and Technology Ability Message code indicates additional abilities that are not accommodated in the Base Page link codeword (see 73.6.2). The Technology Ability and FEC Extension Message code consists of a message code Next Page (see Figure 73–7) with the message code field set to 000 0000 0010 and the unformatted code field used to indicate additional abilities.

Extended Technology Ability bits EA0:EA27 map to bits D16:D43 (U0:U27) and Extended FEC capability bits EF0:EF3 map to bits D44:D47 (U28:31). Reserved fields are sent as zero and ignored on receive.

See Table 73A–1a for more details of the Extended Technology Ability Field bit definitions.

Table 73A–1a—Extended Technology Ability^a

Bit	Technology
EA0	200GBASE-KR1 or 200GBASE-CR1
EA1	400GBASE-KR2 or 400GBASE-CR2
EA2	800GBASE-KR4 or 800GBASE-CR4
EA3	1.6TBASE-KR8 or 1.6TBASE-CR8
EA4 through EA27	Reserved
EF0 through EF3	Reserved for extended FEC ability

^a If the Extended Technology Ability Field is not received or not sent, then its effective value is all zeros.

For message code 2 next pages, ACK2 is set to zero for transmitted next pages and ignored in received next pages.

The 32-bit unformatted code field of Message code 2 Next Page is saved in read only management registers 7.52 to 7.55 (see 45.2.7.13a and 45.2.7.13b).

Annex 90A was added to IEEE Std 802.3-2022 by IEEE Std 802.3cx-2023:

Annex 90A

(informative)

Timestamping accuracy considerations

90A.3 Considerations for use of different data delay measurement points

Change Table 90A–1 (as amended by IEEE Std 802.3df-2024) as follows:

Table 90A–1—Magnitude of potential timestamp accuracy impairments

Ethernet rate	Magnitude of potential timestamp accuracy impairments per transmit or receive port (ns)			
	Mismatched data delay measurement point ^a	Idle insertion / removal ^{b,c}	Alignment marker / codeword marker insertion / removal ^c	PCS lane distribution / merging
10M	800	400	N/A	N/A
100M	80	40		N/A
1G	8	16 ^d , 8 ^e		0 ^e , N/A ^d
2.5G	3.2	12.8		N/A ^g
5G	1.6	6.4		N/A ^g
10G	0.8	3.2		N/A ^{d, f, g} , 0 ^e
25G	0.32	1.28	10.24	N/A
40G	0.2	1.6	6.4	4.8
50G	0.16	1.28	5.12	3.84
100G	0.08	0.64	12.8	12.16
200G	0.04	0.32	<u>5.12</u> 2.56	N/A ^g
400G	0.02	0.16	<u>5.12</u> 2.56	N/A ^g
800G	0.01	0.08	<u>5.12</u> 2.56	N/A ^g
<u>1.6T</u>	<u>0.005</u>	<u>0.04</u>	<u>1.28</u>	<u>N/A</u> ^g

^a The value shown only accounts for the time between the two data delay measurement point options when they are adjacent. See 90A.3 for other factors that can affect some of these values.

^b The value shown corresponds to the effect of a single Idle insertion/removal.

^c The path data delay of a packet can be affected when its data delay measurement point occurs after an alignment marker, codeword marker, or Idle insertion/removal event.

^d For 1000BASE-X or 10GBASE-R

^e For 1000BASE-T or 10GBASE-X

^f For 10GBASE-T

^g For PHYs including FEC, the lane distribution/merging operation belongs only to the forward error correction (FEC) function. The FEC lane distribution/merging operation is not subject to potential timestamp accuracy impairments because its path data delay determination was already clearly defined, and not subject to implementation flexibilities.

Editor's note (to be removed prior to publication): The values for 200/400/800G alignment marker insertion/removal were changed in accordance with maintenance request https://www.ieee802.org/3/maint/requests/maint_1432.pdf

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Annex 120D

(normative)

Chip-to-chip 200 Gb/s four-lane Attachment Unit Interface
(200GAUI-4 C2C) and 400 Gb/s eight-lane Attachment Unit
Interface (400GAUI-8 C2C)

120D.4 200GAUI-4 and 400GAUI-8 chip-to-chip channel characteristics

120D.4.1 Channel Operating Margin

Change Table 120D–8 as follows (unchanged rows not shown):

Table 120D–8—COM parameter values

Parameter	Symbol	Value	Units
...			
Target detector error ratio ^a	DER_0	10^{-5}	—
Case 1		6.7×10^{-6}	
Case 2			

^a If a 200GAUI-4 or 400GAUI-8 C2C is used within a PHY defined in Clause 178 or Clause 179, or within a PHY defined in Clause 180 through Clause 183 with 200GAUI-1 or 400GAUI-2 C2M (see Annex 176D), use Case 2 (see 174A.12) for channel compliance testing. Otherwise, use Case 1.

Change the title of Annex 120F (added by IEEE Std 802.3ck-2022 and modified by IEEE Std 802.3df-2024) as follows:

Annex 120F

(normative)

Chip-to-chip Attachment Unit Interfaces 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~and 800GAUI-8 C2C, and 1.6TAUI-16 C2C~~

120F.1 Overview

Change the text in 120F.1 (as modified by IEEE Std 802.3df-2024) as follows:

This annex defines the functional and electrical characteristics for the optional chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C), ~~and 800 Gb/s eight-lane Attachment Unit Interface (800GAUI-8 C2C), and 1.6 Tb/s sixteen-lane Attachment Unit Interface (1.6TAUI-16 C2C)~~. These interfaces provide electrical characteristics and associated compliance points, which can optionally be used when designing systems with electrical inter-connect of approximately 25 cm in length.

100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~and 800GAUI-8 C2C, and 1.6TAUI-16 C2C~~ are physical instantiations of the PMA service interface between the FEC (for 100GBASE-P PHYs) or PCS (for 200GBASE-R, 400GBASE-R, ~~and 800GBASE-R, and 1.6TBASE-R~~ PHYs) and the PMD, as described in 135.1.4, 120.1.4, ~~and 173.1.4, and 176.2~~. Figure 120F–1 shows example relationships of the C2C interfaces to the ISO/IEC Open System Interconnection (OSI) reference model.

The sublayers, including PCS and FEC (for 100GBASE-P PHYs), for each PHY that can optionally include a 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~or 800GAUI-8 C2C, or 1.6TAUI-16 C2C~~ are summarized in the tables in 80.1.5, 116.1.4, ~~and 169.1.4, and 174.1.4~~, and are specified in the corresponding PMD clause. The positioning of the C2C interfaces relative to other sublayers is shown in 135.1, 120.1, ~~and 173.1, and 176.1~~, with further examples in Annex 135A, Annex 120A, ~~and Annex 173A, and Annex 176B~~.

The C2C bidirectional link is described in terms of a C2C transmitter, a C2C channel, and a C2C receiver. Figure 120F–2 depicts a typical C2C application. The C2C components at both ends of the link have connected ground references.

The C2C interface is composed of independent data paths in each direction. Each 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~or 800GAUI-8 C2C, or 1.6TAUI-16 C2C~~ data path contains one, two, four, ~~or eight, or sixteen~~, respectively, differential lanes, which are AC-coupled.

The C2C transmitter and receiver use PAM4 signaling. The highest differential level corresponds to the tx_symbol or rx_symbol value three, and the lowest differential level corresponds to the tx_symbol or rx_symbol value zero. The nominal signaling rate for each lane is 53.125 GBd.

The C2C channel is specified in 120F.4.

The C2C transmitter on each end of the link is adjusted to an appropriate setting which may be based on channel knowledge. If implemented, the transmitter equalization feedback mechanism described in 120F.3.2.6 may be used to identify an appropriate setting of the link partner transmitter. The adaptive or adjustable receiver performs the remainder of the equalization.

The C2C transmitter supports $1/(1+D) \bmod 4$ precoding, as specified in 135.5.7.2, 120.5.7.2, and ~~173.5.7.2, and 176.7.1.2~~, that may be enabled or disabled as required. The C2C receiver may support $1/(1+D) \bmod 4$ precoding, as specified in ~~135.5.7.2, 120.5.7.2, and 173.5.7.2~~. Precoding may be enabled and disabled using the precoder request mechanism specified in 135F.3.2.1.

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Replace Figure 120F–1 (as modified by IEEE Std 802.3df-2024) with the following figure:

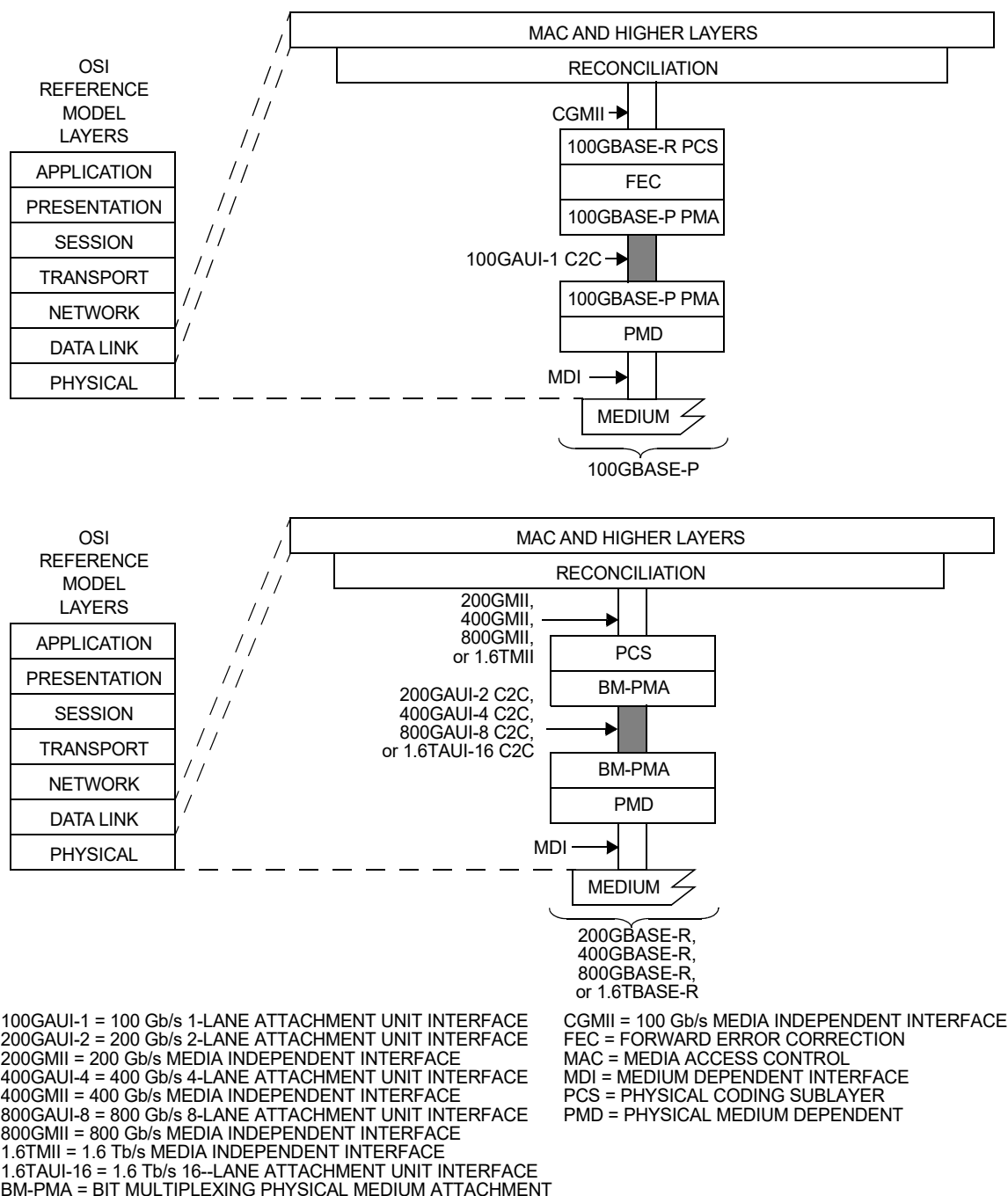
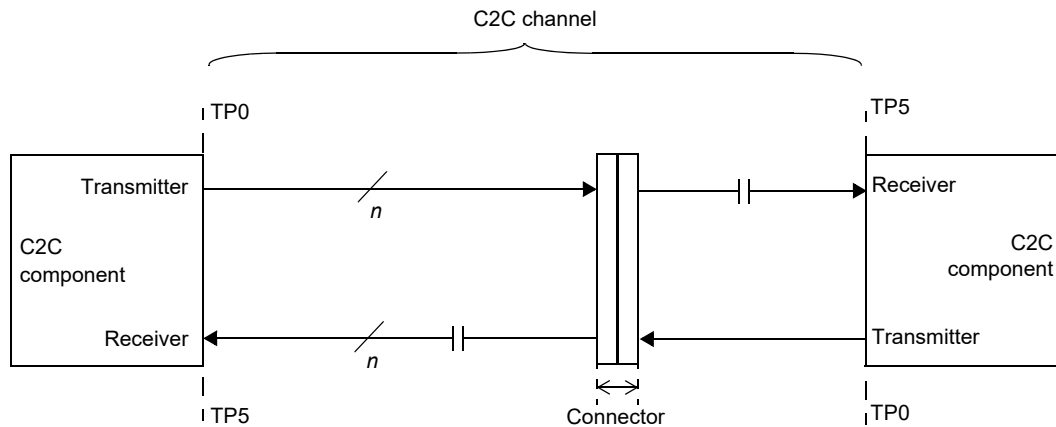


Figure 120F–1—Example 100GAUI-1, 200GAUI-2, 400GAUI-4, 800GAUI-8, and 1.6TAUI-16 C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

Change the NOTE in Figure 120F–2 (as modified by IEEE Std 802.3df-2024) as follows:



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, 4 for 400GAUI-4, and 8 for 800GAUI-8, and 16 for 1.6TAUI-16.

Figure 120F–2—Typical C2C application

120F.3 Electrical characteristics

120F.3.1 Transmitter electrical characteristics

Change footnote a of Table 120F–1 (as modified by IEEE Std 802.3df-2024) as follows (unchanged rows and footnotes not shown):

Table 120F–1—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		$53.125 \pm 50 \text{ ppm}^a$	GBd
...			

^a For 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2C with a PMA in the same package as the PCS sublayer, or for any 800GAUI-8 or 1.6TAUI-16 C2C. In other cases, the signaling rate is derived from the signaling rate presented to the input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMD, PMA, or FEC sublayers.

120F.3.2 Receiver characteristics

Change the first row of Table 120F–4 (as modified by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 120F–4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range) 100GAUI-1, 200GAUI-2, 400GAUI-4 800GAUI-8, <u>1.6TAUI-16</u>	<u>120F.3.2.1</u>	53.125 ± 100 ppm 53.125 ± 50 ppm	GBd
...			

120F.4 Channel characteristics

120F.4.1 Channel operating margin

Change Table 120F–5 as follows (unchanged rows not shown):

Table 120F–5—COM parameter values

Parameter	Symbol	Value	Units
...			
Target detector error ratio ^a <u>Case 1</u> <u>Case 2</u>	DER_0	10^{-5} <u>6.7×10^{-6}</u>	—

^a If a 200GAUI-2, 400GAUI-4, 800GAUI-8, or 1.6TAUI-16 C2C is used within a PHY defined in Clause 178 or Clause 179, or within a PHY defined in Clause 180, Clause 181, Clause 182, Clause 183, or Clause 185 with a 200GAUI-1, 400GAUI-2, 800GAUI-4, 1.6TAUI-8 C2M (see Annex 176D), use Case 2 (see 174A.12) for channel compliance testing. Otherwise, use Case 1.

Change the title of 120F.5 (as modified by IEEE Std 802.3df-2024) as follows:

120F.5 Protocol implementation conformance statement (PICS) proforma for Annex 120F, Chip-to-chip Attachment Unit Interfaces 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~and~~ 800GAUI-8 C2C, and 1.6TAUI-16 C2C¹

¹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120F.5.1 Introduction

Change the first paragraph of 120F.5.1 (as modified by IEEE Std 802.3df-2024) as follows:

The supplier of a protocol implementation that is claimed to conform to Annex 120F, Chip-to-chip Attachment Unit Interfaces 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~and 800GAUI-8 C2C,~~ and 1.6TAUI-16 C2C, shall complete the following protocol implementation conformance statement (PICS) proforma.

120F.5.2 Identification

120F.5.2.2 Protocol summary

Change the table in 120F.5.2.2 (as modified by IEEE Std 802.3df-2024) as follows:

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 120F, Chip-to-chip Attachment Unit Interfaces 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, and 800GAUI-8 C2C, <u>and 1.6TAUI-16 C2C</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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120F.5.3 Major capabilities/options

Change item NOL in the table in 120F.5.3 (as modified by IEEE Std 802.3df-2024) as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120F.3.1	One independent data path in each direction for 100GAUI-1 C2C, two independent data paths in each direction for 200GAUI-2 C2C, four independent data paths in each direction for 400GAUI-4 C2C, or eight independent data paths in each direction for 800GAUI-8 C2C, <u>or sixteen independent data paths in each direction for 1.6TAUI-16 C2C</u>	M	Yes []

Change the title of 120F.5.4 (as modified by IEEE Std 802.3df-2024) as follows:

120F.5.4 PICS proforma tables for Chip-to-chip Attachment Unit Interfaces 100GAUI-1 C2C, 200GAUI-2 C2C, 400GAUI-4 C2C, ~~and 800GAUI-8 C2C, and~~ 1.6TAUI-16 C2C

Change the title of Annex 120G (added by IEEE Std 802.3ck-2022 and modified by IEEE Std 802.3df-2024) as follows:

Annex 120G

(normative)

Chip-to-module Attachment Unit Interfaces 100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, ~~and 800GAUI-8 C2M, and~~ 1.6TAUI-16 C2M

120G.1 Overview

Change the text in 120G.1 (as modified by IEEE Std 802.3df-2024) as follows:

This annex defines the functional and electrical characteristics for the optional chip-to-module (C2M) 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M), ~~and 800 Gb/s eight-lane Attachment Unit Interface (800GAUI-8 C2M), and 1.6 Tb/s sixteen-lane Attachment Unit Interface (1.6TAUI-16 C2M).~~ Figure 120G–1 shows the relationship of the C2M interface to the ISO/IEC Open System Interconnection (OSI) reference model. The C2M interface provides electrical characteristics and associated compliance points, which can optionally be used when designing systems with pluggable module interfaces.

The sublayers, including PCS and FEC (for 100GBASE-P PHYs), for each PHY that can optionally include a 100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, ~~or 800GAUI-8 C2M, or 1.6TAUI-16 C2M~~ are summarized in the tables in 80.1.5, 116.1.4, ~~and 169.1.4, and 174.1.4~~, and are specified in the corresponding PMD clause. The positioning of the C2M interfaces relative to other sublayers is shown in 135.1, 120.1, ~~and 173.1, and 176.1~~, with further examples in Annex 135A, Annex 120A, ~~and Annex 173A, and Annex 176B.~~

The C2M link is described in terms of a host C2M component, a C2M channel with associated insertion loss (ILdd), and a module C2M component. The host C2M component and module C2M component have connected ground references. Figure 120G–3 depicts a typical C2M application and summarizes the ILdd budget associated with the C2M application. The ILdd budget reflects the highest ILdd intended to be supported and is characterized by Equation (120G–5) and illustrated in Figure 120G–12. The C2M interface comprises independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, 400GAUI-4, ~~or 800GAUI-8 C2M, or 1.6TAUI-16 C2M~~ data path contains one, two, four, ~~or eight, or sixteen~~ differential lanes, respectively, using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.

The nominal signaling rate for each lane is 53.125 GBd.

Replace Figure 120G–1 (as modified by IEEE Std 802.3df-2024) with the following figure:

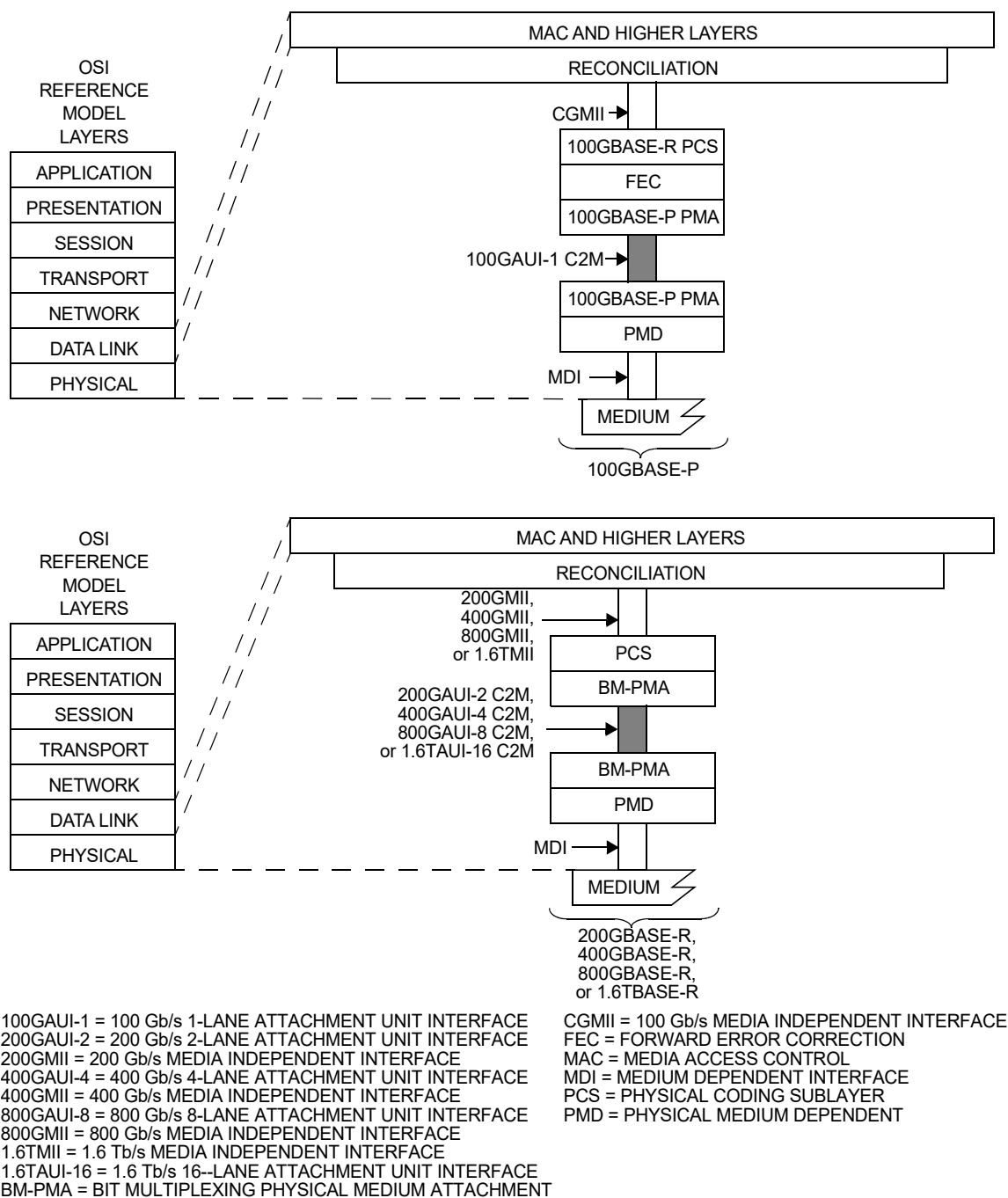
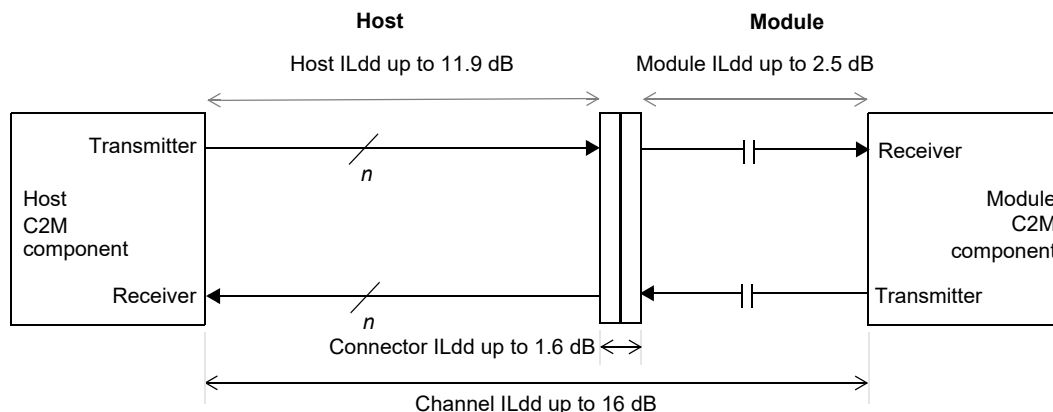


Figure 120G–1—Example 100GAUI-1, 200GAUI-2, 400GAUI-4, 800GAUI-8, and 1.6TAUI-16 C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

Change the title and NOTE in Figure 120G–2 (as modified by IEEE Std 802.3df-2024) as follows:



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, 4 for 400GAUI-4, ~~and 8 for 800GAUI-8, and~~ 16 for 1.6TAUI-16.

Figure 120G–3—100GAUI-1, 200GAUI-2, 400GAUI-4, ~~and 800GAUI-8, and~~ 1.6TAUI-16 C2M insertion loss budget at 26.56 GHz

120G.1.1 Bit error ratio

Change 120G.1.1 (as modified by IEEE Std 802.3df-2024) as follows:

The bit error ratio (BER) when processed by the PMA according to [Clause 135](#) for 100GAUI-1 C2M, according to Clause 120 for 200GAUI-2 or 400GAUI-4 C2M, ~~or~~ according to Clause 173 for 800GAUI-8 C2M, or according to Clause 176 for 1.6TAUI-16 C2M, shall be less than 10^{-5} .

120G.3 Electrical characteristics

120G.3.1 Host output characteristics

Change footnote a in Table 120G–1 (as modified by IEEE Std 802.3df-2024) as follows (unchanged rows and footnotes not shown):

Table 120G–1—Host output characteristics at TP1a

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
...			

^a For 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M with a PMA in the same package as the PCS sublayer₂ or for any 800GAUI-8 or 1.6TAUI-16 C2M. In other cases, the signaling rate is derived from the signaling rate presented to the PMA input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMA or FEC sublayers.

120G.3.1.5 Host output eye height and vertical eye closure (VEC)

Change the second paragraph of 120G.3.1.5 (as modified by IEEE Std 802.3df-2024) as follows:

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q¹ (see 120.5.11.2.1) or PRBS31Q² (see 120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, ~~or~~ 400GBASE-R, ~~or~~ 800GBASE-R, ~~or~~ 1.6TBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 845 mV and transition time (see 120G.3.1.4) of 8.5 ps.

¹PRBS13Q is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see Table 121–9).

²PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see Table 121–9).

120G.3.2 Module output characteristics

120G.3.2.1 Module output modes

Change Table 120G–4 (as modified by IEEE Std 802.3df-2024) as follows:

Table 120G–4—Module output mode mapping

IEEE 802.3 interface type	Module output mode	Host electrical interface
100GAUI-1 C2M	short	100GAUI-1-S C2M
100GAUI-1 C2M	long	100GAUI-1-L C2M
200GAUI-2 C2M	short	200GAUI-2-S C2M
200GAUI-2 C2M	long	200GAUI-2-L C2M
400GAUI-4 C2M	short	400GAUI-4-S C2M
400GAUI-4 C2M	long	400GAUI-4-L C2M
800GAUI-8 C2M	short	800GAUI-8-S C2M
800GAUI-8 C2M	long	800GAUI-8-L C2M
<u>1.6TAUI-16 C2M</u>	<u>short</u>	<u>1.6TAUI-16-S C2M</u>
<u>1.6TAUI-16 C2M</u>	<u>long</u>	<u>1.6TAUI-16-L C2M</u>

120G.3.2.2 Module output eye height and VEC

Change the second paragraph of 120G.3.2.2 (as modified by IEEE Std 802.3df-2024) as follows:

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q³ (see 120.5.11.2.1) or PRBS31Q⁴ (120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or 800GBASE-R, or 1.6TBASE-R~~ signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 750 mV and transition time (see 120G.3.1.4) of 10 ps for short mode and 15 ps for long mode.

³PRBS13Q is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see Table 121–9).

⁴PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see Table 121–9).

120G.3.3 Host input characteristics

Change the first row of Table 120G–7 (as modified by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 120G–7—Host input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range) 100GAUI-1, 200GAUI-2, 400GAUI-4 800GAUI-8, <u>1.6TAUI-16</u>	<u>120G.3.3.1</u>	TP4a	53.125 ± 100 ppm 53.125 ± 50 ppm	GBd
...				

120G.3.3.5 Host stressed input tolerance

120G.3.3.5.2 Host stressed input test calibration

Change item e) in the lettered list in 120G.3.3.5.2 (as modified by IEEE Std 802.3df-2024) as follows:

- e) The counter-propagating crosstalk signals are calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G–8. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11, 119.2.4.9, 172.2.4.11, or 175.2.4.11), or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or~~ 800GBASE-R, or 1.6TBASE-R signal for stressed signal calibration (see step g). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.

120G.3.3.5.3 Host stressed input test procedure

Change the first three paragraphs of 120G.3.3.5.3 (as modified by IEEE Std 802.3df-2024) as follows:

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or~~ 800GBASE-R, or 1.6TBASE-R sequence. The HCB is unplugged from the MCB and is plugged into the host under test. The host electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162–17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or~~ 800GBASE-R, or 1.6TBASE-R sequence, the BER for the AUI under test may be calculated using the

FEC decoder error counters (see 91.6, 119.3.1, and 172.3.1, and 175.8), as the number of FEC symbol errors divided by the number of received bits.

120G.3.4 Module input characteristics

Change the first row in Table 120G–9 as follows (unchanged rows not shown):

Table 120G–9—Module input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range) 100GAUI-1, 200GAUI-2, 400GAUI-4 800GAUI-8, 1.6TAUI-16	120G.3.4.1	TP1	53.125 ± 100 ppm 53.125 ± 50 ppm	GBd
...				

120G.3.4.3 Module stressed input tolerance

120G.3.4.3.2 Module stressed input test calibration

Change item e) in the lettered list in 120G.3.4.3.2 (as modified by IEEE Std 802.3df-2024) as follows:

- e) The counter-propagating crosstalk signals are calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G–10. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11, 119.2.4.9, 172.2.4.11, ~~or~~ 172.2.4.11), or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or~~ 800GBASE-R, ~~or~~ 1.6TBASE-R signal for stressed signal calibration (see step g). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.

120G.3.4.3.3 Module stressed input test procedure

Change the first three paragraphs of 120G.3.4.3.3 (as modified by IEEE Std 802.3df-2024) as follows:

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or~~ 800GBASE-R, ~~or~~ 1.6TBASE-R sequence. The HCB is unplugged from the MCB and the module under test is plugged into the MCB. The module electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162–17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, 400GBASE-R, ~~or 800GBASE-R, or 1.6TBASE-R~~ sequence, the BER for the AUI under test may be calculated by placing the module under test into local loopback (see 120.5.9) and feeding the module output into a compliant host or its equivalent. The BER for the AUI under test is calculated using the FEC decoder error counters (see 91.6, 119.3.1, ~~and 172.3.1~~, and 175.8), as the number of FEC symbol errors divided by the number of received bits.

Change the title of 120G.6 (as modified by IEEE Std 802.3df-2024) as follows:

**120G.6 Protocol implementation conformance statement (PICS) proforma
for Annex 120G, Chip-to-module Attachment Unit Interfaces
100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, ~~and 800GAUI-8 C2M~~, and
1.6TAUI-16 C2M⁵**

120G.6.1 Introduction

Change the first paragraph of 120G.6.1 (as modified by IEEE Std 802.3df-2024) as follows:

The supplier of a protocol implementation that is claimed to conform to Annex 120G, Chip-to-module Attachment Unit Interfaces 100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, ~~and 800GAUI-8 C2M~~, and 1.6TAUI-16 C2M shall complete the following protocol implementation conformance statement (PICS) proforma.

⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120G.6.2 Identification

120G.6.2.2 Protocol summary

Change the table in 120G.6.2.2 (as modified by IEEE Std 802.3df-2024) as follows:

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 120G, Chip-to-module Attachment Unit Interfaces 100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, and 800GAUI-8 C2M, <u>and 1.6TAUI-16 C2M</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	
Date of Statement	

120G.6.3 Major capabilities/options

Change item NOL in the table in 120G.6.3 (as modified by IEEE Std 802.3df-2024) as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120G.1	One, two, four, or eight , <u>or sixteen</u> independent data paths in each direction for 100GAUI-1, 200GAUI-2, 400GAUI-4, and 800GAUI-8, <u>and 1.6TAUI-16</u> C2M, respectively	M	Yes []

Change the title of 120G.6.4 (as modified by IEEE Std 802.3df-2024) as follows:

120G.6.4 PICS proforma tables for Chip-to-module Attachment Unit Interfaces 100GAUI-1 C2M, 200GAUI-2 C2M, 400GAUI-4 C2M, ~~and~~ 800GAUI-8 C2M, and 1.6TAUI-16 C2M

Insert new Annex 174A corresponding to Clause 174 as follows:

Annex 174A

(normative)

Error ratio allocation

174A.1 Scope

This clause defines the error ratio allocation for Physical Layer implementations that include one or more inter-sublayer links (ISLs) (see 178B.3) with data rate of 200 Gb/s or higher per lane.

174A.2 Introduction

Ethernet Physical Layers operating at a data rate of 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s are specified to enable a frame loss ratio (see 1.4.344) lower than 6.2×10^{-11} for 64-octet MAC frames with minimum interpacket gap. This frame loss ratio is equivalent to the frame loss ratio that would result from uncorrelated random bit errors on the MAC/PLS service interface with a bit error ratio (BER) of 10^{-13} . This defines the maximum expected error ratio allocation for these Physical Layers.

The use of forward error correction (FEC) allows higher BER at intermediate points within the Physical Layer, such as at the PMD, FEC, and PMA service interfaces. The acceptable BER specified at each interface depends on the data encoding as well as the distribution of errors over time.

This annex defines the error ratio allocation at each interface that enables the expected maximum error ratio allocation of the Physical Layer.

174A.3 Error ratio allocation for an Ethernet network path

Error ratio allocation of the path between two Ethernet DTEs is defined in terms of the frame loss ratio from the PLS service interface of the Reconciliation Sublayer (RS) of the transmitting DTE to the PLS service interface of the RS of the receiving DTE.

The frame loss ratio for 64-octet MAC frames with minimum interpacket gap is expected to be less than 6.2×10^{-11} .

NOTE—The frame loss ratio is affected by multiple components along the network path and is not a normative requirement of a specific component.

174A.4 Error ratio allocation for an xMII Extender

This subclause defines the error ratio allocation for a 200GMII, 400GMII, 800GMII, or 1.6TMII Extender (see Clause 118 and Clause 171).

Error ratio allocation of the xMII Extender is defined in terms of the frame loss ratio between the transmitting XS (DTE or PHY) and the receiving XS (PHY or DTE, respectively).

The frame loss ratio for 64-octet frames with minimum interpacket gap is expected to be less than 10^{-12} . This is equivalent to a FEC codeword error ratio (see 174A.11.1) lower than 2.4×10^{-13} . If the errors at the input of the RS-FEC are uncorrelated, this is equivalent to a pre-correction BER (BER_{total}) of 2.21×10^{-4} .

NOTE—The frame loss ratio is affected by multiple components within the xMII Extender and is not a normative requirement of a specific component.

174A.5 Error ratio allocation for a PHY-to-PHY link

Error ratio allocation of a PHY-to-PHY link is defined in terms of the frame loss ratio between the service interfaces of the transmitting PCS and the receiving PCS.

The frame loss ratio for 64-octet MAC frames with minimum interpacket gap is expected to be less than 6×10^{-11} .

For 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PHYs that do not include an FEC-to-FEC link, (see 174A.6), the expected frame loss ratio is equivalent to an FEC codeword error ratio (see 174A.11.1), as measured at the PCS, of less than 1.45×10^{-11} . If the errors at the input of the RS-FEC are uncorrelated, this is equivalent to a pre-correction BER (BER_{total}) of 2.92×10^{-4} .

NOTE—The frame loss ratio is affected by multiple components within the PHYs and by the medium, and is not a normative requirement of a specific component.

For an 800GBASE-ER1 or 800GBASE-ER1-20 PHY with no 800GMII extender, the frame loss ratio for 64-octet MAC frames with minimum interpacket gap is expected to be less than 6.2×10^{-11} . The codeword error ratio expectations are defined in 174A.7.

174A.6 Error ratio allocation for an FEC-to-FEC link

An FEC-to-FEC link is between FEC sublayers that fully terminate the RS-FEC from the PCS and replace it with a separate FEC (see 169.2.4b). Error ratio allocation for a FEC-to-FEC link is defined in terms of the frame loss ratio between the service interfaces of the transmitting FEC and the receiving FEC.

The frame loss ratio for 64-octet MAC frames with minimum interpacket gap is expected to be less than 6×10^{-11} .

For PHYs using the 800GBASE-ER1 FEC, the expected frame loss ratio is equivalent to a CRC error ratio (see 174A.11.2) for CRC32 blocks (see 186.2.4.4) of less than 5.903×10^{-11} . If the errors at the input of the FEC are uncorrelated, this is equivalent to a pre-correction BER (BER_{total}) of 2×10^{-2} .

174A.7 Error ratio allocation for a PCS-to-FEC link

A PCS-to-FEC link is between the PCS and FEC sublayers that fully terminate the RS-FEC from the PCS and replace it with a separate FEC (see 169.2.4b). Error ratio allocation of a PCS-to-FEC link is defined in terms of the frame loss ratio between the service interfaces of the transmitting PCS (or FEC) and the receiving FEC (or PCS).

The frame loss ratio for 64-octet frames with minimum interpacket gap is expected to be less than 10^{-12} . This is equivalent to an RS-FEC codeword error ratio (see 174A.11.1) lower than 2.4×10^{-13} . If the errors at the input of the RS-FEC are uncorrelated, this is equivalent to a pre-correction BER (BER_{total}) of 2.21×10^{-4} .

NOTE—The frame loss ratio is affected by multiple components between the PCS and FEC sublayers and is not a normative requirement of a specific component.

174A.8 Error ratio tests for 200 Gb/s per lane ISLs

This subclause defines test methods for an ISL (see 178B.3) with 200 Gb/s per lane signaling between a pair of PMAs including:

- a PMD at each end and a medium between
- an Inner FEC and PMD at each end and a medium between
- an xAUI-n between

These test methods may be used to verify that the error ratio over the ISL is within the error ratio allocation for the ISL. They utilize two parameters: BER_{total} and BER_{added} .

BER_{total} represents the total BER allocation for the PHY-to-PHY link or xMII Extender link. It is derived from the codeword error ratio allocated to the PHY-to-PHY link (see 174A.5) or xMII Extender (see 174A.4) assuming random errors.

BER_{added} represents the BER allocated to account for all other ISLs within the same PHY-to-PHY link or xMII Extender assuming random errors. The value assigned to BER_{added} is defined by the Physical Layer specification that invokes the method.

A method for constraining the error ratio of a PHY based on error masks using PMA measurements is defined in 174A.8.1.5.

Block error ratio is an estimate of the codeword error ratio adjusted for the error ratio allocated to ISLs that are not included in the test. A method for constraining the error ratio of an ISL based on block error ratio using PMA measurements is defined in 174A.8.1.6. A method for constraining the error ratio of an ISL based on block error ratio using PCS measurements is defined in 174A.10.1.

PMA measurements require the error checker and counters defined in 174A.8.1.2. A set of histogram bins are measured and calculated using PMA counters as defined in 174A.8.1.3.

174A.8.1 Block error ratio test methods using PMA-based measurements

174A.8.1.1 PMA block error ratio test configurations

Test configurations for each of the ISL types are illustrated in Figure 174A–1 through Figure 174A–3.

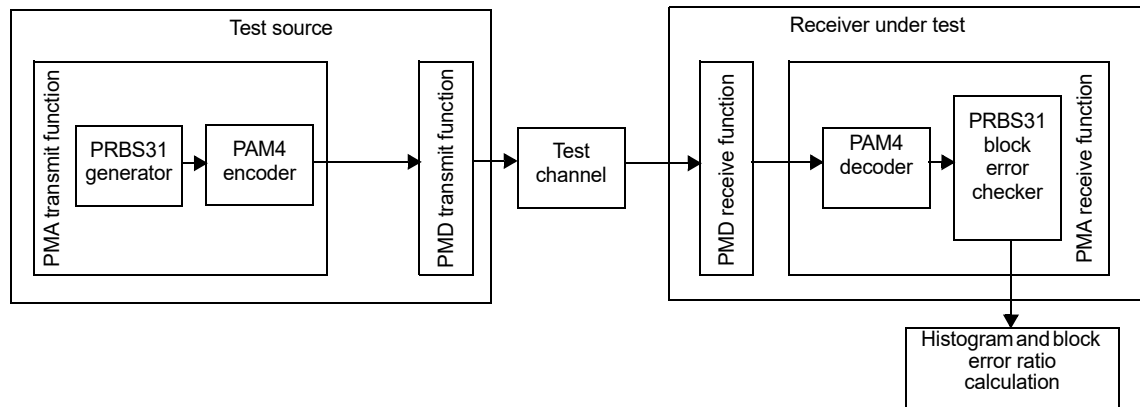


Figure 174A-1—Test configuration for a PMD without Inner FEC

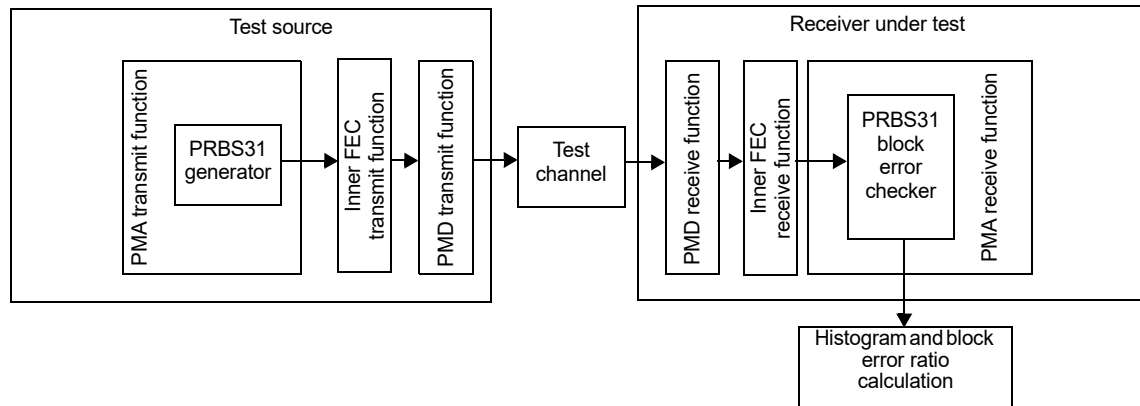


Figure 174A-2—Test configuration for a PMD with Inner FEC

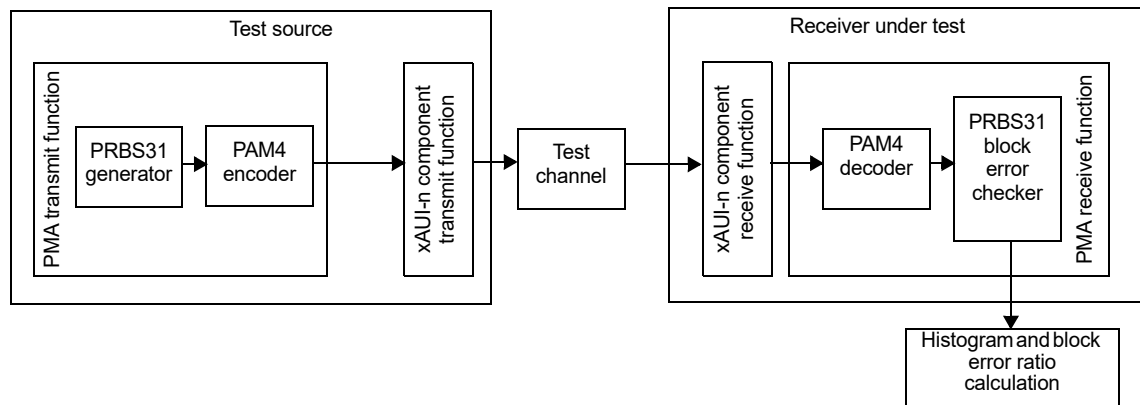


Figure 174A-3—Test configuration for an xAUI-n

174A.8.1.2 PMA block error counters

Test symbols are defined as non-overlapping groups of 5 consecutive PAM4 symbols or, equivalently, 10 consecutive bits.

A test block is defined as a set of $544/p$ test symbols composed of every fourth test symbol in a set of $4 \times 544/p$ consecutive test symbols, where p is the number of physical lanes.

17 bin counters are defined for each lane i in the range 0 through p , as follows (see 176.7.4.7):

- test_block_error_count_i_k, for k in the range 0 through 15, counts test blocks with k test symbol errors
- test_block_error_count_i_16p counts test blocks with 16 or more test symbol errors

174A.8.1.3 PMA error histogram measurement

Using the count accumulated during a test, a set of test block error histograms is calculated.

$H_m^{(i)}(k)$ is a set of p measured 17-bin histograms, one histogram for each lane i , defined as follows:

- $H_m^{(i)}(k)$ where $k < 16$ is the probability of k test symbol errors in a test block for lane i .
- $H_m^{(i)}(16)$ is the probability of more than 15 test symbol errors in a test block for lane i .

Histograms $H_m^{(i)}(k)$ are measured according the following method for each lane i :

- a) At the transmitting device generate a PRBS31 or PRBS31Q test pattern in the PMA.
- b) In the receiving PMA, identify errored bits from the physical link using the PRBS31 or PRBS31Q block error checker.
- c) Divide the stream into a series of test symbols and test blocks as defined in 174A.8.1.2.
- d) For each test block, count the number of test symbols with one or more bit errors and, based on the total number of test symbol errors in the test block, increment the appropriate bin counter.
- e) The total number of test blocks analyzed is determined according to Equation (174A–1). The value of test_block_total_count_i should be sufficiently large to reliably verify that the expected block error ratio is met, either by direct measurement or statistical projection. The projection should provide an accurate prediction of the value of $H_m^{(i)}(k)$ that would be observed over longer-term testing or at least provide an upper bound on the value.
- f) Calculate the histogram bins $H_m^{(i)}(k)$ according to Equation (174A–2).

$$\text{test_block_total_count_i} = \text{test_block_error_bin_i_16p} + \sum_{k=0}^{15} \text{test_block_error_count_i_k} \quad (174A-1)$$

$$H_m^{(i)}(k) = \frac{\text{test_block_error_count_i_k}}{\text{test_block_total_count_i}} \quad (174A-2)$$

174A.8.1.4 Convolution of error histograms

The convolution of a pair of error histograms is accomplished using the function $\text{hconv}(H_x(k), H_y(k))$, which returns histogram $H(k)$ calculated according to Equation (174A–3) and Equation (174A–4). $H(k)$, $H_x(k)$ and $H_y(k)$ are 17-bin error histograms representing a count of the number of test blocks with k test symbol errors for $k < 16$ and a count of the number of test blocks with 16 or more test symbol errors for $k = 16$.

$$H(k) = \sum_{j=0}^k H_x(j)H_y(k-j) \quad 0 \leq k \leq 15 \quad (174A-3)$$

$$H(16) = \sum_{j=0}^{16} H_x(j)H_y(16-j) + \sum_{j=1}^{16} H_x(j) \left(\sum_{i=17-j}^{16} H_y(i) \right) \quad (174A-4)$$

174A.8.1.5 Error mask test method using PMA-based measurements

This test method permits measurement of the performance of each physical lane in a PMD or xAUI-n independently of the others using error checkers and counters in the PMA. If this test passes, then the PHY or xMII Extender will meet the expected codeword error ratio. If this test fails, then the performance may be further verified using the method in 174A.8.1.6.

For each lane i , measure the error histogram $H_m(k)$ (see 174A.8.1.3) and assign $H_m(k)$ to $H_m^{(i)}(k)$.

The upper limit for $H_m^{(i)}(k)$ is defined by the histogram $H_{\max}(k)$. Compute the histogram $H_{\max}(k)$ using Equation (174A-5), where $n = 544 / p$ and $BER = BER_{\text{total}} - BER_{\text{added}}$.

$$H_{\max}(k) = \frac{n!}{k!(n-k)!} RSSER^k (1 - RSSER)^{n-k} \quad (174A-5)$$

$$RSSER = 1 - (1 - 2BER)^5 \quad (174A-6)$$

The expected block error ratio is met if, for each lane i , $H_m^{(i)}(k)$ is less than $H_{\max}(k)$ for all $k > 0$.

174A.8.1.6 Block error ratio method for all lanes using PMA-based measurements

This test method permits measurement of the performance of all physical lanes in a PMD or xAUI-n as a group using error checkers and counters in the PMA. If this test passes, then PHY or xMII Extender will meet the expected codeword error ratio.

Determine the block error ratio as follows:

- For each lane i , measure the error histogram $H_m(k)$ (see 174A.8.1.3) and assign $H_m(k)$ to $H_m^{(i)}(k)$.
- Calculate the error histogram $H_a(k)$ for the added BER using Equation (174A-5) with $BER = BER_{\text{added}}$.
- Initialize $H_e(k)$, the composite error histogram, with $H_a(k)$.
- For each lane i , iteratively assign the result of $\text{hconv}(H_e(k), H_m^{(i)}(k))$ (see 174A.8.1.4) to $H_e(k)$.
- The measured block error ratio is equal to $H_e(16)$.

The expected block error ratio is met if the measured value is less than the codeword error ratio limit specified in 174A.4 for an xMII Extender or 174A.5 for a PHY-to-PHY link.

174A.8.1.7 Block error ratio method for a single lane using PMA-based measurements

This test method permits measurement of the performance of each physical lane in a PMD or xAUI-n with p lanes using error checkers and counters in the PMA. If this test passes for all lanes, then PHY or xMII Extender will meet the expected codeword error ratio.

Determine the block error ratio for each lane i as follows:

- a) For lane i , measure the error histogram $H_m(k)$ (see 174A.8.1.3).
- b) Calculate the error histogram $H_a(k)$ for the added BER using Equation (174A-5) with $BER = BER_{added}$.
- c) Initialize $H_e(k)$, the composite error histogram, with $H_a(k)$.
- d) For p times, iteratively assign the result of $hconv(H_e(k), H_m(k))$ (see 174A.8.1.4) to $H_e(k)$.
- e) The measured block error ratio for lane i is equal to $H_e(16)$.

The expected block error ratio is met if the measured value on each lane is less than the codeword error ratio limit specified in 174A.4 for an xMII Extender or 174A.5 for a PHY-to-PHY link.

174A.9 Error ratio tests for 800GBASE-LR1 ISLs

This subclause defines test methods for an ISL (see 178B.3) with 800 Gb/s per lane signaling between a pair of 200GBASE-LR1 Inner FEC sublayers including a PMD and Inner FEC at each end and the medium between.

These tests are based on those defined in 174A.8.

174A.9.1 Block error ratio test methods using Inner FEC measurements

174A.9.1.1 Inner FEC block error ratio test configuration

Test configuration for an 800GBASE-LR1 ISL is illustrated in Figure 174A-4.

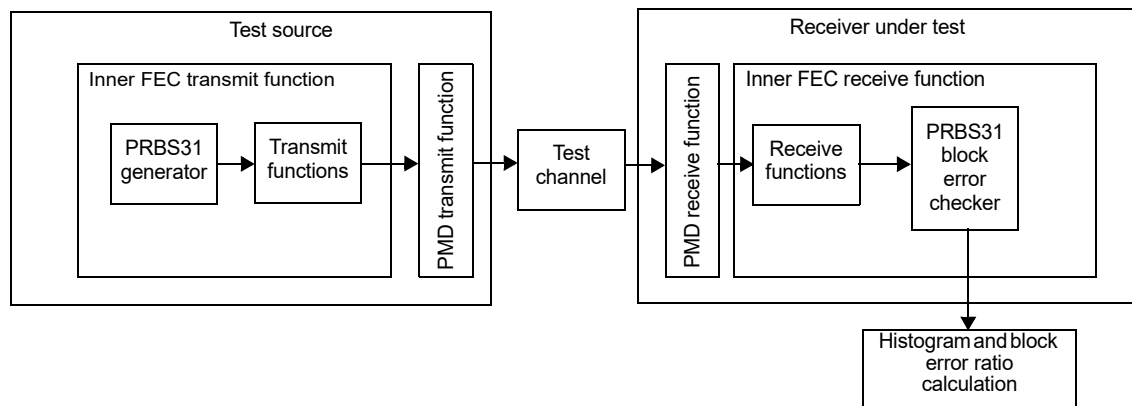


Figure 174A-4—Test configuration for an 800GBASE-LR1 PMD

174A.9.1.2 Inner FEC block error counters

The block counters are defined in 174A.8.1.2 with the number of lanes $p = 1$.

174A.9.1.3 Inner FEC error histogram measurement

Error histograms are measured according 174A.8.1.3.

174A.9.1.4 Inner FEC error mask test method using Inner FEC measurements

Conformance using an error histogram mask is defined in 174A.8.1.5 for a single lane ($p = 1$).

174A.9.1.5 Block error ratio method for using Inner FEC measurements

Conformance using block error ratio is defined in 174A.8.1.6 for a single lane ($p = 1$).

The expected block error ratio is met if the measured value is less than the codeword error ratio limit specified in 174A.5 for a PHY-to-PHY link.

174A.10 Error ratio tests for a PHY

This subclause defines test methods for a PCS-to-PCS path which includes 200 Gb/s per lane signaling on one or more ISLs.

The test method may be used to verify that the error ratio over a PCS-to-PCS path will be within the error ratio allocation for the PHY. This method utilizes the parameter BER_{added} defined in 174A.8.

174A.10.1 Block error ratio method using PCS-based measurements

This test method permits measurement of the performance of all physical lanes in a PHY as a group using FEC error counters in the PCS. If this test passes, then PHY will meet the expected codeword error ratio.

174A.10.1.1 Test configuration

The configuration for a PHY test using the PCS is illustrated in Figure 174A–5.

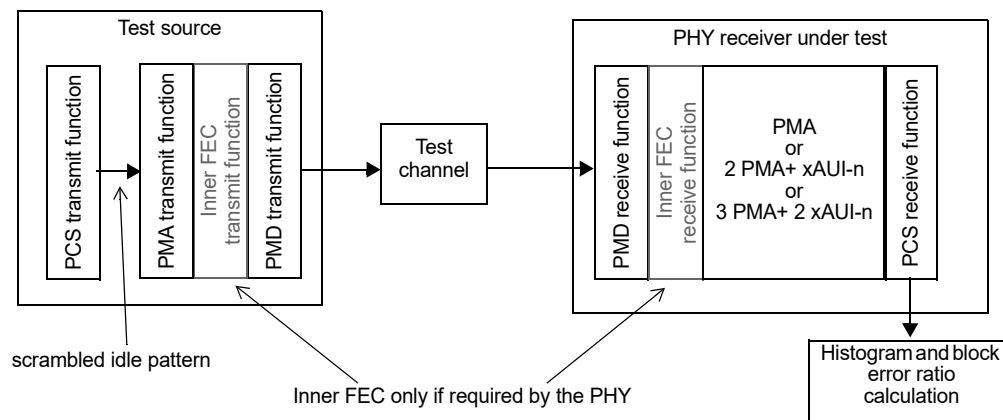


Figure 174A–5—Test configuration for a PHY receiver using PCS counters

174A.10.1.2 PCS Error Counters

The following defined PCS counters are utilized:

- FEC_cw_counter: total codewords
- FEC_codeword_error_bin_k: codewords with k symbol errors, $k = 1$ to 15
- FEC_uncorrected_cw_counter: codewords with more than 15 symbol errors
- FEC_corrected_cw_counter: codewords with 1 to 15 symbol errors

A counter for codewords with no symbol errors, FEC_codeword_error_bin_0 is mathematically derived as FEC_cw_counter – FEC_corrected_cw_counter – FEC_uncorrected_cw_counter.

174A.10.1.3 PCS error histogram measurement

An error histogram using PCS counters is measured using the following method:

- a) At the transmitting device generate a scrambled idle test pattern in the PCS.
- b) At the receiving PCS measure symbol errors using the defined PCS counters. The total number of codewords analyzed, FEC_cw_counter, should be sufficiently large to reliably verify that the expected block error ratio is met, either by direct measurement or statistical projection. The projection should provide an accurate prediction of the value of $H_m(k)$ that would be observed over longer-term testing or at least provide an upper bound on the value.
- c) Calculate the measured histogram $H_m(k)$.

A measured error histogram $H_m(k)$ is calculated as follows:

- For k in the range 0 to 15, $H_m(k) = \text{FEC_codeword_error_bin_}k / \text{FEC_cw_counter}$
- $H_m(16) = \text{FEC_uncorrected_cw_counter} / \text{FEC_cw_counter}$

174A.10.1.4 PCS block error ratio method for a complete PHY receiver

The following method is used to calculate the block error ratio for a complete PHY receiver using FEC bin counters provided in the PCS.

- a) Measure the error histogram $H_m(k)$ (see 174A.10.1.3) with no stress applied to the receive input on any lane and assign $H_m(k)$ to $H_{mu}(k)$.
- b) For each lane i , measure the error histogram $H_m(k)$ (see 174A.10.1.3) with stress, as specified for the PMD or AUI component, applied only to the receive input on lane i and assign $H_m(k)$ to $H_{ms}^{(i)}(k)$.
- c) Calculate the error histogram $H_a(k)$ for the added BER using Equation (174A–5) with $BER = BER_{added}$.
- d) Initialize $H_e(k)$, the composite error histogram, to $H_a(k)$.
- e) Iteratively, for each lane i , assign the result of $\text{hconv}(H_e(k), H_{ms}^{(i)}(k))$ (see 174A.8.1.4) to $H_e(k)$, and optionally (for better accuracy) deconvolve $H_{mu}(k)$ from $H_e(k)$.
- f) The measured block error ratio is equal to $H_e(16)$.

The measured block error ratio is expected be less than 1.45×10^{-11} .

174A.11 Error ratio definitions

174A.11.1 FEC codeword error ratio

FEC codeword error ratio is a ratio of the number of FEC codewords which are deemed to be uncorrectable by the FEC decoder to the total number of FEC codewords processed by the FEC decoder.

174A.11.2 CRC error ratio

CRC error ratio is a ratio of the number of CRC blocks in which errors are detected to the total number of CRC blocks processed by the CRC checker.

174A.12 Summary of error ratio allocations

Allocation of frame loss ratio, codeword error ratio, and BER (assuming uncorrelated bit errors) are summarized in Table 174A–1 through Table 174A–3.

Table 174A–1—Error ratio allocations for optical PHYs with no FEC sublayer or with an Inner FEC sublayer

ISL	Frame loss ratio for entire PCS-to-PCS link	Codeword error ratio for entire PCS-to-PCS link	BER for entire PCS-to-PCS link (BER_{total})	BER per ISL ^a
xAUI-n C2C ^b	6×10^{-11}	1.45×10^{-11}	2.92×10^{-4}	0.08×10^{-4}
xAUI-n C2M				0.24×10^{-4}
PMD-to-PMD				2.28×10^{-4}
xAUI-n C2M				0.24×10^{-4}
xAUI-n C2C ^b				0.08×10^{-4}

^a Measured at the PMA closest to the PMD or AUI component and after Inner FEC decoding, if present, except measured at the Inner FEC for 800GBASE-LR1.

^b If the PMD is a type defined in Clause 180, Clause 181, Clause 182, Clause 183, or Clause 185, and xAUI-n C2C is a type defined in Annex 120D (i.e., 50 Gb/s per lane) or Annex 120F (i.e., 100 Gb/s per lane), the xAUI-n C2C is expected to meet the BER allocations in this table.

Table 174A–2—Error ratio allocations for electrical PHYs

ISL	Frame loss ratio for entire PCS-to-PCS link	Codeword error ratio for entire PCS-to-PCS link	BER for entire PCS-to-PCS link (BER_{total})	BER per ISL ^a
xAUI-n C2C ^b	6×10^{-11}	1.45×10^{-11}	2.92×10^{-4}	0.08×10^{-4}
PMD-to-PMD				2.76×10^{-4}
xAUI-n C2C ^b				0.08×10^{-4}

^a Measured at the PMA closest to the PMD or AUI component.

^b If the PMD is a type defined in Clause 178 or Clause 179 (i.e., 200 Gb/s per lane) and xAUI-n C2C is a type defined in Annex 120D (i.e., 50 Gb/s per lane) or Annex 120F (i.e., 100 Gb/s per lane), the xAUI-n C2C is expected to meet the BER allocations in this table.

Table 174A–3—Error ratio allocations for xMII Extenders and PCS-to-FEC links

ISL	Frame loss ratio for entire XS-to-XS or PCS-to-FEC link	Codeword error ratio for entire XS-to-XS or PCS-to-FEC link	BER for entire XS-to-XS or PCS-to-FEC link (BER_{total})	BER per ISL ^a
xAUI-n C2C	0.1×10^{-11}	2.4×10^{-13}	2.21×10^{-4}	0.1×10^{-4}
xAUI-n C2M				0.24×10^{-4}

^a Measured at the PMA closest to the AUI component.

Insert new Annex 175A corresponding to Clause 175 as follows:

Annex 175A

(informative)

1.6TBASE-R PCS FEC codeword examples

This annex provides an example of the PCS transmit function including the 64B/66B to 256B/257B transcoding, scrambling, pre-FEC distribution, Reed-Solomon encoding, and PCSL formation defined in 175.2.4. This annex presents data in a tabular form. The contents of the tables are hexadecimal representations of the data and are transmitted from left to right within each row starting from the top row and ending at the bottom row. The most significant bit of each hex symbol is transmitted first.

A constant transmission of idle control characters is sent to the 256B/257B transcoder resulting in a continuous stream of 257-bit blocks, $\text{tx_xcoded}\langle 0:256 \rangle$, shown in hex:

$\text{tx_xcoded}\langle 0:256 \rangle = 007000000000000000780000000000000078000000000000007800000000000000$

In this example, an alignment marker group is due for insertion. The scrambler seeds, see Equation (49–1), for each of the two flows, just before the first 257-bit blocks to be scrambled are as follows:

For flow 0, the scrambler seed is $S\langle 0:57 \rangle = 24\text{E}6959\text{D}0\text{F}\text{A}5\text{D}\text{B}\text{D}$

For flow 1, the scrambler seed is $S\langle 0:57 \rangle = 1\text{F}\text{B}58857\text{D}816246$

After scrambling transcoded data and then inserting the alignment marker group, the variables $\text{tx_scrambled_am_f0}$ and $\text{tx_am_scrambled_f1}$ are produced as shown in Table 175A–1 and Table 175A–2. In these tables, italicized characters are alignment markers and bold characters are padding for the alignment markers. In each flow, just prior to insertion, the seed of the PRBS9 generator used for padding is $P\langle 0:8 \rangle = 0\text{x}100$. The PRBS9 padding data is produced from the output of $P\langle 8 \rangle$ as shown in Figure 175A–1, and $\text{tx_am_sf}\langle 2:0 \rangle = \{1,0,0\}$.

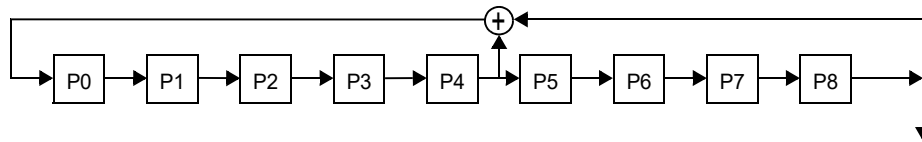


Figure 175A–1—PRBS9 Padding Generation

After distributing data into message blocks and PCS FEC encoding, codewords c_A and c_B are produced in flow 0, and codewords c_C and c_D are produced in flow 1. The codewords are illustrated as four 5440-bit representations cx_A , cx_B , cx_C , and cx_D as follows:

For all $i=0$ to 543:

$$\begin{aligned} \text{cx}_A \langle (10i+9):(10i) \rangle &= c_A \langle i \rangle \langle 9:0 \rangle \\ \text{cx}_B \langle (10i+9):(10i) \rangle &= c_B \langle i \rangle \langle 9:0 \rangle \\ \text{cx}_C \langle (10i+9):(10i) \rangle &= c_A \langle i \rangle \langle 9:0 \rangle \\ \text{cx}_D \langle (10i+9):(10i) \rangle &= c_B \langle i \rangle \langle 9:0 \rangle \end{aligned}$$

The expanded codewords are shown in Table 175A–3, Table 175A–4, Table 175A–5 and Table 175A–6 (bold characters are parity).

The codeword data is distributed as 10-bit RS-FEC symbols to form 16 PCS lanes. The PCSLs are shown in Table 175A–7.

NOTE—A text file containing the patterns in the following tables is available at <https://standards.ieee.org/downloads/802.3/>.

Editor’s note (to be removed prior to publication): The following file will be placed in the 802.3 downloads folder referenced above when this amendment is published:
https://www.ieee802.org/3/dj/public/24_05/opsasnick_3dj_03_2405.txt

**Table 175A–1—Example tx_scrambled with alignment marker group
for 1.6TBASE-R PCS flow 0**

Index <i:j>	tx_scrambled_am_f0<i:j>
<0:256>	0B2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4CB2A4
<257:513>	1965499654996549AB66E6B66FAB66DEB66CAB66E6B66DEB66DEB66D2B66FEB66
<514:770>	1A56CD956CDC56CDED6CD9D6CDC56CDB18324408CACAC03FA6F35833ABA46A3D4
<771:1027>	08425B3651068C3BA8E4DFAE3714562543CCC89A72561956 00846139561BD3722
<1028:1284>	08CCD177010357E8DAB65B6F9160968075341AC958CFC828D8372DC66242646D8
<1285:1541>	026FEAA09736E915DD54C5F7E600ECFCBD7D7C207941C5D7182D8B95FE412ED12
<1542:1798>	014CCE446415F10CDC7BBAFB1CC4B442EF5737AFB4F3BED48D240C9095F5FD3A6
<1799:2055>	10219FBE135EDE0B73C15238C150007886562E30A50CB27DF341954DB8CA1C564
<2056:2312>	16598EAD3C229838CB9BEE7F793139EE1F89FDAD9071687CBEDF89B4E527A03CD
<2313:2569>	1C2044F260A85EE6EB506C25E7C1196298548B4B3576F1B084C83F2E3CB5FC5F6
<2570:2826>	1C9D40EF27471536FE9EB5E3BC3870D46755080EA6D29F09CF8FA697A738DD3EC
<2827:3083>	193FA8A7675CE03706A49019B93544E19DD66CE78E6A94BEBD07ECCAE7DF2076D
<3084:3340>	081919F234C00959E5AFFC9F83C909805378B5F3F2E4917F3E48B5DE4758CEA42
<3341:3597>	1D378C193CDD3D0F781ABC3529BF76A66B7C73A7230B517D96DAFF6A39F9EA9B6
<3598:3854>	05E852B6917E9D1410B6509E778629A516FB2BCEABA79F12241DBC94CDAFFFFF02
<3855:4111>	1D98F5991744C9FC5393EFD6299558CB4F28DF3BF3C8636DBA505E3A296C7BD2E
<4112:4368>	00FAACFBFDBC1C29CED046C73C9AAAFE3DE8849ADAD16E864814846BB8CB889AD
<4369:4625>	04D11BFE0E8DEF5ECFDBE45D1EE4607B4B44DF874F8E976DD828FCFD38158A8FF
<4626:4882>	00EE28B635049BA9506024BA13862A1D6BAE22D0DB5DDB1F4DBE0F80E9EDBBCC6
<4883:5139>	1CD0769BC76446DA04136E7C6D19905DFB81AC3BFD931426901028D729E529F5A
<5140:5396>	14C3FF33243D7D2B5137B63359FD25B82CDE769D0F10D2E60A0386E60820BE8F4

**Table 175A–1—Example tx_scrambled with alignment marker group
for 1.6TBASE-R PCS flow 0 (continued)**

Index <i:j>	tx_scrambled_am_f0<i:j>
<5397:5653>	1993F1FE2D6C60459F26A74BD0265C2959899E4C5B24196AFD51DB241CFCF9091
<5654:5910>	18EE61EA5A0D584DE8AC602618874BBA60A907115513BC960F6BE32C68F1F2458
<5911:6167>	042251FD6EF16662F5C99D9779B20B865B1401C96034C99550401BCB2792E5638
<6168:6424>	0CE3AE06476B769F33E5577CE4BA6D62071CABE3EA9561D6077FD23909568A25A
<6425:6681>	1BBC46AC3DF26A5BB0EA4FDBCBBF42738A84007619C594E24AB396AFB5F05FBFF
<6682:6938>	076C96A407E106D494AA6B0C5168E1F335B812D7BC530CE8C6FC13F6C54EC2165
<6939:7195>	056CC06AFE2BD2A9894C4D1AD9E632F861208AD3894EFF0DE8B0291C486EA3281
<7196:7452>	094534A897031E6307E32C1906C85E99CD4696FC818D2B5EAF26A576DE149AE12
<7453:7709>	085AE9048EDDC5401827FAA937411FF35C10287236FFDC87E7E7E334B038EE3F9
<7710:7966>	014179E0E2359D3B961A1302FA4B7AC384F0564B15D7504DADBE3C65EE8F6F13A
<7967:8223>	08D40F4B092621CCA4919E81D0C15014204622D41814894DBE389C149E5E2257A
<8224:8480>	0E4772C6718DA5361ABFAAD60F5C32D304740DE671045CDAD6617125F4BBF457D
<8481:8737>	071A7D2B02B41F13971A22FE8A29F0A03C1CEC43CA0411D7872F18D12E0A2BD06
<8738:8994>	1344403844B360B4A61968D045941FF821511C2E95455C300EED811D314AD7018
<8995:9251>	0275A5C4A36F6B8B13AFAFFCFDDBD9B13F00488494886CDEB13B3FC9F7941CF
<9252:9508>	168E51B4882712DECD042307B45EBFF549E07C92FD3C3DAB5A655CC419BBA06F1
<9509:9765>	02B88C8D61954691B36009D54637647F750E1B99733303EA76E0805B182D3B9D0
<9766:10022>	1C50996262A56FB5D7631247C2F52B5150C17BA61FEB56A37BE13F2ABC226AA0A
<10023:10279>	1A1F1CF493816E6F69603FF83E895BA5AFF2ECB8E9090CB28AE93C232756FEDAF

**Table 175A–2—Example tx_scrambled with alignment marker group
for 1.6TBASE-R PCS flow 1**

Index <i:j>	tx_scrambled_am_f1<i:j>
<0:256>	08DB4C8414C8C54C8B54C90F4C89F4C9794C8894C80D4C9AD4C8BF4C86D4C8314
<257:513>	190A29902E990B69ADFDC5A9613E0CC067BFA2ADD4ADC35EF71B252465E0C55C7
<514:770>	0E02B1475DCCD5D481B98B6D61054D5104767D3DBF667B508085A4527C794591C
<771:1027>	185B6EDA7CEAEE233C9D71B4661C56FA9CE9C33DF6555D9E00846139561BD3721
<1028:1284>	0D86712D0BB857113C0B3BECCC6DD238DCD620976ACD22761CDDBF9E5F7126F80
<1285:1541>	17D20BB458BCD8129BB394A60A01638FA7712B459F1601BF50F4FD64BB21862EF
<1542:1798>	08E234452B12D312B55B34ECE1DE72C4888E87DDFE8DAC2D9FDCEC27F854B2BFB

**Table 175A–2—Example tx_scrambled with alignment marker group
for 1.6TBASE-R PCS flow 1 (continued)**

Index <i:j>	tx_scrambled_am_fl<i:j>
<1799:2055>	11E74136D4BA915E13F938979283A5F5D6EB22AF4B44AB30E3DE2284B0ED707DF
<2056:2312>	1DA2F79981437B105C3F60967066BAD624B8E8E9B5E7C458FAD1F5E5C804B5D57
<2313:2569>	1FAAE03357A5BF7F8B9747AB1678F16DB47D1B274554A15B564C6093C384CE14B
<2570:2826>	09CE5AC990B496EFE737DB49F2EBD44F599305D4244635DDCB09F962AA1B64800
<2827:3083>	0E24DCC70DAE09BA00AC6DD01F83365B8CA4726B57D4DE07F8333369FAC7E798D
<3084:3340>	1E02AB63008FC939412BC752517019DEE9D334A7E1D5D1D33C8E825BD30D69CB9
<3341:3597>	087D184EA8E41FFA8597DB95CC0CD11649EEFCD12FA7986BDE607B7B3E5ADA016
<3598:3854>	1CB4D6F728D4691AC4E4151800938E9B0E84611D12FEAE019CBDBA18BC92FB131
<3855:4111>	1C5F47B205053700745BE6F41A577D0CFFAF13FC8C201D186ADCBF1938D2FF64
<4112:4368>	10DBCC761FB64CBCDD4C711E9429B7D569B414CAA0BE9D73FFC44F92C858D460D
<4369:4625>	0976AA35AF38DD3E3174FCD1B255A675F9DB9F207E9A2A40319AFC4B26E9A5F4F
<4626:4882>	01958304A711D17E6BAEE28B38A3025FFA49E4CA9734043700D75AADA300A3756
<4883:5139>	1D51545C55EB181FEB7EC1254510FA7892BA68A5B51BEAF5E2701EB8AD3E5D45E
<5140:5396>	1ACBA523889B35BCF2787E895F299FC18B21F1684971B3801F6EB9755C523D753
<5397:5653>	025DF25BFDDA017C270722FD8ECE514C3D5523C12BEEF948D09F2DB86A1F6C6F5
<5654:5910>	0E4EA184B2EAAAE89D4DB4796B90214FA63A3E187B442795B9F90E51FA7A1772D
<5911:6167>	1B1EC0A0F67D84EF81DCD33440628246183408149490F99623AB24D6087933C14
<6168:6424>	0DC38FCB398267BCE110F601195BCED4418F37DBFE70AB7FD3B117A9638DBDDBB
<6425:6681>	1D75E246A824061BD7A8D9A63EAD746A1CB342126F647C03700E4C63DFFF1C44
<6682:6938>	18F59CD9EEC9091F893AA4698D5032AA9A33BA060138CDD2876CF01BEB3DAD41E
<6939:7195>	19A4FB409965E900EF8C1BF458A5580C0C37B7A6314E6C6C460F75548B1705FB2
<7196:7452>	15E6F65267293079DAF3C6F939F9F9FB49024DBD88EC7A493C02B796A96697202
<7453:7709>	1B8E510833FC8CB4F473BB1596BCA4B304352F124921469F51D9861F76F214D8
<7710: 7966>	09618C7E818265DAA2E01B6BD5DCB69E682D63984BFDFD5D3C4F1DE84507C974C
<7967:8223>	06EAE0750E96FB7B39B130B553CDBCAF7EEBB62DAD92FCB399B6AE92038C41BB
<8224:8480>	0945978072B086F556008111468DBC57A52D5F290C254F571ECF80DDA76E4AC60
<8481:8737>	112DD7F8AF0738ABBDABF05A9B57D0FD8517C5FB2EFE10EACBAC82EA6A1363B7
<8738:8994>	078E938BBD54009EF35E4AEE68E69B290CC376AC6BBF4CAE6F8AD583B30C3E30E

[illegible][illegible]

[illegible]

Index <i:j>	$\mathbf{cx_C<i:j>}$
<5439:5120>	D8842189A284B22F4A2280AB2E8AC26094240B42FB65607F98546D4F7A937A4EAA81D7754ECD6D95
<5119:4800>	C41E5CCC08448F17136DCF847AE662DA973EF3AA401C8F6284473BA4479A19B8935ADD227673CC91
<4799:4480>	E83743D25229C14F1D23F9B602F34961B8B445A45AB96F7091C2CBF6879D3FEA69E36C257C8A45C1
<4479:4160>	AE9448B70C11F0DF828B0CDBDE1CD2EB248B8BCFF8F5F00957757ABEFFA78D63CE2F93529AB90487
<4159:3840>	D2AD3A13BB7DA7C4564C855631A634DB300CEF5B177637E1A6896AD1ECC1E5A7E3ECD4F885371DD1
<3839:3520>	BDD99C3D71B897BCB78572209A167633009A3F57DD63817CC165A6772C791314E4E1843FE8C0BB49E
<3519:3200>	91AFA810EC68AF7BABF3F2043863D3C726F9EFF0E98654BFD6CAA6CBFFFF4FA3458155CF5F2F2D36
<3199:2880>	E676EE055198F485943C18047CFE3BA3BD27274A843358B6A28AEFA235D7F20C8A479652B6F503B51
<2879:2560>	AEA6BE259B27A2FCA63047DD200FAE1D5D7BA7F680439BF69CBC0F177EC4B4E15F62B99A7AA6574F
<2559:2240>	0823230E113B538BC9A70DE6FDCB3B08A41B05287C1C4AC89E8A3E30CE7B785315B9E3FEDAA6E52B
<2239:1920>	EDAFB24490BDA6C57D8BCD1923E003C623FE30D91248FC48552AAEE9025DE6EBE56A594097A7DEFD
<1919:1600>	A97037B7288DAEF345FAB3CC9065EB4FBF3497B61BB92F68D20269CCC34CCE26A53C838F612AF986

Table 175A-5—RS(544,514) codeword C

Index <i:j>	cx _C <i:j>
<5439:5120>	D8842189A284B22F4A2280AB2E8AC26094240B42FB65607F98546D4F7A937A4EAA81D7754ECD6D95
<5119:4800>	C41E5CCC08448F17136DCF847AE662DA973EF3AA401C8F6284473BA4479A19B8935ADD227673CC91
<4799:4480>	E83743D25229C14F1D23F9B602F34961B8B445A45AB96F7091C2CBF6879D3FEA69E36C257C8A45C1
<4479:4160>	AE9448B70C11F0DF828B0CDBDE1CD2EB248B8BCFF8F5F00957757ABEFFA78D63CE2F93529AB90487
<4159:3840>	D2AD3A13BB7DA7C4564C855631A634DB300CEF5B177637E1A6896AD1ECC1E5A7E3ECDAF885371DD1
<3839:3520>	BDD99C3D71B897BCB78572209A16763009A3F57DD63817CC165A6772C791314E4E1843FE8C0BB49E
<3519:3200>	91AFA810EC68AF7BABF3F2043863D3C726F9EFF0E98654BFDB6CAA6CBFFFF4FA3458155CF5F2F2D36
<3199:2880>	E676EE055198F485943C18047CFE3BA3BD272F4A84358B6A28AEFA235D7F20C8A479652B6F503B51
<2879:2560>	AEA6BE259B27A2FCA63047DD200FAE1D5D7BA7F680439BF69CBC0F177EC4B4E15F62B99A7AA6574F
<2559:2240>	0823230E113B538BC9A70DE6FDCB3B08A41B05287C1C4AC89E8A3E30CE7B785315B9E3FEDAA6E52B
<2239:1920>	EDAFB24490BDA6C57D8BCD1923E003C623FE30D91248FC48552AAEE9025DE6EBE56A594097A7DEFD
<1919:1600>	A97037B7288DAEF345FAB3CC9065EB4FBF3497B61BB92F68D20269CCC34CCE26A53C838F612AF986

Table 175A–5—RS(544,514) codeword C (continued)

Index <i:j>	$cx_C<i:j>$
<1599:1280>	F6C6C630C0B7403578B66810C7F51E2F5F0864D75C1ED234AA4F5776D3F4664978C52D0F61A55502
<1279:960>	8B3D4D586157BE62DEA44887FCE2F7A0EB5BF0FA74DC235D95C6F8FBA00159FBA2CD84ADFEECE95A
<959:640>	19863DAE286B6D097103DEA163F5D5BECDD5C23D56C4284B978675CBA43827EA65C774C85F6136235
<639:320>	6D02C1417B48CD1F3E20B4DE3385450239A556CF93232E688A904A300AC18A760A903AE28B432EB9
<319:0>	D226E110C8E288A0F40ED4DEBF74E711C8F1551B28FC86C94FB7CB3438B1661F7C5664234FEB0550

Table 175A–6—RS(544,514) codeword D

Index <i:j>	$cx_D<i:j>$
<5439:5120>	659946559565D97651946559565D956519465D968EC860CDFDAEEC363A248C0EE14667095A304115
<5119:4800>	F35FB56F42F2745A196D754F3D8CE195F0EA6DE6621AAECB0D16B50079BE76DD82099A0DFEDDF4FB
<4799:4480>	12DA206CDD19468B962D1A0AFAFC4DD1886A5196D4E6A7222DDEB6FCC86953BF82F4A7AB2749FE9
<4479:4160>	D74BDD52F1217AD7F8F7851040DF07D6B8E5B111169D35AFF301A547DE2799B662EA26A2321F073
<4159:3840>	9CB26495CF96FAED7B4111377FC8AA090910E2415002ECC0C764CAFFE33335C67A8006926A4494C0
<3839:3520>	72F29755E6481863A58BC55FFFA73A22F7922D9E9EDED6B68B5962AC5048032E109173ABA78C3E9
<3519:3200>	3C778A0B8067D600BFD713EE0B5663F4984DC5367644728755506829D51126B15BA5ACEC68C2CDAA
<3199:2880>	7E8F965C02FD764BEB5329117AE9A2060BF99059076B50615DA8AA8819FB2A57CD4A947D91ED79F2
<2879:2560>	7A12E221EDC3FD47F02642E6CDF1748946593DBB0F84EDC7282A948A7F908776F3A830ABB2216D3A
<2559:2240>	F978B6F13C4FFF1742370506F81F32462181024A69935104F387669CC84300DED045F61CDFFD10E3
<2239:1920>	B77AE5586057865ADF8510A6FB0327FEE23CD6EFC4AB98CA9862CC098DC2C0779BC37C6980283068
<1919:1600>	01BB0C64D9060AA0E9534DCA773F1EF3AFC4808D4BE01AD34B769084FEF237536348AC491626E2F8
<1599:1280>	A10D2BF0C8A2ADB9DECCE37FA5D71E511D2576B8F6A73D0ECFEF5B79B73456F01D80D19C3B000D44
<1279:960>	7B65229D52F1D80EDC31AECF5547D5A54BE285FA1FD35D04853B79255EF293E2CA6D86B1A658FB83
<959:640>	E1B4E48C46D0A6B2C1E5B0BDEC61589EF6ADAC2BDBCCF2F4BC9B671866A3643D71826F7172F264E6
<639:320>	A1F555E90B9A1DB1DB77D47A61D43BF4D0089C7494FB34D34D9D9E8BA3694260E5793D0B2C97676F
<319:0>	578C1EBA0E5B385A08212F7A22D5BF061C6A3D3B3E3BFA869291D6A79AB9ED18FB84A228B61EB096

Table 175A–7—Example PCS Lanes for 1.6TBASE-R PCS

PCSL	Index <i:j>	pcsl_x<i:j>
0	<0:271>	5952646DA6AD9B9B7F713064808ECF0084600846C09BF05F489B8F756AB6886565D6
	<272:543>	EBE5F6FC7D687DB9352D393706ABA00AC029BEF74E5F65B1AF79FE05D6263CE14E76
	<544:815>	5E6E81ED299D7EB63E9086E029F5A5D45E18116805F0E97740429F4C9950F996B451
	<816:1087>	26DEEDC20DA92123E632FA5580C695A763D20E3B8DBC85926210E96F7AE19346DE28
	<1088:1359>	104CBBF86295A66187441388DBDB518DD2DB85C3FD6327293275612DEA83A3BC507C
1	<0:271>	59526420A6AD9BE6A58481195A7B7E1395613956AA8252ED1675F6369D9C2E30A22A
	<272:543>	F4C4DA7FAF2EA81DECB59349019C6DD0C5AF999A536BFDA81EDE98F595F47B68236A
	<544:815>	6388D137E76E7CB7C1F1AEB5A61FFD65D27C9A99C1C8C15204C7475040123AB2D6EF
	<816:1087>	1DF5D792954F127586120C0C37F579349E01FE4283612CCCA49B7B39698232BD2975
	<1088:1359>	E1C43AB2E0302C61CB96F66143CC91FDD342ABAD46F4C737FEDAF7660CBB58F13FF7
2	<0:271>	59526462A6AD9B7F833095807CCF6A1BD371BD37CDBA42F360989683BCE550CB2B44
	<272:543>	AB293D04025A4E8E221692B93541F833F9724F0EAE8E7E7CF96B917442050539E4DF
	<544:815>	4A141D3E381FA601D3D5B4609992191C44D2F40BF63BE22AAC30F6BCB274D6081AB0
	<816:1087>	F891AAD618A48D318AD38B7A6352BB65BCB52F3C1318FD19E81B130BA06F36AF94BC
	<1088:1359>	634EB20B75A5CD6F128211874AF44386E0A17A7C27E59CB28CE9D221D725F714DA15
3	<0:271>	5952645AA6AD9B219EFEF4DE61010B22466216C3577554A6EC85DEA891117DF3430E
	<272:543>	3D1E6F0EABFEA6DFDDDFCE4E19D65B8C8BF9F8E99EAA6D86805BC9FC537007557F1D
	<544:815>	BEABA5DB78A900B7798146EAE8E95D9ADE9970A3945327792884F292E567933C7C9A
	<816:1087>	9090182D1C3AA06594EFF14E6CF0A4D4B34BC46B30304CD0C15553CD388228612A4B
	<1088:1359>	828A9A844A36F145883DA2F1E8F465CCCEB4255784512CB0E00C131C1BEFC5F2925
4	<0:271>	59526487A6AD9B98AB75B067548A4F68BB838968317DFE5298E6F5FD0FBB1954DE22
	<272:543>	848113CAB80C3D6BC6FB69D66CEA4726245AE47412BD0A5969AD393EF45BE6EF4424
	<544:815>	DA0A60A3FC66BFDCD07DD515A89BD793C3566260F554C1ED7B73F238671146E16EC3
	<816:1087>	A6F5EAE66B7553460DE8B6C460709219016EA772CBB54501420BCAF7E6D6B7AB8FAF
	<1088:1359>	41AD8EDC6B8B1B590B5FFAAB9AD9C0FA9C48594D541825B9AD0ED51CDA6A632FB789
5	<0:271>	5952644FA6AD9B72B70D748D48F28B081AB5DC2B9803B2805869E77FD1B5B8CA1B0E
	<272:543>	D7982A1D5E96778703E5D778E6AB57D4F23ACDE9866D22FEE51AD6299EF41A4D6D66
	<544:815>	55053F4E012C9B69BC745C55B19ACF44AF793168F04AC658D1CA3FD7032C7E5993F6
	<816:1087>	F36698025AF7740C0291CF75546BA413944234260C036D4622DEEBB630B8967C0668
	<1088:1359>	880F1D273AF0B7594F03ED3A39DB8208B36E5A1F1C505EDBDAF512873A047CD15A

Table 175A–7—Example PCS Lanes for 1.6TBASE-R PCS (continued)

PCSL	Index <i:j>	pcsl_x<i:j>
6	<0:271>	595264BCA6AD9B77BDC67A88423985F46D5889E03F2F5E3E9DDA91A85B3FC564B07D
	<272:543>	FE7B9BAFD2E1A8CA89EB94BEBDE07F67521B4E5CD3A288D235558CB577D08B743F
	<544:815>	4EB9562A3A697D6446DEB181FE92D94CFEC9065FBBE51E3E44F42E3B5BBCC1332FED
	<816:1087>	0FABB578A6102719486EA8B17023B770CFF25F4967ABB9418142DAD92FA5DED3B770
	<1088:1359>	896177AAFFCDF3A434497E90966116C60F3DB8CF4934005D3F9D8BC1055FD127099E
7	<0:271>	59526444A6AD9B4C949125B36B6EDAB2DB759DF6F5F08C4AD148192B9D842CC75D17
	<272:543>	BCD41B0E5D1EEAA103260BD07EC8333374DE3A1F46216CA89C824F28DCFFAF240A4F
	<544:815>	FE22FC1DC3C32BA0413FEB7EC166F0C590ABF54234278B042E5BB14F99FDE7089CE2
	<816:1087>	AD1A879D18DBA50E328145FB2A1500632D3DF58706D3CD894DB2FCB3FA2BE256306C
	<1088:1359>	1698013DDBD9BC0A9EE1ED541F15B4EE7047BB816E68365B97602702106DED68D275
8	<0:271>	59526406A6AD9BF97831510687CEAEC8B046636E1E50767C5812BEBFF0A969E11CC0
	<272:543>	A1979F0AC59E1D4DAA8488CAE7D69FAC064F313AA313CEFA3001F3BF313FC8235DC7
	<544:815>	C964516C6060946E7C6C12543B4E8F8B4276C90CB6E1251FDEC0A02ABBE87B001001
	<816:1087>	D2CD08F827ED9E03A29A5F37B209FEA1CEEC9E0AC05AC7E389C99B6A9C69F44B754C
	<1088:1359>	32DE6BC913F005ED0DAD32A5F826438A1CB62BF6960946429DA74CAEF47A379198B7
9	<0:271>	595264D6A6AD9B4571DCC3BA8E233CB403A91C6E175C606FD4FA74D657F74C1C6BD8
	<272:543>	824658A3C5B653E13C6BBBF20767E79874F43907FE0C534271D3C8636C201D65C442
	<544:815>	C6A3A0937E23A2D1990510FA788694B8D973F3EA87DB6EF16F67D8725D38CADE8671
	<816:1087>	649BD9D8A9D7D67B44B8193394A4DD0565B2962BA3097F149E5E92034AC0AFE2BC1A
	<1088:1359>	08B5DCD148849685EFE620CA67AF32C4CC79973FF8306DC11D7F0D7917260CCEFCB6
10	<0:271>	5952645FA6AD9B205628E4DFA9D71B9A0D66B1040B62E3D3F50219F1E7415CDF7E1F
	<272:543>	B061522D1F469F1F49613FD40C8DF015DE06AA165FA2DF661D08DBA50186ADD69340
	<544:815>	6A5D52A0CFCD755DFB87892B73050C00FB424711BD5C662F54EF816B10376A205389
	<816:1087>	21F00D842CA5A83D8F318983CE47FCDF292CEA09BBFABAE22578C41BD07C41CE2A28
	<1088:1359>	3FFCD3654886CBF063DDD03B8C1854ADF872B8E895B255EA34C26BBEB462B8EA8E1D
11	<0:271>	59526436A6AD9B8EBB99E37144661C4AC67BB56657F9092EC8BE13536D4B3FBC94B3
	<272:543>	83D2CD5C9D15D2F4E2C554CF91A5B180F0D4A6E573579D5C23A25E3A2CBF1946FF8A
	<544:815>	A8D604974DC5161AC3BA68A51C37375CBADCC3D9D430C99D9DCD338E55FC79BEACE5
	<816:1087>	AC039356CC09A4FB3F196D79E37040AC10D45B7C7789E3A723BB4A2CE5C68EF6AF04
	<1088:1359>	2A221986DEB13AB86A788AE32FD96BAECA2B70A5AFF5C2F20F15FE73F625C8A26514

Table 175A–7—Example PCS Lanes for 1.6TBASE-R PCS (*continued*)

PCSL	Index <i:j>	pcsl_x<i:j>
12	<0:271>	59526418A6AD9BDABA905625456FA9E4146913B04BB44618BBEDE0BA915E89CF735D
	<272:543>	6BDBC6C5285671BA736C906004A47E496FDDA0334474F3E5FD5C96C7B38D2F3A37BB
	<544:815>	CE37270C571460FD931B51BE04105AE2914B41A965D5779B2440621F54ADFF38BED7
	<816:1087>	C18F7F6AFE2409960C8367C9CF1C8DBBC4928CBDDBD08A96338BC0398BFA2C16A538
	<1088:1359>	5D2ED58DB31FCDF77923235E868E6248F1032F2ECB851CD018136E20608D5FAB1114
13	<0:271>	59526428A6AD9B3373163CCC8CE9C3C1B96E6EDF82999D1C4673C1513F930FC4F125
	<272:543>	C721320D5931D993F00E24CF2D7CA09599ADF5927B2448303397D2E07F6486D7B3F4
	<544:815>	F8C543AD74BFF442690AF5E2F47A6EBA98B09BD55D130B86582461B0E055BFE17EF
	<816:1087>	FFC711BD2A95E90042F4CCFCFDF7214851A1EDE20F92EC6D29584378A7C2AD5F4A8
	<1088:1359>	AB877E999F794AB15B8655186FD385EA559C02E90902D24D54A68F118EBBF237F21B
14	<0:271>	5952640BA6AD9B8DAC209A7253DF65E3312CF2FBC88C888A56238C189792ED6C8474
	<272:543>	DAFCB8F824F8A8A76DCC70FE4FCE3A921CE9CBF344B7929B7431D567DDE63B6F917D
	<544:815>	3F345C45A93C991028D701EB64FC7977C9158C0A9B68B1401834083BFE99D88BCED9
	<816:1087>	231EB3894C4EF8C1E6A34A4812F9F9F7D476748D4986EAB0D5FAAB0080F073F61460
	<1088:1359>	1DD5CDF11CFB410F5BA46CD1B2646A2A1AAB91CB28AD326E71D3EAA3C5A0A9C834D7
15	<0:271>	5952642DA6AD9B6A9AA26195655D9E132368937C2BE2125A625000783A5F38B43F3E
	<272:543>	222D7F1E133875CE0DAE091E4848B80C8C2D4BE9E69B5FF7925FFEDE00FDB247B916
	<544:815>	C9561B6BB52E68729E58AD3EF8B5B6FF764C310F2D72C9603149491C84AD4B1CD480
	<816:1087>	F9B3DD1AD9BF458B7E406DEC48CD2C6187D0F4B0EB075D56B04088A3B10F5F17EB0
	<1088:1359>	23A5AB07728DAABF5080275AC59C82F74F34B8E93C29D7DB49D9ED8B72828D30A9A4

Insert new Annex 176A through Annex 176D corresponding to Clause 176 as follows:

Annex 176A

(informative)

SM-PMA test vectors

This annex provides test vectors for the 200GBASE-R, 400GBASE-R, 800GBASE-R, 1.6TBASE-R SM-PMA transmit and receive functions defined in 176.4.2 and 176.4.3.

A file containing the test vectors is available at <https://standards.ieee.org/downloads/802.3/>.

Editor's note (to be removed prior to publication):

The test vectors are given as plain-text files, which can be found at the following link:

https://www.ieee802.org/3/dj/public/24_06/loewenthal_3dj_02_2406.zip.

This file will be placed in the 802.3 downloads folder referenced above when this amendment is published.

Editor's note (to be removed prior to publication):

The format and content of the test vectors is described in the presentation at the following link:

https://www.ieee802.org/3/dj/public/24_06/loewenthal_3dj_01a_2406.pdf

In the subsequent draft, this annex will be updated with content describing the test vectors.

Annex 176B

(normative)

PMA instantiation options for 200 Gb/s, 400 Gb/s, 800 Gb/s, 1.6 Tb/s Physical Layer implementations

176B.1 Scope

This annex provides guidelines for implementing one or more 200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n within a PHY or xMII Extender.

176B.2 Introduction

Each 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R PHY is defined within the associated PMD clause to include PCS, PMA, and PMD sublayers. Some PHYs also include an Inner FEC sublayer in place of the PMA above the PMD (e.g., 800GBASE-FR4 defined in Clause 183 and 800GBASE-LR1 defined in Clause 185). Some other PHYs include an FEC (rather than the RS-FEC provided by the PCS) and PMA specific to the PMD (e.g., 800GBASE-ER1 defined in Clause 187). A set of generic examples are shown in Figure 176B–1. The PMA or Inner FEC below the PCS converts the number of PCS lanes (q) to the required number of PMD lanes (p).

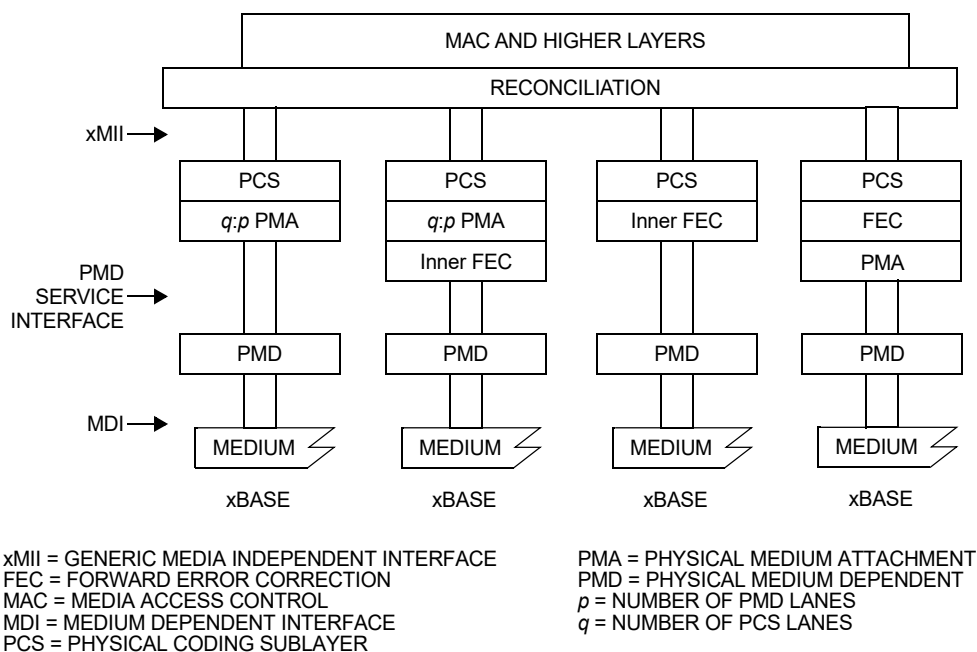


Figure 176B–1—PHYs without physically instantiated interfaces between the PCS and PMD

Within a 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PHY, the interface between the PCS and the PMD may be extended by means of one or two physically instantiated interfaces of

200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n, respectively, where “n” indicates the number of physical lanes. Each xAUI-n is bookended by a pair of PMAs which convert the number of lanes between the sublayer or interface above to the sublayer or interface below.

A set of generic examples of PHYs with one or two xAUI-n are shown in Figure 176B–2. The $q:n$ or $q:m$ PMA below the PCS converts between the number of PCS lanes (q) and the number of xAUI-n lanes (m or n). The $n:p$ PMA above the PMD converts between the number of xAUI-n lanes and the number of PMD lanes (p). The $m:n$ PMA between two xAUI-n converts between the number upper xAUI-n lanes (m) and number of lower xAUI-n lanes (n). This set of PMA and xAUI-n instances, is functionally equivalent to the $q:p$ PMA used in xBASE-R PHYs without xAUI-n as shown in Figure 176B–1.

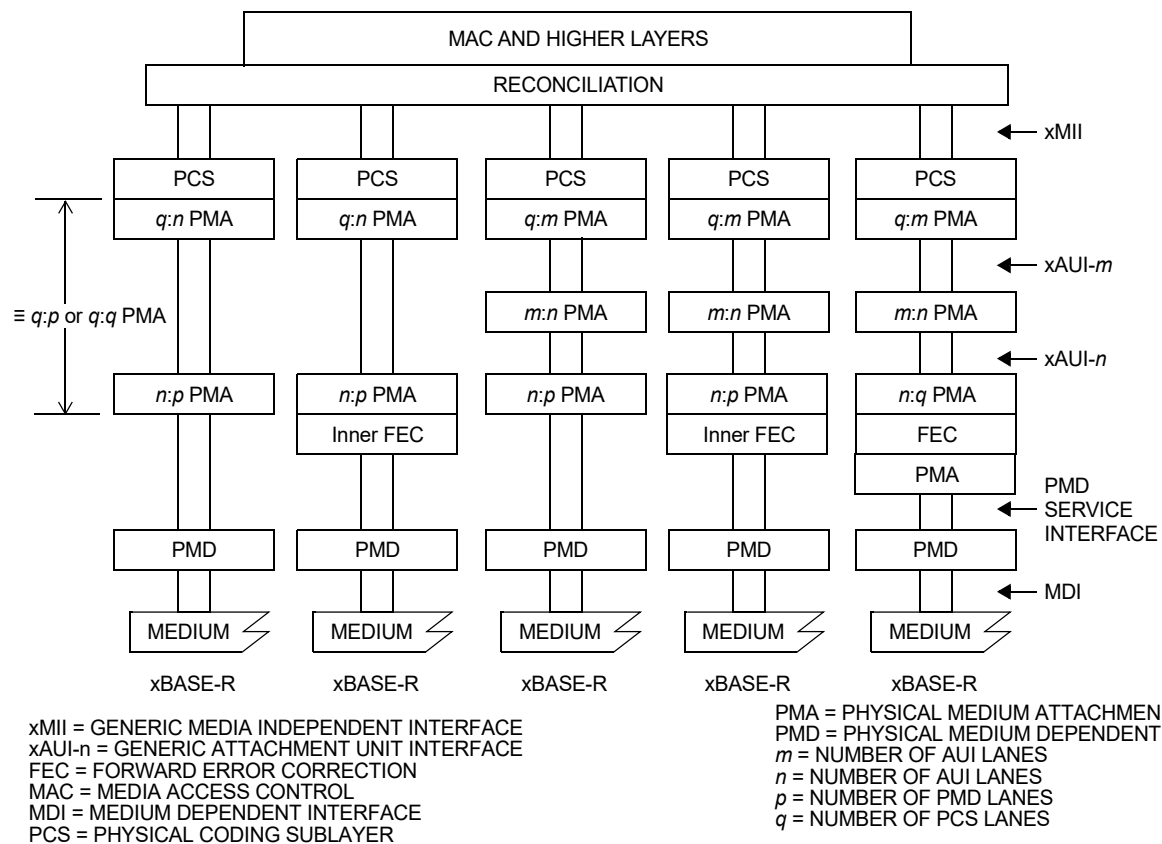
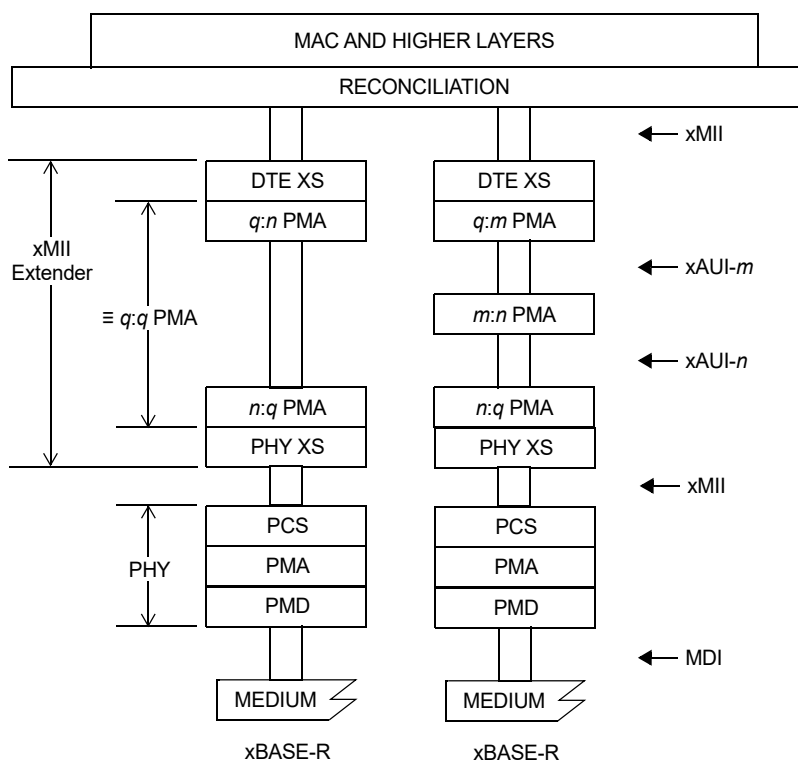


Figure 176B–2—xBASE-R PHYs with one and two physically instantiated interfaces

Within a 200 Gb/s, 400 Gb/s, 800 Gb/s, or 1.6 Tb/s Physical Layer implementation, the interface between the MAC/RS and the PCS may be extended by means of a 200GMII, 400GMII, 800GMII, or 1.6TMII Extender. An xMII Extender includes one or two physically instantiated interfaces 200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n, where “n” indicates the number of physical lanes. Each xAUI-n is bookended by a pair of PMAs, which convert the number of lanes between the sublayer or interface above to the sublayer or interface below.

A set of generic examples of Physical Layer implementations including a xMII Extender with one or two xAUI-n are shown in Figure 176B–3. The $q:n$ or $q:m$ PMA below the DTE XS converts between the number of PCS lanes (q) and the number of xAUI-n lanes (m or n). The $n:q$ PMA above the PHY XS converts

between the number of xAUI- n lanes and the number of PCS lanes (q). The $m:n$ PMA between two xAUI- n converts between the number of upper xAUI- n lanes (m) and number of lower xAUI- n lanes (n). This set of PMA and xAUI- n instances, is functionally equivalent to a $q:q$ PMA.



xMII = GENERIC MEDIA INDEPENDENT INTERFACE
xAUI- n = GENERIC ATTACHMENT UNIT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
 m = NUMBER OF AUI LANES
 n = NUMBER OF AUI LANES
 q = NUMBER OF PCS LANES

Figure 176B-3—xBASE Physical Layer implementations with an xMII Extender with one and two physically instantiated interfaces

Generally, a set of xAUI- n and PMA within an xMII Extender or within a xBASE-R PHY may be referred to as a PMA instantiation. Particular forms of a PMA instantiation may be denoted as follows. A $q:n:p$ (or $q:n:q$) PMA instantiation has q PCS lanes at the top, one xAUI- n with n lanes, and either q PCS lanes or p PMD lanes below. Similarly, a $q:m:n:p$ (or $q:m:n:q$) PMA instantiation has q PCS lanes at the top, one xAUI- n with m lanes, a second xAUI- n with n lanes, and either q PCS lanes or p PMD lanes below.

The $q:n:p$ and $q:m:n:p$ PMA instantiations may take various forms depending upon the number of PCS lanes, the number of lanes on each xAUI- n , and the number of PMD lanes. Details of each supported form are described in 176B.4 through 176B.7.

176B.3 Special Case for 200GBASE-R, 400GBASE-R, and 800GBASE-R PMAs

For 200GBASE-R, 400GBASE-R, and 800GBASE-R PMA, two types of lane multiplexing are defined. Bit multiplexing PMA (BM-PMA) types are defined in Clause 120 and Clause 173. Symbol multiplexing PMA

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interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 200GAUI- n and a B PMD is denoted “SSB”.

Table 176B–1—200 Gb/s PMA 8:{2,4,8}:{2,4} and 8:1:1 (SS or BB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: n PMA	$n = 2$ or 4: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120B (C2C), 120C (C2M) $n = 4$: 120D (C2C), 120E (C2M) $n = 2$: 120F (C2C), 120G (C2M) $n = 1$: 176C (C2C), 176D (C2M)	
200GBASE-R n : p PMA	$p = 2$ or 4: 120 (BM-PMA) $p = 1$: 176 (SM-PMA)	

Table 176B–2—200 Gb/s PMA 8:1:{2,4} and 8:{2,4,8}:1 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: n PMA	$n = 2, 4,$ or 8: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120B (C2C), 120C (C2M) $n = 4$: 120D (C2C), 120E (C2M) $n = 2$: 120F (C2C), 120G (C2M) $n = 1$: 176C (C2C), 176D (C2M)	
200GBASE-R n :8 PMA	$n = 2, 4,$ or 8: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	The combination of n :8 PMA and 8: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: p PMA	$p = 2$ or 4: 120 (BM-PMA) $p = 1$: 176 (SM-PMA)	

Table 176B–3—200 Gb/s PMA 8:{2,4,8}:{2,4,8}:{2,4} and 8:1:1:1 (BBB or SSS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 2, 4, or 8: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 2 or 4: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 1: 176 (SM-PMA)	

Table 176B–4—200 Gb/s PMA 8:1:1:{2,4}, 8:{2,4,8}:{2,4,8}:1 (BBS or SSB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120E (C2M) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 2, 4, or 8: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2 or 4: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	The combination of <i>n</i> :8 PMA and 8: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–5—200 Gb/s PMA 8:1:{2,4,8}:{2,4}, 8:{2,4,8}:1:1 (BSS or SBB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120E (C2M) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2 or 4: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 2 or 4: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 1: 176 (SM-PMA)	

Table 176B–6—200 Gb/s PMA 8:{2,4,8}:1:{2,4} and 8:1:{2,4,8}:1 (SBS or BSB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	The combination of <i>n</i> :8 PMA and 8: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

176B.4.2 8:8 PMA instantiations for 200GMII Extender and 200GBASE-R PHYs

PMA instantiations for use within an 200GMII Extender (see Clause 118) are shown in Figure 176B–2, where the number of PCS lanes q is 8. Instantiations may be of the form $8:n:8$ and $8:m:n:8$. Specific implementations of the PMA instantiations are defined in Table 176B–7 through Table 176B–9.

These instantiations include C2M and C2C 200GAUI- n defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 1, 2, 4, or 8. Interfaces with 2, 4, or 8 lanes require a BM-PMA and interfaces with 1 lane require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 200GAUI- n and one B 200GAUI- n is denoted “SB” or “BS”.

Table 176B–7—200 Gb/s 8:{2,4,8}:8 and 8:1:8 (S or B) PMA instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: n PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120B (C2C), 120C (C2M) $n = 4$: 120D (C2C), 120E (C2M) $n = 2$: 120F (C2C), 120G (C2M) $n = 1$: 176C (C2C), 176D (C2M)	
200GBASE-R $n:8$ PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	

Table 176B–8—200 Gb/s 8:{2,4,8}:{2,4,8}:8 and 8:1:1:8 (SS or BB) PMA instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: m PMA	$m = 2, 4, \text{ or } 8$: 120 (BM-PMA) $m = 1$: 176 (SM-PMA)	
200GAUI- m	$m = 8$: 120B (C2C) $m = 4$: 120D (C2C) $m = 2$: 120F (C2C) $m = 1$: 176C (C2C)	
200GBASE-R $m:n$ PMA	$\{m, n\} = 2, 4, \text{ or } 8$: 120 (BM-PMA) $\{m, n\} = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120C (C2M) $n = 4$: 120E (C2M) $n = 2$: 120G (C2M) $n = 1$: 176D (C2M)	
200GBASE-R $n:8$ PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	

Table 176B–9—200 Gb/s PMA 8:1:{2,4,8}:8 and 8:{2,4,8}:1:8 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	

176B.5 PMA instantiation for 400 Gb/s Physical Layer Implementations

176B.5.1 16:8, 16:4, and 16:2 PMA instantiations for 400GBASE-R PHYs

PMA instantiations for use within a 400GBASE-R PHY are shown in Figure 176B–1, where the number of PCS lanes *q* is 16. Instantiations may be of the form 16:*n*:*p* and 16:*m*:*n*:*p*, where *p* is the number PMD lanes. Specific implementations of the PMA instantiations are defined in Table 176B–10 through Table 176B–15. These instantiations are relevant to the 400GBASE-R PHY types listed in Table 116–3a, Table 116–5, and Table 116–5a, where the number of lanes *p* is 2, 4, or 8.

These instantiations include C2M and C2C 400GAUI-*n* defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (*m* or *n*) is 2, 4, 8, or 16. Interfaces with 4, 8, or 16 lanes require a BM-PMA and interfaces with 2 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 400GAUI-*n* and one B PMD is denoted “SSB”.

Table 176B–10—400 Gb/s PMA 16:{4,8,16}:{4,8}, 16:4:4 and 16:2:2 (SS or BB) instantiation

Sublayer	400 Gb/s Clause/Annex	Notes
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120B (C2C), 120C (C2M) <i>n</i> = 8: 120D (C2C), 120E (C2M) <i>n</i> = 4: 120F (C2C), 120G (C2M) <i>n</i> = 2: 176C (C2C), 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	<i>p</i> = 4: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–11—400 Gb/s PMA 16:2:{4,8} and 16:{4,8,16}:2 (BS or SB) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120B (C2C), 120C (C2M) <i>n</i> = 8: 120D (C2C), 120E (C2M) <i>n</i> = 4: 120F (C2C), 120G (C2M) <i>n</i> = 2: 176C (C2C), 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4: or 8: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–12—400 Gb/s PMA 16:{4,8,16}:{4,8,16} and 16:2:2:2 (BBB or SSS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 2: 176 (SM-PMA)	

Table 176B–13—400 Gb/s PMA 16:2:2:{4,8} and 16:{4,8,16}:{4,8,16}:2 (BBS or SSB) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4 or 8: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–14—400 Gb/s PMA 16:2:{4,8,16}:{4,8} and 16:{4,8,16}:2:2 (SBB or BSS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	{ <i>m</i> , <i>n</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 2 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4 or 8: 120 (BM-PMA) <i>n</i> = 2 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 4 or 8: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 2 176 (SM-PMA)	

Table 176B–15—400 Gb/s PMA and 16:{4,8,16}:2:{4,8} and 16:2:{4,8,16}:2 (BSB or SBS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4 or 8: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

176B.5.2 16:16 PMA instantiations for 400GMII Extender and 400GBASE-R PHYs

PMA instantiations for use within an 400GMII Extender (see Clause 118) are shown in Figure 176B–2, where the number of PCS lanes q is 16. Instantiations may be of the form 16: n :16 and 16: m : n :16. Specific implementations of the PMA instantiations are defined in Table 176B–16 through Table 176B–18.

These instantiations include C2M and C2C 400GAUI- n defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 2, 4, 8, or 16. Interfaces with 4, 8, or 16 lanes require a BM-PMA and interfaces with 2 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 400GAUI- n and one B 400GAUI- n is denoted “SB” or “BS”.

Table 176B–16—400 Gb/s and 16:{4,8,16}:16 and 16:2:16 (B or S) PMA instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: n PMA	$n = 4, 8, \text{ or } 16$: 120 (BM-PMA) $n = 2$: 176 (SM-PMA)	
400GAUI- n	$n = 16$: 120B (C2C), 120C (C2M) $n = 8$: 120D (C2C), 120E (C2M) $n = 4$: 120F (C2C), 120G (C2M) $n = 2$: 176C (C2C), 176D (C2M)	
400GBASE-R n :16 PMA	$n = 4, 8, \text{ or } 16$: 120 (BM-PMA) $n = 2$: 176 (SM-PMA)	

Table 176B–17—400 Gb/s 16:{4,8,16}:{4,8,16}:16 and 16:2:2:16 (SS or BB) PMA instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: m PMA	$m = 4, 8 \text{ or } 16$: 120 (BM-PMA) $m = 2$: 176 (SM-PMA)	
400GAUI- m	$m = 16$: 120B (C2C) $m = 8$: 120D (C2C) $m = 4$: 120F (C2C) $m = 2$: 176C (C2C)	
400GBASE-R m : n PMA	$\{m, n\} = 4, 8, \text{ or } 16$: 120 (BM-PMA) $\{m, n\} = 2$: 176 (SM-PMA)	
400GAUI- n	$n = 16$: 120C (C2M) $n = 8$: 120E (C2M) $n = 4$: 120G (C2M) $n = 2$: 176D (C2M)	
400GBASE-R n :16 PMA	$n = 4, 8, \text{ or } 16$: 120 (BM-PMA) $n = 2$: 176 (SM-PMA)	

Table 176B–18—400 Gb/s PMA 16:2:{4,8,16}:16 and 16:{4,8,16}:2:16 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	

176B.6 PMA instantiation for 800 Gb/s Physical Layer Implementations

The tables in this subclause summarize the forms of 800 Gb/s PMA instantiations and provide references to the clauses that define each sublayer.

176B.6.1 32:4 and 32:8 PMA instantiations for 800GBASE-R PHYs

PMA instantiations for use within a 800GBASE-R PHY are shown in Figure 176B–1, where the number of PCS lanes *q* is 32. Instantiations may be of the form 32:*n*:*p* and 32:*m*:*n*:*p*, where *p* is the number PMD lanes. Specific implementations of the PMA instantiations are defined in Table 176B–19 through Table 176B–24. These instantiations are relevant to the 800GBASE-R PHY types listed in Table 169–2 and Table 169–3, where the number of PMD lanes *p* is 4 or 8. These instantiations are also relevant to the 800GBASE-R PHY types listed in Table 169–3a.

These instantiations include C2M and C2C 800GAUI-*n* defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUl lanes (*m* or *n*) is 4 or 8. Interfaces with 8 lanes require a BM-PMA and interfaces with 4 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 800GAUI-*n* and a B PMD is denoted “SSB”.

Table 176B–19—800 Gb/s PMA 32:4:4 and 32:8:8 ($n = p$, SS or BB) instantiation

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–20—800 Gb/s PMA 32:4:8 and 32:8:4 ($n \neq p$, SB or BS) instantiation

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

Table 176B–21—800 Gb/s PMA 32:4:4:4 and 32:8:8:8 ($m = n$, $n = p$, SSS or BBB) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–22—800 Gb/s PMA 32:4:4:8 and 32:8:8:4 ($m = n$, $n \neq p$, SSB or BBS) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of n :32 PMA and 32: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

Table 176B–23—800 Gb/s PMA 32:4:8:8 and 32:8:4:4 ($m \neq n$, $n = p$, SBB or BSS) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–24—800 Gb/s PMA 32:4:8:4 and 32:8:4:8 ($m \neq n$, $n \neq p$, SBS or BSB) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of n :32 PMA and 32: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

176B.6.2 32:32 PMA instantiations for 800GMII Extender and 800GBASE-R PHYs

PMA instantiations for use within an 800GMII Extender (see Clause 171) are shown in Figure 176B–3, where the number of PCS lanes q is 32. Instantiations may be of the form 32: n :32 and 32: m : n :32. These instantiations are also relevant to the 800GBASE-R PHY type defined in Clause 185 and shown (with Inner FEC) in Figure 176B–2. Specific implementations of the PMA instantiations are defined in Table 176B–25 through Table 176B–27.

These instantiations include C2M and C2C 800GAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 4 or 8. Interfaces with 8 lanes require a BM-PMA and interfaces with 4 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 800GAUI- n and one B PMD is denoted “SB” or “BS”.

Table 176B–25—800 Gb/s 32:4:32 and 32:8:32 PMA instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–26—800 Gb/s 32:8:8:32 and 32:4:4:32 ($n = m$) PMA instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–27—800 Gb/s PMA 32:4:8:32 and 32:8:4:32 ($n \neq m$) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

176B.7 PMA instantiations for 1.6 Tb/s Physical Layer Implementations

The tables in this subclause summarize the forms PMA instantiations for 1.6 Tb/s Physical Layer implementations.

176B.7.1 16:8 PMA instantiations for 1.6TBASE-R PHYs

PMA instantiations for use within a 1.6TBASE-R PHY are shown in Figure 176B–1, where the number of PCS lanes q is 16. Instantiations may be of the form 16: n :8 and 16: m : n :8. Specific implementations of the PMA instantiations are defined in Table 176B–28 and Table 176B–29. These instantiations are relevant to

the 1.6TBASE-R PHY types listed in Table 174–2 and Table 174–3, where the number of PMD lanes p is 8 or 8.

These instantiations include C2M and C2C 1.6TAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 8 or 16. All interfaces require an SM-PMA.

Table 176B–28—1.6 Tb/s 16:16:8 and 16:8:8 PMA instantiations

Sublayer	Clause/Annex	Notes
16: n PMA	176	
1.6TAUI- n	$n = 16$: 120F (C2C), 120G (C2M) $n = 8$: 176C (C2C), 176D (C2M)	
n :8 PMA	176	

Table 176B–29—1.6 Tb/s 16:16:16:8, 16:16:8:8, 16:8:16:8, and 16:8:8:8 PMA instantiations

Sublayer	Clause/Annex	Notes
16: m PMA	176	
1.6TAUI- m	$m = 16$: 120F (C2C) $m = 8$: 176C (C2C)	
m : n PMA	176	
1.6TAUI- n	$n = 16$: 120G (C2M) $n = 8$: 176D (C2M)	
n :8 PMA	176	

176B.7.2 16:16 PMA instantiations for 1.6TMII Extender

PMA instantiations for use within a 1.6TMII Extender (see Clause 171) are shown in Figure 176B–3, where the number of PCS lanes q is 16. Instantiations may be of the form 16: n :16 and 16: m : n :16. Specific implementations of the PMA instantiations are defined in Table 176B–30 and Table 176B–31.

These instantiations include C2M and C2C 1.6TAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 8 or 16. All interfaces require an SM-PMA.

Table 176B–30—1.6 Tb/s 16:16:16 and 16:8:16 PMA instantiations

Sublayer	Clause/Annex	Notes
16: <i>n</i> PMA	176	
1.6TAUI- <i>n</i>	<i>n</i> = 16: 120F (C2C), 120G (C2M) <i>n</i> = 8: 176C (C2C), 176D (C2M)	
<i>n</i> :16 PMA	176	

Table 176B–31—1.6 Tb/s 16:16:16:16, 16:16:8:16, 16:8:16:16, and 16:8:8:16 PMA instantiations

Sublayer	Clause/Annex	Notes
16: <i>m</i> PMA	176	
1.6TAUI- <i>m</i>	<i>n</i> = 16: 120F (C2C) <i>n</i> = 8: 176C (C2C)	
<i>m</i> : <i>n</i> PMA	176	
1.6TAUI- <i>n</i>	<i>n</i> = 16: 120G (C2M) <i>n</i> = 8: 176D (C2M)	
<i>n</i> :16 PMA	176	

Annex 176C

(normative)

Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C

176C.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-chip 200 Gb/s one-lane Attachment Unit Interface (200GAUI-1 C2C), 400 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), 800 Gb/s four-lane Attachment Unit Interface (800GAUI-4 C2C), and 1.6 Tb/s eight-lane Attachment Unit Interface (1.6TAUI-8 C2C). These interfaces have specified electrical characteristics, and may optionally be used when designing systems with electrical interconnects.

200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C and 1.6TAUI-8 C2C are physical instantiations of the PMA service interface, as described in 176.2, which enable partitioning a Physical Layer implementation across multiple devices. Within this annex, these interfaces are collectively referenced as 200 Gb/s per lane AUI-C2C.

Figure 176C–1 shows the relationship of the 200 Gb/s per lane AUI-C2C, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. Physical Layer partitioning options using 200 Gb/s per lane AUI-C2C are provided in Annex 176B.

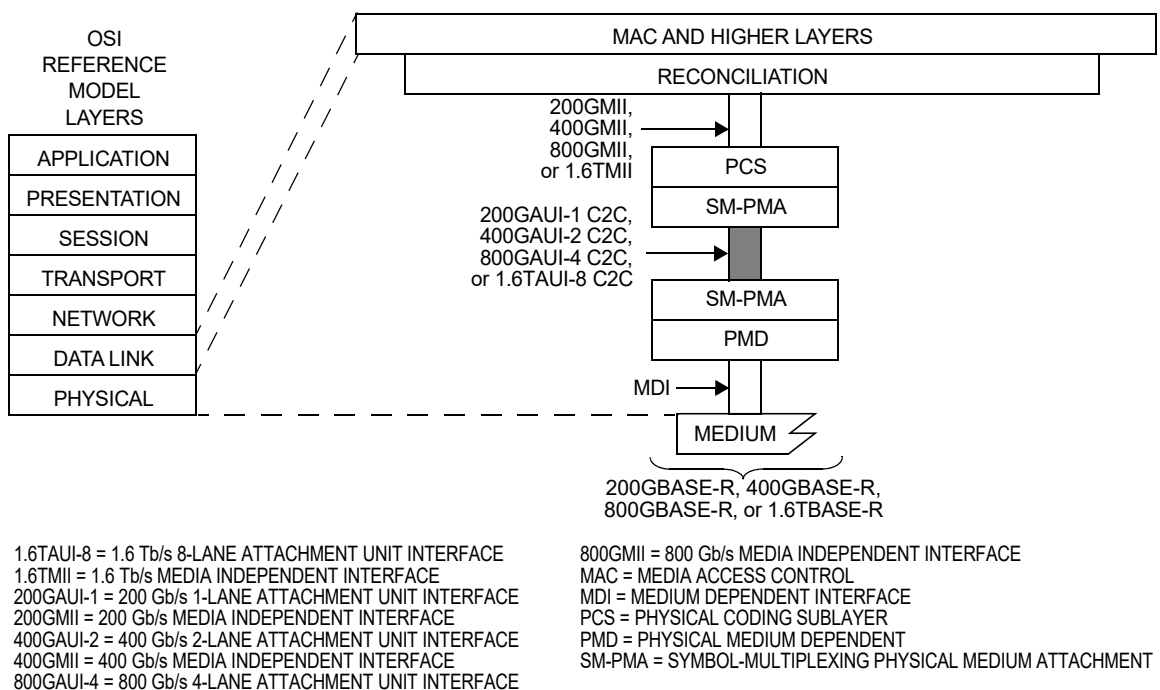


Figure 176C–1—200 Gb/s per lane AUI-C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

176C.2 Error Ratio Allocation

With a compliant input signal, a C2C component receiver (see 176C.6.4) is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the C2C component using the method described in 174A.8, with BER_{added} equal to 2.841×10^{-4} .

176C.3 Functional specification

A 200 Gb/s per lane AUI-C2C is a bidirectional data path. Each direction of a 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C or 1.6TAUI-8 C2C data path contains one, two, four, or eight lanes, respectively

The 200 Gb/s per lane AUI-C2C bidirectional link is described in terms of two C2C components that are connected by a C2C channel.

An n -lane C2C component is functionally equivalent to a corresponding n -lane PMD specified in Clause 178 (see 178.8) using PAM4 signaling at a nominal signaling rate of 106.25 GBd on each lane. The service interfaces are defined in 176C.4. Specifically, a C2C component shall provide the inter-sublayer link training (ILT) function for a Type E1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function is used to request changes to the C2C peer transmitter state (modulation, training pattern, and precoder state), control the transmitter output on each lane, indicate the receiver state, and coordinate the transition to DATA mode.

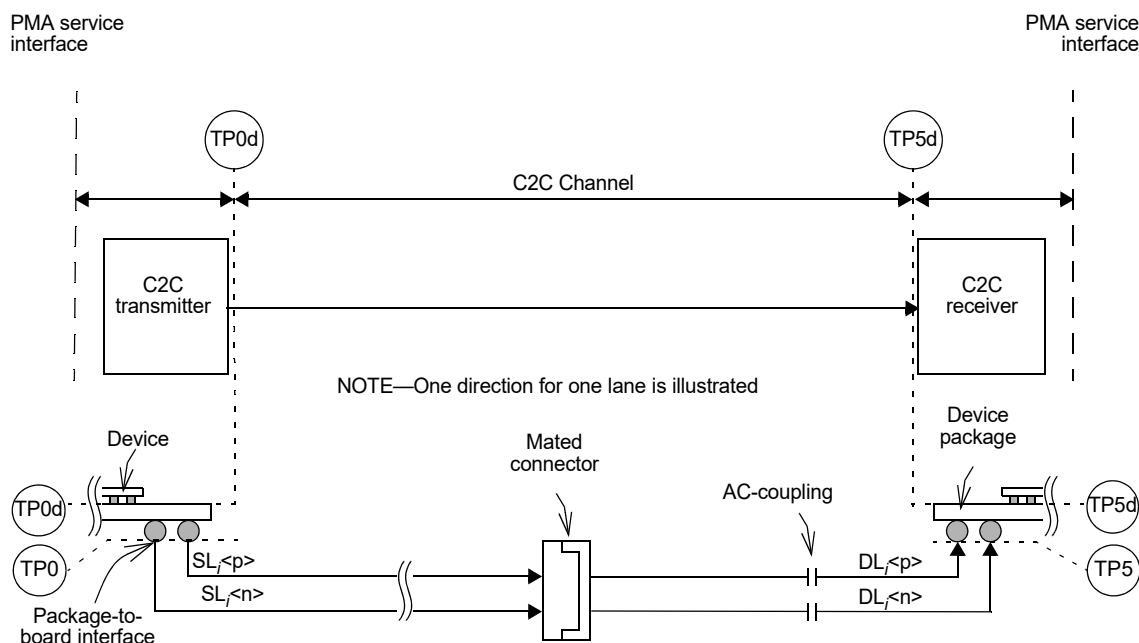
A C2C component transmitter supports the coefficient indexes `k_list` = {−3, −2 −1, 0, 1} and the initial conditions preset 1 through preset 6 and initialize (see Table 176D–9).

Each lane consists of one differential transmission line in each direction between the two C2C components. The transmission lines are AC-coupled within the C2C channel and have a common ground reference. The C2C channel is specified in 176C.7.

The electrical characteristics of the C2C components are specified in 176C.6.3 and 176C.6.4.

NOTE—The electrical specifications of C2C components are not identical to those of the corresponding PMDs defined in Clause 178.

Figure 176C–2 depicts a typical 200 Gb/s per lane AUI-C2C application.



NOTE 1—The number of lanes n is equal to 1 for 200GAUI-1, 2 for 400GAUI-2, 4 for 800 GAUI-4, and 8 for 1.6TAUI-8.
NOTE 2—The C2C channel is defined between TP0d and TP5d, including device packages that are part of the components.
NOTE 3—The source lane (SL) signals $SL_{i<p>}$ and $SL_{i<n>}$ are the positive and negative sides of the transmitter's differential signal pair on lane i and the destination lane (DL) signals $DL_{i<p>}$ and $DL_{i<n>}$ are the positive and negative sides of the receiver's differential signal pair on lane i .

Figure 176C-2—Typical 200 Gb/s per lane AUI-C2C application

This annex defines specifications for two classes of C2C transmitters and two classes of C2C receivers, identified by transmitter package class and receiver package class, respectively. The package is either class A or class B. Devices conform to electrical specifications of either class A or class B. The required characteristics of the electrical interconnect between two devices depend on the transmitter package class on one device and the receiver package class on the other device.

176C.4 Service interfaces

The SM-PMA above the 200 Gb/s per lane AUI-C2C is any $m:1$ PMA for 200GAUI-1, $m:2$ PMA for 400GAUI-2, $m:4$ PMA for 800GAUI-4, and $m:8$ PMA for 1.6TAUI-8, as specified in Clause 176. The PMA below the 200 Gb/s per lane AUI-C2C is any $1:n$ SM-PMA for 200GAUI-1, $2:n$ SM-PMA for 400GAUI-2, $4:n$ SM-PMA for 800GAUI-4, and $8:n$ SM-PMA for 1.6TAUI-8, as specified in Clause 176.

The service interface above and below the 200 Gb/s per lane AUI-C2C is the PMA service interface as specified in 176.2.

The SIGNAL_OK parameter of the PMA:IS_SIGNAL.indication (for an AUI component above the AUI channel) or PMA:IS_SIGNAL.request (for an AUI component below the AUI channel) corresponds to the variable training_status of the inter-sublayer training function, as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes are unspecified.

The service interfaces are illustrated in Figure 176C–3.

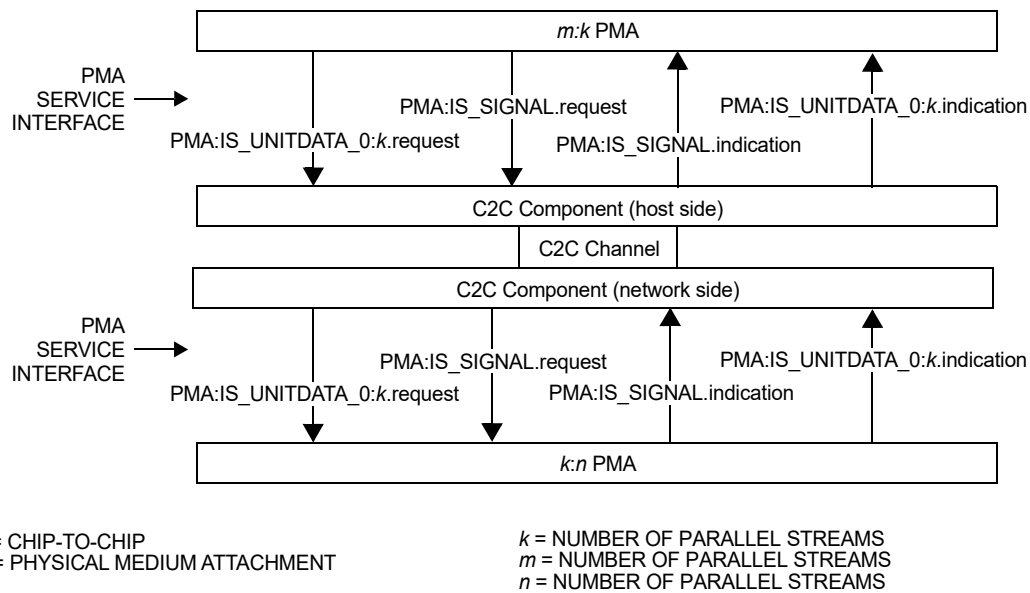


Figure 176C–3—Inter-sublayer service interfaces for a 200GAUI-1, 400GAUI-2, 800GAUI-4, or 1.6TAUI-8 C2C

176C.5 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by a 200 Gb/s per lane AUI-C2M component shall be no more than the maximum delays listed in Table 176C–1.

Table 176C–1—Delay constraints

Component	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GAUI-1	9 216	18	46.08
400GAUI-2	18 432	36	46.08
800GAUI-4	36 864	72	46.08
1.6TAUI-8	73 728	144	46.08

^a One bit time is equal to 5 ps for 200GAUI-1, 2.5 ps for 400GAUI-2, 1.25 ps for 800GAUI-4, and 625 fs for 1.6TAUI-8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GAUI-1, 1.28 ns for 400GAUI-2, 640 ps for 800GAUI-4, and 256 ps for 1.6TAUI-8. See 31B.2 for the definition of pause_quanta.

Descriptions of overall system delay constraints can be found in 116.4 for 200GAUI-1 and 400GAUI-2, in 169.4 for 800GAUI-4, and in 174.4 for 1.6TAUI-8.

176C.6 Electrical characteristics

176C.6.1 Compliance points

The electrical characteristics for C2C components are defined at test points TP0v and TP5v, described in 178.9.2.1 and 178.9.3.1.

Some of the measurements at the compliance points are compared to calculated reference results that account for a reference package, using the procedure in Annex 163A but replacing the COM-related calculation in Annex 93A with those of Annex 178A. The reference package parameters for TP0v and TP5v measurements are selected from Table 176C–7 according to the transmitter package class and receiver package class, respectively, to which a device claims compliance.

176C.6.2 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

176C.6.3 Transmitter electrical characteristics

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used for all transmitter signal measurements, unless otherwise specified.

A C2C component's transmitter on each lane shall meet the specifications given in Table 176C–2.

Table 176C–2—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate (range)		106.25 $\pm$ 50 ppm	GBd
Differential peak-to-peak output voltage (max)	176D.8.1	0.035	V
Transmitter disabled		1	V
Transmitter enabled			
DC common-mode voltage (range)	176D.8.1	0.2 to 1	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	176C.6.3.2	0.032	V
Signal to AC common-mode noise ratio, $SCMR$ (min)	176C.6.3.3	15	dB
Difference effective return loss, $dERL$ (min)	176C.6.3.5	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	178.9.2.3	3.25	dB
Difference steady-state voltage, dv_f (range)	178.9.2.4	0 to 0.1	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163A.3.2.1	0	—
Level separation mismatch ratio, R_{LM} (min)	179.9.4.2	0.95	—

Table 176C–2—Transmitter electrical characteristics at TP0v (*continued*)

Parameter	Reference	Value	Units
Output waveform			
absolute value of step size for all taps (min)	179.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	179.9.4.1.4	0.025	—
value at min state for c(–3) (max)	179.9.4.1.5	–0.05	—
value at max state for c(–3) (min)	179.9.4.1.5	0	—
value at min state for c(–2) (max)	179.9.4.1.5	0	—
value at max state for c(–2) (min)	179.9.4.1.5	0.1	—
value at min state for c(–1) (max)	179.9.4.1.5	–0.3	—
value at max state for c(–1) (min)	179.9.4.1.5	0	—
value at min state for c(0) (max)	179.9.4.1.5	0.5	—
value at min state for c(1) (max)	179.9.4.1.5	–0.1	—
value at max state for c(1) (min)	179.9.4.1.5	0	—
Difference signal-to-noise-and-distortion ratio, dSNDR (min)	178.9.2.7	0	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176C.6.3.4	28	dB
Output jitter (max)	176C.6.3.6		
J_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
$J4u_{03}$			
Tx package Class A		0.118	UI
Tx package Class B		0.12	UI

176C.6.3.1 Transmitter equalization settings

The transmit function includes programmable equalization to compensate for the frequency-dependent loss of the channel and to facilitate data recovery at the receiver. The functional model for the transmit equalizer is the five-tap transversal filter shown in Figure 176C–5. Transmitter output waveform specifications are provided in Table 176C–2. The coefficient initial values in Table 176D–9 are used instead of the values in Table 179–8.

The transmitter output equalization is characterized using the methods specified in 179.9.4, unless specified otherwise. The transmitter output may be manipulated using the Type E inter-sublayer link training (ILT) function as described in 178B.11.



The signal to AC common-mode noise ratio shall meet the specification for *SCMR* (min) in Table 176C-2.

176C.6.3.4 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is defined by the method specified in 179.9.4.3.

Signal-to-residual-intersymbol-interference ratio shall meet the specification for SNR_{ISI} (min) in Table 176C–2.

176C.6.3.5 Transmitter difference effective return loss (ERL)

The transmitter difference ERL at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 176C–3 and Table 176C–7, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v. The reference value, $ERL^{(ref)}$, is calculated based on the transmitter package class to which the device adheres.

Table 176C–3—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Tukey window flag	tw	1	—

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) specified in Table 176C–2.

176C.6.3.6 Output jitter

The output jitter parameters J_{RMS} , J_{4u03} , and EOJ_{03} are defined in 179.9.4.6.

176C.6.4 Receiver characteristics

A receiver on each lane shall meet the specifications given in Table 176C–4

Table 176C–4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate (range)	176C.6.4.1	106.25 ± 50 ppm	GBd
Amplitude tolerance ^a	176C.6.4.2	0.5	V
Difference effective return loss, $dERL$ (min)	176C.6.4.3	–3	dB
Differential-mode to common-mode return loss, RL_{cd}	176C.6.4.4	Equation (176C–1)	dB
Interference tolerance	176C.6.4.5	Table 176C–5	—
Jitter tolerance	176C.6.4.6	Table 179–12	—

^a Specified as the steady-state voltage $v_f^{(meas)}$ (as defined in 178.9.2.4) measured at the test transmitter’s output.

176C.6.4.1 Receiver signaling rate

The receiver shall comply with the requirements of 176C.6.4.5 and 176C.6.4.6 for any signaling rate in the range specified in Table 176C–4.

176C.6.4.2 Receiver amplitude tolerance

When a receiver is connected to a compliant transmitter that has a steady-state voltage (as defined in 179.9.4.1.2) equal to the amplitude tolerance listed in Table 176C–4, using a channel with the minimum insertion loss specified in 178.9.3.4, the receiver operation shall enable a block error ratio as specified in 176C.2.

The receiver is allowed to control the transmitter equalizer coefficients, using the ILT function (see 176C.3) or an equivalent process, to meet these requirements.

NOTE—Steady-state voltage is defined with preset 1. It is not initially generated by a transmitter, due to the initialize setting in Table 176D–9. The receiver is not required to tolerate preset 1 unless it specifically requests it.

176C.6.4.3 Receiver difference ERL

The receiver difference ERL at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 176C–3 and Table 176C–7, and with the value of T_{fx} equal to twice the delay from TP5 to TP5v. The reference value, $ERL^{(ref)}$, is calculated based on the receiver package class to which the device adheres.

Receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 176C–4.

176C.6.4.4 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (176C–1). The differential-mode to common-mode return loss limit is illustrated by Figure 176C–6.

$$RL_{cd}(f) \geq \begin{cases} 25 - 0.36f & 0.05 \leq f \leq 27.8 \\ 15 & 27.8 < f \leq 60 \end{cases} \quad (176C-1)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

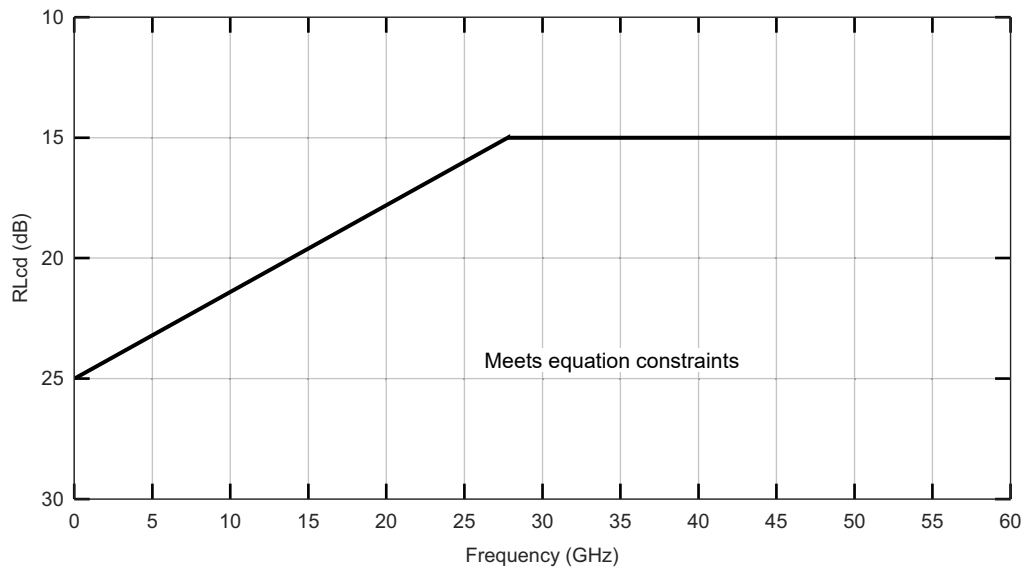


Figure 176C-6—Receiver differential-mode to common-mode return loss limit

176C.6.4.5 Receiver interference tolerance

The receiver on each lane shall meet the expected block error ratio specified in 176C.2 with channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 176C-5.

176C.6.4.5.1 Test setup

The interference tolerance test is performed with the setup shown in Figure 93C-2 or its equivalent. Characterization and calibration of the various elements in the test setup uses the test configurations in Figure 93C-3, Figure 93C-4, Figure 93C-5, and Figure 93C-6. In each figure TP0v replaces TP0a and TP5v replaces TP5a.

The transmitter meets the requirements stated in 176C.6.3 with the added constraint that the differential peak-to-peak voltage (see 93.8.1.3) is less than or equal to 0.8V for any transmitter equalizer setting. The ISI channel emulates the frequency dependent loss of the channel. The channel noise source emulates crosstalk, noise, and any other non-equalizable signal distortions that may be introduced by a transmitter or channel.

The ERL of the test setup in Figure 93C-4 measured at TP5 replica towards TPt meets the requirements in 176C.7.3.

The channel noise source has an adjustable output amplitude that is set according to the calibration procedure in 176C.6.4.5.2. The noise produced by the channel noise source is measured directly at the output of the noise source (see Figure 93C-6). The noise is Gaussian with a crest factor of at least 5. The noise spectral density, $N_s(f)$, is normalized and constrained according to the relations in Equation (93C-1), where f_b is the signaling rate and f_l is 1 GHz. The average noise spectral density, $NSD_{average}$, is defined by Equation (93C-2).

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used to measure the output of the transmitter and the channel noise source.

176C.6.4.5.2 Noise calibration

The output amplitude of channel noise source is set to yield a target test channel Channel Operating Margin (COM). The test channel COM is derived from the S-parameters for the test channel from TPt to TP5 replica $S^{(c)}$ measured using the test setup in Figure 93C-4. COM is calculated according to Annex 178A using the parameter values in Table 176C-7 and Table 176C-8 for the corresponding receiver class but with the following exceptions.

The transmitter model is determined in one of the following ways.

- a) If the transmitter is a device with known S-parameters and transition time T_r , these parameters are used instead of the transmitter device package model in 178A.1.4, and the $ILdd$ in Table 176C-5 is the $ILdd$ of the concatenated S-parameters of the device and the channel between TPt and TP5 replica. T_r should be provided as the value at the input of the device S-parameters network, as defined in 120G.3.1.4 but with no observation filter.
- b) If a calibrated instrument-grade transmitter is used, the TP0 to TP0v trace in Figure 93C-2 and Figure 93C-3 and the TP0 to TP0v replica trace in Figure 93C-4 are omitted and the transmitter model $S^{(t)}$ is omitted from Equation (178A-2). In this case, the $ILdd$ in Table 176C-5 is the $ILdd$ of the channel between TPt and TP5 replica. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A-10) uses the filter $H_i(f)$ defined in 178A.1.6.2, where T_r is the measured transmitter transition time (see 120G.3.1.4).

In addition, the following parameters are derived from measurements of the test transmitter collected using the test setup in Figure 93C-3.

- a) The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 55$.
- b) The parameter R_{LM} is set to the measured value of R_{LM} .
- c) The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{4u03} and J_{RMS} using Equation (178-2) and Equation (178-3) respectively.

The interference noise (see 178A.1.5) is derived from the S-parameters for the noise path from TPn to TP5 replica $S^{(n)}$ measured using the test setup shown in Figure 93C-5. For a given RMS noise amplitude σ_{ns} , the spectral density of the channel noise source $N_s(f)$ may be assumed to be $2\sigma_{ns}^2/f_b$ from DC to $f_b/2$ and 0 for frequencies greater than $f_b/2$. The spectral density measured at the output of the channel noise source may also be used as $N_s(f)$ for the calculation of COM and, in such cases, the value of σ_{ns}^2 is the integral of $N_s(f)$ over frequency.

COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 176C-7 on the receiver side. The value of COM is taken as the lower of the two calculated values. The value of σ_{ns} that yields the target value of COM specified in Table 176C-5 is determined (via iterative calculation or by other means). The amplitude of the channel noise source is then configured so that the measured RMS value at its output matches σ_{ns} .

176C.6.4.5.3 Test procedure

The receiver of each lane is tested using the following procedure.

- 1) Calibrate the amplitude of the channel noise source using the procedure defined in 176C.6.4.5.2.
- 2) Allow the receiver under test to configure the test transmitter equalizer using the ILT function (see 178.8.9) or any other method that yields the same result.
- 3) Configure the test transmitter to transmit either scrambled idle or PRBS31Q.

Verify that the block error ratio requirements are met using the test methods defined in 174A.8.1 or 174A.10.1.

Table 176C–5—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
Block error ratio ^a	$< 1.45 \times 10^{-11}$						—
Insertion loss, <i>ILdd</i> , at 53.125 GHz ^b			—			—	dB
Receiver package class A	9.5	10.5		25.5	26.5		
Receiver package class B	9.5	10.5		22	23		
COM including effects of broadband noise ^c	—	—	3	—	—	3	dB

^a The block error ratio (see 174A.8) is measured with BER_{added} specified in 176C.2

^b *ILdd* measured between TPt and TP5 (see Figure 93C–4) plus *ILdd* of the specific package used by the test transmitter. See 178.9.3.4.2.

^c The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

176C.6.4.6 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 179–12. The test setup shown in Figure 93–12, or its equivalent, is used. The test channel meets the insertion loss requirement for Test 2 in Table 176C–5. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 176C.6.4.5, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items d) through g) in 176C.6.4.5, is at least 3 dB.
- For the COM parameter calibration described in 176C.6.4.5 item e), the test channel transmitter J_{RMS} and J_{4u} values are measured with the jitter frequency and amplitude set according to Case F from Table 179–12.

The receiver under test shall meet the block error ratio in Table 176C–5 for each case in Table 179–12.

176C.7 Channel characteristics

Table 176C–6 provides a summary of the channel characteristics for 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C and 1.6TAUI-8 C2C interfaces, and references to the subclauses addressing each parameter.

Table 176C–6—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	176C.7.1	3	dB
Maximum insertion loss at 53.125 GHz (recommended)	176C.7.2	32	dB
Minimum ERL	176C.7.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	176C.7.4	Equation (178–5)	dB
Maximum AC-coupling 3 dB corner frequency	176C.7.5	50	kHz

176C.7.1 Channel Operating Margin

The Channel Operating Margin (COM) is computed using the procedure in 178A.1 and the parameters in Table 176C–7 and Table 176C–8. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2. This minimum value allocates margin for practical limitations on the receiver implementation as well as the allowed transmitter equalization coefficients. The maximum likelihood sequence detection (MLSD) defined in 178A.1.10 is not included in the calculation of COM.

NOTE—The parameters and values in Table 176C–7 and Table 176C–8 correspond to behavioral models of transmitters and receivers that are compliant to the C2C component specifications in this clause. The purpose of these parameters and values is to compute COM, a channel metric, and they do not represent requirements for transmitter and receiver implementations. It is expected that the C2C component specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation.

Table 176C-7—Device termination and package parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, class A			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1	$z_p^{(1)}$	33	mm
Transmission line 1 length, Test 2	$z_p^{(1)}$	12	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Device package model, class B			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, Test 1, Tx / Rx	$z_p^{(1)}$	45 / 44	mm
Transmission line 1 length, Test 2, Tx / Rx	$z_p^{(1)}$	30 / 29	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Single-ended reference resistance	R_0	50	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package model parameters are based on the reference impedance R_0 . Channel scattering parameters can be measured with different reference impedance and renormalized to match the reference impedance before computation.			

Table 176C–8—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$	0	—
Transmitter equalizer, coefficient –2	$c(-2)$	0	
Minimum value		0.14	—
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient –1	$c(-1)$	–0.34	
Minimum value		0	—
Maximum value		0.02	
Step size			
Transmitter equalizer, coefficient 0	$c(0)$	0.54	—
Minimum value			
Transmitter equalizer, coefficient 1	$c(1)$	–0.2	
Minimum value		0	—
Maximum value		0.02	
Step size			
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		–20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		–6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
One-sided noise spectral density	η_0	10^{-8}	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI

Table 176C–8—COM parameter values (continued)

Parameter	Symbol	Value	Units
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	5	—
Number of fixed-position taps	N_{fix}	14 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	56 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{max}(j)$	0.7 0.05	— —
Normalized lower limit on feed-forward coefficient $w(j)$ $1 \leq j \leq d_w$ and $d_w+2 \leq j \leq N_{fix}$ $N_{fix} < j$	$w_{min}(j)$	–0.7 –0.05	— —
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	0.67×10^{-5}	—
Include maximum likelihood sequence detection		no	—

^a The number of fixed-position post-cursor taps is 8.

^b The latest post-cursor position for a floating tap is 50.

176C.7.2 Channel insertion loss (recommended)

The recommended maximum channel insertion loss, IL_{dd} , is specified in Table 176C–6.

176C.7.3 Channel ERL

The channel ERL at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 176C–9. Parameters that do not appear in Table 176C–9 take values from Table 176C–7. Channel ERL at TP0 and at TP5 shall meet the requirement in Table 176C–6.

Table 176C–9—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	4000	UI
Equalizer length associated with reflection signal	N_{bx}	16	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

176C.7.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet the requirements of 178.10.4.

176C.7.5 Channel AC-Coupling

Channel AC-coupling is specified in 178.10.6.

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**176C.8 Protocol implementation conformance statement (PICS) proforma
for Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C,
400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C⁶**

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

176C.8.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

176C.8.2 Identification

176C.8.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

176C.8.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 176C, Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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176C.8.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	176C.1	One independent data path in each direction for 200GAUI-1 C2C, two independent data paths in each direction for 400GAUI-2 C2C, four independent data paths in each direction for 800GAUI-4 C2C, and eight independent data paths in each direction for 1.6TAUI-8 C2Ce	M	Yes ☐
*CHAN	Channel	176C.7	Items marked with CHAN include channel specifications not applicable to a PHY manufacturer	O	Yes ☐ No ☐
*PCSP KG	PMA in same package as PCS.			O	Yes ☐ No ☐

176C.8.4 PICS proforma tables for Chip-to-chip Attachment Unit Interfaces 200GAUI-1 C2C, 400GAUI-2 C2C, 800GAUI-4 C2C, and 1.6TAUI-8 C2C

176C.8.4.1 Transmitter

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMA is in same package as PCS	176C.6.3	Per Table 176C–2	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMA is not in same package as PCS	176C.6.3	Derived from PMA input lanes	!PCSP KG:M	Yes [] N/A []
TC3	Peak-to-peak differential output voltage, transmitter enabled	176C.6.3	Per Table 176C–2	M	Yes []
TC4	Peak-to-peak differential output voltage, transmitter disabled	176C.6.3	Per Table 176C–2	M	Yes []
TC5	DC common-mode voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC6	Low-frequency peak-to-peak AC common-mode voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC7	Signal to AC common-mode noise ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC8	Difference ERL	176C.6.3	Per Table 176C–2	M	Yes []
TC9	Common-mode to common-mode return loss	176C.6.3	Per Table 176C–2	M	Yes []
TC10	Output waveform coefficient limits	176C.6.3	Per Table 176C–2	M	Yes []
TC11	Output waveform coefficient step size	176C.6.3	Per Table 176C–2	M	Yes []
TC12	Output jitter	176C.6.3	Per Table 176C–2	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC15	Transmitter precoder request	176C.1		O	Yes [] No []
TC16				M	Yes []
TC17	Difference steady-state voltage	176C.6.3	Per Table 176C–2	M	Yes []
TC18	Difference linear fit pulse peak ratio	176C.6.3	Per Table 176C–2	M	Yes []
TC19	Level separation mismatch ratio	176C.6.3	Per Table 176C–2	M	Yes []

176C.8.4.2 Receiver

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Difference ERL	176C.6.4	Per Table 176C–4	M	Yes []
RC2	Differential-mode to common-mode return loss	176C.6.4	Per Table 176C–4	M	Yes []
RC3	Interference tolerance	176C.6.4.5		M	Yes []
RC4	Jitter tolerance	176C.6.4.6		M	Yes []
RC5	Transmitter equalization feedback	176C.2.4.6		O	Yes [] No []

176C.8.4.3 Channel

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Channel operating margin (COM)	176C.7.1	≥ 3 dB	CHAN :M	Yes [] N/A []
CC2	Channel insertion loss	176C.7.2	Meets <equation> constraints	CHAN :O	Yes [] No [] N/A []
CC2	Channel ERL	176C.7.3	≥ 9.7 dB	CHAN :M	Yes [] N/A []
CC4	Channel AC-coupling	176C.7.5	Per 178.10.6	CHAN :M	Yes [] N/A []

Annex 176D

(normative)

Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M

176D.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-module 200 Gb/s one-lane Attachment Unit Interface (200GAUI-1 C2M), 400 Gb/s two-lane Attachment Unit Interface (400GAUI-2 C2M), 800 Gb/s four-lane Attachment Unit Interface (800GAUI-4 C2M), and 1.6 Tb/s eight-lane Attachment Unit Interface (1.6TAUI-8 C2M). These interfaces have specified electrical characteristics at accessible compliance points, which may be used for building and testing systems with pluggable modules.

The 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M are physical instantiations of the PMA service interface, as described in 176.2, which enable partitioning a Physical Layer implementation across multiple devices. Within this annex, these interfaces are collectively referenced as 200 Gb/s per lane AUI-C2M.

Figure 176D–1 shows the relationship of the 200 Gb/s per lane AUI-C2M, along with other Ethernet sublayers, to the ISO/IEC Open System Interconnection (OSI) reference model. Physical Layer partitioning options using 200 Gb/s per lane AUI-C2M are provided in Annex 176B.

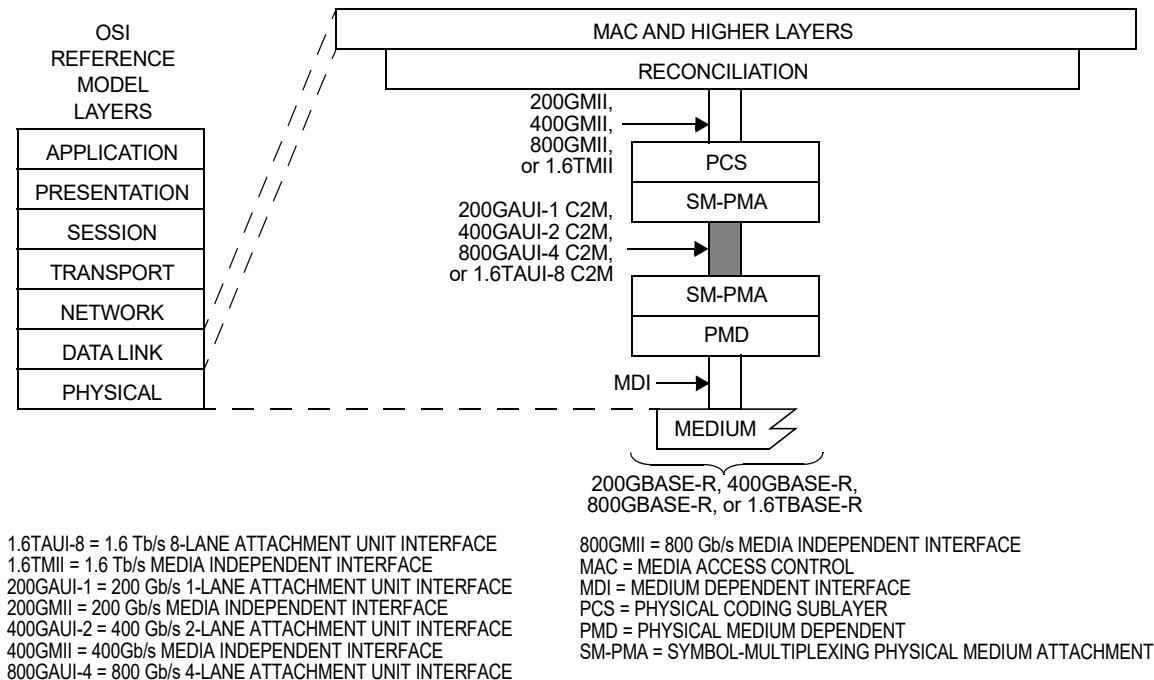


Figure 176D–1—200 Gb/s per lane AUI-C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

176D.2 Error ratio allocation

With a compliant input signal, a C2M component receiver (see 176D.6.5 and 176D.6.6) is expected to meet the block error ratio of 1.45×10^{-11} (see 174A.5), measured at the PMA adjacent to the C2M component using the method described in 174A.8, with BER_{added} equal to 2.681×10^{-4} .

176D.3 Functional specification

A 200 Gb/s per lane AUI-C2M is a bidirectional data path. Each direction of a 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M or 1.6TAUI-8 C2M data path contains one, two, four, or eight lanes, respectively.

A 200 Gb/s per lane AUI-C2M is specified in terms of a host and a module. The host and the module both contain components that implement the interface, referred to as C2M components.

An n -lane C2M component is functionally equivalent to a corresponding n -lane PMD specified in Clause 179 (see 179.8) using PAM4 signaling at a nominal signaling rate of 106.25 GBd on each lane. The service interfaces are defined in 176D.4. Specifically, a C2M component shall provide the inter-sublayer link training (ILT) function for a Type E1 interface, specified in Annex 178B. When the variable `mr_training_enable` is true, the ILT function is used to request changes to the C2M peer transmitter state (modulation, training pattern, and precoder state), control the transmitter output on each lane, indicate the receiver state, and coordinate the transition to DATA mode.

A C2M component transmitter supports the coefficient indexes `k_list` = {−3, −2 −1, 0, 1} and the initial conditions preset 1 through preset 6 and initialize (see Table 176D–9).

Each lane consists of one differential transmission line in each direction between the host C2M component and the module C2M component. The transmission lines are AC-coupled within the module and have a common ground reference.

The electrical characteristics of the host are specified in 176D.6.3 and 176D.6.5. The electrical characteristics of the module are specified in 176D.6.4 and 176D.6.6.

NOTE—The host and module electrical specifications are not equivalent to those of the corresponding PMDs defined in Clause 179. Specifically, the test points at which module compliance is defined are different.

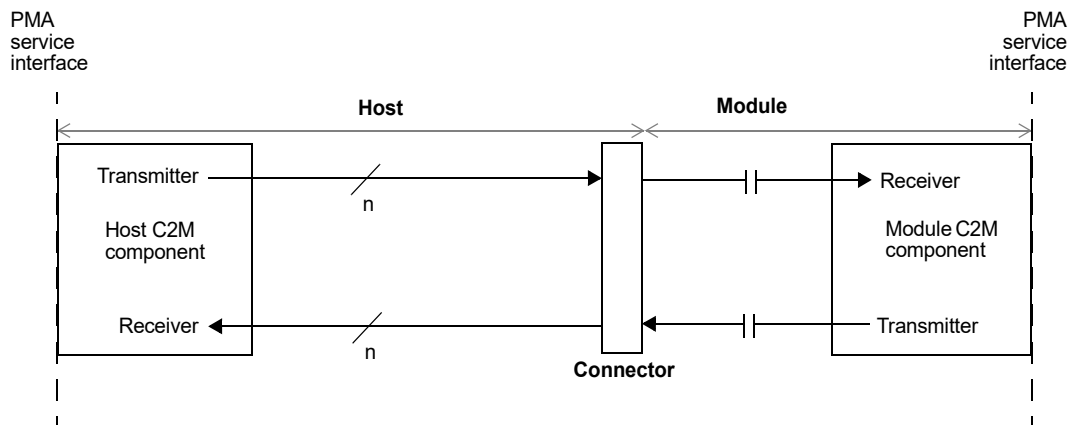
Figure 176D–2 depicts a typical 200 Gb/s per lane AUI-C2M application. The reference characteristics of the channel between the C2M components are provided in 176D.7.

176D.4 Service interfaces

The PMA above the 200 Gb/s per lane AUI-C2M is any $m:1$ PMA for 200GAUI-1, $m:2$ PMA for 400GAUI-2, $m:4$ PMA for 800GAUI-4, and $m:8$ PMA for 1.6TAUI-8, as specified in Clause 176. The PMA below the 200 Gb/s per lane AUI-C2M is any $1:n$ PMA for 200GAUI-1, $2:n$ PMA for 400GAUI-2, $4:n$ PMA for 800GAUI-4, and $8:n$ PMA for 1.6TAUI-8, as specified in Clause 176.

The service interface above and below the 200 Gb/s per lane AUI-C2M is the PMA service interface as specified in 176.2.

The `SIGNAL_OK` parameter of the `PMA:IS_SIGNAL.indication` (for an AUI component above the AUI channel) or `PMA:IS_SIGNAL.request` (for an AUI component below the AUI channel) corresponds to the

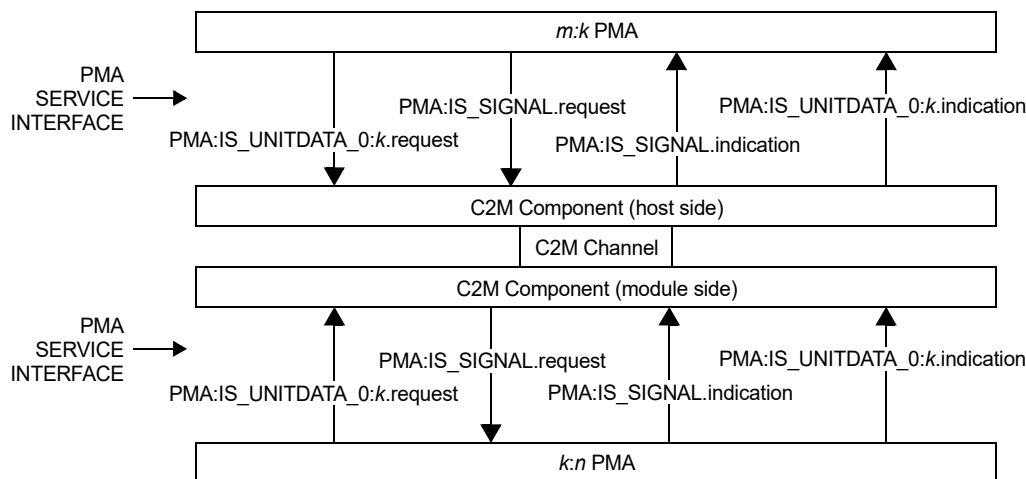


NOTE—The number of lanes n is 1 for 200GAUI-1, 2 for 400GAUI-2, 4 for 800 GAUI-4, and 8 for 1.6TAUI-8.

Figure 176D-2—Components of a 200 Gb/s per lane AUI-C2M

variable training_status of the ILT function, as defined in 178B.14.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the corresponding tx_symbol parameters on all lanes are unspecified.

The service interfaces are illustrated in Figure 176D-3.



C2M = CHIP-TO-MODULE
PMA = PHYSICAL MEDIUM ATTACHMENT

k = NUMBER OF PARALLEL STREAMS
 m = NUMBER OF PARALLEL STREAMS
 n = NUMBER OF PARALLEL STREAMS

Figure 176D-3—Inter-sublayer service interfaces for a 200GAUI-1, 400GAUI-2, 800GAUI-4, or 1.6TAUI-8 C2M

176D.5 Delay constraints

The sum of the transmit and receive delays contributed by a 200 Gb/s per lane AUI-C2M component including the AUI-C2M channel in one direction shall be no more than the maximum value listed in Table 176D–1. It is assumed that the one-way delay through the channel is no more than 5 ns.

Table 176D–1—Delay constraints

Component	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
200GAUI-1	9 216	18	46.08
400GAUI-2	18 432	36	46.08
800GAUI-4	36 864	72	46.08
1.6TAUI-8	73 728	144	46.08

^a One bit time is equal to 5 ps for 200GAUI-1, 2.5 ps for 400GAUI-2, 1.25 ps for 800GAUI-4, and 625 fs for 1.6TAUI-8. See 1.4.160 for the definition of bit time.

^b One pause_quantum is equal to 2.56 ns for 200GAUI-1, 1.28 ns for 400GAUI-2, 640 ps for 800GAUI-4, and 256 ps for 1.6TAUI-8. See 31B.2 for the definition of pause_quanta.

Descriptions of overall system delay constraints can be found in 116.4 for 200GAUI-1 and 400GAUI-2, in 169.4 for 800GAUI-4, and in 174.4 for 1.6TAUI-8.

176D.6 Electrical characteristics

176D.6.1 Compliance points

The electrical characteristics for the C2M host and module are defined at specified compliance points. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters. A Host Compliance Board (HCB) is mechanically equivalent to a module and can be plugged into a host. A Module Compliance Board (MCB) is mechanically equivalent to a host and enables a module to be plugged in.

Figure 176D–4 depicts the location of compliance points in which host characteristics are specified. The test points are separate for each lane. The output of the HCB at TP1a is used to verify the host electrical output compliance. The input of the HCB at TP4a is used to verify the host input compliance.

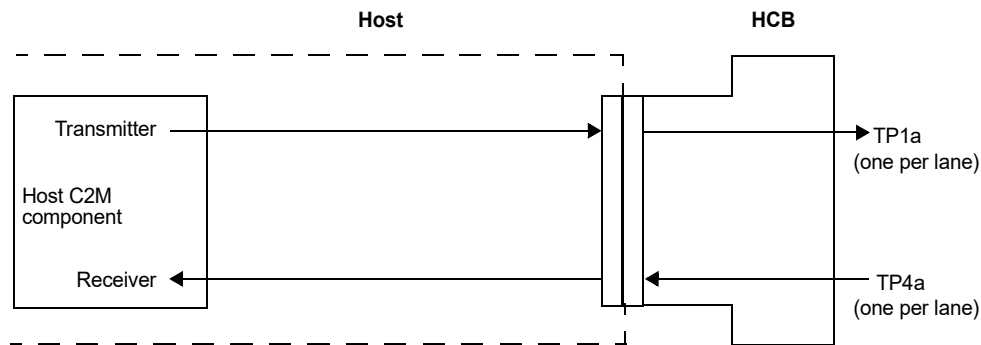


Figure 176D–4—Host compliance points

NOTE 1—TP1a and TP4a correspond to the Clause 179 test points TP2 and TP3, respectively.

Figure 176D–5 depicts the location of compliance points in which module characteristics are specified. The test points are separate for each lane. The output of the MCB at TP4 is used to verify the module electrical output compliance. The input of the MCB at TP1 is used to verify the module input compliance.

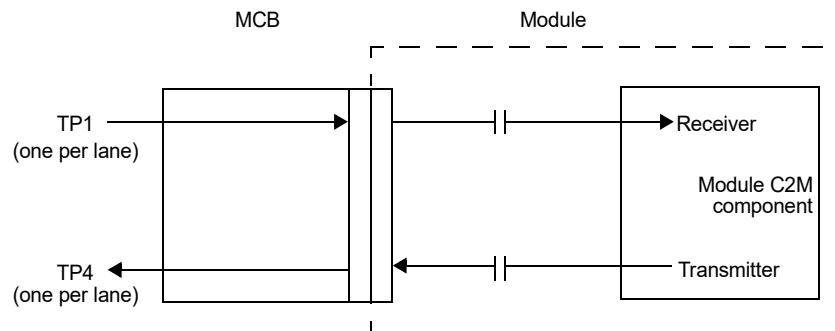


Figure 176D–5—Module compliance points

NOTE 2—The test points defined above are not necessarily equivalent to the outputs of the channel or the inputs of the host and module C2M components.

176D.6.2 HCB/MCB characteristics

HCB characteristics are described in 179B.2 where the HCB performs the equivalent function as the TP2 or TP3 test fixture. The MCB characteristics are described in 179B.3 where the MCB performs the equivalent functionality as the cable assembly test fixture. The mated compliance board characteristics are described in 179B.4.

176D.6.3 Host output characteristics

A host shall meet the host output specifications given in Table 176D–2 on each lane.

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used for all output signal measurements, unless otherwise specified.

Table 176D–2—Summary of host output specifications at TP1a

Parameter	Reference	Value	Units
Signaling rate (range)		106.25 ± 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.1		
Output disabled		0.03	V
Output enabled		1	V
DC common-mode voltage (range)	176D.8.1	0.2 to 1	V
AC common-mode peak-to-peak voltage (max)	176D.8.1		
Low-frequency, $V_{CM_{LF}}$		0.03	V
Full-band, $V_{CM_{FB}}$		0.085	V
Effective return loss, ERL (min)	176D.8.2	7.3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	176D.8.3	Equation (179–19)	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	176D.8.3	Equation (179–20)	dB
Transmitter steady-state voltage, v_f (range)	176D.8.4	0.4 to 0.5	V
Linear fit pulse peak ratio, R_{peak} (min)	176D.8.4	0.123	—
Level separation mismatch ratio, R_{LM} (min)	176D.8.5	0.95	—
Transmitter output waveform			
absolute value of step size for all taps (min)	176D.8.6	0.005	—
absolute value of step size for all taps (max)	176D.8.6	0.025	—
value at minimum state for $c(-3)$ (max)	176D.8.6	–0.06	—
value at maximum state for $c(-2)$ (min)	176D.8.6	0.12	—
value at minimum state for $c(-1)$ (max)	176D.8.6	–0.34	—
value at minimum state for $c(0)$ (max)	176D.8.6	0.5	—
value at minimum state for $c(1)$ (max)	176D.8.6	–0.2	—
Difference signal-to-noise-and-distortion ratio, dSNDR (min)	176D.8.7	0	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176D.8.8	26	dB
Output jitter (max)	176D.8.9		
J_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
$J4u_{03}$		0.135	UI

176D.6.4 Module output characteristics

A module shall meet the module output specifications given in Table 176D–3.

The module output shall be AC-coupled within the module, with low-frequency 3 dB cutoff frequency less than 100 kHz.

A test system with a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz is to be used for all output signal measurements, unless otherwise specified.

Table 176D–3—Summary of module output specifications at TP4

Parameter	Reference	Value	Units
Signaling rate (range) ^a		106.25 ± 50 ppm	GBd
Differential peak-to-peak voltage (max)	176D.8.1		
Output disabled		0.03	V
Output enabled		1	V
DC common-mode voltage tolerance (range)	176D.8.1	0.15 to 1.05	V
AC common-mode peak-to-peak voltage (max)	176D.8.1		
Low-frequency, $V_{CM_{LF}}$		0.015	V
Full-band, $V_{CM_{FB}}$		0.06	V
Effective return loss, ERL (min)	176D.8.2	8.5	dB
Common-mode to common-mode return loss, RL_{cc} (min)	176D.8.3	Equation (179–19)	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	176D.8.3	Equation (179–20)	dB
Transmitter steady-state voltage, v_f (range)	176D.8.4	0.4 to 0.5	V
Linear fit pulse peak ratio, R_{peak} (min)	176D.8.4	0.567	—
Level separation mismatch ratio R_{LM} (min)	176D.8.5	0.95	—
Transmitter output waveform			
absolute value of step size for all taps (min)	176D.8.6	0.005	—
absolute value of step size for all taps (max)	176D.8.6	0.025	—
value at minimum state for $c(-3)$ (max)	176D.8.6	–0.06	—
value at maximum state for $c(-2)$ (min)	176D.8.6	0.12	—
value at minimum state for $c(-1)$ (max)	176D.8.6	–0.34	—
value at minimum state for $c(0)$ (max)	176D.8.6	0.5	—
value at minimum state for $c(1)$ (max)	176D.8.6	–0.2	—
Difference signal-to-noise-and-distortion ratio, dSNDR (min)	176D.8.7	0	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	176D.8.8	28	dB
Output jitter (max)	176D.8.9		
J_{RMS}		0.023	UI
EOJ_{03}		0.025	UI
$J4u_{03}$		0.118	UI

^a The signaling rate is derived from the PMD receiver input.

176D.6.5 Host input characteristics

A host shall meet the host input specifications given in Table 176D–4.

Table 176D–4—Summary of host input specifications at TP4a

Parameter	Reference	Value	Units
Signaling rate (range)		106.25 ± 50 ppm	GBd
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	176D.8.10	0.015 0.08	V V
Effective return loss, ERL (min)	176D.8.2	7.3	dB
Differential-mode to common-mode return loss, RL_{cd} (min)	176D.8.3	Equation (179–27)	dB
Amplitude tolerance ^a	176D.8.11	0.5	V
Interference tolerance	176D.8.12	Table 176D–10	—
Jitter tolerance	176D.8.13	Table 176D–11	—
DC common-mode voltage (range) ^b	176D.8.1	0.2 to 1	V

^a Specified as the steady-state voltage (as defined in 176D.8.4) measured at the test transmitter's output.

^b Generated by host, referred to host ground.

176D.6.6 Module input characteristics

The module input shall be AC-coupled within the module. It is recommended that the low-frequency 3 dB cutoff frequency be less than 100 kHz.

The module input shall meet the specifications given in Table 176D–5.

Table 176D–5—Summary of module input specifications at TP1a

Parameter	Reference	Value	Units
Signaling rate (range)		106.25 ± 50 ppm	GBd
Single-ended voltage tolerance (range)		–0.4 to 1.4	V
DC common-mode voltage tolerance (range)	176D.8.10	0.15 to 1.05	V
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	176D.8.10	0.032 0.08	V V
Differential-mode to common-mode return loss, RL_{cd} (min)	176D.8.3	Equation (179–27)	dB
Effective return loss, ERL (min)	176D.8.2	8.5	dB
Amplitude tolerance ^a	176D.8.11	0.5	V
Interference tolerance	176D.8.12	Table 176D–10	—
Jitter tolerance	176D.8.13	Table 176D–11	—

^a Specified as the steady-state voltage (as defined in 176D.8.4) measured at the test transmitter's output.

176D.7 Expected channel properties

Unlike the related C2C interface specified in Annex 176C, the channel between the C2M components is not specified from end to end, since it is divided between two entities with different compliance requirements. The C2M components in the host and in the module with their respective portions of the channel are specified by their input and output characteristics at the compliance points specified in 176D.6.1. This subclause describes the expected properties of the channels from TP0d to TP1d and from TP4d to TP5d, as depicted in Figure 176D–6. These test points are typically not accessible in an implemented system.

The content of this subclause is a reference model that may be used for host and module design. It is expected that the normative input and output specifications of host and module in this annex can be met with a variety of implementation approaches.

176D.7.1 Reference insertion loss budget

Figure 176D–6 depicts the reference differential insertion loss (ILdd) values at 53.125 GHz for specific parts of the channel between the C2M components. The insertion loss of the host, module, and die-to-die channels is not expected to be measurable.

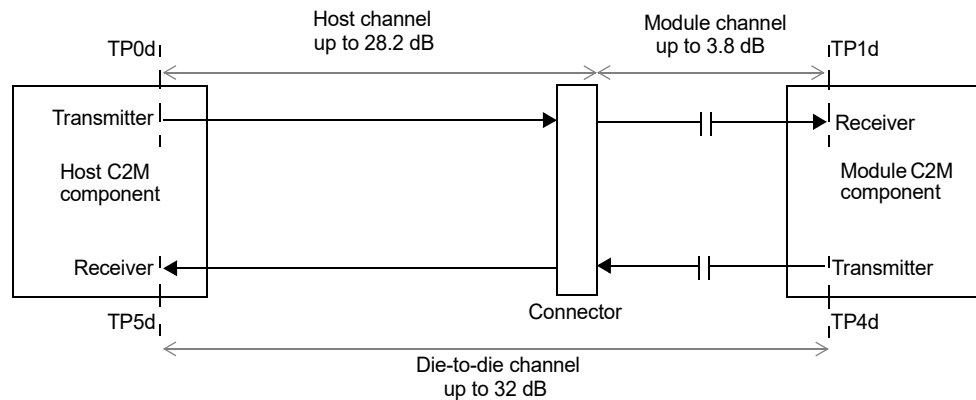


Figure 176D–6—Reference insertion loss budget at 53.125 GHz

176D.7.2 COM reference model

The 200 Gb/s per lane AUI-C2M channel is associated with a channel operating margin (COM) reference model, which defines the assumed capabilities of the transmitter and receiver functions of the C2M components. COM calculation, as defined in 178A.1, is also used for calibration of noise in the interference tolerance test (see 176D.8.12).

Calculation of COM is performed with the parameters in this subclause. The maximum likelihood sequence detection (MLSD) defined in 178A.1.10 is not included in the calculation of COM.

The host parameters in Table 176D–6 represent a host channel that includes a combination of a packaged device and a PCB, using the S-parameter model defined in 178A.1.4 and 178A.1.5. This model is used for host input test calibration and is not a host channel specification.

The parameters in Table 176D–7 correspond to behavioral models of transmitters and receivers that, with suitable host and module channel models, are compliant to the host and module specifications in this clause. The purpose of these parameters is to compute COM, and they do not represent requirements for transmitter and receiver implementations.

It is expected that the specifications in this clause can be met with a variety of approaches to transmitter and receiver implementation and host channel construction.

NOTE—COM is not a specification for hosts and modules. COM analysis (see Annex 178A) may be used informatively for host design.

Table 176D–6—Host and module model parameters

Parameter	Symbol	Value	Units
Device termination model			
Single-ended device capacitance for stage 1	$C_d^{(1)}$	40×10^{-6}	nF
Single-ended device capacitance for stage 2	$C_d^{(2)}$	90×10^{-6}	nF
Single-ended device capacitance for stage 3	$C_d^{(3)}$	110×10^{-6}	nF
Single-ended device series inductance for stage 1	$L_s^{(1)}$	0.13	nH
Single-ended device series inductance for stage 2	$L_s^{(2)}$	0.15	nH
Single-ended device series inductance for stage 3	$L_s^{(3)}$	0.14	nH
Single-ended bump capacitance	C_b	30×10^{-6}	nF
Device package model, host			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	6.5×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2.93×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length	$z_p^{(1)}$	45	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	2	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	95	Ω
Transmission line 3 length	$z_p^{(3)}$	1.3	mm
Transmission line 3 characteristic impedance	$Z_c^{(3)}$	100	Ω
Transmission line 4 length	$z_p^{(4)}$	1.5	mm
Transmission line 4 characteristic impedance	$Z_c^{(4)}$	78	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Device package model, module			
Transmission line parameter γ_0	γ_0	5×10^{-4}	1/mm
Transmission line parameter a_1	a_1	8.9×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	a_2	2×10^{-4}	ns/mm
Transmission line parameter τ	τ	6.141×10^{-3}	ns/mm
Transmission line 1 length, case 1	$z_p^{(1)}$	4	mm
Transmission line 1 length, case 2	$z_p^{(1)}$	10	mm
Transmission line 1 characteristic impedance	$Z_c^{(1)}$	87.5	Ω
Transmission line 2 length	$z_p^{(2)}$	1.8	mm
Transmission line 2 characteristic impedance	$Z_c^{(2)}$	92.5	Ω
Single-ended package capacitance at package-to-board interface	C_p	40×10^{-6}	nF
Partial host channel model			
Single-ended board capacitance at the package-to-board interface	$C_0^{(h)}$	0	nF
Transmission line parameter γ_0	$\gamma_0^{(h)}$	0	1/mm
Transmission line parameter a_1	$a_1^{(h)}$	5.95×10^{-4}	ns ^{1/2} /mm
Transmission line parameter a_2	$a_2^{(h)}$	2.6×10^{-5}	ns/mm
Transmission line parameter τ	$\tau^{(h)}$	5.97×10^{-3}	ns/mm
Transmission line characteristic impedance	$Z_c^{(h)}$	92.5	Ω
Transmission line length	$z_p^{(h)}$	250	mm
Single-ended capacitance at the model-to-measurement interface	C_1	0	nF

Table 176D–6—Host and module model parameters (*continued*)

Parameter	Symbol	Value	Units
Single-ended reference resistance	R_0	50	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	46.25	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	46.25	Ω
NOTE—The device package and partial host channel model parameters are based on the reference impedance R_0 . Channel scattering parameters measured with different reference impedance need to be renormalized to match the reference impedance before computation.			

Table 176D–7—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	106.25	GBd
Transmitter differential peak output voltage			
Victim	A_v	0.385	V
Far-end aggressor	A_{fe}	0.385	V
Near-end aggressor	A_{ne}	0.481	V
Transmitter transition time	T_r	0.004	ns
Receiver 3 dB bandwidth	f_r	$0.55 \times f_b$	GHz
Transmitter equalizer, coefficient –3	$c(-3)$		
Minimum value		0	—
Maximum value		0	—
Step size		0	—
Transmitter equalizer, coefficient –2	$c(-2)$		
Minimum value		0	—
Maximum value		0.14	—
Step size		0.02	—
Transmitter equalizer, coefficient –1	$c(-1)$		
Minimum value		–0.34	—
Maximum value		0	—
Step size		0.02	—
Transmitter equalizer, coefficient 0	$c(0)$		
Minimum value		0.54	—
Transmitter equalizer, coefficient 1	$c(1)$		
Minimum value		–0.2	—
Maximum value		0	—
Step size		0.02	—

Table 176D-7—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter			
Zero 1 frequency for $g_1 = 0$	f_{z1}	$f_b / 2.5$	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	$f_b / 80$	GHz
Pole 1 frequency	f_{p1}	$f_b / 2.5$	GHz
Pole 2 frequency	f_{p2}	f_b	GHz
Pole 3 frequency	f_{p3}	$f_b / 80$	GHz
Gain 1	g_1		
Minimum value		−20	dB
Maximum value		0	dB
Step size		1	dB
Gain 2	g_2		
Minimum value		−6	dB
Maximum value		0	dB
Step size		1	dB
Number of signal levels	L	4	—
Transmitter signal-to-noise ratio	SNR_{TX}	33.5	dB
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
Random jitter, RMS	σ_{RJ}	0.01	UI
One-sided noise spectral density	η_0	1×10^{-8}	V ² /GHz
Receiver discrete-time equalizer parameters			
Number of pre-cursor taps	d_w	5	—
Number of fixed-position taps	N_{fix}	14 ^a	—
Number of floating tap groups	N_{wg}	2	—
Number of taps per floating tap group	N_{wf}	4	—
Highest allowed tap index	N_{max}	56 ^b	—
Normalized upper limit on feed-forward coefficient $w(j)$	$w_{max}(j)$		
$1 \leq j \leq d_w$ and $d_w + 2 \leq j \leq N_{fix}$		0.7	—
$N_{fix} < j$		0.05	—
Normalized lower limit on feed-forward coefficient $w(j)$	$w_{min}(j)$		
$1 \leq j \leq d_w$ and $d_w + 2 \leq j \leq N_{fix}$		−0.7	—
$N_{fix} < j$		−0.05	—
Number of feedback taps	N_b	1	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	0.85	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	0	—
Level separation mismatch ratio	R_{LM}	0.95	—
Target detector error ratio	DER_0	2×10^{-5}	—
Include maximum likelihood sequence detection		no	—

^aThere are 8 fixed-position post-cursor taps.

^bThe latest post-cursor position for a floating tap is 50.

176D.8 Methodology

176D.8.1 Maximum voltages

Differential and common-mode signals are defined in 93.8.1.3.

Peak-to-peak output voltages are defined to a probability P , with respect to the distribution of the output voltage V_{out} sampled at an effective rate of at least two samples per UI. The sampling may be either synchronous or asynchronous.

A maximum output voltage is defined as the value V_{max} such that the probability that $V_{\text{out}} > V_{\text{max}}$ is $P/2$. A minimum output voltage is defined as the value V_{min} such that the probability that $V_{\text{out}} < V_{\text{min}}$ is $P/2$. A peak-to-peak output voltage is defined as $V_{\text{max}} - V_{\text{min}}$.

Differential peak-to-peak output voltage is defined with $P=10^{-7}$ for the differential output signal. For compliance testing, it is sufficient to measure it from a square wave output with a period of at least 128 UI, while lanes not under test transmit PRBS31Q.

Full-band AC common-mode peak-to-peak voltage, V_{CMFB} , is defined with $P=10^{-7}$ for the common-mode output signal. For compliance testing, it is sufficient to measure it from the PRBS13Q test pattern, while lanes not under test transmit PRBS31Q.

Low-frequency AC common-mode peak-to-peak voltage, $V_{\text{CM}_{\text{LF}}}$, is defined with $P=10^{-7}$ for the common-mode output signal filtered by a low-pass filter as defined by Equation (93A–20) with f_r set to 100 MHz. For compliance testing, it is sufficient to measure it from the PRBS13Q test pattern, while lanes not under test transmit PRBS31Q.

NOTE 1—AC common-mode measurement may be sensitive to mismatches between the single-ended paths in the test fixture and the test setup. Careful design and calibration of the test system are recommended.

NOTE 2—AC common-mode measurement should take into consideration frequencies down to the specified AC-coupling frequency.

176D.8.2 Effective return loss (ERL)

ERL is computed using the procedure in 93A.5 with the values in Table 176D–7 and Table 176D–8, and with T_{fx} equal to twice the test fixture delay. T_{fx} is provided by the test fixture provider.

For host input and output ERL, the test fixture delay is defined as the propagation delay between the test connector and the host-facing connection, excluding the mating interface discontinuity. For module input and output ERL, the test fixture delay is defined as the propagation delay between the test connector and the module-facing connection, excluding the mating interface discontinuity.

Table 176D–8—ERL parameter values

Parameter	Symbol	Value		Units
		Host input and output	Module input and output	
Transition time associated with a pulse	T_r	0.005		ns
Incremental available signal loss factor	β_x	0		GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618		—
Length of the reflection signal	N	1600	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	0	UI
Tukey window flag	tw	1		—

176D.8.3 Return loss specifications

Common-mode to common-mode return loss (RL_{cc}) for host and module output is specified by Equation (179–19), illustrated by Figure 179–5.

Common-mode to differential-mode return loss (RL_{dc}) for host and module output is specified by Equation (179–20), illustrated by Figure 179–6.

Differential to common-mode return loss (RL_{cd}) for host and module input is specified by Equation (179–27), illustrated by Figure 179–8.

176D.8.4 Steady-state voltage and linear fit pulse peak ratio

Steady-state voltage (v_p) and linear fit pulse peak ratio (R_{peak}) are defined in 179.9.4.1.2.

176D.8.5 Level separation mismatch ratio

Level separation mismatch ratio (R_{LM}) is defined in 120D.3.1.2.

Host and module output are required to meet the minimum R_{LM} in Table 176D–2 and Table 176D–3, respectively, when the transmit equalization is set to each of the initial conditions defined in Table 176D–9.

176D.8.6 Transmit equalization

Host and module output transmit equalization is controlled by the inter-sublayer link training (ILT) function for a Type E1 interface, specified in Annex 178B, or by other methods.

Coefficient initialization requirements are specified in 179.9.4.1.3, except that the coefficient values listed in Table 176D–9 are used instead of the ones in Table 179–8.

Table 176D–9—Coefficient initial conditions

ic_req	$c(-3)$	$c(-2)$	$c(-1)$	$c(0)$	$c(1)$
initialize	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.75 ± 0.025	0 ± 0.025
preset 1 ^a	0	0	0	1	0
preset 2	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.5 ± 0.025	0 ± 0.025
preset 3	0 ± 0.025	0 ± 0.025	–0.075 ± 0.025	0.75 ± 0.025	0 ± 0.025
preset 4	0 ± 0.025	0.05 ± 0.025	–0.2 ± 0.025	0.75 ± 0.025	0 ± 0.025
preset 5	–0.025 ± 0.025	0.075 ± 0.025	–0.25 ± 0.025	0.65 ± 0.025	0 ± 0.025
preset 6	0 ± 0.025	0 ± 0.025	0 ± 0.025	0.75 ± 0.025	0 ± 0.025

^a Preset 1 is the reference for the calculation of the normalized coefficients of the transmit equalizer (see 179.9.4.1.1). As a result, the normalized coefficients for preset 1 do not include any tolerances.

Coefficient step size requirements are defined in 179.9.4.1.4, except that the minimum and maximum absolute value of step sizes are specified in Table 176D–2 for host output and in Table 176D–3 for module output.

Coefficient range requirements are defined in 179.9.4.1.5, except that the values at minimum and maximum states are specified in Table 176D–2 for host output and in Table 176D–3 for module output.

176D.8.7 Difference signal-to-noise-and-distortion ratio

Difference signal-to-noise-and-distortion ratio (dSNDR) for host output is defined in 179.9.4.5 with the following exceptions:

- Parameter values are taken from Table 176D–7.
- In the calculation of the reference channel transfer function in 179.9.4.5.2, the transmitter S-parameter model $S^{(t)}$ is calculated with the host model parameters in Table 176D–6, and S is the measured S-parameter matrix of the HCB used in the measurement mated with an MCB such that the mated test fixtures comply with the requirements in 179B.4.
- The host inputs at TP4a on each lane are driven by asynchronous signals created by PRBS31Q or PCS data, with transmit equalization (see 176D.8.6) set to preset 1, and calibrated at the generator output with target maximum steady-state voltage as specified in Table 176D–2 and transition time of 6 ps.

NOTE—The input signals are included to excite the NEXT that will be present when a module is plugged into the host under test.

Difference signal-to-noise-and-distortion ratio (dSNDR) for module output is defined in 179.9.4.5 with the following exceptions:

- Parameter values are taken from Table 176D–7.
- In the calculation of the reference channel transfer function in 179.9.4.5.2, the transmitter S-parameter model $S^{(t)}$ is calculated with the module model parameters Table 176D–6, and S is the measured S-parameter matrix of the MCB used in the measurement mated with an HCB such that the mated test fixtures comply with the requirements in 179B.4.

176D.8.8 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio (SNR_{ISI}) is defined in 179.9.4.3.

176D.8.9 Output jitter

The output jitter parameters J_{RMS} , J_{4u03} , and EOJ_{03} are defined in 179.9.4.6.

176D.8.10 Common-mode voltage tolerance

DC common-mode voltage (see 93.8.1.3) is generated by the host. A compliant module meets the module input and output specifications with any DC common-mode voltage, measured at TP4 and TP1, within the ranges specified in Table 176D–3 and Table 176D–5.

Maximum peak-to-peak AC common-mode voltage is specified using the parameters VCM_{LF} and VCM_{FB} (see 176D.8.1) for both host output and module output. A compliant host meets the host input specifications with both VCM_{LF} and VCM_{FB} , measured at TP4, up to the limits specified in Table 176D–4. A compliant module meets the module input specifications with both VCM_{LF} and VCM_{FB} , measured at TP1a, up to the limits specified in Table 176D–5.

NOTE—The specified voltages allow for the effects of ground offset voltage.

176D.8.11 Amplitude tolerance

Amplitude tolerance of a receiver is defined as the maximum steady-state voltage (see 176D.8.4) that the receiver can tolerate at its input, such that it satisfies the error ratio allocation requirements specified in 176D.2 when it operates in DATA mode (see Annex 178B).

The steady-state voltage is measured for the transmitter that is connected to the input of the receiver under test. A receiver under test is allowed to control the transmit equalizer coefficients of the transmitter using the ILT protocol (see 176D.8.6) to create suitable output signal.

NOTE—Steady-state voltage is defined with preset 1. It is not initially generated by a transmitter, due to the initialize setting in Table 176D–9. The receiver is not required to tolerate preset 1 unless it specifically requests it.

For a host, the input signal is applied at TP4a and measured at TP4. For a module, the input signal is applied at TP1 and measured at TP1a.

176D.8.12 Interference tolerance

Interference tolerance is defined by the procedure described in 176D.8.12.1 through 176D.8.12.4. Interference tolerance test requirements are specified in Table 176D–10.

Table 176D–10—Interference tolerance test parameters

Parameter	Host test	Module test 1 (low loss) ^a	Module test 2 (high loss)	Units
Test channel insertion loss at 53.125 GHz	9 ± 2	9 ± 2	32 ± 0.5	dB
Host channel parameters	Table 176D–6	N/A	N/A	
Test pattern	PRBS31Q			
Block error ratio ^b	$< 1.45 \times 10^{-11}$			
COM ^c	3			dB

^a The low loss test channel consists of mated MCB and HCB with no frequency dependent attenuator.

^b Block error ratio (see 174A.8) is measured with BER_{added} specified in 176D.2.

^c COM is calculated as defined in 176D.8.12.2. Meeting the test requirements with a lower value of COM demonstrates margin to the specification but is not required for compliance.

For modules, interference tolerance is defined separately for two tests, corresponding to low-loss and high-loss host channel. For hosts, there is only one test case.

176D.8.12.1 Test setup

Host interference tolerance is defined based on the test setup shown in Figure 176D–7a. The test channel is shown in Figure 176D–7b. The test is performed with addition of noise at the pattern generator, calibrated

such that the channel operating margin (COM) calculated with the test channel and the transmitter's measured parameters matches the value in Table 176D–10.

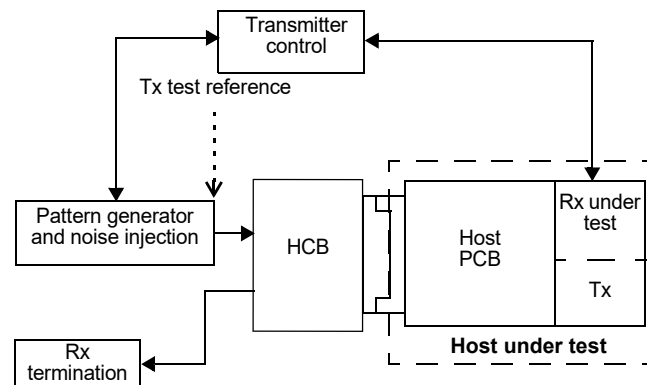


Figure 176D–7a—Host interference tolerance test setup

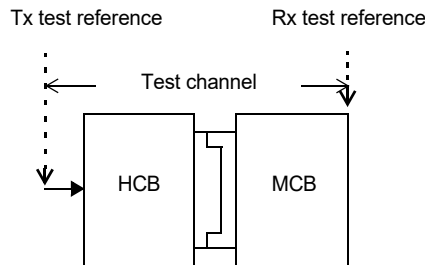


Figure 176D–7b—Host test channel calibration

Module interference tolerance is defined based on the test setup shown in Figure 176D–8a. The test channel is shown in Figure 176D–8b.

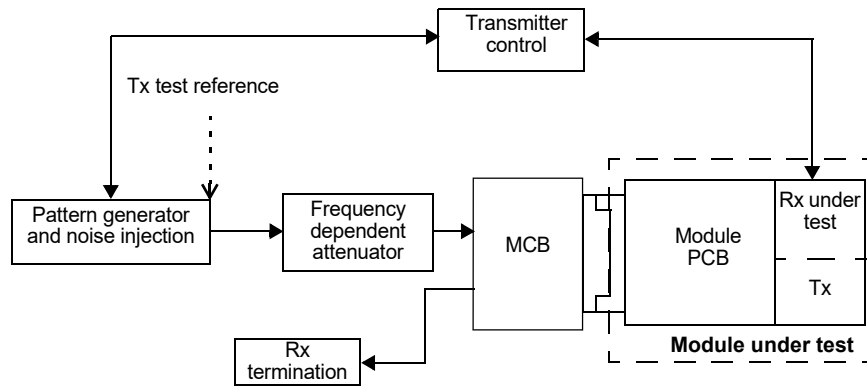


Figure 176D-8a—Module interference tolerance test setup

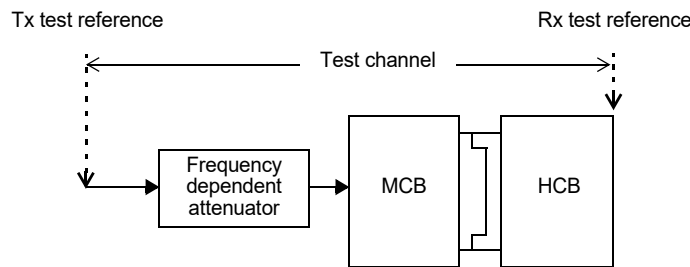


Figure 176D-8b—Module test channel calibration

176D.8.12.2 Test channel calibration

The COM of the test channel is calculated using the method defined in Annex 178A and the parameters of 176D.7.2, with the following considerations:

- For the host test, the test channel is measured between the Tx and Rx test references shown in Figure 176D-7b, and COM is calculated using the host channel, device package, and device termination models in Table 176D-6 for the receiver S-parameter model.
- For the module test, the test channel is measured between the Tx and Rx test references shown in Figure 176D-8b, and COM is calculated using the module device package and device termination models in Table 176D-6 for the receiver S-parameter model. Calculation is performed for both case 1 and case 2 of the package transmission line 1 length, and the value of COM is taken as the lower of the two calculated values.
- The transition time T_r is measured at the Tx test reference with the transmit equalizer set to preset 1 (i.e., equalization off with maximum amplitude), using the method in 120E.3.1.5, with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 60 GHz.
- EOJ_{03} , $J_{4u_{03}}$, and J_{RMS} are measured at the Tx test reference with pattern generator noise injection disabled (see 176D.8.12.3). For host test, the jitter complies with the module output specification in Table 176D-3. For module test, the jitter complies with the host output specification in Table 176D-2. In the calculation of COM, A_{DD} and σ_{RJ} are calculated from the measured values of $J_{4u_{03}}$ and J_{RMS} using Equation (179-21) and Equation (179-22), replacing the values in Table 176D-7. It is recommended to adjust the pattern generator jitter such that $J_{4u_{03}}$ and J_{RMS} are as close as practical to their limits.

- e) In the calculation of the channel S-parameters in 178A.1.2, the transmitter S-parameter model $S^{(t)}$ is omitted. The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (178A–10) uses a rise time filter $H_t(f)$ (see 178A.1.6.2) calculated with T_r measured in step c.
- f) In the calculation of COM, SNR_{TX} is set to the SNDR value with $N_p=55$ measured at the Tx test reference with pattern generator noise disabled.
- g) The interference noise (see 178A.1.7.5) is derived from the S-parameters of the test channel. For a given noise RMS noise voltage σ_{ns} measured at the Tx test reference, the spectral density of the noise source $N_s(f)$ used for calculation of COM may be calculated using Equation (179–23). Alternatively, $N_s(f)$ may be obtained by measurement of the noise spectral density at the output of the noise source.
- h) The value of σ_{ns} that yields the target value of COM specified in Table 176D–10 is determined (e.g., via iterative calculation of COM). The noise level at the Tx test reference is then configured so that the measured RMS voltage matches σ_{ns} .

176D.8.12.3 Pattern generator and noise injection

The pattern generator transmits data to the device under test. At the start of transmitter training, the pattern generator output amplitude is set such that steady-state voltage v_f (see 176D.8.4) at the Tx test reference is the minimum specified in Table 176D–3 for host test, or in Table 176D–2 for module test. v_f is not permitted to exceed that minimum during transmitter training.

Broadband noise is added to the signal before the Tx test reference, with noise level set according to step h) in 176D.8.12.2.

176D.8.12.4 Test procedure

The pattern generator is first configured to transmit training frames as defined in 178B.6.1 with the injected noise as specified in 176D.8.12.3. During this initialization period, the device under test (DUT) may configure the pattern generator transmit equalizer to the coefficient settings it would select using the ILT protocol (see 176D.8.6). The coefficient settings may be communicated via the ILT protocol or by other means.

After the pattern generator equalizer has been configured and the receiver reports that it is ready (using the `signal_ok` variable), the pattern generator is set to generate a PRBS31Q pattern. During the test, the transmitters in the DUT transmit a scrambled idle pattern at preset 1 (maximum amplitude with no equalization).

The device under test is required to meet the error ratio allocation requirements defined in 176D.2.

176D.8.13 Jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 176D.8.13.1 and 176D.8.13.2.

176D.8.13.1 Test setup

For hosts, jitter tolerance is measured with the setup described by Figure 176D–7a. For modules, jitter tolerance is measured with a channel described by Figure 176D–8a that meets the insertion loss of Module test 2 as specified in Table 176D–10.

The pattern generator includes sinusoidal jitter injection but does not include noise injection.

176D.8.13.2 Test procedure

The jitter tolerance test procedure is similar to that of 176D.8.12.4, with the exception that no noise is injected (i.e., step h in 176D.8.12.2 is not performed). Instead, jitter with the specified frequency and amplitude is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency in Table 176D–11 at the Tx test reference (see Figure 176D–7a and Figure 176D–8a). The test channel COM, calculated per 176D.8.12.2 with the jitter-stressed transmitter output, shall not be lower than the value in Table 176D–10.

A host or module shall meet the error ratio allocation requirement specified in 176D.2 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 176D–11 with jitter added to all lanes.

Table 176D–11—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Case G	Units
Jitter frequency	0.04	0.1333	0.4	1.333	4	12	40	MHz
Jitter amplitude (peak-to-peak)	5	1.5	0.5	0.15	0.05	0.05	0.05	UI

NOTE—For a module input test, A_{DD} and σ_{RJ} calculated from pattern generator measurements using Equation (179–21) and Equation (179–22) can be higher than the values in Table 176D–7. In this case, a suitable channel should be chosen in order to meet the COM requirement with these higher values. For a host input test, if the calculated COM is lower than the requirement, a better pattern generator is needed.

176D.9 Protocol implementation conformance statement (PICS) proforma for Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M⁷

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

176D.9.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

176D.9.2 Identification

176D.9.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

176D.9.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 176D, Chip-to-module Attachment Unit Interfaces 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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176D.9.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	176D.1	One, two, or four independent data paths in each direction for 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M, respectively	M	Yes []
BER	Meets BER requirement	176D.2	See 176D.2	M	Yes []
*PCSPKG	PMA in same package as PCS.			O	Yes [] No []

176D.9.4 PICS proforma tables for Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

176D.9.4.1 Host output

Item	Feature	Subclause	Value/Comment	Status	Support
TH1	Signaling rate, PMA in same package as PCS	176D.6.3	Per Table 176D–2	PCSPKG:M	Yes [] N/A []
TH2	Signaling rate, PMA not in same package as PCS	176D.6.3	Derived from PMA input lanes	!PCSPKG:M	Yes [] N/A []
TH3	DC common-mode output voltage	176D.6.3	Per Table 176D–2	M	Yes []
TH4	Single-ended output voltage	176D.6.3	Per Table 176D–2	M	Yes []
TH5	Low-frequency peak-to-peak AC common-mode voltage	176D.6.3	Per Table 176D–2	M	Yes []
TH6	Full-band peak-to-peak AC common-mode voltage	176D.6.3	Per Table 176D–2	M	Yes []
TH7	Differential peak-to-peak voltage, transmitter disabled	176D.6.3	Per Table 176D–2	M	Yes []
TH8	Differential peak-to-peak voltage, transmitter enabled	176D.6.3	Per Table 176D–2	M	Yes []
TH9	Steady-state voltage	176D.6.3	Per Table 176D–2	M	Yes []
TH12	Common-mode to differential-mode return loss	176D.6.3	Per Table 176D–2	M	Yes []
TH13	ERL	176D.6.3	Per Table 176D–2	M	Yes []

176D.9.4.2 Module output

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	Signal rate, nominal	176D.6.4	Per Table 176D–3	M	Yes []
TM2	Signal rate, range	176D.6.4	Derived from PMD receiver input.	M	Yes []
TM3	Low-frequency peak-to-peak AC common-mode voltage	176D.6.4	Per Table 176D–3	M	Yes []
TM4	Full-band peak-to-peak AC common-mode voltage	176D.6.4	Per Table 176D–3	M	Yes []
TM5	Peak-to-peak differential output voltage	176D.6.4	Per Table 176D–3	M	Yes []
TM10	Common-mode to differential-mode return loss	176D.6.4	Per Table 176D–3	M	Yes []
TM11	ERL	176D.6.4	Per Table 176D–3	M	Yes []
TM14	DC common-mode voltage	176D.6.4	Per Table 176D–3	M	Yes []

176D.9.4.3 Host input

Item	Feature	Subclause	Value/Comment	Status	Support
RH2	Signaling rate tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RH4	Common-mode to differential-mode return loss	176D.6.5	Per Table 176D–4	M	Yes []
RH5	ERL	176D.6.5	Per Table 176D–4	M	Yes []
RH6	Common-mode voltage	176D.6.5	Per Table 176D–4	M	Yes []
RH7	Amplitude tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RH8	Interference tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RH8	Jitter tolerance	176D.6.5	Per Table 176D–4	M	Yes []

176D.9.4.4 Module input

Item	Feature	Subclause	Value/Comment	Status	Support
RM2	Signaling rate tolerance	176D.6.6	Per Table 176D–5	M	Yes []
RM3	Differential peak-to-peak input voltage tolerance	176D.6.6	Per Table 176D–5	M	Yes []
RM4	Common-mode to differential-mode return loss	176D.6.6	Per Table 176D–5	M	Yes []
RM5	ERL	176D.6.6	Per Table 176D–5	M	Yes []
RM6	Differential termination mismatch	176D.6.6	Per Table 176D–5	M	Yes []
RM7	Single-ended voltage tolerance	176D.6.6	Per Table 176D–5		
RM8	Common-mode voltage tolerance	176D.6.6	Per Table 176D–5	M	Yes []
RM9	Amplitude tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RM10	Interference tolerance	176D.6.5	Per Table 176D–4	M	Yes []
RM11	Jitter tolerance	176D.6.5	Per Table 176D–4	M	Yes []

Insert new Annex 177A corresponding to Clause 177 as follows:

Annex 177A

(informative)

Test vectors for 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Inner FEC

This annex provides Inner FEC test vectors produced by the convolutional interleaver, circular shift, Inner FEC encode, PAM4 interleaver, and pad insertion as defined in Clause 177.

A file containing the text vectors is available at <https://standards.ieee.org/downloads/802.3/>.

Editor's note (to be removed prior to publication):

The test vectors are given as plain-text files, which can be found at the following link:

https://www.ieee802.org/3/dj/public/24_01/levy_3dj_02a_2401.7z.

This file will be placed in the 802.3 downloads folder referenced above when this amendment is published.

The test vectors are generated as shown in Figure 177A–1, where six test points (TP1 to TP6) are defined. TP1 and TP6 are required for interoperability, and TP2 to TP5 are given as a reference for debug. The pad insertion process is equivalent to the process shown in Figure 177–2, where the 120 bits of the payload of each pad codeword is defined in 177.4.7, and all pad payload will go through the Inner FEC encode and 8:1 PAM4 interleaver.

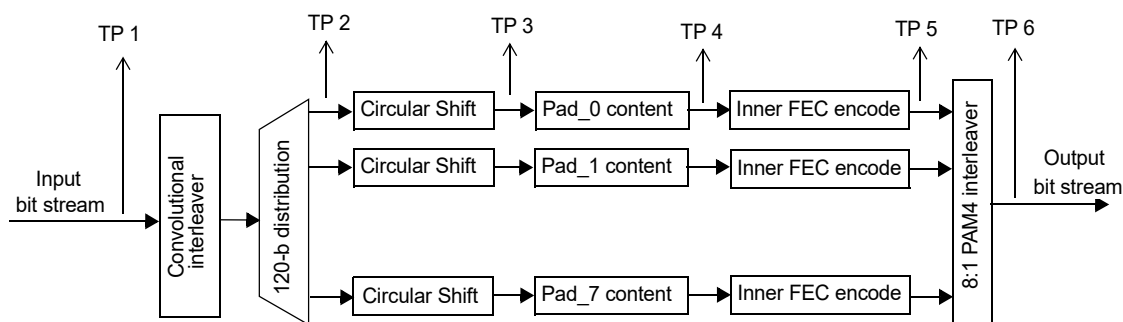


Figure 177A–1—Test points for test vectors

The input bit stream is given by a PRBS31 generator as defined in 49.2.8. The initial seed is 0x7FFF FFFF. In the text files, the bit farthest to the left is the first bit to be transmitted. Before Inner FEC encode each row has 120 bits, and after Inner FEC encode each row has 128 bits. All bits marked with “x” are undetermined values due to delays inside of the convolutional interleaver.

Insert new Annex 178A and Annex 178B corresponding to Clause 178 as follows:

Annex 178A

(normative)

Specification methods for 200 Gb/s per lane electrical channels

178A.1 Channel Operating Margin

178A.1.1 Introduction

The Channel Operating Margin (COM) is a figure of merit for a channel derived from a measurement of its scattering parameters (S-parameters). This annex defines a variant of the calculation defined in 93A.1 to be used by the Physical Layer and interface specifications listed in Table 178A–1. This variant of the calculation includes a different reference receiver and an associated method for parameter optimization. It also enhances the device package and termination models.

Table 178A–1—Physical Layer and interface specifications that employ Annex 178A COM

Physical Layer or interface	Reference	Parameter values
200GAUI-1 C2C 400GAUI-2 C2C 800GAUI-4 C2C 1.6TAUI-8 C2C	Annex 176C	Table 176C–8
200GAUI-1 C2M 400GAUI-2 C2M 800GAUI-4 C2M 1.6TAUI-8 C2M	Annex 176D	Table 176D–7
200GBASE-KR1 400GBASE-KR2 800GBASE-KR4 1.6TBASE-KR8	Clause 178	Table 178–13
200GBASE-CR1 400GBASE-CR2 800GBASE-CR4 1.6TBASE-CR8	Clause 179	Table 179–18

178A.1.2 Channel

The channel is illustrated in Figure 178A–1. It is composed of K signal paths indexed from 0 to $K - 1$ where $k = 0$ corresponds to the victim signal path and other indices correspond to crosstalk paths. The number of

crosstalk aggressors is a function of the structure of the channel. All significant contributors to the channel crosstalk should be included in the calculation of COM.

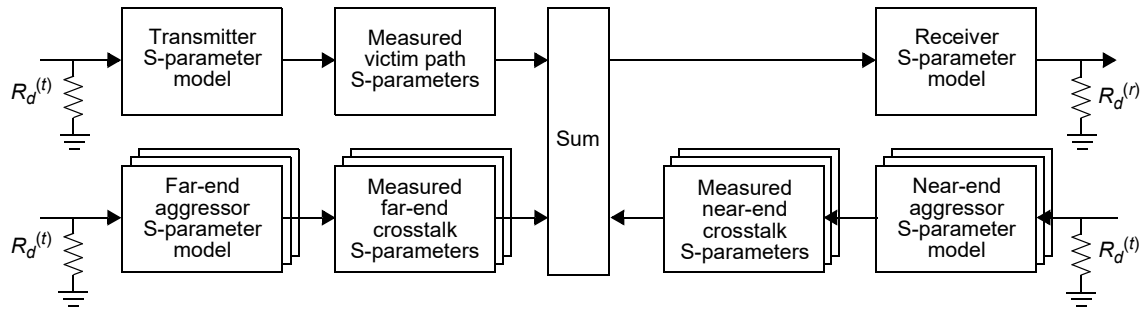


Figure 178A-1—Channel considered for the calculation of COM

Each signal path in the channel is measured according to the guidance given in 178A.1.3 to obtain the differential-mode scattering matrix defined by Equation (178A-1) at each measurement frequency f .

$$S(f) = \begin{bmatrix} s_{11}(f) & s_{12}(f) \\ s_{21}(f) & s_{22}(f) \end{bmatrix} \quad (178A-1)$$

The channel is then extended with the addition of the transmitter S-parameter model defined in 178A.1.4 and the receiver S-parameter model defined in 178A.1.5. The transmitter S-parameter model may vary between the victim transmitter, far-end aggressor transmitter, and near-end aggressor transmitter as defined by the clause or annex that specifies the use of this calculation. The S-parameters of the extended channel for signal path k , $S^{(k)}$, are defined by Equation (178A-2).

$$S^{(k)} = \text{cascade}(S^{(t)}, S, S^{(r)}) \quad (178A-2)$$

where

- $\text{cascade}()$ is the cascade connection of two-port networks defined in 93A.1.2.1
- $S^{(t)}$ is the transmitter S-parameter model
- $S^{(r)}$ is the receiver S-parameter model

The voltage transfer function for signal path k is computed from the S-parameters of the extended channel and the parameters summarized in Table 178A-2 using Equation (178A-3).

Table 178A-2—Summary of voltage transfer function parameters

Parameter	Symbol	Units
Single-ended reference resistance	R_0	Ω
Single-ended transmitter termination resistance	$R_d^{(t)}$	Ω
Single-ended receiver termination resistance	$R_d^{(r)}$	Ω

$$H_{21}^{(k)}(f) = \frac{s_{21}^{(k)}(f)(1 - \Gamma_1)(1 + \Gamma_2)}{1 - s_{11}^{(k)}(f)\Gamma_1 - s_{22}^{(k)}(f)\Gamma_2 + \Gamma_1\Gamma_2\Delta S^{(k)}(f)} \quad (178A-3)$$

where

Γ_1 is the transmitter reflection coefficient at port 1 $(R_d^{(t)} - R_0)/(R_d^{(t)} + R_0)$
 Γ_2 is the receiver reflection coefficient at port 2 $(R_d^{(r)} - R_0)/(R_d^{(r)} + R_0)$
 $\Delta S^{(k)}(f)$ is the determinant of the S-parameter matrix $s_{11}^{(k)}(f)s_{22}^{(k)}(f) - s_{12}^{(k)}(f)s_{21}^{(k)}(f)$

178A.1.3 Measurement of the channel under test

The S-parameters for each signal path are measured between the test points specified by the clause or annex that utilizes this calculation. It is recommended that the scattering parameters be measured with a uniform frequency step from a start frequency no greater than 10 MHz to a stop frequency of at least 67 GHz. The measurement frequency step corresponds to the time span of the pulse response derived from the S-parameters (see 178A.1.6). The frequency step should be chosen to be small enough so that all significant components of the pulse response are included.

The reference impedance for the measurement of differential-mode S-parameters is 100 Ω .

178A.1.4 Transmitter S-parameter model

The transmitter S-parameter model $S^{(t)}$ is the cascade connection of a device termination model, a device package model, and an optional partial host channel model as shown in Figure 178A-2. It is computed using Equation (178A-4).

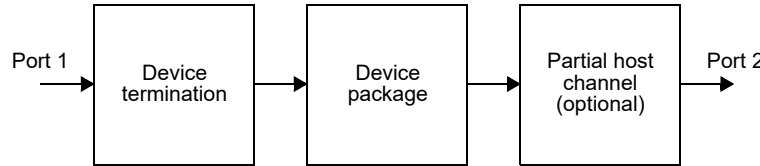


Figure 178A-2—Transmitter S-parameter model

$$S^{(t)} = \text{cascade}(S^{(d)}, S^{(p)}) \quad (178A-4)$$

where

$S^{(d)}$ is the device termination model defined in 178A.1.4.1
 $S^{(p)}$ is the device package model defined in 178A.1.4.2

The partial host channel model may be included to account for the portions of a host that are not included in the channel S-parameter measurement. When the optional partial host channel model is included, $S^{(t)}$ is modified by the assignment defined by Equation (178A-5).

$$S^{(t)} = \text{cascade}(S^{(t)}, S^{(h)}) \quad (178A-5)$$

where $S^{(h)}$ is the partial host channel model defined in 178A.1.4.3.

178A.1.4.1 Device termination

The device termination model is shown in Figure 178A–3 and is a function of the parameters summarized in Table 178A–3. It is the cascade connection of N stages and a shunt capacitor C_b . Each stage consists of a shunt capacitor and series inductor.

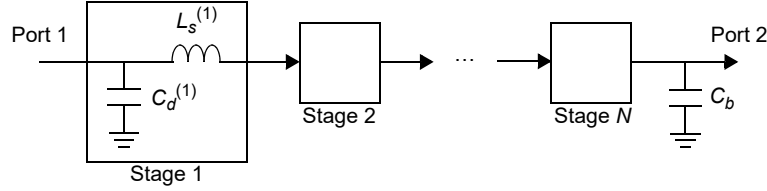


Figure 178A–3—Device termination model

Table 178A–3—Summary of device termination parameters

Parameter	Symbol	Units
Single-ended series inductance for stage n	$L_s^{(n)}$	nH
Single-ended shunt capacitance for stage n	$C_d^{(n)}$	nF
Single-ended bump capacitance	C_b	nF

The S-parameters for each stage $S^{(n)}$ are computed using Equation (178A–6).

$$S^{(n)} = \text{cascade}(S(C_d^{(n)}), S(L_s^{(n)})) \quad (178A-6)$$

where

$S(C_d^{(n)})$ are the S-parameters for shunt capacitor $C_d^{(n)}$ defined in 93A.1.2.2
 $S(L_s^{(n)})$ are the S-parameters for series inductor $L_s^{(n)}$ defined in 93A.1.2.2a

The S-parameters for the device termination model $S^{(d)}$ are initialized to $S(C_b)$. The model for each stage n is then added to the device termination model, in reverse order from $n = N$ to $n = 1$, using the assignment defined by Equation (178A–7).

$$S^{(d)} = \text{cascade}(S^{(n)}, S^{(d)}) \quad (178A-7)$$

178A.1.4.2 Device package

The device package model is shown in Figure 178A–4 and is a function of the parameters summarized in Table 178A–4. It is the cascade connection of N stages and a shunt capacitor C_p . Each stage is a model of a transmission line.

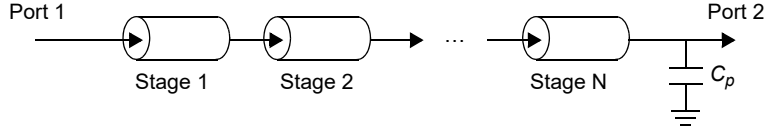


Figure 178A–4—Device package model

Table 178A–4—Summary of device package parameters

Parameter	Symbol	Units
Transmission line parameter, γ_0 , for stage n	$\gamma_0^{(n)}$	1/mm
Transmission line parameter, a_1 , for stage n	$a_1^{(n)}$	ns ^{1/2} /mm
Transmission line parameter, a_2 , for stage n	$a_2^{(n)}$	ns/mm
Transmission line parameter, τ , for stage n	$\tau^{(n)}$	ns/mm
Transmission line differential characteristic impedance for stage n	$Z_c^{(n)}$	Ω
Transmission line length for stage n	$z_p^{(n)}$	mm
Single-ended package capacitance at the package-to-board interface	C_p	nF

The S-parameters for the device package model $S^{(p)}$ are initialized to $S(C_p)$. The stage models $S^{(n)}$ are then added to the device package model, in reverse order from $n = N$ to $n = 1$, using the assignment defined by Equation (178A–8).

$$S^{(p)} = \text{cascade}(S^{(n)}, S^{(p)}) \quad (178A-8)$$

where $S^{(n)}$ are the S-parameters for transmission line stage n defined in 93A.1.2.4.

178A.1.4.3 Partial host channel

The partial host channel model is shown in Figure 178A–5 and is a function of the parameters summarized in Table 178A–5. It is the cascade connection of a first shunt capacitor C_0 , a transmission line, and a second shunt capacitor C_1 .

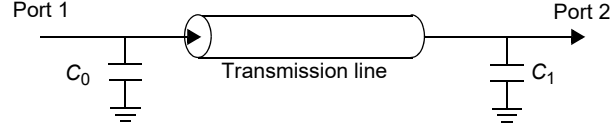


Figure 178A–5—Partial host channel model

Table 178A–5—Summary of partial host channel parameters

Parameter	Symbol	Units
Single-ended board capacitance at the package-to-board interface	C_0	nF
Host transmission line parameter, γ_0	$\gamma_0^{(h)}$	1/mm
Host transmission line parameter, a_1	$a_1^{(h)}$	ns ^{1/2} /mm
Host transmission line parameter, a_2	$a_2^{(h)}$	ns/mm
Host transmission line parameter, τ	$\tau^{(h)}$	ns/mm
Host transmission line differential characteristic impedance	$Z_c^{(h)}$	Ω
Host transmission line length	$z_p^{(h)}$	mm
Single-ended capacitance at the model-to-measurement interface	C_1	nF

The S-parameters for the partial host channel model $S^{(h)}$ are initialized to the S-parameters for a transmission line defined in 93A.1.2.4. They are then augmented by the assignment defined by Equation (178A–9).

$$S^{(h)} = \text{cascade}(S(C_0), S^{(h)}, S(C_1)) \quad (178A-9)$$

where

$S(C_0)$ are the S-parameters for shunt capacitor C_0 defined in 93A.1.2.2
 $S(C_1)$ are the S-parameters for shunt capacitor C_1 defined in 93A.1.2.2

178A.1.5 Receiver S-parameter model

The receiver S-parameter model $S^{(r)}$ is the cascade connection of an optional partial host channel model, a device package model, and a device termination model as shown in Figure 178A–6. It is computed using the procedure defined in 178A.1.4 but with the parameter values specified for the receiver model. The port order of the resulting model is then reversed so that port 1 becomes the input to the optional partial host channel

(or the device package when the partial host channel is not included) and port 2 becomes the output of the device termination.

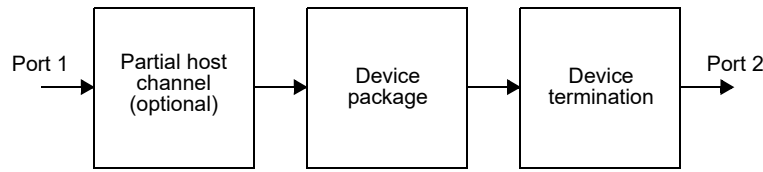


Figure 178A-6—Receiver S-parameter model

178A.1.6 Time-domain responses

For each signal path k , the voltage transfer function computed in 178A.1.2 is subject to additional filtering as defined by Equation (178A-10). The parameters for the various filter transfer functions are summarized in Table 178A-6.

$$H^{(k)}(f) = H_{txffe}(f)H_t(f)H_{21}^{(k)}(f)H_r(f)H_{ctf}(f) \quad (178A-10)$$

where

$H_{txffe}(f)$	is transmitter equalizer transfer function defined in 178A.1.6.1
$H_t(f)$	is the input rise time transfer function defined in 178A.1.6.2
$H_{21}^{(k)}(f)$	is the voltage transfer function for signal path k defined in 178A.1.2
$H_r(f)$	is the receiver noise filter transfer function defined in 178A.1.6.3
$H_{ctf}(f)$	is the receiver continuous-time equalizer transfer function defined in 178A.1.6.4

Table 178A-6—Summary of filter parameters

Parameter	Symbol	Units
Signaling rate	f_b	GBd
Differential peak output voltage, victim transmitter	A_v	V
Differential peak output voltage, far-end aggressor	A_{fe}	V
Differential peak output voltage, near-end aggressor	A_{ne}	V
Input rise time	T_r	ns
Receiver 3 dB bandwidth	f_r	GHz

The filtered voltage transfer function is then converted to a time-domain pulse response as defined by Equation (178A-11).

$$h^{(k)}(t) = \text{IDFT}[A_r T_b \text{sinc}(f T_b) H^{(k)}(f)] \quad (178A-11)$$

where

$\text{IDFT}[x]$	is the inverse discrete Fourier transform of x
------------------	--

A_t is the transmitter peak differential output voltage
 T_b is the unit interval $1/f_b$
 $\text{sinc}(x)$ is the normalized sinc function of x , $\text{sinc}(x) = \sin(\pi x)/(\pi x)$

The value of A_t that is used to calculate $h^{(k)}(t)$ is a function of the path index k . For the victim signal path ($k = 0$), A_t is set to A_v . For a far-end crosstalk signal path, A_t is set to A_{fe} . For a near-end crosstalk signal path, A_t is set to A_{ne} .

NOTE—COM is calculated using measurements at discrete frequencies over a bounded frequency range (see 178A.1.3) and the inverse discrete Fourier transform is used in Equation (178A–11) to derive the corresponding discrete time signal. The filtered voltage transfer function may need to be extrapolated (both to DC and to one half of the sampling frequency) for this calculation. The time step of the discrete time signal should be no larger than $T_b/32$ (corresponding to a sampling frequency of at least $32f_b$). The extrapolation method should be chosen carefully to limit the error in the COM result.

178A.1.6.1 Transmitter equalizer

The parameters of the transmitter equalizer are summarized in Table 178A–7 and the transmitter equalizer transfer function by Equation (178A–12).

Table 178A–7—Summary of transmitter equalizer parameters

Parameter	Symbol	Units
Coefficient n	$c(n)$	
Maximum value		—
Minimum value		—
Step size		—

$$H_{\text{txff}}(f) = \sum_n c(n) e^{-j2\pi n f / f_b} \quad (178A-12)$$

The same parameter values are used to compute the transmit equalizer transfer function for the victim and far-end crosstalk signal paths. The sum of the coefficient magnitudes is 1 and combinations of coefficient values that do not satisfy this constraint are not used. The transmit equalizer transfer function for near-end crosstalk signal paths is set to 1 (no equalization).

178A.1.6.2 Input rise time filter

The input rise time filter $H_i(f)$ corresponding to a rise time T_r is defined by Equation (93A–46).

178A.1.6.3 Receiver noise filter

The receiver noise filter $H_r(f)$ with 3 dB bandwidth f_r is defined by Equation (93A–20).

178A.1.6.4 Receiver continuous-time equalizer

The parameters of the receiver continuous-time equalizer are summarized in Table 178A–8 and the equalizer transfer function is defined by Equation (178A–13).

Table 178A-8—Summary of receiver continuous-time equalizer parameters

Parameter	Symbol	Units
Pole 1 frequency	f_{p1}	GHz
Pole 2 frequency	f_{p2}	GHz
Pole 3 frequency	f_{p3}	GHz
Zero 1 frequency for $g_1 = 0$	f_{z1}	GHz
Zero 2 frequency for $g_2 = 0$	f_{z2}	GHz
Gain 1 Maximum value Minimum value Step size	g_1	dB dB dB
Gain 2 Maximum value Minimum value Step size	g_2	dB dB dB

$$H_{ctf}(f) = \frac{\left(10^{g_1/20} + j\frac{f}{f_{z1}}\right)\left(10^{g_2/20} + j\frac{f}{f_{z2}}\right)}{\left(1 + j\frac{f}{f_{p1}}\right)\left(1 + j\frac{f}{f_{p2}}\right)\left(1 + j\frac{f}{f_{p3}}\right)} \quad (178A-13)$$

178A.1.7 Noise autocorrelation function

The sources of noise considered in the calculation of COM are illustrated in Figure 178A-7. Each source of noise is represented by its power spectral density at the output of the sampler. The parameters used to compute these power spectral densities are summarized in Table 178A-9.

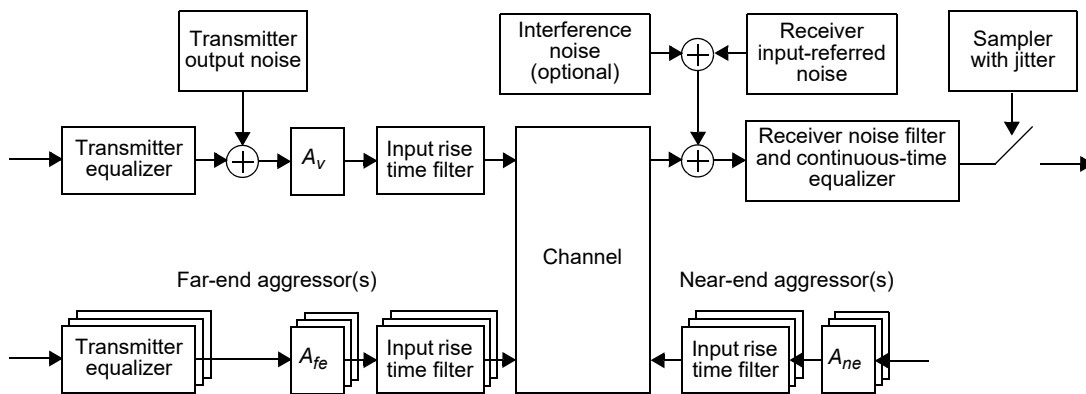


Figure 178A-7—Sources of noise considered in calculation of COM

Table 178A–9—Summary of noise parameters

Parameter	Symbol	Units
Number of signal levels	L	—
One-sided noise spectral density at receiver input	η_0	V ² /GHz
Transmitter signal-to-noise ratio	SNR_{TX}	dB
Random jitter, RMS	σ_{RJ}	UI
Dual-Dirac jitter, peak	A_{DD}	UI

The total noise power spectral density S_n at the sampler output is the sum of the individual noise power spectral densities as defined in Equation (178A–14).

$$S_n(\theta) = S_{rn}(\theta) + \sum_{k=1}^{K-1} S_{xn}^{(k)}(\theta) + S_{in}(\theta) + S_{jn}(\theta) + S_{in}(\theta) \quad (178A-14)$$

where

θ	is the normalized frequency $2\pi f/f_b$ with range $[-\pi, \pi)$
$S_{rn}(\theta)$	is the receiver input-referred noise power spectral density defined in 178A.1.7.1
$S_{xn}^{(k)}(\theta)$	is the crosstalk power spectral density for signal path k defined in 178A.1.7.2
$S_{in}(\theta)$	is the transmitter output noise power spectral density defined in 178A.1.7.3
$S_{jn}(\theta)$	is the power spectral density of the noise due to jitter defined in 178A.1.7.4
$S_{in}(\theta)$	is the interference noise power spectral density defined in 178A.1.7.5

The noise autocorrelation is derived from the total noise power spectral using Equation (178A–15).

$$R_n(n) = \text{IDFT}[S_n(\theta)]f_b \quad (178A-15)$$

178A.1.7.1 Receiver input noise

The power spectral density of the sampled receiver noise at the input to the discrete time receiver equalizer is defined by Equation (178A–16).

$$S_{rn}(\theta) = \frac{\eta_0}{2} \sum_{m \text{ integer}} \left| H_{rn}\left(\frac{\theta + 2\pi m}{2\pi T_b}\right) \right|^2 \quad (178A-16)$$

where $H_{rn}(f)$ is $H_r(f)H_{ctf}(f)$.

178A.1.7.2 Crosstalk

The power spectral density of the sampled crosstalk signal corresponding to signal path k ($k > 0$) at the input to the discrete-time receiver equalizer is defined by Equation (178A–17).

$$S_{xn}^{(k)}(\theta) = \sigma_X^2 \left| \text{DFT}[h_{xn}^{(k)}(n)] \right|^2 / f_b \quad (178A-17)$$

where

DFT[x] is the discrete Fourier transform of x
 σ_X^2 is the power of the PAM- L signal $(L^2 - 1)/[3(L - 1)^2]$
 $h_{xn}^{(k)}(n)$ is $h^{(k)}(t_s^{(k)} + nT_b)$ where $t_s^{(k)}$ is chosen to maximize the sum of $[h_{xn}^{(k)}(n)]^2$ over n

178A.1.7.3 Transmitter output noise

For a given sampling time $t_s^{(0)}$, the power spectral density of the sampled transmitter noise at the input to the receiver discrete-time equalizer is defined by Equation (178A–18).

$$S_{tn}(\theta) = 10^{-SNR_{TX}/10} |\text{DFT}[h_{tn}(n)]|^2 / f_b \quad (178A-18)$$

where $h_{tn}(n)$ are the sampled values $h_{noffe}(t_s^{(0)} + nT_b)$ and the pulse response $h_{noffe}(t)$ is defined by Equation (178A–19).

$$h_{noffe}(t) = \text{IDFT}[A_v T_b \text{sinc}(fT_b) H_{noffe}(f)] \quad (178A-19)$$

where $H_{noffe}(f) = H_t(f) H_{21}^{(0)}(f) H_r(f) H_{ctf}(f)$.

178A.1.7.4 Noise due to transmitter jitter

For a given sampling time $t_s^{(0)}$, the slope of the victim signal path pulse response $h^{(j)}(n)$ around each sampled pulse response value is defined by Equation (178A–20).

$$h^{(j)}(n) = [h^{(0)}(t_s^{(0)} + \Delta t + nT_b) - h^{(0)}(t_s^{(0)} - \Delta t + nT_b)] / (2\Delta t) \quad (178A-20)$$

An estimate of the power spectral density of the noise due to transmitter jitter at the input to the discrete time receiver equalizer is defined by Equation (178A–21).

$$S_{jn}(\theta) = \sigma_X^2 (A_{DD}^2 + \sigma_{RJ}^2) |\text{DFT}[h^{(j)}(n)]|^2 / f_b \quad (178A-21)$$

178A.1.7.5 Interference noise

COM may be used to calibrate a test signal for receiver interference tolerance testing. In these cases, the S-parameters of the noise path are measured from the output of the interference noise source to the reference point specified by the clause or annex that utilizes this calculation. The measured S-parameters $S^{(n)}$ are then augmented by the receiver S-parameter model $S^{(r)}$ as defined by Equation (178A–22) and shown in Figure 178A–8.

$$S^{(i)} = \text{cascade}(S^{(n)}, S^{(r)}) \quad (178A-22)$$

The interference noise voltage transfer function $H_{21}^{(i)}(f)$ is computed from $S^{(i)}$ using Equation (178A–3) with Γ_1 set to 0.

The power spectral density of the sampled interference noise at the input to the discrete time receiver equalizer is defined by Equation (178A–23).

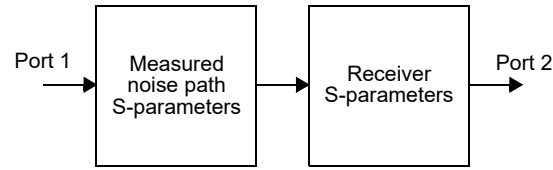


Figure 178A-8—Interference noise path S-parameter model

$$S_{in}(\theta) = \sum_{m \text{ integer}} N_{in}\left(\frac{\theta + 2\pi m}{2\pi T_b}\right) \quad (178A-23)$$

where $N_{in}(f)$ is defined by Equation (178A-24).

$$N_{in}(f) = \frac{N_s(|f|)}{2} |H_{21}^{(n)}(f) H_r(f) H_{ctf}(f)|^2 \quad (178A-24)$$

where $N_s(f)$ is the single-sided power spectral density at the noise generator output in V^2/Hz as defined by the clause or annex that utilizes this calculation.

When COM is not used to calibrate a test signal for receiver interference tolerance testing, $S_{in}(\theta)$ is set to 0.

178A.1.8 Optimization of variable parameters

Some parameters of the COM calculation are not assigned specific values but are instead defined by a range and a step size. These parameters are associated with the transmitter equalizer defined in 178A.1.6.1 and the receiver continuous-time equalizer defined in 178A.1.6.4. For each such parameter, a value within the specified range is chosen and the time-domain responses are computed as defined in 178A.1.6. The sampling time $t_s^{(0)}$ is then chosen and the noise autocorrelation function is computed as defined in 178A.1.7. The results of these calculations are then used to determine the values of the receiver discrete-time equalizer coefficients as defined in 178A.1.8.1 and calculate the mean-squared error as defined in 178A.1.8.2. The values of the variable equalizer parameters, and the sampling time $t_s^{(0)}$, that are used to compute COM are the values that minimize the mean-squared error.

178A.1.8.1 Receiver discrete-time equalizer coefficients

The receiver discrete-time equalizer is illustrated in Figure 178A-9. It is composed of a feed-forward filter and a feedback filter. The input to the feed-forward filter is the output of the sampler. The input to the feedback filter is an estimate of the transmitted PAM- L symbol which is assumed to be correct. The impact

of an incorrect symbol estimate at the input to the feedback filter is considered in the specified target symbol error ratio.

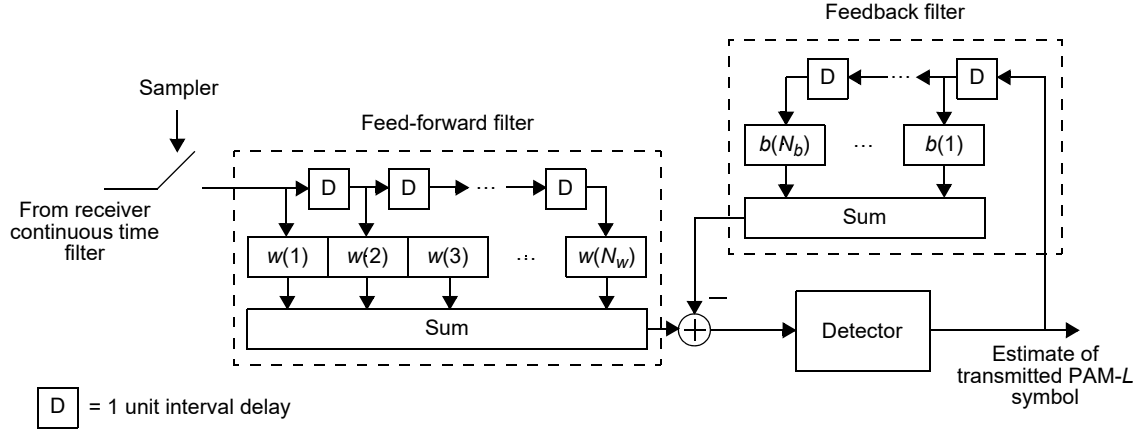


Figure 178A-9—Receiver discrete-time equalizer

The parameters of these filters are summarized in Table 178A-10. The coefficients of the filters are chosen to minimize the mean-squared error using the algorithm described below.

Table 178A-10—Summary of receiver discrete-time equalizer parameters

Parameter	Symbol	Units
Number of pre-cursor taps	d_w	—
Number of fixed-position taps	N_{fix}	—
Number of floating tap groups	N_{wg}	—
Number of taps per floating tap group	N_{wf}	—
Highest allowed tap index	N_{max}	—
Normalized upper limit on feed-forward coefficient $w(j)$	$w_{max}(j)$	—
Normalized lower limit on feed-forward coefficient $w(j)$	$w_{min}(j)$	—
Number of feedback taps	N_b	—
Normalized upper limit on feedback coefficient $b(j)$	$b_{max}(j)$	—
Normalized lower limit on feedback coefficient $b(j)$	$b_{min}(j)$	—

Given the pulse response of the victim signal path $h^{(0)}(t)$ and the sampling time $t_s^{(0)}$, a $1 \times N$ vector of sampled pulse responses values is defined by Equation (178A-25).

$$h = [h(-d_h) \dots h(0) \dots h(N - d_h - 1)] \quad (178A-25)$$

where $h(n)$ is $h^{(0)}(t_s^{(0)} + nT_b)$.

Define $H^{(all)}$ to be an $(N + N_{\max} - 1) \times N_{\max}$ diagonal-constant (Toeplitz) matrix whose first column is h followed by $N_{\max} - 1$ zeros and whose first row is $h(-d_h)$ followed by $N_{\max} - 1$ zeros. When N_{wg} is not defined or set to 0, then $N_{\max}$ is N_{fix} . Define $R_{nn}^{(all)}$ to be an $N_{\max} \times N_{\max}$ Toeplitz matrix whose first row and first column are $R_n(n)$ for $n = 0$ to $N_{\max} - 1$.

The total number of taps in the feed-forward filter, N_w , is $N_{fix} + N_{wg}N_{wf}$. Define i to be a $1 \times N_w$ vector consisting of indices 1 to N_{fix} for fixed-position taps and $N_{wg}N_{wf}$ additional indices for floating taps. Feed-forward filter tap index 1 corresponds to the earliest pre-cursor tap, tap index N_{fix} corresponds to the latest fixed-position post cursor tap, and $N_{\max}$ corresponds to the highest index that can be chosen for a floating tap. This means that the feed-forward filter includes $N_{fix} - d_w - 1$ fixed-position post-cursor taps and $N_{\max} - d_w - 1$ is the latest post-cursor position that can be chosen for a floating tap. Floating tap indices are chosen to minimize the mean-squared error (see 178A.1.8.2) with the constraint that indices within a floating tap group are contiguous and do not overlap with indices of fixed-position taps or other floating tap groups.

Let H be an $(N + N_{\max} - 1) \times N_w$ matrix formed from the columns of $H^{(all)}$ indexed by i . Let R_{nn} be the $N_w \times N_w$ matrix formed from the rows and columns of $R_{nn}^{(all)}$ indexed by i . Let d be $d_h + d_w$. The feed-forward and feedback filter coefficients that minimize the means-squared are determined by Equation (178A-26).

$$\begin{bmatrix} w \\ b \\ \lambda \end{bmatrix} = \begin{bmatrix} R & -H_b^T & -h_0^T \\ -H_b & I_b & z_b^T \\ h_0 & z_b & 0 \end{bmatrix}^{-1} \begin{bmatrix} h_0^T \\ z_b^T \\ 1 \end{bmatrix} \quad (178A-26)$$

where

R	is the $N_w \times N_w$ matrix $H^T H + R_{nn} / \sigma_X^2$ and H^T is the matrix transpose of H
h_0	is row $d + 1$ from H
H_b	is the $N_b \times N_w$ matrix formed from rows $d + 2$ to $d + N_b + 1$ from H
I_b	is a $N_b \times N_b$ identity matrix
z_b	is a $1 \times N_b$ vector of zeros

In Equation (178A-26), the first N_w rows of the left hand side of the equation are the feed-forward filter coefficients w , the next N_b rows are the feedback filter coefficients b , and the scalar λ is ignored. The solution to Equation (178A-26) not only minimizes the mean-squared error but also ensures that the amplitude of the equalized pulse (the value at time $d + 1$) is 1.

Limits on the feedback filter coefficients b are imposed to yield b_{lim} using Equation (178A-27).

$$b_{lim}(j) = \begin{cases} b_{\max}(j) & b(j) \geq b_{\max}(j) \\ b_{\min}(j) & b(j) \leq b_{\min}(j) \\ b(j) & \text{otherwise} \end{cases} \quad (178A-27)$$

If $b_{lim}(j)$ differs from $b(j)$ for any j , then the feed-forward filter coefficients are re-computed using Equation (178A-28).

$$\begin{bmatrix} w \\ \lambda \end{bmatrix} = \begin{bmatrix} R & -h_0^T \\ h_0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} h_0^T + H_b^T b_{lim} \\ 1 \end{bmatrix} \quad (178A-28)$$

In Equation (178A–28), the first N_w rows of the left-hand side of the equation are the updated feed-forward coefficients w and the scalar λ is again ignored.

Limits on the feed-forward coefficients w are imposed to yield w_{lim} using Equation (178A–29).

$$w_{lim}(j) = \begin{cases} w(d_w + 1)w_{\max}(j) & w(j) \geq w(d_w + 1)w_{\max}(j) \\ w(d_w + 1)w_{\min}(j) & w(j) \leq w(d_w + 1)w_{\min}(j) \\ w(j) & \text{otherwise} \end{cases} \quad (178A-29)$$

If $w_{lim}(j)$ differs from $w(j)$ for any j , then w_{lim} is divided by the scalar $h_0 w_{lim}$ where h_0 is a $1 \times N_w$ vector and w_{lim} is a $N_w \times 1$ vector, to ensure that the amplitude of the equalized pulse is 1. The feedback filter coefficients b are then recomputed to be $H_b w_{lim}$ and limits are imposed on the updated b to yield b_{lim} using Equation (178A–27).

178A.1.8.2 Calculation of mean-squared error

Given the receiver discrete-time equalizer coefficients w_{lim} and b_{lim} determined in 178A.1.8.1, and the corresponding values of R , h_0 , and H_b , the mean-squared error is computed using Equation (178A–30).

$$\sigma_e^2 = \sigma_X^2(w_{lim}^T R w_{lim} + 1 + b_{lim}^T b_{lim} - 2w_{lim}^T h_0 - 2w_{lim}^T H_b^T b_{lim}) \quad (178A-30)$$

178A.1.9 Calculation of COM

COM is the ratio, expressed in decibels, of the calculated signal amplitude to the calculated noise and interference amplitude as defined by Equation (178A–31) using the parameters summarized in Table 178A–11.

$$COM = 20\log_{10}(A_s/A_{ni}) \quad (178A-31)$$

Table 178A–11—Additional parameters for the calculation of COM

Parameter	Symbol	Units
Level separation mismatch ratio	R_{LM}	—
Target detector error ratio	DER_0	—

178A.1.9.1 Signal amplitude

The minimum distance between the signal and the detector threshold is defined by Equation (178A–32).

$$A_s = R_{LM}/(L - 1) \quad (178A-32)$$

where R_{LM} is the level separation mismatch ratio.

178A.1.9.2 Noise and interference amplitude

The probability distribution function of the noise and interference amplitude $p(y)$ is calculated using the procedure defined in 178A.1.9.3. The corresponding cumulative distribution function is defined by Equation (178A–33).

$$P(y) = \int_{-\infty}^y p(u) du \quad (178A-33)$$

The noise and interference amplitude, A_{ni} , is a positive value that satisfies the relationship $P(-A_{ni}) = DER_0$ where DER_0 is the target detector error ratio.

NOTE 1—COM is expected to be computed using a quantized amplitude axis y . The amplitude step Δy introduces quantization error in the calculated distribution function that is compounded by subsequent convolutions with other quantized distribution functions. It is recommended that Δy be no larger than 0.1% of A_s .

NOTE 2—The target detector error ratio DER_0 is $SER_1 \times L / (2L - 2)$ where SER_1 is the maximum allowed probability of the initial error in a PAM- L error burst.

178A.1.9.3 Noise and interference probability distribution function

Given the transmitter equalizer, receiver continuous-time equalizer, sampling time $t_s^{(k)}$ for each signal path k , and the coefficients of the receiver discrete-time equalizer that minimize the mean-squared error, the following procedure is used to compute the probability distribution function of noise and interference amplitude $p(y)$ for use in the calculation of COM.

The sampled values of the pulse response corresponding to signal path k at the output of the feed-forward filter are defined by Equation (178A–34).

$$h_w^{(k)}(n) = \sum_{n \in i} w_{lim}(n) h^{(k)}(n - d_w - 1) \quad (178A-34)$$

where

$h^{(k)}(n)$ is $h^{(k)}(t_s^{(k)} + nT_b)$ where $h^{(k)}(t)$ is defined in 178A.1.6
 $w_{lim}(n)$ is the feed-forward filter coefficient vector computed in 178A.1.8.1
 i is the corresponding tap index vector defined in 178A.1.8.1

The notation $n \in i$ represents n assuming each value in the index vector i and hence the summation in Equation (178A–34) is over all of the values in i .

The residual intersymbol interference is computed from $h_w^{(0)}(n)$ and the feedback filter coefficients b_{lim} defined in 178A.1.8.1 using Equation (178A–35).

$$h_{ISI}(n) = \begin{cases} 0 & n = d + 1 \\ h_w^{(0)}(n) - b_{lim}(n - d - 1) & 1 \leq n - d - 1 \leq N_b \\ h_w^{(0)}(n) & \text{otherwise} \end{cases} \quad (178A-35)$$

where N_b and d are defined in 178A.1.8.1.

The probability distribution function $p_{ISF}(y)$ for the intersymbol interference is computed using the procedure defined in 93A.1.7.3 substituting $h_{ISF}(n)$ for $h(n)$. Initialize $p(y)$ to $p_{ISF}(y)$.

The sampled values of the pulse response corresponding to each crosstalk signal path k ($k > 0$) at the output of the feed-forward filter are computed using Equation (178A–34). The probability distribution function $p^{(k)}(y)$ of the interference from crosstalk signal path k is then computed using the procedure defined in 93A.1.7.3 substituting $h_w^{(k)}(n)$ for $h(n)$. The value of $p(y)$ is then assigned the result of $\text{conv}[p(y), p^{(k)}(y)]$ where $\text{conv}[u, v]$ is the convolution of u and v . This assignment is repeated for all k greater than zero to include probability distribution function of the interference from all crosstalk signals.

The slope of the victim signal path pulse response around each sampled pulse response value at the output of the feed-forward filter is computed using Equation (178A–20) substituting $h^{(J)}(n)$, as defined in 178A.1.7.4, for $h^{(k)}(n)$. Let $h_w^{(J)}(n)$ be the result of this calculation. The probability distribution function $p_{DD}(y)$ for the noise due to transmitter dual-Dirac jitter is then computed using the procedure defined in 93A.1.7.3 substituting $A_{DD}h_w^{(J)}(n)$ for $h(n)$. Assign the value of $p(y)$ to the result of $\text{conv}[p(y), p_{DD}(y)]$ in order to include the probability distribution function for the noise due to transmitter dual-Dirac jitter.

The combined variance of the remaining noise terms at the output of the feed-forward filter is defined by Equation (178A–36).

$$\sigma_G^2 = f_b \int_{-\pi}^{\pi} [S_{in}(\theta) + S_{jn}^{(RJ)}(\theta) + S_{rn}(\theta) + S_{in}(\theta)] |H_{rxffe}(\theta)|^2 d\theta \quad (178A–36)$$

where

$S_{jn}^{(RJ)}(\theta)$ is the power spectral density of the noise due only to transmitter random jitter defined by Equation (178A–37)

$H_{rxffe}(\theta)$ is the transfer function of feed-forward filter defined by Equation (178A–38)

$$S_{jn}^{(RJ)}(\theta) = \sigma_X^2 \sigma_{RJ}^2 |\text{DFT}[h^{(J)}(n)]|^2 / f_b \quad (178A–37)$$

$$H_{rxffe}(\theta) = \sum_{n \in i} w(n) \exp[-j\theta(n - d_w - 1)] \quad (178A–38)$$

The probability distribution function $p_G(y)$ of the corresponding Gaussian noise amplitudes is defined by Equation (93A–50). Assign the value of $p(y)$ to the result of $\text{conv}[p(y), p_G(y)]$ in order to include the probability distribution function for Gaussian noise. The result is used for the calculation of COM.

178A.1.10 Maximum likelihood sequence detection

The reference receiver may be specified to include a maximum likelihood sequence detector (MLSD) instead of the decision feedback equalizer (DFE) defined in 178A.1.8.1. In this case, the receiver discrete-time equalizer is modified as shown in Figure 178A–10.

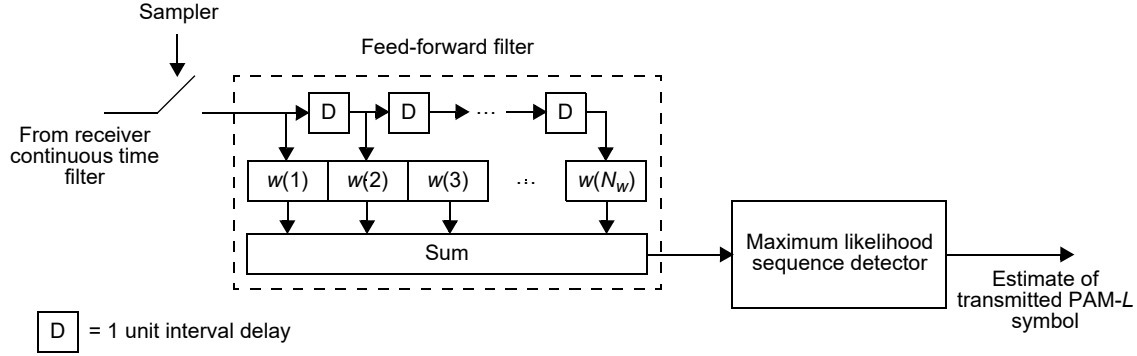


Figure 178A–10—Receiver discrete-time equalizer with MLSD

The receiver discrete-time equalizer coefficients are determined using the procedure defined in 178A.1.8.1 using the parameters defined in Table 178A–10 but with the value of N_b set to 1. COM is then computed as defined in 178A.1.9 and the resulting value is labeled COM_{DFE} . The value of COM_{DFE} and the feedback filter coefficient $b_{lim}(1)$, along with the corresponding noise and residual intersymbol interference computed at the output of the feed-forward filter, are used to calculate a modification to COM_{DFE} that represents the advantage the MLSD-based receiver has over the DFE-based receiver. This modification is defined by Equation (178A–39).

$$COM = COM_{DFE} + 20\log_{10}(-P_{an}^{-1}(DER_{MLSD})/A_s) \quad (178A-39)$$

where $P_{an}(y)$ is defined in 178A.1.10.1 and DER_{MLSD} is defined by Equation (178A–40).

$$DER_{MLSD} = \sum_{j=1}^{\infty} \left(\frac{L-1}{L} \right)^{j-1} P_j \left(-A_s \frac{(u_j^T u_j)^{3/2}}{(u_j^T V_j u_j)^{1/2}} \right) \quad (178A-40)$$

where

$P_j(y)$ is defined in 178A.1.10.2
 u_j is a $(j+1) \times 1$ vector whose elements are defined by Equation (178A–41)
 V_j is defined in 178A.1.10.3

$$u_j(i) = \begin{cases} 1 & i = 1 \\ (-1)^{i-1} (1 - b_{lim}(1)) & 2 \leq i \leq j \\ (-1)^i b_{lim}(1) & i = j + 1 \end{cases} \quad (178A-41)$$

Additional parameters that are used to compute COM for the MLSD-based reference receiver are listed in Table 178A–12. The “Include maximum likelihood sequence detection” parameter is set to “yes” to indicate that the calculations in this subclause are to be included in the determination of the COM value and is set to “no” otherwise.

Table 178A–12—Additional parameter for the MLSD-based reference receiver

Parameter	Symbol	Units
Include maximum likelihood sequence detection	—	—
Penalty for added receiver noise	ΔCOM_{an}	dB

178A.1.10.1 Additional receiver noise

The receiver noise is increased before COM for the MLSD reference receiver is computed. The increased noise represents additional receiver impairments allowed by the minimum COM limit but not explicitly included in the calculation of COM_{DFE} . Let $S_{an}(\theta)$ be the power spectral density of the additional noise as defined by Equation (178A–42).

$$S_{an}(\theta) = g_{an} S_{rn}(\theta) |H_{rxffe}(\theta)|^2 \quad (178A-42)$$

where g_{an} is a scale factor for the added noise with a value greater than or equal to 0.

Let $p_{an}(y)$ be the probability distribution function of noise and interference amplitude that includes the added noise. It is computed as defined in 178A.1.9.2, using the same parameter values that were used to compute COM_{DFE} , but replacing σ_G with σ_{an} where σ_{an} is defined by Equation (178A–43).

$$\sigma_{an}^2 = \sigma_G^2 + f_b \int_{-\pi}^{\pi} S_{an}(\theta) d\theta \quad (178A-43)$$

The corresponding cumulative distribution function $P_{an}(y)$ is defined by Equation (178A–33) substituting $p_{an}(y)$ for $p(y)$.

The value of g_{an} that is used to calculate COM for the MLSD reference receiver is the value that makes the quantity $20 \log_{10}[P_{an}^{-1}(DER_0)/P^{-1}(DER_0)]$ equal to the specified value of ΔCOM_{an} . If the value of ΔCOM_{an} is not specified, it defaults to the minimum COM limit defined by the clause or annex that specifies the use of this calculation.

178A.1.10.2 Sequence noise distribution function

The probability distribution function of the noise corresponding to a sequence of length j is denoted $p_j(y)$. It is initialized by the assignment defined in Equation (178A–44).

$$p_j(y) = \text{conv}\left[p_{an}(y), \frac{1}{|b_{lim}(1)|} p_{an}\left(\frac{y}{b_{lim}(1)}\right)\right] \quad (178A-44)$$

where $p_{an}(y)$ defined in 178A.1.10.1 and $\text{conv}[u, v]$ denotes the convolution of u and v .

It is iteratively updated by repeating the assignment defined by Equation (178A–45) $j - 1$ times.

$$p_j(y) = \text{conv}\left[p_j(y), \frac{1}{|1 - b_{lim}(1)|} p_{an}\left(\frac{y}{1 - b_{lim}(1)}\right)\right] \quad (178A-45)$$

The cumulative distribution function of the noise corresponding to a sequence of length j , $P_j(y)$, is then computed using Equation (178A–33) substituting $p_j(y)$ for $p(y)$.

NOTE—The probability distribution function $p(y/a)/|a|$, where $|a|$ denotes the absolute value of a , represents random variable Y scaled by a factor of a . The integral of the scaled probability distribution function is 1.

178A.1.10.3 Intersymbol interference and noise autocorrelation function

The power spectral density of the noise and residual intersymbol interference at the output of the feed-forward filter is defined by Equation (178A–46).

$$S_{ni}(\theta) = S_n(\theta) |H_{rxffe}(\theta)|^2 + S_{an}(\theta) + \sigma_X^2 |\text{DFT}[h_{ISI}(n)]|^2 / f_b \quad (178A–46)$$

where $S_{an}(\theta)$ is defined in 178A.1.10.1.

The corresponding autocorrelation function $R_{ni}(n)$ is computed using Equation (178A–15) substituting $S_{ni}(\theta)$ for $S_n(\theta)$. The normalized autocorrelation matrix V_j is a $(j+1) \times (j+1)$ Toeplitz matrix whose first row and column are $R_{ni}(n)/R_{ni}(0)$ for $n = 0$ to j .

Annex 178B

(normative)

Inter-sublayer link training for electrical and optical interfaces

178B.1 Scope

This clause defines inter-sublayer link training (ILT) for Physical Layer implementations that include one or more inter-sublayer links (ISLs) (see 178B.3) with data rate of 200 Gb/s or higher per lane.

178B.2 Overview

ILT for electrical and optical interfaces performs the path start-up protocol. ILT facilitates timing recovery and equalization in a ISL or multi-ISL paths while providing a mechanism through which the receiver may configure the transmitter to optimize performance. ILT supports these functions through the continuous exchange of fixed-length training frames between peer interfaces in an ISL and the coordination of the ISLs along a path by transporting end-to-end indications.

178B.3 Conventions

For the purpose of this annex, the following definitions apply. Subclause 1.4 should be referenced for terms not defined in this annex.

AUI component

The interface to an AUI channel. In a device with two AUI components the AUI upper component is the one facing the host and the AUI bottom component is the one facing the medium.

Inter-sublayer link (ISL)

A physically instantiated link between a pair of adjacent sublayers. The ISL may be an xAUI-n between a pair of PMA sublayers within the same Physical Layer implementation or a pair of PMDs and the medium between.

Path

The series of ISLs and sublayers between a pair of PCS sublayers or a pair of Extender Sublayers.

Peer interface

The interface at the other end of the ISL. For a PMD this term is equivalent to link partner.

xAUI-n

This term refers collectively to 200GAUI-1, 400GAUI-2, 800GAUI-4, 1.6TAUI-8, and 1.6TAUI-16, where “n” is the number of physical lanes.

178B.4 ILT function in AUI components and PMDs

The ILT function is used by the AUI component or PMD at each end of a physically instantiated interface.

Devices in a path may include one or two physically instantiated interfaces, specifically PMD or AUI components. An example of the former is a PMA adjacent to a PCS or to a PHY XS with a single AUI-C2M (Annex 176D) or AUI-C2C (Annex 176C) interface (the interface with the PCS or PHY XS is never

physically instantiated). An example of the latter is a retimer with an AUI-C2C (Annex 176C) interface on one side and an AUI-C2M (Annex 176D) on the other side.

The ILT function in AUI components and PMDs is composed of one per-interface function and one or more per-lane functions as shown in Figure 178B–1.

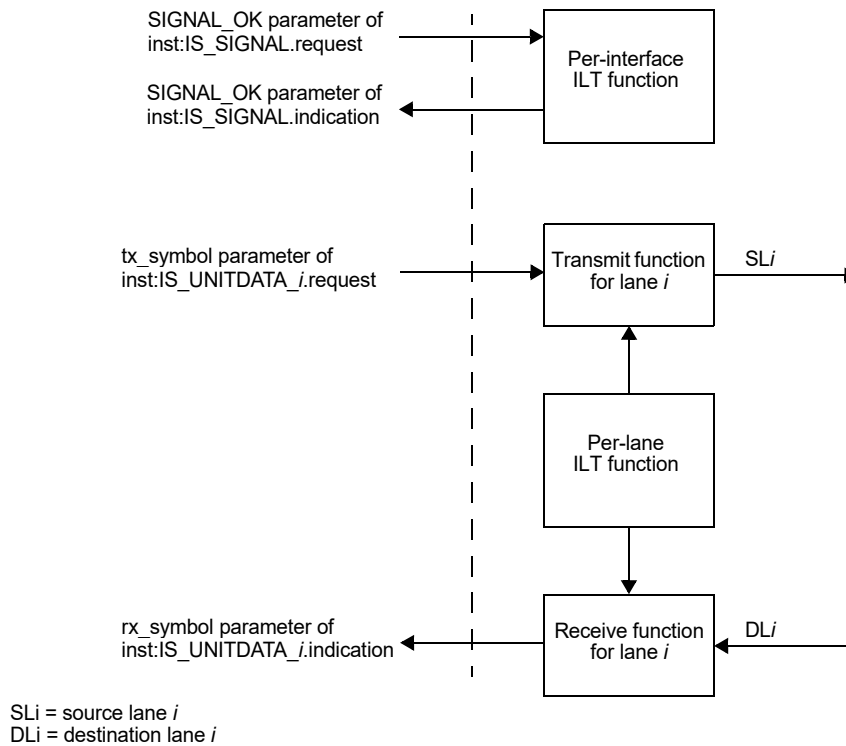


Figure 178B–1—ILT function in AUI components and PMDs

178B.5 Path start-up protocol

ILT enables independent ISL training in a multi-ISL path that includes AUI components and PMDs. It also supports operation over paths that include ISLs that do not implement ILT.

Path start-up is achieved as follows:

- local_rts indicates that an AUI component or PMD is ready to send and receive normal data and propagates from the PCS at one end of the path towards the PCS at the other end of the path.
- remote_rts indicates that a remote AUI component or PMD is ready to send and receive normal data and propagates similarly and independently from PCS to PCS in both directions.
- local_rts and remote_rts are propagated only across an ISL that is ready to send data.
- When a device both sends local_rts and receives remote_rts in both directions, it means all the ISLs in the same path (see 178B.3) are ready and it switches to data mode (tx_mode = data, see 178B.14.3.1).
- When all devices are in data mode, communication on the path is established.

Ready to send (RTS) propagates over ISLs using one of the following methods:

- The Continue training bit in the control field of the training frames (see 178B.7.2) if training is enabled.
- The transmit disable and the AUI component or PMD signal detect function otherwise.

NOTE — Interactions with AUI components and PMDs that do not support ILT as specified in this annex (e.g. those defined in Clause 120 or Clause 173) use the second method.

178B.5.1 Interface behavior

ILT on each interface includes some independent functions on each lane.

If training is available on the interface the behavior is as follows:

- Local variables are sent to the peer interface via the training frames. Remote variables are received from the peer interface.
- When both `local_rts` and `remote_rts` are 1, after a specified delay (`propagation_timer`, see Figure 178B–8), the device switches to its normal functionality (`tx_mode` = data, see 178B.14.3.1), forwarding data between its interfaces in both directions. The condition is shared by all lanes within an ISL, and therefore the switching of all lanes occurs in a period within the limits of `propagation_timer` (see 178B.14.3.3).
- There is no specified timeout when waiting for either `rx_ready` or `remote_rts` to change to the value 1.
- While waiting for `rx_ready` and `remote_rts`, losing frame lock and not recovering it after a specified recovery time (`recovery_timer`, see Figure 178B–8) or losing frame lock a configured number of times (`recovery_event_count`, see Figure 178B–8), would cause training to fail and squelch. Training may be restarted by management after an unspecified time.
- Management may access devices and restart training on a specific ISL if desired.
- The `SIGNAL_OK` status variable of the interface is OK if `local_rts` and `remote_rts` are both 1.

If training is not available on an interface (disabled or not defined for the interface type) the behavior is as follows:

- `local_rx_ready` is generated on each lane based on internal criteria, similar to the signal indication logic (SIL) in existing AUI components or PMDs.
- `remote_rx_ready` is set to 1 (there is no way to get it from the peer interface).
- `local_rts` is independent of `local_rx_ready` and is generated only from the variables of the other interface of the device. It is communicated to the peer interface by squelching (`local_rts` = 0) or unsquelching (`local_rts` = 1) the output.
- `remote_rts` is set to 1 (there is no way to get it from the peer interface).
- The variables `local_tf_lock` and `remote_tf_lock` are undefined and not used.
- The `SIGNAL_OK` status variable of the interface is OK if `local_rts` and `remote_rts` are both 1.

178B.5.2 Training retimers

The retimer reference model is shown in Figure 178B–2. In normal operation, a retimer passes the data received on one of its interfaces to the other using the clock recovered from the received data. During the ILT process, data might not be available from one interface while training the other interface. In this case, the training frames transmission is based on a local clock (`USE_TX_CLOCK(local)`, see 178B.14.2.2) and the transition to the recovered clock is performed when the ISL is ready and recovered clock is available (see Figure 178B–7).

As shown in the RTS control state diagram (Figure 178B–7) `local_rts` is set to true only after the transmit clock is derived from the PCS clock. This guarantees that the transition between clock sources occurs while sending `local_rts` = false. This, in turn, guarantees that the whole path is running with the correct clocks before retimers switch to data mode (`tx_mode` = data, see 178B.14.3.1).

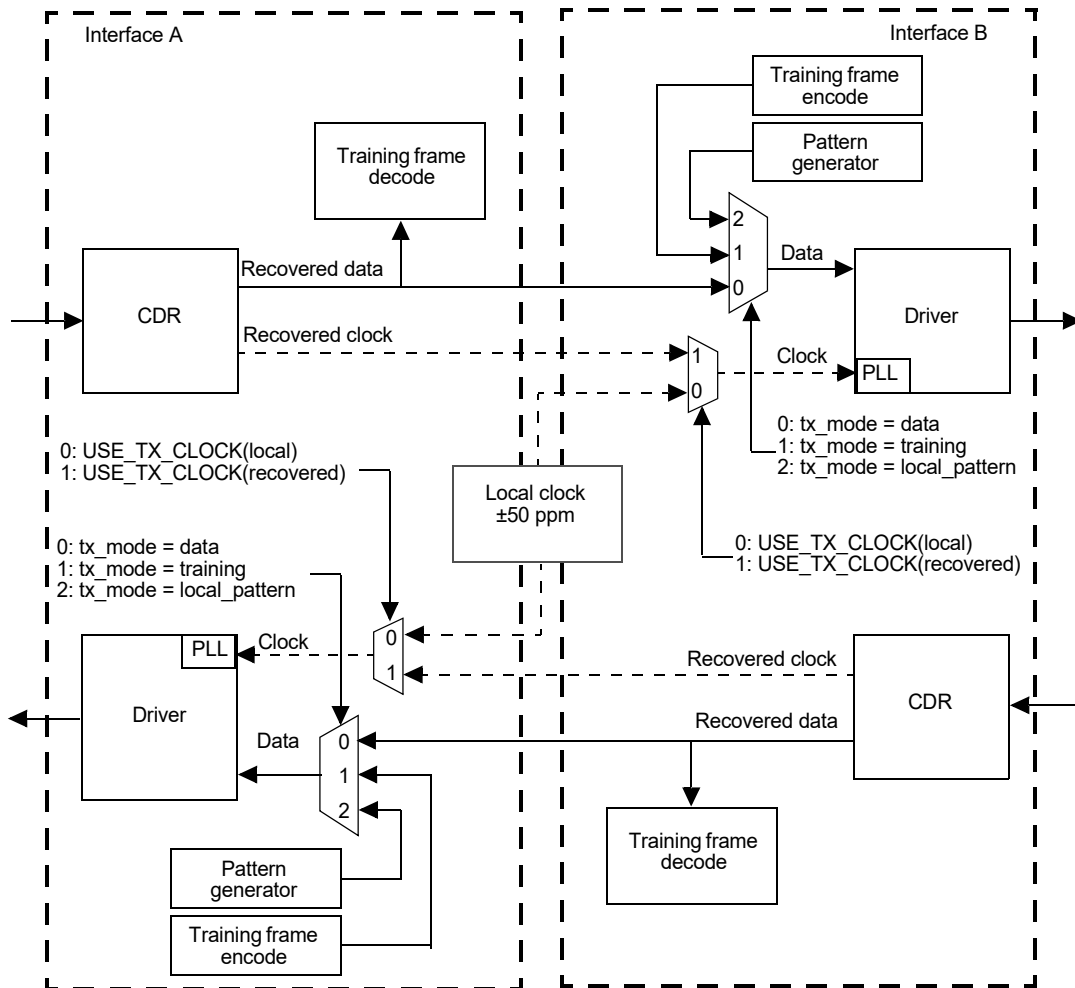


Figure 178B–2—Retimer reference model

178B.5.3 Training xMII Extenders

Training on an ISL within an xMII Extender follows the same process, such that `local_rts` and `remote_rts` are communicated to the PHY XS using its `IS_SIGNAL.indication` and `IS_SIGNAL.request` primitives.

AUI components within an xMII Extender may train before or in parallel with the PCS to PCS path, and training signaling will continue until the main path is ready. This is the same behavior as AUI components within a PHY.

178B.6 Training frame structure

The training frame is a sequence of PAM4 symbols whose values (0, 1, 2, 3) correspond to the possible values of the tx_symbol and rx_symbol variables of the sublayer service interface. The structure of a training frame is shown in Figure 178B–3.

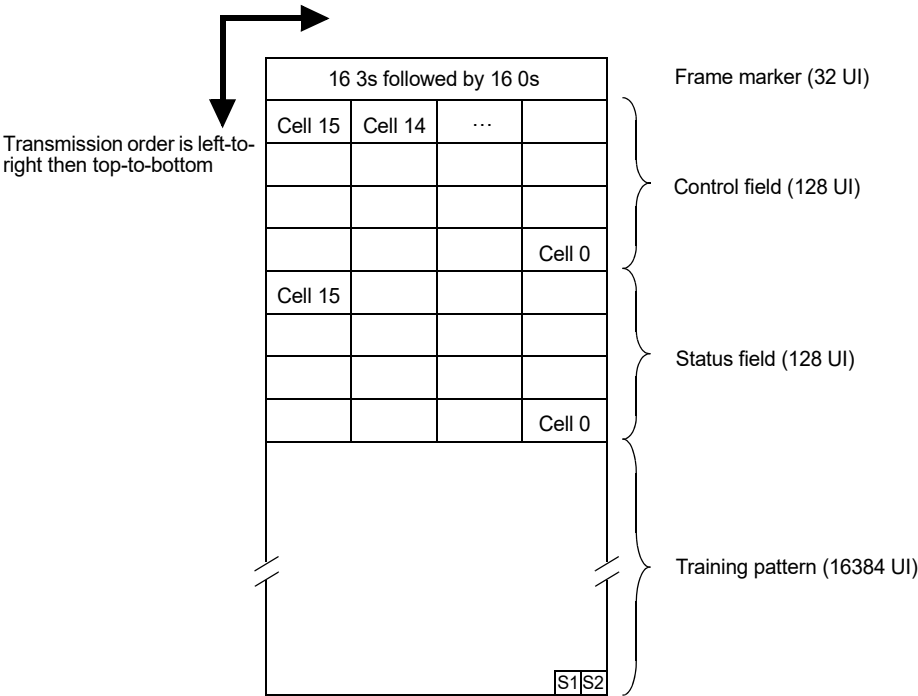


Figure 178B–3—Training frame structure

178B.6.1 Frame marker

Training frames are delimited by a specific sequence of PAM4 symbols. The training frame marker shall be a run of 16 consecutive “3” symbols followed by a run of 16 consecutive “0” symbols. This sequence is not found in the control field, status field, or training pattern and it uniquely identifies the beginning of a training frame.

178B.6.2 Control and status fields

The control field is composed of 16 bits with the structure defined in 178B.7. The status field is composed of 16 bits with the structure defined in 178B.8.

Each bit of the control and status fields is sent as a differential Manchester encoded (DME) cell, where each cell is eight unit intervals in length. The specific rules for this encoding follow.

- A transition from 0 to 3 or from 3 to 0 occurs at the start of each cell.
- A transition from 0 to 3 or from 3 to 0 at the midpoint of a cell, i.e., four unit intervals from the transition at the beginning of the cell, corresponds to a logical one.
- The absence of a transition at the midpoint of a cell corresponds to a logical zero.

The control field is transmitted immediately after the frame marker. The status field is transmitted immediately after the control field. Within each field, the order of transmission is from bit 15 to bit 0.

When a training frame is received, if a violation of the DME encoding rules is detected within the control field or the status field, the contents of both fields in that frame are ignored.

Two formats are defined for the control and status fields, E1 and O1. Each interface using ILT shall identify which format is relevant for it.

178B.6.3 Training pattern

The training pattern is the result of a training pattern generator equivalent to the structure shown in Figure 178B–4.

There are three training pattern generator functions: synchronous PRBS13, free-running PRBS13, and free-running PRBS31.

The contents of the last two symbols of the training frame payload (S1 and S2 in Figure 178B–3) depends on the training pattern. For synchronous PRBS13 these two symbols are “0” and for free-running PRBS13 and free-running PRBS31 these two symbols are taken from the training pattern generator output (see Figure 178B–4).

The training pattern is either PAM2 or PAM4 modulated. For PAM4 modulation, precoding may be enabled or disabled (see 178B.6.4).

The training pattern selector is set to synchronous PRBS13 and the modulation to PAM2 upon entry to the QUIET state of the Training control state diagram (see Figure 178B–8).

NOTE—Exiting TRAINING mode requires both sides to use PAM4 modulation, thus the modulation and precoding request and status are both required to have values other than PAM2.

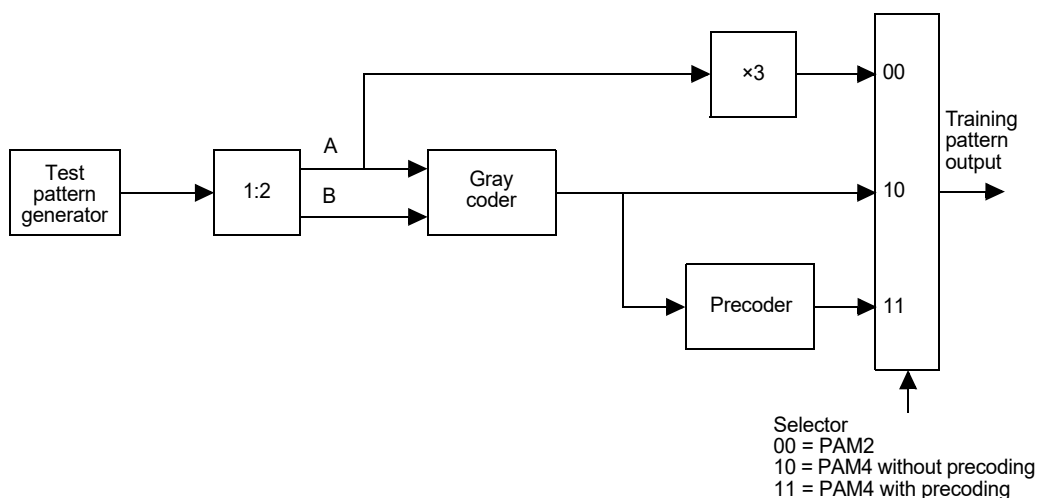


Figure 178B–4—Training pattern generator

178B.6.3.1 Synchronous PRBS13 function

The synchronous PRBS13 function for each lane shall implement four generator polynomials. The polynomial used in each lane i is selected by the variable identifier i .

At the start of the training pattern in each training frame in each lane i , the state of the PRBS13 generator shall be set to the value of the variable prbs13_seed i . A value of all zeros is not valid.

Table 178B–1 specifies the default identifier, the corresponding polynomial, and the recommended default value of prbs13_seed i for each lane i , as well as the first 13 symbols of the training pattern for PAM2, PAM4 without precoding, and PAM4 with precoding, created using the default polynomial and seed.

A sample implementation of the PRBS13 generator for identifier $i = 0$ is shown in Figure 178B–5.

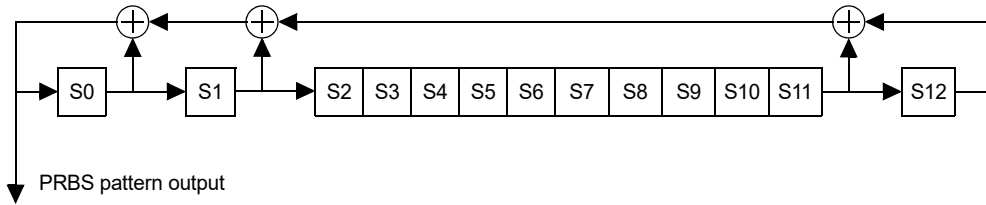


Figure 178B–5—Test pattern PRBS13 generator (identifier $i = 0$)

Table 178B–1—Training pattern default polynomials and seeds

i	Default identifier	Polynomial, $G(x)$	Default prbs13_seed i^a	Initial output, PAM2	Initial output, PAM4 without precoding	Initial output, PAM4 with precoding
0	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	0000010101011	0030330330000	1031320220111	1301200200101
1	1	$1 + x^2 + x^3 + x^7 + x^{13}$	0111010000011 ^c	3330003000330	2220002101230	2022220132031
2	2	$1 + x^2 + x^4 + x^8 + x^{13}$	1001000101100	0303333033030	1212332133031	1102120121301
3	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0100010000010	3330300030330	2231210121221	2032013201110
4	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	1111110100110	3030000303303	3030001313212	3122223012011
5	1	$1 + x^2 + x^3 + x^7 + x^{13}$	1000111011100 ^c	3033030003030	3032020002021	3120022220023
6	2	$1 + x^2 + x^4 + x^8 + x^{13}$	0000001101000	3300303000300	2300212111300	2131102323000
7	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0011000100111	3333000333030	2232000322031	2033131202210

^a The leftmost bit in the sequence corresponds to the initial value of S0 and the rightmost bit corresponds to the initial value of S12 in Figure 178B–5.

^b This training pattern is equivalent to the PRBS13Q test pattern defined in 120.5.11.2.1.

^c The default seed values for $i = 1$ and $i = 5$ are different from the default seed values in Table 162–10a.

178B.6.3.2 Free-running PRBS13 function

A set of four free-running PRBS13 generator polynomials are defined to minimize correlated interference between physical lanes during training. The free-running PRBS13 generator for each lane shall implement each of four generator polynomials defined in Table 178B–1. The polynomial used in each lane i is selectable by identifier $_i$. The default identifier for each lane is the same as that shown in Table 178B–1. A sample implementation of the free-running PRBS13 generator for identifier $_i = 0$ is shown in Figure 178B–5.

The initial value of the free-running PRBS13 generator shall not be all zeros. It may be any other value. For eight-lane interfaces the same polynomial is used for two lanes. The two generators shall be configured such that their relative offsets are large enough to make adjacent lanes uncorrelated within the length of the training frame. For example, this may be achieved by initialization with different seeds or with the same seed at different times.

The free-running PRBS13 generator operates independently of the training frame and is not stopped or reset while training is in progress and this mode is selected. As a result, the state of the free-running PRBS13 generator at the beginning of the training pattern is different in subsequent training frames. The marker, control, and status DME portions of the training frame (see 178B.6) periodically override the free-running PRBS13 generator output (for 288 UI every 16672 UI).

178B.6.3.3 Free-running PRBS31 function

The free-running PRBS31 generator for each lane shall implement the PRBS31 pattern defined by Equation (49-2) and shown in Figure 49–9.

The initial state of the free-running PRBS31 generator shall not be all zeros. It may be any other value. The free-running PRBS31 generators in a multi-lane device shall be configured such that their relative offsets are large enough to make adjacent lanes uncorrelated within the length of the training frame. For example, this may be achieved by initialization with different prbs31_seed $_i$ ($i = 0$ to 7) or with the same seed at different times.

The free-running PRBS31 generator operates independently of the training frame and is not stopped or reset while training is in progress and this mode is selected. As a result, the state of the free-running PRBS31 generator at the beginning of the training pattern is different in subsequent training frames. The marker, control, and status DME portions of the training frame (see 178B.6) periodically override the free-running PRBS31 generator output (for 288 UI every 16672 UI).

178B.6.4 Training pattern modulation and coding

The training pattern generator generates three different training patterns. The construction of each training pattern begins by demultiplexing the test pattern generator output into pairs of bits {A, B} where A corresponds to the 1st, 3rd, 5th, etc. bits output by the test pattern generator and B corresponds to the 2nd, 4th, 6th, etc. bits output by the test pattern generator. There is a new pair of bits each unit interval which implies that the test pattern generator is required to generate bits at twice the signaling rate. Given these pairs of bits, the three different training patterns correspond to three values of the training pattern selector in Figure 178B–4: PAM2, PAM4 without precoding, and PAM4 with precoding.

When the training pattern selector is set to PAM2, the training pattern is the sequence of PAM2 symbols derived by mapping the A bits from the test pattern generator such that logical 0 is transmitted as 0 and logical 1 is transmitted as 3.

When the training pattern selector is set to PAM4 without precoding, the training pattern is the sequence of PAM4 symbols derived by Gray coding the {A, B} pairs as specified for outputs in 120.5.7.1.

When the training pattern selector is set to PAM4 with precoding, the training pattern is the sequence of PAM4 symbols derived by Gray coding the {A, B} pairs as specified for outputs in 120.5.7.1 and precoding the result as specified for outputs in 135.5.7.2.

For synchronous PRBS13, at the beginning of each training pattern the precoder state is set to 0 such that $P(j-1) = 0$ in Equation (135–1) for the first PAM4 symbol of the training pattern. For free-running PRBS13 and PRBS31, it is recommended that the PRBS test pattern generator and the precoder are initialized simultaneously with the seed (for the PRBS test pattern generator) and $P(j-1) = 0$ (for the precoder).

178B.7 Control field structure

The structure of the control field for E1 interfaces shall be as shown in Table 178B–2 and for O1 interfaces as shown in Table 178B–3.

Table 178B–2—Control field structure for E1 interfaces

Bit(s)	Name	Description
15:14	Reserved	Transmit as 0, ignore on receipt
13:11	Initial condition request	13 12 11 1 1 1 = Reserved 1 0 1 = Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control
10	Continue training	1 = Continue training 0 = Switch to data when training is completed
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
7	Reserved	Transmit as 0, ignore on receipt
6:5	Training pattern request	6 5 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
4:2	Coefficient select	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved
1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold

Table 178B-3—Control field structure for O1 interfaces

Bit(s)	Name	Description
15:11	Reserved	Transmit as 0, ignore on receipt
10	Continue training	1 = Continue training 0 = Switch to data when training is completed
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
7	Reserved	Transmit as 0, ignore on receipt
6:5	Training pattern request	6 5 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
4:0	Reserved	Transmit as 0, ignore on receipt

178B.7.1 Initial condition request

The initial condition request bits are used to select one of the up to six predefined transmitter equalizer configurations (presets) specified in the AUI annexes or PMD clauses. When the initial condition request bits are 000, the coefficient select and coefficient request bits are used to update the transmitter equalizer configuration.

178B.7.2 Continue training

The continue training bit carries the inverse value of the local_rts variable (see 178B.14.2.1).

178B.7.3 Modulation and precoding request

The modulation and precoding bits are used to request that the peer interface transmitter modulates and codes the transmitted training pattern using one of the options defined in 178B.6.4.

178B.7.4 Training pattern request

The training pattern bits are used to request that the peer interface transmitter transmits one of the training patterns defined in 178B.6.3.1, 178B.6.3.2, and 178B.6.3.3.

178B.7.5 Coefficient select

The coefficient select bits are used to identify the coefficient that is the target of a coefficient request. The value of the field is the two's complement encoding of the coefficient index. For example, binary value 111 corresponds to a coefficient index of -1.

178B.7.6 Coefficient request

The coefficient request bits are used to change the value of the coefficient specified by the coefficient select bits. A coefficient may be changed by incrementing or decrementing its value or by setting it to “No equalization”. The “No equalization” value is 1 for $c(0)$ and 0 for $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$.

178B.8 Status field structure

The structure of the status field for E1 interfaces shall be as shown in Table 178B–4 and for O1 interfaces as shown in Table 178B–5.

Table 178B–4—Status field structure for E1 interfaces

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14	ILT	Transmit as 1
13:12	Training pattern status	13 12 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Initial condition status	1 = Updated 0 = Not updated
7	Parity	Even parity bit
6	Reserved	Transmit as 0, ignore on receipt
5:3	Coefficient select echo ^a	5 4 3 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$
2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated

^aCoefficient select echo values 010, 011 and 100 are undefined.

Table 178B–5—Status field structure for O1 interfaces

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14	ILT	Transmit as 1
13:12	Training pattern status	13 12 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Reserved	Transmit as 0, ignore on receipt
7	Parity	Even parity bit
6:0	Reserved	Transmit as 0, ignore on receipt

178B.8.1 Receiver ready

The receiver ready bit is used to signal the local receiver state to the peer interface. When this bit is 1, it indicates that the local receiver has completed training and is prepared to receive data (`local_rx_ready = true`). When this bit is 0, it indicates that the local receiver is requesting that training continue (`local_rx_ready = false`).

178B.8.2 ILT

The ILT bit is set to one to signal ILT support.

178B.8.3 Training pattern status

The training pattern status bits encode the value of `local_tp_mode`.

178B.8.4 Modulation and precoding status

The modulation and precoding bits encode the value of `local_mc_mode`.

178B.8.5 Receiver frame lock

When the receiver frame lock bit is set to 1, the receiver is indicating that it has identified training frame marker positions and is in a state where the response time requirements specified in 178B.13 are met.

Receiver frame lock is set to 0 when the variable training is false, and is not set to one until training and local_tf_lock are both true and the response time requirements specified in 178B.13 can be met.

178B.8.6 Initial condition status

The initial condition status bit acknowledges the initial condition request bits received from the peer interface. The acknowledgment reflects the value of ic_sts resulting from the procedure described in 178B.11.1.

178B.8.7 Parity

The parity bit is calculated based on the other bits in the control field and status field to create even parity for these fields. Even parity ensures that the transmitted control and status fields (see 178B.6.2) are DC balanced. This field is ignored on receipt.

178B.8.8 Coefficient select echo

The coefficient select echo bits acknowledge the coefficient select bits received from the peer interface. When a change to the coefficient select bits is detected, the coefficient select echo bits are updated to represent the same coefficient index. This serves as a confirmation to the peer interface that subsequent coefficient requests will act on the targeted coefficient.

178B.8.9 Coefficient status

The coefficient status bits acknowledge the coefficient request bits received from the peer interface. The coefficient status bits reflect the value of the coef_sts variable generated by the coefficient update state diagram (Figure 178B–10).

178B.9 Training frame lock

Training frame lock shall be achieved by looking for the frame marker or the inverted frame marker in consecutive training frames as specified in Figure 178B–9.

178B.10 Polarity detection and correction

When training starts for each lane, the variable polarity_correction is set to false. If inverted frame markers are detected during the frame lock process, the polarity_correction variable is set to true.

The state of the polarity_correction variable persists after training completes, correcting the polarity of the data received when tx_mode = data.

If polarity_correction is true and local_tf_lock is true, the lane input shall be corrected by mapping the received PAM4 symbols 0, 1, 2, and 3 to PAM4 symbols 3, 2, 1, and 0, respectively.

NOTE—Polarity detection and correction is not available when training is disabled.

178B.11 Equalization control

Equalization control is only available for E1 interfaces.

When the training control state diagram (Figure 178B–8) is in the TRAIN_LOCAL state, the device may request its peer interface to change the transmitter equalization coefficients, either to predefined initial conditions or by individual coefficient control. The criteria for initiating such requests are implementation dependent.

When the training control state diagram (Figure 178B–8) is in either the TRAIN_LOCAL or TRAIN_REMOTE states, the device shall respond to requests received from the peer interface.

When the training control state diagram (Figure 178B–8) is in any state other than TRAIN_LOCAL or TRAIN_REMOTE, the device shall not send any request to the peer interface and shall ignore requests from the peer interface.

A new individual coefficient update request or initial condition update request is not initiated until after the prior request has completed.

178B.11.1 Initial condition setting request process

A request to change the initial condition of the peer interface transmitter shall be made by using the following procedure:

- a) Set the initial condition request bits (178B.7.1) to the desired predefined transmitter equalizer configuration (preset) and the coefficient request bits (178B.7.6) to “hold”.
- b) Wait until the initial condition status bits (178B.8.6) indicate “updated” and the coefficient status bits (178B.8.9) indicate “not updated”.
- c) Set the initial condition request bits (178B.7.1) to individual coefficient control and the coefficient request bits (178B.7.6) to “hold”.
- d) Wait until both the initial condition status bits (178B.8.6) and the coefficient status bits (178B.8.9) indicate “not updated”.

178B.11.2 Initial condition setting response process

The handling of incoming requests is specified by the coefficient update state diagram (Figure 178B–10). The behavior of the UPDATE_IC function shall be consistent with the following algorithm.

```
if ic_req = ind_ctl
    ic_sts = not_upd
else if CHECK_REQ(ic_req)
    Set coefficients according to ic_req
    ic_sts = updated
    coef_sts = not_upd
else
    ic_sts = updated
    coef_sts = coefficient not supported
end if
```

The variables ic_req, ic_sts and coef_sts are defined in 178B.14.3.1. The transmitter equalizer coefficients presets shall be within the ranges specified in the AUI annex or PMD clause for the selected preset.

178B.11.3 Coefficient update request process

A request to change an individual equalizer coefficient of the peer interface transmitter shall be made by using the following procedure:

- a) In the transmitted control field, set the initial condition request bits (178B.7.1) to individual control, and set the coefficient select bits (178B.7.5) and coefficient request bits (178B.7.6) to the desired values. This may be done in one step (all fields updated in the same training frame) or sequentially.
- b) Wait until the received coefficient status bits (178B.8.9) do not indicate “not updated” and the coefficient select echo bits (178B.8.8) indicate the requested coefficient select value.
- c) If the subsequent request is to change the same coefficient or request a new initial condition, set the coefficient request bits (178B.7.6) to “hold” and wait until the received coefficient status bits (178B.8.9) indicate “not updated”.

178B.11.4 Coefficient update response process

The handling of incoming requests is specified by the coefficient update state diagram (Figure 178B–10). The behavior of the UPDATE_C(k) function shall be consistent with the following algorithm.

```

if k in k_list
    if coef_req = increment
        ck_ask = c(k) + ck_stp
    else if coef_req = decrement
        ck_ask = c(k) – ck_stp
    else if coef_req = no equalization
        if k = 0
            ck_ask = 1
        else
            ck_ask = 0
        end if
    else
        ck_ask = c(k)
    end if
    if ck_ask > ck_max
        c(k) = ck_max
        if CHECK_EQ(ck_ask,k)
            coef_sts = coefficient at limit and equalization limit
        else
            coef_sts = coefficient at limit
        end if
    else if ck_ask < ck_min
        c(k) = ck_min
        if CHECK_EQ(ck_ask,k)
            coef_sts = coefficient at limit and equalization limit
        else
            coef_sts = coefficient at limit
        end if
    else if CHECK_EQ(ck_ask,k)
        coef_sts = equalization limit
    else
        c(k) = ck_ask
        coef_sts = updated
    end if
else

```

coef_sts = coefficient not supported
end if

The variables coef_req, coef_sts, and k are defined in 178B.14.3.1. The following additional variables and functions are used.

c(k)
Variable that contains the current value of the coefficient $c(k)$.

ck_ask
Variable that contains the value of $c(k)$ that would result from the requested update.

ck_min
Variable that contains the minimum supported value of $c(k)$.

ck_max
Variable that contains the maximum supported value of $c(k)$.

ck_stp
Variable that contains the magnitude of the change in $c(k)$ for one step up or one step down from its current value.

k_list
The set of valid transmitter equalizer coefficient indices as defined by the AUI component or PMD.

CHECK_EQ(ck_ask,k)
Compares the transmitter's steady-state voltage that would result from setting transmit equalization coefficient $c(k)$ value to ck_ask, while keeping all other coefficients unchanged, against the transmitter's equalization capability and coefficient ranges (see 179.9.4.1.5 as an example). Returns true if the resulting combination of coefficients would exceed the transmitter's equalization capability or coefficient ranges. Otherwise returns false.

178B.12 Training pattern setting

When a change to the modulation and precoding request bits or the training pattern request bits is detected, the transmitted training pattern (see 178B.6.3) is chosen accordingly. To confirm that the change to the format of the training pattern was completed, the local_mc_mode variable is set to the value of the modulation and precoding request bits and the local_tp_mode variable to the value of the training pattern request bits. local_mc_mode and local_tp_mode are encoded in status fields (see 178B.8.3 and 178B.8.4).

178B.13 Handshake timing

Changes to the configuration of the transmitter equalizer or the transmitted training pattern may occur at any point between the time the request for the new configuration is received and the time that the request is acknowledged.

When the receiver frame lock bit in the status field of transmitted training frames is set to one, the time from the receipt of a new request to the acknowledgment of that request shall be less than 20 ms. It is recommended that the average response time be less than 2 ms. A new request is defined to be a received training frame whose control field differs from the control field of the preceding training frame. An acknowledgment is defined as the first transmitted training frame that contains a status field encoding that is an appropriate response for the requested action. For example, a change in the coefficient select bits in the control field would be acknowledged by a change in the coefficient select echo bits of the status field. Similarly, a change in the pattern request bits in the control field would be acknowledged by a change in the

pattern status bits in the status field. All timing measurements are referred to a common reference point within the training frame (e.g., the 3 to 0 transition in the training frame marker).

The timing of requests and acknowledgment is illustrated in Figure 178B–6.

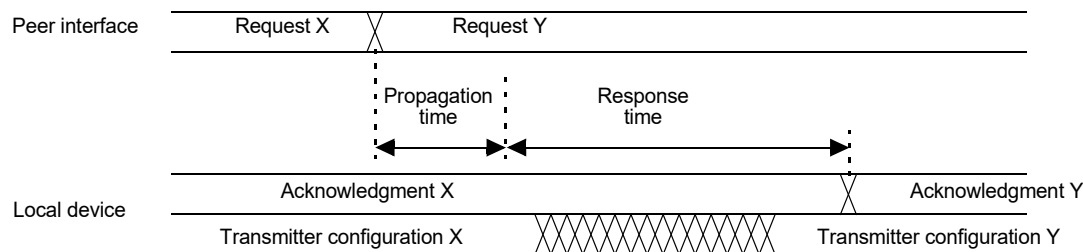


Figure 178B–6—Transmitter update timing

178B.14 State diagrams

This subclause is composed of detailed state diagram processes, including the associated definition of variables, functions, counters and timers.

178B.14.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5. State diagram timers follow the conventions of 14.2.3.2.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

178B.14.2 Per-interface state diagram definitions

An AUI component or PMD implements one instance of the RTS update state diagram, and the set of associated variables, functions, counters and timers defined in this subclause, independently for each of its interfaces (see 178B.5).

178B.14.2.1 Variables

`adjacent_signal_ok`

Enumerated variable derived from the value of the `SIGNAL_OK` parameter on the adjacent service interface. It takes one of the following values: `IN_PROGRESS`, `READY`, `OK`, `FAIL`.

NOTE — For ILT in a PMD or an AUI component below a PMA, the adjacent service interface is the service interface of the PMD or AUI component, and `SIGNAL_OK` is received via the `IS_SIGNAL.request` primitive. For ILT in an AUI component above a PMA, the adjacent service interface is the interface below the AUI component, and `SIGNAL_OK` is received via the `IS_SIGNAL.indication` primitive.

`adjacent_remote_rts`

Boolean variable that indicates the value of `remote_rts` on the adjacent service interface. It is true if `adjacent_signal_ok` is `OK` and false otherwise.

adjacent_isl_ready
Boolean variable that indicates the value of `isl_ready` on the adjacent service interface. It is true if `adjacent_signal_ok` is OK or READY and false otherwise.

isl_ready
Boolean variable that is set to true when `local_rx_ready` and `remote_rx_ready` are true for all lanes of the interface, and to false otherwise.

local_rts
Boolean variable that indicates that an interface is ready to send and receive normal data. It is set by the RTS state diagram (see Figure 178B–7). The logical-NOT of this variable is encoded as the “continue training” bit in the control field of transmitted training frames.

mr_restart
Boolean variable used by system management to restart ILT.

NOTE — There is no specified time limit for the ILT protocol. To avoid live-lock situations, ILT should only be restarted if there is an indication of an unrecoverable fault. The definition of unrecoverable fault is beyond the scope of this annex.

mr_training_enable
Boolean variable that is set by management. When it is true, training is enabled on the interface. When it is false, training is not enabled on the interface.

remote_rts
Boolean variable that indicates the value of `local_rts` in the peer interface. If `mr_training_enable` is true and the “continue training” bit of the control field of received training frames on all lanes of the interface is zero then `remote_rts` is set to true, otherwise it is set to false. If `mr_training` is false then `remote_rts` is always true.

reset
Boolean variable that controls the resetting of the device. It is true whenever a reset is necessary including when initiated by `PMA_reset` for AUI components, `PMD_reset` for PMDs and during power on.

training_status
Enumerated variable that indicates the status of the ILT function. This variable may be assigned one of the following values: IN_PROGRESS, READY, OK, FAIL.

uses_recovered_clock
Boolean variable set to true to indicate that the AUI component or PMD uses a clock recovered from another interface to drive its output when `tx_mode` = data. Otherwise it is set to false.

178B.14.2.2 Functions

USE_TX_CLOCK(source)
Selects the clock source for the transmitter. The source parameter takes one of the values:
local: use a local clock to drive the transmitter output.
recovered: use a clock recovered from the other interface’s receiver (see 178B.5.2) to drive the transmitter output.

NOTE—The details of the clock recovery and forwarding architecture are beyond the scope of this standard.

178B.14.2.3 Timers

forward_rts_timer
This timer is started when the RTS state diagram enters the SWITCH_CLOCK state (see Figure 178B–7). The terminal count of `forward_rts_timer` is between 10 ms and 90 ms.

178B.14.2.4 State diagram figures

The RTS update state diagram (Figure 178B–7) sets the value of the local_rts variable and it shall be implemented for each interface.

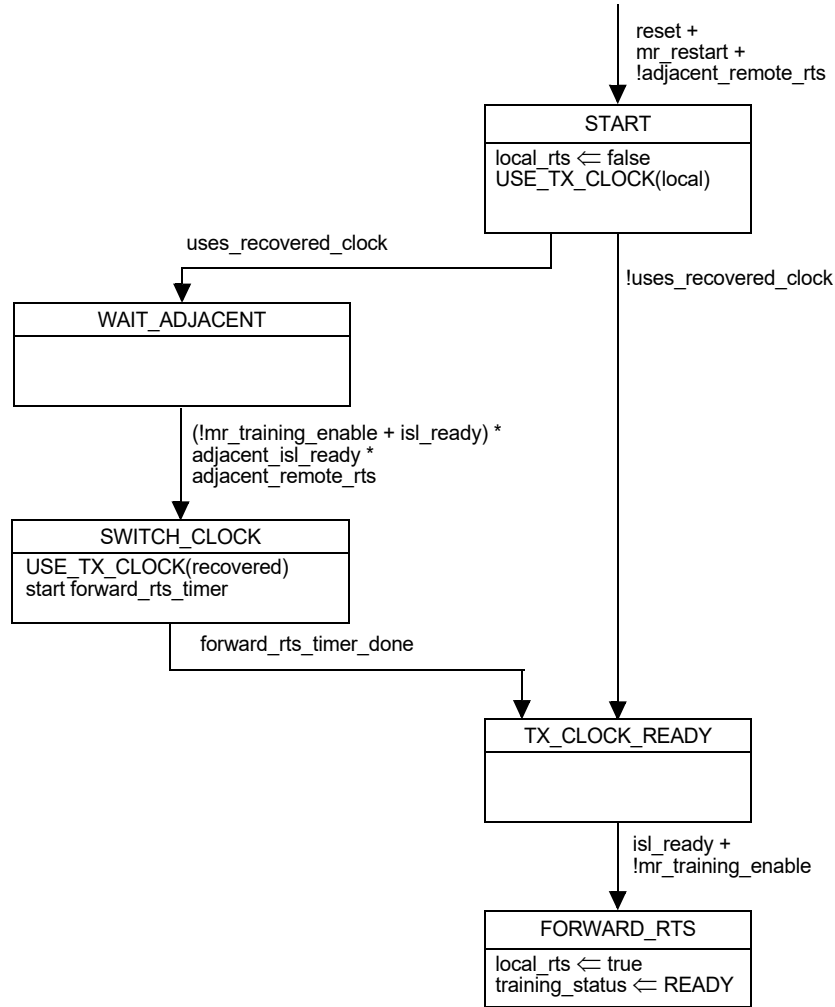


Figure 178B–7—RTS update state diagram

178B.14.3 Per-lane state diagram definitions

An AUI component or PMD implements one instance of each of the Training control and the Training frame lock, and their associated variables, functions, counters and timers defined in this subclause, independently for each of the n physical lanes on each of its interfaces (see 178B.5).

E1 interfaces also implement one instance of the Coefficient update state diagram and its associated variables and functions independently for each of the n physical lanes. For O1 interfaces, this diagram and its associated variables and functions are not used.

If the PATH_READY state in the Training control state diagram (see Figure 178B–8) is entered with local_mc_mode set to “PAM4 with precoding”, then the AUI component or PMD shall cause the adjacent PMA to transmit all subsequent data on the corresponding lane with precoding (see 176.7.1.2) and otherwise cause the adjacent PMA to transmit all subsequent data on the corresponding lane without precoding.

If the PATH_READY state in the Training control state diagram (see Figure 178B–8) is entered with remote_mc_mode set to “PAM4 with precoding”, then the AUI component or PMD shall inform the adjacent PMA that all subsequently received data on the corresponding lane includes precoding (see 176.7.1.2) and otherwise inform the adjacent PMA that all subsequently received data on the corresponding lane does not include precoding.

178B.14.3.1 Variables

coef_req

Enumerated variable derived from the “coefficient request” bits from the control field of the received training frames (see 178B.7). This variable may be one of the following values: hold, decrement, increment, no equalization.

coef_sel

Variable derived from the “coefficient select” bits from the control field of the received training frames (see 178B.7). It is assigned a signed integer value that is the two’s complement interpretation of the bits.

coef_sts

Enumerated variable that may be assigned one of the following values (abbreviations used by the state diagram are included in parentheses): not updated (not_upd), updated, coefficient at limit, coefficient not supported, equalization limit, coefficient at limit and equalization limit. The value is assigned by the UPDATE_C(k) function and coefficient update state diagram (Figure 178B–10), and then encoded in the status field of transmitted training frames via the ENCODE_STS function.

ic_req

Enumerated variable that holds the initial condition setting to be applied to the local transmitter. It may be assigned any of the values assigned to ic_sel, or the value initialize.

ic_sel

Enumerated variable derived from the “initial condition request” bits from the control field of the received training frames (see 178B.7). This variable may be assigned one of the following values (abbreviations used by the state diagram are included in parentheses): individual control (ind_ctl), preset 1, preset 2, preset 3, preset 4, preset 5, preset 6.

ic_sts

Enumerated variable that may be assigned one of the following values (abbreviations used by the state diagram are included in parentheses): not_updated (not_upd), updated.

inverse_marker_valid

Boolean variable that is set to true when the candidate frame marker matches the inverse of the frame marker pattern defined in 178B.6.1 and is set to false otherwise.

k

Variable that stores the most recent value of coef_sel.

local_mc_mode

Enumerated variable that controls the training pattern modulation and coding (see 178B.12), and

may be assigned one of the following values: PAM2, PAM4 without precoding, PAM4 with precoding.	1
	2
local_rx_ready	3
Boolean variable that is set to true when the receiver on a lane of the interface has determined that the peer interface transmitter is transmitting a PAM4 signal, that the remote transmit and local receive equalizers have been optimized, and that no further adjustments are required for normal data transmission. It is set to false otherwise. The exact criteria for setting this variable to true are implementation specific. When mr_training_enable is true, the value of this variable is encoded as the “receiver ready” bit in the status field of transmitted training frames.	4
	5
	6
	7
	8
	9
	10
local_tf_lock	11
Boolean variable that is true when mr_training_enable is true and the training frame marker positions have been identified on a lane of the interface and is false otherwise. The value of this variable is encoded in the receiver frame lock bit in the status field of transmitted training frames.	12
	13
	14
	15
local_tp_mode	16
Enumerated variable that controls the choice of the transmitted training pattern (see 178B.6.3) and may be assigned one of the following values: synchronous PRBS13, free-running PRBS13, free-running PRBS31.	17
	18
	19
marker_valid	20
Boolean variable that is set to true when the candidate frame marker matches the frame marker pattern defined in 178B.6.1 and is set to false otherwise.	21
	22
	23
max_recovery_events	24
Integer variable that controls the maximum allowed number of transitions into the RECOVERY state in the Training control state diagram (Figure 178B–8). A value of zero allows unlimited number of transitions. The value of this variable is implementation dependent.	25
	26
	27
new_marker	28
Boolean variable that is set to true when a new candidate frame marker is available for testing and is set to false when the TEST_MARKER state is entered. A new frame marker is available for testing when the training frame lock process has accumulated 32 consecutive symbols starting at the candidate frame start position.	29
	30
	31
	32
	33
polarity_correction	34
Boolean variable that is set to true when an inverted marker is detected. Otherwise it is set to false.	35
	36
remote_mc_mode	37
Enumerated variable that corresponds to the modulation and coding status of the status field of the received training frames and may be assigned one of the following values: PAM2, PAM4 without precoding, PAM4 with precoding.	38
	39
	40
remote_rx_ready	41
Boolean variable that indicates the value of local_rx_ready on a lane of the peer interface. If mr_training_enable is true, it is derived from the “receiver ready” bit of the status field of received training frames on the correspondent lane of the interface. Otherwise it is set to true.	42
	43
	44
remote_tf_lock	45
Boolean variable that indicates the value of local_tf_lock on a lane of the peer interface. If mr_training_enable is true, it is derived from the “receiver frame lock” bit of the status field of received training frames on the correspondent lane of the interface. Otherwise it is unspecified.	46
	47
	48
	49
remote_tp_mode	50
Enumerated variable that corresponds to the training pattern status of the status field of the received training frames and may be assigned one of the following values: synchronous PRBS13, free-running PRBS13, free-running PRBS31.	51
	52
	53
	54

slip_done	1
Boolean variable that is set to TRUE when the SLIP requested by the Training frame lock state diagram (Figure 178B–9) is completed indicating that the next candidate frame sync position can be tested.	2
	3
	4
	5
tf_offset	6
Boolean variable that is set to true when receiving one full training frame (16 672 symbols, see 178B.6) following the recent frame start position, such that a new frame marker is expected. Otherwise it is set to false	7
	8
	9
training	10
Boolean variable that is set to true when ILT is in progress. Otherwise it is set to false.	11
	12
training_failure	13
Boolean variable that is set to true when training failed to complete. The value of training_failure is set by the training control state diagram (see Figure 178B–8).	14
	15
tx_disable	16
Boolean variable that controls the transmitter’s output on the lane. It is set by the training control state diagram (see Figure 178B–8). When it is true, the transmitter’s output on the lane is disabled. When it is false, tx_mode controls the content of the transmitter’s output on the lane.	17
	18
	19
	20
tx_mode	21
Enumerated variable that controls the content of the transmitter’s output of the lane when tx_disable is false. It is set by the training control state diagram (see Figure 178B–8) to one of the values:	22
training: transmit training frames.	23
local_pattern: transmits a pattern from a valid pattern generator.	24
data: transmit data received from the other interface, after processing by the device’s data path functions.	25
	26
	27
	28

178B.14.3.2 Functions

CHECK_REQ(ic_req)	30
This function compares the value of ic_req against the list of specified presets for the AUI component or PMD, and returns true if the value is supported and false otherwise.	31
	32
	33
ENCODE_STS	34
Encodes portions of the status field of transmitted training frames. k is mapped to the coefficient select echo bits, coef_sts is mapped to the coefficient status bits, and ic_sts is mapped to the initial condition status bit.	35
	36
	37
	38
SLIP	39
Causes the next candidate frame marker position to be tested. Repeated invocations of SLIP cause all possible positions to be evaluated. The precise method for determining the next candidate frame marker position is implementation dependent and beyond the scope of this standard.	40
	41
	42
UPDATE_C(k)	43
Updates the value of $c(k)$ based on the current values of the variables k and coef_req. The result of the update is stored in the variable coef_sts. This function implements the algorithm specified in 178B.11.4.	44
	45
	46
	47
UPDATE_IC	48
Updates all transmit equalizer coefficients based on the current value of the variable ic_req. The result is stored in the variable ic_sts. This function implements the algorithm specified in 178B.11.2.	49
	50
	51
	52
	53
	54

178B.14.3.3 Timers

quiet_timer

This timer is started when the training control state diagram on a lane enters the QUIET state (see Figure 178B–8). The terminal count of this timer is between 100 ms and 200 ms.

propagation_timer

This timer is started when the training control state diagram on a lane enters the PATH_READY state (see Figure 178B–8). The terminal count of this timer is between 100 ms and 200 ms.

recovery_timer

This timer is started when the training control state diagram on a lane enters the RECOVERY state (see Figure 178B–8). The terminal count of this timer is between 20 ms and 30 ms.

178B.14.3.4 Counters

bad_markers

Count of the number of consecutive frame marker mismatches.

good_markers

Count of the number of consecutive frame marker matches.

recovery_event_count

This counter increments each time the Training control state diagram (see Figure 178B–8) transitions into the RECOVERY state.

178B.14.3.5 State diagram figures

The training control state diagram (Figure 178B–8) defines the operation of ILT for AUI components and PMDs

The Training control state diagram is supported by the training frame lock, RTS update and coefficient update state diagrams. The training frame lock state diagram (Figure 178B–9) determines when ILT has detected training frame boundaries. The RTS update state diagram (Figure 178B–7) sets the value of the local_rts variable. The coefficient update state diagram (Figure 178B–10) defines the process for updating the local transmitter equalizer coefficients in response to requests from the peer interface.

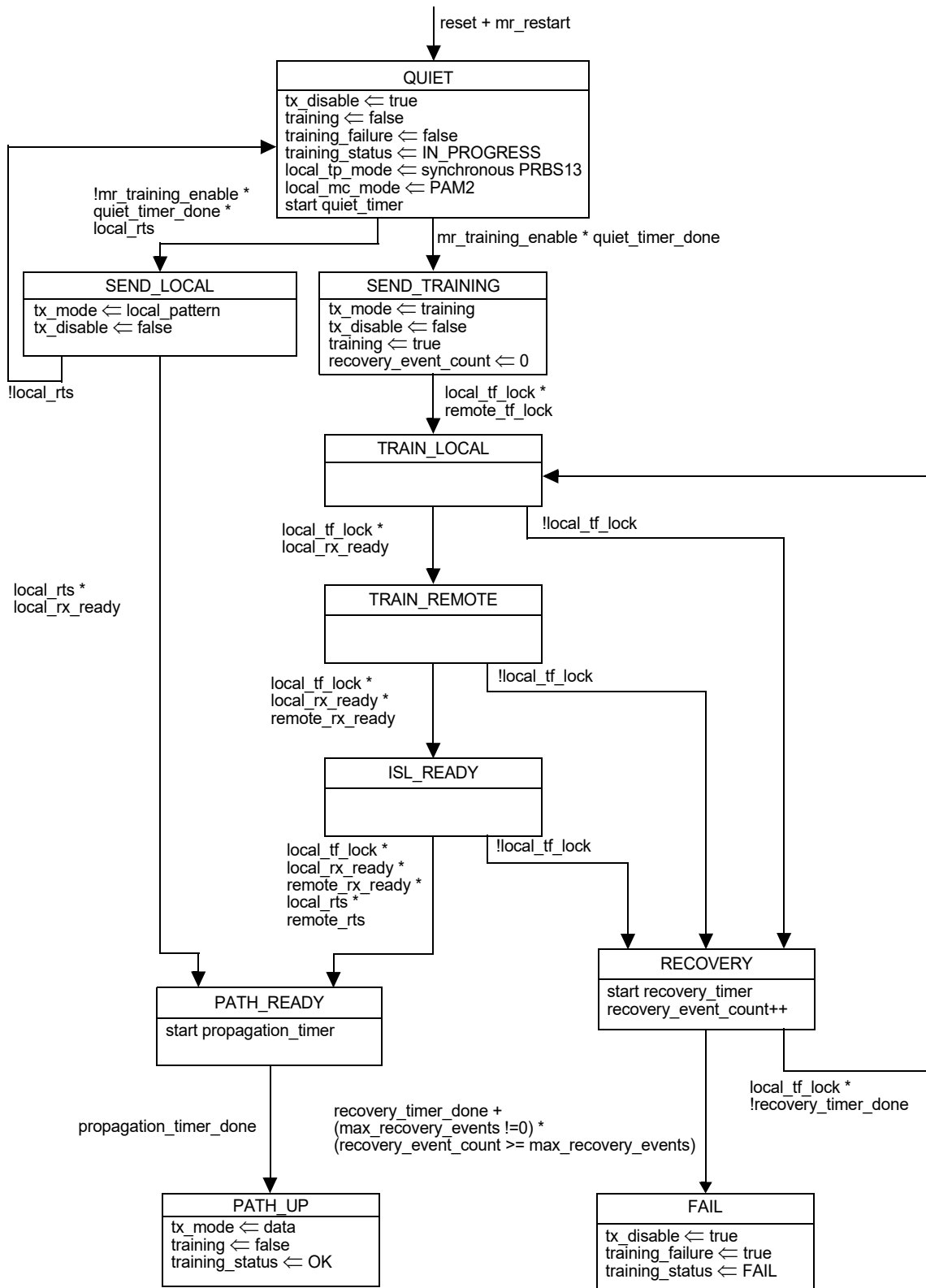


Figure 178B–8—Training control state diagram

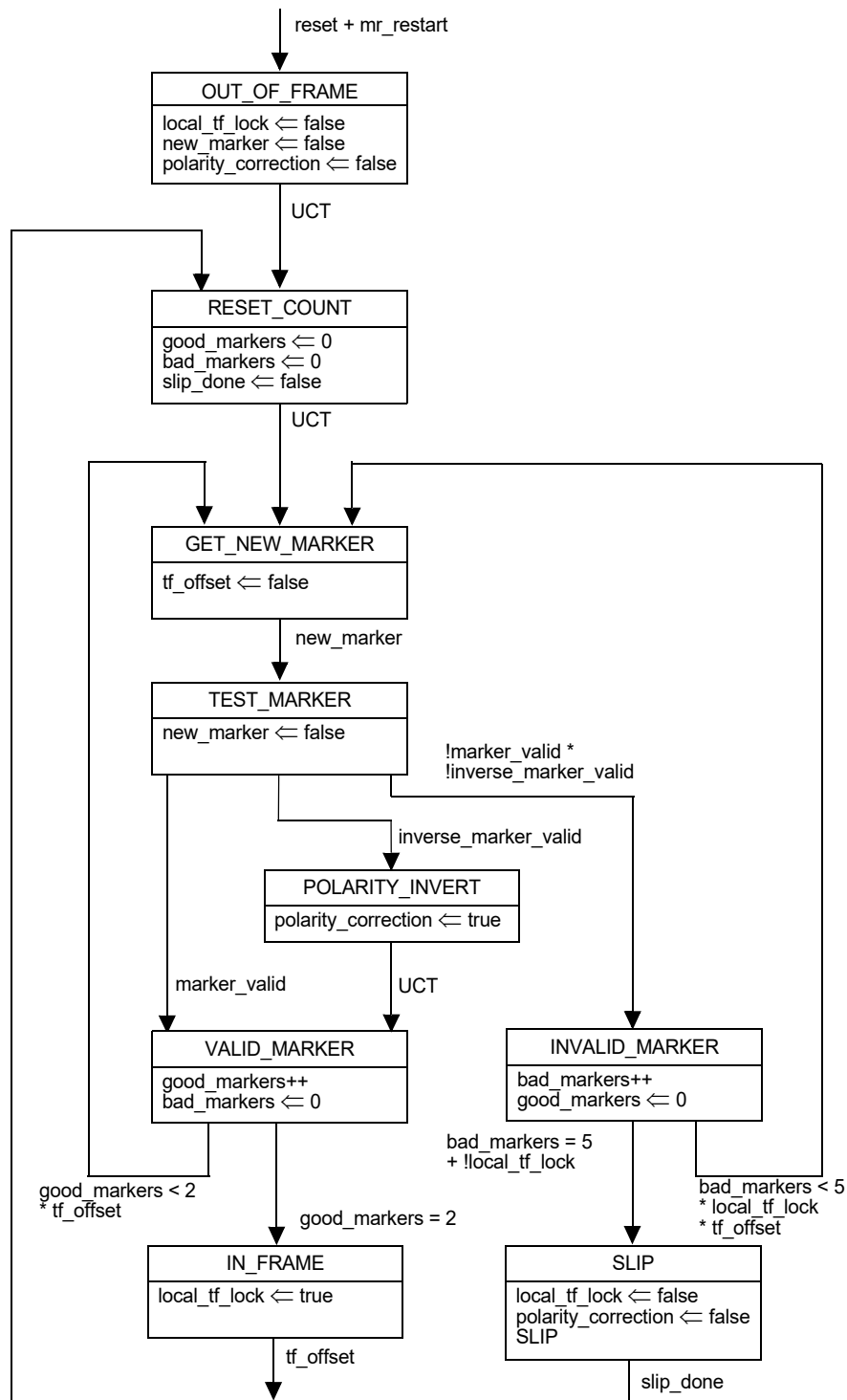


Figure 178B-9—Training frame lock state diagram

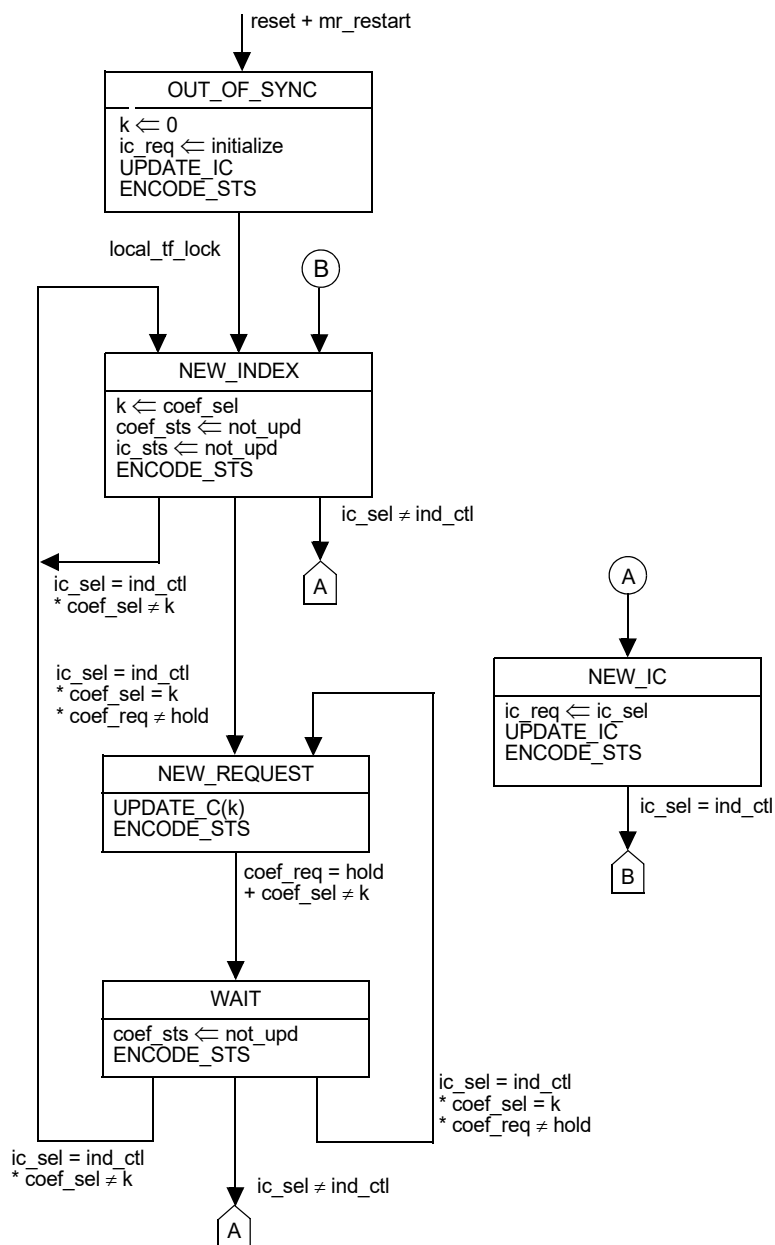


Figure 178B-10—Coefficient update state diagram

178B.15 Management variables

The control and status variables related with this annex and intended to be accessible via a management system are listed in Table 178B-6 and Table 178B-7.

If the MDIO interface (see Clause 45) is implemented, control and status variables shall be mapped to the MDIO registers as shown in Table 178B-6 and Table 178B-7.

If the MDIO Interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 178B–6—Control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number ^a	MDIO register/bit reference
PMA_reset	178B.14.2.1	1.0.15	45.2.1.1
PMD_reset	178B.14.2.1	1.0.15	45.2.1.1
mr_restart	178B.14.2.1	1.150.0	42.2.1.95
mr_training_enable	178B.14.2.1	1.150.1	42.2.1.95
training_status	178B.14.2.1	1.151.2 ^b	42.2.1.96
identifier_i	178B.6.3.1	1.1450.12:11 ^c	45.2.1.168
prbs13_seed_i	178B.6.3.1	1.1450.15:14 1.1450.10:0 ^c	45.2.1.168
prbs31_seed_i	178B.6.3.3	1.1460.15:0 1.1461.14:0 ^c	45.2.1.168a
ic_req	178B.14.3.1	1.1120.13:11 ^c	45.2.1.161
coef_sel	178B.14.3.1	1.1120.4:2 ^c	45.2.1.161
coef_req	178B.14.3.1	1.1120.1:0 ^c	45.2.1.161
coef_sts	178B.14.3.1	1.1420.2:0 ^c	45.2.1.167
ic_sts	178B.14.3.1	1.1420.8 ^c	45.2.1.167
local_tf_lock	178B.14.3.1	1.151.1b	45.2.1.96
remote_tf_lock	178B.14.3.1	1.1220.9 ^c	45.2.1.167
local_rx_ready	178B.14.3.1	1.1420.15 ^c	45.2.1.167
remote_rx_ready	178B.14.3.1	1.1220.15 ^c	45.2.1.167
local_rts	178B.14.2.1	1.1320.10	45.2.1.165
remote_rts	178B.14.2.1	1.1120.10	45.2.1.161
max_recovery_events	178B.14.3.1	1.1477.7:0	45.2.1.168c
uses_recovered_clock	178B.14.2.1	1.1477.8	45.2.1.168c
local_mc_mode	178B.14.3.1	1.1320.9:8 ^c	45.2.1.165
local_tp_mode	178B.14.3.1	1.1320.6:5 ^c	45.2.1.165

^a MDIO register/bit numbers and references are for the bottom AUI component or PMD. The MDIO register/bit numbers and references for the upper AUI component are at +2800 offset.

^b Bit reference is provided for lane 0, bits for lanes 1 to 3 are located in the same register and bits for lanes 4 to 7 are located in BASE-R PMD status 2 register (45.2.1.101).

^c Reference is provided for lane 0. Registers for lanes 1 to 7 are located at an offsets from the lane 0 register.

Table 178B–7—Status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number ^a	MDIO register/bit reference
isl_ready	178B.14.2.1	1.1476.8	45.2.1.168b
training_failure	178B.14.3.1	1.151.3 ^b	45.2.1.96
remote_mc_mode	178B.14.3.1	1.1220.11:10 ^c	45.2.1.163
remote_tp_mode	178B.14.3.1	1.1220.13:12 ^c	45.2.1.163
polarity_correction	178B.14.3.1	1.1476.0 ^d	45.2.1.168b

^aMDIO register/bit numbers and references are for the bottom AUI component or PMD. The MDIO register/bit numbers and references for the upper AUI component are at +4000 offset.

^b Bit reference is provided for lane 0, bits for lanes 1 to 3 are located in the same register and bits for lanes 4 to 7 are located in BASE-R PMD status 2 register (45.2.1.101).

^c Reference is provided for lane 0. Registers for lanes 1 to 7 are located at an offsets from the lane 0 register.

^d Bit reference is provided for lane 0, bits for lanes 1 to 7 are located in the same register.

**178B.16 Protocol implementation conformance statement (PICS) proforma
for Annex 178B, Inter-sublayer link training for electrical interfaces⁸**

178B.16.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 178B, Inter-sublayer link training for electrical interfaces, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

178B.16.2 Identification

178B.16.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

178B.16.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 178B, Inter-sublayer link training for electrical interfaces
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

⁸*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

178B.16.3 ILT function

Item	Feature	Subclause	Value/Comment	Status	Support
IL1	Frame marker	178B.6.1	Transmit frame marker	M	Yes []
IL2	Polarity detection	178B.10	Detect and correct polarity	M	Yes []
IL3	Synchronous PRBS13 training pattern	178B.6.3.1	Each lane implements four generator polynomials defined in Table 178B–1	M	Yes []
IL4	Training pattern seed	178B.6.3.1	State set to the value of prbs13_seed_i at the start of the training pattern	M	Yes []
IL5	PRBS13 free running training pattern	178B.6.3.2	Each lane implements four generator polynomials defined in Table 178B–1	M	Yes []
IL6	PRBS31 free running training pattern	178B.6.3.3	Each lane implements the PRBS31 generator	M	Yes []
IL7	E1 control field structure	178B.7	As shown in Table 178B–2	O<1>	Yes []
IL8	O1 control field structure	178B.7	As shown in Table 178B–3	O<1>	Yes []
IL9	E1 status field structure	178B.8	As shown in Table 178B–4	O<2>	Yes []
IL10	O1 status field structure	178B.8	As shown in Table 178B–5	O<2>	Yes []
IL11	Training frame lock	178B.9	As shown in Table 178B–9	M	Yes []
IL12	Polarity detection	178B.10	Detect and correct polarity	O<1>	Yes []
IL13	Initial condition setting request process	178B.11.1	When requested, follow initial condition setting request process	O<1>	Yes []
IL14	Initial condition setting response process	178B.11.2	Follow initial condition setting response process	O<1>	Yes []
IL15	Coefficient update request process	178B.11.3	When requested, follow coefficient update request process	O<1>	Yes []
IL16	Coefficient update response process	178B.11.4	Follow coefficient update response process	O<1>	Yes []
IL17	Handshake timing	178B.13	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []

178B.16.4 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Mapping of MDIO control variables and MDIO status variables	178B.15	Table 178B–6 and Table 178B–7	MD:M	Yes [] N/A []

178B.16.5 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	RTS state diagram	178B.14.2	As shown in Figure 178B–7	M	Yes []
SM2	Per-lane state diagrams	178B.14.3	As shown in Figure 178B–8, Figure 178B–10	M	Yes []

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Insert new Annex 179A through Annex 179D corresponding to Clause 179 as follows:

Annex 179A

(informative)

Transmitter, receiver, and channel parameters associated with test points TP0d and TP5d for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179A.1 Overview

This annex provides information on transmitter, receiver, and channel parameters associated with test points TP0d and TP5d, which might not be testable in an implemented system. TP0d and TP5d test points are illustrated in the 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 link block diagram of Figure 179–2. This annex also provides information on channel (TP0d-TP5d).

179A.2 Transmitter characteristics at TP0d

The recommended transmitter characteristics at TP0d as measured at TP0v are described in 178.9.2. TP0d is depicted in link block diagram of Figure 179–2.

179A.3 Receiver characteristics at TP5d

The recommended receiver characteristics at TP5d as measured at TP5v are described in 178.9.3. TP5d is depicted in link block diagram of Figure 179–2.

179A.4 Host channel insertion loss

The recommended differential insertion loss at 53.125 GHz for the host channels, consisting of controlled impedance PCB, device package, and host connector, are given in Table 179A–1 and illustrated in Figure 179A–1. The recommended maximum differential insertion loss at 53.125 GHz for (TP0d-to-TP2) or (TP3-to-TP5d) for each of the host channels are given in Table 179A–1, and illustrated in Figure 179A–1. The recommended maximum differential insertion loss (TP0d-to-TP2) or (TP3-to-TP5d) are consistent with the host channels and an assumed mated connector insertion loss of 2.45 dB.

Table 179A–1—Recommended differential insertion loss limits at 53.125 GHz

Host class	Host channels	TP0d to TP2 or TP3 to TP5d
	Range (dB)	Max (dB)
Host-Low (HL)	4.45 to 8.95	12.75
Host-Nominal (HN)	4.45 to 13.95	17.75
Host-High (HH)	4.45 to 18.5	22.75

179A.5 Channel (TP0d-TP5d) insertion loss

This subclause provides information on the channel (TP0d-TP5d) insertion losses for the maximum-loss and minimum-loss cabling topologies.

The differential insertion loss of the channel (TP0d-TP5d) associated with the maximum insertion loss cable assembly and maximum insertion loss of (TP0d-TP2) and (TP3-TP5d) at 53.125 GHz is determined using Equation (179A-1) and illustrated in Figure 179A-3. The differential insertion loss of the channel (TP0d-TP5d) associated with the minimum insertion loss cable assembly and minimum insertion loss of (TP0d-TP2) and (TP3-TP5d) at 53.125 GHz is determined by Equation (179A-2) and illustrated in Figure 179A-3. Loss budget parameters at 53.125 GHz for all link configurations are provided in Table 179A-2 and Table 179A-3. The HN-to-HN link configuration is illustrated in Figure 179A-2.

$$ILdd_{Ch,Max} = ILdd_{CA,Max} + ILdd_{TP0d-TP2,Max} + ILdd_{TP3-TP5d,Max} - 2 \times ILdd_{MTFref} \quad (179A-1)$$

$$ILdd_{Ch,Min} = ILdd_{CA,Min} + ILdd_{TP0d-TP2,Min} + ILdd_{TP3-TP5d,Min} - 2 \times ILdd_{MTFref} \quad (179A-2)$$

where

$ILdd_{Ch,Max}$	is the maximum channel insertion loss between TP0d and TP5d in dB
$ILdd_{Ch,Min}$	is the minimum channel insertion loss between TP0d and TP5d in dB
$ILdd_{CA,Max}$	is the maximum cable assembly insertion loss (TP1 to TP4) in dB, 179.11.2
$ILdd_{CA,Min}$	is the minimum cable assembly insertion loss (TP1 to TP4) in dB, 179.11.2
$ILdd_{TP0d-TP2,Max}$	is the maximum insertion loss from TP0d to TP2 in dB (Table 179A-2)
$ILdd_{TP3-TP5d,Max}$	is the maximum insertion loss from TP3 to TP5d in dB (Table 179A-2)
$ILdd_{TP0d-TP2,Min}$	is the minimum insertion loss from TP0d to TP2 in dB (Table 179A-3)
$ILdd_{TP3-TP5d,Min}$	is the minimum insertion loss from TP3 to TP5d in dB (Table 179A-3)
$ILdd_{MTFref}$	is the reference insertion loss of the mated test fixtures in dB, using Equation (179B-5)

Table 179A-2—Maximum Insertion loss budget values at 53.125 GHz

Link Configuration	$ILdd_{CA,max}$ (dB)	$ILdd_{TP0d-TP2,max}$ (dB)	$ILdd_{TP3-TP5d,max}$ (dB)	$ILdd_{MTFref}$ (dB)	$ILdd_{Ch,max}$ (dB)
HH to HN	19 (CA-A)	22.75	17.75	9.75	40
HH to HL	24 (CA-B)	22.75	12.75	9.75	40
HN to HN	24 (CA-B)	17.75	17.75	9.75	40
HN to HL	29 (CA-C)	17.75	12.75	9.75	40
HL to HL	34 (CA-D)	12.75	12.75	9.75	40

Table 179A-3—Minimum Insertion loss budget values at 53.125 GHz

Link Configuration	$ILdd_{CA,min}$ (dB)	$ILdd_{TP0d-TP2,min}$ (dB)	$ILdd_{TP3-TP5d,min}$ (dB)	$ILdd_{MTFref}$ (dB)	$ILdd_{Ch,min}$ (dB)
Host-Min to Host-Min	16	8.25	8.25	9.75	13

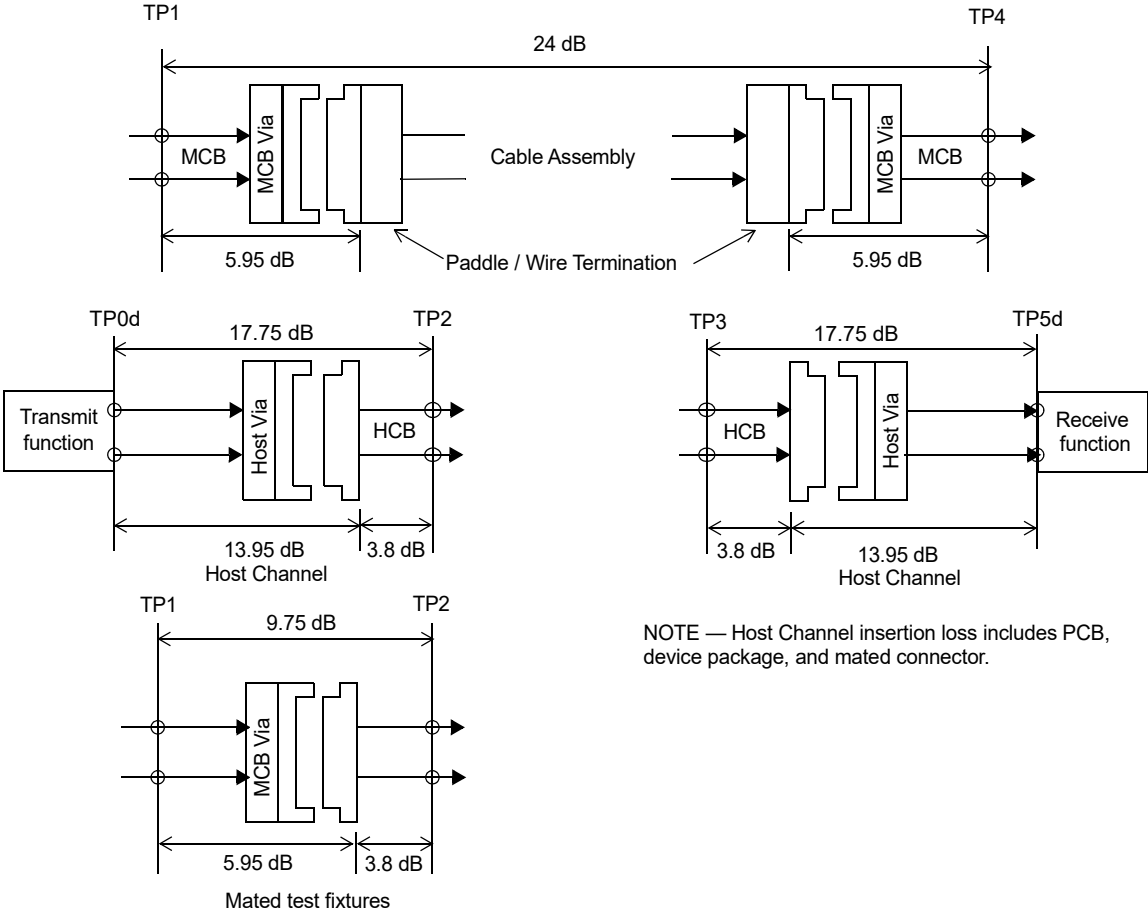


Figure 179A-1—Host-Nominal to Host-Nominal, Cable assembly, and test fixture insertion loss at 53.125 GHz

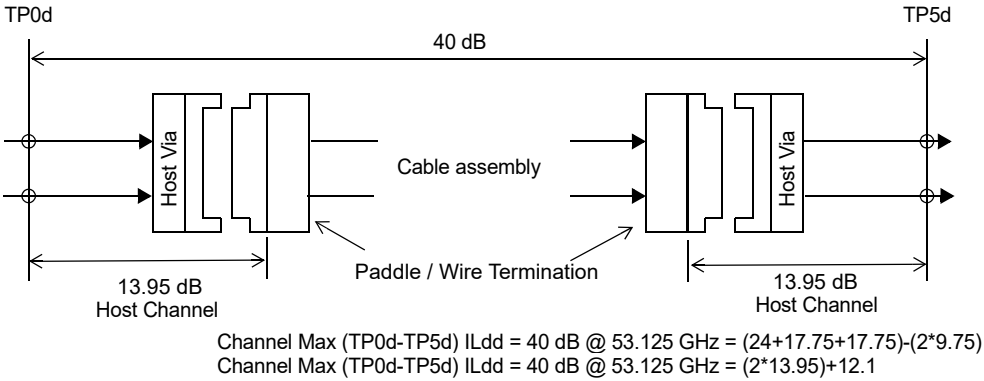


Figure 179A-2—HN-to-HN Channel Max (TP0d-TP5d) at 53.125 GHz

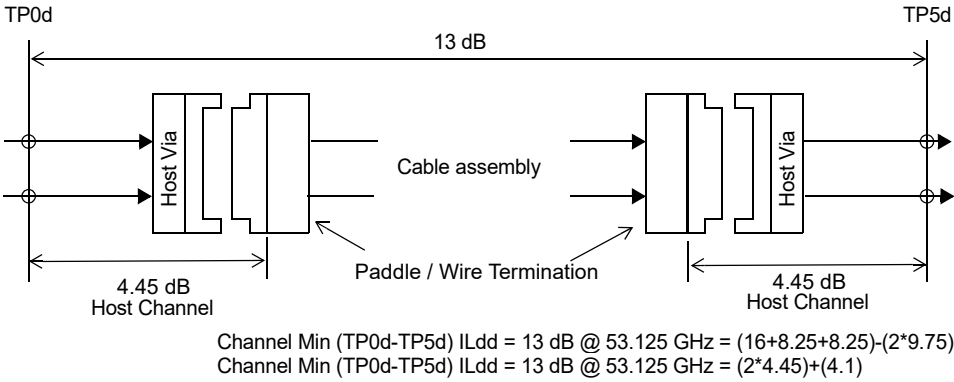


Figure 179A-3— Channel Min (TP0d-TP5d) at 53.125 GHz

Table 179A-4—Channel (TP0d-TP5d) configuration matrix for hosts and cable assemblies

Host class	Cable assembly classes compatible with HL partner	Cable assembly classes compatible with HN partner	Cable assembly classes compatible with HH partner
HL	CA-A,B,C,D	CA-A,B,C	CA-A,B
HN	CA-A,B,C	CA-A,B	CA-A
HH	CA-A,B	CA-A	not supported

179A.6 Channel (TP0d-TP5d) effective return loss (ERL)

The 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4 and 1.6TBASE-CR8 channels are recommended to meet the ERL specified in 178.10.3.

179A.7 Channel (TP0d-TP5d) Operating Margin (COM)

The Channel Operating Margin (COM) for the channel between TP0d and TP5d is computed using the using the maximum likelihood sequence detection (MLSD) defined in 178A.1.10 and the parameters in Table 178–13, and is recommended to be greater than or equal to 3 dB. Device package models are included in the TP0d and TP5d channel (Figure 179A–3); COM (TP0d-TP5d) is computed with the concatenation of device termination models to the TP0d-TP5d channel.

Annex 179B

(normative)

Test fixtures for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8, 200GAUI-1 C2M, 400GAUI-2 C2M, 800GAUI-4 C2M, and 1.6TAUI-8 C2M

179B.1 Test fixtures

Transmitter and receiver measurements at TP2 or TP3 for the 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 hosts (see Annex 179D) and at TP1a or TP4a (see Figure 176D–4) for the 200GAUI-1, 400GAUI-2, 800GAUI-4, and 1.6TAUI-8 C2M hosts (see Annex 176D), are made utilizing the test fixture specified in 179B.2. Cable assembly measurements for the cable assembly types (see Annex 179D) are made between TP1 and TP4 with test fixtures as specified in 179B.3 on both ends. The test fixtures are specified in a mated state to enable connections to measurement equipment. The reference insertion loss of the mated test fixtures is 9.75 dB at 53.125 GHz using Equation (179B–5). Module measurements for modules specified in Annex 176D are made at module compliance points TP1 and TP4 (see Figure 176D–5) with test fixtures as specified in 179B.3. The requirements in the referenced subclauses are not the MDI connector specifications for an implemented design.

179B.2 TP2 or TP3 test fixture

The TP2 or TP3 test fixture (also known as Host Compliance Board) is required for measuring the transmitter and receiver specifications at TP2 and TP3. The TP2 and TP3 test points are illustrated in Figure 179–2.

179B.2.1 TP2 or TP3 test fixture insertion loss

The TP2 or TP3 test fixture printed circuit board (PCB) insertion loss values determined using Equation (179B–1) shall be used as the TP2 or TP3 test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{tfrf}(f) = 0.0051f^2 - 0.0669f^{1.5} + 0.322f - 0.2676\sqrt{f} + 0.1567 \quad (179B-1)$$

where

$ILdd_{tfrf}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The TP2 or TP3 test fixture PCB reference insertion loss is illustrated in Figure 179B–1.

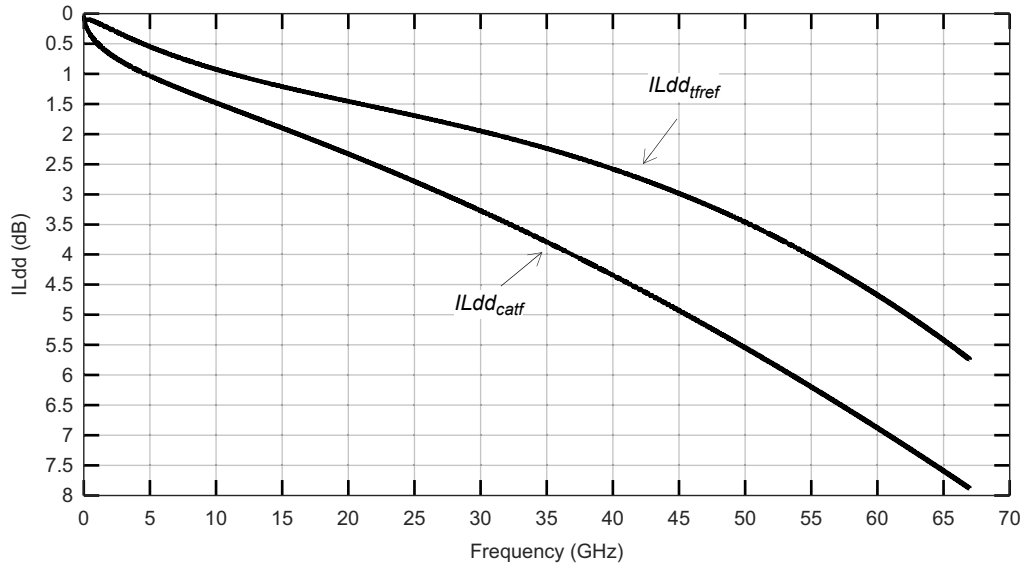


Figure 179B-1—Test fixtures reference insertion losses

179B.3 Cable assembly test fixture

The cable assembly test fixture (also known as Module Compliance Board) is required for measuring the cable assembly specifications in 179.11 at TP1 and TP4. The TP1 and TP4 test points are illustrated in Figure 179-2.

179B.3.1 Cable assembly test fixture insertion loss

The insertion loss of the cable assembly test fixture PCB, test point, connector and any associated vias determined using Equation (179B-2) shall be used as the test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{catfref}(f) = 0.016f^{1.5} - 0.0841f + 0.5829\sqrt{f} - 0.0269 \quad (179B-2)$$

where

$ILdd_{catfref}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The cable assembly test fixture reference insertion loss is illustrated in Figure 179B-1.

179B.4 Mated test fixtures

The TP2 or TP3 test fixture and the cable assembly test fixture are specified in a mated state illustrated in Figure 92–18. The mated test fixtures specifications are given below.

179B.4.1 Mated test fixtures insertion loss

The insertion loss of the mated test fixtures shall meet Equation (179B–3) and Equation (179B–4).

$$ILdd(f) \leq ILdd_{MTFmax}(f) = -0.0014f^2 + 0.0469f^{1.5} - 0.286f + 1.6562\sqrt{f} - 0.03 \quad (179B-3)$$

$$ILdd(f) \geq ILdd_{MTFmin}(f) = 0.00195f^2 - 0.0153f^{1.5} + 0.0813f + 0.436\sqrt{f} - 0.0413 \quad (179B-4)$$

for $0.01 \leq f \leq 67$

where

$ILdd(f)$ is the measured insertion loss in dB at frequency f
 $ILdd_{MTFmax}(f)$ is the maximum mated test fixtures insertion loss in dB at frequency f
 $ILdd_{MTFmin}(f)$ is the minimum mated test fixtures insertion loss in dB at frequency f
 f is the frequency in GHz

The reference insertion loss of the mated test fixtures is determined using Equation (179B–5).

$$ILdd_{MTFref}(f) = 0.0051f^2 - 0.0509f^{1.5} + 0.2379f + 0.3153\sqrt{f} + 0.1298 \quad (179B-5)$$

where

$ILdd_{MTFref}(f)$ is the reference insertion loss of the mated test fixtures in dB at frequency f
 f is the frequency in GHz

The mated test fixtures insertion loss limits are illustrated in Figure 179B–2.

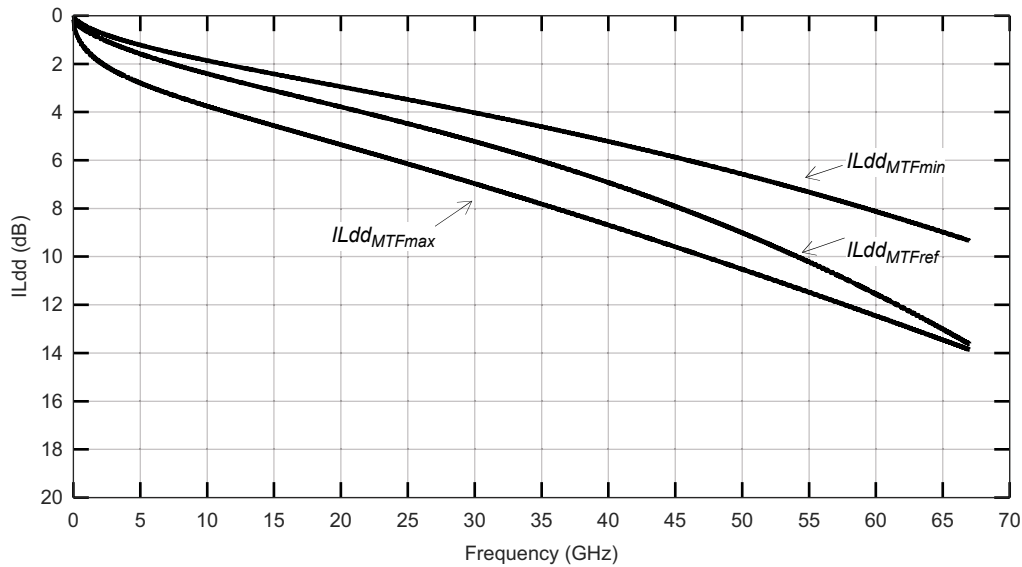


Figure 179B–2—Mated test fixtures insertion loss

The FOM_{ILD} is calculated according to 93A.4 with $f_b = 106.25$ GHz, $T_t = 6$ ps, and $f_r = 0.55 \times f_b$. The fitted insertion loss and insertion loss deviation are computed over the range $f_{min} = 0.05$ GHz to $f_{max} = 67$ GHz. FOM_{ILD} shall be less than or equal to 0.15 dB.

179B.4.2 Mated test fixtures effective return loss (ERL)

The values of the mated test fixtures ERL are computed using the procedure in 93A.5 with the parameter values in Table 179B–1. Parameters that do not appear in Table 179B–1 take values from Table 179–18.

The reference differential impedance for the mated test fixtures ERL computation shall be 92.5 Ω . The mated test fixtures ERL shall be greater than or equal to 10.3 dB.

Table 179B–1—Mated test fixtures ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.005	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	1600	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—
Target detector error ratio	DER_0	2×10^{-5}	—
NOTE—The mated test fixtures test connector and transmission line are not time-gated (by setting T_{fx} to 0) in order to include the entire test fixture.			

179B.4.3 Mated test fixtures common-mode to differential-mode insertion loss

The common-mode to differential-mode insertion loss of the mated test fixtures measured in both directions shall meet Equation (179B–6) as illustrated in Figure 179B–3.

$$ILdc(f) \geq \begin{cases} 30 - 15 \frac{f}{20} & 0.01 \leq f < 20 \\ 15 - 5 \frac{f-20}{30} & 20 \leq f < 50 \\ 10 & 50 \leq f \leq 60 \end{cases} \quad (179B-6)$$

where

$ILdc(f)$ is the common-mode to differential-mode insertion loss in dB at frequency f
 f is the frequency in GHz

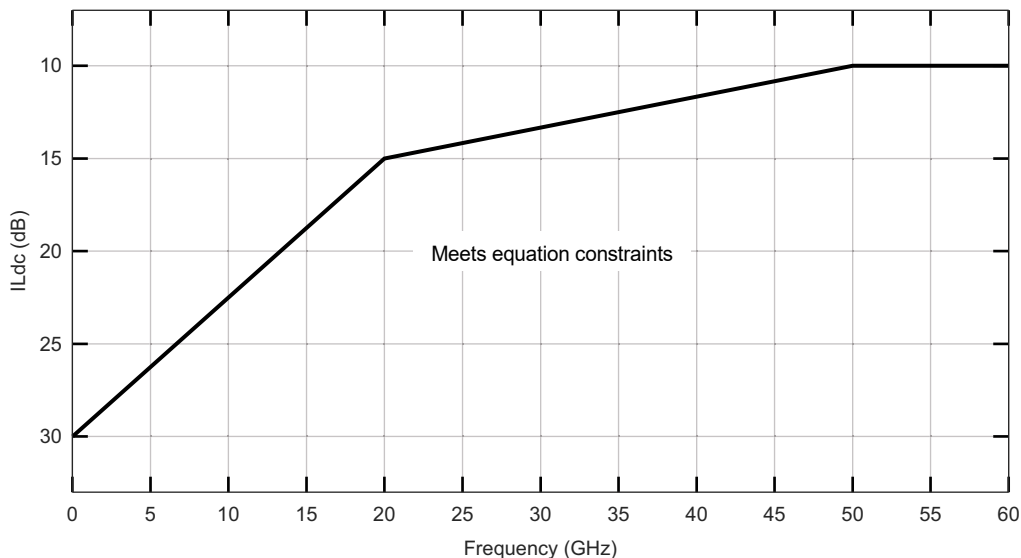


Figure 179B-3—Mated test fixtures common-mode to differential-mode insertion loss requirements

179B.4.4 Mated test fixtures common-mode to common-mode return loss

The common-mode to common-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet Equation (179B-7) as illustrated in Figure 179B-4.

$$RL_{cc}(f) \geq \begin{cases} 3 & 0.05 \leq f < 4 \\ 3 + \frac{2}{40}(f - 4) & 4 \leq f < 44 \\ 5 + \frac{2}{16}(44 - f) & 44 \leq f \leq 60 \end{cases} \quad (179B-7)$$

where

$RL_{cc}(f)$ is the common-mode to common-mode return loss in dB at frequency f
 f is the frequency in GHz

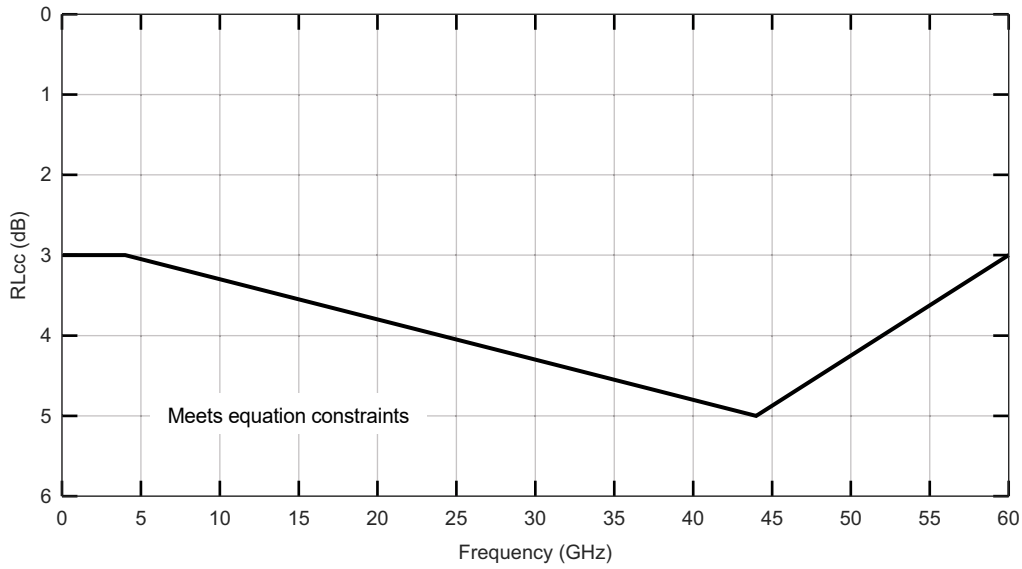


Figure 179B-4—Mated test fixtures common-mode to common-mode return loss requirements

179B.4.5 Mated test fixtures common-mode to differential-mode return loss

The common-mode to differential-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet Equation (179B-8) as illustrated in Figure 179B-5.

$$RLdc(f) \geq \begin{cases} 30 - 15 \frac{f}{12.89} & 0.01 \leq f < 12.89 \\ 15 - 5 \frac{f - 12.89}{35 - 12.89} & 12.89 \leq f < 35 \\ 10 & 35 \leq f \leq 60 \end{cases} \quad (179B-8)$$

where

$RLdc(f)$ is the common-mode to differential-mode return loss in dB at frequency f
 f is the frequency in GHz

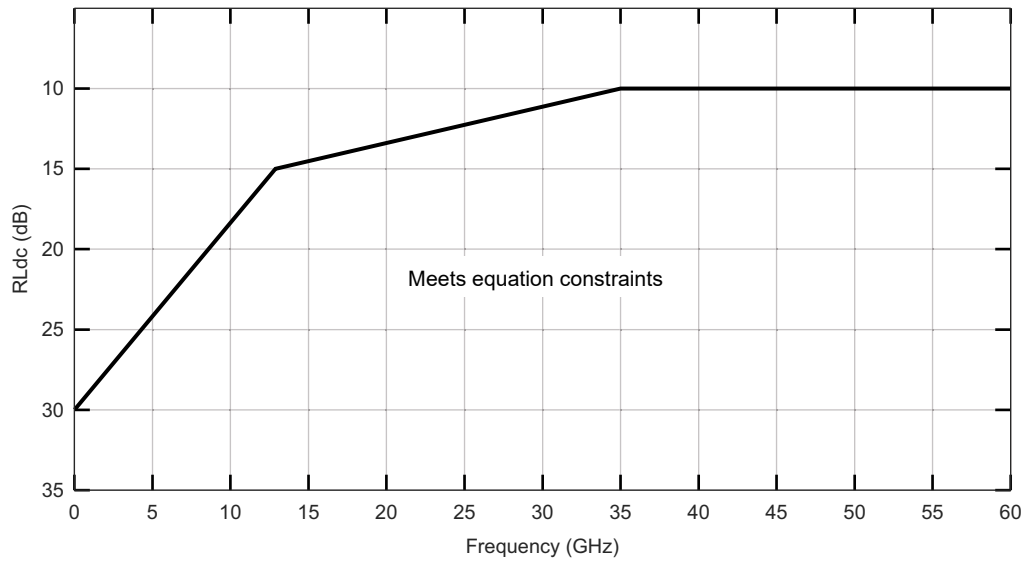


Figure 179B-5—Mated test fixtures common-mode to differential-mode return loss requirements

179B.4.6 Mated test fixtures integrated crosstalk noise

The mated test fixtures integrated crosstalk noise voltage determined from the appropriate disturber loss measured over N uniformly spaced frequencies f_n spanning the frequency range 0.05 GHz to 67 GHz with maximum frequency spacing of 10 MHz, and the parameters shown in Table 179B-2.

Table 179B-2—Mated test fixtures integrated crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	106.25	GBd
3 dB reference receiver bandwidth	f_r	58.4375	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Far-end disturber peak differential output amplitude	A_{ft}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	4.25	ps
Far-end disturber 20% to 80% rise and fall times	T_{ft}	4.25	ps

The SFP224 mated test fixtures integrated near-end crosstalk noise voltage for the disturber near-end crosstalk loss is determined according to the method in 110B.1.3.6, given the disturber near-end crosstalk loss $NEXT_loss(f)$. The mated test fixtures integrated near-end crosstalk noise voltage shall meet the specification in Table 179B-3.

Table 179B–3—SFP224 mated test fixtures integrated near-end crosstalk noise voltage

Parameter	Value	Units
Integrated near-end crosstalk noise voltage (max)	1.6	mV

The multi-lane mated test fixtures integrated crosstalk noise voltages are determined using Equation (92–44) through Equation (92–48), given the multiple disturber near-end crosstalk loss $MDNEXT_{loss}(f)$ and multiple disturber far-end crosstalk loss $MDFEXT_{loss}(f)$. The multi-lane mated test fixtures integrated crosstalk noise shall meet the specifications in Table 179B–4.

Table 179B–4—Multi-lane mated test fixtures integrated crosstalk noise voltage

Parameters	Value	Units
MDFEXT integrated crosstalk noise voltage (max)	4.2	mV
MDNEXT integrated crosstalk noise voltage (max)	1.5	mV
Total integrated crosstalk noise voltage (max)	4.4	mV

179B.5 Protocol implementation conformance statement (PICS) proforma for Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M⁹

179B.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

179B.5.2 Identification

179B.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

179B.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-202x, Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-202x.)	

Date of Statement	
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⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

179B.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
HTF1	TP2 or TP3 test fixture	162B.2		O	Yes [] No []
CATF1	Cable assembly test fixture	162B.3		O	Yes [] No []

179B.5.4 PICS proforma tables for test fixtures for Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Mated test fixtures insertion loss	162B.4.1	Per Equation (162B–3) and Equation (162B–4)	M	Yes []
TF2	Mated test fixtures FOM_{ILD}	162B.4.1	Less than or equal to 0.15 dB	M	Yes []
TF3	Mated test fixtures ERL	162B.4.2	Greater than or equal to 10.3 dB	M	Yes []
TF4	Mated test fixtures common-mode to differential-mode insertion loss	162B.4.3	Per Equation (162B–6)	M	Yes []
TF5	Mated test fixtures common-mode to common-mode return loss	162B.4.4	Per Equation (162B–7)	M	Yes []
TF6	Mated test fixtures common-mode to differential-mode return loss	162B.4.5	Per Equation (162B–8)	M	Yes []
TF7	SFP224 Mated test fixtures integrated near-end crosstalk noise	162B.4.6	Meets the requirements of Table 162B–3	M	Yes []
TF8	Multi-lane mated test fixtures integrated crosstalk noise	162B.4.6	Meets the requirements of Table 162B–5	M	Yes []

Annex 179C

(normative)

MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179C.1 Overview

This annex defines the Medium Dependent Interface (MDI) for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8. The MDI couples the PMD (specified in 179.8 and 179.9) to the cable assembly (specified in 179.11). The PMDs supportable for each MDI connector type are given in Table 179C–1. The SFP-DD224, QSFP224, QSFP-DD1600, and OSFP1600 are also referred to as multi-lane connectors.

In all MDIs, the plug connector is used on the cable assembly and the receptacle is used on the PMD.

Table 179C–1—Number of PMDs supportable for each connector type

MDI types	200GBASE-CR1	400GBASE-CR2	800GBASE-CR4	1.6TBASE-CR8	Reference
SFP224	1	—	—	—	179C.2.1
SFP-DD224	1, 2	1	—	—	179C.2.2
QSFP224	1, 2, 4	1, 2	1	—	179C.2.3
QSFP-DD1600	1, 2, 4, 8	1, 2, 4	1, 2	1	179C.2.4
OSFP1600	1, 2, 4, 8	1, 2, 4	1, 2	1	179C.2.5

For 200GBASE-CR1, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.1 (SFP224), 179C.2.2 (SFP-DD224), 179C.2.3 (QSFP224), 179C.2.4 (QSFP-DD1600), or 179C.2.4 (OSFP1600).

For 400GBASE-CR2, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.2 (SFP-DD224), 179C.2.3 (QSFP224), 179C.2.4 (QSFP-DD1600), or 179C.2.5 (OSFP1600).

For 800GBASE-CR4, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.3 (QSFP224), 179C.2.4 (QSFP-DD1600), or 179C.2.5 (OSFP1600).

For 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 179C.2.4 (QSFP-DD1600) or 179C.2.5 (OSFP1600).

The MDI connector electrical performance is consistent with the signal quality and electrical requirements of 179.9 and 179.11.

An MDI connector type may support one or more PMDs. The assignment of PMD signals to connector signals is specified in Table 179C–2, where as an example 0:DL0n refers to the DL0n signal of the first PMD; see 179.8.2 and 179.8.3 for signal naming definitions. When an MDI connector is not fully utilized the lower PMD numbers in Table 179C–2 should be used.

Table 179C–2—PMD to connector signal assignments

PMD signal <PMD number>:<PMD signal>				Connector signal
200GBASE-CR1	400GBASE-CR2	800GBASE-CR4	1.6TBASE-CR8	
0:DL0n	0:DL0n	0:DL0n	0:DL0n	DL0n
0:DL0p	0:DL0p	0:DL0p	0:DL0p	DL0p
1:DL0n	0:DL1n	0:DL1n	0:DL1n	DL1n
1:DL0p	0:DL1p	0:DL1p	0:DL1p	DL1p
2:DL0n	1:DL0n	0:DL2n	0:DL2n	DL2n
2:DL0p	1:DL0p	0:DL2p	0:DL2p	DL2p
3:DL0n	1:DL1n	0:DL3n	0:DL3n	DL3n
3:DL0p	1:DL1p	0:DL3p	0:DL3p	DL3p
4:DL0n	2:DL0n	1:DL0n	0:DL4n	DL4n
4:DL0p	2:DL0p	1:DL0p	0:DL4p	DL4p
5:DL0n	2:DL1n	1:DL1n	0:DL5n	DL5n
5:DL0p	2:DL1p	1:DL1p	0:DL5p	DL5p
6:DL0n	3:DL0n	1:DL2n	0:DL6n	DL6n
6:DL0p	3:DL0p	1:DL2p	0:DL6p	DL6p
7:DL0n	3:DL1n	1:DL3n	0:DL7n	DL7n
7:DL0p	3:DL1p	1:DL3p	0:DL7p	DL7p
0:SL0n	0:SL0n	0:SL0n	0:SL0n	SL0n
0:SL0p	0:SL0p	0:SL0p	0:SL0p	SL0p
1:SL0n	0:SL1n	0:SL1n	0:SL1n	SL1n
1:SL0p	0:SL1p	0:SL1p	0:SL1p	SL1p
2:SL0n	1:SL0n	0:SL2n	0:SL2n	SL2n
2:SL0p	1:SL0p	0:SL2p	0:SL2p	SL2p
3:SL0n	1:SL1n	0:SL3n	0:SL3n	SL3n
3:SL0p	1:SL1p	0:SL3p	0:SL3p	SL3p
4:SL0n	2:SL0n	1:SL0n	0:SL4n	SL4n
4:SL0p	2:SL0p	1:SL0p	0:SL4p	SL4p
5:SL0n	2:SL1n	1:SL1n	0:SL5n	SL5n
5:SL0p	2:SL1p	1:SL1p	0:SL5p	SL5p
6:SL0n	3:SL0n	1:SL2n	0:SL6n	SL6n
6:SL0p	3:SL0p	1:SL2p	0:SL6p	SL6p
7:SL0n	3:SL1n	1:SL3n	0:SL7n	SL7n
7:SL0p	3:SL1p	1:SL3p	0:SL7p	SL7p

The MDI connectors shall have data signal and signal ground contact assignments as specified in Table 179C–3, Table 179C–4, and Table 179C–5 for the paired transmit and receive lanes (e.g., (SL0, DL0), (SL1, DL1)).

Table 179C–3—MDI connector contact mapping for SFP224, SFP-DD224

SFP224	SFP-DD224	Connector signal name	Description
11	11	GND	Ground
12	12	DL0n	Receiver inverted data output
13	13	DL0p	Receiver non-inverted data output
14	14	GND	Ground
17	17	GND	Ground
18	18	SL0p	Transmitter non-inverted data input
19	19	SL0n	Transmitter inverted data input
20	20	GND	Ground
—	31	GND	Ground
—	32	DL1n	Receiver inverted data output
—	33	DL1p	Receiver non-inverted data output
—	34	GND	Ground
—	37	GND	Ground
—	38	SL1p	Transmitter non-inverted data input
—	39	SL1n	Transmitter inverted data input
—	40	GND	Ground

Table 179C–4—MDI connector contact mapping for QSFP224 and QSFP-DD1600

QSFP224	QSFP-DD1600	Connector signal name	Description
1	1	GND	Ground
2	2	SL1n	Transmitter inverted data input
3	3	SL1p	Transmitter non-inverted data input
4	4	GND	Ground
5	5	SL3n	Transmitter inverted data input

**Table 179C–4—MDI connector contact mapping for QSFP224 and QSFP-DD1600
(continued)**

QSFP224	QSFP-DD1600	Connector signal name	Description
6	6	SL3p	Transmitter non-inverted data input
7	7	GND	Ground
13	13	GND	Ground
14	14	DL2p	Receiver non-inverted data output
15	15	DL2n	Receiver inverted data output
16	16	GND	Ground
17	17	DL0p	Receiver non-inverted data output
18	18	DL0n	Receiver inverted data output
19	19	GND	Ground
20	20	GND	Ground
21	21	DL1n	Receiver inverted data output
22	22	DL1p	Receiver non-inverted data output
23	23	GND	Ground
24	24	DL3n	Receiver inverted data output
25	25	DL3p	Receiver non-inverted data output
26	26	GND	Ground
32	32	GND	Ground
33	33	SL2p	Transmitter non-inverted data input
34	34	SL2n	Transmitter inverted data input
35	35	GND	Ground
36	36	SL0p	Transmitter non-inverted data input
37	37	SL0n	Transmitter inverted data input
38	38	GND	Ground
—	39	GND	Ground
—	40	SL5n	Transmitter inverted data input
—	41	SL5p	Transmitter non-inverted data input
—	42	GND	Ground
—	43	SL7n	Transmitter inverted data input
—	44	SL7p	Transmitter non-inverted data input
—	45	GND	Ground

**Table 179C-4—MDI connector contact mapping for QSFP224 and QSFP-DD1600
(continued)**

QSFP224	QSFP-DD1600	Connector signal name	Description
—	51	GND	Ground
—	52	DL6p	Receiver non-inverted data output
—	53	DL6n	Receiver inverted data output
—	54	GND	Ground
—	55	DL4p	Receiver non-inverted data output
—	56	DL4n	Receiver inverted data output
—	58	GND	Ground
—	59	DL5n	Receiver inverted data output
—	60	DL5p	Receiver non-inverted data output
—	61	GND	Ground
—	62	DL7n	Receiver inverted data output
—	63	DL7p	Receiver non-inverted data output
—	64	GND	Ground
—	70	GND	Ground
—	71	SL6p	Transmitter non-inverted data input
—	72	SL6n	Transmitter inverted data input
—	73	GND	Ground
—	74	SL4p	Transmitter non-inverted data input
—	75	SL4n	Transmitter inverted data input

Table 179C-5—MDI connector contact mapping for OSFP1600

OSFP1600	Connector signal name	Description
1	GND	Ground
2	SL1p	Transmitter non-inverted data input
3	SL1n	Transmitter inverted data input
4	GND	Ground
5	SL3p	Transmitter non-inverted data input

Table 179C–5—MDI connector contact mapping for OSFP1600 (*continued*)

OSFP1600	Connector signal name	Description
6	SL3n	Transmitter inverted data input
7	GND	Ground
8	SL5p	Transmitter non-inverted data input
9	SL5n	Transmitter inverted data input
10	GND	Ground
11	SL7p	Transmitter non-inverted data input
12	SL7n	Transmitter inverted data input
13	GND	Ground
18	GND	Ground
19	DL6n	Receiver inverted data output
20	DL6p	Receiver non-inverted data output
21	GND	Ground
22	DL4n	Receiver inverted data output
23	DL4p	Receiver non-inverted data output
24	GND	Ground
25	DL2n	Receiver inverted data output
26	DL2p	Receiver non-inverted data output
27	GND	Ground
28	DL0n	Receiver inverted data output
29	DL0p	Receiver non-inverted data output
30	GND	Ground
31	GND	Ground
32	DL1p	Receiver non-inverted data output
33	DL1n	Receiver inverted data output
34	GND	Ground
35	DL3p	Receiver non-inverted data output
36	DL3n	Receiver inverted data output
37	GND	Ground
38	DL5p	Receiver non-inverted data output
39	DL5n	Receiver inverted data output
40	GND	Ground

Table 179C–5—MDI connector contact mapping for OSFP1600 (*continued*)

OSFP1600	Connector signal name	Description
41	DL7p	Receiver non-inverted data output
42	DL7n	Receiver inverted data output
43	GND	Ground
48	GND	Ground
49	SL6n	Transmitter inverted data input
50	SL6p	Transmitter non-inverted data input
51	GND	Ground
52	SL4n	Transmitter inverted data input
53	SL4p	Transmitter non-inverted data input
54	GND	Ground
55	SL2n	Transmitter inverted data input
56	SL2p	Transmitter non-inverted data input
57	GND	Ground
58	SL0n	Transmitter inverted data input
59	SL0p	Transmitter non-inverted data input
60	GND	Ground

179C.2 MDI connector types

179C.2.1 SFP224

For 200GBASE-CR1, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of SFF-TA-1031, Rev 1.0. SFP224 supports one lane.

Editor's note: The reference for SFP224 does not currently include 200G per lane specifications but it's expected to include before publication of this standard.

The SFP224 cable assembly connector is the plug illustrated in Figure 179C–1. The SFP224 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C–2. The connectors have data signal and signal ground contact assignments as specified in Table 179C–3. The pin definition for both the plug and receptacle remains unchanged from Annex 162C.

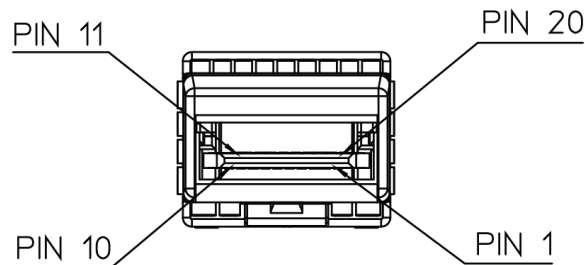


Figure 179C-1—SFP224 cable assembly plug

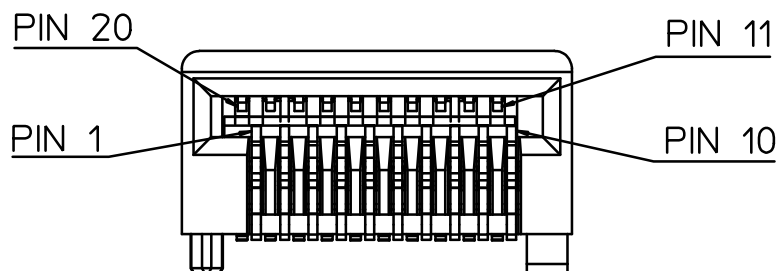


Figure 179C-2—SFP224 PMD receptacle

179C.2.2 SFP-DD224

For 200GBASE-CR1 or 400GBASE-CR2, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the SFP-DD MSA. SFP-DD224 supports up to two lanes.

Editor's note: The reference for SFP-DD224 does not currently include 200G per lane specifications but it's expected to include before publication of this standard.

The SFP-DD224 cable assembly connector is the plug illustrated in Figure 179C-3. The SFP-DD224 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C-4. The connectors have data signal and signal ground contact assignments as specified in Table 179C-3. The pin definition for both the plug and receptacle remains unchanged from Annex 162C.

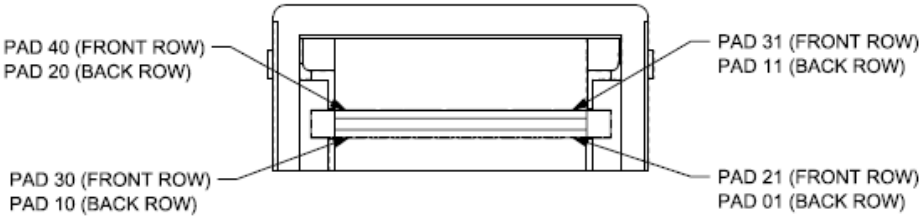


Figure 179C-3—SFP-DD224 cable assembly plug



Figure 179C-4—SFP-DD224 PMD receptacle

179C.2.3 QSFP224

For 200GBASE-CR1, 400GBASE-CR2, or 800GBASE-CR4, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in SFF-TA-1027, Rev 1.0, 2024-04-16, QSFP2 Connector, Cage, & Module Specification. QSFP224 supports up to four lanes.

Editor's note: The reference for QSFP224 does not currently include 200G per lane specifications but it's expected to include before publication of this standard.

The QSFP224 cable assembly connector is the plug illustrated in Figure 179C-5. The QSFP224 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C-6. The connectors have data signal and signal ground contact assignments as specified in Table 179C-4. The pin definition for both the plug and receptacle remains unchanged from Annex 162C.

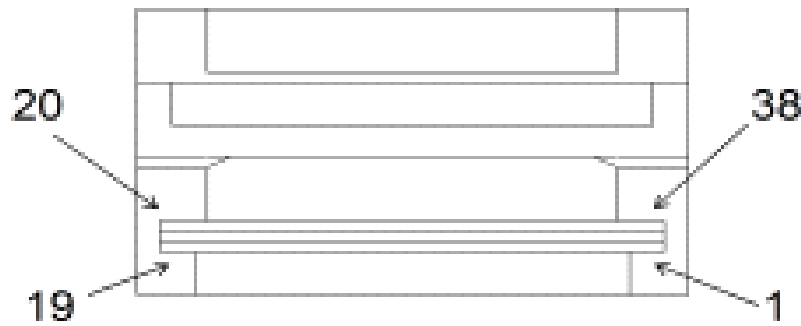


Figure 179C-5—QSFP224 cable assembly plug

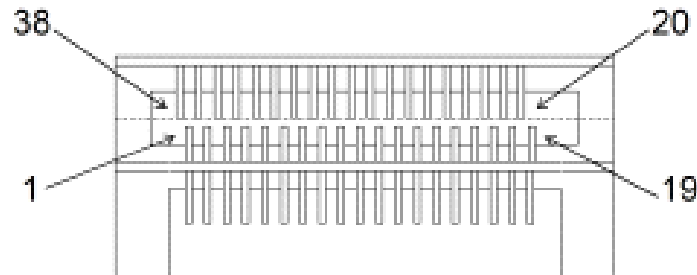


Figure 179C-6—QSFP224 PMD receptacle

179C.2.4 QSFP-DD1600

For 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the QSFP-DD/QSFP-DD800/QSFP-DD1600 Hardware Specification for QSFP Double Density 8x Pluggable Transceivers, Rev 7.1 QSFP-DD1600 supports up to eight lanes.

The QSFP-DD1600 cable assembly connector is the plug illustrated in Figure 179C-7. The QSFP-DD PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C-8. The connectors have data signal and signal ground contact assignments as specified in Table 179C-4.

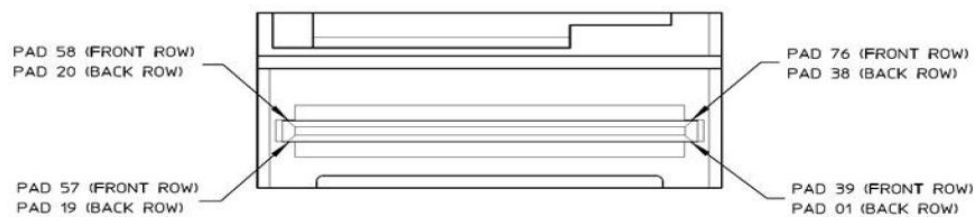


Figure 179C-7—QSFP-DD1600 cable assembly plug

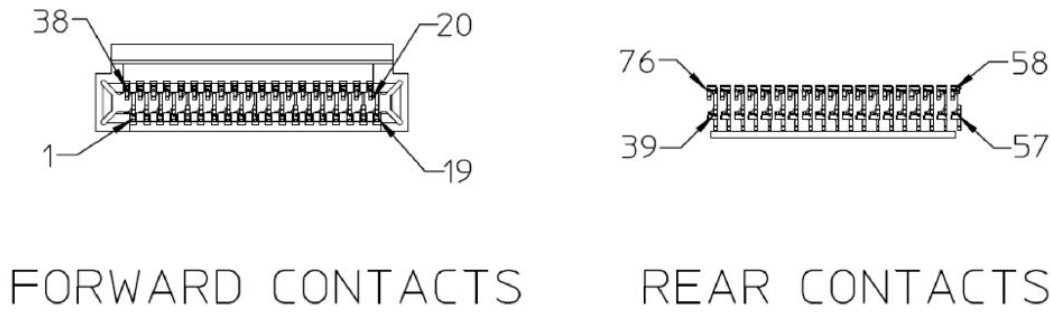


Figure 179C-8—QSFP-DD1600 PMD receptacle

179C.2.5 OSFP1600

For 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the OSFP Octal Small Form Factor Pluggable Module, Rev 5.1 Specification. OSFP1600 supports up to eight lanes.

The OSFP1600 cable assembly connector is the plug illustrated in Figure 179C-9. The OSFP1600 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 179C-10. The connectors have data signal and signal ground contact assignments as specified in Table 179C-5.

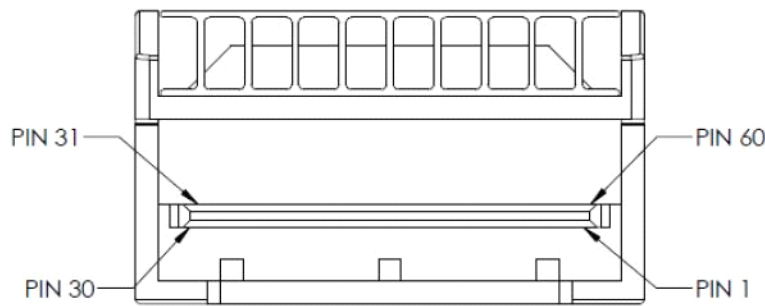


Figure 179C-9—OSFP1600 cable assembly plug

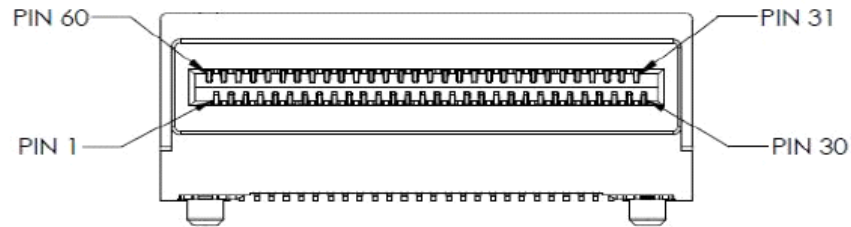


Figure 179C-10—OSFP1600 PMD receptacle

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**179C.3 Protocol implementation conformance statement (PICS) proforma
for Annex 179C, MDIs for 200GBASE-CR1, 400GBASE-CR2,
800GBASE-CR4, and 1.6TBASE-CR8¹⁰**

179C.3.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 179C, MDIs for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

179C.3.2 Identification

179C.3.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier’s terminology (e.g., Type, Series, Model).	

179C.3.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-202x, Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-202x.)	

Date of Statement	
-------------------	--

¹⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

Editor's note: This PICS subclause is incomplete and will be updated in a future draft. Please do not submit comments against this subclause.

179C.3.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MDI1	SFP224 MDI connector	162C.2.1		O	Yes [] No []
*MDI2	SFP-DD224 MDI connector	162C.2.2		O	Yes [] No []
*MDI3	QSFP224 MDI connector	162C.2.4		O	Yes [] No []
*MDI4	QSFP-DD1600 MDI connector	162C.2.5		O	Yes [] No []
*MDI5	OSFP1600 MDI connector	162C.2.6		O	Yes [] No []

179C.3.4 PICS proforma tables for MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

179C.3.4.1 Contact Mapping

Item	Feature	Subclause	Value/Comment	Status	Support
CM1	SFP224 MDI connector contact mapping according to Table 162C–3	162C.2.1		MDI1:M	Yes [] N/A []
CM2	SFP-DD224 MDI connector contact mapping according to Table 162C–3	162C.2.2		MDI2:M	Yes [] N/A []
CM4	QSFP224 MDI connector contact mapping according to Table 162C–4	162C.2.4		MDI4:M	Yes [] N/A []
CM5	QSFP-DD1600 MDI connector contact mapping according to Table 162C–4	162C.2.5		MDI5:M	Yes [] N/A []
CM6	OSFP1600 MDI connector contact mapping according to Table 162C–5	162C.2.6		MDI6:M	Yes [] N/A []

Annex 179D

(informative)

Cable assemblies and hosts for 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8

179D.1 Overview

This annex describes cable assembly types specified in 179.11 for hosts with 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, or 1.6TBASE-CR8 Physical Layers. The five MDI connector receptacles specified are given in Table 179D–1. This enables multiple cable assembly types with different combinations of the plug connectors at each end.

Table 179D–1—Host receptacles and cable assembly plugs

Receptacle/Plugs	Reference
SFP224	179C.2.1
SFP-DD224	179C.2.2
QSFP224	179C.2.3
QSFP-DD1600	179C.2.4
OSFP1600	179C.2.5

179D.1.1 Cable assembly types

Cable assembly types, denoted as 200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8 have a common set of electrical parameters specified in 179.11. Cable assembly types with different combinations of the plug connectors for compatible host receptacles and supportable number of PMDs are given in Table 179D–2 for 200GBASE-CR1, Table 179D–3 for 400GBASE-CR2, and Table 179D–4 for 800GBASE-CR4; other combinations are possible.

Table 179D–2—200GBASE-CR1 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
SFP224	1	SFP224	1	1
SFP-DD224	1	SFP224	2	2
QSFP224	1	SFP224	4	4
QSFP-DD1600	1	SFP224	8	8

Table 179D–2—200GBASE-CR1 cable assembly types and supportable number of PMDs (*continued*)

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
OSFP1600	1	SFP224	8	8
SFP-DD224	1	SFP-DD224	1	2
QSFP224	1	QSFP224	1	4
QSFP-DD1600	1	QSFP-DD1600	1	8
QSFP-DD1600	1	OSFP1600	1	8
OSFP1600	1	OSFP1600	1	8

Table 179D–3—400GBASE-CR2 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
SFP-DD224	1	SFP-DD224	1	1
QSFP224	1	SFP-DD224	2	2
QSFP-DD1600	1	SFP-DD224	4	4
OSFP1600	1	SFP-DD224	4	4
QSFP224	1	QSFP224	1	2
QSFP-DD1600	1	QSFP-DD1600	1	4
QSFP-DD1600	1	OSFP1600	1	4
OSFP1600	1	OSFP1600	1	4

Table 179D–4—800GBASE-CR4 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
QSFP224	1	QSFP224	1	1
QSFP-DD1600	1	QSFP224	2	2
OSFP1600	1	QSFP224	2	2
QSFP-DD1600	1	QSFP-DD1600	1	2
QSFP-DD1600	1	OSFP1600	1	2
OSFP1600	1	OSFP1600	1	2

Table 179D–5—1.6TBASE-CR8 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
QSFP-DD1600	1	QSFP-DD1600	1	1
QSFP-DD1600	1	OSFP1600	1	1
OSFP1600	1	OSFP1600	1	1

Insert new Annex 180A corresponding to Clause 180 as follows:

Annex 180A

(normative)

MDIs for optical PHYs

180A.1 Scope

This annex defines the medium dependent interface (MDI) for 200GBASE-DR1, 200GBASE-DR1-2, 400GBASE-DR2, 400GBASE-DR2-2, 800GBASE-DR4, 800GBASE-DR4-2, 1.6TBASE-DR8, and 1.6TBASE-DR8-2.

180A.2 Overview

The PMDs for 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 are specified in Clause 180. The PMDs for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 are specified in Clause 182.

Table 180A–1 shows the number of PMDs supported by each MDI type.

Table 180A–1—Number of PMDs supported by each MDI type

MDI Type	200GBASE-DR1, 200GBASE-DR1-2	400GBASE-DR2, 400GBASE-DR2-2	800GBASE-DR4, 800GBASE-DR4-2	1.6TBASE-DR8, 1.6TBASE-DR8-2
Single-fiber connector	1	—	—	—
Single-row 12-fiber	1, 2, or 4	1 or 2	1	—
Single-row 16-fiber	1, 2, 4, or 8	1, 2, or 4	1 or 2	1

180A.3 MDI Requirements for multi-fiber interfaces

180A.3.1 Single-row 12-fiber interface

The MDI shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-9: *MPO device receptacle, angled interface*. The plug terminating the optical fiber cabling shall meet the dimensional specifications of IEC 61754-7-1 interface 7-1-1: *MPO female plug connector, down-angled interface for 2 to 12 fibres*. The MDI shall optically mate with the plug on the optical fiber cabling. Figure 180A–1 shows an MPO female plug connector with down-angled interface, and an MDI as an active device receptacle with angled interface.

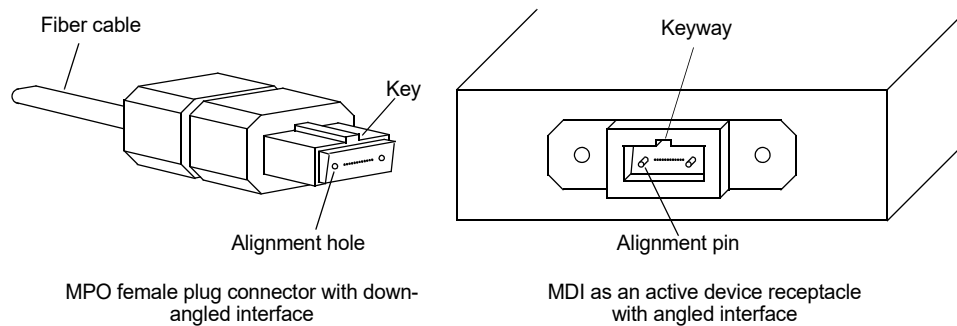


Figure 180A-1—MPO female plug with down-angled interface and MDI active device receptacle with angled interface

The MDI shall meet the interface performance specifications of IEC 61753-021-02 for performance level D/2.

180A.3.2 Single-row 16-fiber interface

The MDI shall optically mate with the compatible plug on the optical fiber cabling. The MDI shall meet the interface performance specifications of IEC 61753-021-02 for performance level D/2. 1.6TBASE-DR8 has a single-row 16-fiber interface optical lane assignment.

If the MDI is constructed with a connectorized fiber pigtail into an adapter, the connectorized pigtail shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-1-1-1 and the adapter shall meet the dimensional specifications of designation FOCIS 18 A-1-0 as defined in ANSI/TIA-604-18-A. If the MDI is constructed with a receptacle, it shall meet the dimensional specifications of designation FOCIS 18 R-1x16-1-8-1-1-2, as defined in ANSI/TIA-604-18-A. The plug terminating the optical fiber cabling that interconnects to either MDI configuration shall meet the dimensional specifications of designation FOCIS 18 P-1x16-1-8-2-1-1, as defined in ANSI/TIA-604-18-A. The MPO-16 female plug connector and MDI are structurally similar to those depicted in Figure 180A-1, but with an angled end facet, 16 fibers, an offset keyway, and different pin diameters and locations.

180A.4 Optical-lane assignments

180A.4.1 Assignments for a single-row 12-fiber interface

Figure 180A-2 shows the generic layout of a single-row 12-position optical connector specified for single-lane, 2-lane and 4-lane optical PMDs. The four transmit and four receive optical lanes shall occupy the positions depicted in Figure 180A-2 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains eight active lanes within twelve total positions. The transmit optical lanes occupy the leftmost four positions. The receive optical lanes occupy the rightmost four positions. The four center positions are unused. Figure 180A-2 also provides details on the ordering of the optical lane assignments.

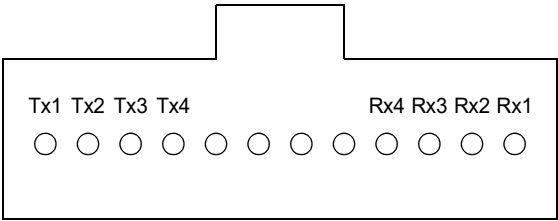


Figure 180A–2—Optical lane assignments for a single-row 12-position connector

Table 180A–2 shows the mapping of PMD signals to optical connector positions for single-row 12-position optical interfaces. Such interfaces support a single 4-lane optical PMD, such as 800GBASE-DR4 or 800GBASE-DR4-2, or alternatively four single lane optical PMDs, such as 200GBASE-DR1 or 200GBASE-DR1-2, or two 2-lane optical PMDs, such as 400GBASE-DR2 or 400GBASE-DR2-2. When an MDI connector is not fully utilized the lower PMD numbers in Table 180A–2 should be used.

Table 180A–2—Mapping PMD signals to positions for 12-position optical connectors

<PMD #>:<PMD signal>			
200GBASE-DR1 ^a	400GBASE-DR2 ^a	800GBASE-DR4 ^a	Connector position
0:SL0	0:SL0	0:SL0	Tx1
1:SL0	0:SL1	0:SL1	Tx2
2:SL0	1:SL0	0:SL2	Tx3
3:SL0	1:SL1	0:SL3	Tx4
0:DL0	0:DL0	0:DL0	Rx1
1:DL0	0:DL1	0:DL1	Rx2
2:DL0	1:DL0	0:DL2	Rx3
3:DL0	1:DL1	0:DL3	Rx4

^a In this table, the mapping of 200GBASE-DR1, 400GBASE-DR2, and 800GBASE-DR4 is shown, however this may equally apply to 200GBASE-DR1-2, 400GBASE-DR2-2, and 800GBASE-DR4-2.

A combination of two single lane optical PMDs, such as 200GBASE-DR2, and one 2-lane optical PMD, such as 400GBASE-DR2, is also supported. Table 180A–3 shows an example of a combination of the mapping of several optical PMDs to a single row, 12-position optical connector.

Table 180A–3—Example mapping on a 12-position optical connector

<PMD #>:<PMD signal>		
200GBASE-DR1	400GBASE-DR2	Connector position
—	0:SL0	Tx1
—	0:SL1	Tx2
2:SL0	—	Tx3
3:SL0	—	Tx4
—	0:DL0	Rx1
—	0:DL1	Rx2
2:DL0	—	Rx3
3:DL0	—	Rx4

180A.4.2 Assignments for a single-row 16-fiber interface

Figure 180A–3 shows the generic layout of a single-row 16-position optical connector specified for single-lane, 2-lane, 4-lane, and 8-lane optical PMDs. The interface contains 16 active lanes within 16 total positions. The transmit optical lanes occupy the leftmost eight positions and the receive optical lanes occupy the rightmost eight positions as depicted in Figure 180A–3 when looking into the MDI receptacle with the connector keyway feature on top. Figure 180A–3 also provides details on the ordering of the optical lane assignments.

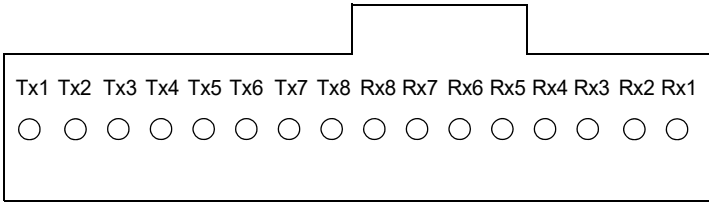


Figure 180A–3—Optical lane assignments for a single-row 16-position connector

Table 180A–4 shows the mapping of PMD signals to positions for a single-row, 16-position optical connector. Such connectors support a single 8-lane optical PMD, such as 1.6TBASE-DR8 or 1.6TBASE-DR8-2, or alternatively eight single lane optical PMDs, such as 200GBASE-DR1 or 200GBASE-DR1-2, four 2-lane optical PMDs, such as 400GBASE-DR2 or 400GBASE-DR2-2, or two 4-lane optical PMDs such as 800GBASE-DR4. When an MDI connector is not fully utilized the lower PMD numbers in Table 180A–4 should be used.

Table 180A–4—Mapping PMD signals to positions for 16-position optical connectors

<PMD #>:<PMD signal>				
200GBASE-DR1 ^a	400GBASE-DR2 ^a	800GBASE-DR4 ^a	1.6TBASE-DR8 ^a	Connector position
0:SL0	0:SL0	0:SL0	0:SL0	Tx1
1:SL0	0:SL1	0:SL1	0:SL1	Tx2
2:SL0	1:SL0	0:SL2	0:SL2	Tx3
3:SL0	1:SL1	0:SL3	0:SL3	Tx4
4:SL0	2:SL0	1:SL0	0:SL4	Tx5
5:SL0	2:SL1	1:SL1	0:SL5	Tx6
6:SL0	3:SL0	1:SL2	0:SL6	Tx7
7:SL0	3:SL1	1:SL3	0:SL7	Tx8
0:DL0	0:DL0	0:DL0	0:DL0	Rx1
1:DL0	0:DL1	0:DL1	0:DL1	Rx2
2:DL0	1:DL0	0:DL2	0:DL2	Rx3
3:DL0	1:DL1	0:DL3	0:DL3	Rx4
4:DL0	2:DL0	1:DL0	0:DL4	Rx5
5:DL0	2:DL1	1:DL1	0:DL5	Rx6
6:DL0	3:DL0	1:DL2	0:DL6	Rx7
7:DL0	3:DL1	1:DL3	0:DL7	Rx8

^a In this table the mapping of 200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8 is shown, however this may equally apply to 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8.

Furthermore, combinations of single lane optical PMDs, such as 200GBASE-DR2, 2-lane optical PMD, such as 400GBASE-DR2, or 800GBASE-DR4, are also supported. Table 180A–5 shows an example of a combination of the mapping of several optical PMDs to a single row, 16-position optical connector.

Table 180A–5—Example mapping on a 16-position optical connector

<PMD #>:<PMD signal>			
200GBASE-DR1	400GBASE-DR2	800GBASE-DR4	Connector position
—	0:SL0	—	Tx1
—	0:SL1	—	Tx2
2:SL0	—	—	Tx3
3:SL0	—	—	Tx4
—	—	1:SL0	Tx5

Table 180A–5—Example mapping on a 16-position optical connector

<PMD #>:<PMD signal>			
200GBASE-DR1	400GBASE-DR2	800GBASE-DR4	Connector position
—	—	1:SL1	Tx6
—	—	1:SL2	Tx7
—	—	1:SL3	Tx8
—	0:DL0	—	Rx1
—	0:DL1	—	Rx2
2:DL0	—	—	Rx3
3:DL0	—	—	Rx4
—	—	1:DL0	Rx5
—	—	1:DL1	Rx6
—	—	1:DL2	Rx7
—	—	1:DL3	Rx8

Insert new Annex 184A corresponding to Clause 184 as follows:

Annex 184A

(informative)

800GBASE-LR1 Inner FEC encoding examples

184A.1 Scope

This annex provides test vectors for the 800GBASE-LR1 Inner FEC transmit function including permutation, convolutional interleaver, BCH encoder, circular shift, BCH interleaver, DSP frame, pilot sequence, DSP frame generation, and DP-16QAM mapper defined in 184.4.

184A.2 Vectors

A file containing the text vectors is available at <https://standards.ieee.org/downloads/802.3/>.

Editor's note (to be removed prior to publication):

The test vectors are given as plain-text files, which can be found at the following link:

https://www.ieee802.org/3/dj/public/24_07/kota_3dj_04_2407.zip

This file will be placed in the 802.3 downloads folder referenced above when this amendment is published.

184A.3 Test vectors reference points

Figure 184A–1 shows the test vectors reference points.

184A.3.1 VP1

VP1 (vp1.txt) input bits are the result of a 800GBASE-R signal generated by feeding IDLE at the 800GMII interface (see Figure 172–2).

The vector file starts with alignment markers (AM). Each row has 32 RS FEC symbols (320 bits) one per PCS lane. The first 10 bits belong to PCS Lane 0, the second 10 bits belong to PCS Lane 1, and so on. The first bit is the oldest one.

184A.3.2 VP2

VP2 (vp2.txt) are the bits after the permutation function. The vector file format is the same as VP1.

184A.3.3 VP3

VP3 (vp3.txt) are the bits after the convolutional interleaver function.

Each row has the 110 bits that are the inputs of each of the 32 BCH encoders. All bits marked with “x” at the beginning of the file are undetermined due to delays inside of the convolutional interleaver.

184A.3.4 VP4

VP4 (vp4.txt) are the bits after the BCH encoder function.

The vector file format is the same file format as VP3, except that 16-bit parity is added to the end of each row.

184A.3.5 VP5

VP5 (vp5_xi.txt, vp5_xq.txt, vp5_yi.txt and vp5_yq.txt) are the bits after the BCH interleaver but prior to pilot insertion.

There are four separate files, one per component (xi, xq, yi and yq). Each row has 126 bits representing 63 symbols, starting from the oldest one.

184A.3.6 VP6

VP6 (vp6_xi.txt, vp6_xq.txt, vp6_yi.txt and vp6_yq.txt) are the symbols after pilots insertion.

There are also four separate files, one per component (xi, xq, yi and yq). Each row has 128 bits representing 64 symbols, and the first symbol of each row is the pilot.

There is one extra file that contains the symbols after mapping the bit pairs to four-level signals (see Table 184–3). There are 4 columns per row that contains xi, xq, yi, and yq symbols.

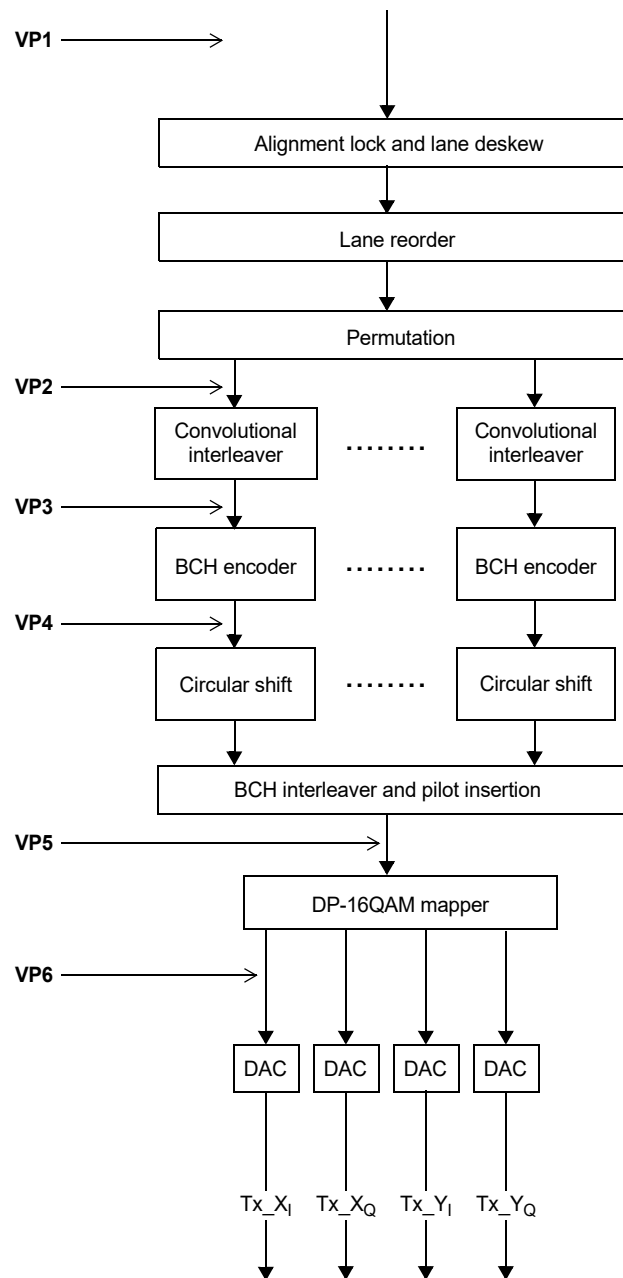


Figure 184A–1—Inner FEC test vectors reference points

Insert new Annex 185A corresponding to Clause 185 as follows:

Annex 185A

(normative)

Measurement methods for coherent optical Physical Layer devices

185A.1 Scope

This annex defines measurement methodologies for coherent optical Physical Layer devices.

185A.2 Extended transmitter constellation closure (ETCC) test setup and calculation

ETCC is a parameter representing the quality of the transmitter output signal used for 800GBASE-LR1, 800GBASE-ER1, and 800GBASE-ER1-20 PMDs. The ETCC parameter is defined in this annex.

185A.2.1 ETCC conformance test setup

A block diagram for the ETCC test setup is shown in Figure 185A–1. In the test setup, the coherent transmitter under test is connected to the ETCC reference receiver using a single-mode fiber patch cord between 2 m and 5 m in length. The test setup consists of the functional blocks described in 185A.2.2.1 through 185A.2.3.7. The ETCC reference receiver is used for the measurement of ETCC.

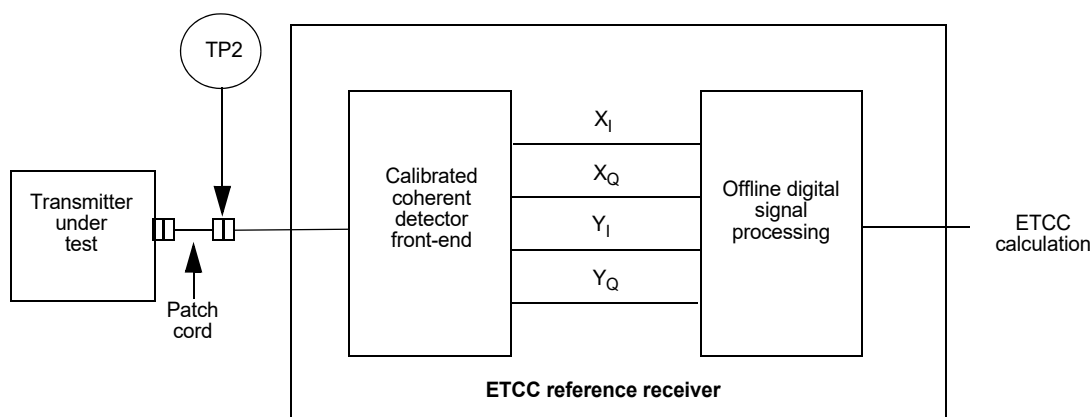


Figure 185A–1—ETCC test setup

185A.2.2 ETCC reference receiver

The ETCC reference receiver shown in Figure 185A–1 is composed of calibrated coherent detector front-end and offline digital signal processing.

185A.2.2.1 Calibrated coherent detector front-end

The calibrated coherent detector front-end is intended to reproduce an undistorted digital equivalent of the incoming optical signal under test.

The calibrated coherent detector front-end is composed of two blocks:

- Coherent detector front-end
- Calibrated front-end sampling

A block diagram for the calibrated coherent detection front-end is shown in Figure 185A–2. The coherent detector front-end generates four electrical signals representing the in-phase (I) and quadrature (Q) components of the baseband of two orthogonal polarizations of the optical input signal. The front-end calibrated sampling performs analog-to-digital conversion and reduces any inherent imperfections of the coherent front-end, such as I-Q mismatches and polarization mismatches, to be acceptably low. Post-calibration residual parameters for the calibrated coherent detector front-end are listed in Table 185A–2. The values assigned to these parameters are defined by the Physical Layer specification that invokes the method.

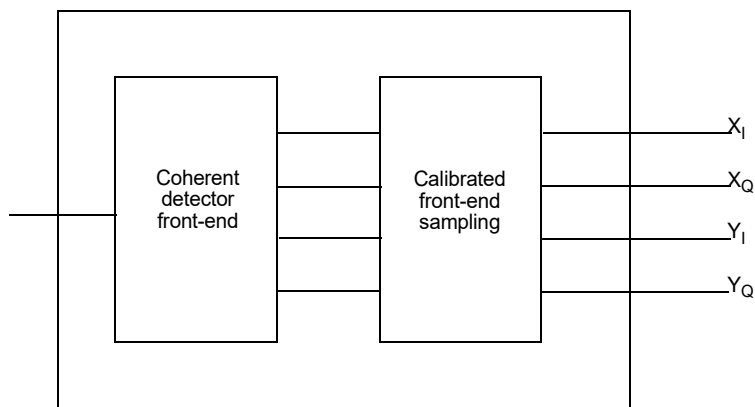


Figure 185A–2—Calibrated coherent detector front-end

185A.2.2.1.1 Coherent detector front-end

The coherent detector front-end is shown in Figure 185A–3. The incoming optical signal is split into two orthogonal polarizations (X and Y) by a polarization beam splitter (PBS). The two polarization components are mixed with their copolarized components from the optical local oscillator (LO) in two 90-degree optical hybrids, followed by four balanced photo-detectors (PD_k), which convert the four components of the optical signals into their corresponding electric signals. Parameters for the coherent detector front-end are listed in Table 185A–1. The values assigned to these parameters are defined by the Physical Layer specification that invokes the method.

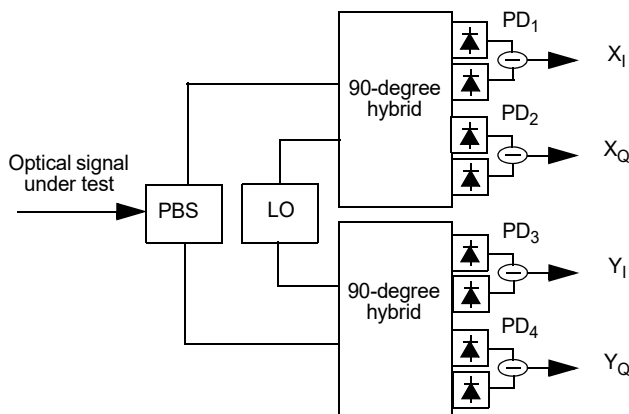


Figure 185A-3—Coherent detector front-end

Table 185A-1—Coherent detector front-end parameters

Description	Unit
3 dB bandwidth (min)	GHz
Effective number of bits (<i>ENOB</i>)	bits
Oversampling ratio	—
Local oscillator linewidth (max)	kHz
BER_{ref}	—

185A.2.2.1.2 Calibrated front-end sampling

The coherent calibrated sampling performs analog-to-digital conversion of the four electric signals obtained by the coherent detector front-end and mitigates the imperfections of the front-end as listed in Table 185A-2. The values assigned to these parameters are defined by the Physical Layer specification that invokes the method.

Table 185A-2—Coherent front-end post calibration residuals

Description	Unit
X-Y gain error (max)	dB
Skew between X-Y polarizations (max)	ps
I-Q phase error for X (max)	degree

Table 185A–2—Coherent front-end post calibration residuals (*continued*)

Description	Unit
I-Q gain error for X (max)	dB
I-Q skew for X (max)	dB
I-Q phase error for Y (max)	degree
I-Q gain error for Y (max)	dB
I-Q skew for Y (max)	dB
Carrier frequency offset (max)	GHz

185A.2.3 Digital signal processing

A block diagram for the offline digital signal processing is shown in Figure 185A–4. The offline digital signal processing recovers the information bits carried by the optical signal from the four digitized data streams representing the I/Q components of the baseband of two orthogonal polarizations of the optical input signal, X_I , X_Q , Y_I , and Y_Q . The BER is obtained by comparing the received information bits with the original bits sent by the transmitter, which are specified and known. This processing is done in a series of steps described in 185A.2.3.1 through 185A.2.3.7. This digital signal processing is then used in combination with virtual digital noise loading for ETCC calculation, which is described in 185A.2.5.

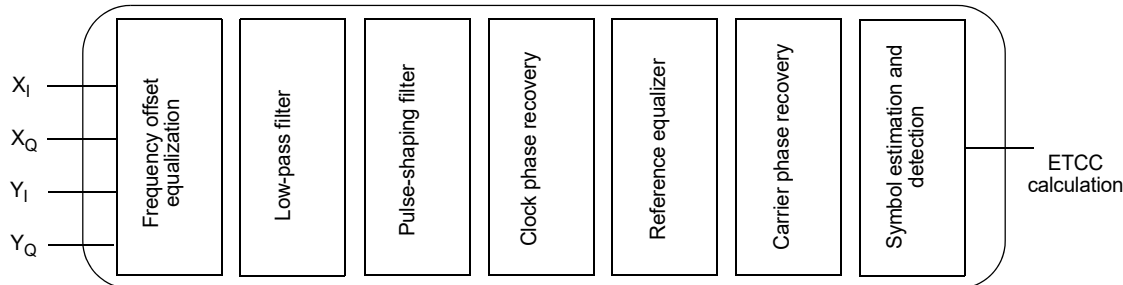


Figure 185A–4— Offline digital signal processing used for ETCC calculation

185A.2.3.1 Frequency offset equalization

A preconditioning procedure is used to perform resampling to 2 samples per symbol and frequency offset equalization of the test signal.

185A.2.3.2 Low-pass filter

A low-pass filter is used to emulate the expected bandwidth limitation of the calibrated coherent receiver front-end. In this specification, the low-pass filter is a 4th order Bessel-Thomson filter with a 3 dB bandwidth equal to 0.55 times the signaling rate, ± 1 GHz.

185A.2.3.3 Pulse-shaping filter

A predetermined pulse-shaping filter is used to perform preliminary matched filtering attempting to match a typical transmitter filter. In this specification, the pulse-shaping filter is a root-raised-cosine filter with a roll-off factor of 0.1.

185A.2.3.4 Clock phase recovery

A clock phase recovery algorithm is used to correct the timing of the oversampled signal.

185A.2.3.5 Reference equalizer

A reference equalizer based on a 2×2 multiple-in multiple-out (MIMO) filter with an adaptive T-spaced feed-forward equalizer, where T is the symbol period, is used for polarization demultiplexing and channel equalization.

185A.2.3.6 Carrier phase recovery

A carrier phase recovery algorithm is used to recover the phase of each equalized signal.

185A.2.3.7 Symbol estimation and detection

Symbol estimation and detection are performed to recover the information bits carried by the optical signal.

185A.2.4 Definition of optical parameters for the test systems

185A.2.4.1 Effective number of bits (ENOB)

ENOB is a measure of the SNR of a real (physical) ADC. ENOB defines the number of bits that a perfect (ideal) ADC would require to produce the same SNR as that of the coherent detector front-end ADC.

185A.2.4.2 Oversampling ratio

The oversampling ratio is defined as the ratio by which the ADC sampling rate exceeds the symbol rate.

185A.2.4.3 Local oscillator linewidth

The local oscillator linewidth defines the full width half maximum spectral width of the receiver local oscillator, following the definition in Recommendation ITU-T G.698.2 (11/2018) section 7.2.8.

185A.2.4.4 X-Y gain error

The X-Y gain error specifies the gain difference in dB between the X and Y polarizations.

185A.2.4.5 Skew between X-Y polarizations

The skew between X and Y polarizations is defined in 185.8.5.

185A.2.4.6 I-Q phase error

The maximum I-Q phase error is the largest phase difference of the in-phase component I and quadrature component Q of the signal, measured relative to local oscillator.

185A.2.4.7 I-Q gain error

The I-Q gain error specifies the gain difference in dB between the I and Q phases, for each polarization.

185A.2.4.8 I-Q skew

The maximum I-Q skew is the maximum relative skew between the in-phase component I and quadrature component Q of the signal.

185A.2.4.9 Carrier frequency offset

The carrier frequency offset specifies the offset between the local oscillator frequency and the central carrier frequency of the received signal from the transmitter under test.

185A.2.5 ETCC calculation

A flow chart of the ETCC calculation steps is shown in Figure 185A–5.

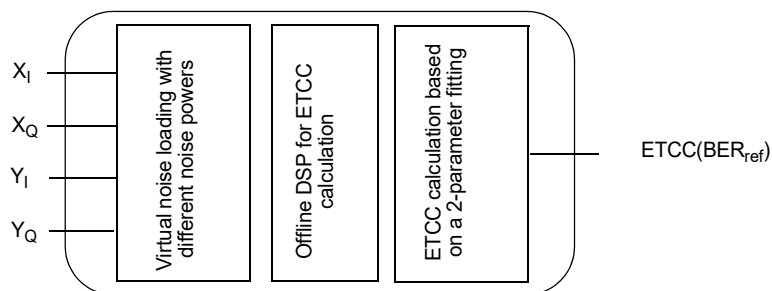


Figure 185A–5— ETCC calculation flowchart

NOTE—The ETCC calculation also provides the informative values EC_{tx} , NSR_{tx} in dB and $ETCC(BER_{ref}/10)$.

The parameter BER_{ref} represents the expected worst case BER at the input to the Inner FEC for 800GBASE-LR1 or the input to the PCS for 800GBASE-ER1 and 800GBASE-ER1-20. The BER_{ref} value is specified by the Physical Layer specification that invokes this method.

185A.2.5.1 Parameter definitions

The following parameters are used in the equations for ETCC derivation and ETCC calculation:

a	is the slope of the derived linear equation
b	is the y-intercept of the derived linear equation
EC_{rx}	is the eye-closure term representing signal loss due to receiver imperfections
EC_{tx}	is the eye-closure term representing signal loss due to transmitter imperfections
EC_{trx}	is the eye-closure term representing signal loss due to transmitter and receiver imperfections
$ENSR_{ref}$	is the theoretical noise-to-signal ratio at BER_{ref} for the modulation format being used

$ESNR$	is the signal-to-noise ratio for the modulation format used at the FEC input	1
$ESNR_{ref}$	is the theoretical signal-to-noise ratio at the BER_{ref} for the modulation format and FEC being used	2
		3
N_{ase}	is the amplified spontaneous emission (ASE) noise power or the non-transmitter related noise power in the signal Nyquist bandwidth	4
		5
N_{trx}	is the noise including contributions from the transmitter and receiver impairments and physical noise sources	6
		7
$N_{vase,i}$	is the virtual ASE noise power of each increment	8
		9
NSR_{ase}	is the ASE noise-to-signal ratio	10
		11
NSR_{rx}	is the intrinsic front-end noise power	12
NSR_{tx}	is the transmitter noise-to-signal ratio including contributions from the transmitter imperfections and physical noise sources	13
		14
NSR_{trx}	is the transmitter and receiver noise-to-signal ratio including contributions from impairments and physical noise sources	15
		16
$RSNR_{ase}$	is the required SNR to meet a specific BER of a device in the presence of virtual ASE	17
		18
$RSNR_{ase,tx}$	is the required SNR to meet a specific BER of the transmitter in the presence of ASE	19
		20
$RSNR_{trx}$	is the required SNR to meet a specific BER of the transmitter and receiver	21
		22
$\Delta RSNR_{tx}$	is the required signal-to-noise ratio penalty due to transmitter impairments or ETCC	23
		24
S	is the signal power of the captured dual-polarization waveform	25

185A.2.5.2 ETCC derivation

ETCC calculation is based on BER and digital noise loading and the process is described in the following steps.

The estimated signal to noise ratio (ESNR) for a signal is related to its eye-closure and noise terms according to Equation (185A-1).

$$ESNR = \frac{EC_{tx}^{-1} S}{N_{ase} + N_{trx}} = \frac{EC_{tx}^{-1}}{NSR_{ase} + NSR_{tx}} \quad (185A-1)$$

The required signal to noise ratio (RSNR) is related to the $ESNR_{ref}$ according to Equation (185A-2).

$$RSNR_{ase} = ((EC_{tx} \cdot ESNR_{ref})^{-1} - NSR_{tx})^{-1} \quad (185A-2)$$

For an ideal device where $NSR_{tx} = 0$ and $EC_{tx} = 1$, the theoretical RSNR is equal to the reference ESNR, $ESNR_{ref}$.

$\Delta RSNR_{trx}$ is related to the eye closure, RSNR, and NSR_{tx} by Equation (185A-3) expressed in dB.

$$\Delta RSNR_{trx} = 10 \times \log_{10} \left(\frac{RSNR_{ase}(EC_{tx} \cdot NSR_{tx})}{ESNR_{ref}} \right) \quad (185A-3)$$

$\Delta RSNR_{trx}$ includes contributions from both the transmitter and receiver. ETCC is defined by the transmitter contribution $\Delta RSNR_{tx}$. The contributions from only the transmitter are defined in Equation (185A-4) and Equation (185A-5) expressed in dB.

$$RSNR_{ase,tx} = ((EC_{tx} \cdot ESNR_{ref})^{-1} - NSR_{tx})^{-1} \quad (185A-4)$$

$$ETCC = \Delta RSNR_{tx} = 10 \log 10 \left(\frac{RSNR_{ase,tx}}{ESNR_{ref}} \right) \quad (185A-5)$$

EC_{tx} and NSR_{tx} are measured using a noise loading procedure based on captured waveforms as waveforms as described in Figure 185A-5.

A set of data relating $ESNR$ to NSR_{ase} is created and used to derive a linear fit with parameters a and b according to Equation (185A-6). From the derived linear fit parameters, $EC_{tx} = a$ and $NSR_{tx} = b / a$.

$$ENSR = EC_{tx} NSR_{ase} + EC_{tx} NSR_{tx} = a NSR_{ase} + b \quad (185A-6)$$

185A.2.5.3 ETCC calculation

ETCC is calculated using the following steps.

A reference receiver and a real-time sampling oscilloscope are used to acquire X_i , X_q , Y_i and Y_q digital waveforms as detailed in 185A.2.2.

The sampled waveforms are processed using the reference receiver DSP algorithm described in 185A.2.3 to estimate the BER with no added noise power, BER_0 , of the preconditioned test waveform from a given transmitter under test.

Add incremental, controlled amounts of additive white Gaussian noise (AWGN) with virtual amplified spontaneous emission noise power of each increment, $N_{vase,i}$, to the transmitter under test waveform and repeat the processing to the estimate BER_i of each increment. Repeat the increments a minimum of 10 times with small enough noise increments such that BER_i is less than BER_{ref} .

For each BER_i calculate $NSR_{vase,i}$ and $ENSR_i$ using Equation (185A-7) and Equation (185A-8).

$$NSR_{vase,i} = \frac{N_{vase,i}}{S} \quad (185A-7)$$

A set of data relating $ESNR_i$ to $NSR_{vase,i}$ is created and used to derive a linear fit with parameters a and b according to Equation (185A-8).

$$ENSR_i = a \times NSR_{vase,i} + b \quad (185A-8)$$

As a result of the fit $EC_{tx} = a$ and $NSR_{tx} = b / EC_{tx}$.

Determine the intrinsic receiver noise power NSR_{rx} and EC_{rx} of the calibrated coherent detector front-end via a measurement or calibration process using a known transmitter.

Determine the NSR_{tx} using Equation (185A-9) and the EC_{tx} using Equation (185A-10).

$$NSR_{tx} = NSR_{trx} - NSR_{rx} \quad (185A-9)$$

$$EC_{tx} = \frac{EC_{trx}}{EC_{rx}} \quad (185A-10)$$

Calculate the $RSNR_{\text{ase,tx}}$ using Equation (185A–4).

Calculate ETCC using Equation (185A–5).

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Insert new Annex 186A corresponding to Clause 186 as follows:

Annex 186A

(informative)

800GBASE-ER1 FEC and PMA test vectors

This annex provides test vectors of the 800GBASE-ER1 FEC transmit function, including data encoding, mapping, and FEC encoding defined in 186.2.3, and 800GBASE-ER1 PMA transmit function defined in 186.3.3.

Editor's note: A contribution proposing test vectors for the 800GBASE-ER1 PCS and PMA transmit functions is encouraged.